

AERIS-10

Pulsed-LFM X-band phased-array radar

Complete Engineering & Assembly Manual

Antidrone Ukraine · antidrone.cc

Edition 2026-10-09 — BETA (reconstruction, proposed designs and beta implementations; not hardware-verified)

Repository: https://github.com/herasymyuks/PLFM_RADAR-aeris · Upstream project files: NawfalMotii79/PLFM_RADAR (ORIGINAL PROJECT FILE where labelled)

AERIS-10 — Complete Engineering & Assembly Manual

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Built 2026-10-09 by tools/build_manual.py from manual/chapters (order: manual/00_OUTLINE.md). Status: BETA documentation — no item is hardware-verified; see the status label on every figure and procedure.

<!-- chapter 0: Title, status statement, how to read, revision table -->

AERIS-10 — Complete Engineering & Assembly Manual

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Status summary: this chapter is editorial (BETA documentation edition). It introduces a manual in which no item is VERIFIED on hardware; every figure and procedure carries one of the labels defined in section 4 below.

Sources: docs/AERIS10_ENGINEERING_BUILD_MANUAL.md (header, v1.3), docs/AERIS10_BETA_ENGINEERING_REPORT.md §1–2, MANUAL_BUILD_SPEC.md, manual/STYLE_GUIDE.md, manual/00_OUTLINE.md, README.md.

Planned figures: none.

1. Title and authorship

Field	Value
Title	AERIS-10 — Pulsed-LFM X-band phased-array radar — Complete Engineering & Assembly Manual
Author (compilation, generated drawings, reconstruction, proposed designs)	Antidrone Ukraine · antidrone.cc
Edition	2026-10-09 — BETA
Compiled from	repository PLFM_RADAR (branch main), reconstruction documents docs/, engineering package engineering/, BETA implementations beta/
Upstream project files	NawfalMotii79/PLFM_RADAR, labelled ORIGINAL PROJECT FILE wherever they appear (source: README.md badges; MANUAL_BUILD_SPEC.md §AUTHORSHIP)
Build	python3 tools/build_manual.py concatenates manual/chapters/*.md in the order of manual/00_OUTLINE.md and writes manual/AERIS10_MANUAL.md, manual/build/AERIS10_MANUAL.html and manual/build/AERIS10_MANUAL.pdf

The manual's authorship covers the compilation, the generated drawings (title blocks carry the `ATtribution` line of `tools/eagle_svg_common.py`), the reconstruction work and the proposed designs. Original upstream files (EAGLE schematics and boards, firmware sources, GUI scripts, photographs, the draw.io block diagram) keep their own authorship and are labelled ORIGINAL PROJECT FILE in captions and tables (source: manual/STYLE_GUIDE.md, rule "Authorship").

2. Revision table

Edition	Date	Author	Scope	Status
2026-10-09	2026-10-09	Antidrone Ukraine · antidrone.cc	First complete edition: system description, theory of operation, architecture and interfaces, four PCBs, antenna, host link and 22 V supply, mechanical design, FPGA/STM32/GUI, manufacturing, assembly and bring-up procedures, open-issue registers, appendices	BETA — not hardware-verified

Lineage of the source documents this edition compiles (each is cited where used):

Source document	Version / date	Content
docs/AERIS10_ENGINEERING_BUILD_MANUAL.md	v1.3, 2026-10-09 (v1.0 2026-10-08; v1.1 drawings package; v1.2 proposed designs; v1.3 BETA)	reconstruction, build preparation and documentation manual, Parts I–XIII
docs/AERIS10_BETA_ENGINEERING_REPORT.md	v0.9 draft, 2026-10-09	consolidated report of everything built, simulated and calculated in the BETA phase, with executed evidence
engineering/DRAWING_REGISTER.md	2026-10-09	89 registered drawings (39 SOURCE-DERIVED, 28 PARTIAL, 16 PROPOSED, 1 CONCEPTUAL, 5 BLOCKED); file check 333 OK, 0 problems
engineering/DESIGN/00_DESIGN_BASIS.md	DSN-00 Rev A, 2026-10-09	design basis and decision log D-01...D-15 for the proposed designs
beta/fpga, beta/stm32, beta/gui, beta/pcb READMEs and CHANGELOGs	2026-10-09	BETA implementations (build/simulate/test on a workstation only)
Upstream repository content	dated 2026-03-14 (source: docs/AERIS10_ENGINEERING_BUILD_MANUAL.md header)	schematics, layouts, firmware, GUI, physics notes, photographs

3. Status statement

The reconstruction manual summarises the repository state in one sentence: "the repository is a documented design study with schematics for four boards, partially routed layouts, non-building FPGA RTL, non-building STM32 firmware, one offline-runnable Python demo, and no mechanical design. No subsystem can be built reproducibly today" (source: docs/AERIS10_ENGINEERING_BUILD_MANUAL.md, header, "Readiness in one sentence"). The BETA phase changed the software state on the authoring workstation but not the hardware state; the executed evidence per area is (source: docs/AERIS10_BETA_ENGINEERING_REPORT.md §1, copied):

AERIS-10 — Engineering & Assembly Manual

Area	State before	State now	Executed evidence
FPGA RTL	did not parse; 5 modules/IP missing; 400 MHz fabric; no host path	parses/lints; all modules present; ISERDES 1:4 capture + 100 MHz polyphase DDC (bit-exact vs legacy); SPI host bridge; register map	beta/fpga/build.sh: 0 failures, 9 testbench runs PASS
STM32 firmware	no build system/HAL; 7+ defects; empty ADAR1000 phase tables	builds and links (93 KB flash); 12 defects fixed; ADAR1000 tables from the datasheet; bridge + REG commands	beta/stm32/build.sh exit 0; host tests 6/6
Python GUI	V6 stub; no tests; wrong packet format	package with firmware-exact protocol, bridge-frame path, register panel, simulator; PyInstaller app	pytest 72 passed; selftests exit 0
PCBs	no Gerbers; Main 15 / Power 308 unconnected; 0 MPN	Main/RF PA 0 unconnected; Power 89 (listed); BOM MPNs; fab notes; full packages; Main rev. B with FT601	KiCad DRC reports, export logs
Host data path	FT601 unwired	option A pin plan (bank 35) + option B SPI bridge implemented end-to-end	tb PASS, firmware build, GUI tests on the RTL vector
Antenna	none	16×8 patch panel, KiCad board, openEMS simulated	S11 −18 dB at f0, 128 MHz band, coupling −20.7 dB
Thermal / power	none	drain-gating requirement, 22 V module design, 2-D plate map	64 °C plate / ≈ 73 °C PA base in case B
Mechanics	none	head + pedestal, 51-part detail model, flat patterns, parts list; torque check	FreeCAD model builds; register file check 0 problems
Radar performance	TBD everywhere	range equation with simulated gain	R(1 m²) ≈ 6 km at B = 50 MHz

What remains impossible on the authoring machine, and therefore is not claimed anywhere in this manual: Vivado synthesis, timing and bitstream generation; flashing and bench tests; PCB fabrication; measurement of component heights; antenna coupon on a VNA; PA thermal test; the 89 Power Board pour connections (manual CAD); EAGLE schematic updates for rev. B; the owner decisions K1–K8 and D-01...D-19 (source: docs/AERIS10_BETA_ENGINEERING_REPORT.md §2).

4. How to read the status labels

Every figure caption and every procedure title carries exactly one of the following labels (source: manual/STYLE_GUIDE.md , "Status vocabulary"; MANUAL_BUILD_SPEC.md , "HARD RULES"). Nothing in this manual is promoted above the status recorded in its source register.

Label	Meaning	Typical content
ORIGINAL PROJECT FILE	as found in the upstream repository; authorship stays with the upstream project; photographs carry no dimensions	8_Utils/*.jpg , 2_Functional Diagram & Interconnection Matrices/RADAR_V6.jpg , EAGLE .sch/.brd , 9_Firmware/
SOURCE-DERIVED	generated from the native design files by a repository tool; some elements may be unverifiable and are listed	schematic renders, KiCad-converted board plots, system diagrams SYS-01...04, netlists
PARTIAL	significant parts missing from the sources	FPGA pipeline SD-02, STM32 diagrams SD-03/04, end-to-end flow SD-07
CONCEPTUAL	documented intent only, no geometry or wiring behind it	exploded view ASM-EXP-01, antenna and host blocks in SYS-01
PROPOSED DESIGN	new engineering content that depends on logged decisions D-xx	antenna panel, head and pedestal, 22 V supply, harness schedule, host link
BETA	builds, simulates or tests on a workstation; not synthesised for the target, not flashed, not on hardware	beta/fpga , beta/stm32 , beta/gui , beta/pcb
BLOCKED — MISSING DATA	cannot be produced from the repository	enclosure dimensions, component heights, cable lengths (G-01...G-12)
VERIFIED	reserved for hardware- or vendor-confirmed items; currently unused in the whole project	—

Supplementary vocabularies used inside copied tables keep the meaning of their source: CONFIRMED / UNVERIFIED / MISSING SPEC (interconnection and power-rail registers, source: engineering/SYSTEM/interfaces/interconnection_table.md), DONE / OPEN / BLOCKED (task tracker, source: docs/04_RECOVERY_TASKS.md), MET / NOT MET / NOT RUN / PARTIALLY MET (acceptance criteria, source: docs/TESTING/ACCEPTANCE_CRITERIA.md), ASSUMED / ESTIMATE (numbers, source: manual/STYLE_GUIDE.md). Citations are inline, in the form (source: path:line) or (source: path §section) ; a table copied from a source states the source once in its caption or lead-in. Where a value is not available in any repository file the text says "UNKNOWN" and names the register item that tracks it.

Identifier families used for cross-references (defined in Appendix D): K1–K8 configuration conflicts, D-01...D-19 design decisions, G-01...G-12 geometry gaps, MDR-01...13 missing-drawing recovery guides, DSN- *proposed-design drawings*, SYS-/SD-/ELEC-/MECH-/ASM-/PCB-registered drawings, CBL- *cables*, AC- acceptance criteria, R- *recovery tasks*, C1–C7 *firmware conflicts*, MAN- observations raised by this manual.

5. Safety

Three hazards are present in the design as documented and are repeated before the first procedure step that needs them (source: manual/STYLE_GUIDE.md , rule "Procedures"): the 22 V PA drain supply (QPA2962 bias-up sequence VG −4 V → VD +22 V, 2 A per board, 16 boards; source: engineering/ELECTRICAL/power_distribution/power_rails.md §1 row +22V0 / VD), RF radiation from the array (up to 16 × 10 W peak for the Extended variant; source: README.md , Technical Specifications), and the rotating pedestal (head mass estimate 10.4 kg, 50 azimuth positions per revolution; source: engineering/DESIGN/CALCS/DESIGN_CALCULATIONS.md §2).

6. Upstream acknowledgement

The AERIS-10 project was published by NawfalMotii79 as PLFM_RADAR under an MIT licence badge (source: README.md, lines 3–6). The upstream repository supplies the four EAGLE schematic/board sets, the FPGA RTL, the STM32 sources, the Python GUI versions, the physics and hardware notes under 00_notation/, 01_physics/, 02_hardware/, the power-management spreadsheet, the draw.io block diagram and the photographs. These files are reproduced or rendered in this manual as ORIGINAL PROJECT FILE and were not modified (source: MANUAL_BUILD_SPEC.md, "Preserve originals"). No LICENSE file exists in the repository root; acceptance criterion AC-D4 records that the licence file consistent with the badge is NOT MET (source: docs/TESTING/ACCEPTANCE_CRITERIA.md, row AC-D4).

7. Document map

Chapter order equals the concatenation order of manual/00_OUTLINE.md (copied):

#	Chapter	Content	Status of content
0	Front matter	title, revision table, status statement, labels, acknowledgement, map	editorial
1	System overview	pulsed-LFM X-band phased array, variants Nexus/Extended, documented vs calculated performance, conflicts K1–K8	SOURCE-DERIVED + PROPOSED (calculations)
2	Theory of operation	waveform and timing, pulse compression, Doppler, beamforming, FPGA chain, data path	ORIGINAL PROJECT FILE (physics notes) + PARTIAL (RTL diagrams) + BETA
3	Architecture and interfaces	boards, connectors, cables, power rails, enable sequence, harness	SOURCE-DERIVED + PROPOSED (harness)
4	Main Board	schematic set, layout, BOM, DRC state, rev. B host interface	SOURCE-DERIVED + BETA
5	Power Supply Board	same pattern	SOURCE-DERIVED + BETA
6	Frequency Synthesizer	same pattern	SOURCE-DERIVED + BETA
7	RF PA board (×16)	same pattern + thermal	SOURCE-DERIVED + BETA + PROPOSED
8	Antenna	proposed 16×8 patch panel, simulation, fabrication data	PROPOSED DESIGN
9	Host link and 22 V supply	FPGA→host options A/B, 22 V PA supply module	PROPOSED DESIGN + BETA
10	Mechanical design	head and pedestal, drawings, parts, flat patterns, thermal map, torque	PROPOSED DESIGN
11	FPGA firmware	hierarchy, build, constraints, tests, synthesis status	BETA
12	STM32 firmware	architecture, build, defects fixed, sequencing, USB protocol	BETA
13	GUI software	install, run, protocol, register panel, packaging	BETA
14	Manufacturing	packages per board, stack-ups, BOMs, fab checklist	SOURCE-DERIVED + BETA
15	Assembly procedure	numbered steps with checks	PROPOSED DESIGN (geometry)
16	Bring-up and test	power-up, flashing, calibration, bench tests, acceptance	procedures OPEN, not executed
17	Open issues and decisions	K, D, G, MDR, R registers	registers
A	Drawing register	copy of engineering/DRAWING_REGISTER.md	register
B	Parts and cables	BOM summaries, mechanical parts, cable schedule	SOURCE-DERIVED + PROPOSED
C	Tooling	generators and checkers under tools/	documentation
D	Glossary	acronyms, symbols, status vocabulary, ID families	documentation

<!-- chapter 1: What AERIS-10 is: pulsed-LFM X-band phased array, variants Nexus/Extended, performance targets vs calculated -->

System overview — what AERIS-10 is

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Status summary: system description SOURCE-DERIVED (reconstructed from the four EAGLE schematics, the firmware and the RTL); variant table and performance targets ORIGINAL PROJECT FILE (README claims, not verified); calculated performance PROPOSED DESIGN / BETA (DSN-CALC-01, with stated assumptions); figures F1.1 SOURCE-DERIVED, F1.2 and F1.3 ORIGINAL PROJECT FILE. Eight configuration conflicts K1–K8 between the CAD and the documentation remain open.

Sources: README.md, 00_notation/parameter_table.md, docs/SYSTEM/BLOCK_DIAGRAM.md, docs/AERIS10_BETA_ENGINEERING_REPORT.md, engineering/DESIGN/CALCS/DESIGN_CALCULATIONS.md §3, engineering/SYSTEM/architecture/README.md, 02_hardware/01_system_overview.md.

Planned figures: F1.1 system block diagram (SYS-01), F1.2 original draw.io block diagram, F1.3 prototype photographs.

1. Purpose and principle

AERIS-10 is described by its upstream README as "an open-source, low-cost 10.5 GHz phased array radar system featuring Pulse Linear Frequency Modulated (LFM) modulation", available in two versions (3 km and 20 km range) for researchers, drone developers and SDR enthusiasts (source: README.md, introduction). The radar transmits linear-frequency-modulated pulses (chirps) at X-band, steers the beam electronically in elevation with four ADAR1000 beamformer ICs across a 16-element linear array, rotates the head mechanically in azimuth with a stepper motor, and processes the echoes in an FPGA (down-conversion, decimation, pulse compression, Doppler FFT, detection) before sending results to a host PC running a Python GUI (source: README.md, "Processing Pipeline"; engineering/DESIGN/00_DESIGN_BASIS.md §1, row "Array").

The carrier frequency $f_c = 10.5$ GHz and wavelength $\lambda = 0.02857$ m are the canonical values (source: 00_notation/parameter_table.md, "Inconsistency Resolutions" §1; 9_Firmware/9_1_Microcontroller/9_1_3_C_Cpp_Code/main.cpp:1133). The pulse timing coded in the firmware is: long chirp 30 μ s at PRI 167 μ s, short chirp 0.5 μ s at PRI 175 μ s, guard 175.4 μ s, 32 chirps per beam position, 31 elevation positions, 50 azimuth positions per revolution (source: main.cpp:178–184, main.cpp:190; 00_notation/parameter_table.md, "Waveform and Timing"). The chirp bandwidth B is **TBD** in the parameter table ("Requires ADF4382 config"); every range-resolution and range-equation number in this manual that depends on B says so explicitly.

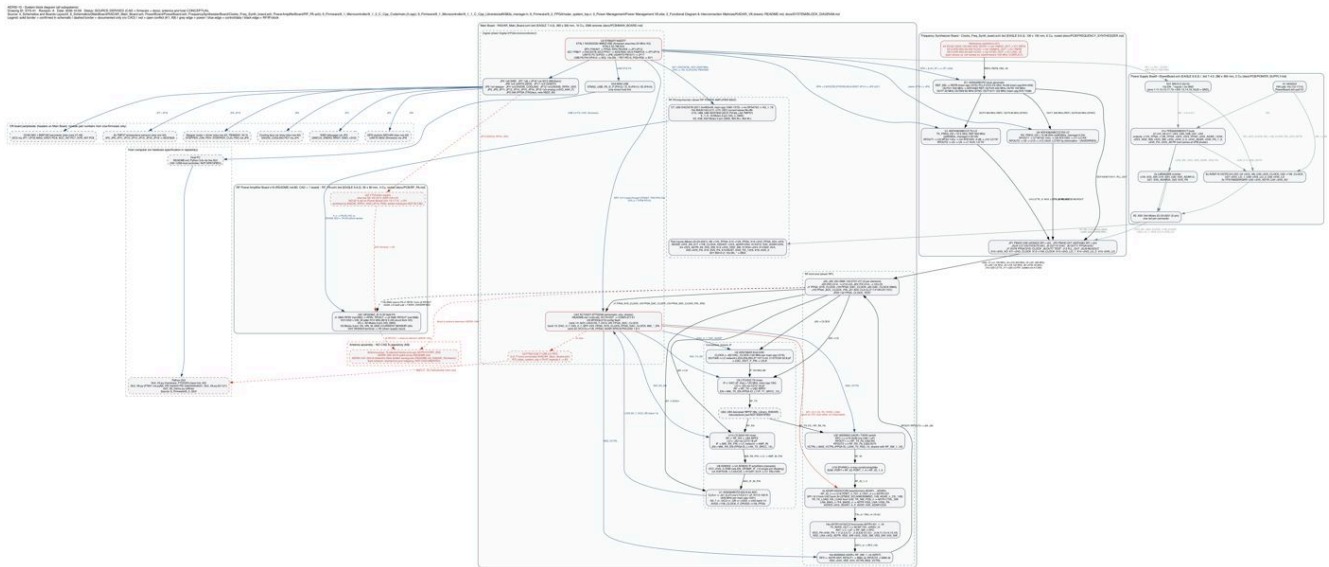


Figure 1 — F1.1 — System block diagram SYS-01, all subsystems with status legend — SOURCE-DERIVED; antenna, host PC, 22 V supply and off-board modules are CONCEPTUAL (source: engineering/SYSTEM/block_diagrams/system_block_diagram.dot; produced by Graphviz dot -Tpng -Gdpi=150)

Figure F1.1 shows what is wired in the CAD, not what the README describes; every box and edge is traceable to a repository file (source: engineering/SYSTEM/architecture/README.md, row SYS-01). The original designer's block diagram is reproduced as F1.2 for comparison; it labels the FPGA XC7A50T-2FTG256 and names the parts FPGA, AD9708, ADF4382 TX/RX, LTC5552 $\times 2$, LPF/BPF, AD9484, EP4RKU+, ADAR1000 $\times 4$, ADTR1107, M3SWA2-34DR+, FT601, STM32F746ZGT7, AD8352, OCXO ECOC-2522-10.000, AD9523, XO CCHD-957-100, VCXO CVHD-950-100.000, MTX2-143+, ATS1005-3DB and "QPA2862 $\times 4$ " (the CAD has QPA2962_B) (source: engineering/SYSTEM/architecture/README.md, row SYS-01, "RADAR_V6.drawio original block names").

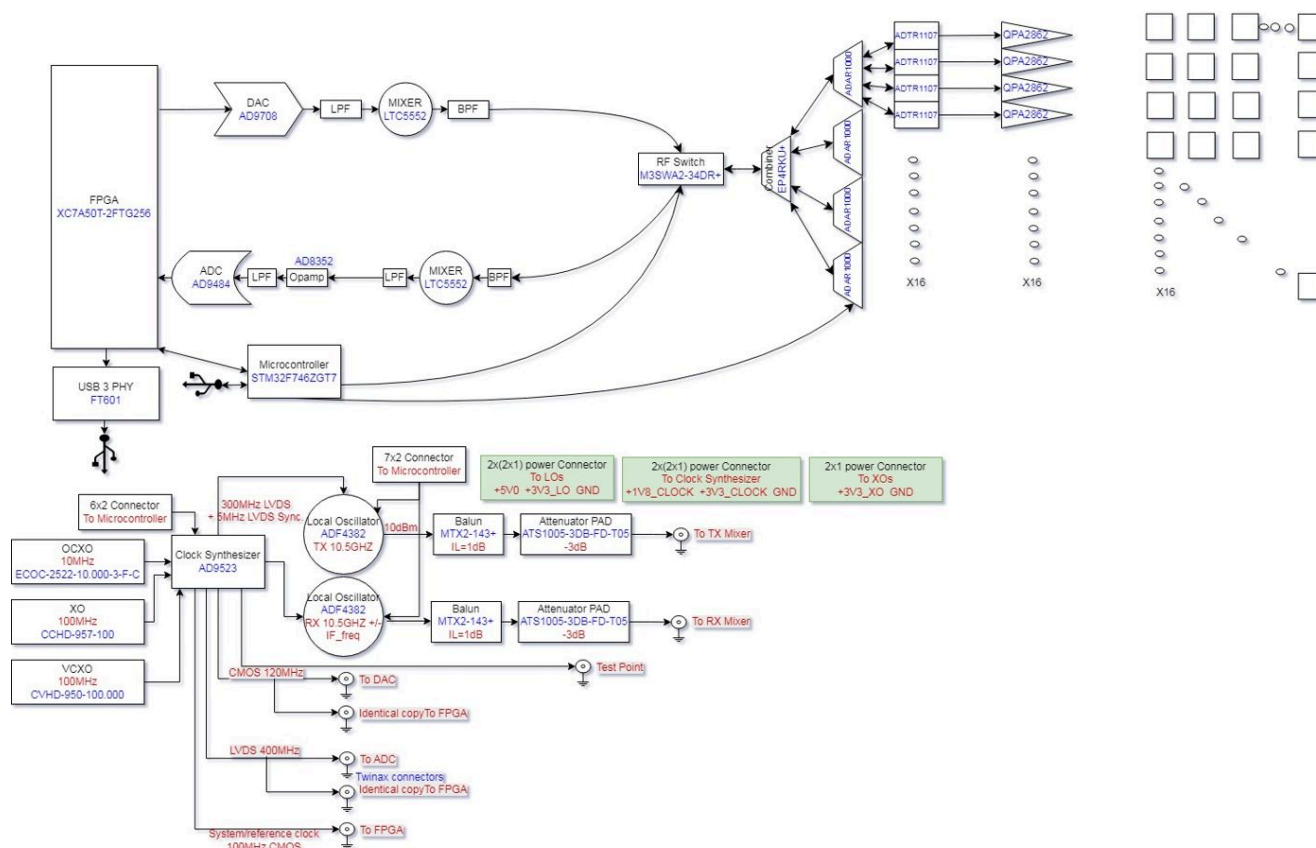


Figure 2 — F1.2 — Original functional block diagram RADAR_V6, draw.io 29.6.1 export — ORIGINAL PROJECT FILE, upstream authorship, not dimensioned (source: 2_Functional Diagram & Interconnection Matrices/RADAR_V6.jpg; as found)

2. Variants

The README specifies two variants (source: `README.md`, "Technical Specifications", copied verbatim):

Parameter	AERIS-10N (Nexus)	AERIS-10X (Extended)
Frequency	10.5 GHz	10.5 GHz
Max Range	3 km	20 km
Antenna	8x16 Patch Array	32x16 Slotted Waveguide
Beam Steering	Electronic ($\pm 45^\circ$)	Electronic ($\pm 45^\circ$)
Mechanical Scan	360° (stepper motor)	360° (stepper motor)
Output Power	~1Wx16	10Wx16 (GaN amplifier)
Processing	FPGA + STM32	FPGA + STM32

The hardware notes add the following per-variant differences (source: 02_hardware/01_system_overview.md §2.1, copied; gains are marked TBD in the source):

	Nexus (AERIS-10N)	Extended (AERIS-10X)
Transmit power P _t per element	1 W (ADTR1107)	10 W (QPA2962 GaN)
Antenna type	8x16 patch array	32x16 slotted waveguide
Antenna gain G	~20 dBi (patch, TBD)	~30 dBi (waveguide, TBD)
Detection range R _{max}	3 km	20 km

Three facts qualify this table. (1) The README names the PA-board variant "AERIS-10E" in one place and "AERIS-10X" elsewhere (source: README.md:79 vs README.md:24,83); this manual uses AERIS-10X. (2) The " $\pm 45^\circ$ " steering figure is a README statement; the firmware phase table spans -160° to $+160^\circ$ and the parameter table derives from it a steering range it states as "approximately $\pm 33^\circ$ " (source: 00_notation/parameter_table.md, "Inconsistency Resolutions" §4) — the arithmetic disagreement between the project's own documents is recorded as observation MAN-01 in chapter 17. (3) No antenna CAD exists for either variant (conflict K8); the only antenna geometry in the repository is the PROPOSED 16-row $\times$ 8-patch panel of engineering/DESIGN/ANTENNA/ (chapter 8).

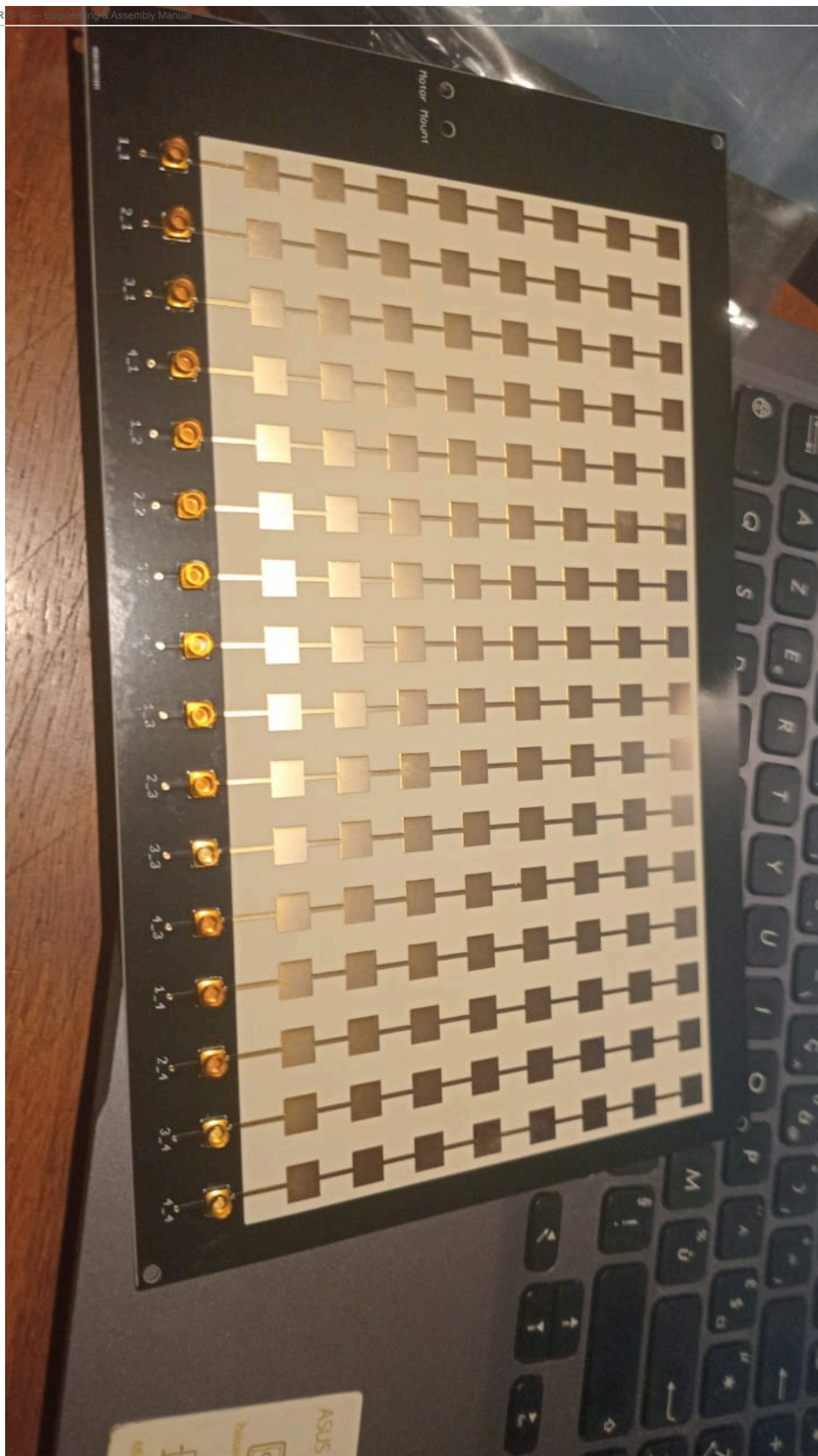


Figure 3 — F1.3 — Prototype antenna array photograph — ORIGINAL PROJECT FILE, undimensioned, not used for any dimension in this manual (source: 8_Utills/Antenna_Array.jpg; as found)

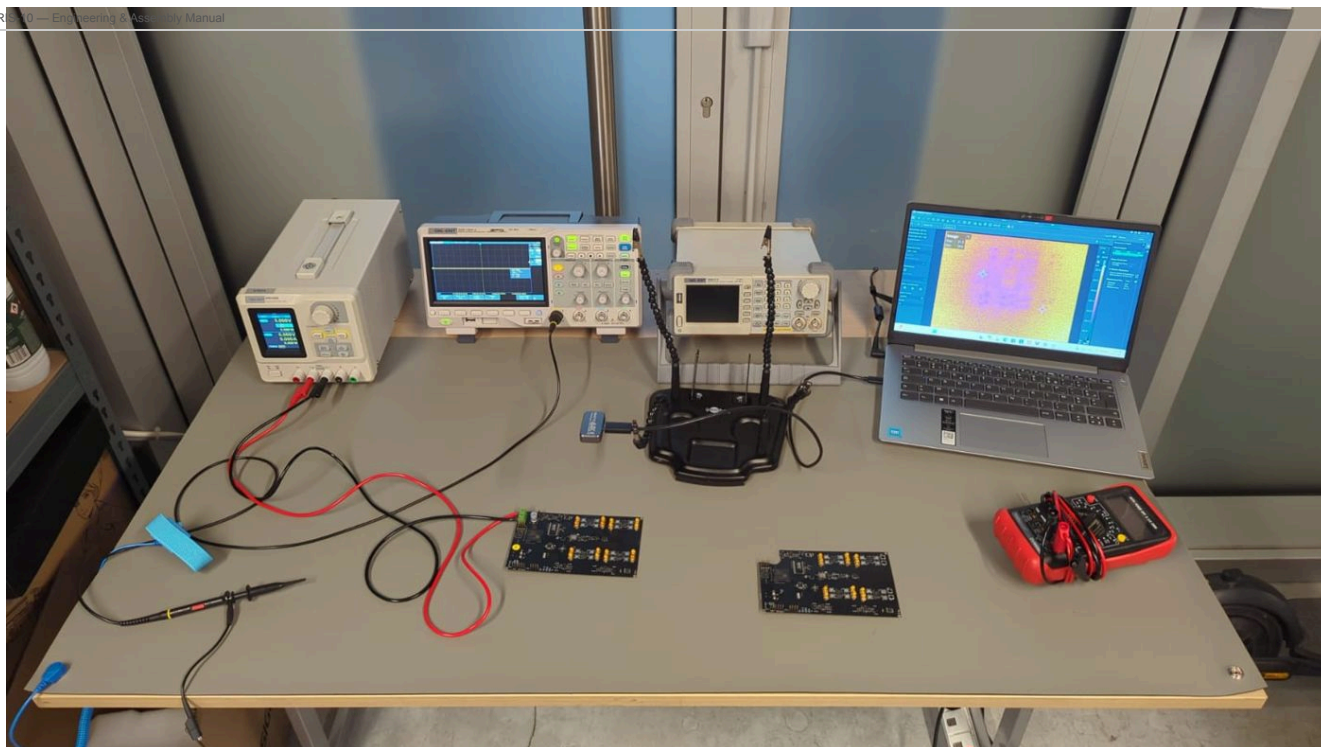


Figure 4 — F1.3b — Prototype electronics photograph — ORIGINAL PROJECT FILE, undimensioned, not used for any dimension in this manual (source: 8_Utils/0044.jpg; as found)

3. Subsystems as evidenced by the CAD

The system consists of four PCB types, an antenna, a pedestal and off-board modules. Board sizes and layer counts come from the EAGLE board files; part counts from the schematics (source: docs/SYSTEM/BLOCK_DIAGRAM.md §1, Mermaid diagram node labels; engineering/DESIGN/00_DESIGN_BASIS.md §1, rows "PA board", "Other boards"):

Subsystem	Size / layers	Key parts (reference designators)	CAD status	Chapter
Power Supply Board	280 × 300 mm, 2 layers, 8 holes	X1 AK300/2 VIN 12–17 V; 21 × TPS562208, 6 × ADM7151, 2 × TPS7A8300, 5 × LM2662; 34 rail outputs X2..X35; enable bus SV1	layout unfinished in the source (308 unconnected before BETA, 89 after)	5
Frequency Synthesizer Board	100 × 100 mm, 6 layers, 4 holes	X4 OCXO 100 MHz, X5/X6 VCXO 50 MHz (schematic values; K7); IC1 AD9523; U1 ADF4382 TX LO; U6 ADF4382 RX LO	complete; only board with a designer P&P/BOM export	6
Main Board	260 × 300 mm, 10 layers, 10 holes	U2 STM32F746ZGT7; U42 XC7A50T-2FTG256I (K1); U1 AD9484 8-bit 400 MSPS; U3 AD9708 8-bit DAC; U5/U13 LTC5552 mixers; 4 × ADAR1000; 16 × ADTR1107; 17 × M3SWA2-34DR+; U6 FT601 (0 of 77 pins connected, K3); X53 mini-USB	layout unfinished in the source (15 unconnected before BETA, 0 after)	4
RF PA board (×16, AERIS-10X only)	35 × 60 mm, 4 layers, 7 × Ø3.2 holes	U\$1 QPA2962 GaN 10 W; J1 RFIN / J2 RFOUT; X2 VG; X3 VD/VIN_M sense; 22V AK300/2 terminal	complete (1 clearance DRC item in the source)	7
Antenna array	—	8×16 patch (Nexus) or 32×16 slotted waveguide (Extended) per README	NO CAD (K8); PROPOSED panel exists	8
Off-board modules	—	GPS (UART5), GY-85 IMU + BMP180 (I2C3), 8 × TMP37 via ADS7830, stepper driver (PD4/PD5), fan relay (PD7), PA drain switch (PD6)	named only; no part numbers, no module schematics	3, 9
Host PC	—	Python GUI; USB 2.0 full-speed CDC on X53 is the only host data link in the CAD	GUI_V5 (hardware) / GUI_V6_Demo (offline) in the source; beta/gui package	13

The README's description of the processing pipeline (waveform generation in the DAC, up/down conversion in the LT5552 mixers, beam steering by the ADAR1000s, FPGA processing of ADC capture → I/Q down-conversion → CIC/FIR decimation → pulse compression → Doppler FFT → MTI/CFAR, STM32 system management, Python visualisation) is the design intent (source: README.md, "Processing Pipeline"). Chapter 2 states what the RTL as committed and the BETA RTL actually implement; the differences are material (no MTI, fixed-threshold "CFAR" placeholder, no wired FPGA→host path).

4. Documented versus verified: conflicts K1–K8

Independent cross-reading of the CAD, the firmware, the RTL, the GUI and the documentation produced eight configuration conflicts that only the hardware owner can close (source: docs/SYSTEM/BLOCK_DIAGRAM.md §4, copied verbatim):

#	Topic	A	B	Owner
K1	FPGA part	CAD/xlsx/drawio XC7A50T-2FTG256I	README/docs/XDC XC7A100T	hardware designer
K2	STM32 HSE	schematic 8 MHz crystal	firmware 25 MHz	hardware designer
K3	Host data path	CAD: STM32 USB-FS only	RTL/GUI V6: FT601 USB 3.0; GUI V2–V5: FT232H	system architect

#	Topic	A	B	Owner
K4	PA supply	xlsx 16 × 22 V / 2 A	Power Board Vin [12–17] V, no 22 V rail	power designer
K5	FPGA packet format	RTL 0xAA ... 0x55	GUI A5C3 ... CRC16	software
K6	ADF4382 pins	main.h PG6..PG15 (= schematic)	adf4382a_manager.h PG0..PG9	firmware
K7	Synth oscillators	schematic OCXO 100 MHz + 2 × VCXO 50 MHz	xlsx VCXO 100 MHz; firmware vcxo_freq = 100 MHz	RF designer
K8	Antenna	README patch 8×16 / slotted 32×16	two different waveguide simulations, no CAD	antenna designer

The BETA work adopted interim assumptions for several of them without closing them: the BETA FPGA project targets xc7a50tftg256–2, the schematic part (K1); the BETA firmware clock tree is written for the real 8 MHz crystal (K2); the host path is implemented as option B (SPI bridge through the STM32 CDC) with option A (FT601 on a Main Board rev. B) prepared as a pin plan (K3); a 22 V supply module DSN-PSU-01 is proposed (K4); the bridge frame replaces both packet formats (K5); the ADF4382 pin collision is fixed in the BETA firmware (K6); K7 and K8 remain open, the antenna being a PROPOSED patch panel per decision D-01 (source: docs/AERIS10_BETA_ENGINEERING_REPORT.md §4, §7, §8, §12; beta/fpga/README.md, "Remaining work" item 2). The full register with these interim positions is in chapter 17.

5. Performance targets versus calculated performance

The README targets (3 km Nexus, 20 km Extended) are not accompanied by a link budget in the upstream repository; the parameter table lists antenna gain, LNA noise figure and chirp bandwidth as TBD (source: 00_notation/parameter_table.md, "TBD Tracking"). The BETA phase produced a single-pulse coherent range-equation estimate for the long chirp, using the simulated gain of the proposed antenna panel and an **assumed** bandwidth (source: engineering/DESIGN/CALCS/DESIGN_CALCULATIONS.md §3, both tables copied verbatim):

Parameter	Value	Basis
f_o / λ	10.5 GHz / 28.57 mm	verified
TX power	16 × 10 W (PSAT) = 160 W peak, coherent	QPA2962 datasheet; array coherence assumed
Antenna gain (TX = RX)	22.6 dBi	one simulated row 11.5 dBi + 10·log10(16) – 1 dB feed/scan loss (proposed panel, D-01)
Chirp	T_c = 30 μs, B = 50 MHz ASSUMED (TBD in parameter_table.md) → pulse-compression gain 31.8 dB, ΔR = 3 m	firmware timing
Coherent integration	16 long chirps → +12.0 dB	main.cpp
Noise figure / system losses	4.0 dB / 6 dB	ADTR1107 2.5 dB NF + chain; losses assumed
Required SNR	13 dB (Pd ≈ 0.9, Pfa 1e-6, Swerling 1)	standard
Unambiguous range (PRI 167 μs)	25.1 km	firmware

Target RCS (m²)	R_max (km)
0.01	1.9
0.1	3.4
1.0	6.0
10.0	10.6

The source's verdict (source: engineering/DESIGN/CALCS/DESIGN_CALCULATIONS.md §3, "Verdict"): with the proposed antenna, 16 × 10 W and 16-chirp integration, R_max(1 m²) ≈ 6.0 km — the 20 km Extended-variant goal is **not** reached and needs ≈ 21 dB more (longer coherent integration across the 31 elevations/azimuth dwell, a narrower B with the same T_c, lower losses, or a higher-gain antenna). The 3 km Nexus goal is met for RCS ≥ 0.1 m² (R ≈ 3.4 km); a 0.01 m² drone-class target falls near 1.9 km. Dominant unknowns: B (assumed), real array gain (16-row coupling not yet simulated), RF chain losses.

Two caveats apply to reading this estimate. First, it combines the Extended variant's PA power (16 × 10 W) with the proposed patch panel's simulated gain, i.e. neither variant of the README exactly; the Nexus variant with ~1 W per element would be 10 dB lower in transmit power, which the source does not evaluate. Second, nothing in the estimate has been measured: the gain comes from an openEMS simulation of one row (source: docs/AERIS10_BETA_ENGINEERING_REPORT.md §8) and the noise figure and losses are assumptions stated in the table.

6. Reconstruction state at a glance

The project is being rebuilt in three layers that this manual keeps separate: the upstream originals (ORIGINAL PROJECT FILE), the SOURCE-DERIVED engineering package generated from them (engineering/, 89 registered drawings, file check 333 OK, 0 problems; source: engineering/DRAWING_REGISTER.md, "Summary" and "File check"), and the PROPOSED / BETA additions (engineering/DESIGN/, beta/). Recovery tasks are tracked in docs/04_RECOVERY_TASKS.md (status summary at its 2026-10-08 date: 7 DONE, 20 OPEN, 9 BLOCKED, plus the R-ENG, R-DSN and R-BETA groups added 2026-10-09; source: docs/04_RECOVERY_TASKS.md, "Status summary" and the three added sections). Acceptance criteria and their current MET / NOT MET / NOT RUN state are in docs/TESTING/ACCEPTANCE_CRITERIA.md; the physical criteria (AC-F9, AC-S6, AC-S7, AC-B7, AC-E7) are all NOT RUN (source: docs/TESTING/ACCEPTANCE_CRITERIA.md, sections A–G).

Theory of operation — waveform, pulse compression, Doppler, beamforming, processing chain

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Status summary: equations ORIGINAL PROJECT FILE (the upstream physics notes `01_physics/*.md`, transcribed to plain text; equation tags kept); firmware timing constants SOURCE-DERIVED (`main.cpp`); frequency plan SOURCE-DERIVED (firmware-programmed values, not measurements); FPGA processing chain as committed PARTIAL (SD-02: missing modules, placeholders) and as implemented in `beta/fpga` BETA (simulated, not synthesised); figures F2.1 SOURCE-DERIVED, F2.2 and F2.3 PARTIAL. The chirp bandwidth B is TBD in the parameter table; the matched-filter reference memories encode a 10→30 MHz baseband sweep whose relation to B is not established.

Sources: `01_physics/01_fmcw_theory.md`, `01_physics/02_lfm_waveform_model.md`, `01_physics/03_beamforming_theory.md`, `01_physics/04_detection_theory.md`, `01_physics/05_noise_analysis.md`, `01_physics/06_calibration_theory.md`, `02_hardware/04_antenna_beamforming.md`, `00_notation/parameter_table.md`, `engineering/SYSTEM/data_flow/signal_and_data_flow.dot`, `engineering/SOFTWARE_DIAGRAMS/FPGA/fpga_data_pipeline.dot`, `engineering/SOFTWARE_DIAGRAMS/DATA_FLOW/end_to_end_data_flow.dot`, `engineering/SYSTEM/interfaces/interconnection_table.md` §6, `beta/fpga/README.md`, `engineering/DESIGN/HOST_LINK/HOST_LINK_DESIGN.md` §2.

Planned figures: F2.1 signal and data flow (SYS-03), F2.2 FPGA pipeline as written in the original RTL (SD-02), F2.3 end-to-end data flow (SD-07).

Notation: equations are written in plain text because the manual build has no LaTeX renderer; symbols follow `00_notation/symbol_table.md` (Appendix D), equation tags follow `00_notation/conventions.md` §1 and refer to the display equation of that tag in the cited file. x^2 is a power, `sqrt()` a square root, *complex conjugation when written as $s(t)$* , $\cdot$ multiplication.

1. Waveform model

1.1 LFM chirp

The transmitted pulse is a linear-frequency-modulated chirp of duration T_c and bandwidth B. In complex baseband (source: `01_physics/02_lfm_waveform_model.md` §1, Eq. LFM-1, LFM-3, LFM-4):

- $s(t) = \text{rect}(t / T_c) \cdot \exp(j \cdot 2\pi \cdot (f_c \cdot t + (\mu/2) \cdot t^2))$ (LFM-1)
- chirp rate $\mu = B / T_c$ (LFM-3)
- instantaneous frequency $f_i(t) = f_c + \mu \cdot t$ (LFM-4)

The time-bandwidth product $TBP = B \cdot T_c$ (LFM-5) equals the pulse-compression gain (section 2) and the ratio of the two chirp modes' TBPs is $T_{c,1} / T_{c,2}$ (LFM-7) (source: `01_physics/02_lfm_waveform_model.md` §2).

1.2 Timing coded in the firmware

The firmware defines the following constants (source: `9_Firmware/9_1_Microcontroller/9_1_3_C_Cpp_Code/main.cpp:178–184,190,195`; symbols per `00_notation/parameter_table.md`, "Waveform and Timing", copied):

Parameter	Symbol	Value	Firmware variable	Source line
Chirps per beam position	M	32	<code>m_max</code>	<code>main.cpp:178</code>
Elevation positions	<code>N_el</code>	31	<code>n_max</code>	<code>main.cpp:179</code>
Long chirp duration	<code>T_c,1</code>	30 μs	<code>T1</code>	<code>main.cpp:180</code>
Long chirp PRI	<code>T_r,1</code>	167 μs	<code>PRI1</code>	<code>main.cpp:181</code>
Short chirp duration	<code>T_c,2</code>	0.5 μs	<code>T2</code>	<code>main.cpp:182</code>
Short chirp PRI	<code>T_r,2</code>	175 μs	<code>PRI2</code>	<code>main.cpp:183</code>
Guard time	<code>T_guard</code>	175.4 μs	<code>Guard</code>	<code>main.cpp:184</code>
Azimuth positions per revolution	<code>N_az</code>	50	<code>y_max</code>	<code>main.cpp:189</code>
IF frequency	<code>f_IF</code>	120 MHz	<code>IF_freq</code>	<code>main.cpp:190</code>
Stepper steps per revolution	—	200	<code>Stepper_steps</code>	<code>main.cpp:195</code>
Centre frequency / wavelength	<code>f_c / \lambda</code>	10.5 GHz / 0.02857 m	<code>wavelength</code>	<code>main.cpp:1133</code>
Chirp bandwidth	B	TBD	—	<code>00_notation/parameter_table.md</code> , "TBD Tracking"

The firmware comment at `main.cpp:186` states the per-position sequence: "m = N° of chirp/position = 16 (made of T1 and PRF1) + Guard = 175 μs + 16 (made of T2 and PRF2)", i.e. 16 long chirps, a guard interval, then 16 short chirps per beam position. The resulting beam-position frame time is 5647.4 μs (source: `engineering/DESIGN/HOST_LINK/HOST_LINK_DESIGN.md` §2, row "Beam-position frame time", basis `main.cpp:180–186`). The parameter table resolves the apparent PRF discrepancy between firmware and GUI: `PRI1 = 167  $\mu\text{s}$` is the chirp-

AERIS10_BETA_ENGINEERING_REPORT.md §3.2.2, "Inconsistency Resolutions" §2).
level PRI ($f_r = 5988$ Hz), whereas the GUI variables $\text{prf1} = 1000$ Hz / $\text{prf2} = 2000$ Hz are display rates, not chirp PRFs (source: `00_notation/parameter_table.md`, "Inconsistency Resolutions" §2).

1.3 Range, resolution and unambiguous range

With round-trip delay $\tau = 2R / c$ (FMCW-2) the dechirped beat frequency of a stationary target is $f_b = 2 \cdot \mu \cdot R / c$ (FMCW-17), giving $R = c \cdot f_b / (2\mu)$ (FMCW-18). Two targets are resolvable when their beat frequencies differ by at least $1/T_c$, which yields the range resolution (source: `01_physics/01_fmcw_theory.md` §5–6, Eq. FMCW-19):

- $\Delta R = c / (2B)$ (FMCW-19) — depends only on B; numerical value not computable while B is TBD (the source says so explicitly).

The maximum unambiguous range follows from the PRI (source: `01_physics/01_fmcw_theory.md` §8, Eq. FMCW-22):

- $R_{\max} = c \cdot T_r / 2 = c / (2 f_r)$ (FMCW-22) → 25.1 km for $T_{r,1} = 167 \mu\text{s}$ (source: `engineering/DESIGN/CALCS/DESIGN_CALCULATIONS.md` §3, row "Unambiguous range").

The radar range equation used for the performance estimate of chapter 1 §5 is (source: `01_physics/01_fmcw_theory.md` §2, Eq. FMCW-11):

- $\text{SNR} = P_t \cdot G^2 \cdot \lambda^2 \cdot \sigma / ((4\pi)^3 \cdot R^4 \cdot k_B \cdot T_0 \cdot B_n \cdot F \cdot L)$ (FMCW-11)

1.4 Doppler

For a radial velocity v (approaching positive) the Doppler shift is $f_d = 2v / \lambda = 2 v f_c / c$ (FMCW-4); the full beat frequency with Doppler coupling is $f_b = 2\mu R/c \pm f_d$ (FMCW-16). Velocity and velocity resolution over M pulses are $v = \lambda f_d / 2$ (FMCW-20) and $\Delta v = \lambda / (2 M T_r)$ (FMCW-21); the unambiguous velocity is $v_{\max} = \lambda f_r / 4$ (FMCW-23) and the range–velocity trade-off $R_{\max} \cdot v_{\max} = c \lambda / 8$ (FMCW-24) (source: `01_physics/01_fmcw_theory.md` §1, §4, §7, §8).

Range–Doppler coupling of an LFM pulse displaces the apparent range by $\Delta R_{\text{Doppler}} = c f_d / (2\mu) = v c T_c / (\lambda B)$ (FMCW-27); the coupling ratio of the long to the short chirp equals $T_{c,1} / T_{c,2}$ (FMCW-28), i.e. 60 for the firmware values, and range migration across a CPI is $\Delta R_{\text{migration}} = v \cdot M \cdot T_r$ (FMCW-30) (source: `01_physics/01_fmcw_theory.md` §9). This is the stated reason for the two chirp modes: the long chirp gives processing gain and finer Doppler resolution, the short chirp smaller coupling and wider unambiguous range (source: `01_physics/02_lfm_waveform_model.md` §7, "Design Tradeoffs").

2. Pulse compression

2.1 Matched filter

The matched filter for $s(t)$ is $h(t) = s^*(-t)$ (LFM-8), or in the frequency domain $H(f) = S^*(f)$ (LFM-9). It maximises the output SNR to $2E / N_0$ (LFM-10) and provides the processing gain $G_p = B \cdot T_c$ (LFM-14); the matched-filter output SNR is the range-equation SNR multiplied by $B \cdot T_c$ (LFM-15). The compressed pulse width is $\tau_c = 1 / B$ (LFM-16), so the compression ratio is $T_c / \tau_c = B T_c$ (LFM-17) and the range resolution is again $\Delta R = c \tau_c / 2 = c / (2B)$ (LFM-18). The compressed envelope is approximately $|y(\tau)| \approx T_c \cdot |\text{sinc}(B \tau)|$ (LFM-20) with a first sidelobe of -13.3 dB for rectangular weighting (LFM-21); windowing trades mainlobe width for sidelobe level (source: `01_physics/02_lfm_waveform_model.md` §3–5).

The LFM ambiguity function $|\chi(\tau, v)| = (1 - |\tau|/T_c) \cdot |\text{sinc}((v + \mu\tau)(T_c - |\tau|))|$ for $|\tau| \leq T_c$ (LFM-23) has its ridge on $v = -\mu\tau$ (LFM-24); the zero-delay Doppler cut gives the single-pulse Doppler resolution $\Delta f_{d,\text{pulse}} = 1 / T_c$ (LFM-27) and $\Delta v_{\text{pulse}} = \lambda / (2 T_c)$ (LFM-28) (source: `01_physics/02_lfm_waveform_model.md` §6). With $B = 50$ MHz **ASSUMED** and $T_c = 30 \mu\text{s}$ the pulse-compression gain is 31.8 dB and $\Delta R = 3$ m (source: `engineering/DESIGN/CALCS/DESIGN_CALCULATIONS.md` §3, row "Chirp"); these two numbers are estimates until B is known.

2.2 What the repository data encodes — the .mem finding

The RTL loads its matched-filter reference from memory files `long_chirp_seg{0,1,2}_{i,q}.mem` and `short_chirp_{i,q}.mem`. The BETA analysis established that the long-chirp files are **conjugate FFT-domain** coefficients, not time-domain samples: `long_chirp_seg{0,1,2}_{i,q}.mem = conj(FFT_1024(u_s)) \cdot 31128 / \max|.|`, where u is a unit-amplitude $10 \rightarrow 30$ MHz linear up-chirp of 3000 samples at 100 MSPS, split into 1024-sample segments (phase-fit residual 0.0018 rad; regenerated to within 1 LSB by `beta/fpga/gen_chirp_mem.py`). Segment 3 lies beyond the chirp and is all zeros — that is the file that was missing from the upstream repository and was generated (source: `beta/fpga/README.md`, "What was found and decided", item 1; `docs/AERIS10_BETA_ENGINEERING_REPORT.md` §3.2).

Consequences recorded in the same source:

- the matched filter is a frequency-domain chain (FFT → multiply by the stored reference → IFFT), as the orphan wrappers `fft_1024_forward/inverse_enhanced` and `frequency_matched_filter` of the original RTL imply; a time-domain FIR would not have used the repository data;
- the stored reference is already conjugated, so the chain must **not** conjugate again (`CONJUGATE_REF = 0` in the BETA RTL); with the extra conjugation neither the numpy model nor the RTL produces a compression peak;

- the same data equals the plain FFT of a 30 → 10 MHz down-chirp, so the actual baseband chirp direction after the RF/IF chain decides whether `CONJUGATE_REF` must be 0 or 1 — an open point for the designer (UNRESOLVED, hardware bring-up item);
- `short_chirp_{i,q}.mem` (50 words) matches neither a time- nor a frequency-domain chirp and cannot be used; the short-chirp processing path is UNRESOLVED.

The 10 → 30 MHz sweep of the reference data is a 20 MHz span at baseband. No repository file states whether this equals the RF chirp bandwidth B (TBD in `00_notation/parameter_table.md`; 50 MHz ASSUMED in `engineering/DESIGN/CALCS/DESIGN_CALCULATIONS.md §3`); the discrepancy is logged as observation MAN-02 in chapter 17.

2.3 Detection

The detection notes derive the cell-averaging CFAR threshold multiplier $\alpha = N_{\text{ref}} \cdot (P_{\text{fa}}^{-(1/N_{\text{ref}})} - 1)$ (DET-20) with $P_{\text{fa}} = (1 + \alpha/N_{\text{ref}})^{-(N_{\text{ref}})}$ (DET-19), the Swerling I detection probability $P_d = P_{\text{fa}}^{(1/(1 + \text{SNR}_{\text{mean}}))}$ (DET-22) and the CFAR loss $L_{\text{CFAR}} \approx (1/N_{\text{ref}}) \cdot P_d / ((1 - P_d) \cdot \ln P_{\text{fa}})$ (DET-24) (source: `01_physics/04_detection_theory.md §6–8`). The RTL as committed does not implement this: the "CFAR" stage is a fixed threshold $|I| + |Q| > 10000$ on the Doppler output, marked PLACEHOLDER by the original source comment at `radar_system_top.v:298–299` (source: `engineering/SOFTWARE_DIAGRAMS/FPGA/fpga_data_pipeline.dot`, node "CFAR"; `beta/fpga/README.md`, "Remaining work" item 7: the detector is still a fixed threshold, default 10000 via `CFAR_THRESHOLD_DEFAULT`, writable over the bridge as `CFAR_THR` in the BETA).

2.4 Noise

The receive-chain noise figure follows Friis' formula $F_{\text{sys}} = F_{\text{LNA}} + (F_{\text{mix}} - 1)/G_{\text{LNA}} + (F_{\text{IF}} - 1)/(G_{\text{LNA}} G_{\text{mix}}) + (F_{\text{ADC}} - 1)/(G_{\text{LNA}} G_{\text{mix}} G_{\text{IF}})$ (NF-8); ADC quantisation adds $\text{SQNR} = 6.02 b + 1.76 \text{ dB}$ (NF-11) for $b = 8$ bits; the CIC decimator has DC gain $G_{\text{CIC}} = D_{\text{CIC}}^{N_{\text{CIC}}}$ (NF-14) and bit growth $b_{\text{out}} = b_{\text{in}} + N_{\text{CIC}} \cdot \text{ceil}(\log_2 D_{\text{CIC}})$ (NF-15) (source: `01_physics/05_noise_analysis.md §3–6`). The numerical budget is "pending parameter resolution" in the source (§7.3): LNA noise figures and the Extended-variant chain are TBD (source: `00_notation/parameter_table.md`, "TBD Tracking").

3. Beamforming

3.1 Array factor and steering

For a uniform linear array of N elements at spacing d , the inter-element propagation phase is $\Delta\phi_{\text{prop}} = k d \sin\theta$ with $k = 2\pi/\lambda$ (BF-1); the electrical angle is $\psi = k d \sin\theta + \Delta\phi$ (BF-2) and the array factor $\text{AF}(\theta) = \sum_{n=0}^{N-1} w_n \cdot \exp(j n \psi)$ (BF-3). Steering to θ_0 requires $\Delta\phi = -k d \sin\theta_0$ (BF-4). With uniform weights $|\text{AF}(\theta)| = |\sin(N\psi/2) / \sin(\psi/2)|$ (BF-8), the half-power beamwidth is $\theta_{3\text{dB}} \approx 0.886 \lambda / (N d)$ (BF-10), broadening by $1/\cos\theta_0$ when scanned (BF-11). Grating lobes appear at $\sin\theta_{\text{GL}} = \sin\theta_0 + m \lambda / d$ (BF-14); with $d = \lambda/2$ they never enter visible space for any scan angle, and for a limited scan $|\theta_0| \leq \theta_{\text{max}}$ the spacing may be relaxed to $d/\lambda < 1 / (1 + \sin\theta_{\text{max}})$ (BF-16) (source: `01_physics/03_beamforming_theory.md §1–6`). The 2-D extension factorises into the product of two 1-D array factors (BF-19) (source: same, §9).

AERIS-10 values: $N = 16$, $d = \lambda/2 \approx 14.3 \text{ mm}$ (`element_spacing = wavelength / 2.0f`, `main.cpp:1134`), four ADAR1000 beamformers of four channels each, 31 elevation positions, azimuth by mechanical rotation (source: `00_notation/parameter_table.md`, "Antenna and Beamforming"; `02_hardware/04_antenna_beamforming.md §1`). The 8×16 patch description of the README is interpreted in the proposed antenna as 16 rows of 8 series-fed patches, each row being one "element" of the elevation-scanned array (decision D-02, source: `engineering/DESIGN/00_DESIGN_BASIS.md §2`); that interpretation is a PROPOSED DESIGN choice, not a verified fact about the prototype.

3.2 Phase table and steering angles

The firmware holds 31 inter-element phase differences $\Delta\phi_n$ (`phase_differences[31]` in `main.cpp`) (source: `02_hardware/04_antenna_beamforming.md §3.1`, copied verbatim):

Index	$\Delta\phi_n$ (deg)	Index	$\Delta\phi_n$ (deg)	Index	$\Delta\phi_n$ (deg)
0	+160.000	11	+13.333	22	-17.778
1	+80.000	12	+12.308	23	-20.000
2	+53.333	13	+11.429	24	-22.857
3	+40.000	14	+10.667	25	-26.667
4	+32.000	15	0.000	26	-32.000
5	+26.667	16	-10.667	27	-40.000
6	+22.857	17	-11.429	28	-53.333
7	+20.000	18	-12.308	29	-80.000
8	+17.778	19	-13.333	30	-160.000
9	+16.000	20	-14.545		
10	+14.545	21	-16.000		

Position 15 is broadside; the table is symmetric. The per-element phase is $\phi_n = n \cdot \Delta\phi_{\text{pos}}$ (HW-ANT-4), quantised to the 7-bit

ADAR1000 register `reg_n = floor(( $\phi_n \bmod 360^\circ$ )/ $360^\circ \cdot 128$ ) \bmod 128 (HW-ANT-5) with a phase step of  $360^\circ/128 = 2.8125^\circ$  (HW`

ANT-1; CAL-6) and a maximum quantisation error of 1.40625° (CAL-7) (source: 02_hardware/04_antenna_beamforming.md §2.4, §3.3–3.4; 01_physics/06_calibration_theory.md §4). The steering angle for $d = \lambda/2$ is $\theta_0 = \arcsin(\Delta\phi_n / 180^\circ)$ (HW-ANT-3) (source: 02_hardware/04_antenna_beamforming.md §3.2).

The project's documents disagree on the resulting scan range: 02_hardware/04_antenna_beamforming.md §3.2 evaluates HW-ANT-3 at $|\Delta\phi_n| = 160^\circ$ to $|\theta_0| \approx \pm 62.7^\circ$ and then calls $\approx \pm 33^\circ$ the "safe scan range" before grating lobes, while 01_physics/03_beamforming_theory.md §6 shows that $d = \lambda/2$ is grating-lobe-free at every scan angle, and 00_notation/parameter_table.md ("Inconsistency Resolutions" §4) states $\approx \pm 33^\circ$ at $\Delta\phi = \pm 160^\circ$ from the same formula. The README states $\pm 45^\circ$. This manual does not resolve the disagreement; it is recorded as MAN-01 in chapter 17 for the antenna designer. The BETA firmware filled the ADAR1000 vector-modulator tables from datasheet Tables 10–13 (128 rows, $\leq 3.1^\circ$ encoding error) and fixed a channel-index defect that "rotated the beam by one element" (source: docs/AERIS10_BETA_ENGINEERING_REPORT.md §4).

3.3 Calibration and errors

Per-element amplitude and phase errors $h_n = (a_n + \delta a_n) \cdot \exp(j \delta \phi_n)$ (CAL-1) raise the RMS sidelobe level to $\sigma_a^2 + \sigma_\phi^2$ (CAL-5); mutual coupling is modelled as $v_{\text{actual}} = C \cdot w$ (CAL-11) and pre-compensated by $w_{\text{applied}} = C^{-1} \cdot w_{\text{desired}}$ (CAL-12). The calibration procedure (measure each element against a reference, compute $c_n = a_n / (h_n, \text{meas} / S_{\text{ref}})$ (CAL-14), apply, verify) leaves a residual phase error bounded by half a quantisation step (CAL-15) (source: 01_physics/06_calibration_theory.md §1–8). No calibration has been performed; per-board phase calibration is listed as open in the BETA report (source: docs/AERIS10_BETA_ENGINEERING_REPORT.md §4, "Open").

3.4 Beam sequence

During each azimuth position the firmware loads the 15 positive-steering patterns (*matrix1*), broadside (*vector_0*) and the 15 negative patterns (*matrix2*) into all four ADAR1000s (TX and RX) and, for each, executes M/2 long chirps at $T_{r,1}$, the guard time, then M/2 short chirps at $T_{r,2}$ (source: 02_hardware/04_antenna_beamforming.md §3.5–3.6). The STM32 signals each new chirp, elevation and azimuth to the FPGA by toggling *DIG_0..2* (PD8..PD10) and enables the mixers with *DIG_3* (PD11); the bit-to-port mapping is inferred from source comments and rated MEDIUM confidence (source: engineering/SOFTWARE_DIAGRAMS/DATA_FLOW/end_to_end_data_flow.dot, edge "handshake DIG_0..4"; docs/SYSTEM/BLOCK_DIAGRAM.md §3).

4. Frequency plan and signal flow

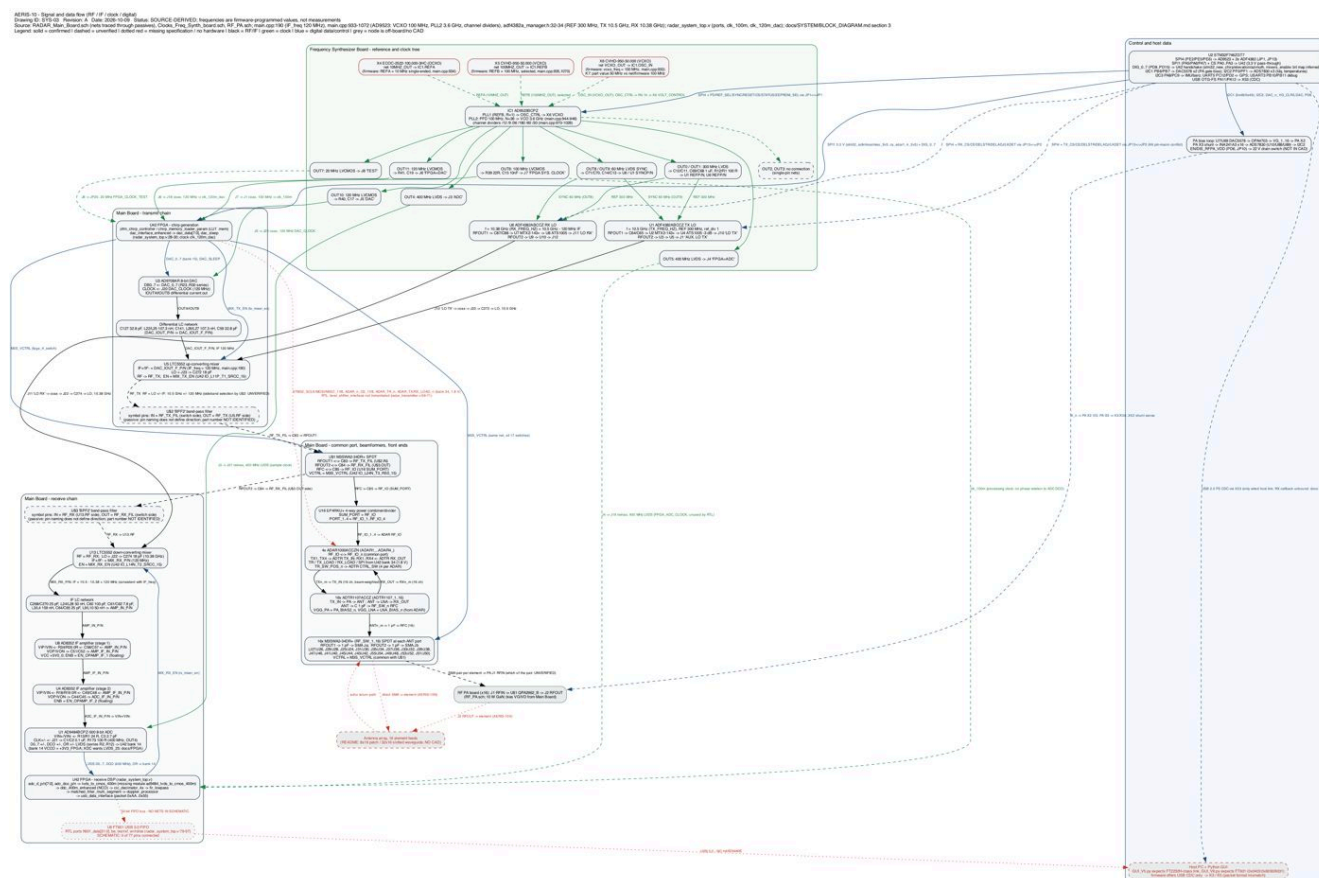


Figure 5 — F2.1 — Signal and data flow SYS-03: reference and clock tree, TX chain, RX chain, control paths; solid = confirmed, dashed = unverified, dotted red = missing specification — SOURCE-DERIVED; frequencies are firmware-programmed values, not measurements (source: engineering/SYSTEM/data_flow/signal_and_data_flow.dot; produced by Graphviz dot -Tpng -Gdpi=150)

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The frequencies below are the values programmed by the firmware, traced through the schematics; none has been measured (source: [engineering/SYSTEM/data_flow/signal_and_data_flow.dot](#), node labels; [engineering/SYSTEM/interfaces/interconnection_table.md](#) §6):

Signal	Value (firmware intent)	Path	Evidence
Reference	REFB 100 MHz selected (X5 net 100MHz_OUT); VCXO X6 on OSC_IN, vcxo_freq = 100 MHz	Synth IC1 AD9523: PLL1 (REFB, R = 1) → VCXO; PLL2 PFD 100 MHz, N = 36 → VCO 3.6 GHz	main.cpp:933–946, 1070; K7: schematic part values are 50 MHz VCXO / 100 MHz OCXO
ADF4382 reference	300 MHz LVDS (OUT0/OUT1)	IC1 → C11/C12, C68/C69, R1/R12 → U1/U6 REFP/N	synth schematic; adf4382a_manager.h:32–34
TX LO	10.5 GHz (TX_FREQ_HZ)	U1 ADF4382 RFOUT1 → MTX2-143+ → ATS1005 –3 dB → J10 "LO TX" → coax → Main J23 → C272 → U5 LTC5552 LO	adf4382a_manager.h:32–34; interconnection §6, CBL-36
RX LO	10.38 GHz (RX_FREQ_HZ) = 10.5 GHz – 120 MHz	U6 ADF4382 RFOUT1 → U7 → U8 → J11 "LO RX" → Main J22 → C274 → U13 LTC5552 LO	same; CBL-37
ADC sample clock	400 MHz LVDS (OUT4)	J3 → J21 twinax → U1 AD9484 CLK±	main.cpp:983–984; CBL-34
FPGA ADC clock	400 MHz LVDS (OUT5)	J4 → J19 → U42 bank 14 MRCC	not used by the RTL; CBL-35
FPGA system clock	100 MHz LVCMOS (OUT6)	J7 → J1 → U42 IO_L13P_T2_MRCC_15 = clk_100m	main.cpp:1004; CBL-30
DAC clock	120 MHz LVCMOS (OUT10)	J5 → J20 → U3 AD9708 CLOCK	main.cpp:1025–1026; CBL-31
FPGA DAC clock	120 MHz LVCMOS (OUT11)	J6 → J18 → U42 = clk_120m_dac	CBL-32
Test clock	20 MHz (OUT7)	J8 → JP20 FPGA_CLOCK_TEST	CBL-33 (adapter UNVERIFIED)
IF	120 MHz	TX: DAC → LC network → U5 IF±; RX: U13 IF± → LC → 2 × AD8352 → AD9484	main.cpp:190; ddc_400m.v:48

TX chain (source: SYS-03 node labels): the FPGA chirp controller (plfm_chirp_controller, LUT-based) drives the 8-bit AD9708 DAC at 120 MHz; the differential LC network (C127 32.8 pF, L22/L25 107.3 nH, C141, L26/L27 107.3 nH, C59 32.8 pF) feeds the IF port of the LTC5552 up-converter U5; the RF output passes band-pass filter U\$2 "BPF2" (part number NOT IDENTIFIED; sideband selection UNVERIFIED), the SPDT switch U\$1 M3SWA2-34DR+, the 4-way combiner/divider U16 EP4RKU+, the four ADAR1000s (TX1..TX4 → ADTR TX_IN), the sixteen ADTR1107 T/R front ends and sixteen M3SWA2-34DR+ element switches to SMA pairs J24..J55, from which the Extended variant goes through one QPA2962 PA board per element to the antenna, and the Nexus variant goes directly to the antenna. RX chain: the echo returns through the same element switch and ADTR1107 LNA path (RX_OUT → ADAR RXn), the combiner, U\$1 to the LTC5552 down-converter U13 (LO 10.38 GHz → IF 120 MHz, "consistent with IF_freq"), the IF LC network, two cascaded AD8352 amplifiers U8/U4 (enable pins EN_OPAMP_IF_1/2 floating) and the AD9484 ADC (VIN± via R13/R1 24 Ω, C3 2.7 pF), whose LVDS D0..7 + DCO go to FPGA bank 14 (source: [engineering/SYSTEM/data_flow/signal_and_data_flow.dot](#), nodes of the "transmit chain", "common port", "receive chain" clusters).

5. FPGA processing chain

5.1 As written in the original RTL (PARTIAL)

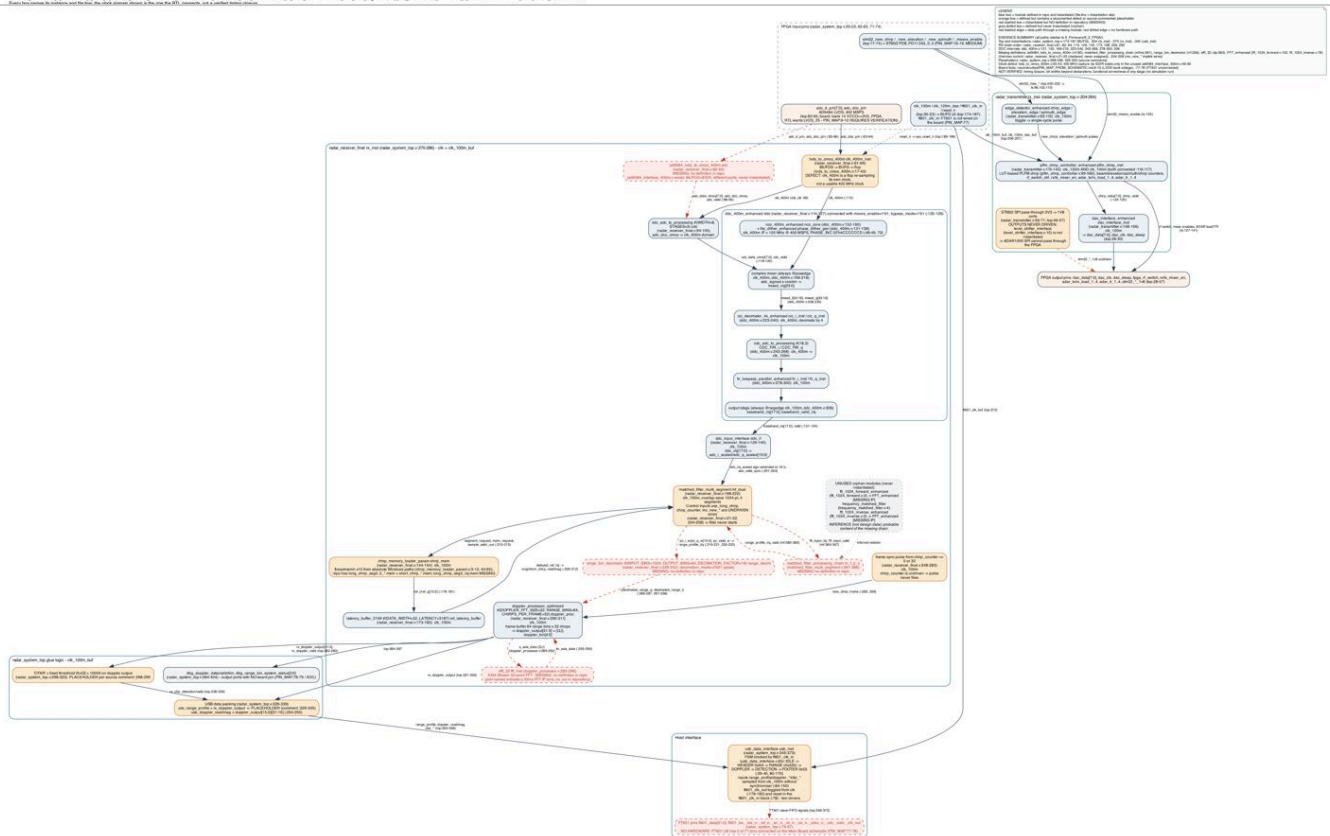


Figure 6 — F2.2 — FPGA signal-processing pipeline as written in the original RTL (SD-02): blue = defined and instantiated, orange = documented defect or placeholder, red dashed = instantiated but missing from the repository, grey dotted = orphan — PARTIAL (source: engineering/SOFTWARE_DIAGRAMS/FPGA/fpga_data_pipeline.dot; produced by Graphviz dot - Tpng -Gdpi=150)

The original receiver chain (radar_receiver_final.v, instantiation order at lines 61, 82, 94, 114, 129, 144, 173, 198, 226, 290) is, per stage (source: engineering/SOFTWARE_DIAGRAMS/FPGA/fpga_data_pipeline.dot, node labels):

Stage	Module (file:line)	Function and parameters	State in the original RTL
LVDS capture	lvds_to_cmos_400m (rnf:61), ad9484_lvds_to_cmos_400m (rnf:82)	IBUFDS → BUFG; 8-bit ADC bus + DCO at 400 MSPS	clock defect (lvds_to_cmos_400m.v:35–43 : a flop re-sampling its own clock); capture module MISSING
CDC	cdc_adc_to_processing #(8,3) (rnf:94)	ADC DCO → clk_400m	present
DDC	ddc_400m_enhanced (rnf:114) with nco_400m_enhanced + lfsr_dither_enhanced (ddc_400m.v:131–160)	NCO IF 120 MHz at 400 MSPS, PHASE_INC 32'h4CCCCCD; complex mixer; cic_decimator_4x_enhanced *2 (decimate by 4, 5 stages); CDC to clk_100m; fir_lowpass_parallel_enhanced *2; baseband_i/q[17:0]	connected with mixers_enable = 1'b1, bypass_mode = 1'b1 (:125–126); 400 MHz fabric logic
Scaling	ddc_input_interface (rnf:129)	18-bit → 16-bit adc_i/q_scaled	present
Reference	chirp_memory_loader_param (rnf:144) + latency_buffer_2159 #(32, 3187) (rnf:173)	\$readmemh *10 from absolute Windows paths; long_chirp_seg3_i/q.mem MISSING	path defect; one file missing
Matched filter	matched_filter_multi_segment (rnf:198) → matched_filter_processing_chain (matched_filter_multi_segment.v:361)	overlap-save 1024-point, 4 segments	control inputs use_long_chirp, chirp_counter, mc_new_* UNDRIVEN (rnf:21–25, 204–208) → filter never starts; processing chain MISSING; FFT wrappers orphaned, IP missing
Range decimation	range_bin_decimator #(1024, 64, 16) (rnf:226)	1024 → 64 bins, peak mode	MISSING
Doppler	doppler_processor_optimized #(32, 64, 32) (rnf:290) → xfft_32 (doppler_processor.v:283)	64 range bins × 32 chirps frame → 32-point FFT	xfft_32 MISSING (Xilinx IP, no .xci); frame-sync pulse never fires (chirp_counter undriven)
Detection	glue in radar_system_top.v:298–323	fixed threshold $abs(I) + abs(Q) > 10000$	PLACEHOLDER per source comment
Host	usb_data_interface (radar_system_top.v:345–373)	FT601 slave FIFO, packet 0xAA header ... 0x55 footer, FSM on ft601_clk_in	NO HARDWARE (FT601 U6 0 of 77 pins connected); inputs sampled across clock domains without synchroniser; ft601_clk_out two drivers
Transmitter	radar_transmitter (radar_system_top.v:204–264)	edge detectors on STM32 toggles; plfm_chirp_controller_enhanced (LUT chirp, beam/elevation/azimuth/chirp counters, RF switch, mixer enables, ADAR load/TR); dac_interface_enhanced at 120 MHz	level_shifter_interface not instantiated → STM32→ADAR1000 SPI pass-through outputs never driven

The register `docs/SYSTEM/BLOCK_DIAGRAM.md §3` summarises the same findings per interface (STM32→FPGA handshake inferred, SPI pass—through BROKEN in RTL, FPGA→host NO HARDWARE, ADC→FPGA capture not implemented and bank-voltage conflict).

5.2 As implemented in `beta/fpga` (BETA)

The BETA RTL keeps the architecture and repairs it so that it parses, lints and simulates with Icarus Verilog 13.0 and Verilator 5.052; it is **not synthesised**, has **no timing closure** and has **not been loaded on hardware**; the two Xilinx FFT IP cores are not generated (source: `beta/fpga/README.md`, "Status"). The functional changes relevant to the theory of operation (source: `beta/fpga/README.md`, "What was found and decided", "ADC capture and DDC front end", "Remaining work"; `docs/AERIS10_BETA_ENGINEERING_REPORT.md §3.2`):

- **ADC capture** (`ADC_CAPTURE_MODE = 1`, default): the AD9484 output is SDR LVDS at the sample rate (datasheet facts: DCO at 400 MHz, data valid on the rising DCO edge, tSKEW ± 0.07 ns). Capture uses IBUFDS(DCO) → BUFIO + BUFR/4, IDELAYE2 per lane with calibration (`adc_capture_calib`: default tap 16, manual tap/bitslip, automatic IDELAY sweep with the ADC test pattern, per-lane lock status, pattern error counter), ISERDESE2 SDR 1:4, and a 32-bit asynchronous FIFO into `clk_100m`. Mode 0 keeps the legacy 400 MHz fabric path for comparison.
- **DDC at 100 MHz** (`ddc_4x_100m`): 4-phase NCO, 8 mixers, CIC as a 16-tap FIR and the unchanged FIRs, all at 100 MHz; bit-exact with the legacy 400 MHz chain (`tb_ddc_4x`: 1855 outputs, max diff 0 LSB).
- **Matched filter**: frequency-domain chain with `CONJUGATE_REF = 0` (section 2.2); reference alignment by address instead of the fixed 3187-cycle latency buffer; `tb_matched_filter` peak at bin 302 for a 300-sample delay (± 4), peak/sidelobe 4.76.
- **Clock-domain crossings**: Gray-pointer FIFOs, reset synchroniser per domain, STM32 toggles synchronised in the consuming domain.
- **Register map** (`radar_control_regs`): the previously floating control inputs have one driver with reset defaults; its write/read port is driven by the SPI bridge (command set v2, 5-bit word addresses 0x00..0x10, 16-bit registers; table in `engineering/DESIGN/HOST_LINK/HOST_LINK_DESIGN.md §7`).
- **Host path option B**: `rd_map_packer` turns each 64×32 Doppler frame into a 2066..2162-byte frame (header, 2048 $\times$ uint8 log-magnitude, up to 32 detections, CRC-16/CCITT-FALSE) and `host_bridge_spi` streams it to the STM32 as an SPI slave on the existing SPI1 nets with `DIG_5..7` as CS/DRDY/spare; the ADAR1000 pass-through is gated during transfers (section 6).

System-level evidence from `tb_system_smoke` (source: `beta/fpga/README.md`, "System smoke test evidence"): for alternating echo delays of 100 and 420 baseband samples the segment-0 peak sat at decimated bins 8 and 28 on every chirp — a 20-bin shift for a 320-sample delay change ($320/16 = 20$) — i.e. pulse compression works end to end through capture → DDC → matched filter → decimator in simulation; 2048 Doppler outputs (one 64×32 frame); 2688 USB packets with 0 header/footer/sequence errors.

Two architectural limitations remain and matter for interpreting the processing chain (source: `beta/fpga/README.md`, "Remaining work" items 4–5): (a) **throughput** — the matched-filter FSM is not pipelined against the sample collector; one long chirp (4 segments) occupies it for $\sim 92 \mu\text{s}$ with the behavioural FFT latency and $\sim 270 \mu\text{s}$ with a realistic IP latency, whereas the transmitter repeats long chirps every 167 μs , so samples arriving outside `ST_COLLECT_DATA` are dropped (original behaviour; the smoke test uses a 300 μs period for this reason); (b) **segment semantics** — each 1024-sample segment is compressed against its own reference segment and yields its own 64-bin profile; the four profiles per chirp are not summed, and the Doppler processor counts each profile as a "chirp", so a 32-"chirp" frame covers 8 real chirps. Whether to sum the partial correlations (true partitioned matched filter) or use a 4096-point transform is a design decision not yet taken.

6. Data path to the host

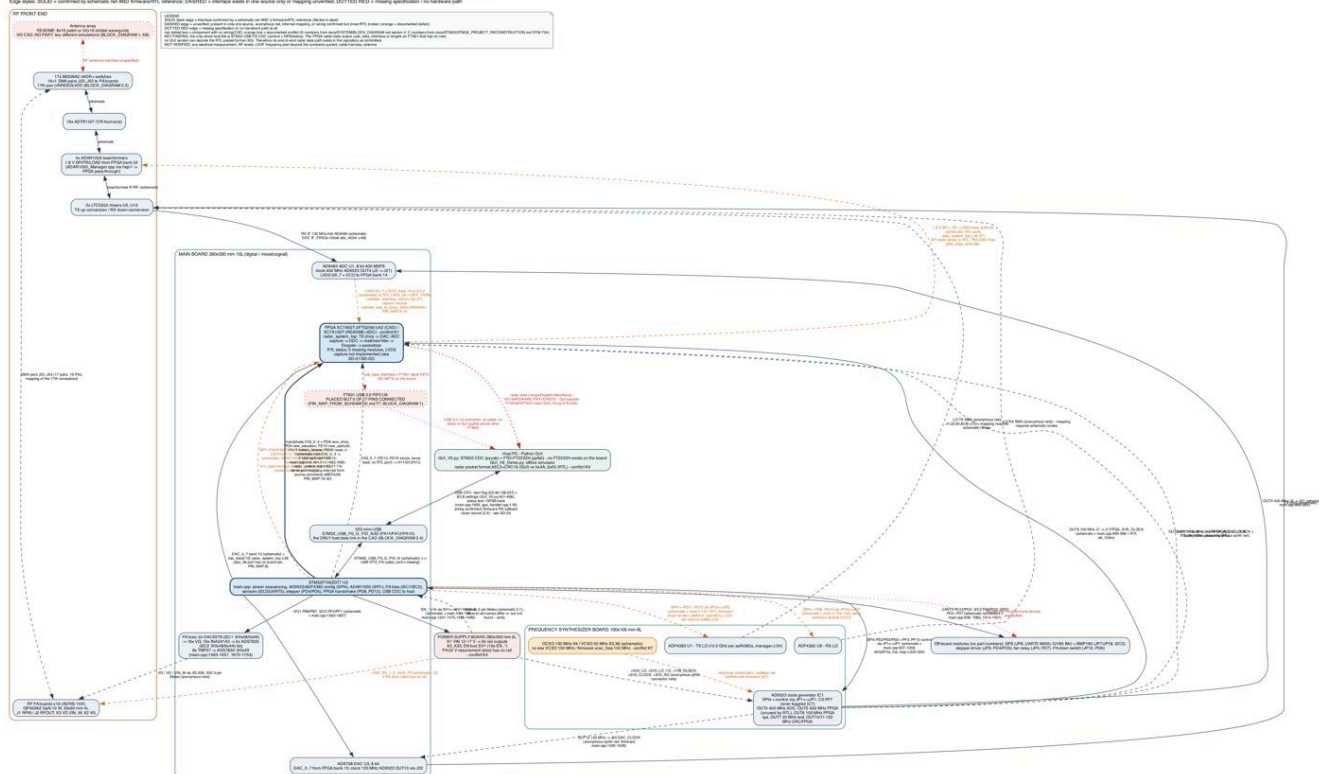


Figure 7 — F2.3 — End-to-end signal and control data flow SD-07: antenna → RF → ADC → FPGA → host and host → STM32 → clock/LO/beamformer → FPGA; solid = confirmed by schematic net and firmware/RTL, dashed = unverified, dotted red = missing specification — PARTIAL; the FPGA→host path has no hardware (source: engineering/SOFTWARE DIAGRAMS/DATA FLOW/end to end data flow.dot; produced by Graphviz dot -Tpng -Gdpi=150)

Key finding of SD-07 (source: `engineering/SOFTWARE_DIAGRAMS/DATA_FLOW/end_to_end_data_flow.dot`, legend): the only wired host link is the STM32 USB-FS CDC on X53 (control, GPS and status); the FPGA radar-data output targets an FT601 that has no nets; no GUI version in the upstream repository can decode the RTL packet format (K5). Therefore no end-to-end radar data path exists in the repository as committed.

The data-rate budget that frames the BETA solution (source: [engineering/DESIGN/HOST_LINK/HOST_LINK_DESIGN.md](#) §2, copied verbatim):

Quantity	Value	Basis
Range-Doppler cells per beam position	$64 \times 32 = 2048$	doppler_processor.v (RANGE_BINS 64, 32 chirps)
Beam-position frame time	5647.4 μ s	main.cpp:180–186
Raw RTL packet stream (44 B per cell)	16.0 MB/s	usb_data_interface.v (11 $\times$ 32-bit words)
Compact map frame (8-bit log-magnitude per cell + header + ≤ 32 detections + CRC)	2164 B $\rightarrow$ 383 kB/s	this design, §5
STM32 USB-FS CDC practical limit	$\approx 0.8\text{--}1.1$ MB/s	USB 2.0 FS bulk (19 $\times$ 64 B per 1 ms frame max)
SPI1 STM32 $\leftrightarrow$ FPGA (existing lines)	27 Mbit/s $\approx$ 3.3 MB/s (DMA)	APB2 108 MHz / 4 (beta clock tree)
FT601 245 sync FIFO, 32 bit @ 100 MHz	up to 400 MB/s	FT601

The source concludes that the raw stream needs the FT601 (option A, Main Board rev. B, decision D-16) while the compact map fits the existing STM32 path with 3× margin (option B, implemented as BETA, decision D-17). The STM32 forwards each bridge frame unchanged over CDC, interleaved with its status strings, and the GUI stream parser resynchronises on the 0xA5 0x5A sync word (source: [engineering/DESIGN/HOST_LINK/HOST_LINK_DESIGN.md §5](#)). Chapter 9 gives the frame layout and the command set; chapter 17 lists the decisions D-16...D-19.

```
<!-- chapter 3: Boards, modules, connectors, cables, power rails -->
```

Architecture and interfaces — boards, connectors, cables, power rails

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Status summary: interconnection diagram and tables SOURCE-DERIVED (connector nets of the four EAGLE schematics, board silkscreen, `main.h`), with every row carrying CONFIRMED / UNVERIFIED / MISSING SPEC; power-rail register SOURCE-DERIVED (voltages from net names, currents UNKNOWN, firmware sequence as coded, not executed); harness schedule PROPOSED DESIGN (DSN-HAR-01, lengths computed from the proposed layout); cable IDs are proposals, not upstream data; figures F3.1 and F3.2 SOURCE-DERIVED.

Sources: `engineering/SYSTEM/interfaces/interconnection_table.md`, `engineering/SYSTEM/architecture/README.md`, `engineering/ELECTRICAL/power_distribution/power_rails.md`, `engineering/DESIGN/HARNESS/HARNESS_SCHEDULE.md`, `docs/SYSTEM/BLOCK_DIAGRAM.md` §2–3.

Planned figures: F3.1 hardware interconnection diagram (SYS-02), F3.2 power distribution diagram (ELEC-PWR-01 / SYS-04).

1. Physical architecture

The radar electronics consist of a Power Supply Board (280 × 300 mm, 2 layers), a Frequency Synthesizer Board (100 × 100 mm, 6 layers), a Main Board (260 × 300 mm, 10 layers) and, for the Extended variant, sixteen RF PA boards (35 × 60 mm, 4 layers), interconnected by 2-pin and 3-pin Molex 22-23-20x1 power cables, one 20-way enable ribbon (SV1), two control ribbons (JP1↔JP1 2×6, JP13↔JP2 2×7), coaxial clock/LO links and 34 SMA RF ports (source: `docs/SYSTEM/BLOCK_DIAGRAM.md` §1; `engineering/SYSTEM/interfaces/interconnection_table.md`, lead-in). The antenna, the host PC, the 22 V PA drain supply and the off-board modules (GPS, IMU, barometer, temperature sensors, stepper driver, fan relay, drain switch) have no CAD in the repository and appear as CONCEPTUAL nodes (source: `engineering/SYSTEM/architecture/README.md`, row SYS-01).

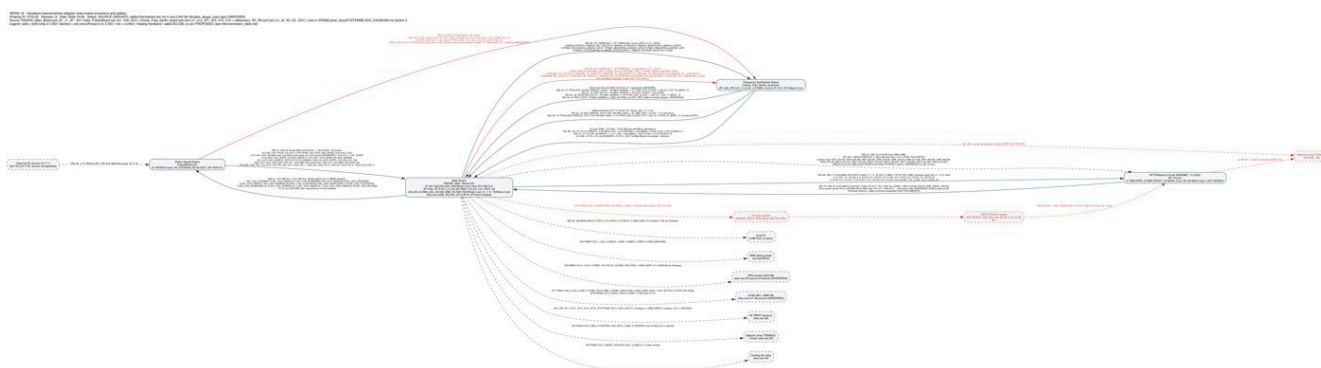


Figure 8 — F3.1 — Hardware interconnection diagram SYS-02: one node per board/module, one edge per connector/cable with connector IDs, pin numbers where the symbol has them, signal names and interface standard — SOURCE-DERIVED; cable types, lengths, Molex pin order, SMA RFIN/RFOUT side and PA-instance mapping are UNVERIFIED (source: `engineering/SYSTEM/interfaces/hardware_interconnection.dot`; produced by Graphviz dot -Tpng -Gdpi=150)

How the diagram was obtained (source: `engineering/SYSTEM/architecture/README.md`, "What was verified against the schematics"): the four `.sch` files were parsed read-only with Python `xml.etree`; parts and `<pinref>` nets were tabulated; anonymous nets (`N$...`) were followed through two-pin passives until a named net or an IC pin was reached; the Synth and PA `.brd` files were parsed for silkscreen text nearest each connector (the Main Board `.brd` contains no free silkscreen text); MCU pins were read from `main.h`, the enable order and clock settings from `main.cpp`, `ADAR1000_Manager.cpp` and `adf4382a_manager.{h,c}`; Power Management `V6.xlsx` was unzipped and its sheet XML parsed.

Corrections to `docs/SYSTEM/BLOCK_DIAGRAM.md` established by SYS-02 (source: `engineering/SYSTEM/architecture/README.md`, same section, copied): (1) SV1 has 15 enable lines; (2) Synth power connectors X10..X15 = `+3V3_X0`, `+3V3_CLOCK`, `+1V8_CLOCK`, `+3V3_L0_1`, `+3V3_L0_2`, `+5V0_L0`; (3) the "17th SMA pair" J22/J23 are the RX-LO and TX-LO inputs of U13/U5 (18 pF coupled); (4) Synth LO SMAs: J10 = L0 TX (U1 RFOUT1 path), J11 = L0 RX (U6 RFOUT1 path), J1 = AUX. L0 TX, J12 = second RX path; (5) `+5V0_ADAR`, `+5V0_ADTR`, `+3V3_Sw` outputs (X13, X26, X29) and `+5V0_1..5` (X2, X9, X17, X25, X28) are intermediate rails with no off-board consumer; (6) the PA gate-bias cables are X_1..X_16 (VG_n), the 3-pin X3/X38..X52 carry the INA241 shunt-sense pair.

2. Inter-board interconnection tables

Conventions of the copied tables (source: `engineering/SYSTEM/interfaces/interconnection_table.md`, lead-in): pins are EAGLE pin names of the connector symbols; Molex 22-23-20x1 symbols name all pads S, so their pin order cannot be read from the schematic and is written S (UNVERIFIED order). Cable IDs `CBL-xx` are a proposal of that document, not from the original project. "Power requirement" is copied from 3 Power Management/Power Management V6.xlsx (column G, mA, per device) only where the rail maps to a single row. Status:

CONFIRMED = both ends found in the schematics with the same net meaning; UNVERIFIED = one end or the pin order cannot be established; MISSING SPEC = no cable/part/number exists in the repository. Evidence shorthand: MB = RADAR_Main_Board.sch, PB = PowerBoard.sch, SY = Clocks_Freq_Synth_board.sch (+ .brd silkscreen), PA = RF_PA.sch, main.h = 9_Firmware/9_1_Microcontroller/9_1_3_C_Cpp_Code/main.h, xlsx = Power Management V6.xlsx sheet Feuil1.

2.1 DC input (source: interconnection_table.md §1)

Source device	Source connector	Pin	Signal	Destination device	Destination connector	Pin	Interface standard	Power requirement	Cable ID	Evidence	Status
External DC source (not specified)	—	—	VIN 12–17 V	Power Board	X1 AK300/2	KL (1)	DC power	UNKNOWN (sum not computed in xlsx)	CBL-00	PB X1 nets VIN, GND; brd silk Vin [12–17]V	MISSING SPEC (source)
External DC source	—	—	GND	Power Board	X1 AK300/2	KL (2)	DC power	—	CBL-00	PB X1	MISSING SPEC

2.2 Power Board → Main Board rails, Molex 22-23-2021 → 22-23-2021, 2 wires each (source: interconnection_table.md §2)

Source device	Source connector	Pin	Signal	Destination device	Destination connector	Pin	Interface standard	Power requirement (xlsx)	Cable ID	Evidence	Status
Power Board	X4	S	+1V0_FPGA	Main Board	X8	S	DC power 1.0 V	30 mA (row 17, VCCINT/VCCBRAM)	CBL-01	PB X4; MB X8	CONFIRMED (pin c UNVERIFIED)
Power Board	X5	S	+1V8_FPGA	Main Board	X10	S	DC power 1.8 V	50+22+100 mA (rows 14,16,26)	CBL-02	PB X5; MB X10	CONFIRMED
Power Board	X27	S	+3V3_FPGA	Main Board	X16	S	DC power 3.3 V	20+200+20+230 mA (rows 15,18,20,27)	CBL-03	PB X27; MB X16	CONFIRMED
Power Board	X16	S	+3V3	Main Board	X24	S	DC power 3.3 V	130+10+1+50+1 mA (rows 13,47-49,55)	CBL-04	PB X16; MB X24	CONFIRMED
Power Board	X12	S	+3V3_AN	Main Board	X9 (X56 is on the same net)	S	DC power 3.3 V	30+2×150+1+2+6+16×6 mA (rows 19,21,50,51,54,56)	CBL-05	PB X12; MB X9, X56	CONFIRMED (whic X9/X56 is used: UNVERIFIED)
Power Board	X10	S	+1V8_CLOCK	Main Board	X17	S	DC power 1.8 V	300 mA (row 25, AD9484 AVDD)	CBL-06	PB X10; MB X17 → L1 → +1V8_CLOCK_F	UNVERIFIED — PE X10 is the only +1V8_CLOCK outp and is also required Synth X12 (CBL-24
Power Board	X14	S	+3V3_ADAR_12	Main Board	X20 (+3V3_ADAR12)	S	DC power 3.3 V	2×700 mA (row 30)	CBL-07	PB X14; MB X20	CONFIRMED (net names differ by one underscore)
Power Board	X15	S	+3V3_ADAR_34	Main Board	X21 (+3V3_ADAR34)	S	DC power 3.3 V	2×700 mA (row 30)	CBL-08	PB X15; MB X21	CONFIRMED (nam differs)
Power Board	X21	S	−5V0_ADAR12	Main Board	X13	S	DC power −5 V	2×60 mA (row 29)	CBL-09	PB X21; MB X13	CONFIRMED
Power Board	X20	S	−5V0_ADAR34	Main Board	X15	S	DC power −5 V	2×60 mA (row 29)	CBL-10	PB X20; MB X15	CONFIRMED
Power Board	X34	S	+3V3_ADTR	Main Board	X4	S	DC power 3.3 V	16×80 mA (row 35)	CBL-11	PB X34; MB X4	CONFIRMED
Power Board	X18	S	−3V3_SW	Main Board	X6	S	DC power −3.3 V	16×0.12 mA (row 37)	CBL-12	PB X18; MB X6	CONFIRMED
Power Board	X30	S	+3V3_VDD_SW	Main Board	X12	S	DC power 3.3 V	16×0.014 mA (row 38)	CBL-13	PB X30; MB X12	CONFIRMED
Power Board	X23	S	+3V4	Main Board	X1 (X54 same net)	S	DC power 3.4 V	17×2.9 mA (row 22)	CBL-14	PB X23; MB X1, X54	CONFIRMED (X1/X choice UNVERIFIED)
Power Board	X24	S	−3V4	Main Board	X11 (X22 same net)	S	DC power −3.4 V	17×1.8 mA (row 23)	CBL-15	PB X24; MB X11, X22	CONFIRMED (X11/ choice UNVERIFIED)
Power Board	X3	S	+5V5_PA	Main Board	X55	S	DC power 5.5 V	2×50 mA (row 52)	CBL-16	PB X3; MB X55	CONFIRMED
Power Board	X19	S	−5V5_PA	Main Board	X19	S	DC power −5.5 V	2×50 mA (row 53)	CBL-17	PB X19; MB X19	CONFIRMED

Source device	Source connector	Pin	Signal	Destination device	Destination connector	Pin	Interface standard	Power requirement (xlsx)	Cable ID	Evidence	Status
Power Board	X31	S	+5V0_PA_1	Main Board	X14	S	DC power 5 V	5×250 mA (row 40; ADTR1107_1,2,3,4,7)	CBL-18	PB X31; MB X14	CONFIRMED
Power Board	X32	S	+5V0_PA_2	Main Board	X5	S	DC power 5 V	5×250 mA (ADTR1107_5,6,8,10,12)	CBL-19	PB X32; MB X5	CONFIRMED
Power Board	X33	S	+5V0_PA_3	Main Board	X7	S	DC power 5 V	6×250 mA (ADTR1107_9,11,13,14,15,16)	CBL-20	PB X33; MB X7	CONFIRMED
Power Board	X22	S	+5V0_0	Main Board	X18	S	DC power 5 V	2×200 mA (row 24, AD8352) — also feeds PB U5 (+3V3_AN) on the Power Board	CBL-20a	PB X22; MB X18	CONFIRMED
Power Board	X2, X9, X17, X25, X28	S	+5V0_1..+5V0_5	— (no counterpart on Main or Synth)	—	—	DC power 5 V	—	—	PB X2,X9,X17,X25,X28; MB/SY: net absent	UNVERIFIED (span test outputs; rails are consumed on-board U23/U25/U27/U29/U30)
Power Board	X13	S	+5V0_ADAR	— (no counterpart)	—	—	DC power 5 V	—	—	PB X13; MB: net absent (ADAR AVDD is −5 V from X13/X15)	UNVERIFIED (consumed on-board U20/U21)
Power Board	X26	S	+5V0_ADTR	— (no counterpart)	—	—	DC power 5 V	—	—	PB X26; MB: net absent	UNVERIFIED (consumed on-board U32)
Power Board	X29	S	+3V3_SW	— (no counterpart)	—	—	DC power 3.3 V	—	—	PB X29; MB: net absent	UNVERIFIED (consumed on-board U18)

2.3 Main Board → Power Board enable bus, SV1 MA10-2 ↔ SV1 MA10-2, 20-way, cable CBL-21 (source: power_rails.md §2, identical pin map on both boards; all rows CONFIRMED in interconnection_table.md §3)

SV1 pin	Net	STM32 pin (main.h)	Power Board EN input
1	EN_+1V0_FPGA	PE7	U1
2	EN_+5V0_PA2	PG1	U15
3	EN_+1V8_FPGA	PE8	U2
4	EN_+5V0_PA3	PG2	U16
5	EN_+3V3_FPGA	PE9	U4
6	EN_+5V5_PA	PG3	U17
7	EN_+5V0_ADAR	PE10	U13
8	EN_+1V8_CLOCK	PG4	U25
9	EN_+3V3_ADAR12	PE11	U6
10	EN_+3V3_CLOCK	PG5	U23
11	EN_+3V3_ADAR34	PE12	U7
13	EN_+3V3_ADTR	PE13	U32
15	EN_+3V3_SW	PE14	U10
17	EN_+3V3_VDD_SW	PE15	U8
19	EN_+5V0_PA1	PG0	U14
12, 14, 16, 18, 20	GND	—	—

15 enable lines (not 16 as stated in docs/SYSTEM/BLOCK_DIAGRAM.md). All enables are driven as push-pull outputs initialised LOW in MX_GPIO_Init (main.cpp:2239–2247); the Power Board has no pull-downs/pull-ups on the EN nets visible in the netlist (not checked exhaustively — REQUIRES VERIFICATION) (source: power_rails.md §2).

2.4 Power Board → Frequency Synthesizer Board rails, Molex 22-23-2021, 2 wires each (source: interconnection_table.md §4; Synth-side nets are anonymous N\$69..N\$77 and reach the named rail through a series inductor SY L9..L13)

Source device	Source connector	Pin	Signal	Destination device	Destination connector	Pin	Interface standard	Power requirement (xlsx)	Cable ID	Evidence	Status
Power Board	X35	S	+3V3_X0	Synth Board	X10 (N\$69 → L9 → +3V3_X0)	S	DC power 3.3 V	1200+25+25 mA (rows 2,4,5)	CBL-22	PB X35; SY X10, L9, X4/X5/X6 VDD	CONFIRMED
Power Board	X11	S	+3V3_CLOCK	Synth Board	X11 (N\$71 → L10 → +3V3_CLOCK)	S	DC power 3.3 V	250 mA (row 6)	CBL-23	PB X11; SY X11, L10, IC1 VDD3_*	CONFIRMED

Source device	Source connector	Pin	Signal	Destination device	Destination connector	Pin	Interface standard	Power requirement (xlsx)	Cable ID	Evidence	Status
Power Board	X10	S	+1V8_CL0CK	Synth Board	X12 (N\$73 → L11 → +1V8_CL0CK)	S	DC power 1.8 V	250 mA (row 7)	CBL-24	PB X10; SY X12, L11, IC1 VDD1.8_*	UNVERIFIED — same PB X10 also needed by Main X17 (CBL-06); a Y-cable or second output is not in CAD
Power Board	X8	S	+3V3_L0_1	Synth Board	X13 (N\$75 → L12 → +3V3_L0_1)	S	DC power 3.3 V	2×240 mA (row 11)	CBL-25	PB X8; SY X13, L12, U1/U6 V3_LDO/LS/NDIV/PFD/REF/SYNC	CONFIRMED
Power Board	X7	S	+3V3_L0_2	Synth Board	X14 (N\$77 → L13 → +3V3_L0_2)	S	DC power 3.3 V	2×340 mA (row 12)	CBL-26	PB X7; SY X14, L13, U1/U6 V3_OUTDIV/RFOUT/COB	CONFIRMED
Power Board	X6	S	+5V0_L0	Synth Board	X15 (+5V0_L0)	S	DC power 5 V	2×(200+0.3+70) mA (rows 8-10)	CBL-27	PB X6; SY X15, FB1..FB4	CONFIRMED

2.5 Main Board ↔ Frequency Synthesizer Board control headers (source: `interconnection_table.md` §5)

Source device	Source connector	Pin	Signal	Destination device	Destination connector	Pin	Interface standard	Power requirement	Cable ID	Evidence	Status
Main Board (U2 PF3)	JP1 PINHD-2X6	1	AD9523_PD	Synth Board	JP1 PINHD-2X6	1	3.3 V CMOS GPIO	—	CBL-28	MB JP1.1, U2.PF3 (<code>main.h:62</code>); SY JP1.1 → IC1	CONFIRMED
Main Board (PF4)	JP1	2	AD9523_REF_SEL	Synth Board	JP1	2	3.3 V CMOS	—	CBL-28	<code>main.h:64</code> ; SY IC1.REF_SEL	CONFIRMED
Main Board (PF5)	JP1	3	AD9523_SYNC	Synth Board	JP1	3	3.3 V CMOS	—	CBL-28	<code>main.h:66</code>	CONFIRMED
Main Board (PF6)	JP1	4	AD9523_RESET	Synth Board	JP1	4	3.3 V CMOS	—	CBL-28	<code>main.h:68</code>	CONFIRMED
Main Board (PF7)	JP1	5	AD9523_CS	Synth Board	JP1	5	SPI CS (3.3 V)	—	CBL-28	<code>main.h:70</code> ; firmware never toggles it (docs C7)	CONFIRMED (hardware)
Main Board (PE2 SPI4_SCK)	JP1	6	STM32_SCLK4 → AD9523_SCLK	Synth Board	JP1	6	SPI 3.3 V (R6 0.65 k series)	—	CBL-28	MB U2.PE2; SY JP1.6, R6, IC1.SCLK	CONFIRMED
Main Board (PE6 SPI4_MOSI)	JP1	7	STM32_MOSI4 → AD9523_SDIO	Synth Board	JP1	7	SPI 3.3 V (R8 0.65 k)	—	CBL-28	MB U2.PE6; SY R8, IC1.SDIO	CONFIRMED
Synth Board (IC1 SDO)	JP1	8	AD9523_SDO → STM32_MIS04	Main Board (PE5)	JP1	8	SPI 3.3 V	—	CBL-28	SY IC1.SDO; MB U2.PE5	CONFIRMED
Synth Board	JP1	9 / 10	AD9523_STATUS0 / STATUS1	Main Board (PF8 / PF9)	JP1	9 / 10	3.3 V CMOS	—	CBL-28	<code>main.h:72-75</code>	CONFIRMED
Main Board (PF10)	JP1	11	AD9523_EEPROM_SEL	Synth Board	JP1	11	3.3 V CMOS	—	CBL-28	<code>main.h:76</code>	CONFIRMED
—	JP1	12	GND	—	JP1	12	—	—	CBL-28	both	CONFIRMED
Synth Board (U1/U6 SDO)	JP2 PINHD-2X7	1	ADF4382_SDO → STM32_MIS04	Main Board (PE5)	JP13 PINHD-2X7	1	SPI 3.3 V (shared SPI4 bus with AD9523)	—	CBL-29	SY JP2.1; MB JP13.1	CONFIRMED
Main Board (PE2)	JP13	2	STM32_SCLK4 → ADF4382_SCLK	Synth Board	JP2	2	SPI 3.3 V	—	CBL-29	MB JP13.2; SY JP2.2, U1/U6.SCLK	CONFIRMED
Main Board (PE6)	JP13	3	STM32_MOSI4 → ADF4382_SDIO	Synth Board	JP2	3	SPI 3.3 V	—	CBL-29	MB JP13.3; SY JP2.3	CONFIRMED
Main Board (PG14)	JP13	4	ADF4382_TX_CS	Synth Board (U1 CSB, 200 k pull-up R24)	JP2	4	SPI CS 3.3 V	—	CBL-29	<code>main.h:154</code> ; SY R24	CONFIRMED (K6: <code>adf4382a_manager.h</code> expects other pins)
Main Board (PG15)	JP13	5	ADF4382_TX_CE	Synth Board (U1 CE, R25)	JP2	5	3.3 V CMOS	—	CBL-29	<code>main.h:156</code>	CONFIRMED
Main Board (PG12)	JP13	6	ADF4382_TX_DELSTR	Synth Board (U1)	JP2	6	3.3 V CMOS	—	CBL-29	<code>main.h:150</code>	CONFIRMED
Main Board (PG13)	JP13	7	ADF4382_TX_DELADJ	Synth Board (U1)	JP2	7	3.3 V CMOS	—	CBL-29	<code>main.h:152</code>	CONFIRMED

Source device	Source connector	Pin	Signal	Destination device	Destination connector	Pin	Interface standard	Power requirement	Cable ID	Evidence	Status
Synth Board (U1 LKDET)	JP2	8	ADF4382_TX_LKDET	Main Board (PG11)	JP13	8	3.3 V CMOS	—	CBL-29	main.h:148	CONFIRMED
Main Board (PG10)	JP13	9	ADF4382_RX_CS	Synth Board (U6 CSB, R37)	JP2	9	SPI CS	—	CBL-29	main.h:146	CONFIRMED
Main Board (PG9)	JP13	10	ADF4382_RX_CE	Synth Board (U6 CE, R38)	JP2	10	3.3 V CMOS	—	CBL-29	main.h:144	CONFIRMED
Main Board (PG8)	JP13	11	ADF4382_RX_DELSTR	Synth Board (U6)	JP2	11	3.3 V CMOS	—	CBL-29	main.h:128	CONFIRMED
Main Board (PG7)	JP13	12	ADF4382_RX_DELADJ	Synth Board (U6)	JP2	12	3.3 V CMOS	—	CBL-29	main.h:126	CONFIRMED
Synth Board (U6 LKDET)	JP2	13	ADF4382_RX_LKDET	Main Board (PG6)	JP13	13	3.3 V CMOS	—	CBL-29	main.h:124	CONFIRMED
—	JP13	14	GND	—	JP2	14	—	—	CBL-29	both	CONFIRMED

2.6 Frequency Synthesizer Board → Main Board clocks and LO, coaxial (source: interconnection_table.md §6; frequencies are the values programmed in main.cpp:970–1028 and adf4382a_manager.h:32–34, firmware intent, not measurements)

Source device	Source connector	Pin	Signal	Destination device	Destination connector	Pin	Interface standard	Power requirement	Cable ID	Evidence	Status
Synth (IC1 OUT6 → R39 22 R → C15 10 nF)	J7 SMA 142-0731-211, silk FPGA SYS. CLOCK	1	AD9523_OUT6+ 100 MHz LVCMOS → FPGA_SYS_CLOCK	Main Board (U42 IO_L13P_T2_MRCC_15)	J1 SMA	1	50 Ω coax, single-ended clock	—	CBL-30	SY J7, R39, C15; MB J1, U42; main.cpp:1004	CONFIRMED
Synth (OUT10 → R40 → C17)	J5 SMA, silk DAC	1	120 MHz LVCMOS → DAC_CLOCK	Main Board (U3 AD9708 CLOCK via R31)	J20 SMA	1	50 Ω coax	—	CBL-31	SY J5; MB J20, R31, U3	CONFIRMED
Synth (OUT11 → R41 → C19)	J6 SMA, silk FPGA=DAC	1	120 MHz LVCMOS → FPGA_DAC_CLOCK	Main Board (U42 IO_L12N_T1_MRCC_15)	J18 SMA	1	50 Ω coax	—	CBL-32	SY J6; MB J18, U42	CONFIRMED
Synth (OUT7)	J8 SMA, silk TEST	1	20 MHz LVCMOS → FPGA_CLOCK_TEST	Main Board (U42 IO_L24P_T3_RS1_15)	JP20 PINHD-1X2	1 (2 = GND)	SMA → 0.1" header (adapter not specified)	—	CBL-33	SY J8; MB JP20	UNVERIFIED (adapter)
Synth (OUT4)	J3 CJT-T-P-HH-ST-TH1, silk ADC	1 P, 2 N	AD9523_OUT4_P/N 400 MHz LVDS → N\$2_P/N → C1/C2 → ADC_CLK_IN_P/N	Main Board (U1 AD9484 CLK+/CLK-, R173 100 Ω)	J21 CJT	1 P, 2 N	twinax differential	—	CBL-34	SY J3; MB J21, C1, C2, R173, U1	CONFIRMED
Synth (OUT5)	J4 CJT, silk FPGA=ADC	1 P, 2 N	400 MHz LVDS → FPGA_ADC_CLOCK_P/N	Main Board (U42 IO_L13P/N_T2_MRCC_14 , R4)	J19 CJT	1 P, 2 N	twinax differential	—	CBL-35	SY J4; MB J19, U42	CONFIRMED (not used by RTL)
Synth (U1 ADF4382 TX RFOUT1 → C64/C65 → U2 MTX2-143+ → U4 ATS1005 -3 dB)	J10 SMA, silk LO TX	1	TX LO 10.5 GHz (TX_FREQ_HZ)	Main Board (U5 LTC5552 LO via C272 18 pF)	J23 SMA (N\$23)	1	50 Ω coax, X-band	—	CBL-36	SY U1, U2, U4, J10 (N\$30), brd silk; MB J23, C272, U5.LO	CONFIRMED (silk-to-net association from brd text proximity)
Synth (U6 ADF4382 RX RFOUT1 → U7 → U8)	J11 SMA, silk LO RX	1	RX LO 10.38 GHz (RX_FREQ_HZ)	Main Board (U13 LTC5552 LO via C274 18 pF)	J22 SMA (N\$24)	1	50 Ω coax, X-band	—	CBL-37	SY U6, U7, U8, J11 (N\$55); MB J22, C274, U13.LO	CONFIRMED
Synth (U1 RFOUT2 → U3 → U5)	J1 SMA, silk AUX. LO TX	1	auxiliary TX LO	—	—	—	—	—	—	SY J1 (N\$35)	UNVERIFIED (no destination)
Synth (U6 RFOUT2 → U5)	J12 SMA (silk by	1	auxiliary RX LO	—	—	—	—	—	—	SY J12 (N\$59)	UNVERIFIED (no

Source device	Source connector	Pin	Signal	Destination device	Destination connector	Pin	Interface standard	Power requirement	Cable ID	Evidence	Status
→ U9 → U10)	elimination AUX. L0 RX)										destination, silk not adjacent)
Synth (U1 MUXOUT / U6 MUXOUT / IC1 PLL_OUT)	J9 / J2 / J13 SMA	1	test outputs	—	—	—	—	—	—	SY J9 (N\$37), J2 (N\$51), J13 (N\$61)	test points, no counterpart

2.7 Main Board ↔ RF PA boards, 16 instances (source: `interconnection_table.md` §7)

Source device	Source connector	Pin	Signal	Destination device	Destination connector	Pin	Interface standard	Power requirement	Cable ID	Evidence	Status
Main Board RF_SW_n (n = 1..16) RFOUT1 / RFOUT2 via 1 pF	J27/J26 (n=1), J29/J28 (2), J25/J24 (3), J31/J30 (4), J35/J34 (5), J37/J36 (6), J33/J32 (7), J39/J38 (8), J47/J46 (9), J41/J40 (10), J45/J44 (11), J43/J42 (12), J55/J54 (13), J49/J48 (14), J53/J52 (15), J51/J50 (16)	1	element RF (TX to PA / RX return)	RF PA board n	J1 RFIN / J2 RFOUT	1	50 Ω coax, 10.5 GHz	—	CBL-40..CBL-55	MB RF_SW_n pins (traced through C126/C128-type 1 pF caps); PA J1 (N\$2), J2 (N\$8), brd silk	UNVERIFIED — which of each p RFIN vs RFOUT a PA-instan mapping not documen
Main Board (U7/U69 DAC5578 → OPA4703 → VG_n)	X_7=VG_1, X_16=VG_2, X_8=VG_3, X_15=VG_4, X_4=VG_5, X_11=VG_6, X_3=VG_7, X_12=VG_8, X_5=VG_9, X_14=VG_10, X_6=VG_11, X_13=VG_12, X_2=VG_13, X_9=VG_14, X_1=VG_15, X_10=VG_16 (Molex 22-23-2021)	S (VG), S (GND)	gate bias VG_n (xlsx: ~4 ... -1.2 V, 10 mA)	RF PA board	X2 Molex 22-23-2021	1 VG, 2 GND	analogue DC	10 mA (row 60)	CBL-56..CBL-71	MB X_n nets VG_n; PA X2	CONFIRMED (VG_n → instance assignme UNVERIFIED
RF PA board (R10 WSL2816 5 mΩ shunt: VD / VIN_M)	X3 Molex 22-23-2031	1 VD, 2 VIN_M, 3 GND	drain current sense pair	Main Board (INA241A3 U11 for X3, U73 for X38, ...)	X3, X38..X52 Molex 22-23-2031	S (GND), S (N\$207 =IN+), S (N\$209 =IN-) (X3 example)	analogue differential sense	—	CBL-72..CBL-87	PA X3, R10; MB X3 → U11.IN+/IN-, X38 → U73 ...	CONFIRMED topology; ORDER UNVERIFIED (pads nar S), PA-instance Main-con map NOT DOCUMENTED
+22 V drain supply (NOT IN CAD)	—	—	VD 22 V (xlsx row 59: 18–22 V, 2000 mA, "Set VD +22 V")	RF PA board	22V AK300/2 terminal	KL1 VD, KL2 GND	DC power	2 A per board (xlsx)	CBL-88	PA part 22V; xlsx row 59; Power Board has no 22 V rail (K4)	MISSING SPEC
Main Board (U2 PD6)	JP10 PINHD-1X3	1 EN/DIS_RFPA_VDD, 3 GND	PA drain enable (set in main.cpp:1601)	PA drain switch — device NOT IN CAD	—	—	3.3 V CMOS	—	—	MB JP10; main.h:140	MISSING SPEC

2.8 Main Board ↔ host and off-board modules (source: `interconnection_table.md` §8)

Source device	Source connector	Pin	Signal	Destination device	Destination connector	Pin	Interface standard	Power requirement	Cable ID	Evidence	Status
Main Board (U2 PA11 / PA12 / PA10)	X53 MINI-USB	2 STM32_USB_FS_D_N , 3 D_P , 4 ID , 5 GND (1 VBUS not connected in sch)	USB 2.0 full-speed, CDC class (firmware usbd_cdc_if references)	Host PC	USB-A/host port	—	USB 2.0 FS	bus-powered: NO (VBUS pin unconnected; VDDUSB = +3V3)	CBL-90	MB X53, U2.PA10-PA12, VDDUSB	CONFIRMED (host side MISSING SPEC)
Main Board (U6 FT601)	—	—	USB 3.0 (RTL radar_system_top.v:79–97 , GUI_V6.py FT601 class)	Host PC	—	—	USB 3.0	—	—	MB U6: 0 nets	MISSING SPEC — no hardware path (K3)
Main Board	JP2 PINHD-1X6	1 +3V3 , 2 STM32_SWCLK , 3 GND , 4 STM32_SWDIO , 5 STM32_Nrst , 6 STM32_SW0	SWD	debug probe (not specified)	—	—	ARM SWD 3.3 V	—	CBL-91	MB JP2	CONFIRMED (probe MISSING SPEC)
Main Board (U2 PC12 / PD2)	JP8 PINHD-1X4	1 +3V3 , 2 STM32_TX5 , 3 STM32_RX5 , 4 GND	UART5 (9600 Bd per firmware GPS driver)	GPS module NEO-6M (xlsx row 49)	module header	—	UART 3.3 V	50 mA (row 49)	CBL-92	MB JP8, U2.PC12/PD2	UNVERIFIED (module pinout)
Main Board (U2 PA8 / PC9 / PC6 / PC7 / PC8)	JP7 PINHD-1X8	2 +3V3 , 3 GND , 4 STM32_SCL3 , 5 STM32_SDA3 , 6 MAG_DRDY , 7 ACC_INT , 8 GYR_INT (pin 1 unconnected)	I2C3 + interrupts	GY-85 IMU (xlsx row 47)	module header	—	I2C 3.3 V	10 mA	CBL-93	MB JP7; main.h:130–135	UNVERIFIED (module pinout)
Main Board	JP18 PINHD-1X4	1 STM32_SDA3 , 2 STM32_SCL3 , 3 GND , 4 +3V3	I2C3	BMP180 barometer (xlsx row 48)	module header	—	I2C 3.3 V	1 mA	CBL-94	MB JP18	UNVERIFIED (module pinout; JP7/JP18 assignment to IMU vs BMP180 is an assumption)
Main Board	JP5, JP6, JP11, JP12, JP14, JP15, JP16, JP19 PINHD-1X3	1 +3V3_AN4_F , 2 analogue (N\$295, N\$296, N\$306, N\$298, N\$299, N\$307, N\$301, N\$297) , 3 GND	TMP37 output → ADS7830 channel	8 × TMP37 (xlsx row 50)	sensor leads	—	analogue 3.3 V	1 mA each	CBL-95..CBL-102	MB JP5..JP19 nets	CONFIRMED (header) / channel order UNVERIFIED
Main Board (U2 PD4 / PD5)	JP9 PINHD-1X4	1 GND, 2 STEPPER_CW+ , 3 GND, 4 STEPPER_CLK+	direction / step	stepper driver "TBS6600" (xlsx row 63; TB6600-class)	driver inputs	—	3.3 V logic	driver 4000 mA @ 20 V from its own supply (row 63)	CBL-103	MB JP9; main.h:136–139	UNVERIFIED (driver pinout, supply not in CAD)
Main Board (U2 PD7)	JP4 PINHD-1X3	1 EN/DIS_COOLING , 3 GND	fan relay control (main.cpp:1767–1770)	cooling relay / fans (xlsx row 64)	—	—	3.3 V logic	UNKNOWN	CBL-104	MB JP4; main.h:142	MISSING SPEC (relay/fan parts)
Main Board (U2 PB10 / PB11)	JP17 PINHD-1X3	1 STM32_TX3 , 2 STM32_RX3 , 3 GND	USART3 debug console (huart3 in main.cpp)	serial adapter (not specified)	—	—	UART 3.3 V	—	CBL-105	MB JP17	CONFIRMED (header)
Main Board	JP3 PINHD-2X4	1 N\$32 , 2 +3V3_FPGA , 3 N\$33 , 5 N\$34 , 7 N\$36 , 8 GND	FPGA JTAG header (nets reach U42 TCK/TMS/TDI/TDO through R62/R49/R64/R63 — see docs/FPGA/)	JTAG programmer	—	—	JTAG 3.3 V	—	CBL-106	MB JP3	UNVERIFIED (pin-to-TAP assignment not traced here)

2.9 Antenna (source: interconnection_table.md §9)

Source device	Source connector	Pin	Signal	Destination device	Destination connector	Pin	Interface standard	Power requirement	Cable ID	Evidence	Status
RF PA board n	J2 RFOUT	1	element n TX (AERIS-10X)	antenna array element	—	—	50 Ω coax / waveguide transition	10 W per element (README)	—	README.md; RADAR_V6.drawio	MISSING SPEC (no antenna CAD, K8)

Source device	Source connector	Pin	Signal	Destination device	Destination connector	Pin	Interface standard	Power requirement	Cable ID	Evidence	Status
Main Board	J24..J55	1	element RF (AERIS-10N direct)	antenna array element	—	—	50 Ω	~1 W per element (README)	—	README.md	MISSING SPEC

2.10 Items left UNVERIFIED by SYS-02 and why (source: engineering/SYSTEM/architecture/README.md, "Items left UNVERIFIED", copied)

Item	Reason
Pin order of all Molex 22-23-2021/2031 connections	EAGLE symbol names every pad S; order not recoverable from the schematic
Which SMA of each element pair (e.g. J27/J26) goes to PA RFIN vs returns from RFOUT, and PA-board-instance → Main-connector assignment	no silkscreen text on the Main Board .brd, no cabling document
Synth J12 function (AUX..LO RX)	silkscreen text not adjacent; assigned by elimination
Oscillator identities (X4 100 MHz OCXO vs net 10MHz_OUT; X5/X6 50 MHz VCXO vs nets 100MHz_OUT / VCXO_OUT and firmware 100 MHz)	contradictory sources (K7)
+1V8_CLOCK distribution to both Main X17 and Synth X12 from the single Power output X10	no cable/Y-splice in CAD
22 V PA drain source and the switch driven by EN/DIS_RFPA_VDD	absent from all CAD (K4)
GPS / IMU / BMP180 / stepper-driver / relay module pinouts and supplies	modules named only in the xlsx; no module schematics
JP3 pin → FPGA TAP signal assignment	nets N\$32..N\$36 not traced here (see docs/FPGA/)
Host-link packet format, USB 3.0 path	no FT601 nets (K3); RTL/GUI packet formats differ (K5)
DIG_0..7 bit meaning	inferred from firmware/RTL names only (docs/SYSTEM/BLOCK_DIAGRAM.md §3)
Output voltages, regulator current capability, VIN total current	dividers listed, no datasheet for TPS562208 in repo; no budget sum in xlsx
Cable types, lengths, coax grade, wire gauge	not in any repository file
Antenna feed network, element-to-port mapping, waveguide transition	no antenna CAD (K8)

3. Interface implementation status

The electrical existence of an interface says nothing about whether the firmware or the RTL drives it. The cross-check per interface (source: docs/SYSTEM/BLOCK_DIAGRAM.md §3, copied verbatim; the RTL column describes the original 9_Firmware/9_2_FPGA, the BETA state is in chapters 2 and 11):

Interface	CAD	Firmware	RTL	Status
STM32 → FPGA handshake (DIG_0..4)	yes	main.cpp:448,484,514,1483,1660	stm32_new_*, stm32_mixers_enable, reset_n	bit mapping inferred (MEDIUM)
STM32 SPI1 → FPGA → ADAR1000 (1.8 V)	yes	ADAR1000_Manager.cpp on hspi1	level shifter not instantiated (radar_transmitter.v:59–71 undriven)	BROKEN in RTL
FPGA → host (FT601)	not wired	—	usb_data_interface.v	NO HARDWARE
STM32 ↔ host (USB CDC)	yes	RX callback never bound (C4); start-flag padding (C5)	—	BROKEN in firmware
STM32 → AD9523 (SPI4)	yes	CS never toggled, 36 MHz (C7)	—	questionable
STM32 → ADF4382 (SPI4)	yes	pin macros collide with PA enables (C2); platform_ops=NULL (C3)	—	BROKEN
AD9523 → FPGA clocks	yes (J1, J18, J19)	OUT6/OUT11/OUT5 programmed	clk_100m, clk_120m_dac; 400 MHz path uses ADC DCO only	partially consistent
ADC → FPGA LVDS	yes (bank 14 at 3.3 V)	—	capture not implemented; XDC wants LVDS_25 + DIFF_TERM	BROKEN / bank-voltage conflict
GPS → host	UART5 + USB	GPS_Init never called (C6)	—	BROKEN

The firmware defects C2–C7 are fixed in beta/stm32 (ADF4382 pin collision, platform_ops = NULL, unbound CDC RX, padded settings frames, GPS_Init, AD9523 CS/SPI speed) and the RTL defects in beta/fpga (level-shifter pass-through instantiated, ISERDES capture, SPI bridge); none of these fixes has been exercised on hardware (source: docs/AERIS10_BETA_ENGINEERING_REPORT.md §3–4).

4. Power distribution

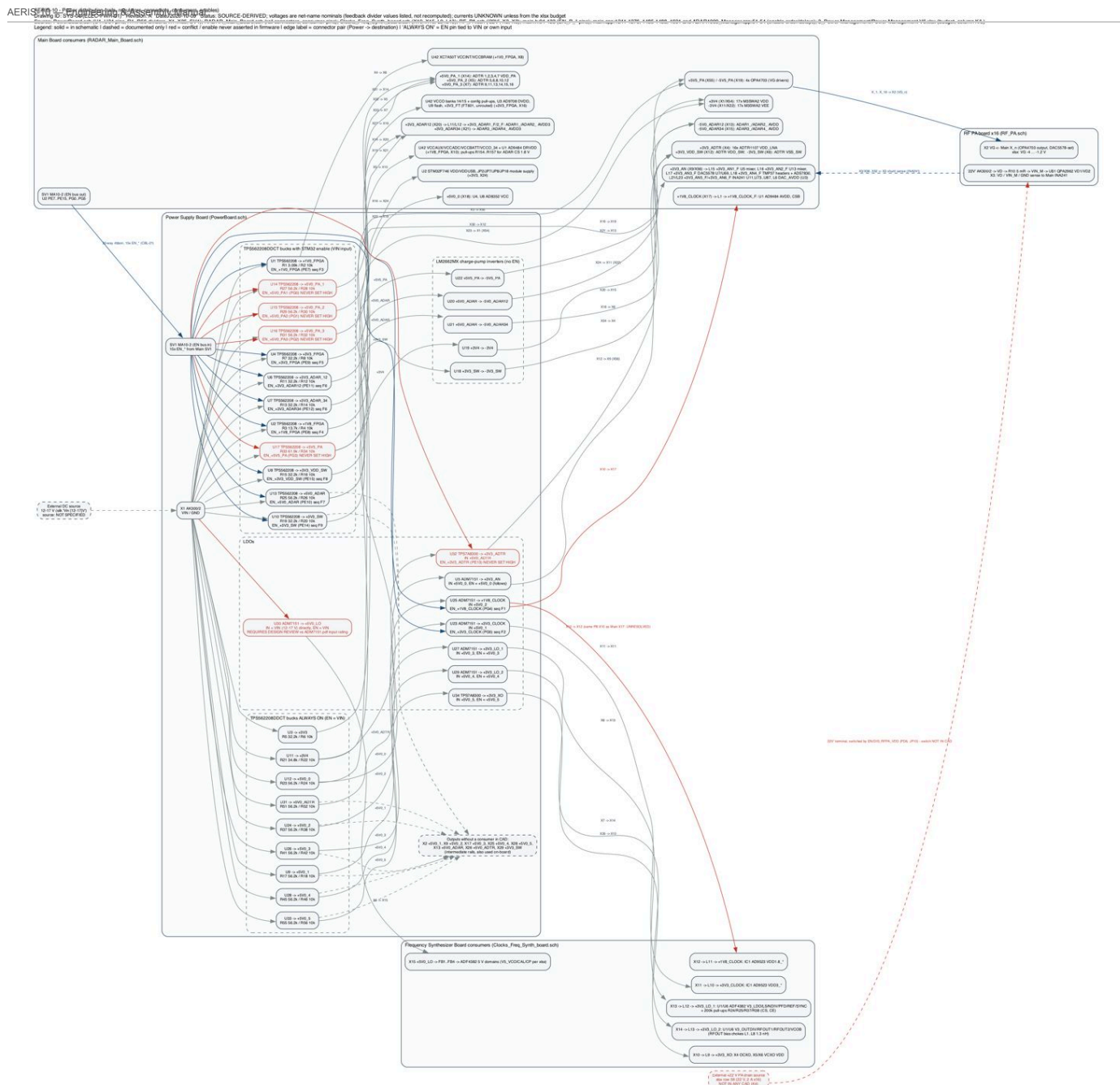


Figure 9 — F3.2 — Power distribution diagram ELEC-PWR-01 / SYS-04: every Power Board rail with regulator, feedback divider, source rail, enable net and MCU pin, firmware sequence step, output connector → destination connector → consumer — SOURCE-DERIVED; output voltages from net names, currents UNKNOWN (source: engineering/ELECTRICAL/power_distribution/power_distribution.dot; produced by Graphviz dot -Tpng -Gdpi=150)

Conventions of the rail register (source: engineering/ELECTRICAL/power_distribution/power_rails.md, lead-in): **Nominal V** is the value encoded in the net name and the xlsx column H "SELECTED VOLTAGE"; feedback resistor values are listed as evidence but output voltages were **not recomputed** (the TPS562208 datasheet is not in the repository — only tps562201.pdf). **Current ratings are UNKNOWN** unless the xlsx gives a per-device budget; regulator capability is not asserted. **Enable** = net on the Power Board EN pin (VIN or own input = always on). Sequence step: xlsx = text of column L/K, FW = firmware order F1..F10 of section 4.2. Status: **CONFIRMED** = regulator, nets and consumer connector all found; **PARTIAL** = found but something inconsistent; **UNVERIFIED** = consumer or cable cannot be established; **CONFLICT** = contradicts another source.

4.1 Rail register (source: power_rails.md §1, copied verbatim)

Rail	Nominal V (source)	Regulator (ref, part)	Source rail	Consumers (board: connector → loads)	Enable signal (MCU pin)	Sequence step	Evidence
VIN	12–17 V (brd silk Vin [12–17]V)	— (input)	external DC source (not specified)	all 21 bucks; U30 ADM7151 directly	—	—	PB X1 AK300/2 (KL: VII)
+1V0_FPGA	1.0 V (net; xlsx row 17 "1")	U1 TPS562208DDCT, FB R1 3.09 k / R2 10 k	VIN	Main: X4→X8 → U42 VCCINT, VCCBRAM	EN_+1V0_FPGA PE7 (main.h:98)	xlsx: FPGA rows order +1V0 → +1V8 → +3V3 (L15-L17), K=YES; FW: F3 (main.cpp:1271, +100.ms)	PB U1 pins; MB X8, U42

Rail	Nominal V (source)	Regulator (ref. part)	Source rail	Consumers (board: connector → loads)	Enable signal (MCU pin)	Sequence step	Evidence
+1V8_FPGA	1.8 V (xlsx rows 14,16,26)	U2 TPS562208, R3 13.7 k / R4 10 k	VIN	Main: X5→X10 → U42 VCCAUX, VCCADC, VCCBATT, VCCO_34; U1 AD9484 DRVDD; R154–R157 pull-ups (ADAR CS 1.8 V)	EN_+1V8_FPGA PE8	K=YES; FW: F4 (main.cpp:1273 , +100 ms)	PB U2; MB X10
+3V3_FPGA	3.3 V (rows 15,18,20,27)	U4 TPS562208, R7 32.2 k / R8 10 k	VIN	Main: X27→X16 → U42 VCCO banks 14/15 + config pull-ups, U3 AD9708 DVDD, U9 flash; L19 → +3V3_FT (FT601, unrouted)	EN_+3V3_FPGA PE9	K=YES; FW: F5 (main.cpp:1275 , +100 ms)	PB U4; MB X16
+3V3	3.3 V (rows 13,47-49,55)	U3 TPS562208, R5 32.2 k / R6 10 k	VIN	Main: X16→X24 → U2 STM32 VDD/VDDUSB, headers JP2/JP7/JP8/JP18 (+3V3 pin for modules), ADS7830 VREF (xlsx row 55)	always on (EN = VIN)	xlsx K=NO	PB U3; MB X24, U2.VDD
+3V3_AN	3.3 V (rows 19,21,50,51,54,56)	U5 ADM7151ACPZ-04, EN tied to its input	+5V0_0 (U12)	Main: X12→X9 (and X56 same net) → L15 +3V3_AN1_F U5 mixer VCC; L16 +3V3_AN2_F U13 mixer; L17 +3V3_AN3_F U7/U69 DAC5578; L18 +3V3_AN4_F TMP37 headers + ADS7830; L21/L23 +3V3_AN5_F / _6_F 16 × INA241; L6 BLM15H DAC_AVDD U3	none (follows +5V0_0)	K=NO	PB U5; MB X9, X56, L6/L
+1V8_CLOCK	1.8 V (rows 7, 25)	U25 ADM7151, EN = EN_+1V8_CLOCK	+5V0_2 (U24)	Synth: X10→X12 → L11 → IC1 AD9523 VDD1.8_OUT*, VDD1.8_PLL2. Main: X10→X17 → L1 → +1V8_CLOCK_F → U1 AD9484 AVDD ×11, CSB	EN_+1V8_CLOCK PG4 (main.h:120)	xlsx L6: "bring the 1.8 V supplies high ... prior to or simultaneously with the 3.3 V"; FW: F1 (main.cpp:1241 , +100 ms)	PB U25, X10; SY X12, L1
+3V3_CLOCK	3.3 V (row 6)	U23 ADM7151, EN = EN_+3V3_CLOCK	+5V0_1 (U9)	Synth: X11→X11 → L10 → IC1 VDD3_OUT*, VDD3_PLL1/2, VDD3_REF	EN_+3V3_CLOCK PG5 (main.h:122)	xlsx L6 (after 1.8 V), K=YES; FW: F2 (main.cpp:1243 , +100 ms, then AD9523 RESET released)	PB U23, X11; SY X11, L1
+3V3_X0	3.3 V (rows 2-5)	U34 TPS7A8300RGR, EN = its input	+5V0_5 (U33)	Synth: X35→X10 → L9 → X4 OCXO SUPPLY_VOLTAGE (+R2/R3 VC divider), X5/X6 VCXO VDD	none (always on when VIN present)	K=NO	PB U34, X35; SY X10, L9
+5V0_L0	5.0 V (rows 8-10)	U30 ADM7151, EN = VIN	VIN directly (12–17 V)	Synth: X6→X15 → FB1..FB4 ferrites → ADF4382 U1/U6 5 V domains	none	K=NO	PB U30 (VIN = VIN), X6
+3V3_L0_1	3.3 V (row 11)	U27 ADM7151, EN = its input	+5V0_3 (U26)	Synth: X8→X13 → L12 → U1/U6 V3_LDO, V3_LS, V3_NDIV, V3_PFD, V3_REF, V3_SYNC; 200 k pull-ups R24/R25/R37/R38	none	K=NO	PB U27, X8; SY X13, L12
+3V3_L0_2	3.3 V (row 12)	U29 ADM7151, EN = its input	+5V0_4 (U28)	Synth: X7→X14 → L13 → U1/U6 V3_OUTDIV, V3_RFOUT1/2, V3_VCOB; RFOUT bias chokes L1..L8	none	K=NO	PB U29, X7; SY X14, L13

Rail	Nominal V (source)	Regulator (ref. part)	Source rail	Consumers (board: connector → loads)	Enable signal (MCU pin)	Sequence step	Evidence
+5V0_ADAR	5.0 V	U13 TPS562208, R25 56.2 k / R26 10 k	VIN	Power Board only: U20, U21 LM2662 inputs; X13 output has no consumer on Main	EN_+5V0_ADAR PE10 (main.h:104)	FW: F7 (main.cpp:1488, +500 ms) — enables the -5 V ADAR rails indirectly	PB U13, U20, U21, X13; I +5V0_ADAR net
-5V0_ADAR12	-5 V (row 29)	U20 LM2662MX/NOPB (charge-pump inverter)	+5V0_ADAR	Main: X21→X13 → ADAR1_, ADAR2_ AVDD	via EN_+5V0_ADAR	xlsx L30: AVDD3 (3.3 V) before or with AVDD1 (-5 V); FW: F7 after F6 (+500 ms) — consistent	PB U20, X21; MB X13
-5V0_ADAR34	-5 V	U21 LM2662	+5V0_ADAR	Main: X20→X15 → ADAR3_, ADAR4_ AVDD	via EN_+5V0_ADAR	as above	PB U21, X20; MB X15
+3V3_ADAR_12 (Main: +3V3_ADAR12)	3.3 V (row 30)	U6 TPS562208, R11 32.2 k / R12 10 k	VIN	Main: X14→X20 → L11 +3V3_ADAR1_F ADAR1_ AVDD3 ×3; L12 +3V3_ADAR2_F ADAR2_	EN_+3V3_ADAR12 PE11 (main.h:106)	K=YES; FW: F6 (main.cpp:1485, +500 ms with ADAR34)	PB U6, X14; MB X20, L11
+3V3_ADAR_34 (Main: +3V3_ADAR34)	3.3 V	U7 TPS562208, R13 32.2 k / R14 10 k	VIN	Main: X15→X21 → ADAR3_, ADAR4_ AVDD3 (filters analogous)	EN_+3V3_ADAR34 PE12 (main.h:108)	FW: F6 (main.cpp:1486)	PB U7, X15; MB X21
+5V0_ADTR	5.0 V	U31 TPS562208, R51 56.2 k / R52 10 k	VIN	Power Board only: U32 TPS7A8300 input; X26 output has no consumer	always on (EN = VIN)	—	PB U31, U32, X26
+3V3_ADTR	3.3 V (row 35, VDD_LNA 16 × 80 mA)	U32 TPS7A8300RGRR, EN = EN_+3V3_ADTR	+5V0_ADTR	Main: X34→X4 → 16 × ADTR1107 VDD_LNA	EN_+3V3_ADTR PE13 (main.h:110)	xlsx RX power-up step 8 "Set VDD_LNA to 3.3 V" (M38), TX step 6 VDD_LNA = 0 V; FW: never set HIGH — only reset at init (main.cpp:2245–2247) and power-down (main.cpp:394)	PB U32, X34; MB X4; gre 9_Firmware/9_1_Micr for EN_P_3V3_ADTR_Pi GPIO_PIN_SET = 0 hits
+3V3_VDD_SW	3.3 V (row 38)	U8 TPS562208, R15 32.2 k / R16 10 k	VIN	Main: X30→X12 → 16 × ADTR1107 VDD_SW	EN_+3V3_VDD_SW PE15 (main.h:114)	xlsx TX/RX step 2 "Set VDD_SW to 3.3 V" (first ADTR rail); FW: F8 (ADAR1000_Manager.cpp:51, +2 ms)	PB U8, X30; MB X12
+3V3_SW	3.3 V	U10 TPS562208, R19 32.2 k / R20 10 k	VIN	Power Board only: U18 LM2662 input; X29 output has no consumer	EN_+3V3_SW PE14 (main.h:112)	FW: F9 (ADAR1000_Manager.cpp:54, +2 ms) — enables -3V3_SW	PB U10, U18, X29
-3V3_SW	-3.3 V (row 37)	U18 LM2662	+3V3_SW	Main: X18→X6 → 16 × ADTR1107 VSS_SW	via EN_+3V3_SW	xlsx step 3 "Set VSS_SW to -3.3 V" after VDD_SW; FW F9 after F8 — consistent	PB U18, X18; MB X6
+5V0_PA_1	5.0 V (row 40, VDD_PA 250 mA each)	U14 TPS562208, R27 56.2 k / R28 10 k	VIN	Main: X31→X14 → ADTR1107_1, _2, _3, _4, _7 VDD_PA	EN_+5V0_PA1 PG0 (main.h:94)	xlsx TX step 8 "Set VDD_PA to 5 V"; FW: never set HIGH (reset at main.cpp:2239–2241, power-down :381)	PB U14, X31; MB X14, AI nets
+5V0_PA_2	5.0 V	U15 TPS562208, R29 56.2 k / R30 10 k	VIN	Main: X32→X5 → ADTR1107_5, _6, _8, _10, _12	EN_+5V0_PA2 PG1	as above	PB U15, X32; MB X5
+5V0_PA_3	5.0 V	U16 TPS562208, R31 56.2 k / R32 10 k	VIN	Main: X33→X7 → ADTR1107_9, _11, _13, _14, _15, _16	EN_+5V0_PA3 PG2	as above	PB U16, X33; MB X7
+5V5_PA	5.5 V (row 52)	U17 TPS562208, R33 61.9 k / R34 10 k	VIN	Main: X3→X55 → 4 × OPA4703 V+ (VG drivers); Power Board: U22 input	EN_+5V5_PA PG3 (main.h:118)	xlsx K=YES; FW: never set HIGH (reset at main.cpp:2239)	PB U17, X3; MB X55
-5V5_PA	-5.5 V (row 53)	U22 LM2662	+5V5_PA	Main: X19→X19 → 4 × OPA4703 V-	via EN_+5V5_PA	as above	PB U22, X19; MB X19
+3V4	3.4 V (row 22)	U11 TPS562208, R21 34.8 k / R22 10 k	VIN	Main: X23→X1 (X54 same net) → 17 × M3SWA2-34DR+ VDD; Power Board: U19 input	always on	K=NO	PB U11, X23; MB X1, X54
-3V4	-3.4 V (row 23)	U19 LM2662	+3V4	Main: X24→X11 (X22 same net) → 17 × M3SWA2 VEE	always on	K=NO	PB U19, X24; MB X11, X2
+5V0_0	5.0 V (row 24)	U12 TPS562208, R23 56.2 k / R24 10 k	VIN	Main: X22→X18 → U4, U8 AD8352 VCC; Power Board: U5 ADM7151 input	always on	K=NO	PB U12, X22; MB X18
+5V0_1	5.0 V	U9 TPS562208, R17 56.2 k / R18 10 k	VIN	Power Board: U23 input; X2 spare	always on	—	PB U9, X2

Rail	Nominal V (source)	Regulator (ref. part)	Source rail	Consumers (board: connector → loads)	Enable signal (MCU pin)	Sequence step	Evidence
+5V0_2	5.0 V	U24 TPS562208, R37 56.2 k / R38 10 k	VIN	Power Board: U25 input; X9 spare	always on	—	PB U24, X9
+5V0_3	5.0 V	U26 TPS562208, R41 56.2 k / R42 10 k	VIN	Power Board: U27 input; X17 spare	always on	—	PB U26, X17
+5V0_4	5.0 V	U28 TPS562208, R45 56.2 k / R46 10 k	VIN	Power Board: U29 input; X25 spare	always on	—	PB U28, X25
+5V0_5	5.0 V	U33 TPS562208, R55 56.2 k / R56 10 k	VIN	Power Board: U34 input; X28 spare	always on	—	PB U33, X28
+22V0 / VD (PA drain)	22 V (xlsx row 59: 18–22 V, 2000 mA × 16)	none in CAD	external	RF PA board: 22V AK300/2 → VD → R10 5 mΩ → VIN_M → QPA2962 VD1/VD2	EN/DIS_RFPA_VDD PD6 → JP10 (main.h:140; set at main.cpp:1601) — switch device NOT IN CAD	xlsx L58-L62 bias-up: ID limit 2840 mA, VG −4 V, then VD +22 V, raise VG until IDQ 1680 mA, then RF; K=YES	PA sch 22V, R10, X3; xls docs/PCB/POWER_SUPP
VG_1..16 (PA gate)	−4 ... −1.2 V (xlsx row 60; main.cpp:1583 sets −3.98 V = code 126)	Main Board OPA4703 (OPA_1.4) driven by U7/U69 DAC5578	+5V5_PA / −5V5_PA	Main X_1..X_16 → PA X2 → QPA2962 VG via R1..R3	DAC codes over I2C1; LDAC/CLR PB4/PB5/PB8/PB9	xlsx: VG before VD	MB VG_n nets, OPA_2.0 R1-R3
Stepper driver supply	20 V (xlsx row 63, 4000 mA)	none in CAD	external	"TBS6600" driver	—	—	xlsx row 63 only
Cooling	— (xlsx row 64, relay)	none in CAD	external	fans via relay, EN/DIS_COOLING PD7 (JP4)	PD7 (main.cpp:1767–1770)	"Apply when temperature reaches a threshold (Relay)"	xlsx row 64; MB JP4

Rails on the Main Board that have no Power-Board source: none (every Main Board rail connector maps to a Power Board output). Main Board rail names that differ from the Power Board: +3V3_ADAR12/34 vs +3V3_ADAR_12/34 (naming only; electrical identity assumed, to be confirmed by the designer) (source: power_rails.md §1, closing paragraph).

4.2 Firmware enable sequence as coded, not executed on hardware (source: power_rails.md §3, copied verbatim)

Step	Action	Delay after	Source
F0	HAL_Delay(180000) (3 min) then AD9523 RESET low	—	main.cpp:1237–1238
F1	EN_+1V8_CLOCK high	100 ms	main.cpp:1241–1242
F2	EN_+3V3_CLOCK high; AD9523 RESET high; configure_ad9523()	100 ms + 100 ms	main.cpp:1243–1267
F3	EN_+1V0_FPGA high	100 ms	main.cpp:1271–1272
F4	EN_+1V8_FPGA high	100 ms	main.cpp:1273–1274
F5	EN_+3V3_FPGA high	100 ms	main.cpp:1275–1276
F6	DIG_3 (PD11) low "mixers off"; EN_+3V3_ADAR12 + EN_+3V3_ADAR34 high	500 ms	main.cpp:1483–1487
F7	EN_+5V0_ADAR high (→ −5 V ADAR rails)	500 ms	main.cpp:1488–1489
F8	EN_+3V3_VDD_SW high	2 ms	ADAR1000_Manager.cpp:51–52 (powerUpSystem(); whether systemPowerUpSequence() reaches it via initializeADTR1107Sequence() is UNVERIFIED)
F9	EN_+3V3_SW high (→ −3V3_SW)	2 ms	ADAR1000_Manager.cpp:54–55
F10	DAC5578 VG codes written, LDAC pulsed, then EN/DIS_RFPA_VDD high (22 V drain)	—	main.cpp:1560–1601
never	EN_+3V3_ADTR, EN_+5V0_PA1/2/3, EN_+5V5_PA	—	no GPIO_PIN_SET write anywhere under 9_Firmware/9_1_Microcontroller/
power-down	ADAR RX mode → EN_+5V0_PA1..3 low → PA bias safe → EN_+3V3_ADTR low → LNA bias 0 → EN_+3V3_VDD_SW, EN_+3V3_SW low (10 ms steps)	10 ms	main.cpp:372–409

Comparison with the xlsx (source: power_rails.md §3): the 1.8 V-before-3.3 V rule for the AD9523 (L6) and the AVDD3-before-AVDD1 rule for the ADAR1000 (L30) are respected by F1→F2 and F6→F7. The ADTR1107 TX/RX procedures (L31-L46) require VDD_SW, VSS_SW, then VDD_LNA / VDD_PA — the firmware performs F8/F9 but never asserts the LNA (+3V3_ADTR) or PA (+5V0_PA_x) enables, so the ADTR1107 bias procedure cannot complete as coded. The QPA2962 bias-up (L58-L62: VG −4 V → VD 22 V → raise VG to IDQ 1680 mA) is partially mirrored by F10 (VG −3.98 V then drain enable), but the VG driver supply (+5V5_PA) is never enabled. The BETA report lists the firmware defects fixed in beta/stm32 (source: docs/AERIS10_BETA_ENGINEERING_REPORT.md §4); whether the five missing enables are among them is documented in chapter 12, not asserted here.

4.3 Open items for the power designer (source: power_rails.md §4, copied)

2. +5V0_L0 LDO U30 input tied to the 12–17 V bus — confirm against the ADM7151 rating or add a pre-regulator.
3. +1V8_CLOCK output X10 must feed both the Main Board (ADC AVDD) and the Synth Board (AD9523) — add a second output or document a Y-cable.
4. Five +5V0_n outputs (X2, X9, X17, X25, X28), X13, X26 and X29 have no destination — mark as spare or remove.
5. Rail current capability per regulator and the VIN total are not documented anywhere; the xlsx gives device budgets only.
6. Firmware: enables for +3V3_ADTR, +5V0_PA_1..3, +5V5_PA are missing (CONFLICT rows above).

5. Harness schedule (PROPOSED DESIGN)

The harness schedule DSN-HAR-01 is generated by `tools/design_mechanical_drawings.py` from the proposed internal layout (`tools/design_layout.py`) and the connector positions in the KiCad pick-and-place files. Length = Manhattan distance between connector positions in the head + service allowance (40 mm coax / 60 mm wire), rounded up to 10 mm; lengths are PROPOSED and are to be cut after a first fit. Totals: 144 cables; coax 55; wire 89; total proposed length ≈ 41.6 m (source: `engineering/DESIGN/HARNESS/HARNESS_SCHEDULE.md`, header). Cable types follow decision D-15: coax RG-405 (0.086") hand-formable equal-length for Main↔PA and Synth↔Main, Molex 22-01-3027/3037 crimp housings for the 2-/3-pin headers, 20-way IDC ribbon for SV1 (source: `engineering/DESIGN/00_DESIGN_BASIS.md` §2, D-15).

Grouping of the 144 rows (source: `HARNESS_SCHEDULE.md`, rows CBL-001...CBL-144; this grouping is a reading aid of the manual, the row content is the source's):

ID range	Count	From → to	Cable	Notes in the source
CBL-001 ... CBL-034	34	Power Supply X2..X35 → Main Board / Synth rail connectors	2-wire 20 AWG, Molex 22-01-2027 both ends	12 rows have destination "? ?" and length TBD: +5V0_1..5, +3V3_L0_1/2, +3V3_CLOCK, +5V0_ADAR, +3V3_ADAR_12/34, +5V0_ADTR, +3V3_SW — "no connector with this net name on Main/Synth (rail renamed through a filter/inductor or spare output) — resolve with <code>interconnection_table.md</code> "
CBL-035	1	Power Supply SV1 → Main Board SV1	enable bus (15 EN + GND), 320 mm	—
CBL-036 ... CBL-042	7	Synth J7/J5/J6/J3/J10/J11/J4 → Main J1/J20/J18/J21/J23/J22/J19	coax: 100 MHz sys clk, 120 MHz DAC ×2, 400 MHz ADC, LO TX, LO RX, test	100–170 mm
CBL-043 ... CBL-044	2	Synth JP1/JP2 → Main JP1/JP13	control headers	210 / 240 mm
CBL-045 ... CBL-140	96	per PA board n = 1..16, six cables each: Main X_n → PA X2 (VG_n, 2-wire 24 AWG shielded), Main X3/X38..X52 → PA X3 (drain current sense, 3-wire 24 AWG twisted), Main J-pair → PA J1 RFIN and PA J2 → Main J-pair (RG-405 SMA-SMA, EQUAL LENGTH set), PA J2 → antenna row n (RG-405 SMA-2.92 mm, EQUAL LENGTH set), DSN-PSU-01 OUTn → PA 22V (2-wire 18 AWG twisted, AK300/2)	"PA instance ↔ VG_n assignment PROPOSED (n = n); not documented in CAD"; "which SMA of the pair is RFIN/RFOUT is UNVERIFIED (<code>interconnection_table</code> §7)"; "antenna row n ↔ PA n PROPOSED; equal length mandatory for elevation phase"	
CBL-141 ... CBL-142	2	slip ring VIN → Power Supply X1; slip ring VIN → DSN-PSU-01 IN	2 × 2-wire 16 AWG	340 / 470 mm
CBL-143	1	slip ring USB → Main Board X53	USB 2.0 shielded, mini-B	360 mm
CBL-144	1	Main Board stepper pins → pedestal TB6600 (via slip ring)	3-wire 24 AWG shielded, TBD	"the driver sits on the fixed base; needs 3 more slip-ring circuits OR move the driver into the head (then only motor phases cross: 4 circuits) — DECISION NEEDED"

The full 144-row schedule is in [engineering/DESIGN/HARNESS/HARNESS_SCHEDULE.md](#) and, machine-readable, in `engineering/DESIGN/HARNESS/harness_schedule.csv`; Appendix B reproduces it.

Two cable-ID schemes coexist in the repository and must not be confused: the interconnection table uses two-digit IDs `CBL-00...CBL-106` assigned per signal group (e.g. `CBL-01` = +1V0_FPGA X4→X8, `CBL-21` = the whole SV1 ribbon, `CBL-40..55` = the RF pairs), whereas the harness schedule uses three-digit IDs `CBL-001...CBL-144` assigned per physical cable in connector order (e.g. `CBL-003` = +1V0_FPGA X4→X8, `CBL-035` = the SV1 ribbon, `CBL-047/048` = PA1 RF pair) (source: `engineering/SYSTEM/interfaces/interconnection_table.md` §2–3, §7; `engineering/DESIGN/HARNESS/HARNESS_SCHEDULE.md` rows CBL-003, CBL-035, CBL-047). The manual keeps each ID in the form of its source and records the duplication as observation MAN-03 in chapter 17. The twelve "? ?" rows of the harness schedule correspond to the rails that the interconnection table resolves either to Synth connectors through series inductors (+3V3_L0_1/2, +3V3_CLOCK → SY X13/X14/X11 via L12/L13/L10; CBL-23, CBL-25, CBL-26) or to spare outputs with no consumer (+5V0_1..5, +5V0_ADAR, +5V0_ADTR, +3V3_SW), and +3V3_ADAR_12/34, which the interconnection table matches to Main X20/X21 despite the one-underscore name difference (CBL-07, CBL-08) (source: `interconnection_table.md` §2, §4).

<!-- chapter 4: Main Board: schematic set, layout, BOM, DRC state, rev. B host interface -->

4. Main Board (RADAR_Main_Board)

Chapter status summary: schematic set — SOURCE-DERIVED (rendered from the EAGLE 7.4.0 file); connector and part inventory — SOURCE-DERIVED; layout — PARTIAL in the source (2 390 airwires), BETA in `beta/pcb/MAIN_BOARD/` (0 unconnected, 1 049 DRC items dispositioned); layer plots and 3-D renders — SOURCE-DERIVED / BETA (KiCad conversion, no component models); stack-up — PARTIAL (layer order from the DRU) with a PROPOSED fabrication proposal; BOM — PARTIAL (0 MPN in the source; BETA proposals with confidence classes); rev. B host interface — PROPOSED DESIGN (chapter 9). Nothing in this chapter is hardware-verified. Compiled by Antidrone Ukraine · antidrone.cc.

Sources: `docs/PCB/MAIN_BOARD.md`; `engineering/ELECTRICAL/connection_diagrams/MAIN_BOARD_connection_report.md` §1–2; `engineering/ELECTRICAL/schematics/MAIN_BOARD/` (`renders + sheets.json`); `engineering/PCB/MAIN_BOARD/README.md`, `STACKUP.md`; `beta/pcb/MAIN_BOARD/README.md`, `FAB_NOTES.md`, `BOM_MAIN_BOARD_beta.csv`; `engineering/SYSTEM/interfaces/interconnection_table.md`; `docs/SYSTEM/BLOCK_DIAGRAM.md` (K1–K8).

4.1 Role and key parts

The Main Board carries the complete digital and RF core of the radar head: the Artix-7 FPGA, the STM32F7 controller, the 500 MSPS ADC and the DAC, the four ADAR1000 beamformer ICs with sixteen ADTR1107 front-end modules, the two LTC5552 mixers, the PA bias/sense circuitry for sixteen external PA boards, and all power-entry connectors from the Power Board. It is the largest board of the set: **260 × 300 mm, 10 copper layers**, EAGLE 7.4.0 schematic and board, 776 physical parts, 625 nets, 2 893 vias (source: `docs/PCB/MAIN_BOARD.md` §1).

Key ICs — deviceset names from the schematic; the MPN status of every line is UNVERIFIED (source: `docs/PCB/MAIN_BOARD.md` §1):

Function	Part (refdes)	Qty	Note
FPGA	XC7A50T-2FTG256I (U42)	1	conflict K1: README/XDC name an XC7A100T (source: <code>docs/SYSTEM/BLOCK_DIAGRAM.md</code> K1)
MCU	STM32F746ZGT7 (U2)	1	HSE crystal NX3225GD-8MHZ on board vs. 25 MHz in firmware — K2
ADC	AD9484BCPZ-500 (U1)	1	
DAC	AD9708AR (U3)	1	
Beamformer	ADAR1000ACCZN	4	
T/R front end	ADTR1107ACCZ	16	
Mixers	LTC5552IUBTRMPBF (U5, U13)	2	
IF amplifiers	AD8352ACPZ-R7	2	EN_OPAMP_IF_1/2 are single-pin nets (ENB floating)
RF switches	M3SWA2-34DR+	17	
PA current sense	INA241A3IDGKR	16	
Temperature ADC	ADS7830IPWR	3	
PA gate-bias DACs	DAC5578SRGET (U7, U69)	2	thermal-pad pins missing from the symbol — see §4.4
Bias op-amps	OPA4703EA/250	4	
Config flash	MT25QL01GBBB8E12 (U9)	1	
Power module	EP4RKU+ (U16)	1	
USB 3.0 FIFO	FT601Q-B-T (U6)	1	0 of 77 pins connected in the schematic — K3, see §4.8
Connectors	SMA 142-0731-211 ×37, Molex 22-23-20xx ×56, mini-USB X53	—	

Parts whose datasheets are in `7_Components Datasheets` but which are *not* on this schematic: FT2232H, FT232RN, MAX20029, STUW81300, QPM1021, QPA1013, TGA2623 (source: `docs/PCB/MAIN_BOARD.md` §1).

Schematic summary (source: `engineering/ELECTRICAL/connection_diagrams/MAIN_BOARD_connection_report.md` §1):

Item	Count
Sheets	4
Parts (all) / physical	1955 / 776
Nets	625
Pin connections	4456
Single-pin nets	3
Unconnected pins on placed gates	280
Physical parts without value	244
Board airwires (unrouted connections, layer 19)	2390
Missing symbol/footprint records	0

4.2 Connectors

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The connection report lists 117 connector parts (source: `engineering/ELECTRICAL/connection_diagrams/MAIN_BOARD_connection_report.md §2`; the table below groups them by function, nets as in the report; pin order of the Molex symbols is UNVERIFIED because all pads are named 5 — source: `engineering/SYSTEM/interfaces/interconnection_table.md` header).

Group	Refdes	Package	Nets (signal pins; GND omitted)	Sheet
Clock inputs (SMA)	J1	142-0731-211	FPGA_SYS_CLOCK	2
	J18	142-0731-211	FPGA_DAC_CLOCK	2
	J20	142-0731-211	DAC_CLOCK	3
Differential clock inputs	J19	CJT-T-P-HH-ST-TH1	FPGA_ADC_CLOCK_P/N	2
	J21	CJT-T-P-HH-ST-TH1	N\$2_P/N (ADC clock)	3
LO inputs (SMA)	J22, J23	142-0731-211	N\$24 (LO RX), N\$23 (LO TX) — assignment per <code>interconnection_table.md §6</code>	3
Element RF ports (SMA, 16 pairs)	J24–J55	142-0731-211	unnamed nets N\$128..N\$204, one pair per RF switch; pairing J27/J26 = PA1 ... J51/J50 = PA16 (source: <code>interconnection_table.md §7</code>); which SMA of a pair is TX-out vs RX-return is UNVERIFIED	3
Synth control	JP1	PINHD-2X6	AD9523_PD, _REF_SEL, _SYNC, _RESET, _CS, STM32_SCLK4/MOSI4/MISO4, AD9523_STATUS0/1, _EEPROM_SEL	2
	JP13	PINHD-2X7	STM32_MISO4/SCLK4/MOSI4, ADF4382_TX_*, ADF4382_RX_* (CS, CE, DELSTR, DELADJ, LKDET)	2
SWD debug	JP2	PINHD-1X6	+3V3, STM32_SWCLK, STM32_SWDI0, STM32_Nrst, STM32_Sw0	2
FPGA JTAG	JP3	PINHD-2X4	N\$32, +3V3_FPGA, N\$33, N\$34, N\$36	2
Cooling / PA enable	JP4, JP10	PINHD-1X3	EN/DIS_COOLING, EN/DIS_RFPA_VDD	2
Temperature sensors	JP5, JP6, JP11, JP12, JP14, JP15, JP16, JP19	PINHD-1X3	+3V3_AN4_F + one analogue net each (N\$295..N\$307)	4
I2C3 modules	JP7 (1×8), JP18 (1×4)	PINHD	+3V3, STM32_SCL3, STM32_SDA3, MAG_DRDY, ACC_INT, GYR_INT / STM32_SDA3, STM32_SCL3, +3V3	2
UART5 (GPS)	JP8	PINHD-1X4	+3V3, STM32_TX5, STM32_RX5	2
Stepper	JP9	PINHD-1X4	STEPPER_CW+, STEPPER_CLK+	2
USART3 console	JP17	PINHD-1X3	STM32_TX3, STM32_RX3	2
Clock test	JP20	PINHD-1X2	FPGA_CLOCK_TEST	2
Enable bus to Power Board	SV1	MA10-2 (20-way)	15 × EN_+... rails (EN_+1V0_FPGA ... EN_+5V0_PA1)	2
Power inputs (2-pin Molex)	X1, X4–X22, X24, X54–X56	22-23-2021	one rail each: +3V4, +3V3_ADTR, +5V0_PA_2, –3V3_SW, +5V0_PA_3, +1V0_FPGA, +3V3_AN, +1V8_FPGA, –3V4, +3V3_VDD_SW, –5V0_ADAR12, +5V0_PA_1, –5V0_ADAR34, +3V3_FPGA, +1V8_CLOCK, +5V0_0, –5V5_PA, +3V3_ADAR12, +3V3_ADAR34, +3V3, +5V5_PA	1
PA gate bias outputs	X_1 ... X_16	22-23-2021	VG_15, VG_13, VG_7, VG_5, VG_9, VG_11, VG_1, VG_3, VG_14, VG_16, VG_6, VG_8, VG_12, VG_10, VG_4, VG_2 (in refdes order X_1...X_16)	4
PA drain-current sense inputs	X3, X38–X52	22-23-2031	one differential pair each (N\$207/N\$209 ... N\$286/N\$287)	4
Host USB	X53	MINI-USB 32005-201	STM32_USB_FS_D_N, _D_P, _ID (VBUS not connected)	2

The cable-level use of these connectors (which Power Board output feeds which input, which Synth SMA drives J1/J18/J19/J20/J22/J23) is in chapter 3 and in `engineering/SYSTEM/interfaces/interconnection_table.md §2–§8`.

4.3 Schematic sheets

The four sheets were rendered from `4_Schematics` and `Boards Layout/4_6_Schematics/MainBoard/RADAR_Main_Board.sch` with `tools/render_eagle_schematic.py` + `tools/svg_sheets_to_pdf.py` (PDF:

`engineering/ELECTRICAL/schematics/MAIN_BOARD/MAIN_BOARD_schematic.pdf`; sheet sizes and part/net counts per sheet in `engineering/ELECTRICAL/schematics/MAIN_BOARD/sheets.json`). Sheet names are those stored in the EAGLE file (source: `docs/PCB/MAIN_BOARD.md §1`). Sheet 3 is 1 904 × 1 652 mm at 1:1 and must be read from the SVG/PDF for pin-level detail.

 Figure 10 — F4.1 — Main Board schematic sheet 1

 Figure 11 — F4.2 — Main Board schematic sheet 2

 Figure 12 — F4.3 — Main Board schematic sheet 3

 Figure 13 — F4.4 — Main Board schematic sheet 4

Schematic-level findings that no layout work can close (source: `docs/PCB/MAIN_BOARD.md §2`; `beta/pcb/MAIN_BOARD/README.md §3`): FT601 U6 supply/control pins unconnected (board-only signals AVDD, VBUS, VCC3..., VCCIO..., VDPA are forward-annotation residue); 280 unconnected symbol pins; single-pin nets N\$44 (AD9484 CMI), FN_OPAMP_TF_1, FN_OPAMP_TF_2; XADC wiring VP/VN/VREFN to GND and

VREPP to 41V0_FPGA (to be checked against AMD UG480); 4 approved ERC entries on STM32_MISO_1V8. The full list is in [engineering/ELECTRICAL/netlists/MAIN_BOARD_unresolved_connections.md](#).

4.4 Layout state: source vs. engineering conversion vs. BETA

Source (EAGLE 7.4.0)

Item	Finding (source: docs/PCB/MAIN_BOARD.md §2)
Airwires	2 390 stored on layer 19: GND 2 354, +3V3_FPGA 33, +3V3_FT 3 (entirely unrouted). GND has polygons on layers 2, 5, 13, 15 — whether RATSNEST clears the GND airwires REQUIRES VERIFICATION IN EAGLE
Parts outside outline	11: C159, C184, C185, C186, L19 (FT601 decoupling), R60, R61, R83, R84, R145, R146 (parked at x = 0.1 mm, y < 0)
DRC	211 <approved> entries (EAGLE stores no text)
ERC	4 approved entries on net STM32_MISO_1V8
Layer-count mismatch	DRU name says 8 layers; 10 copper layers are defined and all carry copper (L1 7 619 objects, L2 1, L3 18, L4 9, L5 1, L12 587, L13 1, L14 395, L15 5, L16 1 605)
Via drills	0.15 mm ×1 045, 0.2 ×1 367, 0.3 ×332, 0.35 ×113, 0.5 ×21, 0.6 ×7, 1.0 ×4, 1.2 ×4 (all through 1-16; no blind/buried)
Min track	0.1 mm (3 408 wires); 0.204 mm RF width ×2 729 (matches the PCBWay 50 Ω note)
USB 3.0	FT601 unconnected; the only host link is STM32 USB-FS (PA11/PA12 → X53)

Engineering conversion ([engineering/PCB/MAIN_BOARD/](#) , KiCad 10.0.6 import)

Cross-check of the EAGLE XML against the KiCad conversion (source: [engineering/PCB/MAIN_BOARD/README.md §3](#)):

Check	EAGLE (XML)	KiCad	Result	Note
Width (mm)	260.0	260.0	OK	
Height (mm)	300.0	300.0	OK	
Footprints = EAGLE elements + free holes	784	784	OK	KiCad imports every free <hole> as a footprint
Vias	2893	2893	OK	
Tracks (signal wires excl. airwires)	10219	10219	OK	
Copper layers	10	10	OK	
NPTH holes (free holes + package holes)	10	10	OK	pads with drill but no copper are NPTH in KiCad too
PTH holes (vias + THT pads)	3314	3314	OK	

DRC on the conversion with EAGLE-DRU-derived rules: 912 violations, 15 unconnected (source: [engineering/PCB/MAIN_BOARD/README.md §6](#)):

Rule	Count	Interpretation
solder_mask_bridge	199	mask web between adjacent pads thinner than KiCad default (EAGLE has no such rule) — fabricator decision
silk_overlap	199	overlapping silkscreen texts/lines — cosmetic
silk_over_copper	199	silk over exposed copper — cosmetic/assembly
track_dangling	136	track end not connected — review (stubs or unfinished routing)
shorting_items	93	copper of different nets touching after conversion (typically EAGLE polygon vs unnamed copper) — REVIEW in EAGLE
clearance	63	copper clearance < DRU — review
via_dangling	15	via connected on one layer only — review
hole_clearance	4	hole to copper clearance (KiCad rule, no EAGLE equivalent)
silk_edge_clearance	3	silk too close to edge
zones_intersect	1	overlapping pours of different priority — conversion artefact or design issue

Counts of exactly 199 are lower bounds (KiCad report cap).

BETA board ([beta/pcb/MAIN_BOARD/](#))

Before/after table (source: [beta/pcb/MAIN_BOARD/README.md §1](#)):

Check	Before (engineering/PCB conversion)	After (BETA)
unconnected_items	15	0
clearance	63	72
hole_clearance	4	87
shorting_items	93	125
silk_edge_clearance	3	3
silk_over_copper	199	199
silk_overlap	199	199
solder_mask_bridge	199	199

Check	Before (engineering/PCB conversion)	After (BETA)
track_dangling	136	130
via_dangling	15	35
zones_intersect	1	0
DRC violations total	912	1049

What was changed in BETA — nothing in the netlist, every step logged (source: `beta/pcb/MAIN_BOARD/README.md §2`): 7 arc tails where EAGLE arcs end on a pad centre (`fixes_stubs.json`); the `+3V3_FT` input filter L19/C159/C184/C185/C186 moved from outside the outline to (5...12, -232...-240) mm next to X16 (`placement_moves.json`); hand routing of that cluster with 4 GND vias 0.45/0.20 mm, a 0.3 mm `+3V3_FT` bus and a 0.5 mm `+3V3_FPGA` feed to X16.1 (`manual_routing.json`, `bridges.json`); 14 of 30 net-less footprint copper polygons (BPF2 filters, U13, U5) re-created as board-level polygons with the net they touch (`POLYGON_NET_DISPOSITION.md`); same-net zone priorities. R60, R61, R83, R84, R145, R146 (33 Ω 0201, no net on either pin) stay outside the outline and are `dnp = yes` in the BETA BOM.

Why the total went *up* although 15 $\rightarrow$ 0 unconnected: the remaining 1 049 items are dominated by copper that the schematic cannot describe (source: `beta/pcb/MAIN_BOARD/README.md §3`):

Type	Count	Disposition
shorting_items	125	REAL (schematic): GND tracks/arcs run through the thermal-via pads <code>V...V_8</code> of U69 (DAC5578) and U7 — those pads have no net because the symbols have no paddle pin; EAGLE connected them with the GND polygon. Also the 4 ambiguous BPF2 polygons (GND + RF net) and KK headers JP7/JP8/JP10/JP17. Not changed in BETA $\rightarrow$ designer must add the paddle pins to the symbols or confirm the intended net
hole_clearance / clearance	87 / 72	same U69/U7 no-net via pads and the 4 ambiguous BPF2 polygons; plus 3 at ADAR1_0/ADAR3_0/J33 from the source
track_dangling	130	EAGLE stub ends inside pours/pads — cosmetic, copper not modified
via_dangling	35	20 vias of RF_TX/RF_RX/RF_TX_FIL/RF_RX_FIL inside the BPF2 pads + 15 from the source (ADAR load/enable nets)
silk_overlap / silk_over_copper / solder_mask_bridge	199+ each	cosmetic / fab CAM (0201 density)
silk_edge_clearance	3	X53 outline and JP3 reference at the board edge — cosmetic

4.5 Layer plots and 3-D renders

Composite top and bottom views (copper + silkscreen + outline) of the engineering conversion. Per-layer plots (F.Cu, In1...In8, B.Cu) are listed as files in chapter 14, not reproduced here.

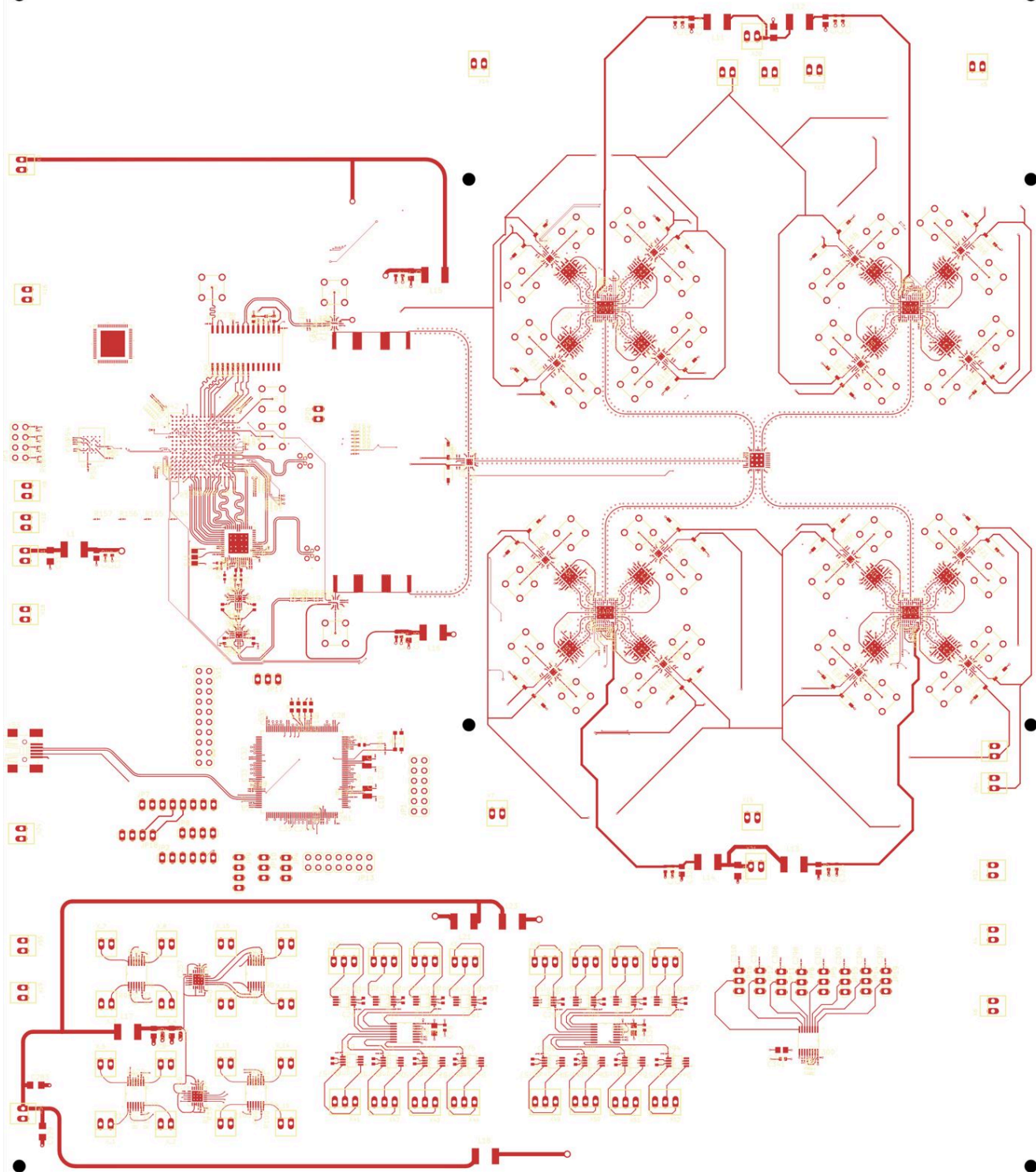


Figure 14 — F4.5a — Main Board top composite (F.Cu + F.Silks + Edge.Cuts) of the KiCad conversion of RADAR_Main_Board.brd, 260 × 300 mm (status: SOURCE-DERIVED; source: engineering/PCB/MAIN_BOARD/svg/MAIN_BOARD_top_composite.svg; produced by tools/svg_sheets_to_pdf.py --png-dir manual/figures)

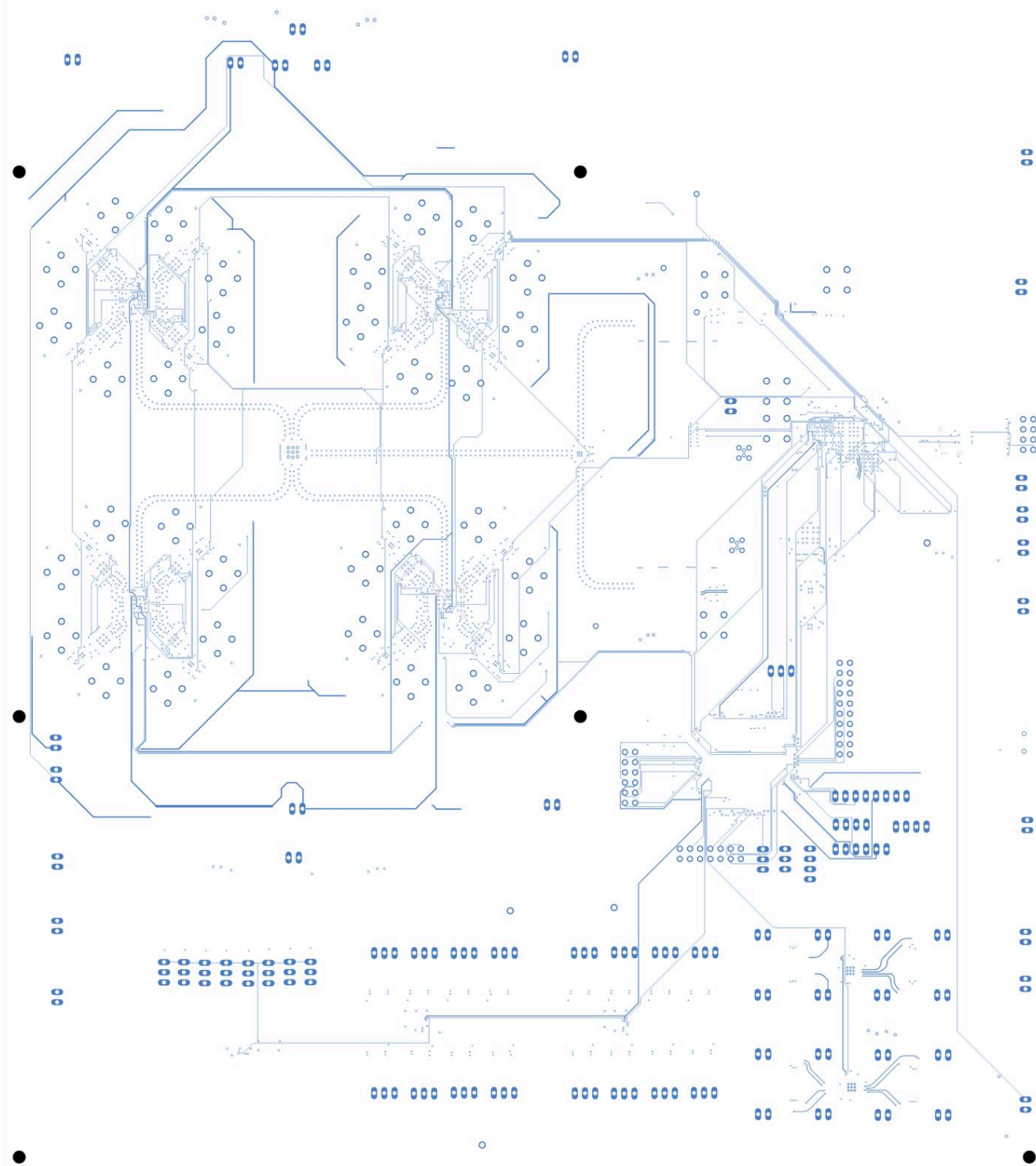


Figure 15 — F4.5b — Main Board bottom composite, mirrored (B.Cu + B.SilkS + Edge.Cuts) (status: SOURCE-DERIVED; source: engineering/PCB/MAIN_BOARD/svg/MAIN_BOARD_bottom_composite_mirrored.svg; produced by tools/svg_sheets_to_pdf.py --png-dir manual/figures)

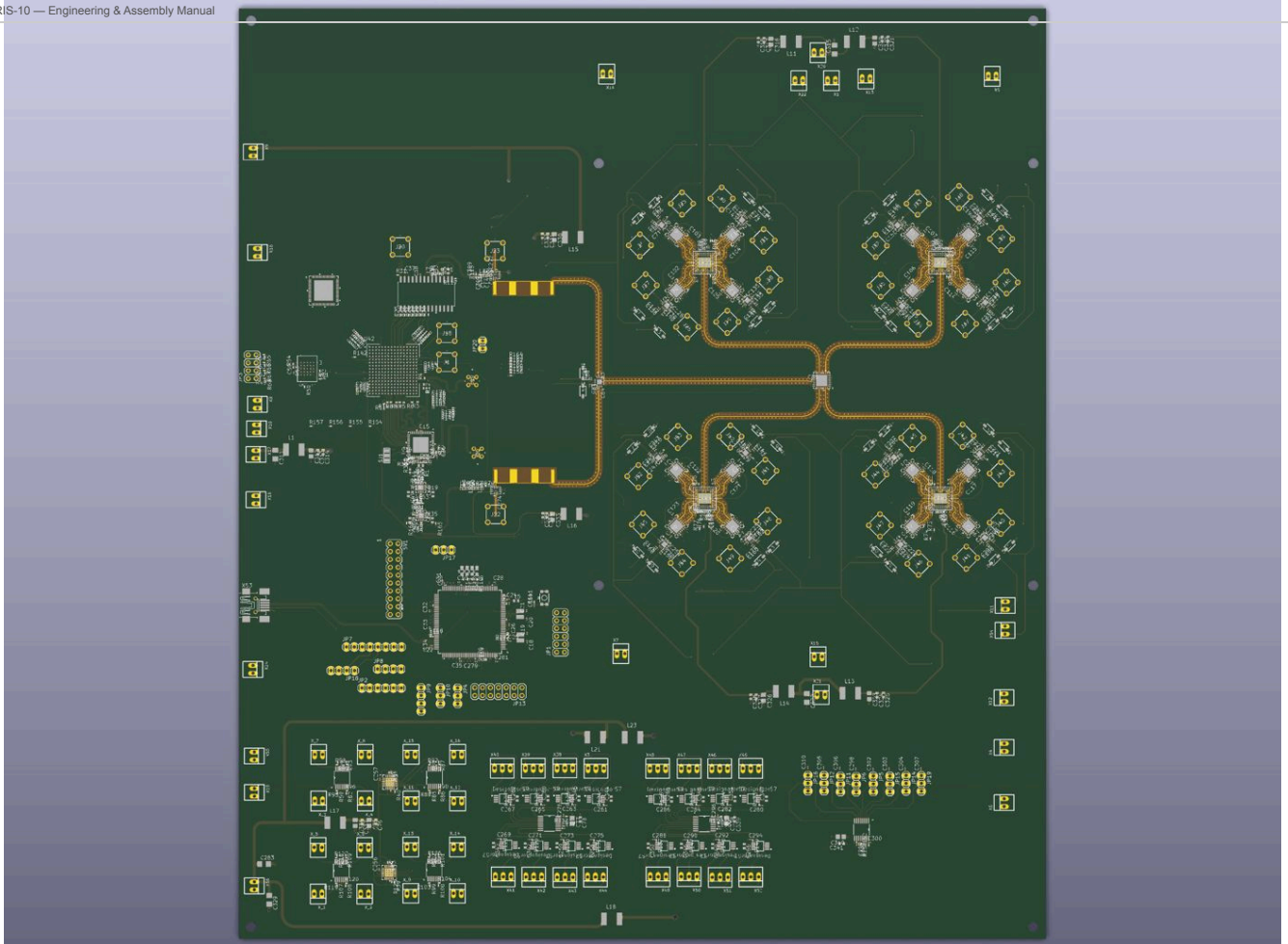


Figure 16 — F4.6a — Main Board 3-D render, top, KiCad conversion; no component models (STEP is board-only) (status: SOURCE-DERIVED; source: engineering/PCB/MAIN_BOARD/3d/MAIN_BOARD_render_top.png; produced by kicad-cli pcb render, tools/kicad_pcb_pipeline.sh)

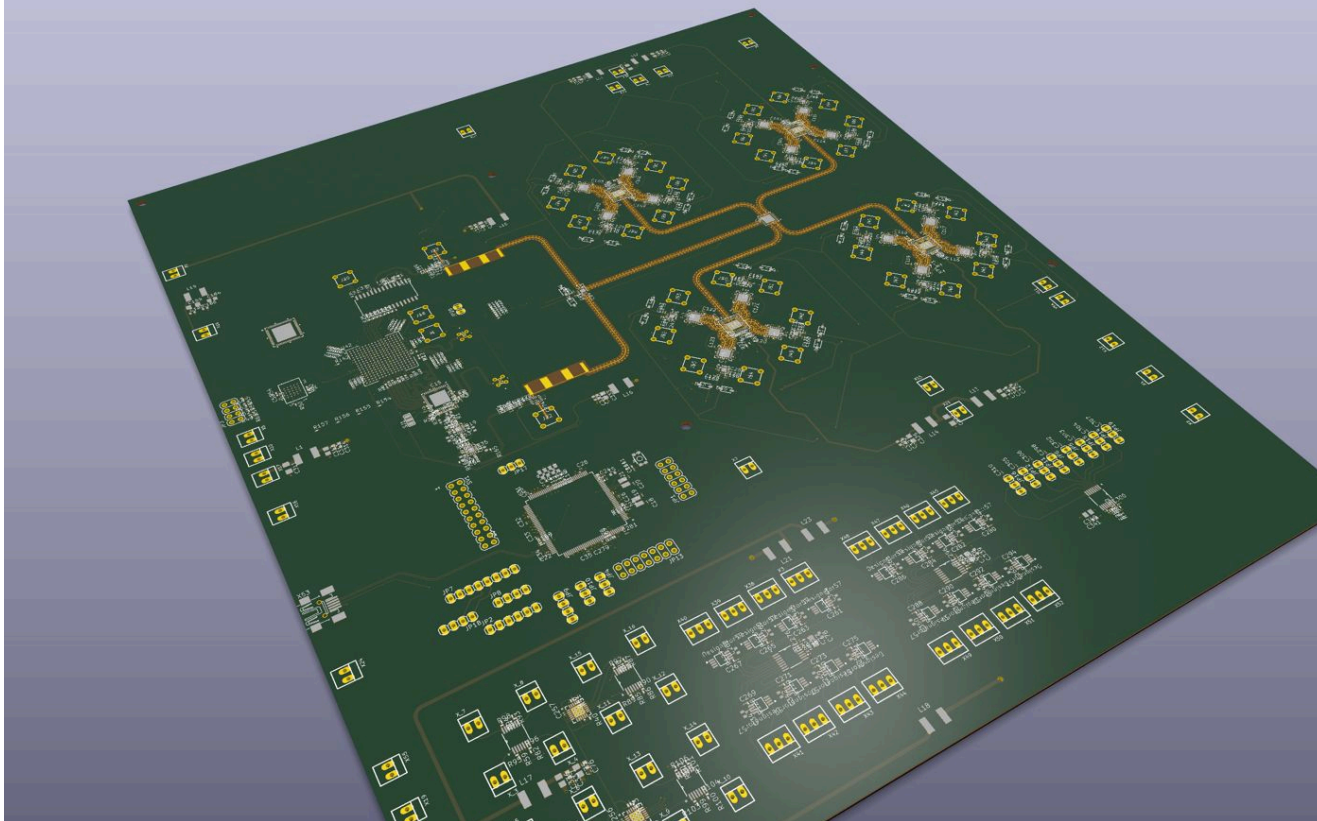


Figure 17 — F4.6b — Main Board BETA board, isometric render after the +3V3_FT cluster placement and routing (status: BETA; source: beta/pcb/MAIN_BOARD/exports/3d/MAIN_BOARD_render_isometric.png; produced by beta/pcb/tools/beta_export_package.sh)

Assembly drawings: the generated `engineering/PCB/MAIN_BOARD/drawings/MAIN_BOARD_assembly_top.pdf` and `beta/pcb/MAIN_BOARD/exports/drawings/MAIN_BOARD_assembly_top.pdf` exist (F.Fab + F.Silks + Edge.Cuts, black and white), but when rendered with `pdftoppm -png -r 110` the A4 page frame plotted by `kicad-cli pcb export pdf --sp` contains only the parts parked outside the outline; the board body itself (at negative y in the KiCad coordinate frame) lies outside the page. The same applies to the Power Supply, Frequency Synthesizer and RF PA assembly PDFs. These PDFs are therefore not used as figures in this manual; the planned figure F4.7 is replaced by the top composite F4.5a (which carries F.Silks and F.Fab reference designators) until the pipeline plots the drawing without the page frame (open item in chapter 17).

4.6 Stack-up

Layer order and copper thickness come from the EAGLE DRU `PCBWay_8L_100um-Track` (the name says 8 layers; the board uses 10). Dielectric thicknesses are the EAGLE DRU table, not a vendor stack-up; total thickness is not defined in the source (KiCad assumed 1.6 mm) (source: `engineering/PCB/MAIN_BOARD/STACKUP.md`, status PARTIAL):

#	EAGLE layer	KiCad layer	Copper (DRU mtCopper)	Dielectric below (DRU mtIsolate)
1	1	F.Cu	0.035mm	0.102mm
2	2	In1.Cu	0.035mm	0.2mm
3	3	In2.Cu	0.035mm	0.2mm
4	4	In3.Cu	0.035mm	0.2mm
5	5	In4.Cu	0.035mm	0.2mm
6	12	In5.Cu	0.035mm	0.15mm
7	13	In6.Cu	0.035mm	0.2mm
8	14	In7.Cu	0.035mm	0.2mm
9	15	In8.Cu	0.035mm	0.102mm
10	16	B.Cu	0.035mm	—

PROPOSED fabrication stack-up (BETA, nothing confirmed by the designer or PCBWay; source: `beta/pcb/MAIN_BOARD/FAB_NOTES.md` §2): outer layers 1 and 9 on **Rogers RO4350B 4 mil (0.102 mm)** as the impedance layers, FR-4 cores/prepregs inside; dielectric sum 1.554 mm + 10 × 35 μm copper ≈ 1.9 mm → PROPOSED finished thickness **2.0 mm ± 10 %** (the DRU sum does not fit 1.6 mm). Materials/finish

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proposal (FAB_NOTES.md §3): 1 oz copper all layers, ENIG, green LPI mask with **no mask over RF traces** on the RF layer, white silk, min track/space 0.10/0.10 mm, min drill 0.15 mm, copper-to-edge 0.3 mm, IPC-A-600 Class 2, 100 % electrical test with the supplied IPC-D-356 netlist. Controlled impedance targets with the geometry measured on the KiCad board (FAB_NOTES.md §4): 50 Ω single-ended at w = 0.204 mm (2 729 segments); 100 Ω differential at w = 0.204 mm / s = 0.26 mm (FPGA_ADC_CLOCK, MIX_RX), ADC_CLK_IN s = 0.288 mm, AMP_IF_IN uncoupled (s = 1.58 mm). The impedance note itself (4_Schematics and Boards Layout/4_7_Production Files/Frequency_Synthesizer/PCBWay_Impedance_Note_R04350B_h0p102mm.pdf) is a Frequency Synthesizer document; its applicability to this board is an assumption of the proposal — if the Main Board is built on plain FR-4 the 0.204 mm lines are not 50 Ω.

4.7 BOM summary

Source BOM (docs/BOM/BOM_MAIN_BOARD.csv, generated from the schematic): 776 references, 98 line items, **0 MPN attributes**, 244 references without value (source: docs/BOM/README.md §Summary). The BETA BOM beta/pcb/MAIN_BOARD/BOM_MAIN_BOARD_beta.csv adds manufacturer, mpn, mpn_confidence, dnp, note columns with proposed part numbers (beta/pcb/tools/beta_bom_mpn.py). Confidence count obtained with

```
python3 -c "import csv,collections;print(collections.Counter(r['mpn_confidence'] for r in csv.DictReader(open('beta/pcb/MAIN_BOARD/BOM_MAIN_BOARD_beta.csv'))))"
```

mpn_confidence	Lines	Meaning (source: beta/pcb/README.md §BOM confidence)
HIGH	23	deviceset name is an orderable MPN
MEDIUM	41	standard passive proposed from value + package
LOW	30	guess / non-standard value / package-value conflict
EMPTY	4	no value in the source
Total	98	quantities 204 / 461 / 93 / 18 (same source)

Flagged conflicts that the designer must resolve before purchase (source: beta/pcb/README.md): non-E-series values (2.443 kΩ, 103 pF, 107.3 nH), 47 μF in 0201, NX3225 footprint with a 32.768 kHz value, 5 mΩ shunt with a 0.1 Ω part number; the 37 SMA and 56 Molex connectors to be confirmed as production parts (many are test/interconnect points); FT601 and its 7 parked passives marked DNP or wired; INA241A3 ×16 and OPA4703 ×4 confirmed for the Nexus variant (source: docs/PCB/MAIN_BOARD.md §8).

4.8 Rev. B host interface (FT601)

The source Main Board has no usable high-speed host path: U6 (FT601Q-B-T) is placed but none of its 77 pins is connected, and the only host link is the STM32 USB full-speed port on X53 (source: docs/PCB/MAIN_BOARD.md §1–2; conflict K3 in docs/SYSTEM/BLOCK_DIAGRAM.md). Chapter 9 documents the two proposed options: option A — wire the FT601 to free FPGA bank-35 pins (47 signals, pin plan engineering/DESIGN/HOST_LINK/ft601_pin_assignment.csv, XDC fragment ft601_bank35.xdc, added parts ft601_added_parts_BOM.csv) in a **Main Board rev. B**; option B — the SPI bridge frame through the STM32 CDC link, implemented end-to-end in simulation (source: docs/AERIS10_BETA_ENGINEERING_REPORT.md §7).

The rev. B board package is beta/pcb/MAIN_BOARD_REVB/ (status **BETA PROPOSAL — explicit netlist change, partially routed, DRC-checked, not reviewed by the original designer, not fabricated**; source: beta/pcb/MAIN_BOARD_REVB/README.md). It was derived from the rev. A BETA board with beta/pcb/tools/beta_revb_ft601.py and the design input of engineering/DESIGN/HOST_LINK/HOST_LINK_DESIGN.md §4. **The EAGLE schematic has not been changed and no longer matches this board**: the designer must enter the connections of NETLIST_DELTA.csv (README §3) and the added parts (README §4) in the schematic, verify them against the FT601 datasheet (not in the repository) and re-annotate before any rev. B layout is released (same source, header note).

State of the rev. B board (source: beta/pcb/MAIN_BOARD_REVB/README.md §1):

Check	Rev. A BETA (MAIN_BOARD/)	Rev. B after netlist change, before routing	Rev. B after routing (exports)
unconnected_items	0	129	59
DRC total	1049	1055	1071

Of the 64 new nets, 17 are routed and connected (USB D±, both SuperSpeed RX lines, SSTX_N and both SSTX_C lines, CC1/CC2, USB_VBUS, FT_VBUS_DET, FT_RREF, FT_XI, FT_GPIO1, partly +3V3_FT / FT_VD10 / FT_AVDD / FT_XO). **The 32-bit FIFO bus and its control lines (46 of 47 FPGA-side signals) are not routed**: 36 of the 47 bank-35 balls of U42 (XC7A50T FTG256, 1.0 mm pitch) have no free position for an escape via because the dog-bone positions are occupied by the existing BGA fan-out; routing the bus requires re-doing the U42 bank-35 breakout (same source; list in beta/pcb/MAIN_BOARD_REVB/UNROUTED.md, 59 open items all on rev. B nets). New DRC items from rev. B copper: 9 track_width neck-downs (0.075 mm, to be widened to 0.1 mm), 2 courtyard overlaps (Y_FT ↔ C_XI/C_XO), 1 copper-edge clearance at the J_USB3 NPTH peg, 5 dangling vias; the USB differential pairs were routed as single traces by Freerouting and must be re-routed coupled (0.204/0.18 mm) before release (README §1–§2). Added parts (README §4): USB-C receptacle J_USB3 Amphenol 12401610E4#2A (SS lane wired in one orientation only — VERIFY), 2 × TPD4E05U06 ESD arrays, 2 × 100 nF SSTX coupling capacitors, 2 × 5.1 kΩ CC resistors, VBUS divider 10 k/3.3 k, RREF 3.24 kΩ, 4 × 4.7 μF VD10 decoupling, AVDD ferrite + 100 nF + 1 μF, 30 MHz crystal ABM8 with 2 × 18 pF — every value carries a VERIFY note against the FT60x datasheet. BOM_MAIN_BOARD_REVB_beta.csv has 110 lines (97 rev. A, 12 new rev. B, 1 rev. A part now wired; counted from the CSV revision column). Full export package in

beta/pcb/MAIN_BOARD_REVB/exports/ (all 24 export steps exit 0 per exports/EXPORT_LOG.md). Procedures, pin plan and the bridge-frame alternative: chapter 9.

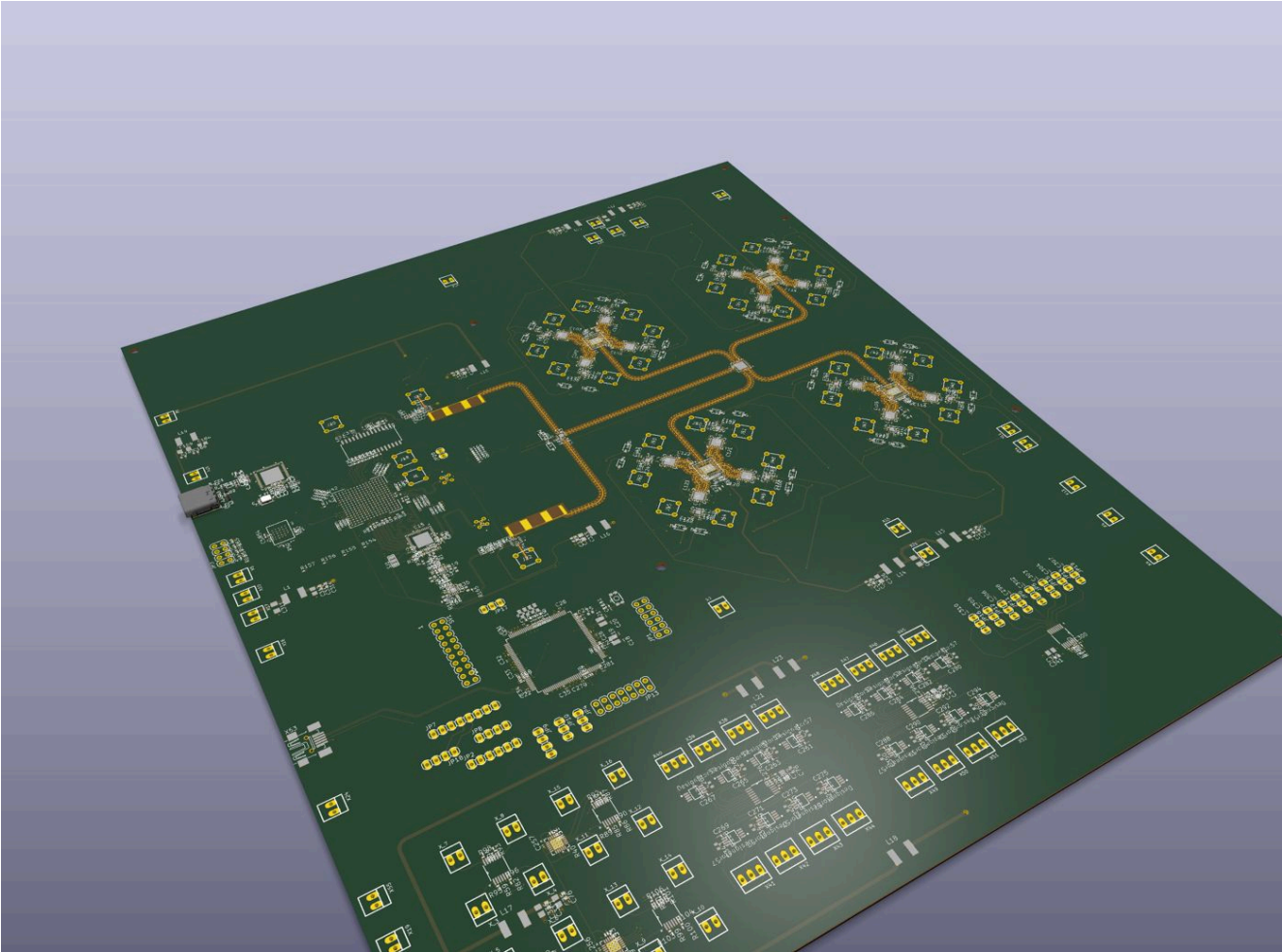


Figure 18 — F4.7 — Main Board rev. B proposal, isometric render: USB-C receptacle J_USB3 on the left edge next to U6 (FT601), ESD arrays, crystal and decoupling in the band below U6; FIFO bus to bank 35 unrouted (status: BETA; source: beta/pcb/MAIN_BOARD_REVB/exports/3d/MAIN_BOARD_REVB_render_isometric.png; produced by beta/pcb/tools/beta_export_package.sh)

The rev. B netlist delta is the only schematic-level change proposed for this board; it is a PROPOSED DESIGN / BETA and not part of the ORIGINAL PROJECT FILE set.

4.9 Open items

ID	Item	Status	Source
K1	FPGA part: XC7A50T-2FTG256I in CAD vs XC7A100T in README/XDC	OPEN — hardware designer	docs/SYSTEM/BLOCK_DIAGRAM.md
K2	STM32 HSE: 8 MHz crystal on board vs 25 MHz in firmware (beta firmware assumes 8 MHz)	OPEN	same; docs/AERIS10_BETA_ENGINEERING_REPORT.md
K3	Host data path: FT601 unwired (option A rev. B / option B bridge)	OPEN — system architect	chapter 9
—	Rev. B: FIFO bus (46 of 47 FPGA-side signals) unrouted — U42 bank-35 breakout must be redone; USB pairs to be re-routed coupled; EAGLE schematic not updated	OPEN — designer	beta/pcb/MAIN_BOARD_REVB/README.md §1, UNROUTED.md
—	2 390 EAGLE airwires; RATSNEST' behaviour on the GND polygons REQUIRES VERIFICATION IN EAGLE	OPEN	docs/PCB/MAIN_BOARD.md §2
—	125 shorting items from net-less thermal-via pads of U7/U69 and 4 ambiguous BPF2 polygons — symbol/footprint fix needed	OPEN — designer	beta/pcb/MAIN_BOARD/README.md §3
—	10-layer stack-up: DRU named for 8 layers; material, thickness, finish, impedance UNVERIFIED (G-01)	BLOCKED — MISSING DATA	engineering/PCB/MAIN_BOARD/STACKUP.md; MDR-07
—	BOM: 0/98 lines with a verified MPN; 244 value-less references	OPEN	docs/BOM/README.md
—	XADC reference wiring; single-pin nets N\$44, EN_OPAMP_IF_1/2; 280 unconnected symbol pins	OPEN	engineering/ELECTRICAL/netlists/MAIN_BOARD_unresolved_connections.md
—	Generated assembly-drawing PDFs clipped by the A4 page frame (§4.5)	OPEN — tooling	this chapter

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—		211 approved DRC entries in the source without justification text	OPEN	docs/PCB/MAIN_BOARD.md §2

Acceptance criteria for calling this board layout-complete are copied in chapter 14 (docs/PCB/MAIN_BOARD.md §10). Do not declare this board fabrication-ready on the basis of the existing .sch/.brd files or of the BETA package.

<!-- chapter 5: Power Board -->

5. Power Supply Board (PowerBoard)

Chapter status summary: schematic — SOURCE-DERIVED (EAGLE 9.6.2 schematic rendered); connector inventory — SOURCE-DERIVED; layout — PARTIAL in the source (309 airwires, 132 of 312 parts parked outside the outline, sch/brd version mismatch) and BETA in beta/pcb/POWER_SUPPLY/ (308 → 89 unconnected, placement by script not reviewed); layer plots and renders — SOURCE-DERIVED / BETA; stack-up — PARTIAL (2-layer, 1.5 mm core from the DRU) with a PROPOSED fabrication note; BOM — PARTIAL (0 MPN in the source). The 22 V PA drain supply does not exist on this board (K4) — see chapter 9. Nothing is hardware-verified. Compiled by Antidrone Ukraine · antidrone.cc.

Sources: docs/PCB/POWER_SUPPLY.md; engineering/ELECTRICAL/connection_diagrams/POWER_SUPPLY_connection_report.md §1–2; engineering/ELECTRICAL/schematics/POWER_SUPPLY/; engineering/PCB/POWER_SUPPLY/README.md, STACKUP.md; beta/pcb/POWER_SUPPLY/README.md, UNROUTED.md, FAB_NOTES.md, BOM_POWER_SUPPLY_beta.csv; engineering/VALIDATION/UNRESOLVED_GEOMETRY.md (G-10).

5.1 Role and key parts

The Power Board converts the single DC input V_{in} [12–17] V (silkscreen label; source: docs/PCB/POWER_SUPPLY.md §1) into the 24 rails consumed by the Main Board and the Frequency Synthesizer, each rail leaving the board on its own 2-pin Molex KK connector, and receives the 15-line enable bus from the STM32 on the 20-way header SV1. Board: **280 × 300 mm, 2 copper layers**, EAGLE schematic 9.6.2 / board 7.4.0 (version mismatch), 312 physical parts, 156 nets, 346 vias, DRU PCBWay_2L_100um-Track with a 1.5 mm core (source: docs/PCB/POWER_SUPPLY.md §1).

Key ICs (deviceset names, MPN UNVERIFIED; source: docs/PCB/POWER_SUPPLY.md §1):

Function	Part	Qty	Note
Buck regulators	TPS562208DDCT	21	datasheet folder contains TPS562201 instead — datasheet to be added
LDO	ADM7151ACPZ-04-R7	6	
LDO	TPS7A8300RGR	2	feedback-select pins without net (24 unconnected pins, see §5.3)
Inverters	LM2662MX/NOPB	5	datasheet folder has LM2663
Tantalum capacitors	T521W476M020ATE045	10	
Rail outputs	Molex 22-23-2021	34	
DC input	AK300/2 (X1)	1	VIN, GND
Not on the board	MAX20029	—	datasheet present, part absent

Rails present as silkscreen labels (layer 51; source: docs/PCB/POWER_SUPPLY.md §1): +5V5_PA, −5V5_PA, +1V8_CLOCK, +3V3_CLOCK, +3V3_VDD_SW, +5V0_PA_1/2/3, −3V3_SW, +3V3_SW, +3V3_ADTR, −5V0_ADAR12, −5V0_ADAR34, +3V4, −3V4, +3V3_L0, +5V0_ADAR, +3V3_ADAR_12, +3V3_ADAR_34, +5V0_0, +1V8_FPGA, +1V0_FPGA, +3V3, +3V3_FPGA, +3V3_AN, 12 × EN. The 16 EN_* nets match the STM32 enable GPIOs PE7...PE15, PG0...PG5 (same source). The design input for rail currents is 3_Power Management/Power Management V6.xlsx (sheet Feuil1); it budgets 16 × QPA2962 at 22 V / 2 A, a rail this board does not have — K4 (source: docs/PCB/POWER_SUPPLY.md §2; docs/SYSTEM/BLOCK_DIAGRAM.md).

Schematic summary (source: engineering/ELECTRICAL/connection_diagrams/POWER_SUPPLY_connection_report.md §1):

Item	Count
Sheets	1
Parts (all) / physical	562 / 312
Nets	156
Pin connections	1061
Single-pin nets	0
Unconnected pins on placed gates	24
Physical parts without value	80
Board airwires (unrouted connections, layer 19)	309
Missing symbol/footprint records	0

5.2 Connectors

36 connector parts (source: engineering/ELECTRICAL/connection_diagrams/POWER_SUPPLY_connection_report.md §2; every Molex output carries the named rail on one pad and GND on the other, pad order UNVERIFIED):

Refdes	Package	Rail / signal
X1	AK300/2	VIN (12–17 V input)
SV1	MA10-2 (20-way)	EN_+1V0_FPGA, EN_+5V0_PA2, EN_+1V8_FPGA, EN_+5V0_PA3, EN_+3V3_FPGA, EN_+5V5_PA, EN_+5V0_ADAR, EN_+1V8_CLOCK, EN_+3V3_ADAR12, EN_+3V3_CLOCK, EN_+3V3_ADAR34, EN_+3V3_ADTR, EN_+3V3_SW, EN_+3V3_VDD_SW, EN_+5V0_PA1 (15 enables + GND)

Refdes	Package	Rail / signal
X2	22-23-2021	+5V0_1
X3	22-23-2021	+5V5_PA
X4	22-23-2021	+1V0_FPGA
X5	22-23-2021	+1V8_FPGA
X6	22-23-2021	+5V0_L0
X7	22-23-2021	+3V3_L0_2
X8	22-23-2021	+3V3_L0_1
X9	22-23-2021	+5V0_2
X10	22-23-2021	+1V8_CLOCK
X11	22-23-2021	+3V3_CLOCK
X12	22-23-2021	+3V3_AN
X13	22-23-2021	+5V0_ADAR
X14	22-23-2021	+3V3_ADAR_12
X15	22-23-2021	+3V3_ADAR_34
X16	22-23-2021	+3V3
X17	22-23-2021	+5V0_3
X18	22-23-2021	-3V3_SW
X19	22-23-2021	-5V5_PA
X20	22-23-2021	-5V0_ADAR34
X21	22-23-2021	-5V0_ADAR12
X22	22-23-2021	+5V0_0
X23	22-23-2021	+3V4
X24	22-23-2021	-3V4
X25	22-23-2021	+5V0_4
X26	22-23-2021	+5V0_ADTR
X27	22-23-2021	+3V3_FPGA
X28	22-23-2021	+5V0_5
X29	22-23-2021	+3V3_SW
X30	22-23-2021	+3V3_VDD_SW
X31	22-23-2021	+5V0_PA_1
X32	22-23-2021	+5V0_PA_2
X33	22-23-2021	+5V0_PA_3
X34	22-23-2021	+3V3_ADTR
X35	22-23-2021	+3V3_X0

Which output feeds which Main Board / Synthesizer input is the subject of `engineering/SYSTEM/interfaces/interconnection_table.md` §2–§4 (chapter 3). Thirteen outputs (`+5V0_1`...`+5V0_5` , `+3V3_L0_1/2` , `+3V3_CLOCK` , `+5V0_ADAR` , `+3V3_ADAR_12/34` , `+5V0_ADTR` , `+3V3_SW`) have no connector with the same net name on the Main or Synth board and are marked TBD in the harness schedule (source: `engineering/DESIGN/HARNESS/HARNESS_SCHEDULE.md` , CBL-001...CBL-028 rows with "no connector with this net name"); some are matched by meaning in the interconnection table (e.g. `+3V3_ADAR_12` → Main `+3V3_ADAR12` , CBL-07 there).

5.3 Schematic sheet

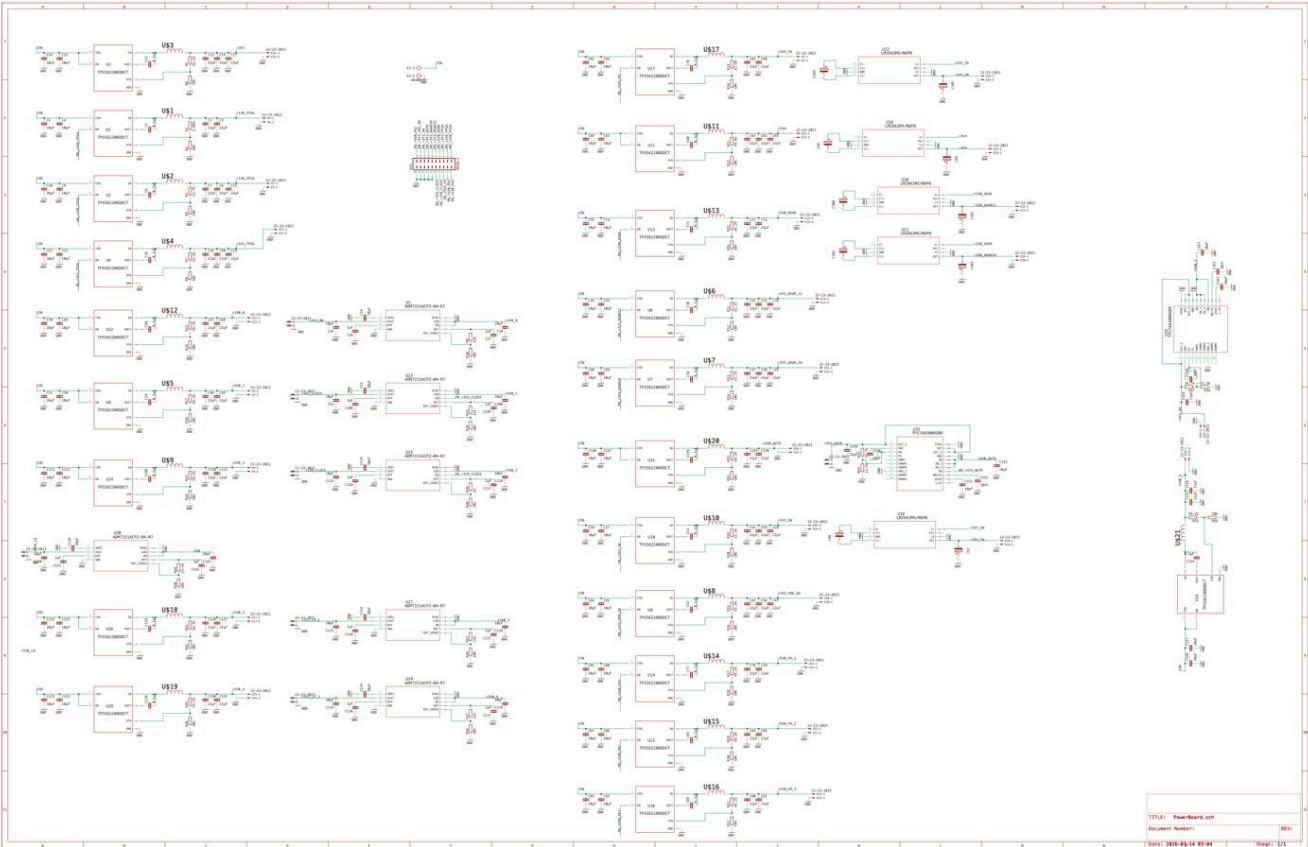


Figure 19 — F5.1 — Power Board schematic, single sheet: 21 × TPS562208 buck stages, 6 × ADM7151 and 2 × TPS7A8300 LDOs, 5 × LM2662 inverters, 34 rail outputs, enable header SV1 (status: SOURCE-DERIVED; source: engineering/ELECTRICAL/schematics/POWER_SUPPLY/png/POWER_SUPPLY_schematic_sheet1.png; produced by tools/render_eagle_schematic.py)

Schematic-level items that no layout work closes: 24 pins without net (TPS7A8300 feedback-select pins, LM2662 FC/OSC — source: beta/pcb/POWER_SUPPLY/README.md §3, detail in engineering/ELECTRICAL/netlists/POWER_SUPPLY_unresolved_connections.md); 80 references without value — the buck-stage inductors and capacitors must carry values for the TPS562208 compensation to be reproducible (source: docs/PCB/POWER_SUPPLY.md §8); one board-only signal +3V3_L0 (sch/brd mismatch, source: docs/PCB/POWER_SUPPLY.md §2).

5.4 Layout state: source vs. engineering conversion vs. BETA

Source (EAGLE 7.4.0 board, 9.6.2 schematic)

Item	Finding (source: docs/PCB/POWER_SUPPLY.md §2)
Routing	309 airwires across 86 of 157 signals; 72 signals entirely unrouted (all 16 EN_*, several +5V0_*, +3V3_X0, +3V3_L0_1/2)
Placement	132 of 312 elements outside the 280 × 300 mm outline; mounting holes lie within 10..270 × 10..230 (+ one at y = 267.85) → the intended final outline is UNRESOLVED (G-10)
Version mismatch	sch 9.6.2 vs brd 7.4.0 — annotation link cannot be trusted; one board-only signal +3V3_L0
DRC	0 approved errors (meaningless while unrouted)
ERC	0 approved entries; 0 single-pin nets
Design-input discrepancies	xlsx budgets 16 × QPA2962 at +22 V (2 A each); the Power Board has no 22 V rail label (Vin [12–17] V) and the RF PA board holds one QPA2962 — PA supply architecture UNRESOLVED

Engineering conversion (engineering/PCB/POWER_SUPPLY/)

Cross-check (source: engineering/PCB/POWER_SUPPLY/README.md §3):

Check	EAGLE (XML)	KiCad	Result	Note
Width (mm)	280.0	280.0	OK	
Height (mm)	300.0	300.0	OK	
Footprints = EAGLE elements + free holes	320	320	OK	KiCad imports every free <hole> as a footprint
Vias	346	346	OK	
Tracks (signal wires excl. airwires)	570	570	OK	
Copper layers	2	2	OK	

Check	EAGLE (XML)	KiCad	Result	Note
NPTH holes (free holes + package holes)	8	8	OK	pads with drill but no copper are NPTH in KiCad too
PTH holes (vias + THT pads)	444	444	OK	

DRC on the conversion: 160 violations, 308 unconnected (source: `engineering/PCB/POWER_SUPPLY/README.md §6`):

Rule	Count	Interpretation
clearance	53	copper clearance < DRU — review
silk_over_copper	40	silk over exposed copper — cosmetic/assembly
silk_overlap	22	overlapping silkscreen texts/lines — cosmetic
zones_intersect	21	overlapping pours of different priority — conversion artefact or design issue
drill_out_of_range	8	drill smaller than DRU msDrill
solder_mask_bridge	8	mask web between adjacent pads thinner than KiCad default (EAGLE has no such rule) — fabricator decision
track_dangling	4	track end not connected — review (stubs or unfinished routing)
isolated_copper	4	review

BETA board (`beta/pcb/POWER_SUPPLY/`)

Before/after (source: `beta/pcb/POWER_SUPPLY/README.md §1`):

Check	Before (engineering/PCB conversion)	After (BETA)
unconnected_items	308	89
clearance	53	56
drill_out_of_range	8	8
isolated_copper	4	0
silk_over_copper	40	168
silk_overlap	22	65
solder_mask_bridge	8	8
track_dangling	4	2
via_dangling	0	20
zones_intersect	21	1
DRC violations total	160	328

Unconnected items 308 → 89 (−71 %); footprints outside the outline 132 → 0. What was done, in order (source: `beta/pcb/POWER_SUPPLY/README.md §2`): netclass `Default` set to track 0.25 mm / clearance 0.2 mm / via 0.5/0.3 mm for new copper only; 110 parts placed inside the outline in 29 net clusters and 22 KK connectors placed on the nearest board edge (`beta/pcb/tools/beta_place_outside.py`, `placement_moves*.json`); Freerouting 2.5.0 pass 1 with the existing 916 tracks/vias fixed (+1 336 tracks, +168 vias; 308 → 102); 45 same-net zones re-prioritised and a board-wide B.Cu GND zone added (zones_intersect 21 → 1); 29 straight same-net bridges kept out of 102 tried (102 → 96); Freerouting pass 2 (+36 tracks, +3 vias; 96 → 96); GND stitching, 22 of 50 vias kept (96 → 89); refill, DRC, export.

The placement by script is topological, not thermal/EMC — regulator clusters sit where free space was nearest to their connectors; a designer must review inductor orientation, input/output capacitor proximity and heat spreading before any routing clean-up (source: `beta/pcb/POWER_SUPPLY/README.md §3`).

The 89 open connections (`beta/pcb/POWER_SUPPLY/UNROUTED.md`)

None of the 89 could be completed without a routing decision that a script should not take (crossing other copper, moving existing tracks, re-shaping the EAGLE pours); the netlist was never changed (source: `UNROUTED.md` header). By net: **GND 49**, **VIN 10**, +5V0_ADAR 4, +3V3_ADAR_12 3, +3V3_FPGA 2, +5V0_PA_3 2, +5V5_PA 2, +1V0_FPGA 2, +3V3_SW 2, +3V3_VDD_SW 2, +3V3_ADAR_34 2, +5V0_0 2, +5V0_PA_1 2, +3V4 1, +1V8_FPGA 1, +3V3 1, +3V3_AN 1, +5V0_PA_2 1. Freerouting's own verdict was 45 unrouted / 23 violations after pass 1; KiCad counts more because each isolated pour island counts separately (same source).

The reasons fall into three classes (wording of the `UNROUTED.md` table):

Class	Typical entry	What is needed
GND pad ↔ GND pad	e.g. U34 pads 8/18/21 at (121.0, −90...−94)	the B.Cu GND plane is fragmented by the autorouted bottom tracks; the stitching via next to the pad was reverted by DRC → manual via placement or move the bottom tracks
zone island ↔ zone island (same net)	e.g. VIN zones priority 22 ↔ 28 at (71.2, −175.9) ↔ (215.6, −208.3)	the EAGLE polygons are separate local pours joined only by airwires; Freerouting routes pins, not pours → a manual 2 mm VIN / GND bus or a redesign of the pour layout
pin ↔ pin unroutable	e.g. +1V8_FPGA tracks at (129.6, −52.7) ↔ (152.5, −49.2)	Freerouting reported the connection unroutable in both passes (2-layer, local congestion) → manual routing
pad/track ↔ pour	e.g. U27 pad 4 ↔ GND zone priority 23	no DRC-clean straight path and the autorouter escape failed

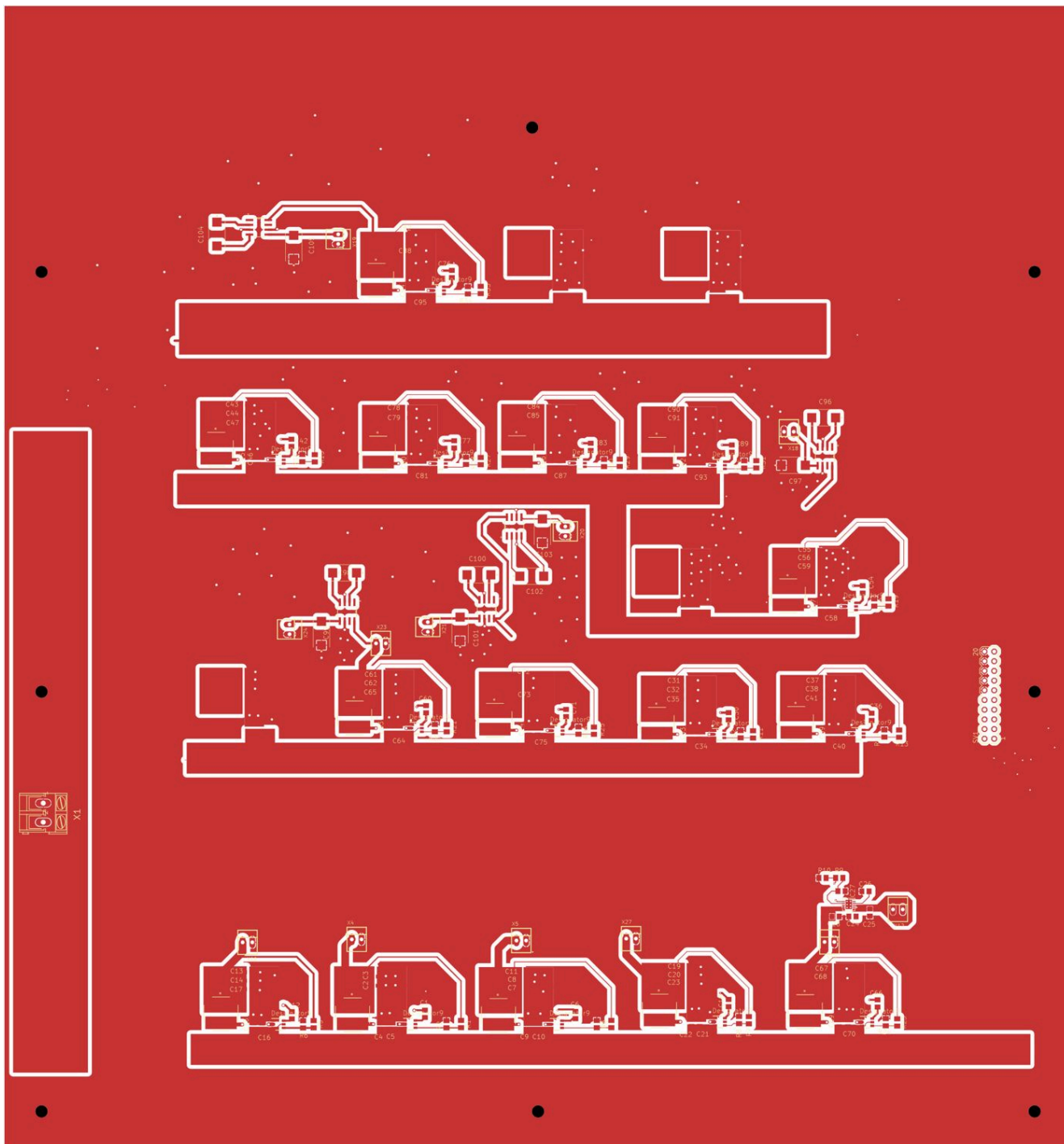
Manual closing procedure (source: `UNROUTED.md §How to close them`): open `POWER_SUPPLY.kicad_pcb`, Inspect → Design Rules Checker →

Run DRC, use the Unconnected Items tab: route 2.0 mm {VIN} / 1.0 mm {other rails} tracks between pours or extend a pour outline, then

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Refill zones, add 0.6/0.3 mm GND vias into the B.Cu plane where clearance allows, else move the autorouted 0.25 mm B.Cu track that cuts the island (all autorouted tracks are listed in `ses_merge.json` / `ses_merge_pass2.json`); re-run DRC until *Unconnected items* = 0, then `bash beta/pcb/tools/beta_export_package.sh POWER_SUPPLY`.

Remaining DRC after BETA (source: `beta/pcb/POWER_SUPPLY/README.md §3`): clearance 56 (0.15 mm pad-to-pad inside the ADM7151/TPS7A83 footprints and 3 Ns tracks of the source — footprint level), drill_out_of_range 8 (0.254 mm via-pads in the TPS7A8300 footprint vs DRU 0.3 mm min drill — fab to confirm), silk 168/65 (reference texts of script-placed parts on their own pads — cosmetic), via_dangling 20 (autorouter vias whose second-side wire was dropped — remove with *Cleanup tracks & vias* once routing is final), solder_mask_bridge 8, track_dangling 2, zones_intersect 1.

5.5 Layer plots and 3-D renders



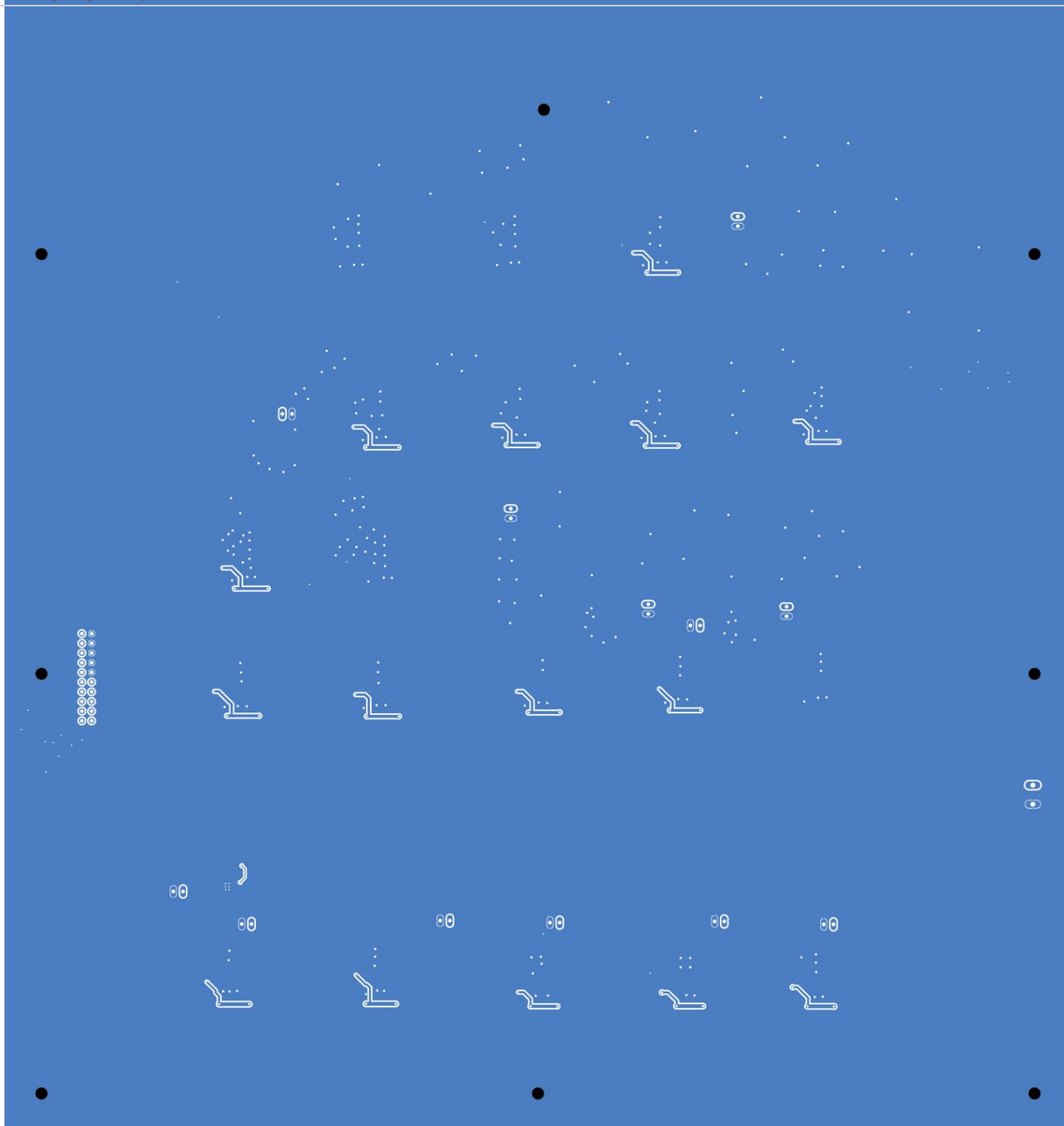


Figure 21 — F5.2b — Power Board bottom composite, mirrored (B.Cu + B.Silks + Edge.Cuts) (status: SOURCE-DERIVED; source: engineering/PCB/POWER_SUPPLY/svg/POWER_SUPPLY_bottom_composite_mirrored.svg; produced by tools/svg_sheets_to_pdf.py --png-dir manual/figures)

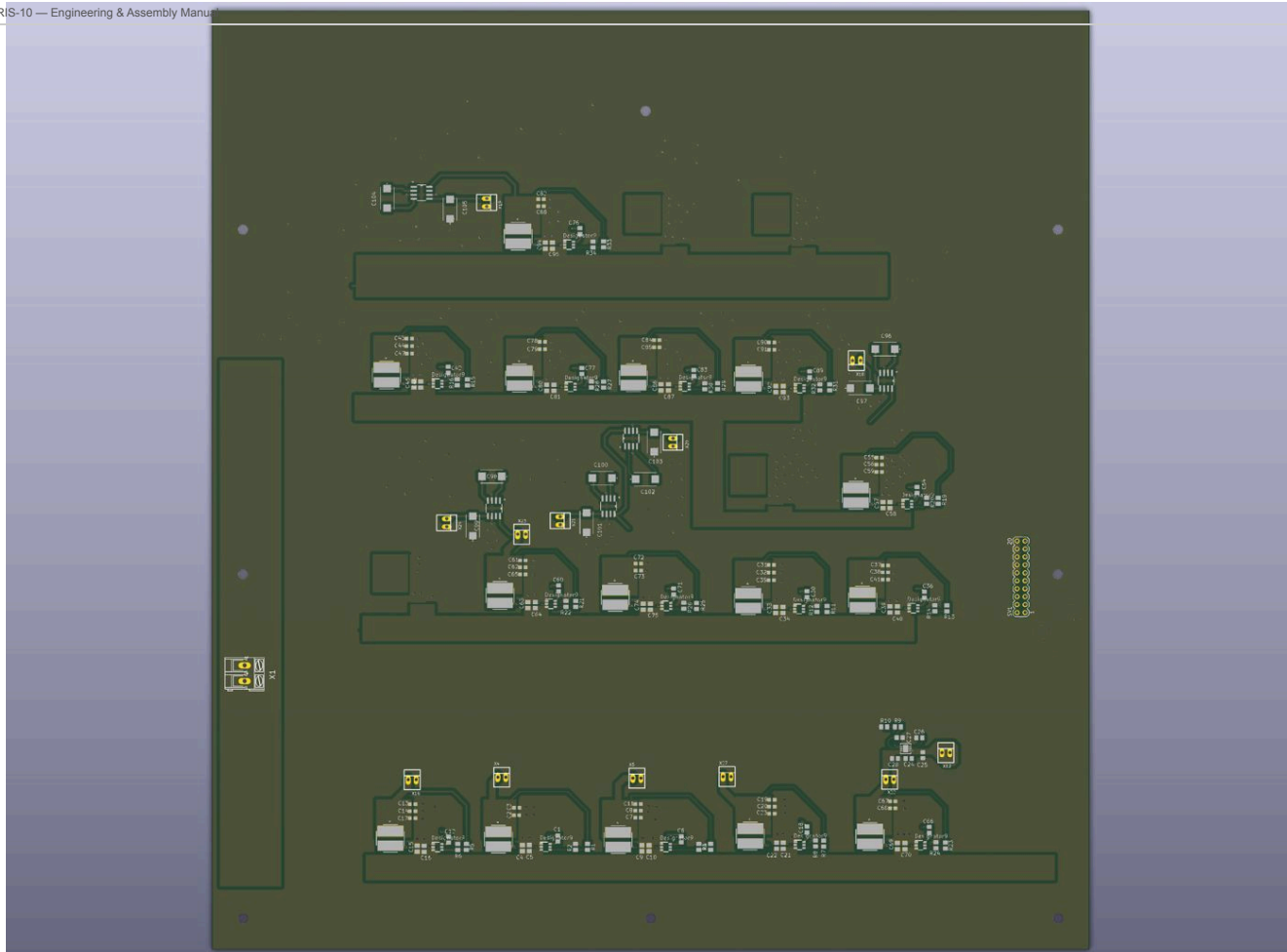


Figure 22 — F5.3a — Power Board 3-D render, top, KiCad conversion of the source layout (unplaced parts outside the outline) (status: SOURCE-DERIVED; source: engineering/PCB/POWER_SUPPLY/3d/POWER_SUPPLY_render_top.png; produced by kicad-cli pcb render, tools/kicad_pcb_pipeline.sh)

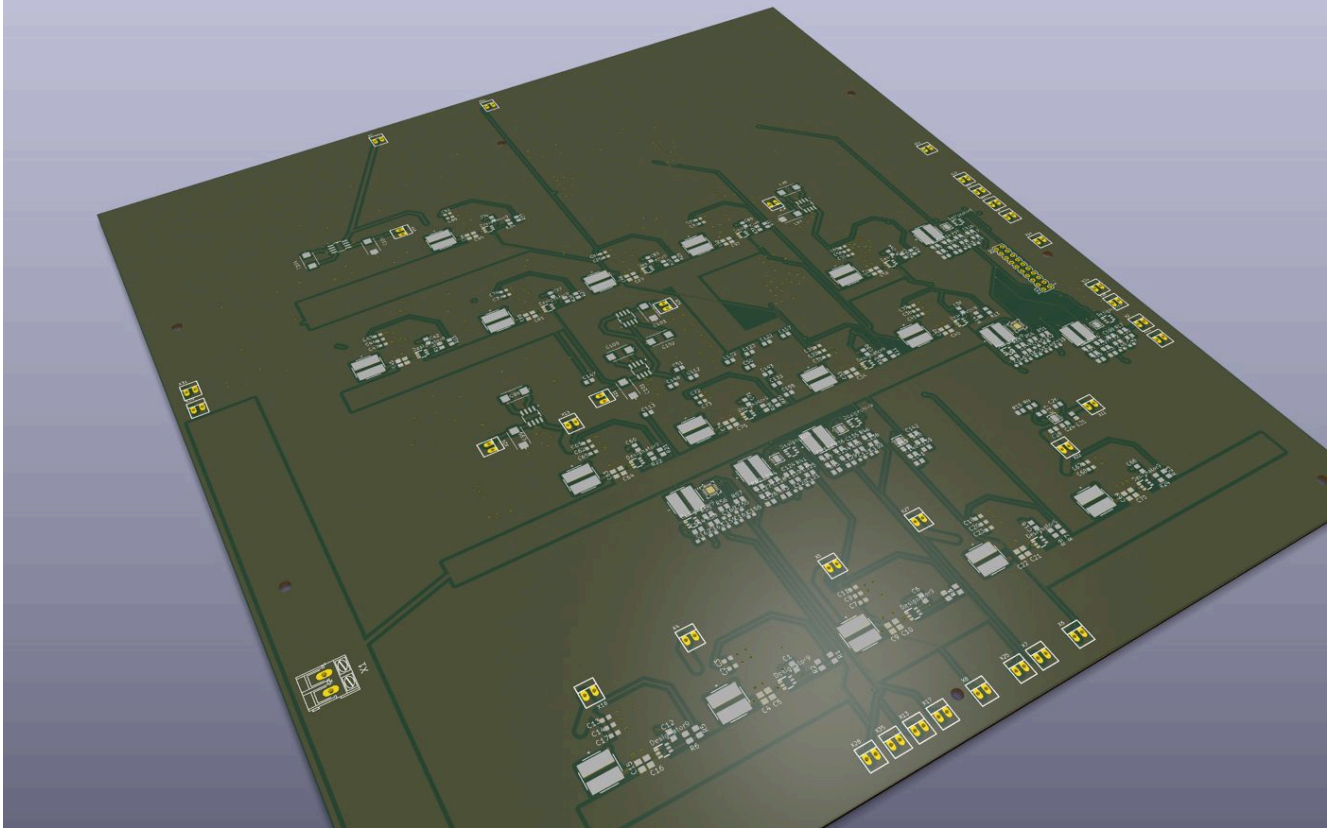


Figure 23 — F5.3b — Power Board BETA board after scripted placement, two Freerouting passes, bridges and GND stitching; 89 connections still open (status: BETA; source: beta/pcb/POWER_SUPPLY/exports/3d/POWER_SUPPLY_render_isometric.png; produced by beta/pcb/tools/beta_export_package.sh)

The generated assembly-drawing PDFs of this board are clipped by the A4 page frame in the same way as described for the Main Board (chapter 4 §4.5) and are not used as figures.

5.6 Stack-up

Source DRU PCBWay_2L_100um-Track (source: engineering/PCB/POWER_SUPPLY/STACKUP.md , status PARTIAL):

#	EAGLE layer	KiCad layer	Copper (DRU mtCopper)	Dielectric below (DRU mtIsolate)
1	1	F.Cu	0.035mm	1.5mm
2	16	B.Cu	0.035mm	—

PROPOSED fabrication note (source: beta/pcb/POWER_SUPPLY/FAB_NOTES.md §2–§4): FR-4 Tg 150 °C core 1.5 mm, finished **1.6 mm ± 10 %**, no controlled impedance; copper **35 μm, PROPOSED 70 μm / 2 oz** because the VIN rails are 2 mm polygons on a power-distribution board (designer to decide); ENIG; min track/space 0.10 (BETA routing 0.25) / 0.20 mm; min drill 0.30 mm (0.5 mm via); copper-to-edge 0.3 mm; IPC-A-600 Class 2; 100 % e-test with exports/ipc/POWER_SUPPLY_netlist.d356 . Note that §3 of that FAB_NOTES file repeats the RF-board template text (RO4350B, maskless RF traces) — it does not apply to this board, which has no RF or LVDS nets (FAB_NOTES.md §4). Current capacity of the highest-current traces (QPA2962 bias ≥ 1.68 A each per the xlsx; FPGA +1V0) is not checkable from XML and REQUIRES DESIGN REVIEW against IPC-2221 (source: docs/PCB/POWER_SUPPLY.md §7).

5.7 BOM summary

Source BOM docs/BOM/BOM_POWER_SUPPLY.csv : 312 references, 28 line items, 0 MPN attributes, 80 references without value (source: docs/BOM/README.md). BETA BOM confidence, counted with python3 -c "import csv,collections;print(collections.Counter(r['mpn_confidence'] for r in csv.DictReader(open('beta/pcb/POWER_SUPPLY/BOM_POWER_SUPPLY_beta.csv'))))":

mpn_confidence	Lines	Quantity (source: beta/pcb/README.md)
HIGH	7	79
MEDIUM	18	225
LOW	3	6

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mpn	confidence	Lines	Quantity (source: beta/pcb/README.md)
EMPTY		0	0
Total		28	312

Before purchase (source: docs/PCB/POWER_SUPPLY.md §8): confirm 21 × TPS562208 against the xlsx rail count; confirm the AK300/2 input connector rating for the total current (sum not computed in the xlsx); add the TPS562208 and LM2662 datasheets to 7_Components Datasheets .

5.8 Open items

ID	Item	Status	Source
K4	PA supply: xlsx needs 16 × 22 V / 2 A; this board has Vin [12–17] V only → DSN-PSU-01 proposal (chapter 9, D-14)	OPEN — power designer	docs/SYSTEM/BLOCK_DIAGRAM.md ; engineering/DESIGN/THERMAL/THERMAL_AND_PA_SUPPLY.md §5
G-10	Final outline: hole pattern (10..270 × 10..230, one at y = 267.85) suggests a smaller board than 280 × 300	UNRESOLVED	engineering/VALIDATION/UNRESOLVED_GEOMETRY.md
—	sch 9.6.2 / brd 7.4.0 mismatch; consistency check (P-EAGLE-01) not yet run in EAGLE	OPEN	docs/PCB/POWER_SUPPLY.md §5
—	89 open connections in BETA; placement review by a designer	OPEN	beta/pcb/POWER_SUPPLY/UNROUTED.md
—	24 pins without net (TPS7A8300 feedback select, LM2662 FC/OSC)	OPEN	engineering/ELECTRICAL/netlists/POWER_SUPPLY_unresolved_connections.md
—	80 value-less references (buck compensation parts)	OPEN	docs/BOM/README.md
—	13 rail outputs without a same-name destination connector	OPEN	engineering/DESIGN/HARNESS/HARNESS_SCHEDULE.md
—	Copper weight 1 oz vs 2 oz; trace current capacity review	OPEN — designer	beta/pcb/POWER_SUPPLY/FAB_NOTES.md §2
—	Datasheets of the parts actually used (TPS562208, LM2662) missing	OPEN	docs/PCB/POWER_SUPPLY.md §1

Acceptance criteria for this board (0 airwires, 0 off-board parts, outline recorded, DRC/ERC 0, Gerber/drill matching POWER_SUPPLY_outline.svg , 100 % MPN, PA-supply architecture decided) are copied from docs/PCB/POWER_SUPPLY.md §10 in chapter 14.

<!-- chapter 6: Synth board -->

6. Frequency Synthesizer Board (Clocks_Freq_Synth_board)

Chapter status summary: schematic — SOURCE-DERIVED (EAGLE 9.6.2 rendered); connector inventory — SOURCE-DERIVED; layout — complete in the source (0 airwires, 0 approved DRC) and SOURCE-DERIVED in the KiCad conversion; BETA copy with silkscreen clean-up only (copper untouched); layer plots and renders — SOURCE-DERIVED / BETA; stack-up — PARTIAL with three contradictory sources (DRU, `Stack_Hybrid.png`, PCBWay impedance note) and a PROPOSED fabrication note; pick-and-place — ORIGINAL PROJECT FILE (the only production artefact the upstream project delivered); BOM — PARTIAL (0 MPN; oscillator part-number conflict K7). Nothing is hardware-verified. Compiled by Antidrone Ukraine · antidrone.cc.

Sources: `docs/PCB/FREQUENCY_SYNTHESIZER.md`;

`engineering/ELECTRICAL/connection_diagrams/FREQUENCY_SYNTHESIZER_connection_report.md §1–2`;

`engineering/ELECTRICAL/schematics/FREQUENCY_SYNTHESIZER/`; `engineering/PCB/FREQUENCY_SYNTHESIZER/README.md`, `STACKUP.md`;

`beta/pcb/FREQUENCY_SYNTHESIZER/README.md`, `DRC_DISPOSITION.md`, `FAB_NOTES.md`, `BOM_FREQUENY_SYNTHESIZER_beta.csv`; `4_Schematics and Boards Layout/4_7_Production Files/Frequency_Synthesizer/` (ORIGINAL PROJECT FILE).

6.1 Role and key parts

The synthesizer board generates every clock and local oscillator of the radar: the AD9523 clock generator distributes 100 MHz (FPGA system clock), 400 MHz (ADC and FPGA ADC-side clock), 120 MHz (DAC and FPGA DAC-side clock), 300 MHz references to the two ADF4382 PLLs and a 20 MHz test output; the two ADF4382 synthesizers produce the TX and RX local oscillators. Board: **100 × 100 mm, 6 copper layers**, EAGLE 9.6.2, 184 physical parts, 139 nets, 769 vias, DRU `PCBWay_6L_100um-Track` (source: `docs/PCB/FREQUENCY_SYNTHESIZER.md §1`).

Key parts (deviceset names, MPN UNVERIFIED; source: `docs/PCB/FREQUENCY_SYNTHESIZER.md §1`):

Function	Part (refdes)	Qty	Note
Clock generator	AD9523BCPZ (IC1)	1	channel programming in firmware <code>main.cpp:965–1030</code>
TX / RX LO synthesizers	ADF4382ABCCZ (U1 TX, U6 RX)	2	
Baluns / transformers	MTX2-143+	4	
3 dB attenuators	ATS1005-3DB-FD-T05	4	
OCXO 100 MHz	ECOC-2522-100.000-3HC (X4)	1	xlsx lists <code>–3FC</code> suffix — CONFLICT
VCXO 50 MHz	CVHD-950-50.000 (X5, X6)	2	xlsx lists CVHD-950-100.000; firmware <code>vcxo_freq = 100 MHz</code> — K7
Ferrites	FBMH1608HL601-T	4	
SMA jacks	142-0731-211	11	silk names: L0 TX, L0 RX, AUX. L0 TX, AUX. L0 RX, ADC, FPGA=ADC, FPGA=DAC, DAC, FPGA SYS. CLOCK, TEST, AD9523 PLL_OUT, TX_LO MUXOUT, RX_LO MUXOUT
Differential outputs	CJT-T-P-HH-ST-TH1 (J3, J4)	2	
Power inputs	Molex 22-23-2021 (X10–X15)	6	
Control headers	JP1 (2×6), JP2 (2×7)	2	mate with Main Board JP1 / JP13

Firmware cross-reference of the AD9523 outputs (source: `docs/PCB/FREQUENCY_SYNTHESIZER.md §1`, `main.cpp:965–1030`): OUT0/1 300 MHz → ADF4382 TX/RX reference; OUT4 400 MHz → ADC; OUT5 400 MHz → FPGA (FPGA=ADC); OUT6 100 MHz → FPGA SYS. CLOCK; OUT7 20 MHz → TEST; OUT8/9 60 MHz SYNC; OUT10 120 MHz → DAC; OUT11 120 MHz → FPGA=DAC. Which oscillator (OCXO X4 or VCXO X5/X6) drives which AD9523 input REQUIRES VERIFICATION against the schematic nets (same source; K7).

Schematic summary (source: `engineering/ELECTRICAL/connection_diagrams/FREQUENCY_SYNTHESIZER_connection_report.md §1`):

Item	Count
Sheets	1
Parts (all) / physical	318 / 184
Nets	139
Pin connections	752
Single-pin nets	5
Unconnected pins on placed gates	34
Physical parts without value	47
Board airwires (unrouted connections, layer 19)	0
Missing symbol/footprint records	0

The 5 single-pin nets are N\$29, AD9523_OUT2±, AD9523_OUT3± — unused clock outputs (source: `docs/PCB/FREQUENCY_SYNTHESIZER.md §1`); they may stay unterminated only if the AD9523 datasheet permits, and `main.cpp:961–968` (unused channel divider = 0) must be checked for

6.2 Connectors

24 connector parts (source: engineering/ELECTRICAL/connection_diagrams/FREQUENCY_SYNTHESIZER_connection_report.md §2; SMA signal nets are mostly unnamed N\$xx — the silkscreen name and the interconnection table §6 give their function):

Refdes	Package	Signal nets (GND omitted)	Function (silk / interconnection table §6)
J1	142-0731-211	N\$35	SMA — see silk label on the board
J2	142-0731-211	N\$51	SMA
J3	CJT-T-P-HH-ST-TH1	AD9523_OUT4_P/N	400 MHz ADC clock → Main J21 (CBL-039 in DSN-HAR-01)
J4	CJT-T-P-HH-ST-TH1	AD9523_OUT5_P/N	OUT5 = 400 MHz FPGA=ADC per the firmware cross-reference → Main J19 (FPGA_ADC_CLOCK_P/N); the harness row CBL-042 labels this cable "test" — discrepancy to resolve
J5	142-0731-211	N\$64	120 MHz DAC → Main J20
J6	142-0731-211	N\$67	120 MHz FPGA=DAC → Main J18
J7	142-0731-211	AD9523_OUT6+	100 MHz FPGA SYS. CLOCK → Main J1
J8	142-0731-211	AD9523_OUT7+	20 MHz TEST
J9	142-0731-211	N\$37	SMA
J10	142-0731-211	N\$30	LO TX → Main J23
J11	142-0731-211	N\$55	LO RX → Main J22
J12	142-0731-211	N\$59	SMA
J13	142-0731-211	N\$61	SMA
JP1	PINHD-2X6	AD9523_PD, _REF_SEL, _SYNC, _RESET, _CS, _SCLK, _SDIO, _SD0, _STATUS0, _STATUS1, _EEPROM_SEL	AD9523 control ↔ Main JP1
JP2	PINHD-2X7	ADF4382_SD0, _SCLK, _SDIO, ADF4382_TX_CS/CE/DELSTR/DELADJ/LKDET, ADF4382_RX_CS/CE/DELSTR/DELADJ/LKDET	ADF4382 control ↔ Main JP13
X4	ECOC-2522 (OCXO)	N\$1, +3V3_X0, 10MHZ_OUT	100 MHz OCXO
X5	CVHD-950 (VCXO)	100MHZ_OUT, +3V3_X0	VCXO
X6	CVHD-950 (VCXO)	N\$2, VCX0_OUT, +3V3_X0	VCXO
X10–X14	22-23-2021	N\$69, N\$71, N\$73, N\$75, N\$77	power inputs (rail names on the Power Board side — interconnection table §4)
X15	22-23-2021	+5V0_L0	power input ← Power Board X6 (CBL-005)

The SMA ↔ silk-name assignment for J1, J2, J9, J12, J13 (AUX. L0 TX, AUX. L0 RX, AD9523 PLL_OUT, TX_L0 MUXOUT, RX_L0 MUXOUT) is in engineering/SYSTEM/interfaces/interconnection_table.md §6; it is not repeated here because the report lists only the net names.

6.3 Schematic sheet

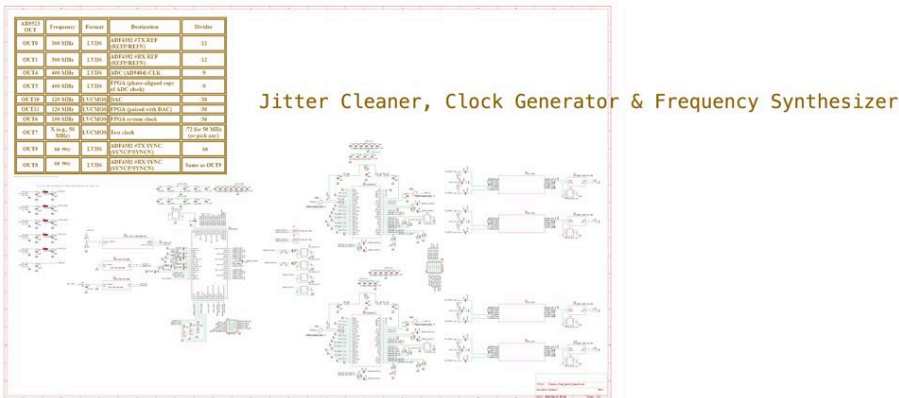


Figure 24 — F6.1 — Frequency Synthesizer schematic, single sheet: AD9523 clock generator, 2 × ADF4382 LO synthesizers, OCXO/VCXO, 11 SMA outputs, control headers JP1/JP2 (status: SOURCE-DERIVED; source: engineering/ELECTRICAL/schematics/FREQUENCY_SYNTHESIZER/png/FREQUENCY_SYNTHESIZER_schematic_sheet1.png; produced by tools/render_eagle_schematic.py)

6.4 Layout state: source vs. engineering conversion vs. BETA

Source (EAGLE 9.6.2)

Item	Finding (source: docs/PCB/FREQUENCY_SYNTHESIZER.md §2)
Routing / DRC / ERC	complete; 0 airwires; 0 approved DRC; 0 approved ERC; sch/brd part and net sets identical
Min track	0.1 mm; RF widths 0.204 mm (×329) and 0.22 mm (×414)
DRU	mdWireWire 0.1, mdCopperDimension 0.3, msDrill 0.15, rvViaOuter 0.25, mtIsolate = 0.11, 0.6, 0.11, 0.36, 0.2 ..., 0.6, 0.11 mm, thermals for vias off
Stack-up	EAGLE: 6 Cu, prepreg-outer/foil construction. Stack_Hybrid.png: 6 Cu with RO4350B 0.102 mm outer cores, FR-4 0.100 centre, prepregs 0.100. PCBWay note: h = 0.102 mm. Three sources disagree on construction and thickness → UNRESOLVED — REQUIRES BOARD DESIGN VERIFICATION
2.0 mm via	one 2.0 mm drill via — check purpose (mounting/thermal)

Production files delivered by the upstream project (ORIGINAL PROJECT FILE; source: docs/PCB/FREQUENCY_SYNTHESIZER.md §1):
Clocks_Freq_Synth_board_BOM.xlsx (41 rows, EAGLE bom.tbl CSV pasted into one column, empty MPN columns, mojibake 0.1µF); -
smd.mnt (173 rows, all TOP) and -tht.mnt (10 rows: JP1, JP2, X10–X15, J3, J4) with duplicates mnt.csv, -tht.csv, smd.xlsx,
to.xlsx; PCBWay_Impedance_Note_RO4350B_h0p102mm.pdf (RO4350B, h = 0.102 mm, Dk 3.48/3.66, 35 µm Cu, no mask on RF, 50 Ω w =

AERIAL 10-09 Engineering/PCB/FREQUENCY_SYNTHESIZER/ 0.204 mm, 100.0 diff w = 0.204 / s = 0.26 mm, via fence ≤ 1 mm pitch, coupons requested). No Gerber, drill, fab or assembly drawing was delivered.

Engineering conversion (engineering/PCB/FREQUENCY_SYNTHESIZER/)

Cross-check (source: engineering/PCB/FREQUENCY_SYNTHESIZER/README.md §3):

Check	EAGLE (XML)	KiCad	Result	Note
Width (mm)	100.0	100.0	OK	
Height (mm)	100.0	100.0	OK	
Footprints = EAGLE elements + free holes	188	188	OK	KiCad imports every free <hole> as a footprint
Vias	769	769	OK	
Tracks (signal wires excl. airwires)	1374	1374	OK	
Copper layers	6	6	OK	
NPTH holes (free holes + package holes)	4	4	OK	pads with drill but no copper are NPTH in KiCad too
PTH holes (vias + THT pads)	863	863	OK	

DRC on the conversion: 639 violations, 0 unconnected (source: engineering/PCB/FREQUENCY_SYNTHESIZER/README.md §6):

Rule	Count	Interpretation
solder_mask_bridge	199	mask web between adjacent pads thinner than KiCad default (EAGLE has no such rule) — fabricator decision
silk_overlap	199	overlapping silkscreen texts/lines — cosmetic
silk_over_copper	199	silk over exposed copper — cosmetic/assembly
track_dangling	35	track end not connected — review (stubs or unfinished routing)
courtyards_overlap	7	footprint courtyards overlap — imported courtyard approximation

BETA board (beta/pcb/FREQUENCY_SYNTHESIZER/)

Copper was **not modified** (source: beta/pcb/FREQUENCY_SYNTHESIZER/README.md). Before/after (same source §1):

Check	Before (engineering/PCB conversion)	After (BETA)
unconnected_items	0	0
courtyards_overlap	7	7
silk_over_copper	199	199
silk_overlap	199	199
solder_mask_bridge	199	199
track_dangling	35	35
DRC violations total	639	639

Change: 43 silkscreen reference texts moved to the nearest free position (≤ 3 mm, 0.05 mm step; beta/pcb/tools/beta_silk_nudge.py, silk_nudges.json); 103 texts remain in conflict because the 0201-dense board has no free spot within 3 mm (silk_conflicts_remaining.json). Disposition of all 639 items (source: beta/pcb/FREQUENCY_SYNTHESIZER/DRC_DISPOSITION.md §1): 7 courtyard overlaps REAL (review) — pairs C25↔L10, L12↔C45, L12↔C33, C27↔L10, L11↔C29, L11↔C31, L9↔C21; the pads do not touch, designer to confirm that the L5650M inductor body does not collide with the adjacent capacitors at assembly; 35 dangling GND arc stubs 0.002–0.212 mm inside the GND pour — COSMETIC; silkscreen 147 → 103 texts in conflict (measured uncapped) — COSMETIC, fab clips silk on pads; solder-mask bridges — FAB REVIEW (PCBWay minimum mask dam 0.1 mm, CAM merges apertures). Recommendation for the production revision: hide 0201/0402 reference designators on silk.

6.5 Layer plots and 3-D renders

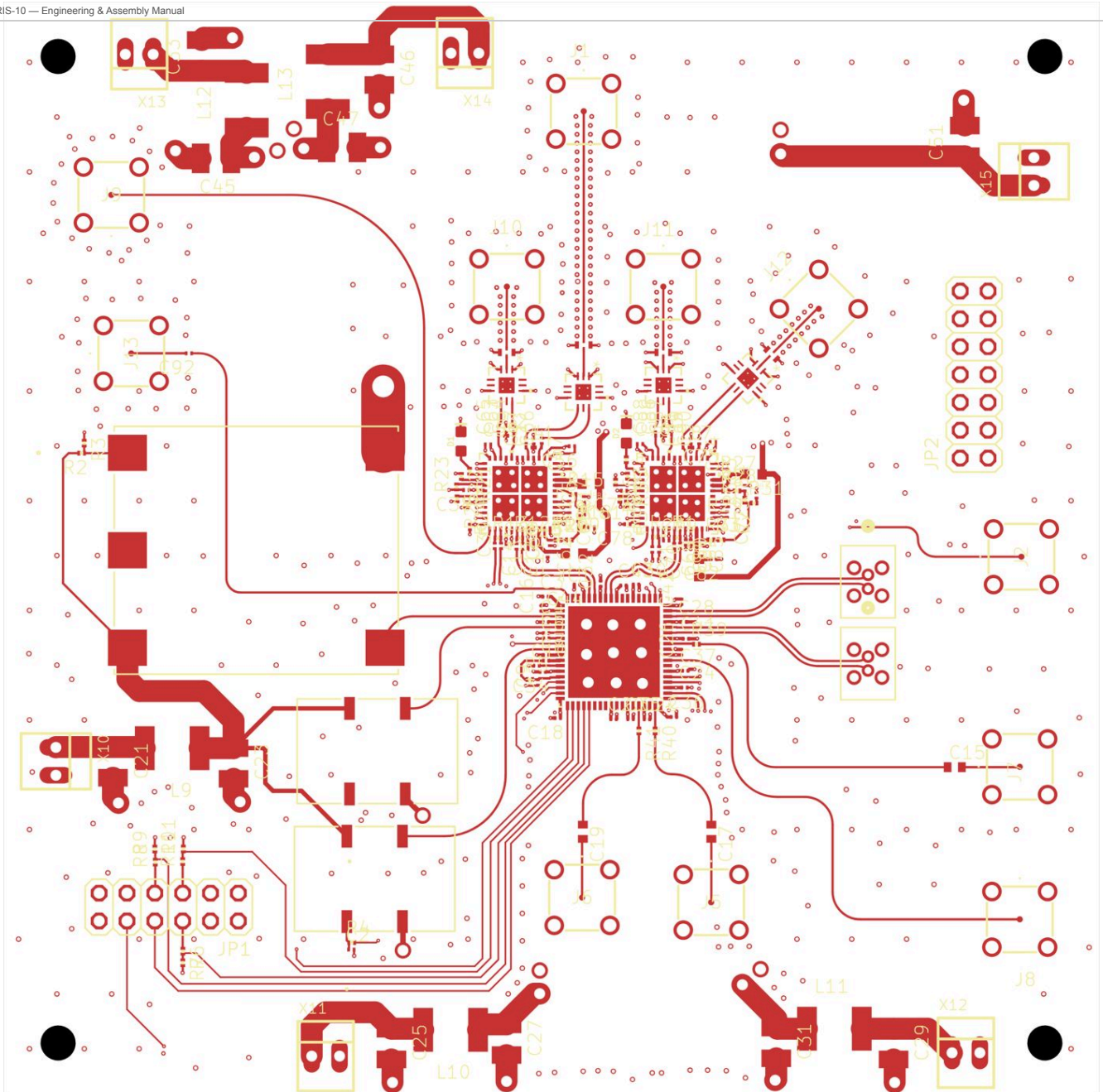


Figure 25 — F6.2a — Frequency Synthesizer top composite (F.Cu + F.Silks + Edge.Cuts), 100 × 100 mm (status: SOURCE-DERIVED; source: engineering/PCB/FREQUENCY_SYNTHESIZER/svg/FREQUENCY_SYNTHESIZER_top_composite.svg; produced by tools/svg_sheets_to_pdf.py --png-dir manual/figures)

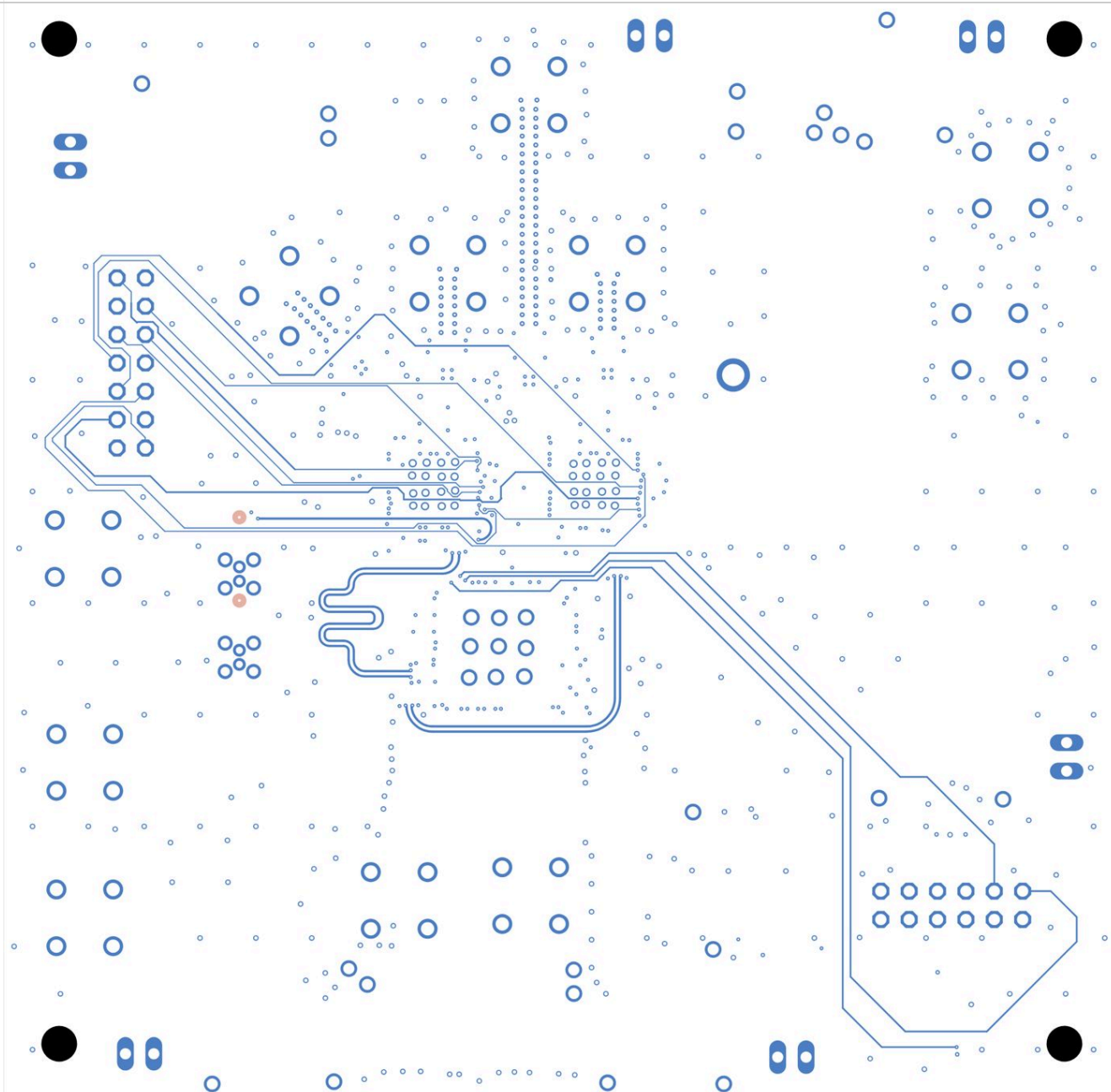


Figure 26 — F6.2b — Frequency Synthesizer bottom composite, mirrored (B.Cu + B.Silks + Edge.Cuts); the AD9523_OUT8/9 pairs run on B.Cu (status: SOURCE-DERIVED; source: engineering/PCB/FREQUENCY_SYNTHESIZER/svg/FREQUENCY_SYNTHESIZER_bottom_composite_mirrored.svg; produced by tools/svg_sheets_to_pdf.py --png-dir manual/figures)

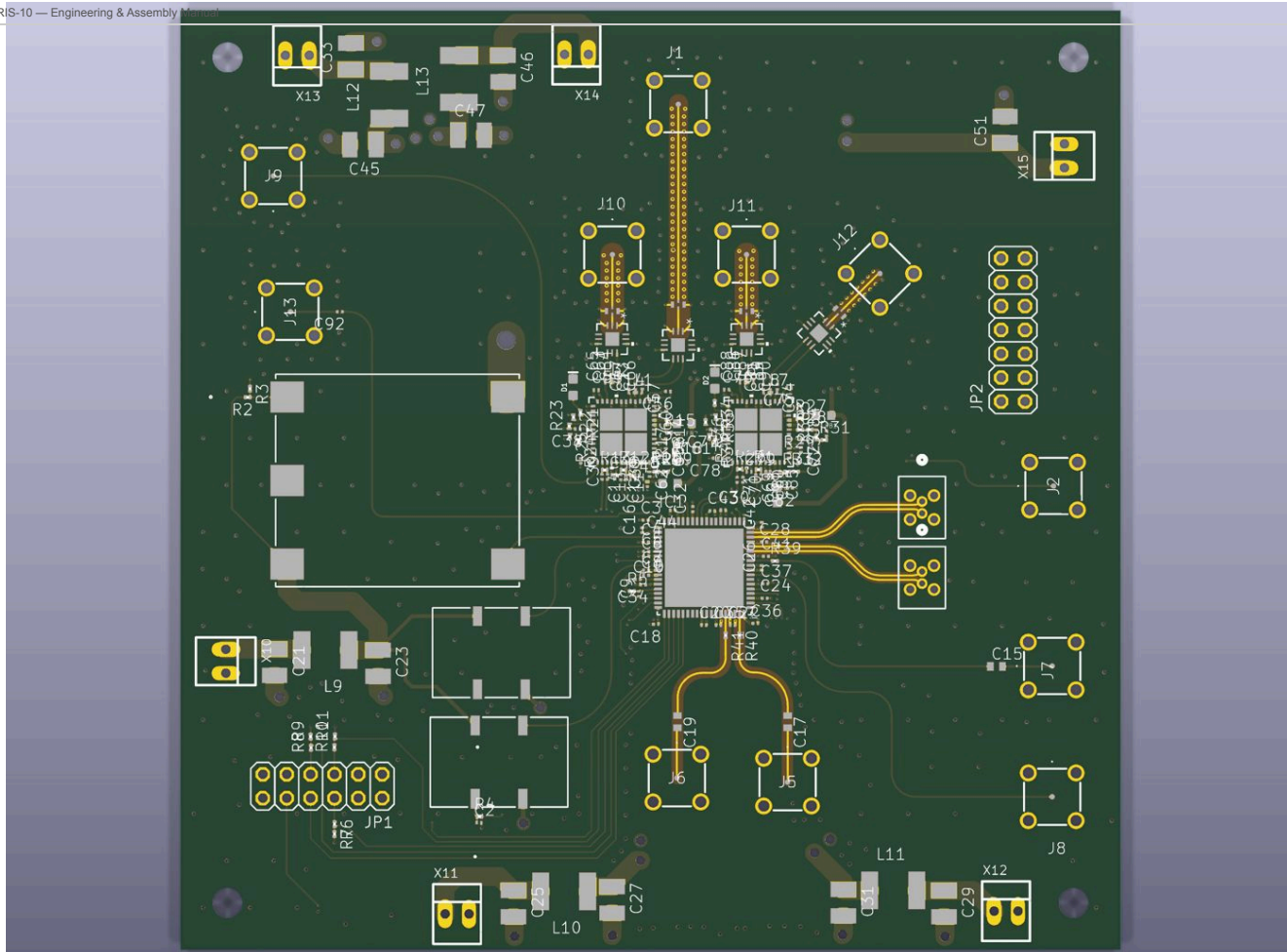


Figure 27 — F6.3a — Frequency Synthesizer 3-D render, top, KiCad conversion; no component models (status: SOURCE-DERIVED; source: engineering/PCB/FREQUENCY_SYNTHESIZER/3d/FREQUENCY_SYNTHESIZER_render_top.png; produced by kicad-cli pcb render, tools/kicad_pcb_pipeline.sh)

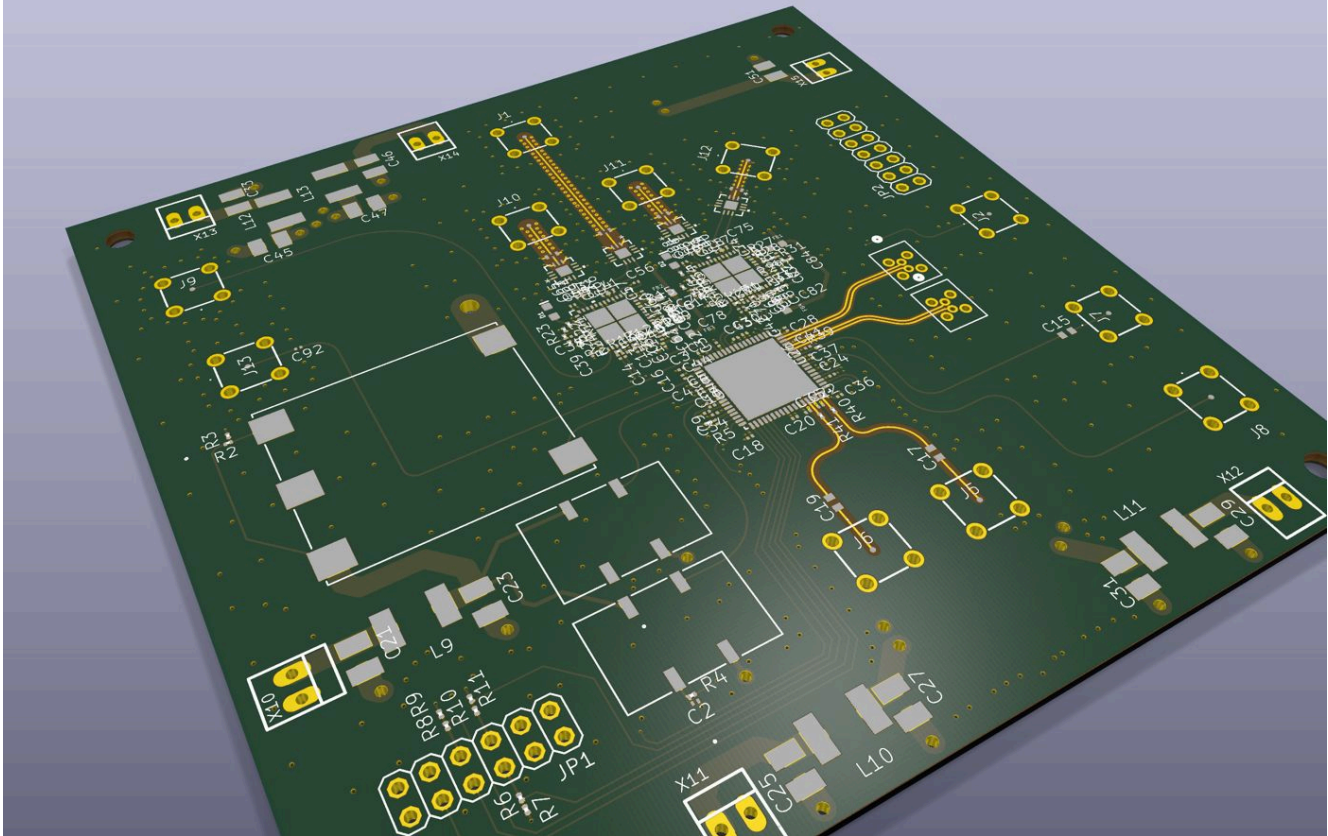


Figure 28 — F6.3b — Frequency Synthesizer BETA board, isometric render (silkscreen nudged, copper identical to the source) (status: BETA; source: beta/pcb/FREQUENCY_SYNTHESIZER/exports/3d/FREQUENCY_SYNTHESIZER_render_isometric.png; produced by beta/pcb/tools/beta_export_package.sh)

The generated assembly-drawing PDFs are clipped by the A4 page frame (chapter 4 §4.5) and are not used as figures; the original `-smd.mnt` / `-tht.mnt` pick-and-place files remain the upstream placement reference.

6.6 Stack-up

Source DRU `PCBWay_6L_100um-Track`, `layerSetup (1+2*3+14*15+16)` (source: `engineering/PCB/FREQUENCY_SYNTHESIZER/STACKUP.md`, status PARTIAL):

#	EAGLE layer	KiCad layer	Copper (DRU mtCopper)	Dielectric below (DRU mtIsolate)
1	1	F.Cu	0.035mm	0.11mm
2	2	In1.Cu	0.035mm	0.6mm
3	3	In2.Cu	0.035mm	0.11mm
4	14	In3.Cu	0.035mm	0.6mm
5	15	In4.Cu	0.035mm	0.11mm
6	16	B.Cu	0.035mm	—

PROPOSED fabrication stack-up (source: `beta/pcb/FREQUENCY_SYNTHESIZER/FAB_NOTES.md §2`): layer 1 on **RO4350B 4 mil** (DRU 0.11 mm vs. note 0.102 mm — to be reconciled; RO4350B 4 mil = 0.101 mm), FR-4 cores 0.6 mm and prepregs 0.11 mm inside, layer 5/6 dielectric PROPOSED as RO4350B as well because B.Cu carries the AD9523_OUT8/9 pairs at 0.204 mm; dielectric sum 1.53 mm + 6 × 35 μm ≈ 1.74 mm → finished **1.6 mm ± 10 %** with thinned cores, or 2.0 mm — to be decided with PCBWay. Controlled impedance (same source §4): 100 Ω differential at w = 0.204 / s = 0.26 mm for AD9523_OUT0...9_P/N, ADF4382_TX/RX_SYNC_P/N (RX_SYNC s = 0.296 mm), TX/RX_RFOUT_1/2_P/N run uncoupled (s ≈ 0.8 mm); 50 Ω at w = 0.204 mm (0.22 mm on some OUT+ stubs) for 100MHZ_OUT, 10MHZ_OUT, VCX0_OUT, AD9523_OUT6/7/10/11+; SPI/control not controlled. The PCBWay impedance note belongs to this board: continuous GND under the RF layer, no solder mask on RF traces, coupons for 50 Ω and 100 Ω, PCBWay may tune w by ±0.02–0.04 mm and s by ±0.03–0.05 mm.

6.7 BOM summary

Source BOM `docs/BOM/BOM_FREQUENCY_SYNTHESIZER.csv`: 184 references, 40 line items, 0 MPN attributes, 47 references without value — consistent with the upstream 40-line xlsx (source: `docs/BOM/README.md`). BETA confidence, counted with `python3 -c "import`

```
csv,collections:print(collections.Counter(r['mpn_confidence'] for r in
csv.DictReader(open('beta/pcb/FREQUENCY_SYNTHESIZER/BOM_FREQUENCY_SYNTHESIZER_beta.csv'))))":
```

mpn_confidence	Lines	Quantity (source: beta/pcb/README.md)
HIGH	10	37
MEDIUM	22	110
LOW	6	29
EMPTY	2	8
Total	40	184

BOM validation before purchase (source: docs/PCB/FREQUENCY_SYNTHESIZER.md §8): add MPN and manufacturer to all 40 lines; fill the 47 missing values; fix the encoding (UTF-8 μ); mark the 11 SMA connectors populated/DNP per variant; resolve the oscillator conflict (ECOC-2522-100.000-3FC vs -3HC suffix; CVHD-950-100.000 in the xlsx vs CVHD-950-50.000 in the schematic — K7).

6.8 Open items

ID	Item	Status	Source
K7	Oscillator topology and part numbers (OCXO 100 MHz + 2 × VCXO 50 MHz in CAD vs VCXO 100 MHz in xlsx/firmware); which oscillator feeds AD9523 REF/OSC_IN	OPEN — RF designer	docs/SYSTEM/BLOCK_DIAGRAM.md ; docs/PCB/FREQUENCY_SYNTHESIZER.md §1
—	Stack-up: DRU / Stack_Hybrid.png / PCBWay note disagree; one construction to be chosen and the DRU mtIsolate updated (G-01, MDR-07)	BLOCKED — MISSING DATA	docs/PCB/FREQUENCY_SYNTHESIZER.md §2
—	5 single-pin nets (unused AD9523 outputs) — datasheet/firmware output_dis check	OPEN	same §7
—	7 courtyard overlaps (L10/L11/L12/L9 vs 0201 capacitors) — body-height check	OPEN — designer	beta/pcb/FREQUENCY_SYNTHESIZER/DRC_DISPOSITION.md §2
—	2.0 mm via — purpose unknown	OPEN	docs/PCB/FREQUENCY_SYNTHESIZER.md §2
—	Duplicate P&P files (mnt.csv, -tnt.csv, smd_.xlsx, th_.xlsx) to be superseded by one regenerated pair	OPEN	same §3
—	BOM: 0/40 MPN verified, 47 value-less references, encoding	OPEN	docs/BOM/README.md
—	103 silkscreen conflicts on 0201 parts; hide small refdes for production	OPEN	beta/pcb/FREQUENCY_SYNTHESIZER/README.md §3

Post-fabrication functional acceptance (hardware test, not a file check; source: docs/PCB/FREQUENCY_SYNTHESIZER.md §10): AD9523 lock (STATUS0/1 high), OUT6 = 100 MHz, OUT4/5 = 400 MHz, OUT10/11 = 120 MHz measured — see chapter 16.

<!-- chapter 7: RF PA board (×16) -->

7. RF Power Amplifier Board (RF_PA, ×16)

Chapter status summary: schematic — SOURCE-DERIVED (EAGLE 9.6.2 rendered); layout — complete in the source (0 airwires, 0 approved DRC), SOURCE-DERIVED in the KiCad conversion, BETA copy with one GND strap and the paddle polygon given its net (RF tracks untouched); layer plots and renders — SOURCE-DERIVED / BETA; stack-up — PARTIAL (4-layer DRU, 70 µm copper slot unexplained) with a PROPOSED fabrication note; BOM — PARTIAL (0 MPN; QPA2962_B is a deviceset name); thermal budget and drain gating — PROPOSED DESIGN (DSN-THM-01, D-10/D-11/D-14, first-order estimates). The 22 V drain supply, the drain-gating switch and the heat path do not exist in the original CAD. RF performance (gain, P1dB at 10.5 GHz) is unverified. Compiled by Antidrone Ukraine · antidrone.cc.

Sources: docs/PCB/RF_PA.md; engineering/ELECTRICAL/connection_diagrams/RF_PA_connection_report.md; engineering/ELECTRICAL/schematics/RF_PA/; engineering/PCB/RF_PA/README.md, STACKUP.md; beta/pcb/RF_PA/README.md, FAB_NOTES.md, BOM_RF_PA_beta.csv; engineering/DESIGN/THERMAL/THERMAL_AND_PA_SUPPLY.md; engineering/SYSTEM/interfaces/interconnection_table.md §7.

7.1 Role and key parts

One RF_PA board amplifies the transmit signal of one antenna row; the AERIS-10X variant uses sixteen boards, 10 W each (source: docs/PCB/RF_PA.md §2, citing README.md:80). Each board holds a single Qorvo QPA2962 GaN PA with its gate-bias input (VG, from the Main Board DAC5578/OPA4703 chain), a drain input (VD) through a 5 mΩ shunt whose two ends (VD, VIN_M) are returned to the Main Board INA241A3 current-sense amplifier, and two SMA jacks. Board: **35 × 60 mm, 4 copper layers**, EAGLE 9.6.2, 25 physical parts, 15 nets, 77 wires, 342 vias (of which 215 × 0.35 mm stitching/thermal), DRU PCBWay_4L_100um-Track (source: docs/PCB/RF_PA.md §1).

Parts (source: docs/PCB/RF_PA.md §1; engineering/ELECTRICAL/connection_diagrams/RF_PA_connection_report.md §4):

Refdes	Value / device	Connections (pad:net)	Note
U\$1	QPA2962_B (Qorvo 10 W GaN PA)	GND, RFIN: N\$2, RFOUT: N\$8, VD1/VD2: VIN_M, VG: VG	deviceset name is not the orderable MPN; datasheet and S-parameters at 22 V / 1 680 mA in 7_Components Datasheets
J1, J2	SMA 142-0731-211	J1 pin 1 N\$2 (RFIN per silk), J2 pin 1 N\$8 (RFOUT per silk)	
X2	Molex 22-23-2021	1 VG, 2 GND	gate bias from Main X_n
X3	Molex 22-23-2031	1 VD, 2 VIN_M, 3 GND	drain-current sense pair to Main X3/X38...X52
22V	AK300/2 screw terminal	1 VD, 2 GND	drain supply input — no source in CAD (K4)
R10	5 mΩ WSL2816	1 VIN_M, 2 VD	shunt; BETA BOM flags a 0.1 Ω part number for a 5 mΩ value
R1, R4	10R 0402	VG / VIN_M decoupling	
R2, R3, R5–R9	0R 0402/0603	VG / VIN_M decoupling links	
C1, C4, C5	10 µF 1206	decoupling	
C2, C3, C6–C9	0.1 µF 0402	decoupling	

Schematic summary (source: engineering/ELECTRICAL/connection_diagrams/RF_PA_connection_report.md §1): 1 sheet; 43 parts / 25 physical; 15 nets; 78 pin connections; 0 single-pin nets; 0 unconnected pins; 6 physical parts without value; 0 airwires; 0 missing symbol/footprint records. Net fan-out: GND 38 connections, VIN_M 10 (same source §3). Silk labels on the board: CURRENT_SENSOR, VIN+, VIN–, VG (source: docs/PCB/RF_PA.md §1).

System connections of one PA instance n (source: engineering/SYSTEM/interfaces/interconnection_table.md §7): element RF from Main Board SMA pair (J27/J26 for n = 1 ... J51/J50 for n = 16) to J1/J2 — which SMA of the Main pair is TX-to-PA and which is RX-return is UNVERIFIED; gate bias VG_n from Main X_7 = VG_1, X_16 = VG_2, X_8 = VG_3, X_15 = VG_4, X_4 = VG_5, X_11 = VG_6, X_3 = VG_7, X_12 = VG_8, X_5 = VG_9, X_14 = VG_10, X_6 = VG_11, X_13 = VG_12, X_2 = VG_13, X_9 = VG_14, X_1 = VG_15, X_10 = VG_16 (xlsx: –4 ... –1.2 V, 10 mA); sense pair to Main X3, X38...X52 (INA241A3); drain VD 22 V from a supply that is NOT IN CAD (xlsx row 59: 18–22 V, 2 000 mA, "Set VD +22 V"); drain enable EN/DIS_RFPA_VDD on Main JP10 (STM32 PD6, main.cpp:1601) to a switch that is NOT IN CAD.

7.2 Schematic sheet

AERIS-10 — RF_PA — Schematic sheet 1/1

Drawing ID: ELEC-SCH-PA-S1 Revision: A Date: 2026-10-09 Units: mm Scale: 1:1 at the stated page size

Source: 4_Schematics and Boards Layout/4_6_Schematics/PowerAmplifierBoard/RF_PA.sch (EAGLE 9.6.2 XML; file revision: none recorded in the source) Status:

Generated by tools/render_eagle_schematic.py — documentation rendering of the native EAGLE file; fonts differ from EAGLE vector font.

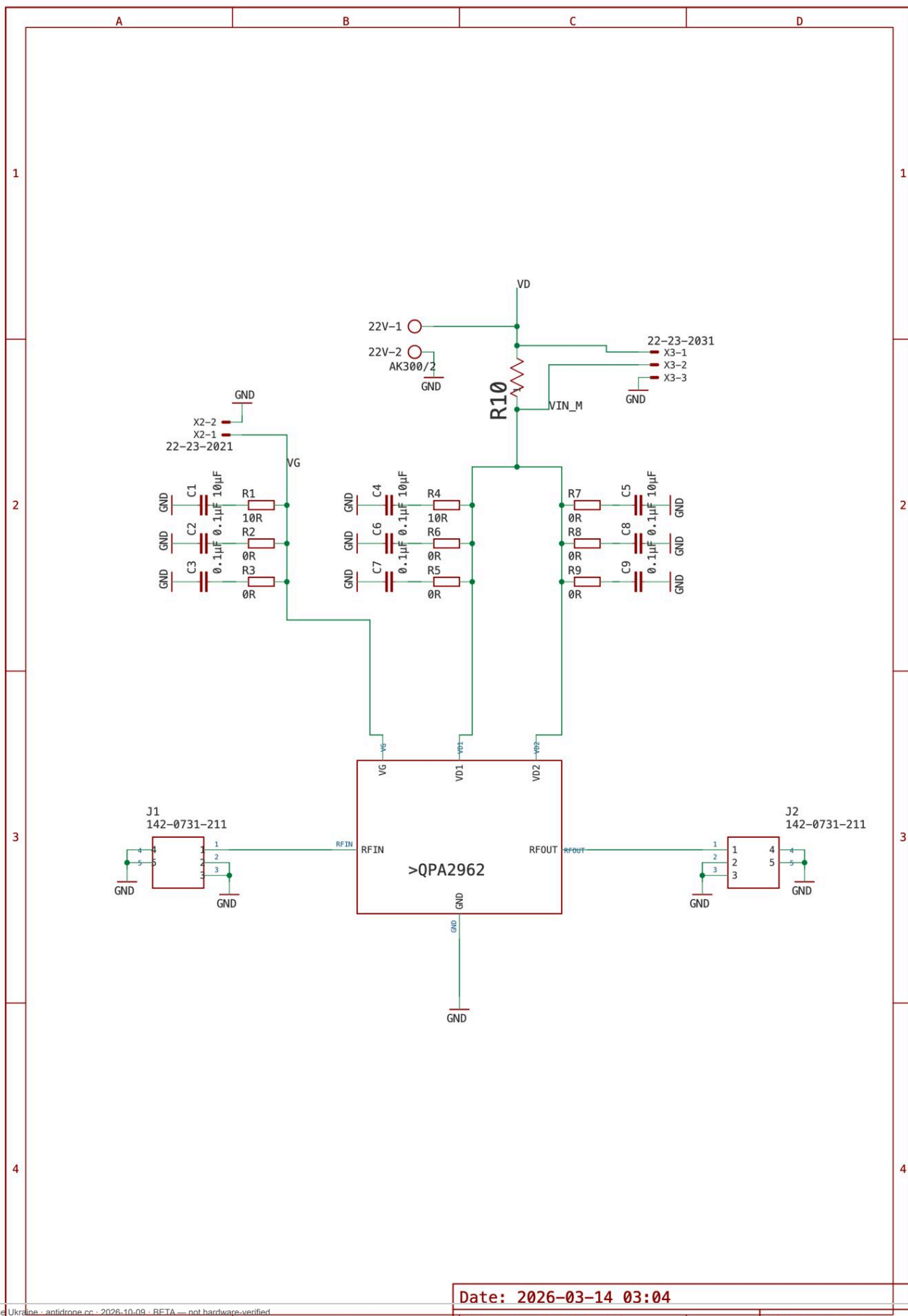


Figure 29 — F7.1 — RF PA schematic, single sheet: QPA2962, SMA in/out, VG and VD/VIN_M decoupling, 5 mΩ sense shunt, 22 V terminal (status: SOURCE-DERIVED; source: engineering/ELECTRICAL/schematics/RF_PA/png/RF_PA_schematic_sheet1.png; produced by tools/render_eagle_schematic.py)

7.3 Layout state: source vs. engineering conversion vs. BETA

Source (EAGLE 9.6.2)

Item	Finding (source: docs/PCB/RF_PA.md §2)
Routing	complete (0 airwires)
DRC/ERC	0 approved entries; sch/brd consistent
Min track	0.204 mm (50 Ω microstrip width of the PCBWay note)
DRU	mdWireWire 0.15, mdCopperDimension 0.3, msDrill 0.15 mm; mtCopper: third slot 0.07 mm (70 μm) although slot 3 is not an active layer — interpretation UNRESOLVED; mtIsolate 0.11 / 1.2 / 0.36 ... mm
Stack-up	4 layers; Stack_Hybrid.png (6 layers) does not describe this board; no vendor stack-up → UNRESOLVED — REQUIRES BOARD DESIGN VERIFICATION
Thermal	215 × 0.35 mm vias under/around the PA: thermal path to the enclosure/heatsink is undocumented (no mechanical data)
System fit	16 PA boards at 10 W each; xlsx budgets +22 V / 2 A × 16; the Power Board has no 22 V rail — PA supply UNRESOLVED

Engineering conversion (engineering/PCB/RF_PA/)

Cross-check (source: engineering/PCB/RF_PA/README.md §3):

Check	EAGLE (XML)	KiCad	Result	Note
Width (mm)	35.0	35.0	OK	
Height (mm)	60.0	60.0	OK	
Footprints = EAGLE elements + free holes	32	32	OK	KiCad imports every free <hole> as a footprint
Vias	342	342	OK	
Tracks (signal wires excl. airwires)	77	77	OK	
Copper layers	4	4	OK	
NPTH holes (free holes + package holes)	7	7	OK	pads with drill but no copper are NPTH in KiCad too
PTH holes (vias + THT pads)	357	357	OK	

DRC on the conversion: 48 violations, 1 unconnected (source: engineering/PCB/RF_PA/README.md §6):

Rule	Count	Interpretation
clearance	16	copper clearance < DRU — review
zones_intersect	8	overlapping pours of different priority — conversion artefact or design issue
silk_overlap	8	overlapping silkscreen texts/lines — cosmetic
silk_over_copper	8	silk over exposed copper — cosmetic/assembly
shorting_items	7	copper of different nets touching after conversion (typically EAGLE polygon vs unnamed copper) — REVIEW in EAGLE
silk_edge_clearance	1	silk too close to edge

BETA board (beta/pcb/RF_PA/)

Before/after (source: beta/pcb/RF_PA/README.md §1):

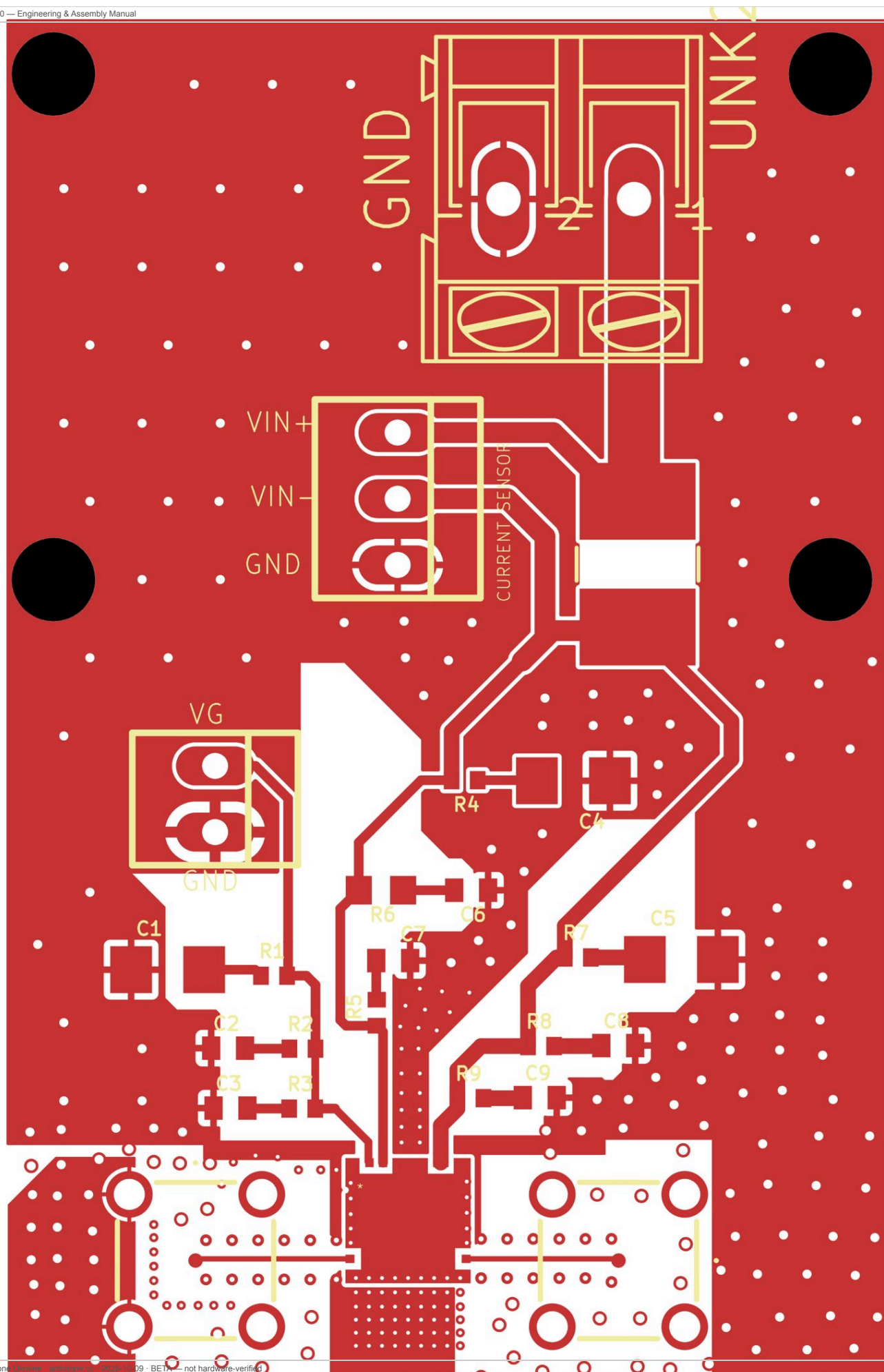
Check	Before (engineering/PCB conversion)	After (BETA)
unconnected_items	1	0
clearance	16	1
shorting_items	7	0
silk_edge_clearance	1	1
silk_over_copper	8	8
silk_overlap	8	8
zones_intersect	8	0
DRC violations total	48	18

Changes, nothing in the netlist (source: beta/pcb/RF_PA/README.md §2): (1) GND strap F.Cu 0.5 mm from the via at (16.20, -11.20) to the QPA2962 ground paddle centre (16.20, -13.40) — the only unconnected item: the paddle was connected to the GND via field only through

a net-less footprint polygon, which KiCad does not count as copper of the net; (2) the U\$1 paddle polygon re-created as a board-level copper polygon with net GND, identical geometry (it touched 21 GND items and nothing else; as a net-less graphic it produced 7 shorting + 19 hole-clearance + 15 clearance false errors); (3) 6 overlapping same-net GND zones given distinct priorities; (4) refill, DRC, export. RF tracks (N\$2, N\$8 at 0.204 mm) and bias lines were **not** touched.

Remaining 18 items (same source §3): 1 clearance — GND paddle polygon ↔ VIN_M track stub at (17.46, -15.63): 0.123 mm < 0.15 mm DRU, exists in the source → designer to confirm or nudge the VIN_M stub; 8 silk_overlap (VIN+ / VIN- texts over the X3 outline, UNK22V0 texts) and 8 silk_over_copper (X2/X3 outlines over THT pads, J1/J2/U\$1 silk on mask-defined areas) — cosmetic; 1 silk_edge_clearance (UNK22V0 reference 0.1 mm from the edge) — cosmetic.

7.4 Layer plots and 3-D renders



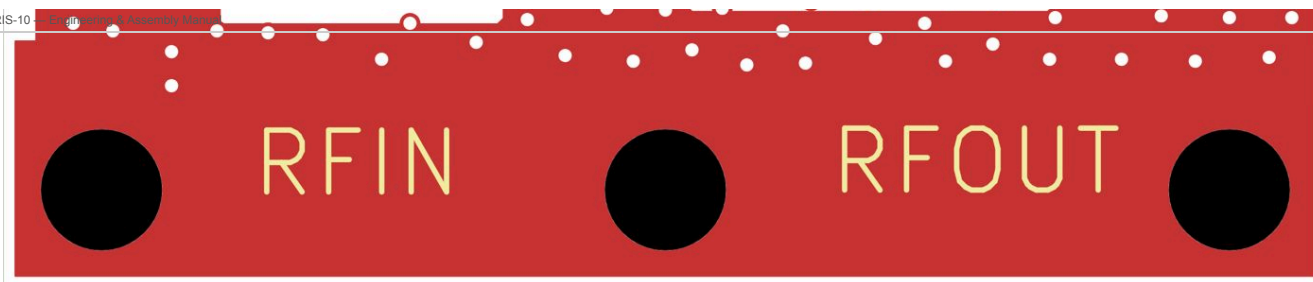
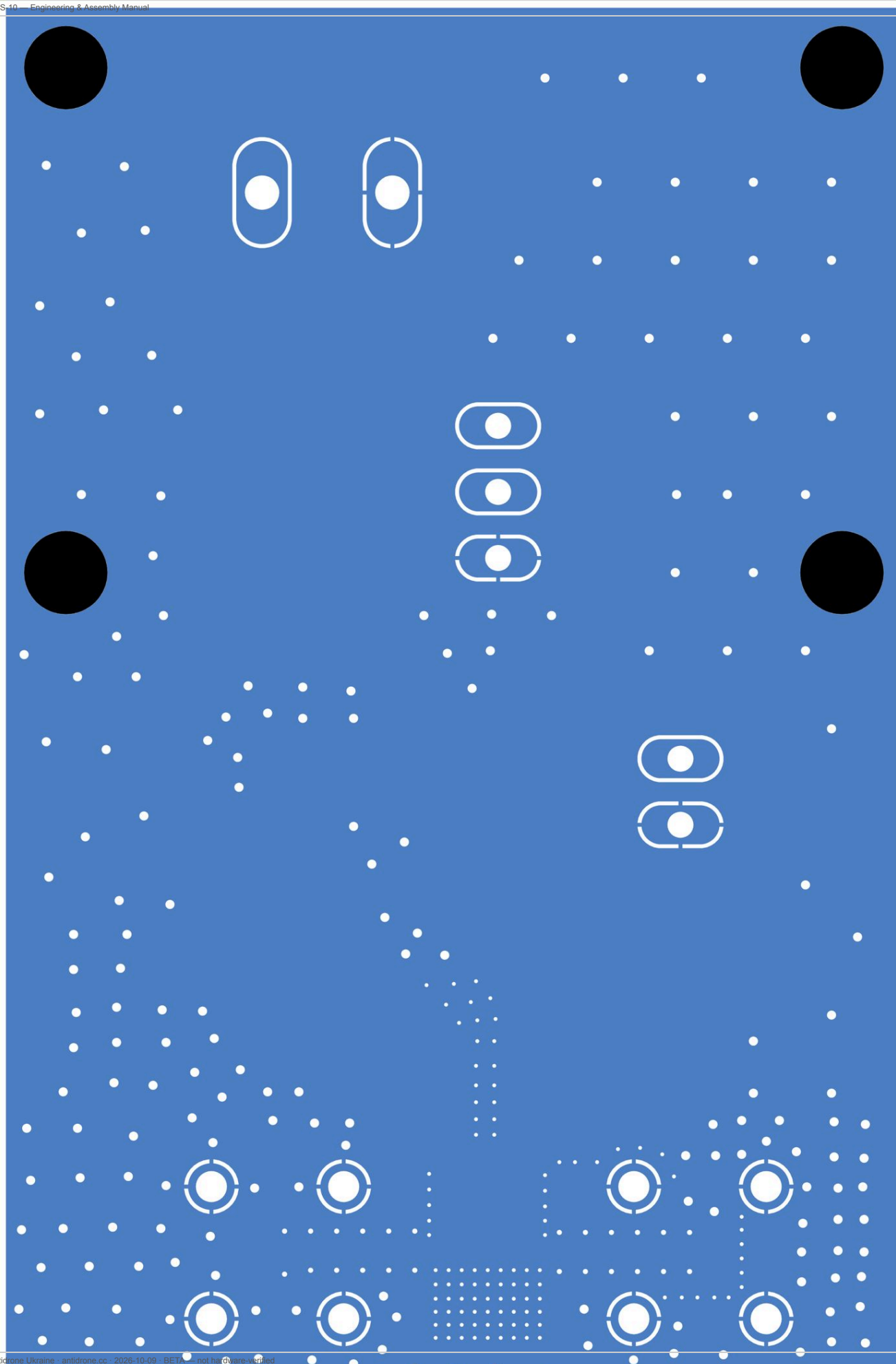


Figure 30 — F7.2a — RF PA top composite (F.Cu + F.Silks + Edge.Cuts), 35 × 60 mm: 0.204 mm RF microstrips J1 → U\$1 → J2 and the thermal-via field under the PA (status: SOURCE-
DERIVED; source: engineering/PCB/RF_PA/svg/RF_PA_top_composite.svg; produced by tools/svg_sheets_to_pdf.py --png-dir manual/figures)



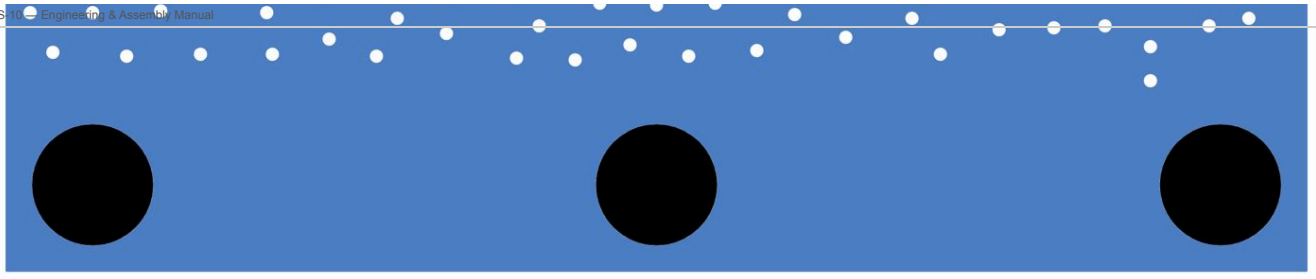


Figure 31 — F7.2b — RF PA bottom composite, mirrored (B.Cu + B.Silks + Edge.Cuts) (status: SOURCE-DERIVED; source: engineering/PCB/RF_PA/svg/RF_PA_bottom_composite_mirrored.svg; produced by tools/svg_sheets_to_pdf.py --png-dir manual/figures)

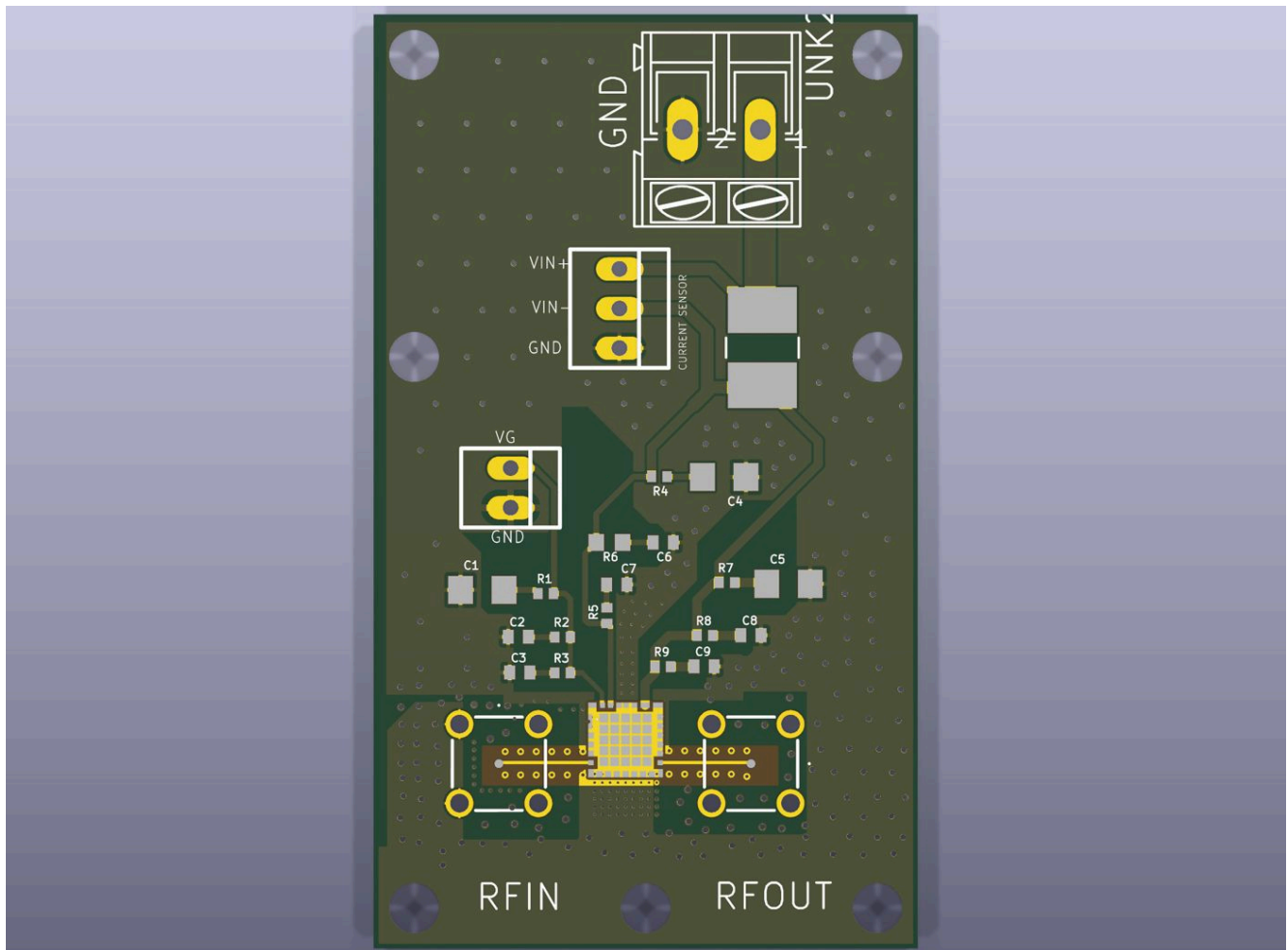


Figure 32 — F7.3a — RF PA 3-D render, top, KiCad conversion; no component models (status: SOURCE-DERIVED; source: engineering/PCB/RF_PA/3d/RF_PA_render_top.png; produced by kicad-cli pcb render, tools/kicad_pcb_pipeline.sh)

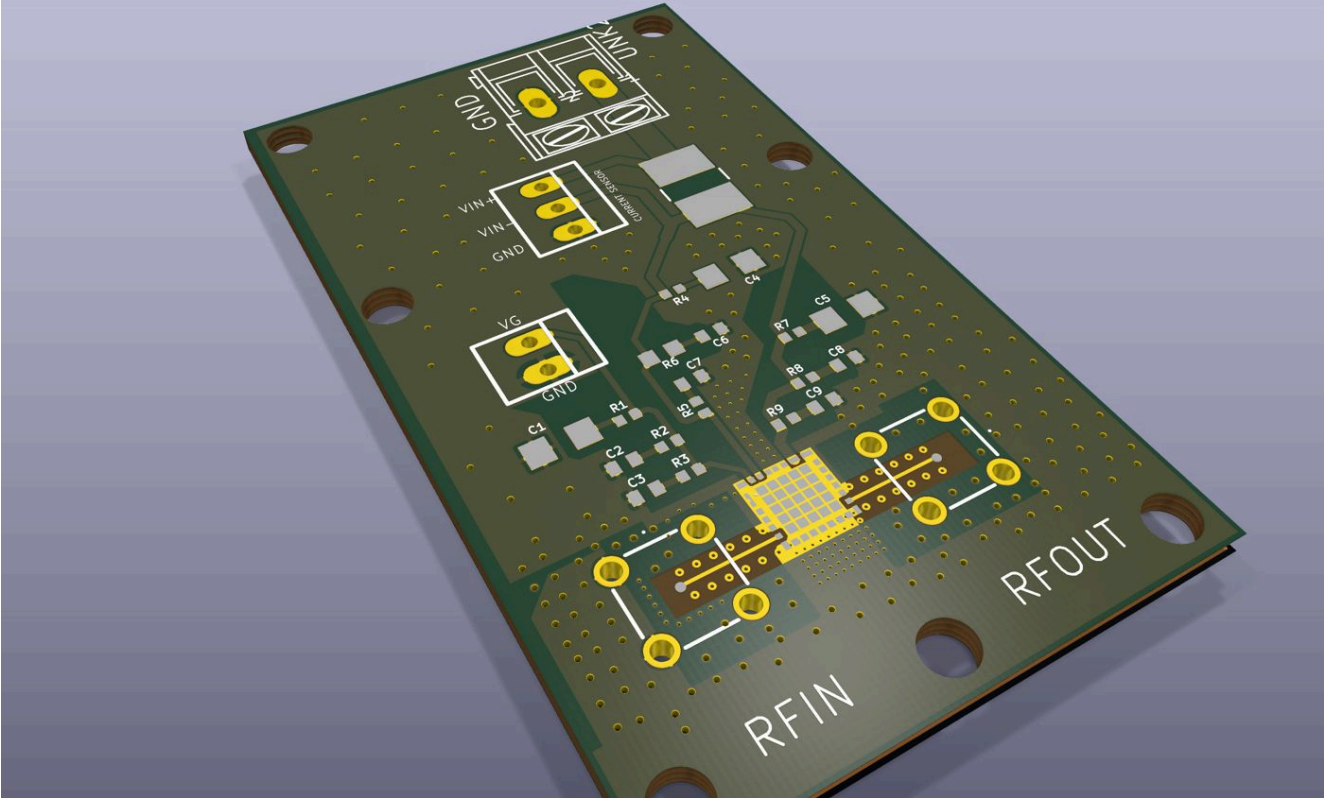


Figure 33 — F7.3b — RF PA BETA board, isometric render (GND strap and paddle polygon net added, RF copper unchanged) (status: BETA; source: beta/pcb/RF_PA/exports/3d/RF_PA_render_isometric.png; produced by beta/pcb/tools/beta_export_package.sh)

The generated assembly-drawing PDFs are clipped by the A4 page frame (chapter 4 §4.5) and are not used as figures.

7.5 Stack-up

Source DRU PCBWay_4L_100um-Track , layerSetup (1+2*15+16) (source: engineering/PCB/RF_PA/STACKUP.md , status PARTIAL):

#	EAGLE layer	KiCad layer	Copper (DRU mtCopper)	Dielectric below (DRU mtIsolate)
1	1	F.Cu	0.035mm	0.11mm
2	2	In1.Cu	0.035mm	1.2mm
3	15	In2.Cu	0.035mm	0.11mm
4	16	B.Cu	0.035mm	—

PROPOSED fabrication note (source: beta/pcb/RF_PA/FAB_NOTES.md §2–§4): F.Cu on **RO4350B 4 mil** (DRU 0.11 mm vs. note 0.102 mm), FR-4 core 1.2 mm, prepreg 0.11 mm (RO4350B only if bottom RF existed — none routed on B.Cu); finished **1.6 mm ± 10 %** (DRU sum 1.42 mm + copper ≈ 1.56 mm); the QPA2962 ground paddle carries 21 GND vias 0.35/0.15 mm — PROPOSED via-in-pad, filled and capped, because the paddle is a solder surface (designer/fab to confirm); 50 Ω ± 10 % on N\$2 / N\$8 at w = 0.204 mm (2 segments each, same geometry as the impedance note); bias nets VIN_M , VD , VG 0.37 / 0.6 / 0.8 / 2.0 mm DC. ENIG, maskless RF traces, min track/space 0.10/0.15 mm, min drill 0.15 mm, IPC-A-600 Class 2 — all PROPOSED. The fab drawing may carry the text of the PCBWay impedance note only after the designer confirms it applies to this 4-layer stack (source: docs/PCB/RF_PA.md §5).

7.6 Thermal budget and drain gating (DSN-THM-01, PROPOSED DESIGN)

First-order estimates from engineering/DESIGN/THERMAL/THERMAL_AND_PA_SUPPLY.md (generator tools/design_thermal.py , inputs engineering/DESIGN/design_parameters.json , decisions D-10, D-11, D-14; no measurement).

Duty cycle from the firmware timing (main.cpp:180–186 ; source: §1 of the thermal note):

Quantity	Value
TX time per beam position	16 × 30 μs + 16 × 0.5 μs = 488.0 μs
Frame per beam position	16 × 167 + 175.4 + 16 × 175 = 5647.4 μs

Quantity	Value
RF duty	8.64 %
Drain-gate duty (switch on 5 μ s around each chirp, assumption)	11.47 %

Dissipation per QPA2962 (datasheet: VD 22 V, IDQ 1.68 A, PSAT 40 dBm, PAE 22 %; source: §2): quiescent, no RF — 37.0 W DC / **37.0 W** dissipated; at PSAT (PIN 27 dBm) — 43.2 W / 33.7 W; design value while the drain is on — **37.0 W**.

System cases, 16 PAs (source: §3):

Case	Total PA dissipation	Verdict
A continuous drain bias (firmware as coded: VD left on)	591 W	INFEASIBLE in a sealed rotating head without liquid cooling (needs $\approx$ 142 CFM at ΔT_{air} 12 K)
B drain gated per chirp (D-14), duty 11.5 %	68 W	design case
C drain gated per CPI frame only (on during the 5.6 ms frame, off while the stepper moves)	532 W	INFEASIBLE in a sealed rotating head without liquid cooling (needs $\approx$ 142 CFM at ΔT_{air} 12 K)

Conclusion (D-10/D-14, source: §3): the firmware as coded (VD left on after bias-up, `main.cpp:1560–1601`) puts the head in case A; the proposal therefore requires per-chirp drain gating — a hardware function of the 22 V switch module driven by a timing line — after which the thermal design is case B at **68 W** average.

Heat path, case B (source: §4): 4.24 W average per PA; $R(\text{PA board thermal-via field})$ 1.5 $^{\circ}\text{C}/\text{W}$ — **ASSUMPTION, the RF_PA stack-up and via field under U\$1 must be checked**; $R(\text{TIM}) + \text{spreading}$ 0.3 + 0.2 $^{\circ}\text{C}/\text{W}$ (assumption, thermal pad 1–3 W/mK, 5 $\times$ 5 mm); ΔT PA base $\rightarrow$ plate 8.5 $^{\circ}\text{C}$; plate $\leq$ 77 $^{\circ}\text{C}$ for TBASE $\leq$ 85 $^{\circ}\text{C}$; required plate-to-air resistance $\leq$ **0.46 $^{\circ}\text{C}/\text{W}$** at 45 $^{\circ}\text{C}$ ambient; heat spreader (D-11) 300 $\times$ 300 $\times$ 10 mm Al with PAs 4 $\times$ 4 on the rear centre; two fin fields 55 $\times$ 280 mm, 26 fins 25 $\times$ 2 mm at 4 mm pitch, 0.39 m²; $R(\text{lateral spreading})$ 0.09 $^{\circ}\text{C}/\text{W}$, $R(\text{fins} \rightarrow \text{air}, h = 25 \text{ W}/\text{m}^2\text{K ASSUMPTION})$ 0.10 $^{\circ}\text{C}/\text{W}$; resulting plate / PA base **58 $^{\circ}\text{C}$ / 66 $^{\circ}\text{C}$** at 45 $^{\circ}\text{C}$ ambient (margin 19 $^{\circ}\text{C}$); airflow **16 CFM** (7.7 L/s) $\rightarrow$ 2 $\times$ 60 mm fans ($\geq$ 15 CFM each) controlled by the existing fan relay. Not included: Power Board regulator losses (currents UNKNOWN), Main Board ($\approx$ 10–20 W estimate), solar load — add 30 % margin when selecting the fans.

22 V drain supply sizing (D-14, source: §5; the block schematic, netlist and BOM are DSN-PSU-01 in chapter 9): peak drain current all 16 PAs at ID_max 2.848 A = **45.6 A** during each 30 μ s chirp; average case B **3.47 A** $\rightarrow$ 76 W (case A 27.3 A $\rightarrow$ 600 W, not supported); bulk capacitance **2734 μ F** total for $\Delta V \leq 0.5 \text{ V}$ over a chirp $\rightarrow \geq 220 \mu\text{F}$ low-ESR polymer per PA board (171 μF each) + 2 $\times$ 1000 $\mu\text{F}/35 \text{ V}$ at the switch module; input 6.9 A average at 12 V, η 92 %; per-PA pulse gating 16 $\times$ high-side P-FET (-40 V , 30 A pulsed) with fast driver (e.g. LTC7003), common TX_GATE TTL input from the FPGA — **spare I/O to be allocated, UNRESOLVED**; sequencing VG (-4 V via DAC5578) before VD (firmware already does this), gate switch only after EN/DIS_RFPA_VDD, power-down in reverse.

Consequence for this board: the " $\geq 220 \mu\text{F}$ low-ESR polymer per PA board" of the proposal is not on the RF_PA schematic (which has 3 $\times$ 10 μF + 6 $\times$ 0.1 μF); whether it is added on a rev. B of this board or at the switch module is an open decision under D-14.

7.7 BOM summary

Source BOM `docs/BOM/BOM_RF_PA.csv`: 25 references, 11 line items, 0 MPN attributes, 6 references without value (source:

```
docs/BOM/README.md). BETA confidence, counted with python3 -c "import csv,collections;print(collections.Counter(r['mpn_confidence'] for r in csv.DictReader(open('beta/pcb/RF_PA/BOM_RF_PA_beta.csv'))))":
```

mpn_confidence	Lines	Quantity (source: beta/pcb/README.md)
HIGH	5	6
MEDIUM	6	19
LOW	0	0
EMPTY	0	0
Total	11	25

Before purchase (source: `docs/PCB/RF_PA.md` §8): enter the QPA2962 orderable part number (package suffix); gate-bias network values consistent with the DAC5578-driven VG range ($-4 \dots -1.2 \text{ V}$ per `Power Management V6.xlsx`); the 5 m Ω shunt vs 0.1 Ω MPN conflict flagged in the BETA `note` column (source: `beta/pcb/README.md`). Multiply all quantities by 16 for the AERIS-10X variant.

7.8 Open items

ID	Item	Status	Source
K4	22 V drain supply and drain switch not in CAD; DSN-PSU-01 proposal (chapter 9)	OPEN — power designer	<code>docs/SYSTEM/BLOCK_DIAGRAM.md</code>
D-14	TX_GATE timing line — spare FPGA I/O not allocated	UNRESOLVED	<code>engineering/DESIGN/THERMAL/THERMAL_AND_PA_SUPPLY.md</code> §5
G-12	Keep-out zones around the RF connectors and the QPA2962	BLOCKED	<code>engineering/VALIDATION/UNRESOLVED_GEOMETRY.md</code>

AERIS-16

Engineering & Assembly Manual	Item	Status	Source
—	Stack-up 4-layer construction, outer dielectric, 70 μm DRU slot, 50 Ω width confirmation	BLOCKED — MISSING DATA	docs/PCB/RF_PA.md §7; MDR-07
—	PA mounting (7 × Ø3.2 mm NPTH), heatsink contact, via-field thermal resistance (1.5 °C/W ASSUMPTION)	OPEN — mechanical/thermal	docs/PCB/RF_PA.md §7; thermal note §4
—	Which SMA of each Main Board pair is RFIN/RFOUT; PA n ↔ antenna row n assignment	UNVERIFIED / PROPOSED	interconnection_table.md §7; HARNESS_SCHEDULE.md
—	1 clearance 0.123 mm (VIN_M stub) in the source	OPEN — designer	beta/pcb/RF_PA/README.md §3
—	Local 220 μF bulk capacitance per PA board (D-14) not on the schematic	OPEN	this chapter §7.6
—	BOM: QPA2962 MPN, 6 value-less references, shunt MPN conflict	OPEN	docs/PCB/RF_PA.md §8
—	RF performance (gain, P1dB at 10.5 GHz)	unverified until measured	docs/PCB/RF_PA.md §10

<!-- chapter 8: Proposed 16×8 patch panel: design, simulation, fabrication data -->

8. Antenna — proposed 16 × 8 microstrip patch panel (DSN-ANT-01)

Chapter status summary: everything in this chapter is **PROPOSED DESIGN** (decisions D-01...D-06): first-order analytical dimensions, a native KiCad board with a fabrication export set, and openEMS simulations of one row and of three adjacent rows. Nothing has been built or measured; the original project contains no antenna CAD (K8, G-06, MECH-ANT-01 BLOCKED — MISSING DATA). The chirp bandwidth B, which decides whether the series-fed row is adequate, is TBD in the parameter table. Compiled by Antidrone Ukraine · antidrone.cc.

Sources: engineering/DESIGN/ANTENNA/ANTENNA_DESIGN_CALC.md; engineering/DESIGN/ANTENNA/simulation/TUNING_LOG.md, tuning_result.json, s11.csv, coupling_3rows.csv; engineering/DESIGN/ANTENNA/kicad_exports/board_statistics.md, DRC_report.txt, EXPORT_LOG.md; engineering/DESIGN/00_DESIGN_BASIS.md (D-01...D-06); engineering/DRAWING_REGISTER.md (DSN-ANT-01, DSN-ANT-01-SIM).

8.1 Why a patch array

The upstream README names two antenna variants — an 8 × 16 patch array ("Nexus") and a 32 × 16 slotted-waveguide array ("Extended") — without CAD for either (K8; source: docs/SYSTEM/BLOCK_DIAGRAM.md). Decision D-01 designs the patch array as the primary antenna because it is manufacturable with the same PCB workflow and verifiable with the openEMS tools already used in 5_Simulations, and gives only a sizing sheet for the waveguide variant; D-02 makes the 16 radiating "elements" of the firmware's elevation-scanned ULA sixteen horizontal rows at 14.3 mm pitch, each a series-fed resonant array of 8 patches (D-03); D-04 fixes the substrate (RO4350B, h = 0.508 mm, 35 μm Cu, full back ground); D-05 the feed (50 Ω end-launch 2.92 mm connectors on the left edge, one per row); D-06 equal feed-line length on every row so that the ADAR1000 calibration tables stay valid (source: engineering/DESIGN/00_DESIGN_BASIS.md D-01...D-06).

The waveguide sizing (CONCEPTUAL, source: ANTENNA_DESIGN_CALC.md §6): WR-90 λ_g at 10.5 GHz = 36.56 mm, resonant shunt slots at λ_g/2 = 18.28 mm, 32 slots → stick ≈ 585 mm; 16 sticks at 14.3 mm cannot be stacked (WR-90 broad wall 22.86 mm + wall), so the Extended variant would need reduced-height or ridged guide or a 2-row interleave — one reason D-01 selects the patch array.

8.2 Design calculation (transmission-line model)

Inputs (source: ANTENNA_DESIGN_CALC.md §1):

Parameter	Value	Basis
f ₀	10.500 GHz, λ ₀ = 28.552 mm	VERIFIED (parameter table)
Rows (elements) × patches per row	16 × 8	VERIFIED / D-03
Row pitch (elevation)	14.3 mm = 0.501 λ ₀	VERIFIED
Substrate	RO4350B ε _r = 3.66, tanδ = 0.0037, h = 0.508 mm, Cu 35 μm	D-04

Patch (Balanis ch. 14; source: §2):

Quantity	Value
Width W = (λ ₀ /2) · √(2/(ε _r +1))	9.352 mm
ε _{eff} (patch)	3.3648
ΔL (fringing)	0.2400 mm
Length L = λ ₀ /(2√ε _{eff}) − 2ΔL	7.303 mm
Slot conductance G1 / mutual G12	1.1922 mS / 0.5701 mS
Edge resonant resistance R _{edge} = 1/(2(G1+G12))	283.7 Ω
Fractional bandwidth (VSWR 2, Balanis approx.)	≈ 1.7 % (≈ 179 MHz) — chirp bandwidth B is TBD in the parameter table; verify B fits

Feed network per row (source: §3):

Element	Z	Width	Length	Note
Inter-patch link	100 Ω	0.278 mm	8.841 mm (λ _g /2, ε _{eff} 2.608)	patches in phase (resonant series feed)
Centre-to-centre patch spacing along the row	—	—	16.143 mm = 0.565 λ ₀	fixed azimuth beam (no scan along the row)
Row input resistance ≈ R _{edge} /M	35.5 Ω	—	—	standing-wave array, in-phase patches
Quarter-wave transformer √(50·R _{in})	42.1 Ω	1.451 mm	4.182 mm	
50 Ω lead to connector	50 Ω	1.112 mm	10.0 mm (equal on all rows, D-06)	ε _{eff} 2.852, λ _g 16.91 mm

Panel (source: §4):

Item	Value
Board outline	165 × 248 mm (rows start x = 36.18 mm; row 1 centre y = 16.68 mm)

Item	Value
Row length (8 patches + 7 links)	120.30 mm
Mounting	6 × Ø3.2 mm (M3 inferred) at 5 mm from the edges
Connectors	16 × end-launch 2.92 mm on the left edge at 14.3 mm pitch (body width ≤ 12 mm — e.g. Southwest 1092-series, Amphenol 901-10510; verify footprint)
Estimated HPBW azimuth (row, 8 × 16.1 mm)	≈ 11.2° (uniform)
Estimated HPBW elevation (16 × 14.3 mm)	≈ 6.3° (matches HW-ANT-10: 6.3°)
Estimated directivity (aperture 129 × 229 mm, η _{ap} 0.7 assumed)	≈ 25.0 dBi (parameter table says ~20 dBi TBD)
Series-feed frequency squint	the row beam tilts with frequency; with B TBD this must be checked in simulation (corporate feed is the fallback, D-02)

8.3 Board and fabrication data

Native editable board `engineering/DESIGN/ANTENNA/kicad/aeris10_patch_array.kicad_pcb` (KiCad 8+/10, RO4350B stack-up entered); exports in `kicad_exports/` generated with `kicad-cli` (all steps exit 0 on 2026-10-09: `drc`, `gerbers`, `drill`, `pdf`, `svg`, `step`, `stats`, `render` — source: `kicad_exports/EXPORT_LOG.md`). Gerber set: `aeris10_patch_array-F_Cu.gtl`, `-B_Cu.gbl`, `-F_Mask.gts`, `-B_Mask.gbs`, `-F_Silkscreen.gto`, `-Edge_Cuts.gm1`, `-job.gbrjob`; drill files in `kicad_exports/drill/`; `aeris10_patch_array.step`; `aeris10_patch_array_top.pdf`; `aeris10_patch_array_F_Cu.svg`.

Board statistics (source: `kicad_exports/board_statistics.md`, KiCad report of 2026-10-09):

Item	Value
Width × height	165.0000 × 248.0000 mm, area 40920.00 mm²
Front / back copper area	9867.896 mm² / 40475.063 mm²
Board stackup thickness	0.5980 mm
Min drill diameter	0.3000 mm
Pads: through hole / SMD / NPTH	32 / 48 / 6
Vias	0 (no through, blind, buried or micro vias)
Components	22 (16 SMD, 6 "unspecified" — the count matches the 6 NPTH mounting holes), all front side
Drill holes	32 × Ø0.3 mm PTH (connector pads), 6 × Ø3.2 mm NPTH

The statistics report also prints "Min track clearance / width 2147.4836 mm"; this equals 2 147 483 647 nm (INT32_MAX) and is read here as KiCad's placeholder for a board whose copper is made of polygons, not tracks — an interpretation of this manual, not a statement of the source file. DRC on the board (source: `kicad_exports/DRC_report.txt`): 46 items — 22 `lib_footprint_issues`, 17 `silk_over_copper`, 7 `silk_edge_clearance` (connector reference fields clipped by the board edge); none is a copper violation.

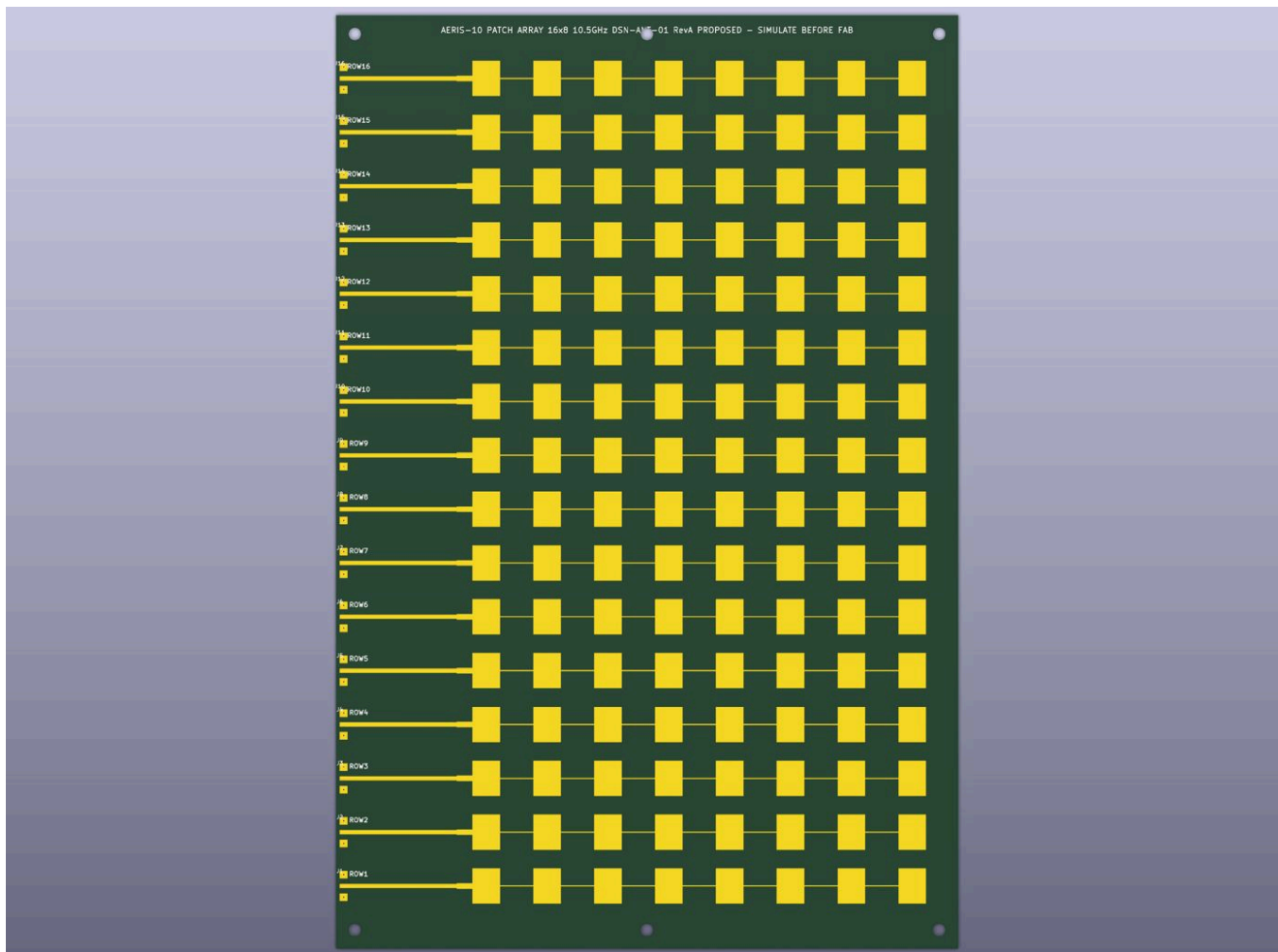


Figure 35 — F8.2 — DSN-ANT-01 KiCad 3-D render of the patch panel, top side (status: PROPOSED DESIGN; source: engineering/DESIGN/ANTENNA/kicad_exports/aeris10_patch_array_render_top.png; produced by kicad-cli pcb render)

8.4 Simulation of one row (DSN-ANT-01-SIM)

openEMS 0.0.36+ built from source (python-openEMS in a Python 3.12 venv), model `openems_patch_row.py`, runner `tools/design_antenna_tune.py`, 1.56 M cells, ~73 s per run, end criterion -40 dB (source: `simulation/TUNING_LOG.md`). Tuning iterations:

iter	L_SCALE	f_res (GHz)	S11 at f_res (dB)	S11 at 10.5 GHz (dB)	-10 dB band (MHz)	D_row (dBi)
0	1.0000	11.092	-37.6	-18.0	2362	11.51
1	1.0564	10.830	-30.6	-10.8	1958	11.35
2	1.1274	10.560	-27.6	-6.3	1162	11.23

Selected: **L_SCALE = 1.0000** (the transmission-line patch length): S11 = -18.0 dB at 10.5 GHz, but the **contiguous -10 dB band around f_0 is only ≈ 128 MHz** — the 2362 MHz span in the table is the non-contiguous extent of all dips and must not be read as bandwidth; between the dips the match is -6...-8 dB, worst -6.2 dB within 10.3–10.7 GHz (source: `TUNING_LOG.md`; `tuning_result.json`: `BW_10dB_MHz_contiguous = 128`, `worst_S11_10.3_10.7_GHz_dB = -6.2`). Single-row directivity 11.51 dBi (16 rows → ≈ 23.5 dBi estimated). Lengthening the patches moves the deepest dip towards 10.5 GHz but narrows the band below the carrier, because the input match of this series-fed row is set by the transformer/feed rather than by a single patch resonance.

Model limits: PEC copper, lossy RO4350B, no connector, one row only, MUR boundaries, lumped 50 Ω port. **Design consequence:** the series-fed resonant row is inherently narrow-band; if the chirp bandwidth B (TBD) exceeds ≈ 100 MHz the row needs a travelling-wave (matched-load) or corporate feed, or a thicker substrate — open item for DSN-ANT-01 rev B. Not evaluated: beam squint vs. frequency, pattern at the band edges, ADAR1000 phase-calibration impact (source: `TUNING_LOG.md`).

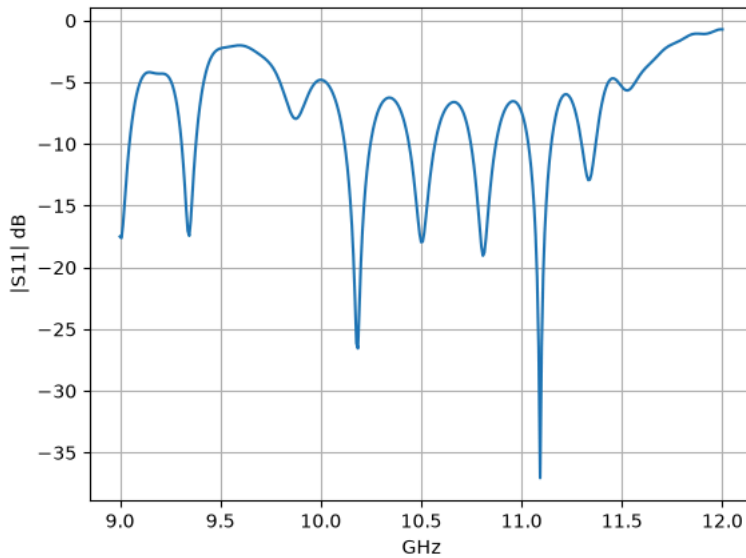


Figure 36 — F8.3 — One-row $|S_{11}|$ vs frequency from openEMS (L_SCALE 1.0): comb of narrow resonances, -18.0 dB at 10.5 GHz, ≈ 128 MHz contiguous -10 dB band (status: PROPOSED DESIGN (simulated); source: engineering/DESIGN/ANTENNA/simulation/s11_row.png; produced by tools/design_antenna_tune.py)

8.5 Three-row mutual coupling

Model `openems_three_rows.py` (2026-10-09): centre row driven, neighbours at ± 14.3 mm terminated in $50\ \Omega$. At 10.5 GHz: S_{22} (centre row) -16.3 dB, $S_{12} = S_{32} \approx -20.7$ dB; worst coupling over $9\text{--}12$ GHz ≈ -18.8 dB (source: `TUNING_LOG.md` §Three-row mutual coupling; data `coupling_3rows.csv`, 401 samples $9\text{--}12$ GHz — the figure below reads $-16.3 / -20.7 / -20.7$ dB at the sample nearest 10.5 GHz and the worst S_{12}/S_{32} of -18.8 dB at 11.3475 GHz, both taken from the CSV by `manual/figures/plot_coupling.py`). Verdict (same source): coupling is at the -20 dB target of `ANTENNA_DESIGN_CALC.md` §7 at the carrier and slightly above it at the band edges — acceptable for a first panel, but the ADAR1000 phase calibration must be done with all rows terminated (array calibration), and the 16-row full-panel simulation remains open.

AERIS-10 — DSN-ANT-01 three-row mutual coupling (openEMS, centre row driven, neighbours $50\ \Omega$)

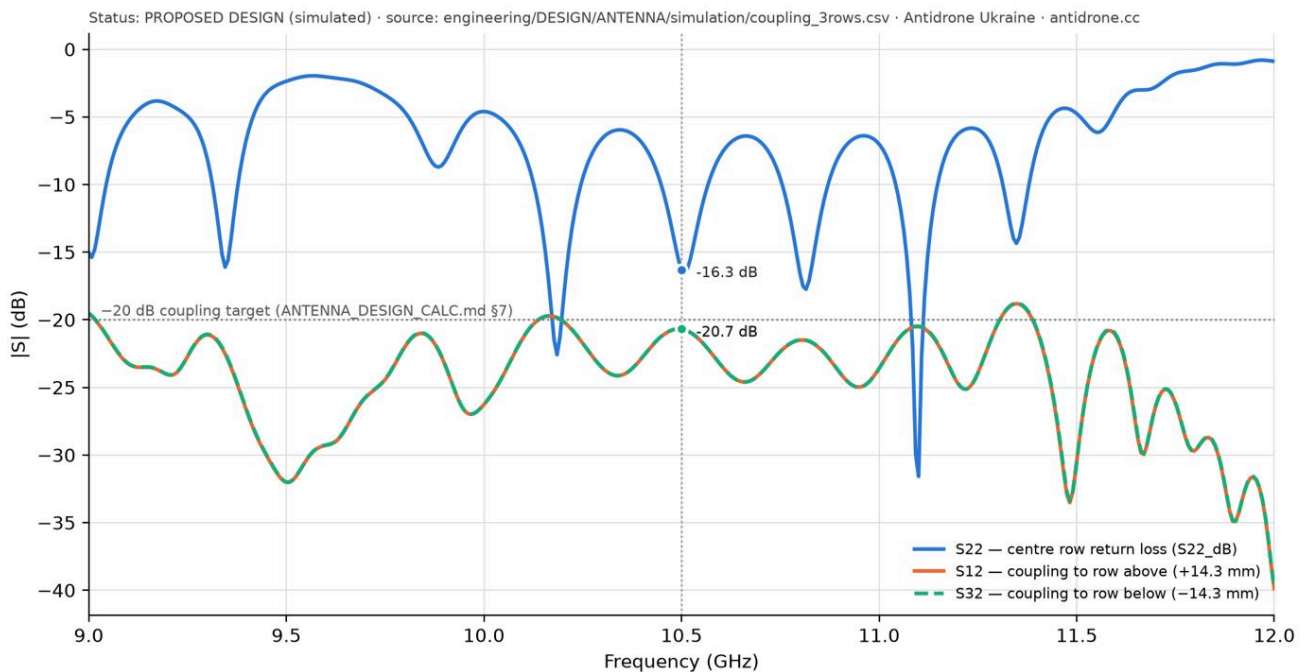


Figure 37 — F8.4 — Three-row mutual coupling S_{12}/S_{32} and centre-row S_{22} vs frequency, $9\text{--}12$ GHz, from openEMS (status: PROPOSED DESIGN (simulated); source: engineering/DESIGN/ANTENNA/simulation/coupling_3rows.csv; produced by `beta/gui/.venv/bin/python manual/figures/plot_coupling.py`, `matplotlib` Agg)

8.6 Verification plan before fabrication

Copied from `ANTENNA_DESIGN_CALC.md` §7 with the current state:

- AERIS-101
- Engineering & Assembly Manual
1. Run `openems_patch_row.py` (`openEMS ≥ 0.0.36 + python-openEMS`): sweep 9.5–11.5 GHz; tune L (± 0.3 mm) and `qwl_en` until $|S_{11}| < -10$ dB at 10.5 GHz $\pm B/2$ — **done for L (three iterations, §8.4); the $\pm B/2$ criterion cannot be applied while B is TBD; the ≈ 128 MHz contiguous band is the result to compare against B.**
2. Simulate 3 adjacent rows for mutual coupling (S_{21} between row ports < -20 dB target) — **done (§8.5): -20.7 dB at f_o , -18.8 dB worst in 9–12 GHz.**
3. Fabricate one 3-row coupon; measure S_{11}/S_{21} on a VNA; compare with simulation; update `design_parameters.json` — **not done.**
4. Only then release the 16-row panel (`kicad_exports/ Gerbers`) and record the result in `engineering/VALIDATION/DRAWING_CHECKS.md` — **not done; the Gerbers exist but are not released.**

8.7 Open items

ID	Item	Status	Source
K8	Antenna variant (patch vs slotted waveguide) — proposal selects patch (D-01); owner decision pending	OPEN	docs/SYSTEM/BLOCK_DIAGRAM.md ; D-01
G-06	Element geometry, substrate, feed, radome of the <i>original</i> design — no source data; the proposal replaces, not recovers, it	BLOCKED — MISSING DATA	engineering/VALIDATION/UNRESOLVED_GEOMETRY.md
—	Chirp bandwidth B TBD → feed topology decision (series resonant vs travelling-wave/corporate, D-02 fallback)	OPEN	TUNING_LOG.md ; ANTENNA_DESIGN_CALC.md §2
—	2.92 mm end-launch connector footprint (body ≤ 12 mm) to verify	OPEN	ANTENNA_DESIGN_CALC.md §4
—	Beam squint vs frequency, band-edge pattern, 16-row full-panel simulation	OPEN	TUNING_LOG.md
—	3-row coupon fabrication and VNA measurement; array calibration procedure with all rows terminated	OPEN	§8.6
—	Mounting of the panel on the heat spreader ($6 \times M3$, 2.4 mm nylon spacers) and radome window — chapter 10	PROPOSED DESIGN	engineering/DESIGN/MECHANICAL/MECHANICAL_PARTS_LIST.md rows 3, 8
—	PA $n \leftrightarrow$ antenna row n cable assignment and equal-length coax set (CBL-049...CBL-139)	PROPOSED	engineering/DESIGN/HARNESS/HARNESS_SCHEDULE.md

<!-- chapter 9: FPGA→host path (options A/B), 22 V PA supply module -->

Host link (FPGA → host, options A/B) and 22 V PA supply module

Author: Antidrone Ukraine · antidrone.cc

Status summary: host data path DSN-LINK-01 PROPOSED DESIGN; option B (SPI bridge through the STM32 CDC) implemented as BETA in `beta/fpga`, `beta/stm32`, `beta/gui` (simulated and unit-tested, never run on hardware); option A (FT601 on Main Board rev. B) is a pin plan plus a partially routed KiCad proposal whose 32-bit bus is blocked by the existing bank-35 fan-out (layout-owner decision required); 22 V PA supply DSN-PSU-01 PROPOSED DESIGN (block schematic, BOM, requirement table; no component-level schematic, no bench test). Figure F9.1 PROPOSED DESIGN; figure F9.2 BETA. Decisions D-16...D-19 and D-14 are not approved by the owner (chapter 17 §2).

Sources: `engineering/DESIGN/HOST_LINK/HOST_LINK_DESIGN.md`, `engineering/DESIGN/HOST_LINK/README.md`, `ft601_pin_assignment.csv`, `ft601_added_parts_BOM.csv`, `option_b_signal_map.csv`, `ft601_bank35.xdc`; `beta/fpga/README.md` (host path, results), `beta/fpga/CHANGELOG.md`; `beta/stm32/README.md`, `beta/stm32/CHANGELOG.md`, `beta/stm32/DECISIONS.md` D-17/D-18; `beta/gui/README.md`, `beta/gui/CHANGELOG.md`; `beta/pcb/MAIN_BOARD_REVB/README.md`, `UNROUTED.md`, `NETLIST_DELTA.csv`; `engineering/DESIGN/ELECTRICAL/PA_SUPPLY_22V/README.md`, `DSN-PSU-01_BOM.csv`; `engineering/DESIGN/THERMAL/THERMAL_AND_PA_SUPPLY.md` §5.

Planned figures: F9.1 DSN-PSU-01 block schematic; F9.2 Main Board rev. B isometric render.

1. Problem (conflict K3)

The RTL streams radar data through an FT601 USB 3.0 FIFO (`usb_data_interface.v`), but on the Main Board **U6 (FT601Q) has 0 of 77 pins connected** — only the decoupling of its +3V3_FT rail exists (L19, C184–C186). The only wired host link is the STM32 USB-FS CDC (X53) (source: `engineering/DESIGN/HOST_LINK/HOST_LINK_DESIGN.md` §1).

1.1 Data-rate budget (source: `HOST_LINK_DESIGN.md` §2, copied verbatim)

Quantity	Value	Basis
Range-Doppler cells per beam position	64 × 32 = 2048	<code>doppler_processor.v</code> (RANGE_BINS 64, 32 chirps)
Beam-position frame time	5647.4 μs	<code>main.cpp</code> :180–186
Raw RTL packet stream (44 B per cell)	16.0 MB/s	<code>usb_data_interface.v</code> (11 × 32-bit words)
Compact map frame (8-bit log-magnitude per cell + header + ≤ 32 detections + CRC)	2164 B → 383 kB/s	this design, §5
STM32 USB-FS CDC practical limit	≈ 0.8–1.1 MB/s	USB 2.0 FS bulk (19 × 64 B per 1 ms frame max)
SPI1 STM32 ↔ FPGA (existing lines)	27 Mbit/s ≈ 3.3 MB/s (DMA)	APB2 108 MHz / 4 (beta clock tree)
FT601 245 sync FIFO, 32 bit @ 100 MHz	up to 400 MB/s	FT601

Conclusion of the source: the raw stream needs the FT601 (option A); the compact map fits the existing STM32 path with 3× margin (option B).

1.2 Options and decisions (source: `HOST_LINK_DESIGN.md` §3, copied verbatim)

	A — FT601 on Main Board rev. B	B — SPI bridge via STM32 (no PCB change)	C — Ethernet mezzanine on bank 35 (future)
Hardware change	route U6 to bank 35 (46 I/Os), add USB 3 connector, crystal, RREF, ESD (<code>ft601_added_parts_BOM.csv</code>)	none: DIG_5/6/7 + SPI1 are already routed to the FPGA	new PCB with RGMII PHY on the 50 free bank-35 pins
Throughput	400 MB/s	≤ 1 MB/s (CDC-bound)	100 MB/s
Firmware/RTL	RTL already written (fix 2-bit BE → 4-bit; honour TXE_N); host driver FTDI D3XX	new RTL <code>host_bridge_spi.v</code> + packer; STM32 <code>host_bridge.c</code> ; GUI parser	new MAC/UDP stack
Risk	10-layer board respin; USB 3 SI	protocol only; SPI1 shared with ADAR1000 (time-multiplexed)	highest
Decision	D-16: target for rev. B	D-17: implement now (BETA)	D-18: documented only

Decision D-19 for the RTL: widen `ft601_be` to 4 bits, drive `BE = 4'b1111` for full words, respect `TXE_N` back-pressure, add `ft601_reset_n/wakeup_n/siwu_n` as outputs — recorded for `beta/fpga`, not yet applied there (source: `HOST_LINK_DESIGN.md` §4).

2. Option A — FT601 on Main Board rev. B (PROPOSED DESIGN)

2.1 Pin plan

`ft601_pin_assignment.csv` maps every FT601 signal to a free bank-35 pad, with CLK on the MRCC pin C4 = `I0_L12N_T1_MRCC_35`; `ft601_bank35.xdc` (60 lines) is the matching constraint fragment. The table has 47 signal rows: CLK, DATA_0..31, BE_0..3, TXE_N, RXF_N, WR_N, RD_N, OE_N, SIWU_N, RESET_N, WAKEUP_N, GPIO0, GPIO1 — all LVCMOS33, bank 35 VCCO = +3V3_FPGA on the schematic (source: `engineering/DESIGN/HOST_LINK/ft601_pin_assignment.csv`, 48 lines including header; `ft601_bank35.xdc` header).

FT601 pad numbers come from the EAGLE library symbol used in the schematic (U6-FT601Q-B-T); the **FT601 datasheet is not in the**

repository AC timing, RREF value, VBUS limits and the 1.0 V core supply arrangement (VD10 pins) must be verified against it before the schematic is edited. Rev. B schematic work per the source: connect U6 VCC33 (pads 20/24/38) and VCCIO (14/49/59/68) to +3V3_FT, GND pads, the 46 signals per the CSV, XI/XO crystal, RREF, VBUS divider, D±/SS pairs to the new connector through the ESD array; route the 32-bit bus as a length-matched group (±25 mm, 100 MHz single-ended, 50 Ω) on the two bank-35 side layers (source: HOST_LINK_DESIGN.md §4). The full pin table is reproduced in section 9 of this chapter.

2.2 Parts to add for rev. B (source: ft601_added_parts_B0M.csv, copied verbatim)

ref	qty	description	proposed_part	note
J_USB3	1	USB 3.1 Gen1 receptacle (Type-C, USB 2.0 + one SuperSpeed pair used) or USB 3.0 micro-B	GCT USB4085-GF-A (Type-C) / Amphenol GSB4211111WEU (micro-B 3.0)	VERIFY pin-out; SS pairs TODP/TODN ↔ RIDP/RIDN per FT601 datasheet
Y_FT	1	Crystal 30 MHz ±30 ppm, 18 pF	Abracon ABM8-30.000MHZ-B2-T	FT601 XI/XO (pads 21/22) + 2 × 18 pF (VERIFY load per datasheet)
R_RREF	1	Resistor 3.24 kΩ 1 % 0402	Yageo RC0402FR-073K24L	FT601 RREF (pad 27) — value per FT60x datasheet, VERIFY
C_VD10	4	Capacitor 4.7 μF 6.3 V 0402 + 100 nF	Murata GRM155R60J475ME47D	1.0 V core pins VD10/VD10_2..4/DV10 (pads 3,30,33,39,48): FT60x internal regulator output, decouple each — VERIFY
C_AVDD	2	Capacitor 100 nF / 1 μF 0402	Murata GRM155R71C104KA88D	AVDD/VDDA (pads 2, 28) analogue 3.3 V via ferrite from +3V3_FT
FB_A	1	Ferrite bead 600 Ω@100 MHz 0603	Murata BLM18PG601SN1D	+3V3_FT → AVDD
D_ESD	1	USB 3.0 ESD array (SS + HS)	TI TPD4E05U06	on connector side of the SS and D± pairs
R_VBUS	2	Resistor divider 10 k / 3.3 k (VBUS detect, pad 37)	Yageo RC0402	VERIFY VBUS pin voltage limit in datasheet
L19/C184-C186	0	already in the schematic: +3V3_FT = +3V3_FPGA via L19; 10 μF + 100 nF + 1 nF	—	present (U6 pads 20, 24, 38 VCC33; 14, 49, 59, 68 VCCIO) — nets to be connected

2.3 Rev. B layout outcome (BETA PROPOSAL, beta/pcb/MAIN_BOARD_REVB/)

Status of the directory: "BETA PROPOSAL — explicit netlist change (rev. B); partially routed; DRC-checked; not reviewed by the original designer; not fabricated." The EAGLE schematic has **not** been changed and no longer matches this board; the designer must enter the same connections and parts in the schematic, verify them against the FT601 datasheet and re-annotate before any rev. B layout is released. Source board: beta/pcb/MAIN_BOARD/MAIN_BOARD.kicad_pcb (rev. A BETA, unchanged); reproducible with beta/pcb/tools/beta_revb_ft601.py (source: beta/pcb/MAIN_BOARD_REVB/README.md , header).

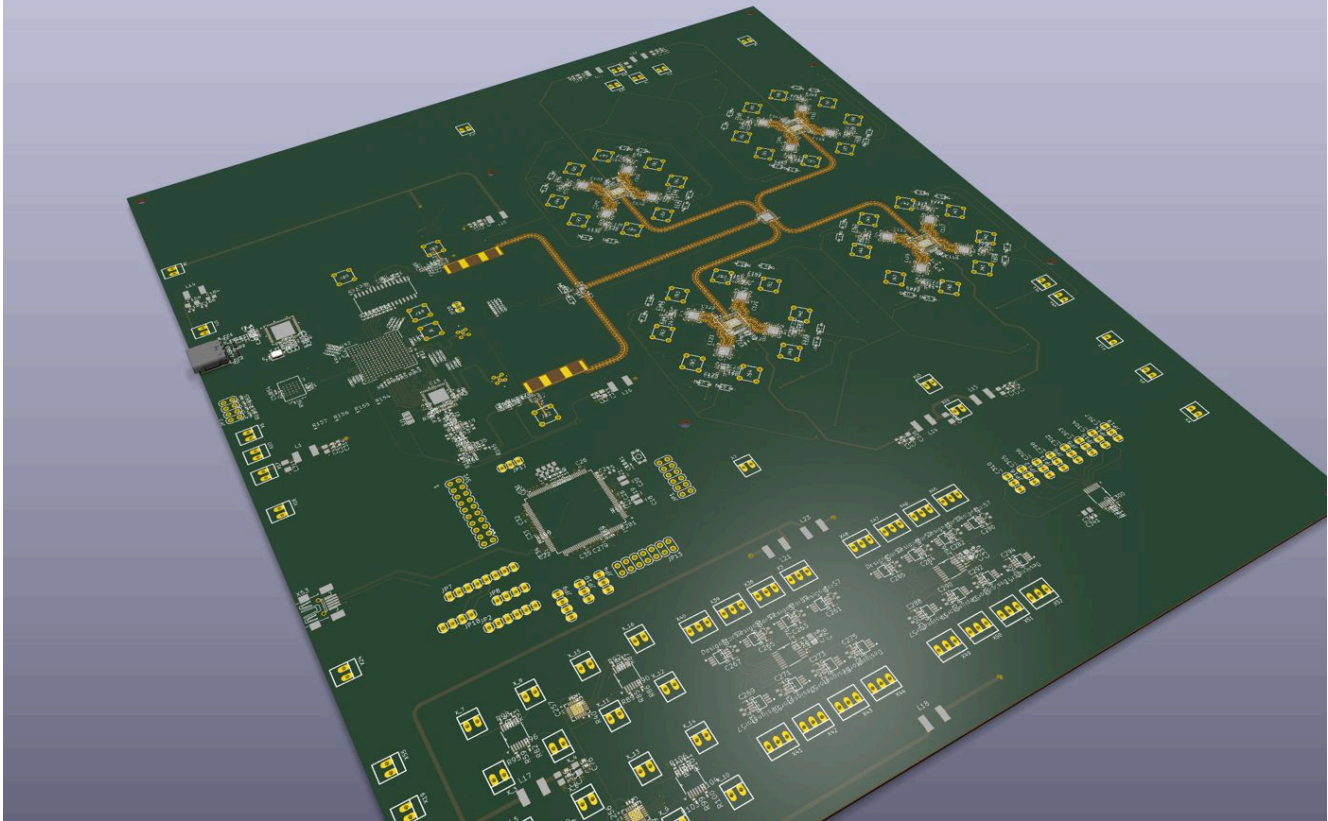


Figure 38 — F9.2 — Main Board rev. B isometric render with the FT601 cluster and USB-C receptacle placed at the left board edge; the 32-bit FIFO bus is unrouted — BETA (source: beta/pcb/MAIN_BOARD_REVB/MAIN_BOARD_REVB.kicad_pcb; produced by kicad-cli pcb render, exports/3d)

DRC result (source: MAIN_BOARD_REVB/README.md §1, copied verbatim):

Check	Rev. A BETA (MAIN_BOARD/)	Rev. B after netlist change, before routing	Rev. B after routing (exports)
unconnected_items	0	129	59
clearance	72	76	72
copper_edge_clearance	0	0	1
courtyards_overlap	0	2	2
hole_clearance	87	87	87
shorting_items	125	125	125
silk_edge_clearance	3	3	3
silk_over_copper	199	199	199
silk_overlap	199	199	199
solder_mask_bridge	199	199	199
track_dangling	130	130	130
track_width	0	0	14
via_dangling	35	35	40
DRC total	1049	1055	1071

Routing result (source: MAIN_BOARD_REVB/README.md §1 bullets; UNROUTED.md §1):

- Routed and connected: 17 of the 64 new nets (USB D±, both SuperSpeed RX lines, SSTX_N and both SSTX_C lines, CC1/CC2, USB_VBUS, FT_VBUS_DET, FT_RREF, FT_XI, FT_GPIO1, partly +3V3_FT/FT_VD10/FT_AVDD/FT_XO).
- **The 32-bit FIFO bus and its control lines (46 of 47 FPGA-side signals) are NOT routed.** Root cause, measured: 36 of the 47 bank-35 balls of U42 (XC7A50T FTG256, 1.0 mm pitch) have **no free position for an escape via** — the four dog-bone positions around each ball are already occupied by the existing BGA fan-out vias/traces of neighbouring balls. Balls with no free position: A2, A3, A7, B1, B2, B6, C1, C2, C3, C4, C6, D1, D3, D4, D6, E1, E2, E3, E5, E6, F2, F3, F4, F5, G4, G5, H4, H5, J1, J3, J5, K1, K2, K3, K5, L2; one free position for A4, A5, B4, B5, B7, C7, G1, H1, H3; two for G2, H2. Freerouting (1 pass, 25 min, existing copper locked) fanned out the FT601 side but could not reach these balls; its partial copper on the 47 open bus nets (127 items) was removed again so the board is left clean.

- **What is needed (layout owner):** re-do the bank-35 corner breakout of U42 (outer two rows on F.Cu, inner rows by dog-bone vias to In2/In3/In5/In7, which requires moving existing vias of adjacent nets), then route the 46 nets as one group at 0.204 mm, length-matched to ±25 mm; alternatively choose bank-35 balls on the two outer rows only, which would require revising `ft601_pin_assignment.csv` together with the XDC. This is "a layout-owner decision outside this BETA's 'existing tracks locked' rule"; no re-done fan-out exists in the repository at the time of writing.
- New DRC items caused by rev. B copper: 9 `track_width` (Freerouting neck-down 0.075 mm on +3V3_FT/CC1/CC2 at 0.4 mm-pitch pads; to be widened to 0.1 mm), 2 courtyard overlaps (Y_FT ↔ C_XI/C_XO — move the load caps 0.5 mm), 1 copper-edge (USB_CC2 inner track 0.2 mm from the J_USB3 NPTH peg), 5 dangling vias (+3V3_FT ×4, USB_SSTX_C_P). Two further scripted clean-ups (second restricted Freerouting pass on the open power nets; widening the neck-downs) were prepared but not executed in that session.

Routing geometry (source: `MAIN_BOARD_REVB/README.md §2`, copied verbatim):

Group	Rule used	Achieved
FT_BUS (FT_DATA_0..31, FT_BE_0..3, FT_CLK, control, GPIO)	netclass FT_BUS 0.204 mm (50 Ω microstrip on the 0.102 mm RO4350B outer layer per <code>./MAIN_BOARD/FAB_NOTES.md</code> ; on inner layers ≈ 42 Ω stripline estimate, FR-4, not field-solved), clearance 0.1 mm (DRU), length-match target ±25 mm	only FT_GPI01 routed: 37.57 mm, 3 vias, F.Cu/In2/In7 — no skew figure exists for the bus because it is unrouted ; FT601→bank-35 Manhattan distance is 20–35 mm, so ±25 mm is achievable once the breakout exists
USB SuperSpeed + D±	netclass USB_DIFF width 0.204 mm, pair gap 0.18 mm (≈ 90 Ω edge-coupled microstrip on RO4350B h = 0.102 mm, closed-form estimate 91 Ω, fab to solve)	Freerouting routes pair members as single traces (coupling not enforced): SSRX_P/N 29.28 / 29.41 mm (skew 0.13 mm), D+/D− 23.72 / 27.12 mm (skew 3.4 mm), SSTX_C_P/N 9.08 / 8.72 mm, SSTX_N 16.08 mm (SSTX_P open); neck-down to 0.153 mm at pads. Pairs must be re-routed coupled (KiCad diff-pair router, 0.204/0.18 mm) before release
Power FT_PWR (FT_VD10, FT_AVDD, USB_VBUS)	0.3 mm	USB_VBUS 34.5 mm, FT_AVDD 18.6 mm; FT_VD10 only partially connected

Netlist delta: 64 new nets, 123 pad connections on existing parts (U6, U42), 71 pad connections on 20 added parts; U6 pads 2, 14, 20, 24, 28, 37, 38, 49, 59, 68 were on single-pin nets in rev. A and were moved to the new nets. Every connection is listed in `MAIN_BOARD_REVB/README.md §3` and `NETLIST_DELTA.csv` (258 rows) (source: `MAIN_BOARD_REVB/README.md §3`, lead-in).

Added parts as placed in rev. B (source: `MAIN_BOARD_REVB/README.md §4`, copied verbatim):

Ref	Footprint (KiCad 10 library)	Value	MPN	Position (mm), rot	Note
J_USB3	Connector_USB:USB_C_Receptacle_Amphenol_12401610E4-2A	USB-C 3.1 receptacle	Amphenol ICC 12401610E4#2A	(4.00, -208.00), 270°	only the TX1/RX1 SuperSpeed lane is wired (no orientation mux): SS works in one plug orientation, USB 2.0 in both — VERIFY acceptable
D_ESD1	Package_SON:US0N-10_2.5x1.0mm_P0.5mm	TPD4E05U06	Texas Instruments TPD4E05U06DQAR	(12.50, -210.00), 0°	pin-out per TPD4E05U06 DQA flow-through (1,2,4,5 IO; 6,7,9,10 NC pass-through; 3,8 GND) — VERIFY
D_ESD2	Package_SON:US0N-10_2.5x1.0mm_P0.5mm	TPD4E05U06	Texas Instruments TPD4E05U06DQAR	(12.50, -206.00), 0°	second array for D+/D- (the design BOM lists one 4-channel array for SS+HS = 6 lines) — channels 3/4 unused
C_SSTX_P	Capacitor_SMD:C_0402_1005Metric	100nF	Murata GRM155R71C104KA88D	(16.50, -211.20), 0°	USB 3 TX AC coupling — added (spec requirement), VERIFY
C_SSTX_N	Capacitor_SMD:C_0402_1005Metric	100nF	Murata GRM155R71C104KA88D	(16.50, -210.00), 0°	USB 3 TX AC coupling — added (spec requirement), VERIFY
R_CC1	Resistor_SMD:R_0402_1005Metric	5.1k	Yageo RC0402FR-075K1L	(11.00, -214.00), 0°	Type-C Rd 5.1 kΩ (device) — added because a USB-C receptacle was chosen
R_CC2	Resistor_SMD:R_0402_1005Metric	5.1k	Yageo RC0402FR-075K1L	(11.00, -202.50), 0°	Type-C Rd 5.1 kΩ (device)
R_VBUS_1	Resistor_SMD:R_0402_1005Metric	10k	Yageo RC0402FR-0710KL	(32.50, -201.90), 90°	VERIFY VBUS pin limit
R_VBUS_2	Resistor_SMD:R_0402_1005Metric	3.3k	Yageo RC0402FR-073K3L	(33.70, -201.90), 90°	VERIFY VBUS pin limit
R_RREF	Resistor_SMD:R_0402_1005Metric	3.24k 1%	Yageo RC0402FR-073K24L	(26.30, -201.90), 90°	value per FT60x datasheet — VERIFY
C_VD10_1	Capacitor_SMD:C_0402_1005Metric	4.7uF 6.3V	Murata GRM155R60J475ME47D	(20.30, -210.40), 90°	VD10/DV10 decoupling — VERIFY
C_VD10_2	Capacitor_SMD:C_0402_1005Metric	4.7uF 6.3V	Murata GRM155R60J475ME47D	(27.50, -201.90), 90°	VD10/DV10 decoupling — VERIFY

Ref	Footprint (KiCad 10 library)	Value	MPN	Position (mm), rot	Note
				90°	
C_VD10_3	Capacitor_SMD:C_0402_1005Metric	4.7uF 6.3V	Murata GRM155R60J475ME47D	(28.70, -201.90), 90°	VD10/DV10 decoupling — VERIFY
C_VD10_4	Capacitor_SMD:C_0402_1005Metric	4.7uF 6.3V	Murata GRM155R60J475ME47D	(29.90, -201.90), 90°	VD10/DV10 decoupling — VERIFY
FB_A	Inductor_SMD:L_0603_1608Metric	600R@100MHz	Murata BLM18PG601SN1D	(18.00, -214.60), 0°	+3V3_FT -> AVDD/VDDA
C_AVDD_1	Capacitor_SMD:C_0402_1005Metric	100nF	Murata GRM155R71C104KA88D	(20.30, -212.60), 90°	
C_AVDD_2	Capacitor_SMD:C_0402_1005Metric	1uF	Murata GRM155R61A105KE15D	(19.20, -212.60), 90°	
Y_FT	Crystal:Crystal_SMD_Abracon_ABM8G-4Pin_3.2x2.5mm	30MHz 18pF	Abracon ABM8-30.000MHZ-B2-T	(22.50, -199.60), 0°	ABM8 land pattern (3.2x2.5, 4 pads; ABM8G footprint of the KiCad library) — VERIFY
C_XI	Capacitor_SMD:C_0402_1005Metric	18pF C0G	Murata GRM1555C1H180JA01D	(19.60, -198.75), 0°	load cap — VERIFY against crystal CL
C_XO	Capacitor_SMD:C_0402_1005Metric	18pF C0G	Murata GRM1555C1H180JA01D	(25.20, -199.20), 0°	load cap — VERIFY against crystal CL

Deviations from ft601_added_parts_BOM.csv (source: MAIN_BOARD_REVB/README.md §4, closing paragraph): USB-C Amphenol 12401610E4#2A (24-pin, has SS pins) instead of GCT USB4085 (the KiCad USB4085 footprint is USB 2.0-only); a second TPD4E05U06 for D± (one 4-channel array cannot cover 6 lines); added 2 × 100 nF SSTX AC-coupling capacitors (USB 3 requirement) and 2 × 5.1 kΩ CC pull-downs (required for a Type-C device receptacle); only the TX1/RX1 SuperSpeed lane is wired (no orientation mux → SS in one plug orientation only); C_VD10 as 4 × 4.7 μF (100 nF companions not placed). Placement: J_USB3 on the left board edge nearest U6, ESD arrays at x = 12.5, crystal/RREF/VBUS divider/VD10 capacitors in the free band below U6, all on F.Cu; placement is a proposal, thermal/EMC not assessed (source: same, §6).

Open items on the routed power/clock/USB nets (source: MAIN_BOARD_REVB/UNROUTED.md §2, copied verbatim):

Net	Open connections	What is needed
FT_VD10	5	connect U6 pads 3/30/33/39/48 together and to C_VD10_1..4 — the QFN pads face other-net pads on all sides; route on F.Cu around the exposed pad corners or via-in-pad to an inner pour
+3V3_FT	5	feed from the L19/C184-186 filter (y ≈ -232) to U6 VCC33/VCCIO pads; 4 autorouter vias are dangling (remove or connect)
FT_AVDD	1	U6 pad 2 to C_AVDD_1 (0.6 mm stub)
FT_XO	1	U6 pad 22 to Y_FT pad 3
USB_SSTX_P	1	U6 pad 32 to C_SSTX_P — route coupled with USB_SSTX_N (0.204/0.18 mm)

Prepared but not executed (source: UNROUTED.md §3): (1) second Freerouting pass restricted to FT_VD10, +3V3_FT, FT_AVDD, FT_XO, USB_SSTX_P; (2) widen the 14 Freerouting neck-downs below 0.1 mm to 0.1 mm; (3) move C_XI / C_XO 0.5 mm away from Y_FT. The export package of rev. B (all 24 export steps exit 0, exports/EXPORT_LOG.md) exists but describes an unfinished board (source: MAIN_BOARD_REVB/README.md §7).

3. Option B — SPI bridge through the STM32 (BETA)

3.1 Signals (source: option_b_signal_map.csv, copied verbatim)

signal	schematic_net	stm32_pin	fpga_pin	direction	note
FPGA_CS_N (option B)	DIG_5	STM32 PD13 (today configured INPUT in main.cpp:2313-2317 → becomes OUTPUT)	U42 H11 (IO_L19P_T3_A22_15)	STM32 → FPGA	active-low chip select for the bridge; ADAR1000 pass-through is gated off while low
DRDY (option B)	DIG_6	STM32 PD14 (INPUT, EXTI14)	U42 G12 (IO_L19N_T3_A21_VREF_15)	FPGA → STM32	a complete frame is in the FPGA TX FIFO
spare / ACK	DIG_7	STM32 PD15	U42 H12 (IO_L20P_T3_A20_15)	FPGA → STM32	reserved (frame dropped / overflow flag)
SCLK	STM32_SCLK1	STM32 PA5 (SPI1_SCK)	U42 J16 (IO_L23N_T3_FWE_B_15)	STM32 → FPGA	existing net, 3.3 V; ≤ 27 MHz (SPI1 on APB2 108 MHz, prescaler 4)
MOSI	STM32_MOSI1	STM32 PA7 (SPI1_MOSI)	U42 H13 (IO_L20N_T3_A19_15)	STM32 → FPGA	existing; carries the bridge command byte
MISO	STM32_MISO1	STM32 PA6 (SPI1_MISO)	U42 G14 (IO_L21P_T3_DQS_15)	FPGA → STM32	existing; frame bytes, MSB first, mode 0

signal	schematic_net	stm32_pin	fpga_pin	direction	note
ADAR_n_CS_3V3	ADAR_1..4_CS_3V3	STM32 GPIO	U42 bank 15	STM32 → FPGA	unchanged; must all be HIGH during a bridge transfer (firmware guarantees; RTL also checks)

Transfer (source: `HOST_LINK_DESIGN.md §5`): the STM32 waits for DRDY (EXTI on PD14), pulls `FPGA_CS_N` low, clocks one command byte (0x01 = read frame) and then reads the frame over MISO with DMA (SPI1 mode 0, MSB first, ≤ 27 MHz); the FPGA holds the ADAR1000 pass-through idle while `FPGA_CS_N` is low; the firmware never starts an ADAR1000 SPI transaction while a bridge read is in progress (both share SPI1). The STM32 forwards each frame unchanged over CDC (`AERIS_USB_SendBridgeFrame`), interleaved with the existing status strings; the GUI stream parser resyncs on the sync word (status strings never contain `0xA5 0x5A`).

3.2 Frame format (source: `HOST_LINK_DESIGN.md §5`, copied verbatim; little-endian; reference parser `engineering/DESIGN/HOST_LINK/gui/bridge_frame.py`)

Offset	Size	Field
0	2	sync 0xA5 0x5A
2	1	version = 1
3	1	flags (bit0 = long-chirp set, bit1 = overflow since last frame)
4	2	sequence number
6	1	azimuth index (1..50)
7	1	elevation index (1..31)
8	2	chirp count
10	1	n_range = 64
11	1	n_doppler = 32
12	2	n_det (≤ 32)
14	2	reserved
16	2048	magnitude map, uint8 = $8 \cdot \log_2(\text{abs}(I) + \text{abs}(Q))$ saturated, range-major
2064	3·n_det	detections: range u8, doppler u8, mag u8
end	2	CRC-16/CCITT-FALSE over bytes 0..end-1

Files (source: `HOST_LINK_DESIGN.md §5`): `rtl/host_bridge_spi.v` (SPI slave + frame FIFO, 1 BRAM), `rtl/rd_map_packer.v` (cell → frame builder, CRC), `rtl/tb_host_bridge.v` (self-checking iverilog test), `stm32/host_bridge.c/h` (SPI1 DMA + EXTI + CDC forward), `gui/bridge_frame.py` (+ tests); integration into `beta/fpga`, `beta/stm32`, `beta/gui` is recorded in their CHANGELOGs.

3.3 Bridge command set v2 — register access (source: `HOST_LINK_DESIGN.md §7`, copied verbatim; RTL implemented 2026-10-09)

All transfers: `FPGA_CS_N` low, SPI mode 0, MSB first; first byte = command. Bytes marked ← are driven by the FPGA on MISO (the master clocks dummy 0x00). Implemented in `beta/fpga/rtl/host_bridge_spi.v` (copy in `engineering/DESIGN/HOST_LINK/rtl/`), firmware counterpart `beta/stm32/Core/Src/host_bridge_proto.c`, verified by `beta/fpga/tb/tb_host_bridge_top.v` (through `radar_system_top`) and `rtl/tb_host_bridge.v` (unit).

Cmd	Total bytes	Bytes after the command	Reply	Meaning
0x01	1 + frame + 2	—	frame + CRC (as §3.2); all zeros when no frame is pending (no sync word)	read the pending range-Doppler frame (unchanged)
0x02	8	a0 = addr[7:0], a1 = addr[15:8], d0..d3 = data[7:0]..[31:24], xx	← byte 7 = 0xA2 (ack = command accepted; the write commits in the clk domain within ~5 clk cycles)	write register word <code>addr</code>
0x03	7	a0, a1, xx, xx, xx, xx	← bytes 3..6 = d0 d1 d2 d3 (little-endian). No turnaround byte: the read is launched when a0 is complete; a1 is accepted but not decoded (the map has 5 address bits, so a1 must be 0)	read register word <code>addr</code>
0x04	9	xx × 8	← bytes 1..8 = four little-endian u16: status word, RTL version (0x0002), frames produced, 0x0000	status without touching the frame
other (incl. 0x00)	any	—	← 0xEE on every following byte	unknown command (ignored)

Status word (assembled in `radar_system_top.v`): bit0 frame ready (= DRDY), bit1 ADAR CS conflict (sticky: an ADAR1000 CS was low while `FPGA_CS_N` was low), bit2 ADC capture FIFO overflow (sticky, `ADC_CAPTURE_MODE = 1`), bit3 calibration lock (all 8 lanes locked), bit4 packer overflow (a frame was dropped since reset), bits 5..15 = 0. The status is sampled when the command byte completes (quasi-static values; frames produced may be one behind).

Register map — word addresses, **16-bit registers** (data[31:16] are ignored on write and read as 0). Source of truth:

`beta/fpga/rtl/radar_control_regs.v` (address-map comment and the two `case` statements); the table is kept identical to it. `toggle` = write 1 to the bit to pulse the action, reads as 0; `level` = stored bit.

Addr	Name	Access	Reset	Bits
0x00	CONTROL	rw	0x0005	bit0 use_long_chirp, bit1 adc_pwdn, bit2 usb_enable
0x01	CFAR_THR	rw	10000	[15:0] abs(l)+abs(Q) detection threshold
0x02	DECIM	rw	0x0001	[1:0] range decimation mode (01 = peak)
0x03	START_BIN	rw	0	[9:0] first range bin passed to the decimator
0x04	CAL_CTRL	rw	0	bit0 start auto calibration (toggle), bit1 manual tap load (toggle), bit2 bitslip load (toggle), bit3 pattern-check enable (level), bit4 blind method (level: 0 = ADC test pattern, 1 = CW tone at the IF)
0x05	CAL_LANE	rw	0	[2:0] lane for CAL_TAP / CAL_SLIP writes and CAL_LANE_INFO / CAL_BLIND_MIN reads
0x06	CAL_TAP	rw	16	[4:0] manual IDELAY tap
0x07	CAL_SLIP	rw	0	[1:0] BITSLLIP pulses for a manual bitslip load
0x08	CAL_PATT	rw	0x55AA	{pattern_b[7:0], pattern_a[7:0]} expected alternating ADC test codes
0x09	CAL_STAT	ro	—	{fifo_ovf, 4'b0, align_fail, busy, done, lock[7:0]}
0x0A	CAL_LANE_INFO	ro	—	{1'b0, win_hi[4:0], win_lo[4:0], tap[4:0]} of CAL_LANE
0x0B	CAL_ERR	ro	—	pattern-check error counter (saturating)
0x0C	CAL_UNDET	ro	—	{8'b0, undetermined[7:0]}
0x0D	CAL_BLIND_COEF	rw	0xEC39	signed Q1.14 cos(2 π -f_IF/f_S) for the blind notch (0xEC39 = -5063 = 120 MHz at 400 MSPS)
0x0E	CAL_BLIND_MARGIN	rw	0x0040	absolute part of the blind pass margin (a tap passes when metric $\leq$ min + margin + min/16)
0x0F	ID	ro	0xBE7A	beta build identifier
0x10	CAL_BLIND_MIN	ro	—	minimum blind metric (sum of abs(r) over the window, $\gg$ 4, saturated) of CAL_LANE

Registers that earlier revisions of this section listed but that do **not** exist in the RTL (run bit, mixers enable, NCO tuning word) have been removed from the table; use_long_chirp is CONTROL bit0. STM32 API: HostBridge_WriteReg(addr, value), HostBridge_ReadReg(addr, &value), HostBridge_Status(&st); exposed to the GUI through the existing settings path as a text command REG W <addr> <value> / REG R <addr> $\rightarrow$ reply REG <addr> <value> in the status stream (ASCII, so the bridge-frame parser passes it through) (source: HOST_LINK_DESIGN.md §7, closing paragraphs).

3.4 BETA implementation status per subsystem

FPGA (beta/fpga, BETA). rd_map_packer turns each 64 $\times$ 32 Doppler frame into a 2066..2162-byte frame and host_bridge_spi streams it to the STM32 as an SPI slave on the existing SCLK/MOSI/MISO nets with DIG_5 = spi_bridge_cs_n (H11), DIG_6 = spi_bridge_drdy (G12), DIG_7 = spi_bridge_spare (H12, packer overflow flag). While the bridge CS is low the ADAR1000 pass-through is gated (CS high, SCLK/MOSI idle on the 1.8 V side) and MISO is driven by the bridge; system_status[1] latches a conflict if any ADAR CS is low during a transfer. Beam indices come from the transmitter's STM32-toggle counters, the chirp count from the receiver's chirp pulses, long_chirp from the register map. The bridge drives the register map's write/read port (5-bit addresses, 16-bit data); register toggles cross SCLK $\rightarrow$ clk through 3-flop synchronisers; a read needs $\geq$ 8 SCLK periods between a0 and d0 ($\geq$ 296 ns at 27 MHz, the fetch takes $\leq$ 5 clk cycles). Executed tests: tb_host_bridge — 3 detections, 32 detections (det_wr regression), command set v2 against a 4-word register model: PASS, 2075- and 2162-byte frames, CRC ok, v2 write/read-back/status/unknown ok; tb_host_bridge_top — v2 register commands (CFAR_THR 10000 $\rightarrow$ 150 written over SPI, 5 read-backs, 0x04 status before/with/after a frame, 0xEE), then one 64 $\times$ 32 frame after DRDY: PASS (~77 s), 2162-byte frame, 32 detections (only possible because the threshold write took effect), sync/dims/az-el/chirp-count/map/CRC ok, ADAR pass-through gated (source: beta/fpga/README.md, "What passed" rows 6e/6f and "Host path"). One defect was found and fixed in the copied packer: det_wr widened from 6 to 7 bits — with 6 bits the packer never left its header/detection state once n_det $\geq$ 22, so the frame never completed and DRDY never asserted (source: beta/fpga/CHANGELOG.md, "Host-link option B integration", row rd_map_packer.v). Unresolved for option B in the RTL: STM32 SPI1 timing versus the FPGA pins (XDC placeholders), BRAM inference of the SCLK-domain frame RAM read (asynchronous read registered on falling SCLK; Yosys maps it to 1,536 LUTs as RAM64M, chapter 11 §12), and the firmware rule that no ADAR1000 transaction overlaps a bridge read (source: beta/fpga/README.md, "Host path"; beta/fpga_synth/README.md, "Reading the numbers" item 2).

STM32 (beta/stm32, BETA). New Core/Src/host_bridge.c / Core/Inc/host_bridge.h (copies of engineering/DESIGN/HOST_LINK/stm32/): SPI1 bridge to the FPGA using the already-routed DIG_5 (PD13 $\rightarrow$ FPGA_CS_N, reconfigured as output), DIG_6 (PD14 $\rightarrow$ DRDY, EXTI14 rising), DIG_7 (PD15 spare); Core/Src/host_bridge_proto.c implements the HAL-free command set v2 (0x02 write/ack 0xA2, 0x03 read, 0x04 status) and the ASCII REG W/R parser/executor/reply formatter; HostBridge_WriteReg/ReadReg/Status/ExecuteTextCommand run over a transport that refuses while a frame read is active or any ADAR CS is low; the main loop executes a pending REG command after HostBridge_Poll() and replies with CDC_Transmit_FS (bounded 50 ms busy wait) (source: beta/stm32/CHANGELOG.md, "ADAR1000 vector-modulator tables + bridge command set v2"). Decision D-17: register access runs in the main loop, never in the USB ISR; one command slot; reply format REG 0x%04X 0x%08X\r\n (a write echoes the written value after the ack), REG ERR\r\n on syntax error, NACK, busy or SPI error. Decision D-18 records the framing assumptions (8-byte write with ack in byte 8, 7-byte read with data little-endian in bytes 3..6, 9-byte status with four u16) (source: beta/stm32/DECISIONS.md D-17, D-18). Build: FLASH 93 276 B, RAM 17 480 B, 0 errors; host tests 6/6 PASSED including test_host_bridge_cmds.c (bridge v2 byte sequences with a mock SPI: 0x02 + ack 0xA2 / no ack / 0xEE / transfer error, 0x03 LE decode, 0x04 status fields; ASCII REG W/R parser and reply formatting) (source: beta/stm32/README.md §3–4).

GUI (beta/gui, BETA). The default hardware path is the SPI bridge: the FPGA serves 64 $\times$ 32 log-magnitude frames over SPI, the STM32 forwards them unchanged over USB CDC, interleaved with status strings and REG replies; the raw 35-byte RTL packet path (option A, FT601) stays available with `—raw-ft601`. The "FPGA registers / ADC calibration" tab issues REG W/R over CDC with the same text protocol as the firmware (one command per USB transfer, timeout + retransmit, no request ID). In demo mode the simulator answers REG

commands from an in-memory model of `radar_control_regs.v`. The GUI register map was brought to the final RTL map (RTL version 0x0002): `protocol/register_map.py` uses 5-bit addresses (`ADDR_MASK = 0x1F`, unmapped 0x11..0x1F), `CAL_CTRL` bit4 blind level, registers 0x0D `CAL_BLIND_COEF`, 0x0E `CAL_BLIND_MARGIN`, 0x10 `CAL_BLIND_MIN`; the register panel gained the blind-method controls and a per-lane `blind_min` read-out; `read_all` issues 3 commands per lane (client pacing test: 52 commands, 0 drops) (source: `beta/gui/CHANGELOG.md`, "2026-10-09 (final register map, RTL 0x0002)"; `beta/gui/aeris10_gui/protocol/register_map.py:18`; `beta/gui/README.md`, limitation 12). 72 tests pass; `--selftest` on the bridge link: 3 frames, 0 CRC errors, 31 REG replies, read-all OK; `--selftest --raw-ft601`: 6144 packets, 0 drops; PyInstaller bundle passes both self-tests (source: `beta/gui/README.md`, "BETA statement", "Run", "Test"; `beta/gui/CHANGELOG.md`, "Verification performed").

Documentation state between the three BETA trees (observation MAN-04, chapter 17 §6). The FPGA README and CHANGELOG state that command set v2 is implemented in the RTL, that the register write/read port is wired to `ctl_regs`, and that `HOST_LINK_DESIGN.md` §7 is kept identical to `radar_control_regs.v` (source: `beta/fpga/README.md`, "What was found and decided" item 5, "Host path"; `beta/fpga/CHANGELOG.md`, "Command set v2..."). The GUI tree has been brought to the same map and its earlier discrepancy items 1–3 are struck through as resolved (source: `beta/gui/CHANGELOG.md`, "Discrepancies / unresolved" and "2026-10-09 (final register map, RTL 0x0002)"). Two firmware documents still carry the earlier state: `beta/stm32/README.md` §6a ("The FPGA side of 0x02..0x04 is not implemented in `beta/fpga` yet", line 120) and `beta/stm32/DECISIONS.md` D-18 (same statement, "grep 2026-10-09"); the FPGA README's remaining-work item 8 still names the GUI `register_map.py` 4-bit mask as open although the file now has `ADDR_MASK = 0x1F` (source: `beta/fpga/README.md`, "Remaining work" item 8; `beta/gui/aeris10_gui/protocol/register_map.py:18`). These are stale statements, not functional gaps; the register path is consistent between the RTL, `HOST_LINK_DESIGN.md` §7, the firmware byte sequences and the GUI map, and it has not been exercised on hardware in any combination.

3.5 What remains for the host link (source: `HOST_LINK_DESIGN.md` §6; `engineering/DESIGN/HOST_LINK/README.md`, "Not done")

- Option A: Main Board rev. B schematic/layout (MDR-13) — the bank-35 breakout decision above; FT601 datasheet checks; FTDI D3XX host driver test; RTL changes of D-19.
- Option B: bench test of the SPI timing (level shifter path is 3.3 V, no translation needed), CDC throughput measurement, firmware arbitration of SPI1 with the ADAR1000 writes; end-to-end REG test on hardware.

4. 22 V PA drain supply module DSN-PSU-01 (PROPOSED DESIGN)

The RF PA boards need a 22 V drain supply (`VD`, `xlxs` row 59: 18–22 V, 2000 mA per board) that exists in no CAD file — conflict K4; the Power Board's input is 12–17 V and it has no 22 V rail (chapter 3 §4.1, rail $+22V_0 / VD$). DSN-PSU-01 closes K4 per decision D-14 with a block schematic, a BOM and a net summary; it has no component-level schematic, no layout and no bench test (source: `engineering/DESIGN/ELECTRICAL/PA_SUPPLY_22V/README.md`, header; `engineering/DESIGN/00_DESIGN_BASIS.md` §2, D-14).

AERIS-10 — 22 V PA drain supply, enable switch and per-PA pulse gating — DSN-PSU-01 Rev A — STATUS: PROPOSED DESIGN (block level)

Date 2026-10-09 · generated by `tools/design_pa_supply_schematic.py` · decision D-14 (00_DESIGN_BASIS.md) · sizing from `tools/design_thermal.py`
Block level only · component-level schematic, layout and protection detail must be captured in KCad (MDR-13). Part numbers are proposals (availability to be verified).

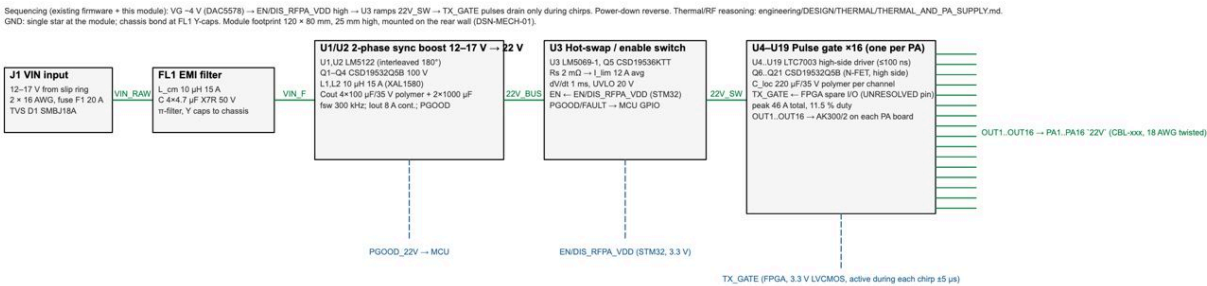


Figure 39 — F9.1 — DSN-PSU-01 block schematic: 2-phase synchronous boost 12–17 V → 22 V, LM5069 hot-swap enable from `EN/IS_RFPA_VDD`, 16 high-side pulse gates from `TX_GATE`, bulk capacitance, 16 output terminals to the PA boards — PROPOSED DESIGN, no component-level schematic, not built (source: `engineering/DESIGN/ELECTRICAL/PA_SUPPLY_22V/DSN-PSU-01_block_schematic.svg`; produced by `tools/design_pa_supply_schematic.py`)

4.1 Requirements (source: `PA_SUPPLY_22V/README.md`, copied verbatim)

Requirement	Value	Source
Input	12–17 V (system VIN via slip ring)	xlxs VIN
Output	22 V ± 2 %, 8 A continuous, 46 A pulsed (11.5 % duty)	QPA2962 datasheet, timing
Bulk energy	≥ 2734 µF total for $\Delta V \leq 0.5$ V per 30 µs chirp	THERMAL_AND_PA_SUPPLY.md §5
Enable	<code>EN/IS_RFPA_VDD</code> (existing STM32 pin) → hot-swap switch	main.h
Pulse gate	<code>TX_GATE</code> from the FPGA — spare pin to allocate (UNRESOLVED); without it the system runs case A (591 W)	D-14
Telemetry	<code>PGOOD/FAULT</code> to the MCU (pin to allocate); per-PA drain current already measured by INA241 on the Main Board	schematic
Mechanical	120 × 80 × 25 mm on the head rear wall (DSN-MECH-01)	layout

4.2 Sizing (source: `engineering/DESIGN/THERMAL/THERMAL_AND_PA_SUPPLY.md` §5, copied verbatim)

Quantity	Value
Peak drain current (all 16 PAs at ID_max 2.848 A)	45.6 A during each 30 μs chirp
Average current, case B (IDQ × gate duty + RF increment × RF duty)	3.47 A → 76 W
Average current, case A (continuous bias)	27.3 A → 600 W (not supported by the proposal)
Bulk capacitance for ΔV ≤ 0.5 V over a chirp (total)	2734 μF → ≥ 220 μF low-ESR polymer per PA board (local, 171 μF each) + 2 × 1000 μF/35 V at the switch module
Input current at VIN_min 12 V, η 92%	6.9 A average (case B)
Converter	synchronous boost 12–17 V → 22 V, 2-phase interleaved (LM5122 ×2 or equivalent), 150 W continuous rating, 300 kHz, 2 × 10 μH / 15 A inductors, output ripple < 100 mV
Protection / enable	LM5069 hot-swap controller + N-FET high-side switch on the 22 V bus, EN from EN/DIS_RFPA_VDD (STM32), current limit 12 A average, dv/dt-limited turn-on; status to the MCU
Per-PA pulse gating	16 × high-side P-FET (–40 V, 30 A pulsed) with fast high-side driver (e.g. LTC7003, ≤ 100 ns), common TX_GATE TTL input from the FPGA (spare I/O to be allocated — UNRESOLVED), local 220 μF per channel
Sequencing	VG (–4 V via DAC5578) before VD (firmware already does this); gate switch only after EN/DIS_RFPA_VDD ; power-down reverse

The drain-gating requirement follows from the thermal analysis: with the drain continuously biased (case A) the 16 QPA2962 dissipate 591 W and the 2-D plate model gives ≈ 213 °C; with per-chirp gating at 11.5 % duty (case B) 68 W, plate 64 °C and PA base ≈ 73 °C at 45 °C ambient (source: docs/AERIS10_BETA_ENGINEERING_REPORT.md §9; engineering/DESIGN/CALCS/DESIGN_CALCULATIONS.md §1). The BOM's gate switch (N-MOSFET with LTC7003) and the thermal document's "high-side P-FET" describe the same function with different device polarity; the BOM is the later, part-level statement, and the component-level design (MDR-12) must settle it.

4.3 Bill of materials (source: PA_SUPPLY_22V/DSN-PSU-01_BOM.csv , copied verbatim)

ref	qty	description	proposed_part	section
J1	1	DC input terminal 2-pole 16 A	Phoenix 1792270 or AK300/2	input
F1	1	Fuse 20 A 32 V automotive	Littelfuse 0297020	input
D1	1	TVS 18 V SMB	SMBJ18A	input
FL1	1	CM choke 10 μH 15 A + 4×4.7 μF/50 V X7R	Würth 744 823 110 / GRM32ER71H475K	filter
U1,U2	2	Sync boost controller	TI LM5122MH	boost
Q1–Q4	4	N-MOSFET 100 V 100 A	TI CSD19532Q5B	boost
L1,L2	2	Inductor 10 μH 15 A	Coilcraft XAL1580-103	boost
C_out	4	Polymer 100 μF/35 V	Panasonic 35SVPF100M	boost
C_bulk	2	Electrolytic 1000 μF/35 V low ESR	Nichicon UHE1V102MHD	boost/bulk
U3	1	Hot-swap controller 9–80 V	TI LM5069MM-1	switch
Q5	1	N-MOSFET 100 V 200 A D2PAK	TI CSD19536KTT	switch
Rs	1	Sense 2 mΩ 3 W	Bourns CSS2H-2512R-L200F	switch
U4–U19	16	High-side gate driver ≤ 100 ns	ADI LTC7003	gate
Q6–Q21	16	N-MOSFET 100 V	TI CSD19532Q5B	gate
C_loc	16	Polymer 220 μF/35 V	Panasonic 35SVPF220M (or on each PA board)	gate
OUT1–16	16	Screw terminal 2-pole	AK300/2 (matches the PA board)	output

4.4 Interfaces and next steps

Cabling: slip ring VIN → DSN-PSU-01 IN (CBL-142, 2 × 2-wire 16 AWG) and DSN-PSU-01 OUT1..16 → PA 22v terminals (CBL-050, 056, ..., 140; 2-wire 18 AWG twisted, AK300/2) in the proposed harness schedule (source: engineering/DESIGN/HARNESS/HARNESS_SCHEDULE.md , rows CBL-050...CBL-142; chapter 3 §5). The module occupies 120 × 80 × 25 mm on the head rear wall in the proposed mechanical layout (DSN-MECH-01). Next steps (MDR-12, source: PA_SUPPLY_22V/README.md): KiCad component-level schematic → layout (4-layer, 2 oz) → bench test of one gate channel with a 30 μs / 45 A dummy load → EMC pre-check of the boost. Decision needed from the owner: D-14, in particular the TX_GATE line (FPGA spare pin, UNRESOLVED — none of the 116 unconstrained RTL ports has a board net; chapter 11 §7) and the telemetry pin (chapter 17 §2).

5. Appendix — FT601 pin assignment, option A (source: engineering/DESIGN/HOST_LINK/ft601_pin_assignment.csv , copied verbatim; identical to ft601_bank35.xdc and beta/pcb/MAIN_BOARD_REVB/README.md §5)

ft601_signal	ft601_pad (QFN76, from EAGLE lib)	proposed_net	fpga_pad	fpga_pin_name	direction (FT601 view)	iostandard
CLK	58	FT_CLK	C4	IO_L12N_T1_MRCC_35	out	LVCMOS33
DATA_0	40	FT_DATA_0	A2	IO_L8N_T1_AD14N_35	bidir	LVCMOS33
DATA_1	41	FT_DATA_1	A3	IO_L4N_T0_35	bidir	LVCMOS33
DATA_2	42	FT_DATA_2	A4	IO_L3N_T0_DQS_AD5N_35	bidir	LVCMOS33
DATA_3	43	FT_DATA_3	A5	IO_L3P_T0_DQS_AD5P_35	bidir	LVCMOS33

ft601_signal	ft601_pad (QFN76, from EAGLE lib)	proposed_net	fpqa_pad	fpqa_pin_name	direction (FT601 view)	iostandard
DATA_4	44	FT_DATA_4	A7	IO_L1N_T0_AD4N_35	bidir	LVCNOS33
DATA_5	45	FT_DATA_5	B1	IO_L9N_T1_DQS_AD7N_35	bidir	LVCNOS33
DATA_6	46	FT_DATA_6	B2	IO_L8P_T1_AD14P_35	bidir	LVCNOS33
DATA_7	47	FT_DATA_7	B4	IO_L4P_T0_35	bidir	LVCNOS33
DATA_8	50	FT_DATA_8	B5	IO_L2N_T0_AD12N_35	bidir	LVCNOS33
DATA_9	51	FT_DATA_9	B6	IO_L2P_T0_AD12P_35	bidir	LVCNOS33
DATA_10	52	FT_DATA_10	B7	IO_L1P_T0_AD4P_35	bidir	LVCNOS33
DATA_11	53	FT_DATA_11	C1	IO_L9P_T1_DQS_AD7P_35	bidir	LVCNOS33
DATA_12	54	FT_DATA_12	C2	IO_L7N_T1_AD6N_35	bidir	LVCNOS33
DATA_13	55	FT_DATA_13	C3	IO_L7P_T1_AD6P_35	bidir	LVCNOS33
DATA_14	56	FT_DATA_14	C6	IO_L5N_T0_AD13N_35	bidir	LVCNOS33
DATA_15	57	FT_DATA_15	C7	IO_L5P_T0_AD13P_35	bidir	LVCNOS33
DATA_16	60	FT_DATA_16	D1	IO_L10N_T1_AD15N_35	bidir	LVCNOS33
DATA_17	61	FT_DATA_17	D3	IO_L11N_T1_SRCC_35	bidir	LVCNOS33
DATA_18	62	FT_DATA_18	D4	IO_L12P_T1_MRCC_35	bidir	LVCNOS33
DATA_19	63	FT_DATA_19	D6	IO_L6P_T0_35	bidir	LVCNOS33
DATA_20	64	FT_DATA_20	E1	IO_L15N_T2_DQS_35	bidir	LVCNOS33
DATA_21	65	FT_DATA_21	E2	IO_L10P_T1_AD15P_35	bidir	LVCNOS33
DATA_22	66	FT_DATA_22	E3	IO_L11P_T1_SRCC_35	bidir	LVCNOS33
DATA_23	67	FT_DATA_23	E5	IO_L13N_T2_MRCC_35	bidir	LVCNOS33
DATA_24	69	FT_DATA_24	E6	IO_0_35	bidir	LVCNOS33
DATA_25	70	FT_DATA_25	F2	IO_L15P_T2_DQS_35	bidir	LVCNOS33
DATA_26	71	FT_DATA_26	F3	IO_L14N_T2_SRCC_35	bidir	LVCNOS33
DATA_27	72	FT_DATA_27	F4	IO_L14P_T2_SRCC_35	bidir	LVCNOS33
DATA_28	73	FT_DATA_28	F5	IO_L13P_T2_MRCC_35	bidir	LVCNOS33
DATA_29	74	FT_DATA_29	G1	IO_L17N_T2_35	bidir	LVCNOS33
DATA_30	75	FT_DATA_30	G2	IO_L17P_T2_35	bidir	LVCNOS33
DATA_31	76	FT_DATA_31	G4	IO_L16N_T2_35	bidir	LVCNOS33
BE_0	4	FT_BE_0	G5	IO_L16P_T2_35	bidir	LVCNOS33
BE_1	5	FT_BE_1	H1	IO_L20N_T3_35	bidir	LVCNOS33
BE_2	6	FT_BE_2	H2	IO_L20P_T3_35	bidir	LVCNOS33
BE_3	7	FT_BE_3	H3	IO_L21N_T3_DQS_35	bidir	LVCNOS33
TXE_N	8	FT_TXE_N	H4	IO_L18N_T2_35	out	LVCNOS33
RXF_N	9	FT_RXF_N	H5	IO_L18P_T2_35	out	LVCNOS33
WR_N	11	FT_WR_N	J1	IO_L22N_T3_35	in	LVCNOS33
RD_N	12	FT_RD_N	J3	IO_L21P_T3_DQS_35	in	LVCNOS33
OE_N	13	FT_OE_N	J5	IO_L19P_T3_35	in	LVCNOS33
SIWU_N	10	FT_SIWU_N	K1	IO_L22P_T3_35	in	LVCNOS33
RESET_N	15	FT_RESET_N	K2	IO_L24N_T3_35	in	LVCNOS33
WAKEUP_N	16	FT_WAKEUP_N	K3	IO_L24P_T3_35	in	LVCNOS33
GPIO0	17	FT_GPIO0	K5	IO_25_35	out	LVCNOS33
GPIO1	18	FT_GPIO1	L2	IO_L23N_T3_35	out	LVCNOS33

Of these 47 balls, 36 have no free escape-via position in the rev. A fan-out and 9 have one, 2 have two (section 2.3); the pin plan therefore cannot be routed on the rev. A breakout as it stands (source: `beta/pcb/MAIN_BOARD_REV/UNROUTED.md §1`).

<!-- chapter 10: Head and pedestal: layout, drawings, parts, flat patterns, thermal map, torque -->

Mechanical design — radar head and pedestal

Author of this chapter: Antidrone Ukraine · antidrone.cc (compilation of the PROPOSED mechanical design; the PCB geometry it is built on is SOURCE-DERIVED from the upstream EAGLE files).

Status summary: every enclosure, layout, pedestal, sheet-metal, fastener, thermal and torque item in this chapter is **PROPOSED DESIGN** (decisions D-07...D-13 of `engineering/DESIGN/00_DESIGN_BASIS.md`, D-12 as revised by DSN-CALC-01 §2). Only the four PCB outlines and hole patterns are **SOURCE-DERIVED** (from the `.brd` files). PCB thickness, component heights, board masses and the mast interface are **BLOCKED — MISSING DATA / ASSUMED** (G-01, G-02, G-03, G-11). Nothing has been built, measured or hardware-verified. The thermal map and the drive-torque check are calculations (BETA, no measurement).

Sources: `engineering/DESIGN/00_DESIGN_BASIS.md` (DSN-00 Rev A), `engineering/DESIGN/MECHANICAL/MECHANICAL_PARTS_LIST.md` (DSN-MECH-PL Rev A), `engineering/DESIGN/CALCS/DESIGN_CALCULATIONS.md` (DSN-CALC-01), `engineering/MECHANICAL/dimensions/.md` (*MECH-DIM-*), `tools/design_layout.py` (station table, printed by `python3 tools/design_layout.py`), `engineering/VALIDATION/UNRESOLVED_GEOMETRY.md` (VAL-GEO-01), `engineering/DESIGN/README.md`, `engineering/DRAWING_REGISTER.md` (DSN-MECH-01...07, DSN-MECH-3D, DSN-MECH-3D-DETAIL, MECH-PLAN-01, ASM-EXP-01).

Figures in this chapter: F10.1–F10.5 (DSN-MECH-01...05), F10.6 (DSN-MECH-06 flat patterns), F10.7 (DSN-MECH-07 assembly section), F10.8 (isometrics of the FreeCAD models), F10.9 (thermal map, two cases), F10.10 (PCB plan view and conceptual exploded view).

10.1 Scope and what is verified

The upstream repository contains no enclosure, antenna, pedestal, cooling or harness design: the drawing register lists MECH-ENC-01, MECH-INT-01, MECH-ANT-01, MECH-COOL-01 and MECH-PED-01 as **BLOCKED — MISSING DATA** (source: `engineering/DRAWING_REGISTER.md`, rows MECH-ENC-01...MECH-PED-01; `README.md` references to `10_docs/Hardware/Enclosure` resolve to nothing). The content of this chapter therefore closes those gaps with **proposed** geometry derived from (a) the verified PCB outlines and hole tables and (b) the explicit design decisions logged in DSN-00. The repository states the rule that applies to all of it: "Everything under `engineering/DESIGN/` is *proposed* ... Status vocabulary for these drawings: **PROPOSED DESIGN** (with the decisions it depends on) — never **SOURCE-DERIVED** or **VERIFIED**" (source: `engineering/DESIGN/00_DESIGN_BASIS.md`, preamble).

Verified mechanical inputs (source: `engineering/MECHANICAL/dimensions/README.md`, copied; generated by `tools/gen_mechanical_package.py` from layer 20 of each `.brd`):

Board	Outline W × H (mm)	Holes/mounting pads	Bare-board mass ESTIMATE (g, 1.6 mm assumed)	Document
Main Board	260.00 × 300.00	10	≈ 231	MAIN_BOARD_dimensions.md
Power Supply Board	280.00 × 300.00	8	≈ 249	POWER_SUPPLY_dimensions.md
Frequency Synthesizer	100.00 × 100.00	4	≈ 30	FREQUENCY_SYNTHESIZER_dimensions.md
RF Power Amplifier	35.00 × 60.00	7	≈ 6	RF_PA_dimensions.md

Mounting-hole coordinates (mm, origin = lower-left corner of the outline bounding box, all Ø3.20 NPTH) (source: the four `engineering/MECHANICAL/dimensions/<BOARD>_dimensions.md` §2):

Board	Hole positions (X, Y)	Extreme pitch
Main Board (EAGLE 7.4.0 file, <code>RADAR_Main_Board.brd</code>)	(4, 4), (256, 4), (116, 114), (256, 114), (116, 250), (256, 250), (4, 296), (256, 296); plus two Ø0.90 package holes of X53 at (5.33, 105.36) and (5.33, 109.76)	X 252.00, Y 292.00
Power Supply (EAGLE 7.4.0, <code>PowerBoard.brd</code>)	(10, 10), (140, 10), (270, 10), (10, 120), (270, 120), (10, 230), (270, 230), (138.45, 267.85)	X 260.00, Y 257.85
Frequency Synthesizer (EAGLE 9.6.2, <code>Clocks_Freq_Synth_board.brd</code>)	(5, 5), (95, 5), (5, 95), (95, 95)	X 90.00, Y 90.00
RF PA (EAGLE 9.6.2, <code>RF_PA.brd</code>)	(2.6, 2.6), (17.5, 2.6), (32.4, 2.6), (2.6, 38), (32.4, 38), (2.6, 57.4), (32.4, 57.4)	X 29.80, Y 54.80

Board thickness is **ASSUMED 1.6 mm** ("not in source; KiCad default. Must be confirmed with the stack-up"), maximum component height is **UNKNOWN** ("no 3-D models in the CAD (Autodesk URNs only)") and the Power Board hole pattern "suggests a smaller board than 280 × 300" (G-10) (source: `engineering/MECHANICAL/dimensions/*_dimensions.md` §3; `engineering/VALIDATION/UNRESOLVED_GEOMETRY.md` rows G-01, G-02, G-10).

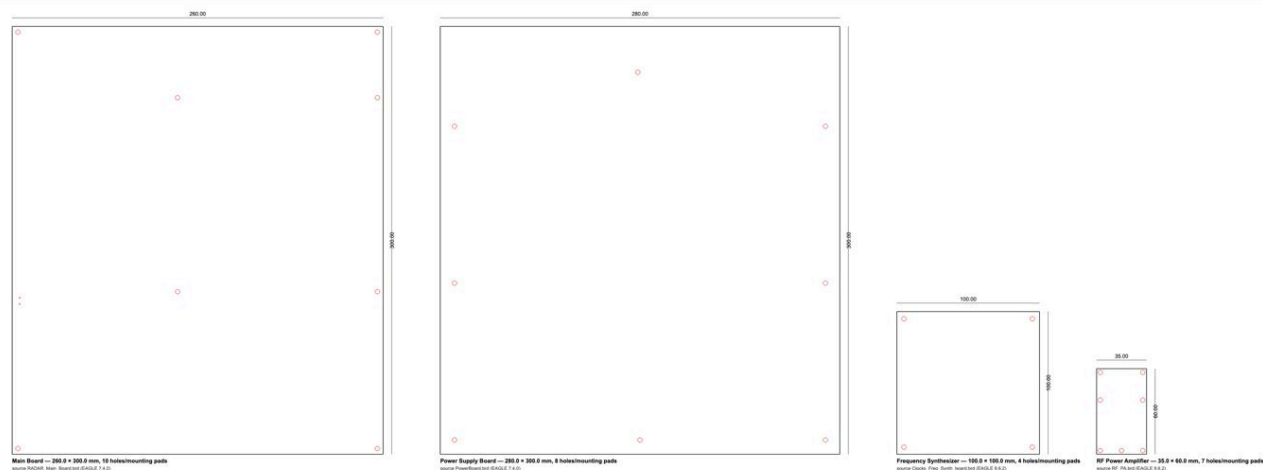


Figure 40 — F10.10a — Plan view of the four PCB outlines with holes, scale 1:1 on the native SVG, placement for comparison only (MECH-PLAN-01) — SOURCE-DERIVED (source: engineering/MECHANICAL/CAD/pcb_set_plan_view.png; produced by tools/gen_mechanical_package.py from the four .brd files)

AERIS-10 — Exploded assembly view, electronics set — ASM-EXP-01 Rev A — STATUS: CONCEPTUAL

Date 2026-10-09 · Units: drawing scale 0.35 (board sizes true to source) · Source: 4 × EAGLE .brd, 02 .hardware/04_antenna_beamforming.md, engineering/SYSTEM/interfaces/interconnection_table.md
CONCEPTUAL ILLUSTRATION — the stacking order, spacing, antenna panel, host and fasteners are NOT from any design file. Not a production drawing. Generated by tools/gen_assembly_exploded_view.py

Legend: solid outline = board geometry VERIFIED from EAGLE .brd (scale 0.35); dashed = CONCEPTUAL item without CAD; red circles = NPTH mounting holes ≥ 2 mm; balloons → PARTS_LIST.md; vertical spacing is arbitrary.

- CBL-40..55 RF, CBL-56..71 VG, CBL-72..87 sense: Main→PA (cable IDs are proposals; types/lengths UNVERIFIED)
- CBL-30.. clocks/LO Synth→Main (SMA coax) (cable IDs are proposals; types/lengths UNVERIFIED)
- CBL-20 enable bus SV1→SV1 (20-way) (cable IDs are proposals; types/lengths UNVERIFIED)
- CBL-01..CBL-15 rails Power→Main (Molex 22-23-2021) (cable IDs are proposals; types/lengths UNVERIFIED)

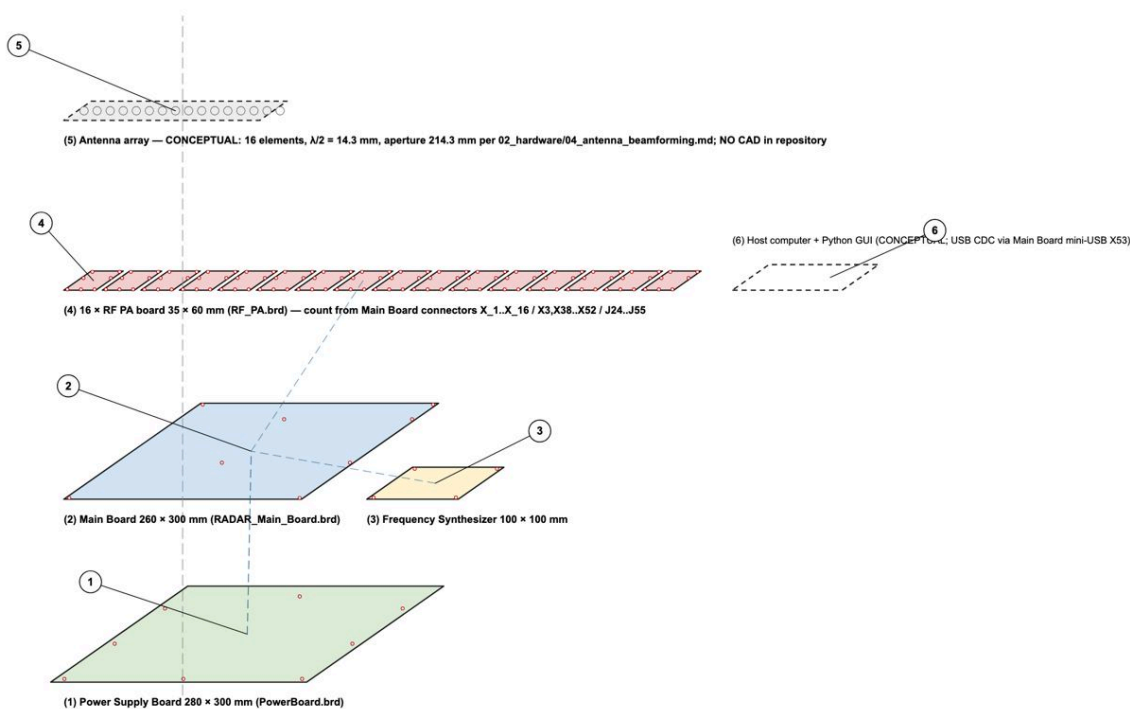


Figure 41 — F10.10b — Conceptual exploded view of the electronics set; board outlines are verified, the vertical arrangement, spacing, antenna panel and host are conceptual (ASM-EXP-01) — CONCEPTUAL (source: engineering/ASSEMBLY/EXPLODED_VIEWS/aeris10_exploded_conceptual.png; produced by tools/gen_assembly_exploded_view.py)

10.2 Design decisions that define the mechanics

Copied from the decision log (source: engineering/DESIGN/00_DESIGN_BASIS.md §2, rows D-07...D-13; D-14/D-15 are listed because the harness and the 22 V module occupy space in the head):

ID	Decision	Rationale	Alternatives
D-07	Board stack pitch 25 mm (component heights assumed 15 mm top / 4 mm bottom until measured)	no 3-D models; standard 25 mm M3 standoffs	measure, then shrink
D-08	Enclosure: folded 2.5 mm aluminium chassis + lid, 10 mm board-to-wall clearance, IP54 target, antenna panel in the front face behind a PTFE/ABS radome window	aluminium gives the heat path for the PA spreader; sheet metal keeps cost low	machined frame
D-09	Board arrangement: Power Board on the base, Main Board above it (25 mm), Synth on the Main-Board tier beside the SMA field; PA spreader vertical behind the antenna panel at the front; stepper/slip ring below the base	shortest coax from Synth J-ports to Main J1/J18–J23; rails cables short; RF to the front	side-by-side single tier (larger footprint)

Engineering & Assembly Manual		Decision	Rationale	Alternatives
D-10	Thermal design case: ambient 45 °C, TBASE(PA) ≤ 85 °C, drain gated with the pulse train (duty computed from the timing); continuous-bias case reported as infeasible without liquid cooling	16 × 37 W = 591 W quiescent is not coolable in a sealed rotating head; gating requires a hardware change (D-14)	—	
D-11	10 mm aluminium heat spreader common to all 16 PA boards, fan-cooled fin stack at the rear, 1.5× airflow margin	conduction from 16 hot spots into one plate, then forced air	per-PA heatsinks	
D-12	Azimuth drive: stepper → GT3 belt 1:3 → turntable; revised by DSN-CALC-01 §2 : with a ≈ 10 kg head a NEMA 23 needs ≥ 200 ms per 7.2° step (revolution ≈ 19 s); a NEMA 34 (3 N·m class) allows ≈ 100 ms; firmware Stepper_steps = 600 for 1:3	torque calculation from the detailed mass table; belt isolates the motor from the slewing ring	ratio 1:6 (Stepper_steps = 1200), lighter head	
D-13	Slip ring 12 channels (VIN ×4 @10 A, 22 V ×4 @10 A, USB 2.0 ×4) through a 60 mm bore slewing bearing; everything else rotates with the head	only DC in and the host link cross the rotation	Ethernet/wireless host link (removes 4 channels)	
D-14	22 V PA supply: synchronous boost 12–17 V → 22 V, 150 W average / bulk capacitance for 45 A pulse peaks, high-side eFuse/MOSFET switch from EN/DIS_RFPA_VDD , plus a per-pulse drain-gating FET per PA board (TTL from the FPGA/MCU DIG lines — pin to be allocated)	closes conflict K4; the firmware already sequences VG before VD	separate 22 V mains PSU on the pedestal (bigger slip ring)	
D-15	Harness: cable IDs from interconnection_table.md; coax RG-405 (0.086") hand-formable equal-length for Main↔PA and Synth↔Main; Molex 22-01-3027/3037 crimp housings for the 2-/3-pin headers; 20-way IDC ribbon for SV1	standard parts matching the footprints in the schematics	—	

The D-09 wording ("Main Board above it (25 mm)") describes the first layout; the generators now place **all boards vertical, parallel to the antenna, in depth order** radome gap → antenna panel → PA heat-spreader plate → 16 PA boards → Main Board → Synth → Power Board → rear wall (source: `tools/design_layout.py` module docstring). The 25 mm pitch of D-07 is kept between tiers (§10.3).

Decisions still open for the owner (source: `engineering/DESIGN/00_DESIGN_BASIS.md §3`; `engineering/DESIGN/README.md` , last paragraph): D-01 antenna variant, D-07 component heights to be measured, D-12 firmware constant / gear ratio, D-13 whether the host link stays USB, D-14 where `TX_GATE` comes from, G-10 Power Board outline, and conflicts K1–K8.

10.3 Radar head — layout and stations

All dimensions below are the values printed by `python3 tools/design_layout.py` (executed 2026-10-09, exit 0; the script reads `engineering/DESIGN/design_parameters.json`). Coordinate system of the head: X = width (0 at the inner left wall, looking at the antenna), Y = depth (0 = inner face of the front wall, antenna side), Z = height (0 = top of the base plate) (source: `tools/design_layout.py` docstring).

Item	Value (mm)	Source key
Outer envelope W × H × D	315.0 × 315.0 × 132.6	<code>outer_mm</code>
Inner envelope W × H × D	310.0 × 310.0 × 127.6	<code>inner_mm</code>
Station Y: antenna panel	8.0	<code>stations.y.antenna</code>
Station Y: PA plate front face	11.0	<code>stations.y.plate_front</code>
Station Y: PA plate rear face	21.0	<code>stations.y.plate_rear</code>
Station Y: Main Board tier	56.0	<code>stations.y.main</code>
Station Y: Synth tier	81.0	<code>stations.y.synth</code>
Station Y: Power Board tier	106.0	<code>stations.y.power</code>
Station Y: rear inner wall	127.6	<code>stations.y.rear_inner</code>
PA heat-spreader plate W × H × t	300.0 × 300.0 × 10	<code>plate</code>
Fin fields: n / height / thickness / pitch / length / strip width	26 / 25.0 / 2.0 / 4.0 / 280.0 / 55.0	<code>fins</code>
Antenna panel W × H × t, gap to plate, radome gap, position (x, z)	165.0 × 248.0 × 0.6, 2.4, 8.0, (72.5, 31.0)	<code>antenna</code>
PA field (x, z, w, h)	(77.5, 23.0, 155.0, 264.0)	<code>pa_field</code>
PA grid cols × rows, pitch x / z	4 × 4, 40.0 / 68.0	<code>pa_grid</code>
Main Board position (x, z)	(25.0, 5.0)	<code>main_pos</code>
Synth position (x, z)	(46.36, 141.60) — basis: "centred on Main Board J1,J18,J20,J21,J22,J23 (P&P)"	<code>synth_pos</code>
Power Board position (x, z)	(15.0, 5.0)	<code>power_pos</code>

The tier pitch Main → Synth → Power is 25 mm (56 → 81 → 106), i.e. decision D-07; the antenna sits 2.4 mm in front of the plate on nylon spacers and 8 mm behind the radome window (source: same table). The envelope 315 × 315 × 133 mm is also the headline number of `engineering/DESIGN/README.md` ("head 315 × 315 × 133 mm (W × H × D), all boards vertical").

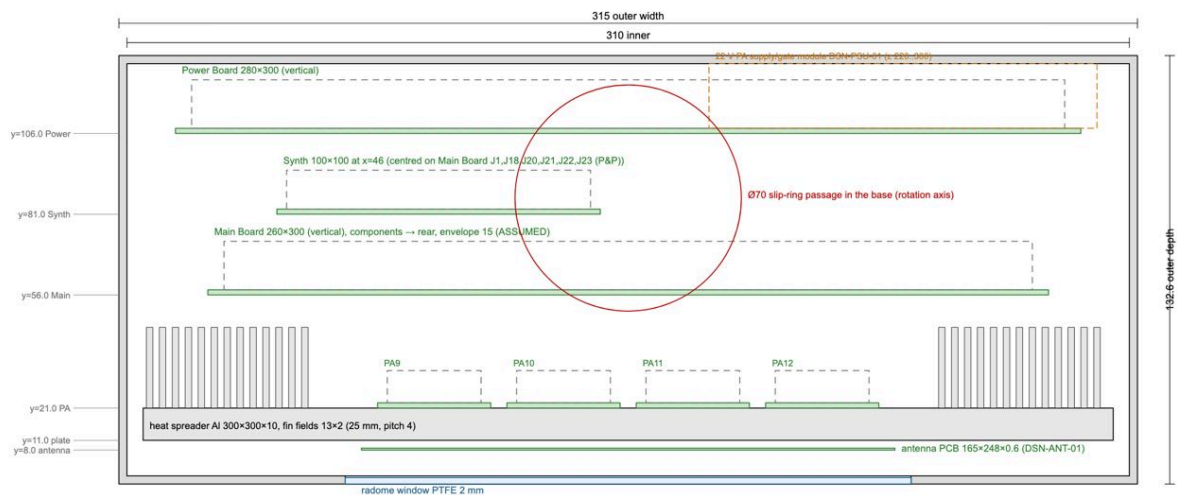


Figure 42 — F10.1 — Head plan section at mid height (X–Y), DSN-MECH-01 Rev A, scale 1:1 on the native SVG — PROPOSED DESIGN, D-07...D-13 (source: engineering/DESIGN/MECHANICAL/drawings/png/DSN-MECH-01_head_plan_section.png; produced by tools/design_mechanical_drawings.py)

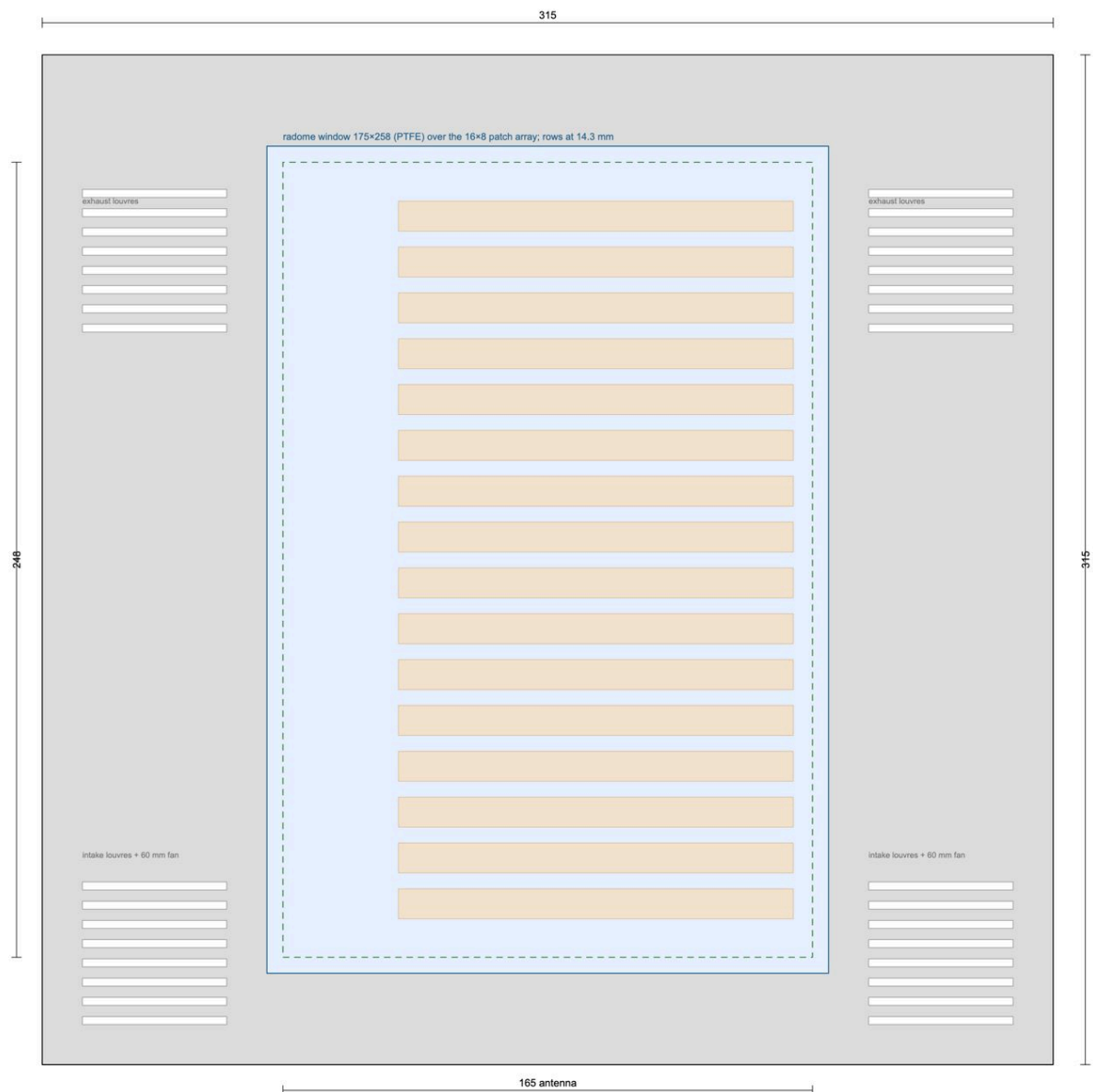


Figure 43 — F10.2 — Head front elevation (X-Z) with radome window and intake ducts, DSN-MECH-02 Rev A — PROPOSED DESIGN, D-07...D-13 (source: engineering/DESIGN/MECHANICAL/drawings/png/DSN-MECH-02_head_front_elevation.png; produced by tools/design_mechanical_drawings.py)

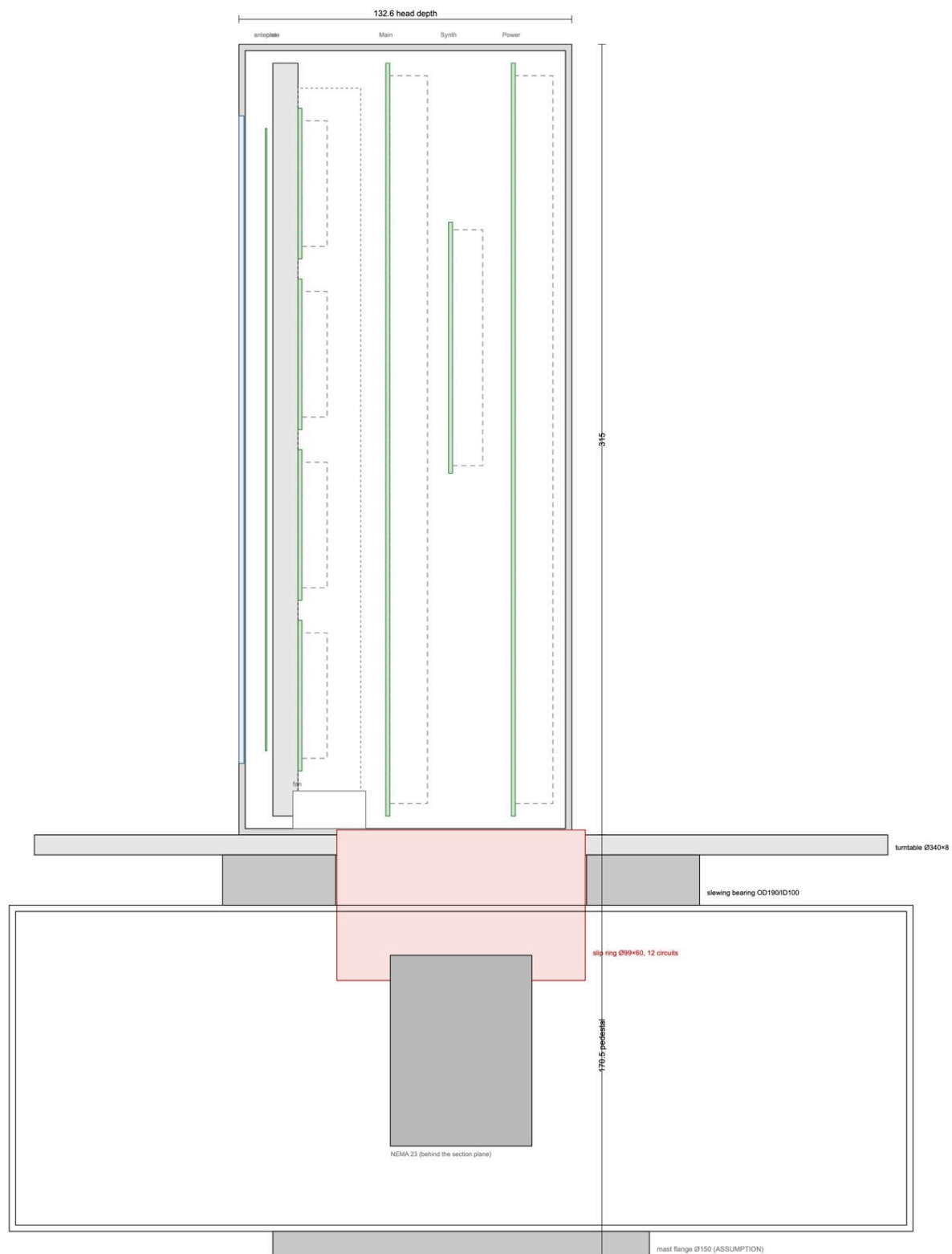


Figure 44 — F10.3 — Head side section including the pedestal, DSN-MECH-03 Rev A — PROPOSED DESIGN, D-07...D-13 (source: engineering/DESIGN/MECHANICAL/drawings/png/DSN-MECH-03_head_side_section.png; produced by tools/design_mechanical_drawings.py)



Figure 45 — F10.4 — Internal layout, each tier seen from the rear, with connector positions: tier 1 PA heat spreader with PA1...PA16 and fin fields, tier 2 Main Board (rear = component side, J1...J55, X_n, JP, SV1), Synth and Power tiers; board outlines and connector positions are from the EAGLE .brd / KiCad P&P files, enclosure, spacing and fasteners are proposed, component heights ASSUMED 15 mm — DSN-MECH-04 Rev A — PROPOSED DESIGN, D-07...D-13 (source: engineering/DESIGN/MECHANICAL/drawings/png/DSN-MECH-04_internal_layout_rear.png; produced by tools/design_mechanical_drawings.py)

The drawing's own title block states the mixed status: "Board outlines and connector positions are VERIFIED (EAGLE .brd / KiCad P&P); enclosure, spacing, pedestal and fasteners are PROPOSED; component heights ASSUMED 15 mm" (source: engineering/DESIGN/MECHANICAL/drawings/DSN-MECH-04_internal_layout_rear.svg , header text). In the vocabulary of this manual the word "VERIFIED" in that title block means SOURCE-DERIVED from the native CAD files, not hardware-verified (manual/STYLE_GUIDE.md).

10.4 Pedestal and azimuth drive

Parameters printed by `python3 tools/design_layout.py`, key `pedestal` (copied):

Item	Value
Turntable	Ø340.0 × 8.0 mm Al
Slewing bearing	OD 190.0 / ID 100.0 × 20.0 mm
Ring pulley	Ø172.0 × 12.0 mm, 180 teeth GT3
Motor	"NEMA 23 (56.4 mm square, 76 mm long, 6.35 mm shaft), 200 steps/rev, ≥ 1.9 N·m"
Motor pulley	Ø57.3 mm, 60 teeth; belt GT3 9 mm; ratio 3
Firmware change	"Stepper_steps = 600 (main.cpp:195) so that 4×3 = 12 pulses give 7.2° at the table; TB6600 at full step"
Slip ring	"through-bore slip ring, bore ≥ 60 mm, 12 circuits (4 × 10 A VIN, 4 × 10 A 22 V, 4 × signal USB 2.0)", OD 99.0, length 60.0 mm
Pedestal base W × D × H	360.0 × 360.0 × 130.0 mm
Mast flange	"Ø 150 mm, 4 × M10 on PCD 110 mm (ASSUMPTION — mast interface not defined)"

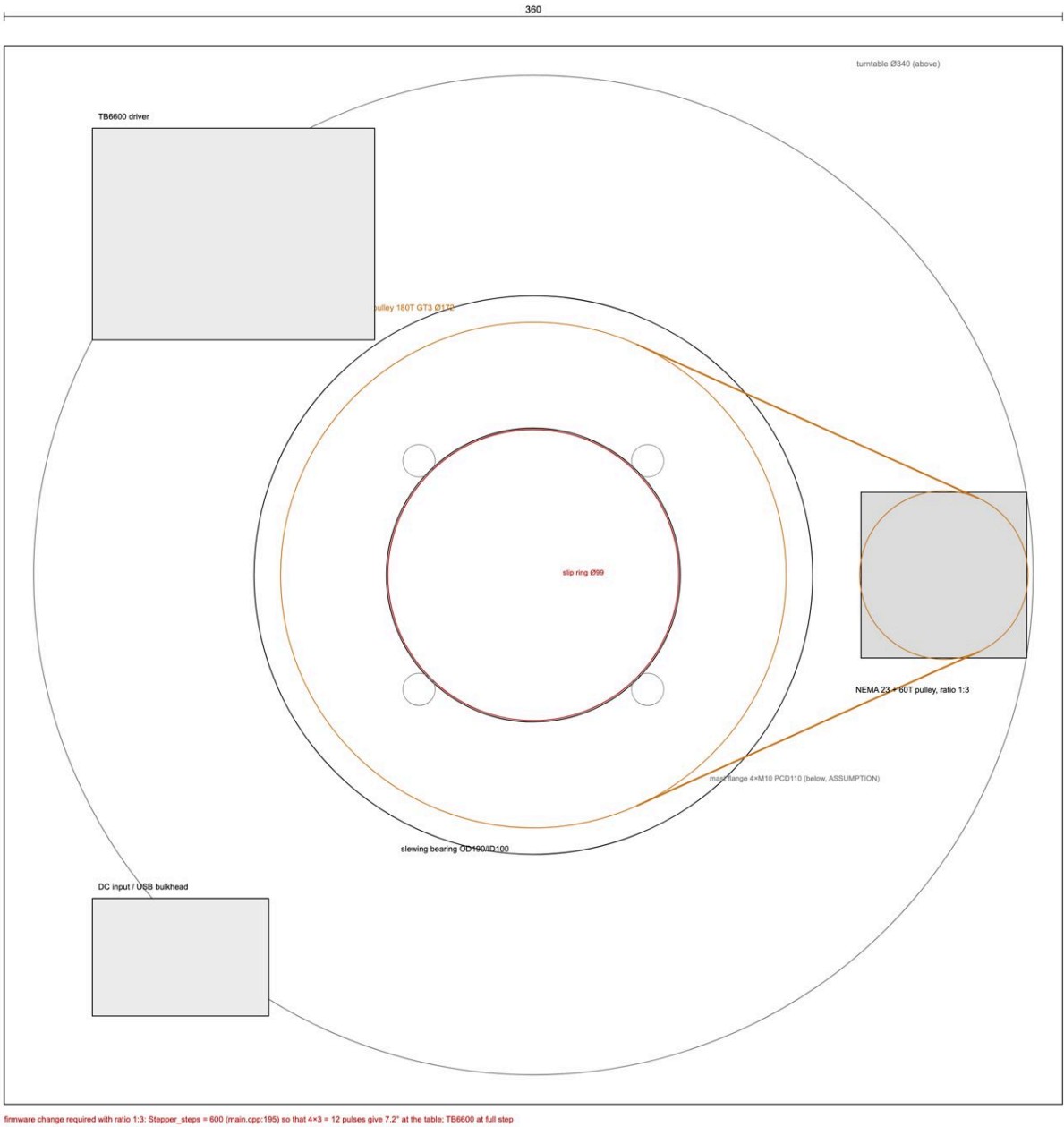


Figure 46 — F10.5 — Pedestal plan: turntable, slewing bearing, ring pulley, NEMA 23 with motor bracket, slip ring, DSN-MECH-05 Rev A — PROPOSED DESIGN, D-12 (as revised), D-13 (source: engineering/DESIGN/MECHANICAL/drawings/png/DSN-MECH-05_pedestal_plan.png; produced by tools/design_mechanical_drawings.py)

Drive-torque check (revises D-12)

Copied from DSN-CALC-01 §2 (source: engineering/DESIGN/CALCS/DESIGN_CALCULATIONS.md §2; inputs MECHANICAL/CAD/detail/parts_list.json; no measurement):

Quantity	Value	Basis
Rotating mass	10.4 kg	detailed head structure 6.4 kg + 4 kg PCBs/antenna/cables (estimate)
Moment of inertia (head as a box 315 × 133 mm about its centre + turntable)	0.128 kg·m²	solid-box formula; head centred on the axis
Move: 7.2° in 50 ms, triangular profile	$\alpha = 201 \text{ rad/s}^2$, peak $\omega = 5.03 \text{ rad/s}$	per azimuth step (50 positions/rev; 5.65 ms dwell per elevation, 31 elevations ≈ 175 ms per azimuth, so a 50 ms move costs 22 % of the scan time)
Table torque	25.69 N·m	$J \cdot \alpha$
Motor torque with 1:3 belt, 80 % efficiency	10.70 N·m at ≈ 144 rpm peak	
NEMA 23 (1.9 N·m holding, ≈ 60 % at that speed)	1.14 N·m available → margin ×0.1	assumption on the torque curve — check the chosen motor's curve at 24 V with the TB6600

Move time versus motor torque (same mass, triangular profile, 1:3 belt) (source: same section):

Move time per 7.2° step	Motor torque needed	NEMA 23 (≈ 1.14 N·m)	NEMA 34 (≈ 3.0 N·m at speed)	Revolution time (50 × (175 ms dwell + move))
50 ms	10.70 N·m	NO	NO	11.2 s
100 ms	2.68 N·m	NO	NO	13.8 s
150 ms	1.19 N·m	NO	OK	16.2 s
200 ms	0.67 N·m	OK	OK	18.8 s
300 ms	0.30 N·m	OK	OK	23.8 s

Verdict as recorded: "the earlier assumption 'NEMA 23, 1:3, fast steps' does not hold for a 10 kg head: NEMA 23 needs ≥ 200 ms per step (revolution ≈ 19 s), NEMA 34 (3 N·m class, 86 mm frame) allows ≈ 100 ms (revolution ≈ 14 s). Alternatives: ratio 1:6 with `Stepper_steps = 1200`, or a lighter head (lid/tray in 2 mm, no PA plate side strips). Bearing moment load from wind is not included (no enclosure wind spec)" (source: `engineering/DESIGN/CALCS/DESIGN_CALCULATIONS.md` §2, "Verdict"). The parts list still models a NEMA 23 (part #45) and the layout file still names a NEMA 23: the motor class is an **open owner decision (D-12)** and the assembly procedure (chapter 15, Step 15.21) carries the $\triangle$ marker for it.

10.5 Enclosure detail: sheet metal, fasteners, sealing

The detailed model (`tools/design_enclosure_detail_freecad.py`, FreeCAD 1.1) builds 51 parts; the flat patterns and the fastener section are drawn from its `parts_list.json` by `tools/design_enclosure_drawings.py` with bend allowance "90°", inside radius $R = t$, $K = 0.4 \rightarrow BA = \pi/2 \cdot (R + K \cdot t)$ (source: `tools/design_enclosure_drawings.py` docstring; `engineering/DRAWING_REGISTER.md` rows DSN-MECH-3D-DETAIL, DSN-MECH-06, DSN-MECH-07).

AERIO 6 — Radar head sheet-metal parts: Flat patterns with bend lines — DSN-MECH-06 Rev A

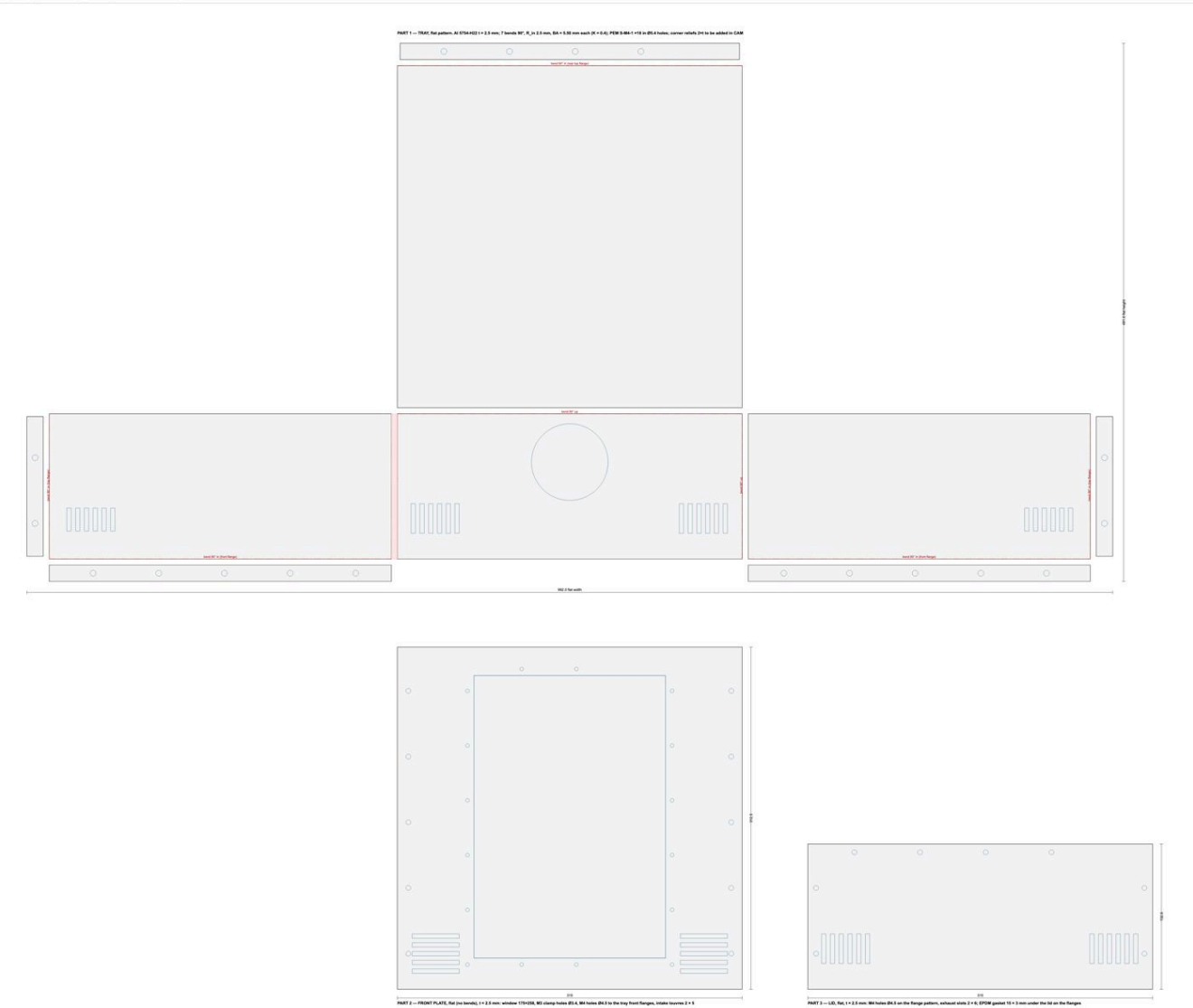


Figure 47 — F10.6 — Sheet-metal flat patterns with bend lines for the tray, front plate and lid (2.5 mm Al 5754), DSN-MECH-06 Rev A; bend allowance per the K-factor formula stated in the generator, CAM check of bend reliefs still open — PROPOSED DESIGN, D-08 (source: `engineering/DESIGN/MECHANICAL/drawings/png/DSN-MECH-06_sheet_metal_flat_patterns.png`; produced by `tools/design_enclosure_drawings.py`)

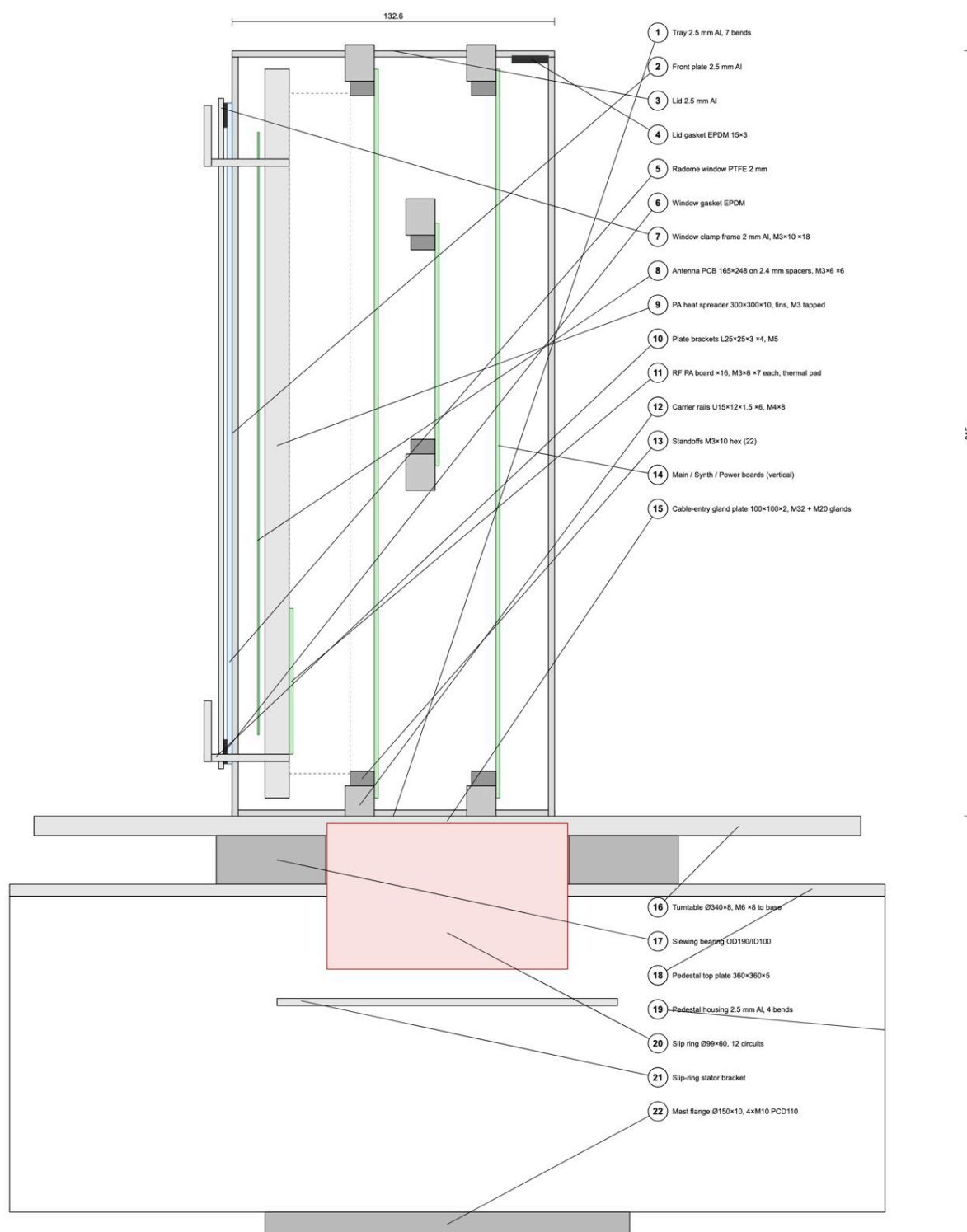


Figure 48 — F10.7 — Assembly section with fastener balloons (balloon numbers = part # of the mechanical parts list), DSN-MECH-07 Rev A — PROPOSED DESIGN, D-07...D-13
(source: engineering/DESIGN/MECHANICAL/drawings/png/DSN-MECH-07_assembly_section_fasteners.png; produced by tools/design_enclosure_drawings.py)

Mechanical parts list (DSN-MECH-PL Rev A)

Copied verbatim (source: `engineering/DESIGN/MECHANICAL/MECHANICAL_PARTS_LIST.md`; "from `CAD/detail/parts_list.json` ... Status: PROPOSED DESIGN · masses are volume × density estimates"). Head ≈ 6.4 kg (structure only, PCBs/antenna not included except as noted), pedestal ≈ 12.4 kg (incl. stepper 1.1 kg, bearing/slip ring as modelled).

#	Group	Part	Material	Mass (g)	Fasteners	Note
1	HEAD	Tray — base + sides + rear, 2.5 mm Al 5754, 3 bends R2.5, flanges 15 mm with M4 PEM nuts	Al	1559	PEM S-M4-1 ×18	8 lid holes M4, 10 front-plate holes M4, Ø70 cable entry, intake slots
2	HEAD	Front plate 2.5 mm Al with radome window opening and intake louvres	Al	346	M4×8 ×10 to tray front flanges	window 175×258, 16 M3 clamp holes
3	HEAD	Radome window PTFE 2 mm (outside the front plate)	PTFE	246		RF loss ≈ 0.1 dB at 10.5 GHz (PTFE ϵ_r 2.1, $\tan\delta$ 0.0002 — to be confirmed)
4	HEAD	Window gasket EPDM 1.5 mm (ring 10 mm)	EPDM	17		
5	HEAD	Window clamp frame 2 mm Al, 14 mm wide	Al	69	M3×10 ×16 + nyloc	
6	HEAD	Lid 2.5 mm Al with exhaust slots	Al	272	M4×8 ×8	
7	HEAD	Lid gasket EPDM 3 mm self-adhesive, 15 mm wide on the flanges	EPDM	44		compressed to 2 mm → IP54 target
8	HEAD	PA heat spreader 300×300×10 Al 6061, machined fin fields, M3 tapped (16×7 PA + 6 antenna)	Al	3400	M3×6 ×118 (PA 112 + antenna 6)	thermal pads 5×5 under each QPA2962; antenna on 2.4 mm nylon spacers
9	HEAD	Plate bracket L 25×25×3 Al at x=5 z=20	Al	26	M5×12 ×2 + M5×16 ×1	bolts the plate to the side wall and to the front flange
10	HEAD	Plate bracket L 25×25×3 Al at x=5 z=265.0	Al	26	M5×12 ×2 + M5×16 ×1	bolts the plate to the side wall and to the front flange
11	HEAD	Plate bracket L 25×25×3 Al at x=280 z=20	Al	26	M5×12 ×2 + M5×16 ×1	bolts the plate to the side wall and to the front flange
12	HEAD	Plate bracket L 25×25×3 Al at x=280 z=265.0	Al	26	M5×12 ×2 + M5×16 ×1	bolts the plate to the side wall and to the front flange
13	HEAD	Main Board carrier rail bottom U15×12×1.5 Al, 280 mm	Al	42	M4×8 ×2	bolted to the side walls (M4) — spans the full width when the board is narrower
14	HEAD	Main Board carrier rail top U15×12×1.5 Al, 280 mm	Al	42	M4×8 ×2	bolted to the side walls (M4) — spans the full width when the board is narrower
15	HEAD	Main Board standoff M3×10 hex at (4,4)	steel	2	M3×6 ×1	
16	HEAD	Main Board standoff M3×10 hex at (256,4)	steel	2	M3×6 ×1	
17	HEAD	Main Board standoff M3×10 hex at (116,114)	steel	2	M3×6 ×1	
18	HEAD	Main Board standoff M3×10 hex at (256,114)	steel	2	M3×6 ×1	
19	HEAD	Main Board standoff M3×10 hex at (116,250)	steel	2	M3×6 ×1	
20	HEAD	Main Board standoff M3×10 hex at (256,250)	steel	2	M3×6 ×1	
21	HEAD	Main Board standoff M3×10 hex at (4,296)	steel	2	M3×6 ×1	
22	HEAD	Main Board standoff M3×10 hex at (256,296)	steel	2	M3×6 ×1	
23	HEAD	Synth carrier rail bottom U15×12×1.5 Al, 120 mm	Al	18	M4×8 ×2	bolted to the side walls (M4) — spans the full width when the board is narrower
24	HEAD	Synth carrier rail top U15×12×1.5 Al, 120 mm	Al	18	M4×8 ×2	bolted to the side walls (M4) — spans the full width when the board is narrower
25	HEAD	Synth standoff M3×10 hex at (5,5)	steel	2	M3×6 ×1	
26	HEAD	Synth standoff M3×10 hex at (95,5)	steel	2	M3×6 ×1	
27	HEAD	Synth standoff M3×10 hex at (5,95)	steel	2	M3×6 ×1	
28	HEAD	Synth standoff M3×10 hex at (95,95)	steel	2	M3×6 ×1	
29	HEAD	Power Board carrier rail bottom U15×12×1.5 Al, 300 mm	Al	44	M4×8 ×2	bolted to the side walls (M4) — spans the full width when the board is narrower
30	HEAD	Power Board carrier rail top U15×12×1.5 Al, 300 mm	Al	44	M4×8 ×2	bolted to the side walls (M4) — spans the full width when the board is narrower
31	HEAD	Power Board standoff M3×10 hex at (10,10)	steel	2	M3×6 ×1	
32	HEAD	Power Board standoff M3×10 hex at (140,10)	steel	2	M3×6 ×1	
33	HEAD	Power Board standoff M3×10 hex at (270,10)	steel	2	M3×6 ×1	
34	HEAD	Power Board standoff M3×10 hex at (10,120)	steel	2	M3×6 ×1	
35	HEAD	Power Board standoff M3×10 hex at (270,120)	steel	2	M3×6 ×1	
36	HEAD	Power Board standoff M3×10 hex at (10,230)	steel	2	M3×6 ×1	
37	HEAD	Power Board standoff M3×10 hex at (270,230)	steel	2	M3×6 ×1	
38	HEAD	Power Board standoff M3×10 hex at (138,268)	steel	2	M3×6 ×1	
39	HEAD	Cable-entry gland plate 100×100×2 Al under the base (M32 + M20 glands)	Al	48	M4×8 ×4; glands M32 + M20 IP68	harness from the slip ring: VIN, 22 V, USB
40	PEDESTAL	Turntable plate Ø340×8 Al, 8×M6 to the head base, 12×M6 to the bearing outer ring	Al	1864	M6×16 ×8, M6×25 ×12	
41	PEDESTAL	Slewing bearing OD190/ID100×20 (4-point contact, e.g. igus PRT-04-100 class)	steel	3218		part number to be selected (axial ≥ 50 kg, moment ≥ 60 N·m)
42	PEDESTAL	Ring pulley GT3 180T Ø172 Al (machined/3D-printed), clamped under the turntable	Al	0	M4×10 ×6	
43	PEDESTAL	Pedestal top plate 360×360×5 Al, 12×M6 to the bearing inner ring, bore Ø90	Al	1658	M6×20 ×12	
44	PEDESTAL	Pedestal housing 360×360×125 folded 2.5 mm Al (4 bends), open top, connector panel cut-out	Al	1575	M5×10 ×12 to the top plate	DC input (XT60/M12), USB-B bulkhead, vent

#	Group	Part	Material	Mass (g)	Fasteners	Note
45	PEDESTAL	Stepper NEMA 23 76 mm	steel	1898	M5×12 ×4	
46	PEDESTAL	Motor bracket 80×80×3 Al with slotted belt-tension holes	Al	41	M5×10 ×4 to the top plate	slots ±5 mm for GT3 tension
47	PEDESTAL	Motor pulley GT3 60T Ø57, bore 6.35	Al	84	grub M4 ×2	
48	PEDESTAL	Through-bore slip ring Ø99×60, bore 60, 12 circuits (4×10 A, 4×10 A, 4 signal)	plastic	351		e.g. Senring H3899 class — select
49	PEDESTAL	Slip-ring stator bracket 140×140×3 Al	Al	95	M4×8 ×4	stator fixed to the pedestal, rotor flange to the turntable
50	PEDESTAL	Stepper driver TB6600 on DIN rail	plastic	299		
51	PEDESTAL	Mast flange Ø150×10 steel, 4×M10 PCD 110 (ASSUMPTION — mast interface undefined)	steel	1360	M10×30 ×4	

Fastener totals (source: same file, "Fastener totals (from the per-part lists)":

Fastener	Qty
M10×30	4
M3×10	16
M3×6	138
M4	2
M4×10	6
M4×8	38
M5×10	16
M5×12	12
M5×16	4
M6×16	8
M6×20	12
M6×25	12
S-M4-1	18

Note on fastener torque: the repository gives no torque table for the fasteners above; the assembly-step source quotes 0.5 N·m for the M3 PA-board screws and 0.9 N·m for the SMA couplings as check values (source: `manual/ASSEMBLY_STEPS_SOURCE.md`, rows B3 and C3) — these values have no deeper source in the repository and are marked ASSUMED in chapter 15.

Sealing and finish

Copied (source: `engineering/DESIGN/MECHANICAL/MECHANICAL_PARTS_LIST.md`, "Sealing and finish"):

- Lid: EPDM 15 × 3 mm self-adhesive gasket on the three top flanges, compressed to 2 mm by the M4 screws (pitch 60 mm). Front plate: 1.5 mm EPDM strip on the two front flanges (same pattern).
- Radome window: PTFE 2 mm clamped by the 2 mm Al frame with EPDM 1.5 mm gasket, M3 × 18. Alternative: 2 mm Rogers/ABS radome if PTFE loss is acceptable but cost is not.
- Cable entry: 100 × 100 gland plate under the base with M32 (power) and M20 (USB) IP68 glands; the Ø70 base hole lets the harness reach the slip ring rotor.
- Finish: chromate conversion (Alodine) + powder coat RAL 7035 outside; bare chromate inside for grounding at the flanges; PA plate bare 6061 with thermal pads.
- Open: bend reliefs and corner welds of the tray, stiffeners of the 315 mm lid (2.5 mm may need a 20 mm hem), fan mounting brackets, antenna spacer material, earthing stud, lifting points, mast interface.

Observation MAN-10-1: the window clamp is listed as "M3×10 ×16" in part #5 and in the fastener totals, but as "M3 × 18" in the sealing text of the same document; the per-part list (16) is taken as the count for chapter 15 until the owner reconciles the text.

10.6 Three-dimensional models

Two FreeCAD models exist: the parametric head + pedestal (`aeris10_head_pedestal.FCStd`, DSN-MECH-3D, STEP/STL/DXF exports and a mass table) and the 51-part detail model (`aeris10_enclosure_detail.FCStd`, DSN-MECH-3D-DETAIL) (source: `engineering/DRAWING_REGISTER.md`, rows DSN-MECH-3D and DSN-MECH-3D-DETAIL). The isometric renders below are flat-shaded projections of the STL exports (`tools/stl_to_svg_iso.py`: "Intended for quick visual checks ... not a CAD drawing").

AERIS-10 — detailed head + pedestal, x-ray isometric from the front-left — DSN-MECH-3D-DETAIL — PROPOSED DESIGN

51 parts: tray, front plate with radome window and louvres, lid, window frame, PA plate with brackets, carrier rails, standoffs, gland plate; pedestal with bearing, pulleys, slip ring, motor bracket
141596 triangles from 3 STL file(s); azimuth 35.0°, elevation 25.0°, scale 0.5; rendered by tools/stl_to_svg_iso.py

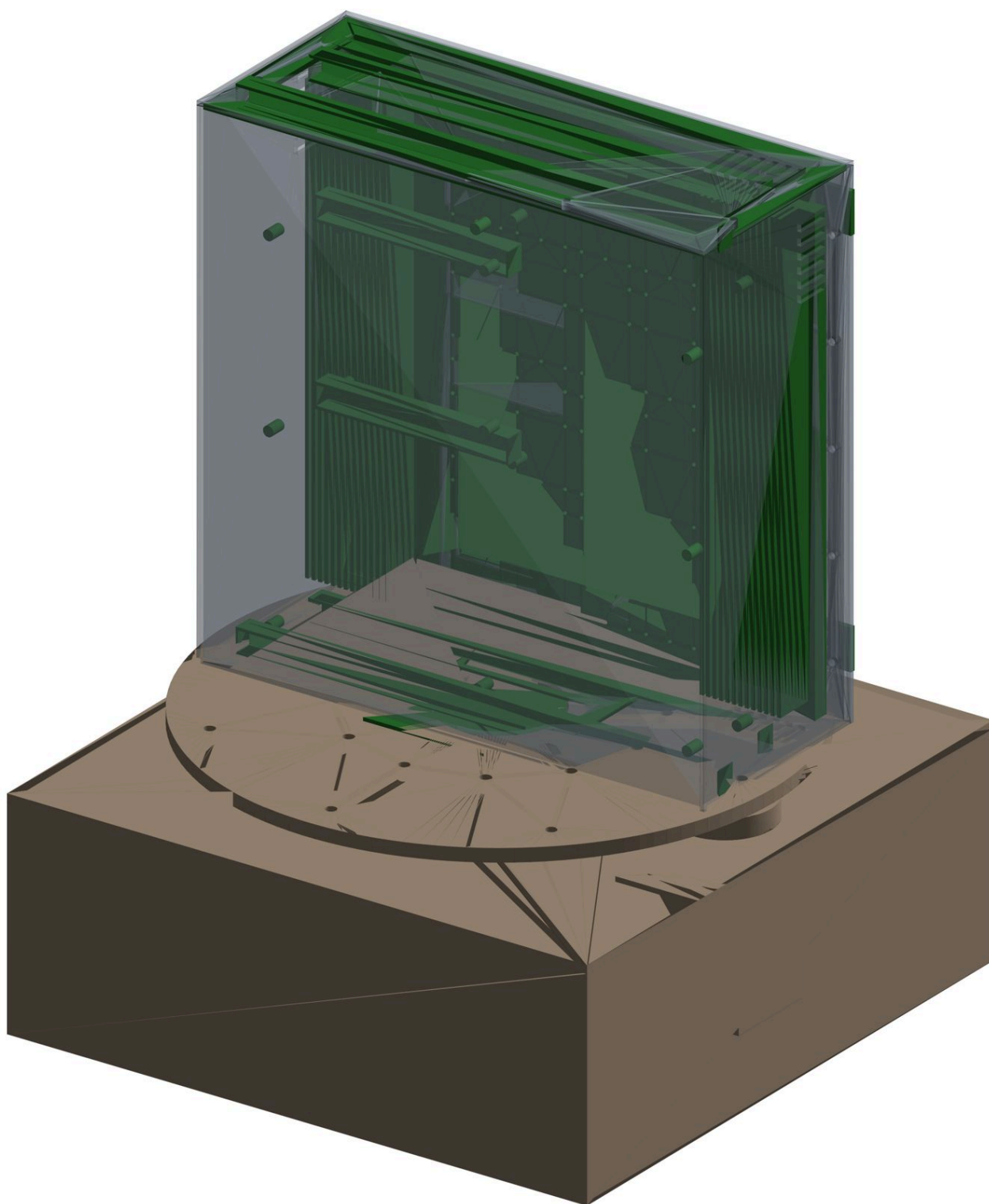


Figure 49 — F10.8a — Detail model, X-ray front isometric: tray, front plate with window, PA plate, carrier rails, boards — PROPOSED DESIGN, DSN-MECH-3D-DETAIL (source: engineering/DESIGN/MECHANICAL/CAD/detail/aeris10_detail_xray_front_iso.png; produced by tools/design_enclosure_detail_freecad.py (FreeCAD 1.1) + tools/stl_to_svg_iso.py)

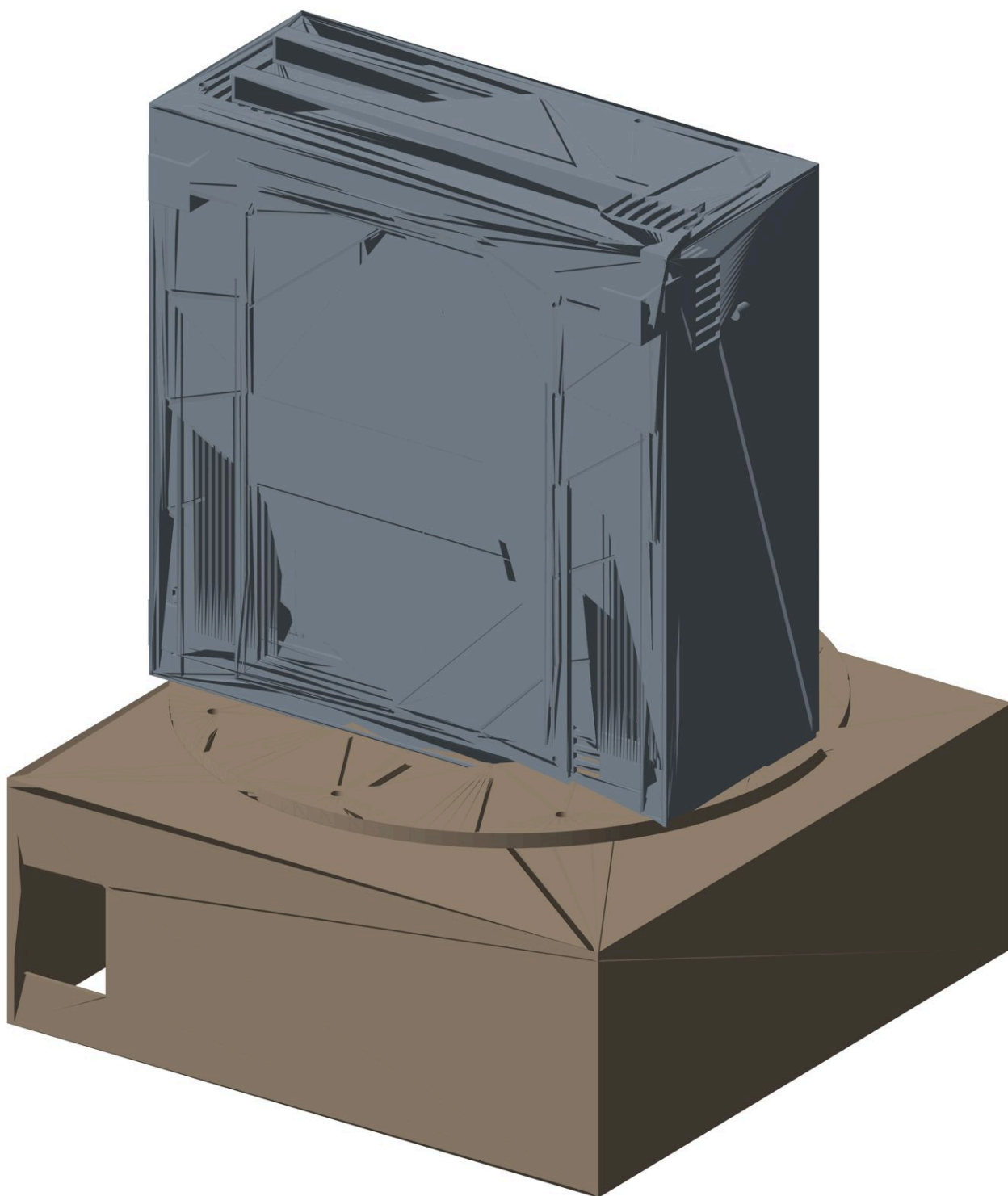


Figure 50 — F10.8b — Detail model, rear isometric: lid, rear wall, pedestal housing, motor bracket — PROPOSED DESIGN, DSN-MECH-3D-DETAIL (source: engineering/DESIGN/MECHANICAL/CAD/detail/aeris10_detail_iso_rear.png; produced by tools/design_enclosure_detail_freecad.py + tools/stl_to_svg_iso.py)

AERIS-10 — Radar head, x-ray isometric from the rear-left (chassis 25 % opaque) — PROPOSED DESIGN

Boards (green), PA plate with fins and 16 PA boards, 22 V module; pedestal below

6188 triangles from 3 STL file(s); azimuth 215.0°, elevation 25.0°, scale 0.5; rendered by tools/stl_to_svg_iso.py

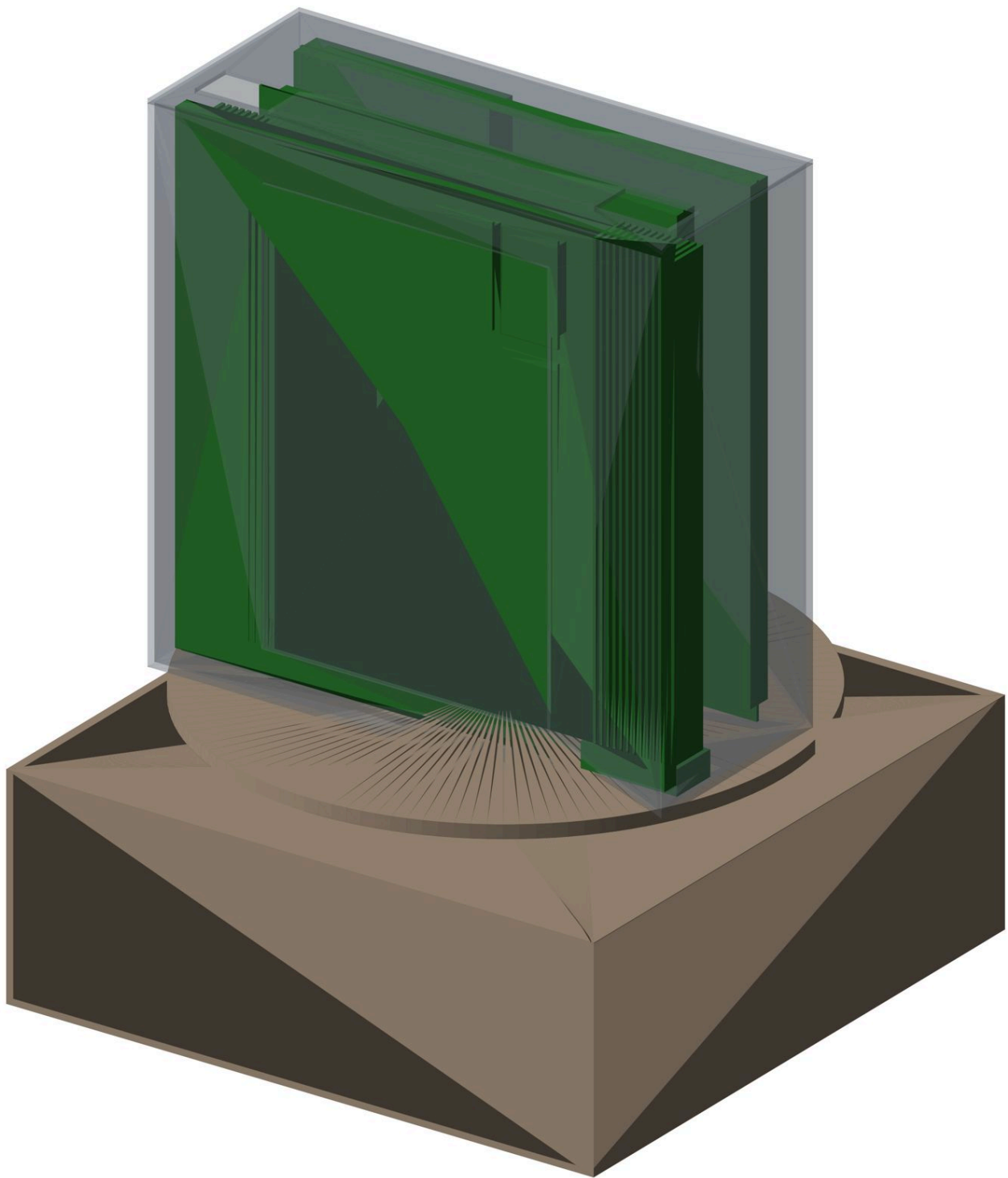


Figure 51 — F10.8c — Parametric head model, X-ray isometric with board envelopes (component heights ASSUMED 15 mm top / 4 mm bottom, D-07) — PROPOSED DESIGN, DSN-MECH-3D (source: engineering/DESIGN/MECHANICAL/CAD/aeris10_head_xray_iso.png; produced by tools/design_mechanical_freecad.py (freecadcmd) + tools/stl_to_svg_iso.py)

Mass figures from the two models disagree by construction and both are estimates: the detail parts list gives head structure ≈ 6.4 kg and pedestal ≈ 12.4 kg; the parametric model's volume-based table gives "head ≈ 9.9 kg (includes component envelopes as solid plastic — overestimate), pedestal ≈ 12 kg + 1.1 kg stepper" (source: engineering/DESIGN/MECHANICAL/MECHANICAL_PARTS_LIST.md header; engineering/VALIDATION/UNRESOLVED_GEOMETRY.md, "Update 2026-10-09", G-03). The torque check of §10.4 uses 10.4 kg.

10.7 PA heat-spreader thermal map

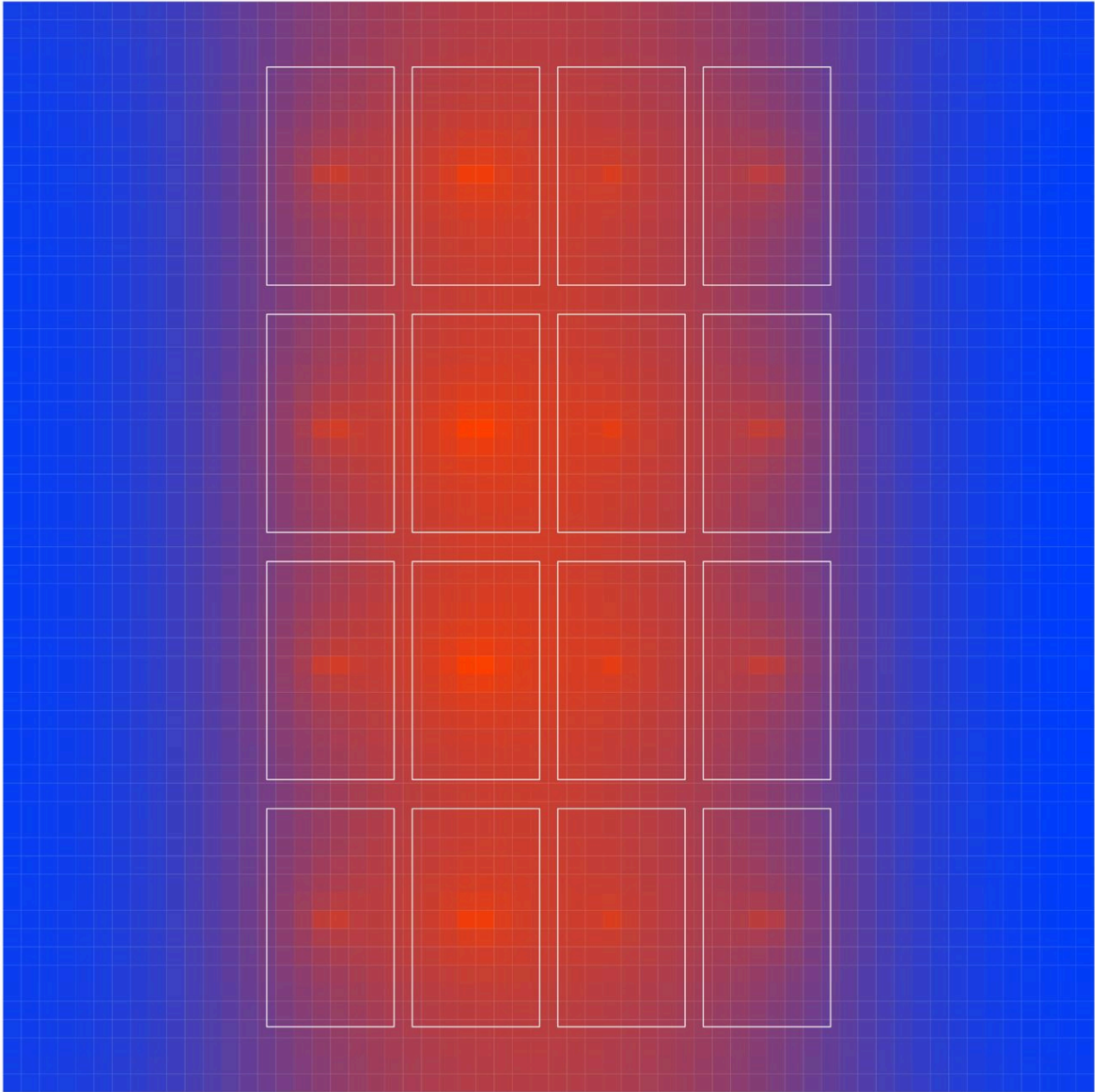
Model statement copied (source: engineering/DESIGN/CALCS/DESIGN_CALCULATIONS.md §1): "300 × 300 × 10 mm aluminium plate ($k = 200$ W/mK). 60 × 60 cells, steady state, 16 heat sources 5 × 5 mm at the PA-board centres (QPA2962 position on the PA board is assumed at

AERIS-10 — Engineering & Assembly Manual
the board centre verify in `engineering/PCB/RF_PA`), two fin strips as an effective convective sink ($h_{\text{eff}} \approx 288 \text{ W/m}^2\text{K}$ over the strip footprint = $25 \text{ W/m}^2\text{K} \times \text{fin area ratio} \times 0.9 \text{ efficiency}$), $5 \text{ W/m}^2\text{K}$ elsewhere; ambient 45°C . Conduction through the PA PCB via field and the thermal pad ($\approx 2^\circ\text{C/W} \rightarrow +8^\circ\text{C}$ in case B) is added analytically."

Case	Per PA	Plate max / min ($^\circ\text{C}$)	PA base estimate ($^\circ\text{C}$)	Verdict
A continuous drain bias	37.0 W	213 / —	> 287	far beyond TBASE 85°C — confirms case A is infeasible
B drain gated per chirp	4.24 W	64.3 / 55.3	≈ 73	below 85°C with margin 12°C

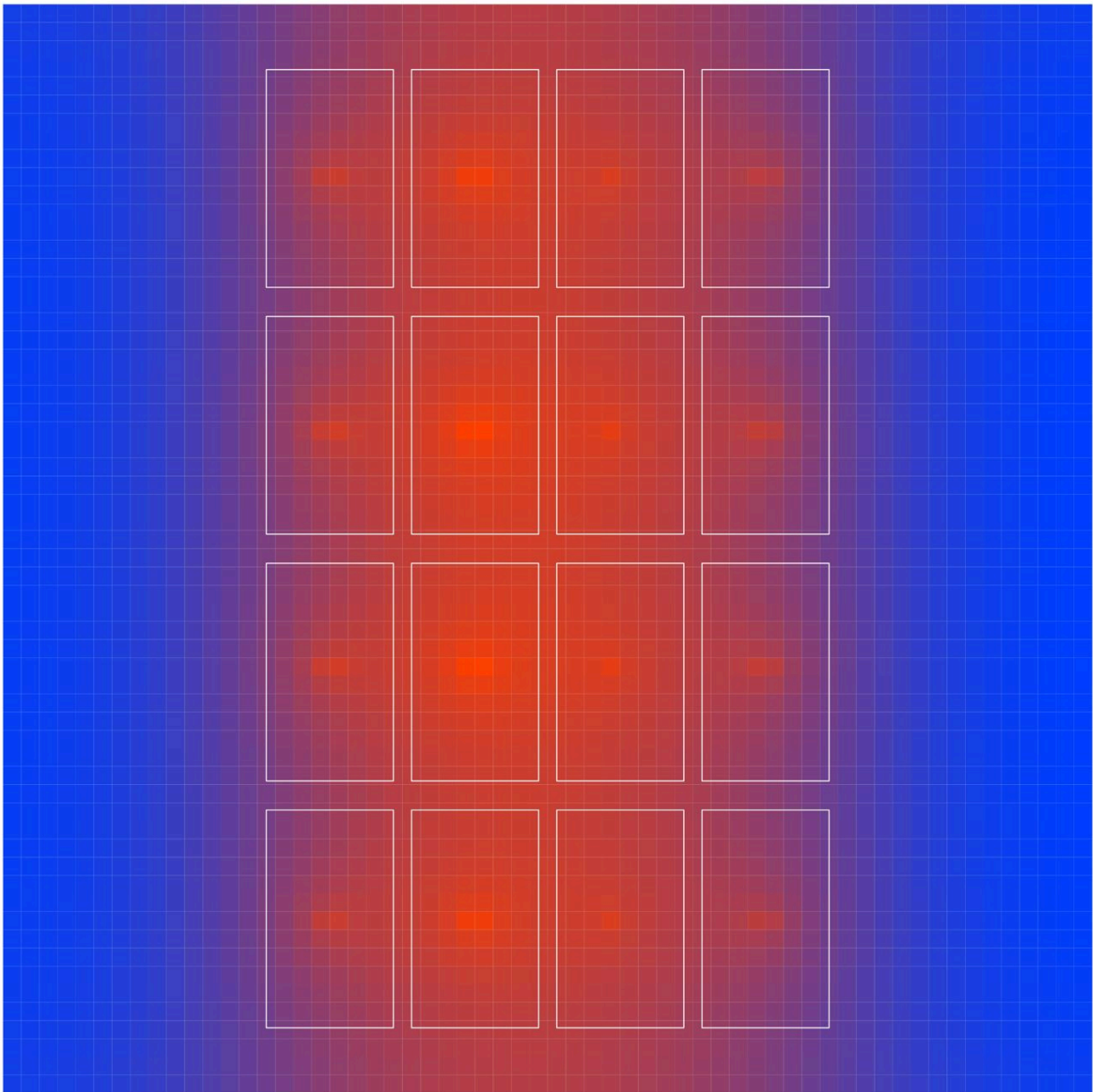
Limits stated by the source: "2-D (through-thickness gradient of a 10 mm plate at these fluxes $< 1^\circ\text{C}$), uniform h on the strips, no radiation, no PCB/antenna heat"; both maps "converged in 3999/3999 iterations" (source: same section). Case B is the design case of D-10 and requires the per-pulse drain gating of D-14, which is not in any CAD file today (K4).

AERIS-10 — PA heat-spreader temperature map, 2-D finite differences — DSN-CALC-01 (B-gated-drain)
Date 2026-10-09 · PROPOSED DESIGN calculation · plate $300 \times 300 \times 10 \text{ Al}$ $k=200 \text{ W/mK}$, 60×60 cells, $16 \times 5 \times 5 \text{ mm}$ sources at the PA board centres, fin strips $h_{\text{eff}} 288 \text{ W/m}^2\text{K}$, elsewhere $5 \text{ W/m}^2\text{K}$. Conduction through the PA board and TIM ($\approx 2^\circ\text{C/W} \rightarrow +8^\circ\text{C}$ at the die base in case B) is NOT included in the map; see `THERMAL_AND_PA_SUPPLY.md`



T min 55.3°C (blue) ... max 64.3°C (red); ambient 45°C ; B-gated-drain

Figure 52 — F10.9a — PA heat-spreader temperature map, case B (drain gated per chirp, 4.24 W per PA): T min 55.3°C ... max 64.3°C at 45°C ambient; white rectangles = PA board outlines in the 4×4 field; conduction through PA board and TIM ($+8^\circ\text{C}$) not included in the map — DSN-CALC-01 — PROPOSED DESIGN (calculated), BETA (source: `engineering/DESIGN/CALCS/thermal_map_B-gated-drain.svg`; produced by `tools/design_calcs.py`, rendered to PNG by `python3 tools/svg_sheets_to_pdf.py -o manual/figures/thermal_map.pdf --png-dir manual/figures ...`)



T min 134.9 °C (blue) ... max 212.9 °C (red); ambient 45 °C; A-continuous-bias

Figure 53 — F10.9b — PA heat-spreader temperature map, case A (continuous drain bias, 37.0 W per PA): plate max 213 °C, PA base > 287 °C — shown only to document why continuous bias is infeasible in the sealed head — DSN-CALC-01 — PROPOSED DESIGN (calculated), BETA (source: engineering/DESIGN/CALCS/thermal_map_A-continuous-bias.svg; produced by tools/design_calcs.py, rendered by tools/svg_sheets_to_pdf.py)

10.8 Unresolved geometry register (VAL-GEO-01)

Copied (source: engineering/VALIDATION/UNRESOLVED_GEOMETRY.md): "Every mechanical quantity that the drawings need but no repository file provides. Nothing below was estimated into a drawing; where a planning estimate exists it is labelled ESTIMATE in the source document."

ID	Quantity	Needed by	Available evidence	Missing	Status
G-01	PCB thickness (all 4 boards)	STEP bodies, stack-up, stand-offs	none (KiCad assumed 1.6 mm)	vendor stack-up / designer decision	BLOCKED — MISSING DATA
G-02	Component heights (both sides)	enclosure clearances, stand-off heights, exploded view	packages only; 3-D URNs not embedded	3-D models or measurement of assembled boards	BLOCKED
G-03	Board masses (bare and assembled)	pedestal/motor sizing, mass table	area × assumed thickness (ESTIMATE in MECHANICAL/dimensions/)	weighing or CAD with densities	BLOCKED (estimate only)

Item	Quantity	Needed by	Available evidence	Missing	Status
G-04	Enclosure: envelope, wall thickness, mounting bosses, connector cut-outs	ASM-EXP-01, M1-M6	README text references only (10_docs/Hardware/Enclosure absent)	enclosure CAD	BLOCKED
G-05	Relative board placement and stacking height	internal layout drawing	none	layout decision	BLOCKED (CONCEPTUAL drawing only)
G-06	Antenna element geometry, substrate, feed network, radome	antenna assembly drawing	02_hardware/04_antenna_beamforming.md ($\lambda/2 = 14.3$ mm, aperture 214.3 mm), 8_Utils/Antenna_Array.jpg (undimensioned photo), two contradictory waveguide simulations	antenna CAD and measurements	BLOCKED
G-07	Pedestal, stepper, slip ring, bearing envelope	pedestal drawing	firmware constants only (200 steps/rev, 50 positions)	mechanical design	BLOCKED
G-08	Heatsink / fan geometry and airflow	cooling drawing	PA dissipation implied by xlsx IDQ	thermal design	BLOCKED
G-09	Cable lengths, routing, bend radii (SMA coax $\times 34$, Molex $\times \sim 50$)	harness drawing	connector list only	harness design	BLOCKED
G-10	Power Board final outline (hole pattern suggests a smaller board than 280×300)	fab drawing	POWER_SUPPLY_dimensions.md (holes at inner positions)	designer confirmation	UNRESOLVED
G-11	Fastener sizes ($\varnothing 3.2$ holes $\rightarrow$ M3 inferred)	parts list	hole diameters	specification	UNRESOLVED (inferred)
G-12	Keep-out zones around RF connectors and the QPA2962	layout/enclosure	none	designer input	BLOCKED

Update recorded in the same register: "G-04, G-05, G-07, G-08, G-09, G-12 now have PROPOSED values in engineering/DESIGN/ (D-07... D-15); they stay open until the owner accepts the decisions and the assumed inputs (G-01 thickness, G-02 component heights, G-11 fasteners) are measured/specified. ... G-06 antenna: PROPOSED patch panel 165×248 mm; G-10 Power Board outline unchanged" (source: engineering/VALIDATION/UNRESOLVED_GEOMETRY.md, "Update 2026-10-09 — proposals"). Photographs in 8_Utils/ "show a prototype but carry no scale reference ... they were **not** used to derive dimensions" (same file).

10.9 Regenerating this chapter's artefacts

All drawings read one parameter file, so a changed decision regenerates every drawing (source: engineering/DESIGN/00_DESIGN_BASIS.md, preamble; engineering/DESIGN/README.md, "Regenerate"):

```
python3 tools/design_layout.py # prints the station table of §10.3
python3 tools/design_mechanical_drawings.py # DSN-MECH-01...05 SVG/PDF/PNG + harness schedule
python3 tools/design_calcs.py # DSN-CALC-01 (+ thermal_map*.svg)
python3 tools/design_enclosure_drawings.py # DSN-MECH-06/07 + MECHANICAL_PARTS_LIST.md
~/Applications/FreeCAD.app/Contents/Resources/bin/freecadcmd -c "exec(open('tools/design_mechanical_freecad.py').read())"
~/Applications/FreeCAD.app/Contents/Resources/bin/freecadcmd -c "exec(open('tools/design_enclosure_detail_freecad.py').read())"
python3 tools/svg_sheets_to_pdf.py -o manual/figures/thermal_map.pdf --png-dir manual/figures \
    engineering/DESIGN/CALCS/thermal_map_B-gated-drain.svg engineering/DESIGN/CALCS/thermal_map_A-continuous-bias.svg
python3 tools/gen_drawing_register.py --check # 89 drawings, file check 0 problems (run 2026-10-09)
```

Executed for this edition: python3 tools/design_layout.py (exit 0, values quoted in §10.3), the svg_sheets_to_pdf.py render of the two thermal maps (2 pages, PNGs written to manual/figures/), and python3 tools/gen_drawing_register.py --check ("register: 89 drawings ... file check: 0 problems"). The FreeCAD generators were not re-run for this edition; their existing exports were used as found.

<!-- chapter 11: FPGA design: module hierarchy, build, constraints, tests, synthesis status -->

FPGA design — module hierarchy, build, constraints, tests, synthesis status

Author: Antidrone Ukraine · antidrone.cc

Status summary: the original RTL (9_Firmware/9_2_FPGA/, ORIGINAL PROJECT FILE) does not elaborate; the BETA project beta/fpga/ parses, lints and simulates with open-source tools (BETA: 0 build failures, 9 testbench runs PASS), is **not synthesised with Vivado**, has no timing closure, has not been loaded on hardware, and its two Xilinx FFT IP cores are not generated. An open-source Yosys/nextpnr estimate exists (beta/fpga_synth/, OPEN-SOURCE ESTIMATE, not a Vivado result; placement did not complete). Figures: F11.1 original hierarchy SOURCE-DERIVED; F11.1b BETA hierarchy SOURCE-DERIVED / BETA (regenerated for this manual); F11.2 original pipeline PARTIAL.

Sources: beta/fpga/README.md, beta/fpga/CHANGELOG.md, beta/fpga/ip/README.md, beta/fpga/constraints/radar_system_top_beta.xdc, beta/fpga_synth/README.md, docs/FPGA/FPGA_PROJECT_RECONSTRUCTION.md, engineering/SOFTWARE_DIAGRAMS/FPGA/fpga_module_hierarchy.md, manual/figures/fpga_beta/fpga_module_hierarchy.md (generated), tools/check_fpga_constraints.py (executed).

Planned figures: F11.1 module hierarchy (SD-01, original), F11.1b module hierarchy of beta/fpga/rtl, F11.2 signal-processing pipeline (SD-02).

1. Starting point — the original RTL as committed

Bottom line of the reconstruction (source: docs/FPGA/FPGA_PROJECT_RECONSTRUCTION.md, header): "the FPGA sources cannot be elaborated by any tool today. Before a Vivado project can even reach synthesis, four user modules and two Xilinx FFT IP configurations must be recovered, three syntax/declaration defects must be fixed, and the device part number must be confirmed (schematic says XC7A50T-2FTG256I; README/XDC say XC7A100T). Pin assignment can be reconstructed for 64 of 67 top-level ports from the Main Board schematic, but 25 ports (FT601 USB 3.0, status and debug outputs) have no board counterpart because the FT601 was never wired."

Device identification (source: docs/FPGA/FPGA_PROJECT_RECONSTRUCTION.md §1, copied verbatim):

Source	Device statement	Evidence
Main Board schematic	XC7A50T-2FTG256I, package BGA256C100P16X16_1700X1700X155	4_Schematics and Boards Layout/4_6_Schematics/MainBoard/RADAR_Main_Board.sch part U42, library/deviceset XC7A50T-2FTG256I
Power budget workbook	XC7A50T-2FTG256	3_Power Management/Power Management V6.xlsx (sharedStrings entry 25)
Block diagram	XC7A50T-2FTG256	2_Functional Diagram & Interconnection Matrices/RADAR_V6.drawio
Constraint file	# Device: [XC7A100T] (placeholder brackets, no package, no speed grade)	cntrt.xdc:4
README / hardware docs	"XC7A100T FPGA"	README.md:52; 02_hardware/05_fpga_board.md:1,15,30,34,262

The source concludes that the only CAD evidence is XC7A50T-2FTG256I (Vivado part string xc7a50tftg256-2); whether the design fits a 50T (52 160 logic cells, 120 DSP48E1, 2 700 Kbit BRAM per AMD DS180) is UNRESOLVED until a real synthesis run (conflict K1).

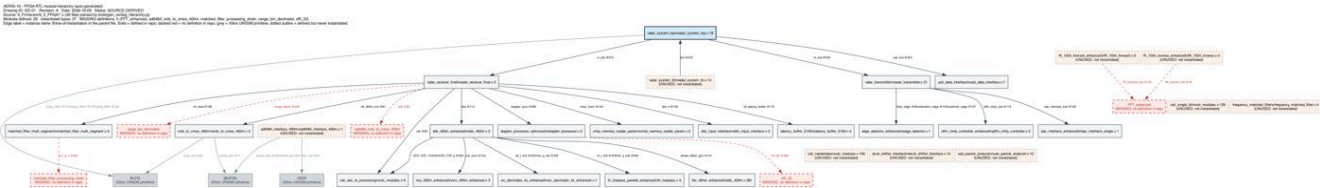


Figure 54 — F11.1 — FPGA RTL module hierarchy of the original 9_Firmware/9_2_FPGA (SD-01): 19 defined, 5 missing, 3 primitives, 9 unused — SOURCE-DERIVED, lexical scan not elaboration (source: engineering/SOFTWARE_DIAGRAMS/FPGA/fpga_module_hierarchy.dot; produced by tools/gen_verilog_hierarchy.py and Graphviz dot)

Hierarchy counts of the original (source: engineering/SOFTWARE_DIAGRAMS/FPGA/fpga_module_hierarchy.md): 26 Verilog files parsed; DEFINED 19, MISSING 5 (fft_enhanced, xfft_32 — Xilinx IP without .xci; ad9484_lvds_to_cmos_400m, matched_filter_processing_chain, range_bin_decimator — RTL never committed), PRIMITIVE 3 (BUFG, IBUFDS, IDDR), UNUSED 9 (ad9484_interface_400m, cdc_handshake, cdc_single_bit, fft_1024_forward_enhanced, fft_1024_inverse_enhanced, frequency_matched_filter, level_shifter_interface, usb_packet_analyzer, the testbench root).

Static verification actually performed on the original (source: docs/FPGA/FPGA_PROJECT_RECONSTRUCTION.md §4, copied verbatim; tools Icarus Verilog 13.0, Verilator 5.052):

Run	Command (abridged)	Exit	Result
A	iverilog -g2012 -s radar_system_top on all .v except testbench, including chirp_lut_init.v	2	chirp_lut_init.v:6: syntax error (initial outside a module)

Run	Command (abridged)	Exit	Result
B	same without chirp_lut_init.v	2	radar_system_top.v:312: syntax error / Syntax in assignment statement l-value
C	run B + -I 9_Firmware/9_2_FPGA	2	identical (never reaches \$readmemh)
D	iverilog -g2012 -s radar_system_tb	20	18 SVA syntax errors radar_system_tb.v:528-543 + error B
lint	verilator --lint-only -Wall -Wno-fatal --top-module radar_system_top	1	radar_system_top.v:312:13: syntax error, unexpected wire; 23 EOFNEWLINE
E-J (scratch copy only, repo untouched)	after hoisting the wire at:312, wire→reg at :155-156, removing .ref_i/.ref_q, and stubbing the missing modules	10/11	iverilog: "modules were missing: BUFG, IBUFDS, ad9484_lvds_to_cmos_400m, matched_filter_processing_chain, range_bin_decimator, xfft_32"; then use-before-declare errors ddc_400m.v:252-266 , radar_receiver_final.v:150 , radar_system_tb.v:329 . Verilator with stubs: 0 errors, 207 warnings
sim	vvp	not run	no executable could be produced

"No simulation or synthesis test PASSED. All results above are FAIL or informational." (source: same). Top-level ports with no board counterpart (source: docs/FPGA/FPGA_PROJECT_RECONSTRUCTION.md §3.2, copied verbatim):

Ports	Bits	Reason
dac_clk	1	RTL forwards clk_120m to a pin (dac_interface_single.v:23); the board clocks the AD9708 from AD9523 OUT10 via SMA J20 (main.cpp:1019-1020); no FPGA net
ft601_clk_in , ft601_data[31:0] , ft601_be[1:0] , ft601_txe_n , ft601_rxf_n , ft601_txe , ft601_rxf , ft601_wr_n , ft601_rd_n , ft601_oe_n , ft601_siwu_n , ft601_srb[1:0] , ft601_swb[1:0] , ft601_clk_out	49	FT601Q-B-T (U6) is placed on the schematic with 0 of 77 pins connected; decoupling parts parked outside the outline; no FPGA pin carries a USB net. The RTL's USB 3.0 path has no hardware
current_elevation[5:0] , current_azimuth[5:0] , current_chirp[5:0] , new_chirp_frame	19	no status nets on the schematic
dbg_doppler_data[31:0] , dbg_doppler_valid , dbg_doppler_bin[4:0] , dbg_range_bin[5:0] , system_status[3:0]	48	debug outputs; no nets

Schematic FPGA nets with no RTL port: ADC_0R_P/N (M6/N6, over-range), FPGA_ADC_CLOCK_P/N (N11/N12, 400 MHz LVDS from AD9523 OUT5 — the RTL uses the ADC DCO instead), FPGA_CLOCK_TEST (H14, 20 MHz test clock), DIG_5..7 (H11/G12/H12, STM32 PD13..PD15 configured as inputs), QSPI flash FPGA_FLASH_* (bank 14, MT25QL01G), JTAG (source: same §3.2). The original cntprt.xdc has 140 PACKAGE_PIN [PIN_NUMBER*] placeholders, LVCMOS33 on 1.8 V bank-34 signals, LVDS_25 + DIFF_TERM on a 3.3 V bank, no set_clock_groups , and no CONFIG_VOLTAGE / CFGBVS (source: same §3.3). The pipeline as written is drawn in F11.2 (chapter 2 §5.1 lists the stage-by-stage findings).

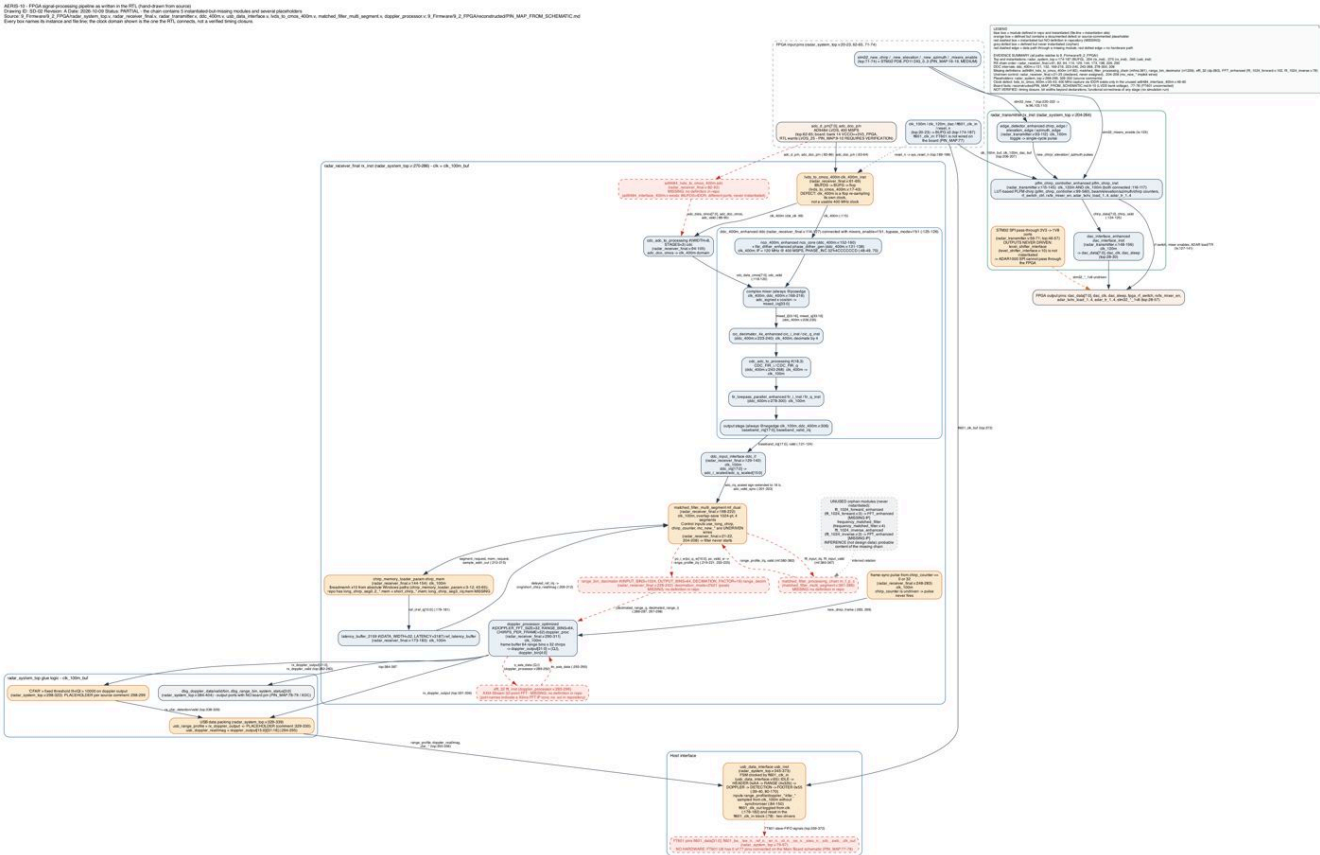


Figure 55 — F11.2 — FPGA signal-processing pipeline as written in the original RTL (SD-02) — PARTIAL; 5 instantiated-but-missing modules, placeholders and undriven controls marked (source: engineering/SOFTWARE_DIAGRAMS/FPGA/fpga_data_pipeline.dot; produced by Graphviz dot -Tpng -Gdpi=150)

Step	Command (from build.sh)	Result	Log
6b	tb_matched_filter - chain + memory, reference chirp delayed 300 samples	PASS: peak at bin 302 (300 +/-4), peak/sidelobe 4.76	logs/tb_matched_filter.run.log
6c	tb_range_bin_decimator - 2 x 1024 bins (one with input gaps) vs numpy, peak mode	PASS: 128/128 bins	logs/tb_range_bin_decimator.run.log
6d	tb_system_smoke - full top, 10 chirps, synthetic IF echoes, 3.3 ms (see 6i for the two capture modes)	PASS (~65 s each): see below	logs/tb_system_smoke_mode1.run.log
6e	tb_host_bridge - HOST_LINK unit test (packer + SPI slave): 3 detections, 32 detections (det_wr regression), command set v2 against a 4-word register model	PASS: 2075- and 2162-byte frames, CRC ok; v2 write/read-back/status/unknown ok	logs/tb_host_bridge.run.log
6f	tb_host_bridge_top - option B bridge through the top: v2 register commands (CFAR_THR 10000 -> 150 written over SPI, 5 read-backs, 0x04 status before/with/after a frame, 0xEE), then the SPI master reads one 64x32 frame after DRDY	PASS (~77 s): 2162-byte frame, 32 detections (only possible because the threshold write took effect), sync/dims/az-el/chirp-count/map/CRC ok, ADAR pass-through gated	logs/tb_host_bridge_top.run.log
6g	tb_adc_iserdes_capture - ISERDES capture + IDELAY calibration (pattern and blind) + FIFO	PASS (~30 s): pattern method 8/8 lanes locked, centre tap 13, skewed lane 5 -> 5 (8 taps = 0.6 ns compensated), bit-slip detected/realigned, 4000 samples exact; blind method on a 120 MHz tone + 0.7 LSB rms noise with lane 2 parked on a metastable tap and lane 0 mis-framed: 8/8 lanes locked, every lane within +/-0 taps of the pattern centre, framing restored, 4000 tone samples exact	logs/tb_adc_iserdes_capture.run.log
6h	tb_ddc_4x - legacy 400 MHz DDC vs polyphase DDC, same input	PASS: 1855 outputs, max diff 0 LSB	logs/tb_ddc_4x.run.log
6i	tb_system_smoke runs twice: ADC_CAPTURE_MODE=1 (default) and =0	both PASS	logs/tb_system_smoke_mode1.run.log , _mode0

build.sh summary line: 0 failure(s), 324 verilog warning line(s) (both views) — 165 synth view + 159 sim view; 219 for the first beta, 241 after the host link, 324 after the ISERDES capture path, unchanged 324 after command set v2 + blind calibration (the only new lint findings were two BLKSEQ on block-local temporaries in the sim-only ISERDESE2 model, marked as intended) (source: beta/fpga/README.md).

System smoke test evidence (source: beta/fpga/README.md, logs/tb_system_smoke.run.log): DAC left mid-scale (57540 samples); 40 range profiles (4 segments x 10 chirps); for alternating echo delays of 100 and 420 baseband samples the segment-0 peak sat at decimated bin 8 and 28 on every chirp — a shift of 20 bins for a 320-sample delay change (320/16 = 20), i.e. pulse compression works end to end through capture → DDC → matched filter → decimator; 2048 Doppler outputs (one 64 x 32 frame); 2688 USB packets with 0 header/footer/sequence errors; no X on outputs; the matched-filter FSM was idle at every toggle.

Verilog warning breakdown (synthesis view, 165; source: beta/fpga/README.md): 59 PINCONNECTEMPTY (monitor/unused IP outputs left unconnected on purpose), 32 UNUSED SIGNAL (diagnostic nets of the original, unused IP steady/tlast), 22 WIDTHTRUNC / 16 WIDTHEXPAND (original arithmetic widths, the copied host-link modules, the UNISIM models), 13 PROCASSINIT + 2 BLKSEQ + 1 ZERODLY (simulation-only UNISIM models in rtl/sim/), 9 DECLFILENAME (original file names differ from module names — kept so the docs stay valid), 7 UNUSEDPARAM, 3 GENUNNAMED (original generate), 1 CMPCONST (host_bridge_spi). None was hidden with a global -Wno-*.

Additional checks (source: beta/fpga/README.md; tools/check_fpga_constraints.py re-executed for this manual on 2026-10-09 with identical output): tools/check_fpga_constraints.py --top beta/fpga/rtl/radar_system_top.v --xdc beta/fpga/constraints/radar_system_top_beta.xdc → 0 placeholders, 0 invalid properties, 0 unknown port refs, 67/183 port bits constrained (the 116 unconstrained bits are exactly the UNRESOLVED ports of section 7); tools/gen_verilog_hierarchy.py --rtl beta/fpga/rtl → 0 missing RTL modules, the only "missing" names are the two deliberate IP placeholders.

4. Engineering findings recorded in the BETA (source: beta/fpga/README.md, "What was found and decided")

- The reference memories are frequency-domain matched-filter coefficients.** long_chirp_seg{0,1,2}_{i,q}.mem = conj(FFT_1024(u_s)) * 31128/max, where u is a unit-amplitude 10 → 30 MHz linear up-chirp, 3000 samples at 100 MSPS (phase fit residual 0.0018 rad; regenerated to within 1 LSB by gen_chirp_mem.py); segment 3 lies beyond the chirp and is all zeros — the missing file. Consequence: matched_filter_processing_chain is the frequency-domain chain the original architecture implies, and the reference is multiplied **without** the extra conjugation (CONJUGATE_REF = 0) — with the original conjugating form neither the numpy model nor the RTL produces a compression peak. Open point for the designer: the same data equals the FFT of a 30 → 10 MHz down-chirp, so the actual baseband chirp direction decides whether CONJUGATE_REF must be 0 or 1. short_chirp_{i,q}.mem (50 words) matches neither a time- nor a frequency-domain chirp and cannot be used — the short-chirp path is UNRESOLVED (see also chapter 2 §2.2 and observation MAN-02).
- AD9484 output is SDR LVDS at the sample rate** (datasheet: "LVDS SDR output", data captured on the rising edge of the DCO, tSKEW ±0.07 ns). The legacy capture module (mode 0) uses the DCO as a 400 MHz clock (IBUFDS → BUFG) and an IDDR as a dual-edge sampler taking the falling-edge sample by default (CAPTURE_FALLING); this assumption must be confirmed by Vivado timing analysis and an ADC test pattern on hardware.

3. **Reference alignment by address, not by delay line.** The chain requests reference bin k when FFT output bin k appears, so the fixed 3187-cycle `latency_buffer_2159` is not needed; works with the behavioural model (latency 160) and with any IP latency.
4. **Clock-domain crossings:** `cdc_adc_to_processing` (Gray-coded data words) replaced by Gray-pointer FIFOs; a reset synchroniser per non-100 MHz domain; STM32 toggles synchronised in the clock domain that consumes them (TX: 120 MHz FSM, RX: 100 MHz).
5. **Register map** (`radar_control_regs`): the receiver's control inputs that were floating wires now have one driver with documented reset defaults; its write/read port is driven by the SPI bridge (command set v2); 5-bit word addresses 0x00..0x10, 16-bit registers; table in `engineering/DESIGN/HOST_LINK/HOST_LINK_DESIGN.md` §7 (kept identical to the RTL; reproduced in chapter 9 §3.3).

5. ADC capture and DDC front end (ADC_CAPTURE_MODE, default 1)

Source: `beta/fpga/README.md`, "ADC capture and DDC front end", table copied verbatim:

	mode 0 (legacy)	mode 1 (default)
Capture	<code>ad9484_lvds_to_cmos_400m</code> : IBUFDS -> BUFG (400 MHz global clock) -> IDDR as dual-edge sampler	<code>ad9484_iserdes_capture</code> : IBUFDS(DCO) -> BUFIO + BUFR/4; IDELAYE2 (VAR_LOAD) -> ISERDESE2 SDR 1:4 per lane; IDELAYCTRL on 200 MHz from <code>clk_gen</code> (MMCM); 32-bit <code>async_fifo</code> into <code>clk_100m</code>
DDC	<code>ddc_400m_enhanced</code> : NCO, mixer, 5-stage CIC at 400 MHz in fabric (timing closure unrealistic), FIFO, FIR	<code>ddc_4x_100m</code> : 4-phase NCO + 8 mixers + CIC as 16-tap FIR + the same FIRs, all at 100 MHz; bit-exact with the legacy path (<code>tb_ddc_4x</code> : max diff 0 LSB)
Calibration	none (<code>CAPTURE_FALLING</code> edge choice)	<code>adc_capture_calib</code> : default tap 16; manual tap/bitslip per lane; pattern method (<code>CAL_CTRL</code> bit4 = 0): auto IDELAY sweep with the ADC in a 2-code test pattern (register 0x0D = 0x48, P1/P2 = 0x19..0x1C = pattern A/B; or 0x04 checkerboard / 0x07 toggle), centre tap per lane, lane rotation alignment, lock/undetermined/align_fail status, pattern error counter; blind method (<code>CAL_CTRL</code> bit4 = 1, no ADC SPI needed): a CW tone at the IF on the live input, notch residual $r[n] = x[n] - 2\cos(w)x[n-1] + x[n-2]$ (<code>CAL_BLIND_COEF</code> = Q1.14 $\cos(w)$), metric = sum of abs(r) over 512 words per tap, one lane swept at a time MSB first, pass = metric <= min + <code>CAL_BLIND_MARGIN</code> + min/16, longest linear run (1-tap holes closed), eye truncated at tap 0 extrapolated with the 32-taps-per-bit period, blind framing alignment (best of 4 framings per lane) before and after each of 2 passes; <code>CAL_BLIND_MIN</code> read-back (register map 0x4..0x10)

Datasheet facts used (AD9484.pdf, re-checked): LVDS SDR, DCO at the sample rate (400 MHz), data valid on the rising DCO edge, tSKEW -0.07..+0.07 ns (tPD 0.85 / tCPD 0.6 ns typ), offset binary. Resource estimate for mode 1 (XC7A50T): 8 ISERDESE2 + 8 IDELAYE2 + 1 IDELAYCTRL (bank 14), 1 BUFIO + 1 BUFR, 1 MMCME2 (of 5) + 1 BUFG, 8 DSP48E1 for the mixers (9×16) + 64 DSP48E1 for the two unchanged 32-tap FIRs (of 120), CIC adder trees in LUTs (~16 constant multiplies per I/Q), 1 BRAM18 for the 32-bit FIFO (or distributed RAM), small ROM for the 65-entry sine table. The 400 MHz fabric path (mode 0) is kept only for comparison (source: same section).

Remaining risks / what to check in Vivado for mode 1 (source: same section):

- BUFR (DCO/4) versus `clk_100m` (AD9523 OUT6): same nominal frequency, unknown phase; the FIFO absorbs phase/jitter, `cal_status[15]` (FIFO overflow) must stay 0 on hardware. If the two clocks are not frequency-locked the design needs a re-sampler (not present).
- IDELAYCTRL placement: all IDELAYE2 and the IDELAYCTRL must be in bank 14 (`IDELAY_GROUP` set in the XDC); the 200 MHz reference comes from the MMCM through a BUFG.
- Bank 14 is 3.3 V: LVDS_25 inputs only with `DIFF_TERM FALSE` and external 100 Ω termination (design conflict kept visible in the XDC; section 6 item 3).
- ISERDESE2 Q1..Q4 bit order: resolved from UG471 (section 6) — `Q1_IS_OLDEST = 0` in RTL, receiver and TB, and the simulation model was corrected to the same order; still confirm on hardware with the ADC PN9 pattern (0x0D = 0x06) against a PN9 generator. BUFR framing is irrelevant for the data path and absorbed by the per-lane alignment.
- Blind calibration limits (`adc_capture_calib.v` header): single CW tone at the programmed frequency with every bit toggling ($\geq \sim 64$ LSB peak for bit 7); the LSB lanes need $\leq \sim 1$ LSB rms input noise at the default 512-word window (0.7 LSB rms in the TB; AD9484 ~ 47 dB SNR = ~ 0.5 LSB rms) — at 1.4 LSB rms lane 0 reports `undetermined` and keeps its tap; the eye-centre extrapolation for runs cut at tap 0 assumes exactly 32 taps per bit (400 MSPS, 78.125 ps/tap) and a 5-tap metastable region (`FAIL_HALF = 2`, the simulation model's 0.4 ns; the hardware value is unknown, an error of e taps moves the centre by e); the blind framing alignment takes the minimum of four metrics without a significance test (`align_fail` stays 0). Run time ~ 3.3 ms (2 passes $\times$ 8 lanes $\times$ 32 taps $\times$ 528 `clk_div` cycles + 3 alignments).
- Checks after implementation: `report_clock_interaction` (`clk_div` $\leftrightarrow$ `clk_100m` only through the FIFO, `adc_dco/clk_div/clk_200m` as derived clocks), `report_timing_summary` for the `clk_div` and `clk_200m` paths, `report_cdc`, `report_io`; on hardware: run the auto calibration with the ADC test pattern, read `CAL_STAT` / `CAL_LANE_INFO` per lane (windows should be ~ 27 of 32 taps wide at 400 MSPS), then switch to normal data and check `CAL_ERR` stays 0 while the pattern check is off.

6. UG471 documentary checks

UG471 v1.10 (`build/docs_ext/ug471.txt` = pdftotext of the web-archive copy, 2026-10-09) was used to settle two open points; the simulation model and the RTL default were changed where they disagreed (CHANGELOG "Command set v2, blind calibration, UG471") (source: `beta/fpga/README.md`, "UG471 checks", quotes copied verbatim):

- ISERDESE2 output order, section "ISERDESE2 Ports - Registered Outputs - Q1 to Q8" (p. 146): *"The first data bit received appears on the highest order Q output. The bit ordering at the input of an OSERDESE2 is the opposite of the bit ordering at the output of an ISERDESE2 block, as shown in Figure 3-3. For example, the least significant bit A of the word FEDCBA is placed at the D1 input of an*

OSERDESE2 but the same bit A emerges from the ISERDESE2 block at the Q8 output. In other words, D1 is the least significant input to the OSERDESE2, while Q8 is the least significant output of the ISERDESE2 block." (OSERDESE2 section, p. 161: "data on the D1 input pin is the first bit transmitted"). For the 1:4 SDR configuration the first (oldest) bit is therefore on **Q4**, the newest on **Q1** → **Q1_IS_OLDEST = 0**, **word[7:0] = Q4** (s0 = oldest). The sim model had Q1 = oldest and was corrected.

2. BITSLLIP in SDR mode, section "BITSLLIP Submodule - Bitslip Operation" (p. 158): "By asserting the Bitslip pin of the ISERDESE2 block, the incoming serial data stream is reordered at the parallel side. ... (Bit 8 of an input ISERDESE2 is the first bit received.) ... The Bitslip operation is synchronous to CLKDIV. In SDR mode, every Bitslip operation causes the output pattern to shift left by one. In DDR mode, every Bitslip operation causes the output pattern to alternate between a shift right by one and shift left by three." and "Although the repeating pattern seems to show that bitslip is a barrel shifting operation, this is not the case. A bitslip operation adds one bit to the input data stream and loses the nth bit in the input data stream." Figure 3-11 (SDR): initial 10010011 → after one Bitslip 00100111 (Q8..Q1 notation), i.e. the word boundary moves one bit later in the serial stream. "Guidelines for Using the Bitslip Submodule" (p. 159): "In NETWORKING mode the Bitslip submodule is available. ... the BITSLLIP port must be asserted High for one CLKDIV cycle. Bitslip cannot be asserted for two consecutive CLKDIV cycles; ... the total latency ... is two CLKDIV cycles. ... The user logic should wait for at least two CLKDIV cycles in SDR mode ... before analyzing the received data pattern" — `adc_capture_calib` pulses BITSLLIP for one cycle with ≥ 1 idle cycle between pulses and waits SETTLE = 16 cycles; the sim model's slip direction was corrected to Fig. 3-11.

3. LVDS_25 inputs in a 3.3 V HR bank, section "LVDS and LVDS_25 (Low Voltage Differential Signaling)" (p. 92): "It is acceptable to have differential inputs such as LVDS and LVDS_25 in I/O banks that are powered at voltage levels other than the nominal voltages required for the outputs of those standards (1.8V for LVDS outputs, and 2.5V for LVDS_25 outputs). However, these criteria must be met: - The optional internal differential termination is not used (`DIFF_TERM = FALSE`, which is the default value). - The differential signals at the input pins meet the VIN requirements in the Recommended Operating Conditions table of the specific device family data sheet. - The differential signals at the input pins meet the VIDIFF (min) requirements in the corresponding LVDS or LVDS_25 DC specifications tables of the specific device family data sheet. - For HR I/O banks in bidirectional configuration, internal differential termination is always used.", Figure 1-72 text: "RDIFF provides the 100 Ohm differential receiver termination because the internal `DIFF_TERM` is set to `FALSE`."; Table 1-55 note 1a: "The optional internal differential termination is not used (`DIFF_TERM = FALSE`, which is the default value) unless the VCCO voltage is at the level required for outputs."; "Differential Termination Attribute" (p. 49): "The VCCO of the I/O bank must be connected to 1.8V for LVDS, and 2.5V for the other differential I/O standards to provide 100 Ohm of effective differential termination. `DIFF_TERM` is only available for inputs and can only be used the appropriate VCCO voltage." → the XDC keeps `LVDS_25 + DIFF_TERM FALSE` on the nine bank-14 pairs and states that an external 100 Ω termination at the FPGA (absent on the schematic) and a VIN/VIDIFF/ICM check of the AD9484 output against DS181 are REQUIRED (board change / verification).

7. Constraints (beta/fpga/constraints/radar_system_top_beta.xdc, 348 lines)

Clocks as read by `tools/check_fpga_constraints.py` (executed 2026-10-09; source: tool output and the XDC lines quoted):

XDC line	Clock	Period	Port	Comment in the XDC
18	<code>clk_100m</code>	10.000 ns	<code>clk_100m</code>	AD9523 OUT6 "FPGA_SYSTEM_CLOCK"
19	<code>clk_120m_dac</code>	8.333 ns	<code>clk_120m_dac</code>	AD9523 OUT11 "FPGA_DAC"
20	<code>adc_dco</code>	2.500 ns	<code>adc_dco_p</code>	AD9484 DCO, 400 MHz SDR
23	<code>ft601_clk_in</code>	10.000 ns	<code>ft601_clk_in</code>	no pin (FT601 unwired)
28	<code>spi_sclk</code>	37.000 ns	<code>stm32_sclk_3v3</code>	27 MHz max (STM32 SPI1, APB2 108 MHz / 4)

Content of the file (source: `beta/fpga/README.md`, "Constraints"; XDC lines 36, 60–61, 70–71, 83, 96–98, 140–146):

- Pins: verbatim schematic-derived candidates for 64 ports / 64 bits (`PIN_MAP_FROM_SCHEMATIC.md`), every ball still to be cross-checked against the AMD FTG256 package file and the final layout.
- `set_clock_groups -asynchronous` for all clock groups (line 36); false paths for the asynchronous STM32 lines and the SPI pass-through; `set_input_delay` for the ADC bus from the datasheet skew plus an **unverified** ±0.25 ns board allowance.
- `IDELAY_GROUP adc_idelay_grp` on the IDELAYCTRL and the per-lane IDELAYE2 cells (lines 60–61).
- DAC output delays: UNRESOLVED placeholder — `dac_clk` has no FPGA pin, the AD9708 is clocked by AD9523 OUT10 (lines 70–71); STM32F7 SPI1 master timing versus SCLK at the FPGA pins UNRESOLVED (line 83).
- `CFGBVS VCC0 / CONFIG_VOLTAGE 3.3` from the bank-0 supply (lines 96–98; CFGBVS pull direction via R to be verified).
- **Design conflict kept visible:** bank 14 is 3.3 V on the schematic; the LVDS_25 ADC inputs (9 pairs: `adc_dco_p/n`, `adc_d_p/n[7:0]`) are constrained `IOSTANDARD LVDS_25 + DIFF_TERM FALSE` (option A — external 100 Ω termination REQUIRED, REQUIRES VERIFICATION; the UG471 criteria are quoted in the XDC header) and the `DIFF_TERM TRUE` lines are left commented as option B (bank VCCO change to 2.5 V, a board change) (lines 140–146 and following).
- Host-link option B pins H11/G12/H12 (`DIG_5..7`, bank 15, LVCMOS33, PULLUP on CS) added; `spi_sclk` is a fifth asynchronous clock group.
- UNRESOLVED (116 bits, listed at the end of the file, unconstrained): `dac_clk`, all `ft601_*` (FT601 U6 has 0/77 pins connected), `current_*`, `new_chirp_frame`, `dbg_*`, `system_status` (no board nets). `write_bitstream` will refuse the unconstrained I/O unless they are removed from the top or `UNCONSTRAINEDPINS ALLOW` is set for a resource/timing trial only.

8. Host path

Option B (SPI bridge on the STM32 SPI1 bus, command set v2, register map) is implemented in the BETA RTL and verified by `tb_host_bridge` and `tb_host_bridge_top`; option A (FT601) is kept unchanged as the original `usb_data_interface` packetiser running on `clk_100m` so the data path can be simulated — a real FT601 needs the FSM re-timed to `ft601_clk` through `async_fifo` and the D-19 changes (source: `beta/fpga/README.md`, "Host path"). Signals, frame format, command set and register map are in chapter 9 §3; the unresolved RTL items are: STM32 SPI1 timing versus the FPGA pins (XDC placeholders), BRAM inference of the SCLK-domain frame RAM read (`host_bridge_spi.v`, asynchronous read registered on falling SCLK), and the firmware rule that no ADAR1000 transaction overlaps a bridge read.

9. Xilinx FFT IP settings (not generated)

Status (source: `beta/fpga/ip/README.md`): IP **not generated** (no Vivado on the authoring machine). `rtl/xfft_32.v` and `rtl/FFT_enhanced.v` contain simulation-only behavioural models (`axis_fft_behav.v`) and, under SYNTHESIS, an instantiation of a deliberately missing module so that synthesis fails loudly until the IP exists; `vivado/create_project.tcl` imports `ip/<name>/<name>.xci` when present. Settings are derived from how the RTL drives the ports; rounding mode, latency and output order are recorded choices so that model and IP agree.

Common settings (source: `beta/fpga/ip/README.md`, copied verbatim):

Setting (Vivado FFT v9.1 GUI)	Value	Evidence
Vendor / library / version	xilinx.com:ip:xfft:9.1	docs/FPGA/FPGA_PROJECT_RECONSTRUCTION.md FPGA-T04
Number of channels	1	one <code>s_axis_data</code> stream
Architecture	Pipelined Streaming I/O	back-to-back frames assumed by <code>matched_filter_processing_chain.v</code> (queue) and continuous output in the model
Data format	Fixed point, Scaled	config word carries a scaling schedule (see below); block floating point would need <code>m_axis_data_tuser</code> BLK_EXP handling that the RTL does not have
Input data width	16	<code>s_axis_data_tdata[31:0] = {Q[15:0], I[15:0]}</code> (<code>fft_1024_forward.v:18</code> , <code>doppler_processor.v:289</code>)
Phase factor width	16	choice (model uses double precision; 16 keeps the +/-2 LSB tolerance of <code>tb_fft_wrappers.v</code> realistic - to be confirmed by running the TB against the IP's simulation model)
Rounding	Convergent rounding	choice; the model uses round-half-up (not bit exact, +/-1 LSB)
Output ordering	Natural order	<code>matched_filter_processing_chain.v</code> addresses the reference memory with a sequential bin counter
Cyclic prefix	none	not used
ACLKEN / ARESETn	ARESETn enabled (active low, >= 2 cycles)	<code>.aresetn(reset_n)</code>
XK_INDEX / OVFL0 in tuser	optional (not connected)	not used by the RTL

`xfft_32` (Doppler, `doppler_processor.v:283–296`; source: same, copied verbatim):

Setting	Value	Evidence
Module name	<code>xfft_32</code>	instance name in the RTL
Transform length	32	<code>DOPPLER_FFT_SIZE = 32</code> , <code>fft_input_last</code> after 32 samples
Throttle scheme	Non Real Time (has <code>m_axis_data_tready</code>)	<code>.m_axis_data_tready(1'b1)</code> is connected
Config word	8 bits: bit0 FWD_INV, bits[5:1] SCALE_SCH (2+2+1 bits for radix-4, radix-4, radix-2 stages)	<code>s_axis_config_tdata(8'h..)</code> ; RTL default <code>FFT_CONFIG_WORD = 8'h35</code> = forward, shift 2+2+1 = 2 ⁵
Transform direction	run-time configurable (forward used)	config bit0 = 1
Model latency (<code>axis_fft_behav</code> LATENCY)	72 clocks from the last input	placeholder; the real IP latency is reported by the GUI and must be below <code>FFT_WAIT_TIMEOUT</code> (1000) in <code>doppler_processor.v</code>

`FFT_enhanced` (pulse compression, `fft_1024_forward.v:102`, `fft_1024_inverse.v:78`; source: same, copied verbatim):

Setting	Value	Evidence
Module name	<code>FFT_enhanced</code>	"This must match the name in your project" (<code>fft_1024_forward.v:101</code>)
Transform length	1024	<code>tlast</code> after 1024 samples (<code>fft_1024_forward.v:58</code>), reference memory 1024 bins per segment
Throttle scheme	Real Time (no <code>m_axis_data_tready</code>)	the instantiations do not connect <code>m_axis_data_tready</code> ; the wrappers tie their internal <code>tready</code> to 1
Config word	16 bits: bit0 FWD_INV, bits[10:1] SCALE_SCH (2 bits x 5 radix-4 stages), upper bits padding	<code>s_axis_config_tdata[15:0]</code> ; forward default <code>SCALE_SCH = 10'h255</code> (shifts 1,1,1,1,2 = 2 ⁶), inverse <code>10'h155</code> (2 ⁵)
Transform direction	run-time configurable	same core used forward (<code>16'h...1</code>) and inverse (<code>16'h...0</code>)
Model latency	160 clocks (beta model value, the IP is ~2-3k clocks for 1024 points)	the chain tolerates any latency (reference fetched by address, frames queued)

The `create_ip / set_property -dict Tcl` for both cores is given in `beta/fpga/ip/README.md` (property names follow the xfft v9.1 `CONFIG.*` set; to be verified with `report_property`, not executed). Scaling-schedule rationale (source: same, "Scaling-schedule arithmetic"): a 1024-sample segment of the 10–30 MHz chirp occupies ~68 bins; with 2⁶ a full-scale input gives ~15.5k without saturation; the inverse

transforms 2¹⁵ puts the compressed peak at ~32k, i.e. the full 16-bit range; the Doppler FFT uses 2¹⁵ (= 1/N) so that any 16-bit input cannot overflow. Verification of a generated IP against the model: run `tb_fft_wrappers` with the IP simulation model (SYNTHESIS not defined), expect PASS at ±2 LSB (raise `TOL` and record the measured difference if the IP rounds differently), record the IP latency and check `FFT_WAIT_TIMEOUT`.

10. Change classes versus the original (beta/fpga/CHANGELOG.md)

Legend of the changelog: **SYN** syntax/elaboration fix, **BUG** functional defect fix, **ARCH** architecture/interface change (documented in the README), **NEW** file written for the beta, **CLEAN** lint/readability only. Counting the typed rows of its tables gives 28 ARCH, 28 BUG, 12 CLEAN, 11 NEW, 9 SYN (count made for this manual with `grep` over `beta/fpga/CHANGELOG.md`; the file has 242 lines). Files copied unchanged except line endings: `dac_interface_single.v`, `fir_lowpass.v`, `level_shifter_interface.v`, `ddc_input_interface.v`; `plfm_chirp_controller.v` LUT contents unchanged. Files moved to `rtl/unused_orig/` (not compiled): `ad9484_interface_400m.v`, `lvds_to_cmos_400m.v`, `cdc_modules.v`, `latency_buffer_2159.v`, `usb_packet_analyzer.v`; `chirp_lut_init.v` and `radar_system_tb.v` were not copied into `rtl/` (the original testbench is kept as `tb/radar_system_tb_original.v`, not run). Per-file sections exist for `radar_system_top.v`, `radar_receiver_final.v` (rewritten; instance names and chain order kept), `radar_transmitter.v`, `plfm_chirp_controller.v`, `edge_detector.v`, `ddc_400m.v`, `cic_decimator_4x_enhanced.v`, `nco_400m_enhanced.v`, `chirp_memory_loader_param.v`, `matched_filter_multi_segment.v`, `frequency_matched_filter.v`, `fft_1024_forward/inverse.v`, `doppler_processor.v`, `usb_data_interface.v`, plus three follow-up sections (host-link option B integration; ISERDES capture + polyphase DDC; command set v2, blind calibration, UG471 checks) (source: `beta/fpga/CHANGELOG.md`, section headings).

New files (source: `beta/fpga/CHANGELOG.md`, "New files", copied verbatim):

File	Content
<code>rtl/ad9484_lvds_to_cmos_400m.v</code>	AD9484 SDR LVDS capture (IBUFDS + BUFG + IDDR as dual-edge sampler, <code>CAPTURE_FALLING</code>), <code>ifdef SIM</code> behavioural model, reset synchroniser, <code>adc_valid</code> , <code>adc_pwn</code> from <code>pwn_req</code>
<code>rtl/matched_filter_processing_chain.v</code>	FFT1024 -> x conj(ref) -> IFFT1024 with address-driven reference fetch and segment queue
<code>rtl/range_bin_decimator.v</code>	1024 -> 64 bins, modes first/peak/mean/sum, <code>start_bin</code>
<code>rtl/xfft_32.v</code> , <code>rtl/FFT_enhanced.v</code>	IP-named AXI4-Stream wrappers: behavioural model under <code>ifndef SYNTHESIS</code> , fail-loud placeholder under <code>SYNTHESIS</code> (see <code>ip/README.md</code>)
<code>rtl/axis_fft_behav.v</code>	behavioural AXI4-Stream FFT (simulation only)
<code>rtl/async_fifo.v</code> , <code>rtl/reset_synchronizer.v</code> , <code>rtl/radar_control_regs.v</code>	CDC FIFO, reset synchroniser, register map
<code>rtl/sim/unisim_sim_models.v</code>	BUFG / IBUFDS / IDDR models for simulation and lint (excluded from Vivado)
<code>mem/long_chirp_seg3_{i,q}.mem</code>	generated (all zero - segment beyond the 3000-sample chirp), <code>gen_chirp_mem.py</code>
<code>tb/tb_*.v</code> , <code>tb/gen_vectors.py</code> , <code>tb/vectors/</code>	self-checking testbenches and numpy references
<code>constraints/radar_system_top_beta.xdc</code> , <code>vivado/*.tcl</code> , <code>ip/README.md</code> , <code>build.sh</code>	constraints, Vivado scripts (not executed), IP settings, open-source flow

The follow-ups added `rtl/host_bridge_spi.v`, `rtl/rd_map_packer.v` (with the `det_wr` width fix), `tb/tb_host_bridge.v`, `tb/tb_host_bridge_top.v`, `rtl/ad9484_iserdes_capture.v`, `rtl/adc_capture_calib.v`, `rtl/ddc_4x_100m.v`, `rtl/clk_gen.v`, `tb/tb_adc_iserdes_capture.v`, `tb/tb_ddc_4x.v`, extended `rtl/sim/unisim_sim_models.v` (BUFIO, BUFR, IDELAYE2, IDELAYCTRL, ISERDESE2, MMCME2_BASE), and one legacy bug fix in `ddc_400m.v` (NCO `phase_valid` gated by `adc_data_valid`) (source: `beta/fpga/CHANGELOG.md`, the three follow-up sections).

11. Synthesis without Vivado — open-source estimate (beta/fpga_synth/)

Status: OPEN-SOURCE ESTIMATE. Not a Vivado result. Every number comes from Yosys (`synth_xilinx -family xc7`) and, where stated, `nextpnr-xilinx`; Vivado's synthesis, retiming, DSP/BRAM inference and timing models differ; nothing here replaces `beta/fpga/vivado/build.tcl` (source: `beta/fpga_synth/README.md`, header).

What was synthesised (source: same, "What was synthesised"): git snapshot **65cd160** of `beta/fpga` (the working tree had uncommitted edits by another session — undriven `host_bridge.reg_rdata`, `calib.blind_*`, `lane_metric`, implicit `inherit_tap/inherit_valid` — so the committed snapshot was used; **re-run REV=HEAD ./run_synth.sh after those edits are committed**, which the command-set-v2 follow-up has since done in `beta/fpga` but the synthesis directory has not been re-run at the time of writing); file list = `build.sh` step 2 minus `rtl/axis_fft_behav.v` and `rtl/sim/unisim_sim_models.v`; top `radar_system_top`, default parameters (`ADC_CAPTURE_MODE = 1`); part `xc7a50tfg256-2`; the two FFT IP cores declared (`blackbox`) — 3 instances excluded from the counts. Tools: Yosys 0.69+post (Homebrew), `nextpnr-xilinx bc9b234` (openXC7, archived fork), `prjxray-db a90f27c`, chip database `xc7a50t.bin` 93 MB, `prjxray tools c9f02d8`.

Utilisation (source: `beta/fpga_synth/README.md`, copied verbatim; budget XC7A50T per AMD DS180 Table 3: 8,150 slices = 32,600 LUT / 65,200 FF, 120 DSP48E1, 75 × 36 kb BRAM, 5 CMT):

Resource	Yosys count	XC7A50T budget	Utilisation
LUT total (logic + distributed RAM + SRL)	12793	32,600	39.2 %
of which logic LUT1..LUT6	9693	-	-

Resource	Yosys count	XC7A50T budget	Utilisation
of which distributed RAM (RAM64M/RAM32M x 4 LUTs)	3100	-	-
of which shift register (SRL16E/SRLC32E)	0	-	-
INV cells (absorbed into LUTs by P&R, not counted)	246	-	-
Flip-flops (FDRE/FDSE/FDCE/FDPE, incl. <code>_1</code> falling-edge)	4726	65,200	7.2 %
Latches (LDCE/LDPE)	0	-	-
DSP48E1	106	120	88.3 %
RAMB36E1	0	-	-
RAMB18E1	2	-	-
BRAM in 36 kb equivalents (RAMB36 + RAMB18/2)	1	75	1.3 %
CARRY4	333	8,150	4.1 %
MUXF7 / MUXF8	0	-	-
BUFG / BUFGCTRL	4	32	12.5 %
BUFIO / BUFR	2	-	-
MMCME2_BASE/ADV, PLLE2	1	5	20.0 %
IBUF / IBUFG / IBUFDS	31	-	-
OBUF / OBUFT / IOBUF	143	-	-
IDELAYE2 / IDELAYCTRL	9	-	-
ISERDESE2 / IDDR	8	-	-
FFT IP black boxes (not counted above)	3	-	-

Per module (source: same, hierarchical run, local cells only; `fir_lowpass_parallel_enhanced` is instantiated twice so its 32 DSP count twice; flat total $36 + 2 \times 32 + 4 + 2 = 106$ DSP):

Module (hierarchical run, -noflatten)	LUT logic	LUT as RAM	FF	DSP48E1	RAMB18E1
<code>host_bridge_spi</code>	809	1536	85	0	0
<code>plfm_chirp_controller_enhanced</code>	2287	0	52	0	0
<code>doppler_processor_optimized</code>	630	1536	128	2	0
<code>chirp_memory_loader_param</code>	1847	0	33	0	0
<code>adc_capture_calib</code>	961	0	572	0	0
<code>ddc_4x_100m</code>	889	0	1118	36	0
<code>rd_map_packer</code>	610	0	726	0	0
<code>range_bin_decimator</code>	318	0	180	0	0
<code>matched_filter_multi_segment</code>	177	0	76	0	2
<code>fir_lowpass_parallel_enhanced</code>	129	0	614	32	0
<code>frequency_matched_filter</code>	126	0	292	4	0
<code>radar_control_regs</code>	84	0	61	0	0
<code>radar_system_top</code>	63	0	98	0	0
<code>usb_data_interface</code>	61	0	118	0	0
<code>async_fifo</code>	22	24	89	0	0
<code>matched_filter_processing_chain</code>	41	4	57	0	0
<code>ad9484_iserdes_capture</code>	7	0	38	0	0
<code>ddc_input_interface</code>	4	0	36	0	0

The four design consequences (source: `beta/fpga_synth/README.md`, "Reading the numbers"):

- DSP48E1 is the binding resource: 106 of 120 (88 %) without the FFT IP.** 64 are the two 32-tap fully parallel FIRs (`fir_lowpass.v`), 36 the polyphase DDC (`ddc_4x_100m.v`: 8 mixers + the CIC-as-FIR), 4 the matched-filter complex multiply, 2 the Doppler window. Only **14 DSP48E1 remain for three FFT cores** ($2 \times 1024\text{-point} + 1 \times 32\text{-point}$, Pipelined Streaming, 16-bit). The `xfft v9.1` DSP count must be read from the Vivado IP GUI before anything else; if it exceeds 14 the design does not fit the XC7A50T as written (options: FIR symmetry folding / time-multiplexing at 100 MHz vs. 25 MSPS output, Radix-2 Lite / Burst I/O FFT architecture, or LUT multipliers via `use_dsp = "no"`). Not decided.
- Block RAM is almost unused ($2 \times$ RAMB18) because the RTL memories cannot map to BRAM**, not because the design is small: `chirp_memory_loader_param.v:32-35` (`ram_style = "block"`) muxes the read data before the output register, which has an asynchronous reset $\rightarrow$ no synchronous read port $\rightarrow$ ~1,850 LUTs of logic; `doppler_processor.v:74-75` feeds the multiplier combinatorially $\rightarrow$ 1,536 LUTs as RAM64M (Yosys-specific trap: by default the frontend turned the memory into 65,611 flip-flops, more than the whole device; `run_synth.sh` reads this file with `-nomem2reg`); the `host_bridge_spi` frame RAM has an asynchronous read in the SCLK domain $\rightarrow$ 1,536 LUTs as RAM64M; only the `matched_filter_multi_segment.v:83-84` input buffers map to RAMB18 (2). Recommended RTL change (designer decision, not made): register each memory output in its own clocked block without reset, mux afterwards — moves ~4,900 LUTs into roughly 7–8 RAMB36.
- LUTs (39 %) and FFs (7 %) leave room; the FFT IP adds BRAM/LUT/DSP on top.
- I/O: 183 port bits need 192 package pins** (9 LVDS pairs) — more than the 170 user I/O of the FTG256 package (DS180 Table 3 / package table — verify against UG475), even before pin conflicts. The 116 UNRESOLVED debug/FT601/status bits must be removed

Synthesis warnings that indicate real problems (source: same section): latches none, multi-driven nets none, post-synthesis check 0 problems; `chirp_memory_loader_param.v:112 $time` inside a `DEBUG`-gated `$display` leaves a `$print` cell that `nextpnr` cannot place (wrap in `translate_off`); forced `ram_style = "block"` impossible on 6 memories (above); tri-state logic on the FT601 bus and MISO mapped to IOBUF/OBUFT (fine); **clocking: Yosys inserted a 4th BUFG on `host_bridge.sclk = stm32_sclk_3v3`, XDC pin J16 = I0_L23N_T3_FWE_B_15**, not a clock-capable MRCC/SRCC pin — Vivado will refuse a non-CC pin driving a BUFG unless the net gets `set_property CLOCK_DEDICATED_ROUTE FALSE` (not in the XDC) or `SCLK` is oversampled in `clk_100m` instead.

P&R trial, `nextpnr-xilinx` (source: `beta/fpga_synth/README.md`, "P&R trial", copied verbatim): **NOT COMPLETED — no placed/routed design, no Fmax, no bitstream**. Inputs: the 67 `PACKAGE_PIN` + `IOSTANDARD` assignments and `create_clock` lines of the beta XDC; netlist with the 116 unconstrained port bits demoted to internal nets, the FFT IP replaced by 1-register pass-through stubs (NOT an FFT), `$print` removed — 12,747 LUT, 4,667 FF, 106 DSP48E1, 2 RAMB18E1.

Step	Result	Log
<code>nextpnr-xilinx</code> build (openXC7 bc9b234)	PASS after <code>-DUSE_OPENMP=OFF</code> (Apple clang: unsupported option <code>'-fopenmp'</code>)	<code>~/cache/aeris10_work/nextpnr_make.log</code>
Chip database <code>xc7a50tfg256-2</code> (<code>bbaexport.py</code> + <code>bbasm</code>)	PASS, 93 MB <code>xc7a50t.bin</code>	<code>~/cache/aeris10_work/nextpnr-src/chipdb/*.log</code>
<code>prjxray xc7frames2bit + fasm2frames.py</code> (venv)	PASS (built/installed, never reached)	<code>~/cache/aeris10_work/prjxray_make.log</code>
<code>nextpnr</code> , first try	FAIL at placement: Unable to place cell ... <code>\$display</code> ...: no Bels remaining of type <code>'\$print'</code> (DEBUG <code>\$display</code> in <code>chirp_memory_loader_param.v:113</code>) -> fixed in <code>synth_yosys_pnr.py</code>	(overwritten; reproducible)
Packing (all primitives)	PASS : IBUFDS/IBUF/OBUF, BUFGCTRL 4/32, BUFIO 1/20, BUFR 1/20, IDELAYE2 8, ISERDESE2 8, IDELAYCTRL 1/5, MMCME2_ADV 1/5 (from MMCME2_BASE), DSP48E1 106/120, RAMB18E1 2/150, CARRY4 344/8150, SLICE_LUTX 15,182/65,200 (LUT sites incl. LUTRAM), SLICE_FFX 4,667	<code>logs/nextpnr_heap_seed*.log.gz</code> (utilisation block)
Placement, HeAP (default) placer, seeds 2-6	FAIL (deterministic, < 5 s): Unable to find legal placement for cell <code>'...ddc_4x_100m.v:175\$12861'</code> - a standalone DSP48E1 (DDC mixer) that cannot be legalised after the 10 cascade chains (8 x 8 DSP FIR adder chains <code>fir_lowpass.v:53</code> , 2 x 7 DDC chains <code>ddc_4x_100m.v:204</code> ; 78 of the 106 DSPs) are fixed in the two 60-site DSP columns (x = 28, 86 in <code>prjxray</code> tilegrid)	<code>logs/nextpnr_heap_seed{2..6}.log.gz</code>
Placement, SA placer, seed 1 (unpatched)	HUNG : annealing converged to iteration 255 (wirelen 743,990) after ~16 min, then 20+ min at 100 % CPU without output; sample shows <code>SAPlacer::random_bel_for_cell</code> spinning in its unbounded <code>while (true)</code> retry loop (<code>common/placer1.cc:876-895</code>) when moving a DSP chain base with <code>force_z</code> . Killed.	<code>logs/nextpnr_sa_livelock.log.gz</code>
Placement, SA placer, seed 1, locally patched retry bound (<code>pnr/nextpnr_placer1_retry_bound.patch</code>)	ABORTED by time-box at iteration ~170 (10.5 min, still placing, same trajectory); routing never started	<code>logs/nextpnr_sa_patched_seed1.log.gz</code>
Routing, timing (Fmax), FASM, <code>fasm2frames</code> , <code>xc7frames2bit</code>	NOT REACHED - no Fmax figure exists from any run; no <code>.fasm</code> , no <code>.bit</code>	-

Blocking items for the open-source P&R (tool side, not design errors; source: same): (1) DSP48E1 cascade chains at 88 % DSP utilisation — HeAP cannot legalise them, SA livelocks; next things to try (not done): the patched SA run to completion (~30–60 min estimate, unverified) or the maintained openXC7/nextpnr himbaechel xilinx flow; (2) even with a routed result the timing would be incomplete: `set_clock_groups`, `set_false_path`, `set_input_delay`, `IDELAY_GROUP`, `DIFF_TERM`, `CFGVBVS/CONFIG_VOLTAGE` are not parsed by `nextpnr-xilinx`, the ADC ISERDES/IDELAY input window is not analysed, and the FFT stubs remove the FFT critical paths. Any bitstream from this flow would contain the stubs (named `..._OPENSOURCE_STUBFFT.bit` by `run_pnr.sh`) and **must never be loaded on the radar hardware**.

What must be run in Vivado to confirm (source: same, "What must be run in Vivado"): generate the FFT IP per section 9; `vivado -mode batch -source beta/fpga/vivado/create_project.tcl`; `vivado -mode batch -source beta/fpga/vivado/build.tcl -tclargs <path>/aeris10_beta.xpr 8`; `inspect report_utilization -hierarchical` (DSP48E1 total with the 3 FFT cores ≤ 120 ; BRAM use of `chirp_mem`, `doppler_proc`, `host_bridge`; RAM/ROM inference tables), `report_timing_summary` (WNS/WH0 for `clk_100m`, `clk_120m_dac`, `adc_dco/BUFR`, `clk_200m`, `spi_sclk`), `report_clock_interaction`, `report_cdc`, `report_methodology`, `report_io` / `report_drc` (non-CC pin J16 → BUFG; bank-14 LVDS_25 at 3.3 V VCCO; IDELAY_GROUP and IDELAYCTRL placement; unconstrained ports — remove the 116 UNRESOLVED bits from the top first); `write_bitstream` only after all of the above (acceptance criteria AC-F5...AC-F8, all NOT MET; source: `docs/TESTING/ACCEPTANCE_CRITERIA.md`).

12. Remaining work (ordered) and known limitations

Remaining work (source: `beta/fpga/README.md`, "Remaining work (ordered)"):

1. **Generate the two FFT IP cores** per `ip/README.md`, run `tb_fft_wrappers` against the IP simulation models (adjust `T0L` if the IP's per-stage rounding differs), record the IP latency and check `FFT_WAIT_TIMEOUT` in `doppler_processor.v`.

README/XDC claim XC7A100T — resolve first, K1). Expect resource pressure: two 1024-point FFT IPs, 2×32 parallel 18×18 multipliers in the FIR (64 DSP48E1 of the 50T's 120); the open-source estimate of section 11 puts the non-FFT design at 106 of 120 DSP48E1.

3. **400 MHz fabric path replaced** (`ADC_CAPTURE_MODE = 1`): remaining are Vivado timing on the `clk_div/clk_200m` paths, `IDELAYCTRL/IODELAY_GROUP` placement, hardware calibration with the ADC test pattern, Q1..Q4 order on hardware.
4. **Matched-filter throughput.** The chain is not pipelined against the collector: one long chirp (4 segments) occupies the FSM for $\sim 92 \mu\text{s}$ with the behavioural FFT latency (160 clocks) and $\sim 270 \mu\text{s}$ with a realistic IP latency ($\sim 2.3\text{k}$ clocks per transform), while the TX repeats long chirps every $167 \mu\text{s}$ (`plfm_chirp_controller.v`: $30 \mu\text{s}$ chirp + $137 \mu\text{s}$ listen). Samples arriving while the FSM is not in `ST_COLLECT_DATA` are dropped (original behaviour). Needs ping-pong buffering or overlapping collect/process; the smoke test uses a $300 \mu\text{s}$ period for this reason.
5. **Segment semantics.** Each 1024-sample segment is compressed against its own reference segment and produces its own 64-bin profile; the four profiles per chirp are not summed, and the Doppler processor counts each profile as a "chirp" (a 32-"chirp" frame = 8 real chirps). Decide whether to sum the partial correlations (true partitioned matched filter) or to use a 4096-point transform.
6. **Hardware bring-up items** (all UNRESOLVED): confirm the chirp direction / `CONJUGATE_REF`; AD9484 capture edge and trace skew; bank-14 LVDS termination; ADC test pattern check; DAC data timing versus the externally clocked AD9708 (`dac_clk` has no pin); STM32 SPI1 clock rate versus the 1-cycle SPI pass-through re-timing; the short-chirp reference; the FT601/host decision.
7. CFAR (the detector is still a fixed threshold, default 10000 via `CFAR_THRESHOLD_DEFAULT`, now writable over the bridge as `CFAR_THR`); removal of the 116 unconstrained debug/status bits from the top before a board build. Register-map host interface: done (command set v2); `det_wr` back-port to `engineering/DESIGN/HOST_LINK/rtl/`: done.
8. GUI beta/`gui/aeris10_gui/protocol/register_map.py` follow-up (5-bit address mask, 0x04 bit4, 0x0D, 0x0E, 0x10) — recorded in the FPGA README; the GUI changelog's "final register map, RTL 0x0002" entry records the alignment (`ADDR_MASK = 0x1F`, registers 0x0D/0x0E/0x10) (source: beta/`gui/CHANGELOG.md`, "2026-10-09 (final register map, RTL 0x0002)"; beta/`gui/aeris10_gui/protocol/register_map.py`:18).

Known limitations of the beta (source: beta/`fpga/README.md`, "Known limitations"): not synthesised, no timing, no hardware test; FFT IP not generated (fail-loud placeholders); behavioural FFT models are not bit-exact with the Xilinx IP (total-shift scaling, round-half-up; TB tolerance ± 2 LSB) and their latencies (72 / 160 clocks) are placeholders; `rtl/sim/unisim_sim_models.v` models only the behaviour the design uses (no timing, IB ignored); the USB packetiser runs on `clk_100m` and drops records that arrive while a packet is in flight (counted internally); the 11-word packet format is a beta definition; the level-shifter pass-through adds one `clk_100m` cycle to SCLK/MOSI/CS and two to MISO; `use_long_chirp` is a register bit (default 1) and the TX sequence's long/short alternation is not mirrored automatically in the receiver.

Acceptance state for the FPGA (source: `docs/TESTING/ACCEPTANCE_CRITERIA.md` §A and §H): AC-F1...AC-F8 NOT MET on the original tree, AC-F9 (physical ADC capture with a CW tone) NOT RUN; AC-X1 (beta RTL parses/elaborates/lints with 0 errors and all testbenches pass) MET (BETA).

<!-- chapter 12: Firmware: architecture, build, defects fixed, sequencing, USB protocol -->

STM32 firmware — architecture, build, defects fixed, power sequencing, USB protocol

Author of this chapter: Antidrone Ukraine · antidrone.cc (compilation of the BETA firmware tree and its decision records; the upstream sources under 9_Firmware/9_1_Microcontroller/ are ORIGINAL PROJECT FILE and untouched).

Status summary: **BETA** — "compiles and links with the ST HAL/USB stack. Not flashed. Not run on hardware. No peripheral, RF or power-sequence behaviour has been observed" (source: beta/stm32/README.md , header). Architecture diagrams SD-03/SD-04 are **PARTIAL** (drawn from the original sources, in which the HAL, CubeMX files and USB middleware were absent). Clock tree, I2C timing, SPI prescalers, USB descriptors and the power-sequencing delays are **engineering decisions** (D-01...D-19 of beta/stm32/DECISIONS.md), not measurements.

Sources: beta/stm32/README.md , beta/stm32/DECISIONS.md , beta/stm32/CHANGELOG.md , beta/stm32/CUBEMX_SETTINGS.md , docs/STM32/STM32_PROJECT_RECONSTRUCTION.md , engineering/ELECTRICAL/power_distribution/power_rails.md §2–§3, engineering/SOFTWARE_DIAGRAMS/STM32/stm32_architecture.md , engineering/DESIGN/HOST_LINK/HOST_LINK_DESIGN.md §5, §7, beta/gui/aeris10_gui/protocol/settings_packet.py and status_text.py (docstrings that quote the firmware line numbers).

Figures in this chapter: F12.1 firmware architecture (SD-03), F12.2 USB CDC flow (SD-04).

12.1 Target and what the firmware does

MCU: **STM32F746ZGTx (LQFP-144)**, part **U2** on **RADAR_Main_Board.sch** (source: beta/stm32/CUBEMX_SETTINGS.md §1). The application is a CubeMX-style main.c renamed to C++ (main.cpp , 2411 lines in the original) that owns every peripheral handle, all MX_*_Init functions, the clock tree and the whole application sequence; drivers live in LIB/ as three families (source: engineering/SOFTWARE_DIAGRAMS/STM32/stm32_architecture.md §1, copied):

Family	Files	Transport	Evidence
C++ board classes	ADAR1000_Manager.cpp/.h , USBHandler.cpp/.h , RadarSettings.cpp/.h , BMP180.cpp/.h , gps_handler.cpp/.h , TinyGPS++.cpp	HAL directly (hspi1 , huart3 , hi2c3) , CDC	ADAR1000_Manager.cpp:8–9,619–621 ; BMP180.cpp:405–409 ; gps_handler.cpp:60,118
ADI no-OS drivers + core	ad9523.c/.h , adf4382.c/.h , adf4382a_manager.c/.h , adar1000.c/.h , no_os_*.c/.h (60 files) , iio*.c/.h	no_os_spi -> platform_ops -> stm32_spi.c -> hspi4	CODE/main.cpp:1039–1040 ; adf4382a_manager.c:42–43,51–52
C HAL sensor drivers	DA5578.c / DAC5578.H , ADS7830.c/.H , GY_85_HAL.c/.h	HAL I2C (hi2c1 , hi2c2 , hi2c3)	DA5578.c:69 ; ADS7830.c:111–121 ; GY_85_HAL.c:35–116 ; handles passed at CODE/main.cpp:1563,1570,1605,1611,1670

Peripheral-to-device map (source: same note §1): I2C1 PB6/PB7 → 2 × DAC5578 (PA gate bias); I2C2 PF0/PF1 → 3 × ADS7830 (IDQ sense, temperatures); I2C3 PA8/PC9 → GY-85 + BMP180; SPI1 PA5/PA6/PA7 + CS PA0..PA3 → ADAR1000 ×4 through the FPGA level shift; SPI4 PE2/PE5/PE6 → AD9523 (CS PF7) and ADF4382 TX/RX (CS PG14/PG10); UART5 PC12/PD2 → GPS NMEA; USART3 PB10/PB11 → debug text; TIM1 → delay_us() ; GPIO PD8..PD12 → FPGA handshake; USB OTG_FS → host CDC.

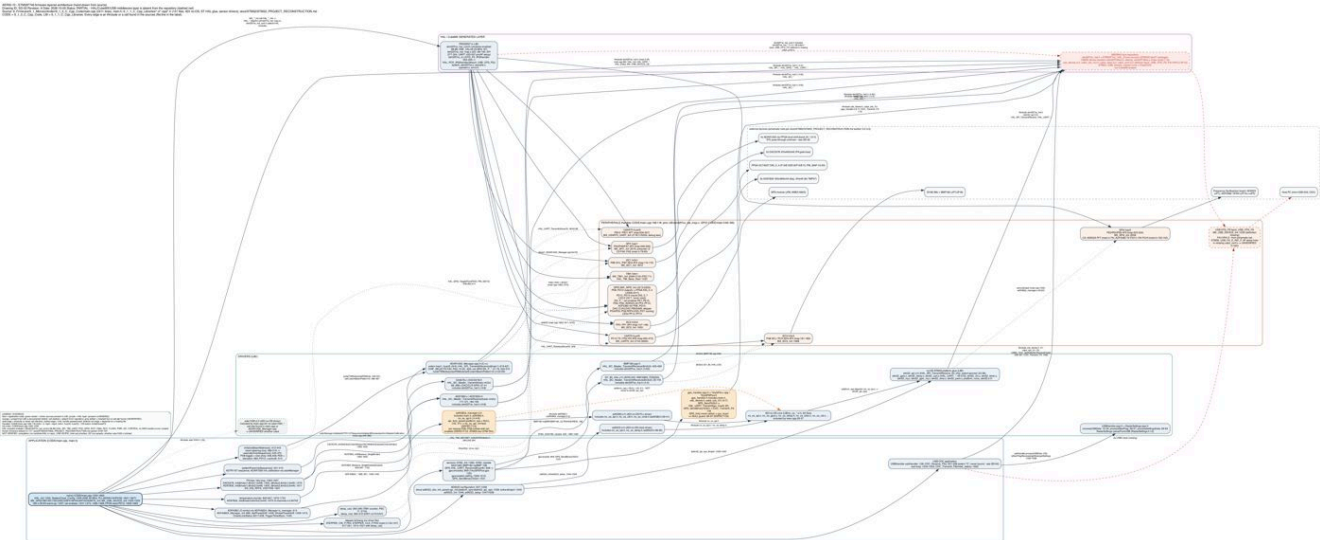


Figure 57 — F12.1 — STM32 firmware architecture: main.cpp application sequence, driver families, peripheral handles; dashed red = HAL/CMSIS/startup/linker/USB middleware absent from the upstream repository (SD-03) — PARTIAL (source: engineering/SOFTWARE_DIAGRAMS/STM32/stm32_firmware_architecture.png; produced by hand-authored DOT + Graphviz dot -Tpng -Gdpi=150)

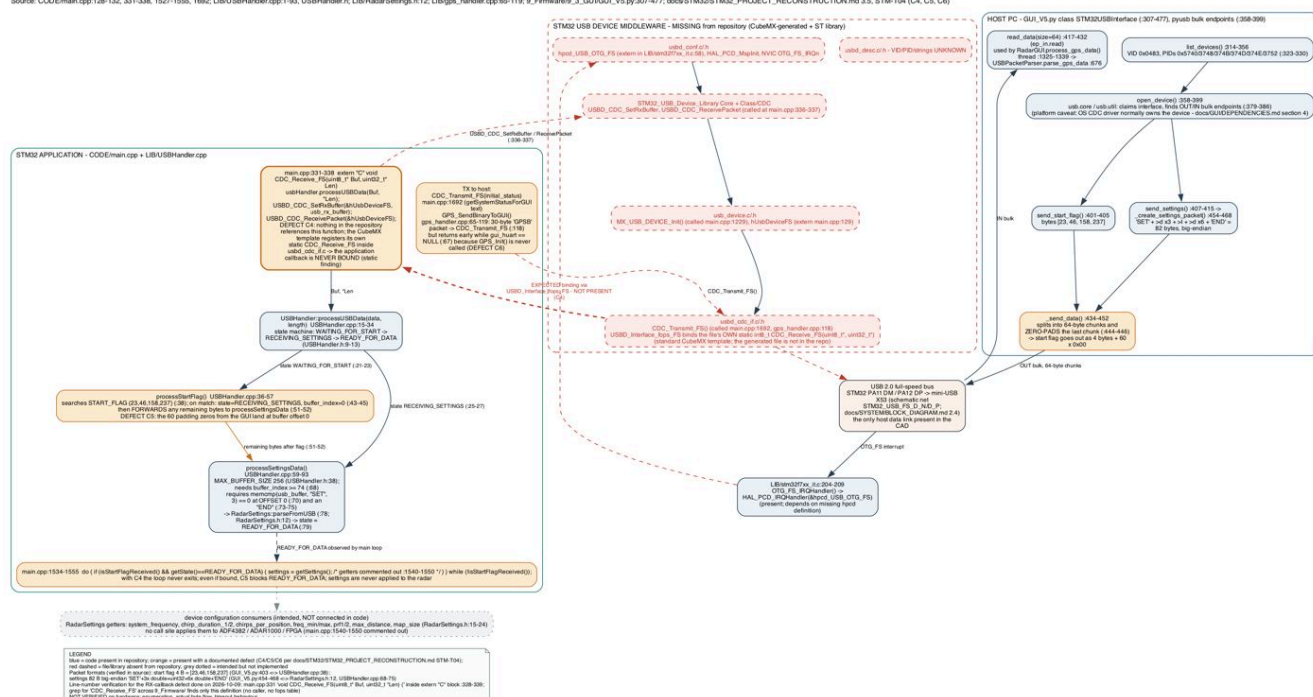


Figure 58 — F12.2 — USB CDC flow host ↔ STM32 as found in the original sources: start flag, zero-padding defect C5, settings packet, unbound receive callback C4, status/GPS transmit (SD-04); the beta tree fixes C4/C5 (§12.4) — PARTIAL (source: engineering/SOFTWARE_DIAGRAMS/STM32/stm32_usb_cdc_flow.png; produced by hand-authored DOT + Graphviz)

12.2 BETA tree layout

(source: beta/stm32/README.md, "Layout", copied)

```
beta/stm32/
├── Core/Inc, Core/Src, Core/Startup    main.cpp/main.h + CubeMX glue (copied), startup_stm32f746xx.s (ST CMSIS template)
├── LIB/                               copy of 9_1_1_C_Cpp_Libraries (+ aeris_beam.*, _excluded/)
├── USB_DEVICE/App, USB_DEVICE/Target  hand-written CubeMX-equivalent USB CDC files
├── STM32F746ZGTx_FLASH.ld             ST Nucleo-F746ZG template (AXIM flash 1 MB @0x08000000, RAM 320 KB @0x20000000)
├── CMakeLists.txt, cmake/arm-none-eabi.cmake, build.sh, setup_cube.sh
├── cube/                              STM32CubeF7 sparse checkout (git-ignored; pinned commits in DECISIONS.md D-15)
├── tests/                             host-side unit tests (run_tests.sh)
├── build/ (ignored), build_out/        artefacts: aeris10_fw.elf/.hex/.bin/.map
├── logs/                              build, size, warning, include-check and test logs
├── CHANGELOG.md, DECISIONS.md, CUBEMX_SETTINGS.md, README.md
```

What was missing upstream and how the beta fills it (source: engineering/SOFTWARE_DIAGRAMS/STM32/stm32_architecture.md §2; beta/stm32/CHANGELOG.md §3): no `stm32f7xx_hal.h/HAL` driver, no CMSIS device header, no `startup_stm32f746xx.s`, no linker script, no `.ioc`, no `usb_device.c/.h`, `usbd_cdc_if.c/.h`, `usbd_desc.c/.h`, `usbd_conf.c/.h`, no USB Device Library. The beta adds the HAL/CMSIS/USB library as a pinned sparse checkout (`cube/`), the ST startup and linker templates verbatim, and hand-written CubeMX-equivalent USB files built from the ST templates (D-07...D-10). The `.ioc` itself cannot be generated without CubeMX; `CUBEMX_SETTINGS.md` lists every setting "so an engineer can rebuild it in ~20 minutes and then diff the generated code against the hand-written files".

12.3 Build procedure

Step 12.1 — Toolchain [BETA, executed on the authoring machine]

- **Purpose:** provide the cross-compiler and build tools.
- **Parts & tools:** Arm GNU Toolchain 14.2.Rel1 (`arm-none-eabi-gcc 14.2.1 20241119`); CMake ≥ 3.20 (4.4.3 used); Ninja optional; Python 3; a host C/C++ compiler for the tests (source: beta/stm32/README.md §1).
- **Action:** install the Arm tarball (on the authoring machine extracted to `~/opt/arm-gnu-toolchain` and symlinked into `/opt/homebrew/bin/arm-none-eabi-`; the Homebrew `cask gcc-arm-embedded` failed because its installer needs `sudo`). The CMake toolchain file searches `/opt/homebrew/bin`, `~/opt/arm-gnu-toolchain/bin` and `/Applications/ArmGNUToolchain/` `/arm-none-eabi/bin`, or pass `-DT00LCHAIN_PREFIX=/path/arm-none-eabi-`.
- **Check:** `arm-none-eabi-gcc --version` prints 14.2.1 (verified on 2026-10-09 on the authoring machine: "Arm GNU Toolchain 14.2.Rel1 (Build arm-14.52) 14.2.1 20241119"). "Any 12.x-14.x arm-none-eabi GCC should work" (README §1 — unverified claim of the source).
- **Figure:** —

AERIS-10 — Embedded STM32CUBE7 (BETA) — **Decision D-13** — compiler/linker flags are **ASSUMED** CubeIDE defaults (`-mcpu=cortex-m7 -mthumb -mfpv5-sp-d16 -mfloat-abi=hard, -Og -g3 Debug, -ffunction-sections -fdata-sections, C++ -fno-exceptions -fno-rtti -fno-use-cxa-atexit -fno-threadsafe-statics, link --specs=nano.specs -u _printf_float --gc-sections, libs c m stdc++ nosys`).

Step 12.2 — STM32CubeF7 package [BETA]

- **Purpose:** obtain the HAL, CMSIS and USB Device Library at the pinned commits.
- **Parts & tools:** `bash beta/stm32/setup_cube.sh` (sparse clone + check with `tools/stm32_check_cube_package.sh`).
- **Action:** run the script; it clones `STMicroelectronics/STM32CubeF7 @ 79165e26...` (sparse: `Drivers/CMSIS, Drivers/STM32F7xx_HAL_Driver, Middlewares/ST/STM32_USB_Device_Library, Nucleo-F746ZG templates and USB_Device applications`) with submodules `stm32f7xx_hal_driver @ e860c4ff...` (HAL 1.3.3), `cmsis_device_f7 @ 2352e888...` (v1.2.10), `stm32-mw-usb-device @ 2a0a3521...` (source: `beta/stm32/DECISIONS.md D-15`).
- **Check:** "Package: `beta/stm32/cube` - missing 0 of 34 required files (exit 0)" (source: `beta/stm32/README.md §2`).
- **Figure:** —
- **Decision:** D-15 (pinned commits).

Step 12.3 — Compile and link [BETA, result recorded]

- **Purpose:** produce `aeris10_fw.elf/.hex/.bin/.map`.
- **Parts & tools:** `bash beta/stm32/build.sh` (Debug, `-Og -g3`) or `bash beta/stm32/build.sh Release`.
- **Action:** run the script; it configures CMake, builds, prints `arm-none-eabi-size`, writes `logs/build_full.log`, `logs/build_warnings.log`, `logs/build_errors.log`, `logs/size.log` and copies the artefacts to `build_out/`. "Exit code 0 only on a clean link."
- **Check:** result recorded on 2026-10-09 (source: `beta/stm32/README.md §3`, copied):

Memory region	Used Size	Region Size	%age Used
RAM:	17480 B	320 KB	5.33%
FLASH:	93276 B	1 MB	8.90%
text data bss dec hex filename			
92480 788 16704 109972 1ad94 aeris10_fw.elf			

0 errors; 5 unique warnings, all pre-existing code (`main.cpp` unused `settings` reference, `BMP180.cpp` misleading indentation `×3`, `TinyGPS++.cpp` implicit fall-through); third-party HAL/USB sources compiled with `-w`. Include audit `python3 tools/check_stm32_includes.py`: 0 case mismatches (was 6); the 11 remaining UNKNOWN headers are all in `io*.c/iiod.c`, which are not built.

- **Figure:** —
- **Decision:** D-14 — sources excluded from the build: `platform_noos_stm32.c`, `adar1000.c`, `io.c`, `io_app.c`, `iiod.c`, `io_trigger.c`, and every `no_os_*.c` not required by `ad9523.c/adf4382.c/no_os_spi.c`.

Step 12.4 — Host unit tests [BETA, 6/6 PASSED on 2026-10-09]

- **Purpose:** test the protocol and driver logic on the host without hardware.
- **Parts & tools:** `bash beta/stm32/tests/run_tests.sh` (needs `cc/c++ + python3` only).
- **Check:** (source: `beta/stm32/README.md §4`, copied)

Test	What it checks
<code>test_settings_parser.cpp</code>	<code>real USBHandler.cpp + RadarSettings.cpp</code> : start flag <code>[23,46,158,237]</code> , 82-byte big-endian <code>SET...END</code> packet, unpadded and GUI-style 64-byte zero-padded framing, flag+settings in one packet, <code>SET</code> straddling a packet boundary, byte-at-a-time, short packets, invalid-value rejection and recovery, big-endian decode against a hand-encoded constant, <code>REG</code> text commands coexisting with the binary path (10 cases)
<code>test_beam_matrix.c</code>	<code>aeris_beam.c</code> (verbatim arithmetic of <code>initializeBeamMatrices / degreesTo7BitPhase</code>): 7-bit phase conversion, reference element, range, hand-computed samples, mirror symmetry, linearity
<code>test_ad9523_regs.c</code>	<code>real ad9523.c + no_os_spi.c</code> against a mock SPI register file: full <code>ad9523_setup()</code> path, channel distribution registers for <code>OUT0/1/4/5/6/7/8/9/10/11</code> (dividers 12/9/36/180/60/30, LVDS/CMOS modes), unused outputs powered down, <code>IO_UPDATE/SYNC/status</code> . This test found defect C8
<code>test_adar_vm_tables.c</code>	ADAR1000 VM tables vs data sheet Tables 10-13: 128 entries, bits 7:6 clear, quadrant signs, spot rows <code>0/45/90/180/230.625/270/312.1875/357.1875°</code> , strictly monotonic decoded phase (max error 3.12°), <code>degreesTo7BitPhase</code> → table round trip
<code>test_host_bridge_cmds.c</code>	bridge v2 byte sequences with a mock SPI (<code>0x02 + ack 0xA2 / no ack / 0xEE</code> / transfer error, <code>0x03</code> LE decode, <code>0x04</code> status fields), ASCII <code>REG W/R</code> parser (hex/decimal, errors) and reply formatting
<code>check_i2c_timing.py</code>	decodes <code>TIMINGR</code> : original <code>0x00808CD2 @36 MHz ≈ 100 kHz</code> ; beta <code>0x10916EA0 @54 MHz ≈ 97.6 kHz</code> , I2C Standard-mode minima met

Not host-testable as-is (need HAL types): `stm32_spi_prescaler_for()`, the USB glue, power sequencing.

12.4 Defects fixed in the BETA tree

Defect C1 is the HSE conflict (firmware 25 MHz vs the 8 MHz crystal XTAL1 NX3225GD-8MHZ on PH0/PH1 of the schematic; "with 8 MHz HSE the PLL input would be 0.32 MHz (below the 0.95 MHz minimum of RM0385)") and is closed by decision D-01 (source:

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docs/STM32/STM32_PROJECT_RECONSTRUCTION.md §5 row C1; beta/stm32/DECISIONS.md D-01) · C2...C10 and the further defects (source: beta/stm32/README.md §5, copied; file:line details in CHANGELOG.md):

ID	Defect in the original	Fix
C1	HSE 25 MHz (firmware) vs 8 MHz crystal (schematic) — <code>hal_conf.h:97</code> , <code>main.cpp:1823</code>	<code>HSE_VALUE = 8000000</code> ; PLL M8/N432/P2/Q9 → 216 MHz (D-01, hardware-truth decision)
C2	<code>adf4382a_manager.h</code> drove PG0..PG9 (PA/clock power enables) as ADF4382 CE/CS/lock-detect	macros alias <code>main.h</code> PG6..PG15; CE parameter widened to 16 bit
C3	<code>ADF4382_platform_ops = NULL</code> → <code>no_os_spi_init</code> returns <code>-EINVAL</code> → <code>Error_Handler()</code>	<code>&stm32_spi_ops</code>
C4	USB RX callback never bound; start-flag wait loop could never exit	<code>usbdc_cdc_if.c: CDC_Receive_FS</code> → <code>AERIS_USB_OnReceive()</code> → <code>USBHandler</code>
C5	GUI zero-padding after the start flag landed in the settings buffer; "SET" never at offset 0	parser synchronises on <code>SET</code> , handles padded/unpadded/split frames (tested)
C6	<code>GPS_Init()</code> never called → <code>GPS_SendBinaryToGUI()</code> returned early	<code>GPS_Init(&huart3)</code>
C7	SPI chip selects driven low permanently and never toggled; SPI clock PCLK/2 regardless of the 10 MHz request	CS index table + toggling in <code>stm32_spi.c</code> , CS parked high, prescaler from <code>max_speed_hz</code>
C8	<code>ad9523_init()</code> called after <code>pdata</code> was configured → whole AD9523 channel/PLL configuration reset to defaults	call removed
C9	<code>ad9523_setup()</code> executed twice (first device leaked)	single call after reset release
C10	ADAR1000 phase/gain setters used (<code>channel & 3</code>) with 1-based channels → element pattern rotated by one channel per device	<code>((channel - 1) & 3)</code>
—	<code>VM_I/VM_Q/VM_GAIN</code> empty → every phase write was <code>I = Q = 0</code>	tables from the data sheet (D-19)
U10	IDQ servo for DAC2 read ADC2 into <code>adc1_readings</code> and computed from stale data	<code>adc2_readings</code>
—	<code>printf()</code> → weak <code>__io_putchar</code> undefined → call to address 0	retargeted to USART3
—	<code>USBHandler::processStartFlag</code> unsigned underflow on packets < 4 bytes	guarded
—	<code>stm32_spi_write_and_read</code> prototype mismatch (<code>uint32_t</code> vs <code>uint16_t</code>), hard error on GCC 14	fixed
—	<code>LIB/errno.h</code> <code>#include_next</code> self-inclusion → <code>EINVAL</code> undefined	shim excluded
—	<code>.H</code> include-case mismatches	files renamed <code>.h</code>

The C2 defect is safety-relevant: with the original macros `ADF4382A_Manager_Init()` "would have driven the PA 5 V enables as chip-enables ... and read 'lock detect' from the clock-enable outputs" (source: beta/stm32/DECISIONS.md D-04).

12.5 Hardware-truth decisions (D-01...D-19)

One-line index (source: beta/stm32/README.md §6, copied; full text in beta/stm32/DECISIONS.md): D-01 8 MHz HSE → PLL M8/N432/P2/Q9, 216 MHz, over-drive, 7 WS · D-02 APB1 54 / APB2 108 MHz, TIM1 PSC 215 · D-03 I2C TIMINGR 0x10916EA0 (computed, not CubeMX) · D-04 ADF4382 pins per schematic · D-05 CS index/idle-high · D-06 SPI 6.75 MHz (≤ 10 MHz) · D-07 OTG_FS device-only PA11/PA12, no VBUS sensing · D-08 USB IRQ priority 0 · D-09 VID/PID 0x0483:0x5740 PLACEHOLDER · D-10 CDC RX hook · D-11 PA sequencing: +5V5_PA before VG DAC programming, VG before VD, VD removed first on power-down (delays are estimates) · D-12 AD9523 single setup · D-13 compiler flags = CubeIDE defaults (assumed) · D-14 excluded sources · D-15 pinned Cube commits · D-16 `GPS_Init` · D-17 REG commands executed from the main loop · D-18 bridge v2 framing assumptions · D-19 VM tables / `VM_GAIN` = 0.

Clock tree as decided (source: beta/stm32/CUBEMX_SETTINGS.md §2, copied):

Item	Value
HSE	Crystal/Ceramic resonator, 8 MHz (XTAL1 NX3225GD-8MHZ on PH0/PH1)
LSE	crystal 32.768 kHz present on PC14/PC15 (XTAL3) — not used (leave disabled)
PLL source	HSE; M = 8, N = 432, P = 2, Q = 9 → VCO 432 MHz
SYSCLK / HCLK	PLLCLK, 216 MHz ; AHB /1
APB1 / APB2	/4 → 54 MHz; /2 → 108 MHz (timer clocks: TIMPRE activated → TIM1CLK = 216 MHz)
USB (CLK48)	PLLQ = 48 MHz
Power	Voltage scale 1, Over-Drive enabled ; Flash latency 7 WS
Original (for reference)	25 MHz HSE, M25 N144 P2 Q3, 72 MHz, APB1 /2, APB2 /1, scale 3, 2 WS — invalid with the 8 MHz crystal

Unverified items named by the decisions: crystal load capacitors / drive level at 8 MHz (D-01); the I2C TIMINGR is "not a CubeMX output — regenerate with CubeMX ... and compare before trusting it on hardware" (D-03); ADAR1000 SCLK maximum and FPGA level-shifter bandwidth at 6.75 MHz (D-06); whether VBUS is wired to PA9 (D-07); a product VID/PID (D-09); the sequencing delays "are conservative estimates, not measured" (D-11); ADAR1000 phase accuracy (data-sheet tables realise the nominal phase within $\approx 3^\circ$, per-board calibration is a hardware task — D-19).

12.6 Power-enable sequencing

Enable-bus pin map (SV1 MA10-2, identical on both boards; 15 enable lines, "not 16 as stated in docs/SYSTEM/BLOCK_DIAGRAM.md") (source: engineering/ELECTRICAL/power_distribution/power_rails.md §2, copied):

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SVI pin	Net	STM32 pin (main.h)	Power Board EN input
1	EN_+1V0_FPGA	PE7	U1
2	EN_+5V0_PA2	PG1	U15
3	EN_+1V8_FPGA	PE8	U2
4	EN_+5V0_PA3	PG2	U16
5	EN_+3V3_FPGA	PE9	U4
6	EN_+5V5_PA	PG3	U17
7	EN_+5V0_ADAR	PE10	U13
8	EN_+1V8_CLOCK	PG4	U25
9	EN_+3V3_ADAR12	PE11	U6
10	EN_+3V3_CLOCK	PG5	U23
11	EN_+3V3_ADAR34	PE12	U7
13	EN_+3V3_ADTR	PE13	U32
15	EN_+3V3_SW	PE14	U10
17	EN_+3V3_VDD_SW	PE15	U8
19	EN_+5V0_PA1	PG0	U14
12, 14, 16, 18, 20	GND	—	—

Firmware enable sequence **as coded in the original** ("not executed on hardware") (source: `power_rails.md` §3, copied):

Step	Action	Delay after	Source
F0	HAL_Delay(180000) (3 min) then AD9523 RESET low	—	main.cpp:1237–1238
F1	EN_+1V8_CLOCK high	100 ms	main.cpp:1241–1242
F2	EN_+3V3_CLOCK high; AD9523 RESET high; configure_ad9523()	100 ms + 100 ms	main.cpp:1243–1267
F3	EN_+1V0_FPGA high	100 ms	main.cpp:1271–1272
F4	EN_+1V8_FPGA high	100 ms	main.cpp:1273–1274
F5	EN_+3V3_FPGA high	100 ms	main.cpp:1275–1276
F6	DIG_3 (PD11) low "mixers off"; EN_+3V3_ADAR12 + EN_+3V3_ADAR34 high	500 ms	main.cpp:1483–1487
F7	EN_+5V0_ADAR high (→ -5 V ADAR rails)	500 ms	main.cpp:1488–1489
F8	EN_+3V3_VDD_SW high	2 ms	ADAR1000_Manager.cpp:51–52 (powerUpSystem(); whether systemPowerUpSequence() reaches it via initializeADTR1107Sequence() is UNVERIFIED)
F9	EN_+3V3_SW high (→ -3V3_SW)	2 ms	ADAR1000_Manager.cpp:54–55
F10	DAC5578 VG codes written, LDAC pulsed, then EN/DIS_RFPA_VDD high (22 V drain)	—	main.cpp:1560–1601
never	EN_+3V3_ADTR, EN_+5V0_PA1/2/3, EN_+5V5_PA	—	no GPIO_PIN_SET write anywhere under 9_Firmware/9_1_Microcontroller/
power-down	ADAR RX mode → EN_+5V0_PA1..3 low → PA bias safe → EN_+3V3_ADTR low → LNA bias 0 → EN_+3V3_VDD_SW, EN_+3V3_SW low (10 ms steps)	10 ms	main.cpp:372–409

What the BETA changes in this sequence (D-11; source: `beta/stm32/DECISIONS.md` D-11 and `CHANGELOG.md` §2 `main.cpp` rows 380–382, 403–406, 1560–1562, 1600–1601): (a) finding — `EN_+5V5_PA` (PG3) "was **never** driven high — the OPA4703 VG buffers were unpowered while `main.cpp:1583–1601` 'programmed' VG and then enabled the 22 V drain"; (b) `EN_+3V3_ADTR` and `EN_+5V0_PA1/2/3` **are** asserted, but through raw pin numbers in `ADAR1000_Manager.cpp` (`initializeADTR1107Sequence()`, `setADTR1107Mode()`, `enable/disablePASupplies()`, `main.h` macros; (c) power-down, in the `xlsx` order `VDD_SW` → `VSS_SW` → `CTRL` → `VGG` → `VDD_PA/VDD_LNA` — kept, re-expressed with `main.h` macros; (c) power-down, never removed the 22 V drain. Decisions: enable `+5V5_PA` 100 ms before the first DAC write; 20 ms VG settle before `EN/DIS_RFPA_VDD`; on power-down drop VD first (`EN_DIS_RFPA_VDD` LOW + 10 ms), then PA 5 V, LNA, switch rails, and `+5V5_PA` last. "Delays are conservative estimates, **not measured**. The IDQ servo loop (`main.cpp:1628–1656`) and the 1.68 A target were not touched. The external 22 V source and its switch are not in CAD (K4)."

The "never" row of the original therefore reads, in the beta: `+5V5_PA` is now enabled before VG programming (D-11), while `+3V3_ADTR/+5V0_PA_x` were already enabled by `ADAR1000_Manager` (finding b). Observation MAN-12-1: the CONFLICT rows of `power_rails.md` §1 (`+3V3_ADTR`, `+5V0_PA_1..3`, `+5V5_PA`) were written before finding (b); the register has not been updated and still states "firmware never enables the LNA supply". The bench verification in chapter 16 (Step 16.2) is the only way to settle the actual order.

12.7 USB protocol

Settings packet (host → firmware)

Start flag 17 2E 9E ED = [23, 46, 158, 237] (source: `USBHandler.cpp:38`, quoted in `beta/gui/README.md` "Protocol facts"), followed by the 82-byte packet SET + 3 × >d + >I + 6 × >d + END, all big-endian (source: `RadarSettings.cpp:23–120`;

beta/gui/aeris10_gui/protocol/settings_packet.py lines 18-32, PACKET_LENGTH = 82). Field order (source: settings_packet.py FIELDS, lines 56-66; defaults from beta/gui/aeris10_gui/model.py RadarSettings):

Offset	Size	Field	Format	Firmware limit (RadarSettings.cpp:80-95 validateSettings)	Default (model.py)
0	3	marker	SET	must be at buffer offset 0 after the flag (USBHandler.cpp:70)	—
3	8	system_frequency	>d	1e9 ... 100e9 Hz	10.0e9
11	8	chirp_duration_1	>d	1e-6 ... 1000e-6 s	30.0e-6
19	8	chirp_duration_2	>d	0.1e-6 ... 10e-6 s	0.5e-6
27	4	chirps_per_position	>I	1 ... 256	32
31	8	freq_min	>d	1e6 ... 100e6 Hz	10.0e6
39	8	freq_max	>d	> freq_min, ≤ 100e6 Hz	30.0e6
47	8	prf1	>d	100 ... 10000 Hz	1000.0
55	8	prf2	>d	100 ... 10000 Hz	2000.0
63	8	max_distance	>d	100 ... 100000 m	50000.0
71	8	map_size	>d	1000 ... 200000 m	50000.0
79	3	marker	END	—	—

Framing rule (C5): the original GUI zero-padded every write to 64 bytes and the firmware copied the zeros into the settings buffer, so SET was never at offset 0; the beta parser "synchronises on SET, handles padded/unpadded/split frames" (source: beta/stm32/README.md §5 row C5; CHANGELOG.md USBHandler.cpp rows 59-93). There is no acknowledgement for a settings packet (source: beta/gui/README.md, assumption 7).

Status and GPS (firmware → host)

Status string built by getSystemStatusForGUI (main.cpp:807-877), sent once with CDC_Transmit_FS at main.cpp:1692, no terminator (source: beta/gui/aeris10_gui/protocol/status_text.py docstring, copied):

```
System Status: NORMAL|LastError:%d|ErrorCount:%lu|
IMU:%.1f,%.1f,%.1f|GPS:%.6f,%.6f|ALT:%.1f|LO_TX:LOCKED|LO_RX:UNLOCKED|
T1:%.1f|...|T8:%.1f|PA_AvgCurrent:%.2f|PA_Enabled:%d|
BeamPos:%d|Azimuth:%d|ChirpCount:%d|
```

(EMERGENCY_STOP replaces NORMAL; the PA block appears only when PowerAmplifier is non-zero; ChirpCount:%d| is always the last field.) GPS: text GPS:%.8f,%.8f,%.2f\r\n on UART3 (gps_handler.cpp:45-62) and the 30-byte binary GPSB frame over CDC — 'GPSB' + lat (>d) + lon (>d) + alt (>f) + pitch (>f) + 16-bit sum of the first 28 bytes, big-endian (gps_handler.cpp:65-119). Known limitation carried into the status fields: getSystemStatusForGUI() "reports raw ADC codes as temperatures (no 0.64705 scale)" (source: beta/stm32/README.md §7 item 6).

REG register commands (host → firmware → FPGA)

Text commands over the same CDC stream (source: beta/stm32/DECISIONS.md D-17, copied): REG W <addr> <value> / REG R <addr> → reply REG 0x%04X 0x%08X\r\n (a write echoes the written value after the ack) or REG ERR\r\n on syntax error, NACK, busy or SPI error. Numbers 0x hex or decimal; keyword REG upper-case at byte 0 of the transfer, W/R either case; **one command per USB transfer**; the firmware holds **one command slot** — a second REG before execution is dropped, so the host waits for the reply. The line is captured in the OTG_FS ISR (CDC_Receive_FS → USBHandler::captureTextCommand) and executed from the main loop right after HostBridge_Poll() (main.cpp:1713-1726 in the beta) because SPI1 is shared with the ADAR1000 path; the transport refuses (REG ERR) while a frame read is active or any ADAR1000 CS is low. Text detection is suppressed while a synchronised binary settings packet is being assembled (test T10).

SPI side towards the FPGA, bridge command set v2 (source: beta/stm32/DECISIONS.md D-18): write = 8 clocked bytes 02 a0 a1 d0 d1 d2 d3 00, ack 0xA2 expected on MISO during the 8th byte; read = 7 bytes 03 a0 a1 00 00 00 00, data little-endian in bytes 3..6; status = 9 bytes 04 + 8 reply bytes (status u16, version u16, frames u16, reserved u16, LE; u16 widths **assumed**); 0xEE in the ack position = unknown command → NACK. **"The beta FPGA RTL (beta/fpga/rtl/host_bridge_spi.v) does not implement 0x02..0x04 yet ... STM32 side is ready, end-to-end untested."**

Bridge-frame forwarding (FPGA → firmware → host, option B)

The firmware waits for DRDY (EXTI on PD14 = DIG_6), pulls FPGA_CS_N (PD13 = DIG_5, reconfigured as output) low, clocks command byte 0x01 and reads one frame over SPI1 (mode 0, MSB first; the beta keeps the ADAR1000 setting of 6.75 MHz → ≈ 2.5 ms per frame), CRC-checks it and forwards it **unchanged** over CDC with CDC_Transmit_FS, interleaved with the status strings (source: beta/stm32/CHANGELOG.md, "host-link option B (DSN-LINK-01)"; engineering/DESIGN/HOST_LINK/HOST_LINK_DESIGN.md §5). Frame layout (little-endian) (source: HOST_LINK_DESIGN.md §5, copied):

Offset	Size	Field
0	2	sync 0xA5 0x5A
2	1	version = 1

Offset	Size	Field			
3	1	flags (bit0 = long-chirp set, bit1 = overflow since last frame)			
4	2	sequence number			
6	1	azimuth index (1..50)			
7	1	elevation index (1..31)			
8	2	chirp count			
10	1	n_range = 64			
11	1	n_doppler = 32			
12	2	n_det (≤ 32)			
14	2	reserved			
16	2048	magnitude map, uint8 = 8·log2(<table><tr><td>I</td><td>+</td><td>Q</td></tr></table>) saturated, range-major	I	+	Q
I	+	Q			
2064	3·n_det	detections: range u8, doppler u8, mag u8			
end	2	CRC-16/CCITT-FALSE over bytes 0..end-1			

Firmware rule: "the firmware must not start an ADAR1000 transaction while `HostBridge_Busy()` " (source: `beta/stm32/CHANGELOG.md`, option B entry). Build after this change: 0 errors, RAM 17 480 B (5.33 %), FLASH 93 276 B (8.90 %).

12.8 USB descriptors and CDC configuration

(source: `beta/stm32/CUBEMX_SETTINGS.md §4–§5`): USB_OTG_FS **Device_Only**, PA11 DM / PA12 DP (AF10, very high speed), VBUS sensing off, SOF off, low-power off, LPM off, PA10 (ID) unassigned (D-07); CDC (VCP) with `USBD_MAX_NUM_INTERFACES 1`, `USBD_SELF_POWERED 1`; descriptors VID 0x0483, PID 0x5740, LANGID 0x409 — **PLACEHOLDER (D-09)**, chosen "only because `GUI_V5.py:323–330` enumerates on that list"; serial number = STM32 96-bit UID; FIFO Rx 0x80 / Tx EP0 0x40 / Tx EP1 0x80 words; `APP_RX_DATA_SIZE 2048`, `APP_TX_DATA_SIZE 2048`; OTG_FS interrupt priority 0/0 (D-08); user code to re-insert after a CubeMX regeneration: `AERIS_USB_OnReceive(Buf, *Len);` in `usbd_cdc_if.c CDC_Receive_FS()` **USER CODE 6** (D-10).

12.9 Flashing and what remains

Flash command recorded for the bring-up (source: `beta/stm32/README.md §7` item 2): `STM32_Programmer_CLI -c port=SWD -w beta/stm32/build_out/aeris10_fw.elf -v -rst` (SWD on PA13/PA14; SWO PB3 on the schematic — `CUBEMX_SETTINGS.md §3`). The bring-up checks themselves are Steps 16.1–16.2 and 16.8.

Unresolved / remaining work (source: `beta/stm32/README.md §7`, condensed, numbering kept): (1) regenerate the CubeMX project from `CUBEMX_SETTINGS.md` and diff the generated `usbd_conf.c`, `usbd_desc.c`, `usbd_cdc_if.c`, `usb_device.c`, startup and linker script against the hand-written files; let CubeMX recompute the I2C TIMINGR (D-03) and confirm the clock tree (D-01/D-02); (2) hardware bring-up per chapter 16; (3) ADAR1000 phase accuracy on hardware; FPGA implementation of bridge commands 0x02..0x04 and an end-to-end `REG` test; (4) ADAR1000 SCLK limit and FPGA level-shifter bandwidth; AD9523/ADF4382 register-level correctness; ADF4382 `DELADJ` "PWM" is a stub (`adf4382a_manager.c:433–460`); (5) VBUS wiring to PA9 (D-07); production VID/PID (D-09); USB interrupt priority policy (D-08); (6) `main.cpp` logic not touched: 180 s OCXO wait at boot; `last_check` reused for the temperature timer (`main.cpp:1774`); raw ADC codes reported as temperatures; health-check thresholds; IDQ servo exit conditions; (7) I/D cache stays disabled, MPU background region as generated; (8) PA drain 22 V source/switch and stepper supply are not in CAD (K4); (9) static analysis (cppcheck) and `-fstack-usage` review not yet done.

Acceptance criteria touched by this chapter (source: `docs/TESTING/ACCEPTANCE_CRITERIA.md`): AC-S1...AC-S5 are recorded NOT MET against the **original** tree; AC-X2 ("beta firmware compiles and links; host tests pass") is **MET (BETA)**; AC-S6/AC-S7 (physical) are NOT RUN.

<!-- chapter 13: GUI: install, run, protocol, register panel, packaging -->

Host GUI software — install, run, protocol, register panel, packaging

Author of this chapter: Antidrone Ukraine · antidrone.cc (compilation of the BETA package `beta/gui`; the upstream scripts `9_Firmware/9_3_GUI/GUI_V1...V6*.py` are ORIGINAL PROJECT FILE and untouched).

Status summary: BETA — "This package has **never been run against hardware**" (source: `beta/gui/README.md`, "BETA statement"). Module and runtime diagrams SD-05/SD-06 are SOURCE-DERIVED from the original scripts. The test suite, the headless self-test and the PyInstaller bundle were executed on the authoring workstation (CPython 3.14, Tk 9.x, macOS arm64); Windows/Linux and older interpreters are unverified. The demo figure F13.3 is a render of the plot canvas only (see its caption).

Sources: `beta/gui/README.md`, `beta/gui/CHANGELOG.md`, `docs/GUI/INSTALLATION.md`, `docs/GUI/DEPENDENCIES.md`, `engineering/SOFTWARE_DIAGRAMS/PYTHON/python_gui_modules.md`, `beta/gui/aeris10_gui/protocol/register_map.py`, `beta/fpga/rtl/radar_control_regs.v`, `beta/gui/pyproject.toml`.

Figures in this chapter: F13.1 module graph (SD-05), F13.2 runtime architecture (SD-06), F13.3 demo-mode plot render.

13.1 Which program is the application

The upstream directory holds nine GUI versions; only one is runnable offline and only one is hardware-capable (source: `docs/GUI/DEPENDENCIES.md` §1, copied):

File	Lines	State	Runnable?
<code>GUI_V1.py</code>	41	fragment of one method with a leading indent — <code>IndentationError</code> line 2	no
<code>GUI_V2.py</code>	1 059	complete (STM32 CDC, FTDI, DBSCAN, Kalman, matplotlib)	parses, imports OK; hardware needed
<code>GUI_V3.py</code>	1 146	complete; pitch in GPS packet	parses; hardware
<code>GUI_V4.py</code>	1 427	complete; Google Maps HTML with placeholder API key (: 229)	parses; hardware; network
<code>GUI_V4_2_CSV.py</code>	678	offline CSV replay; two defects (: 354–368 derives columns before <code>read_csv</code> ; : 430 <code>lambda</code> captures unbound <code>e</code>)	parses; offline
<code>GUI_V5.py</code>	1 542	last complete hardware GUI ; V5 buffer-advance bug : 1313 vs : 809–816 (packet length always 6)	parses; hardware
<code>GUI_V5_Demo.py</code>	1 356	truncated (: 696 , : 1339 "Rest of the methods remain the same"); <code>RadarGUI</code> calls 9 undefined methods → <code>AttributeError</code> at : 955	no
<code>GUI_V6.py</code>	617	partial rewrite for FT601: <code>RadarProcessor</code> , <code>USBPacketParser</code> , <code>RadarPacketParser</code> , <code>MapGenerator</code> are pass stubs (: 90–92 , : 365–375); <code>create_gui/configure_dark_theme</code> pass (: 419–425); <code>STM32USBInterface</code> undefined (: 392); <code>get_packet_length</code> returns constant 64 (: 597–600)	no
<code>GUI_V6_Demo.py</code>	1 221	self-contained simulator (<code>SimulatedRadarProcessor</code> : 61–231), only numpy + matplotlib + tkinter	yes (offline) — module import verified; window not launched

No GUI version could decode the RTL's actual packet (A5C3 +CRC16 expected vs the RTL's `0xAA ... 0x55` without CRC — `docs/GUI/DEPENDENCIES.md` §6 row 5), and both FTDI paths (FT2232H in V2–V5, FT601 in V6) have no hardware on the Main Board (same §4). The BETA package `beta/gui/aeris10_gui` therefore re-implements the host from the firmware and RTL sources; the provenance of every beta module is tabulated in `beta/gui/README.md` ("Mapping: original file -> beta module") and `CHANGELOG.md` ("Provenance").

Example: `from math import sqrt` in `gui_v5.py` (SD-05) is derived from `math` module in `python_gui_modules.png` (SD-05) — SOURCE-DERIVED

Files with syntax errors: 1. Solid edge = unconditional import, dashed edge = import inside try/except (optional). `if __name__ == '__main__':` are listed inside each file node, not drawn as edges. No file imports another GUI file (no local edges found).

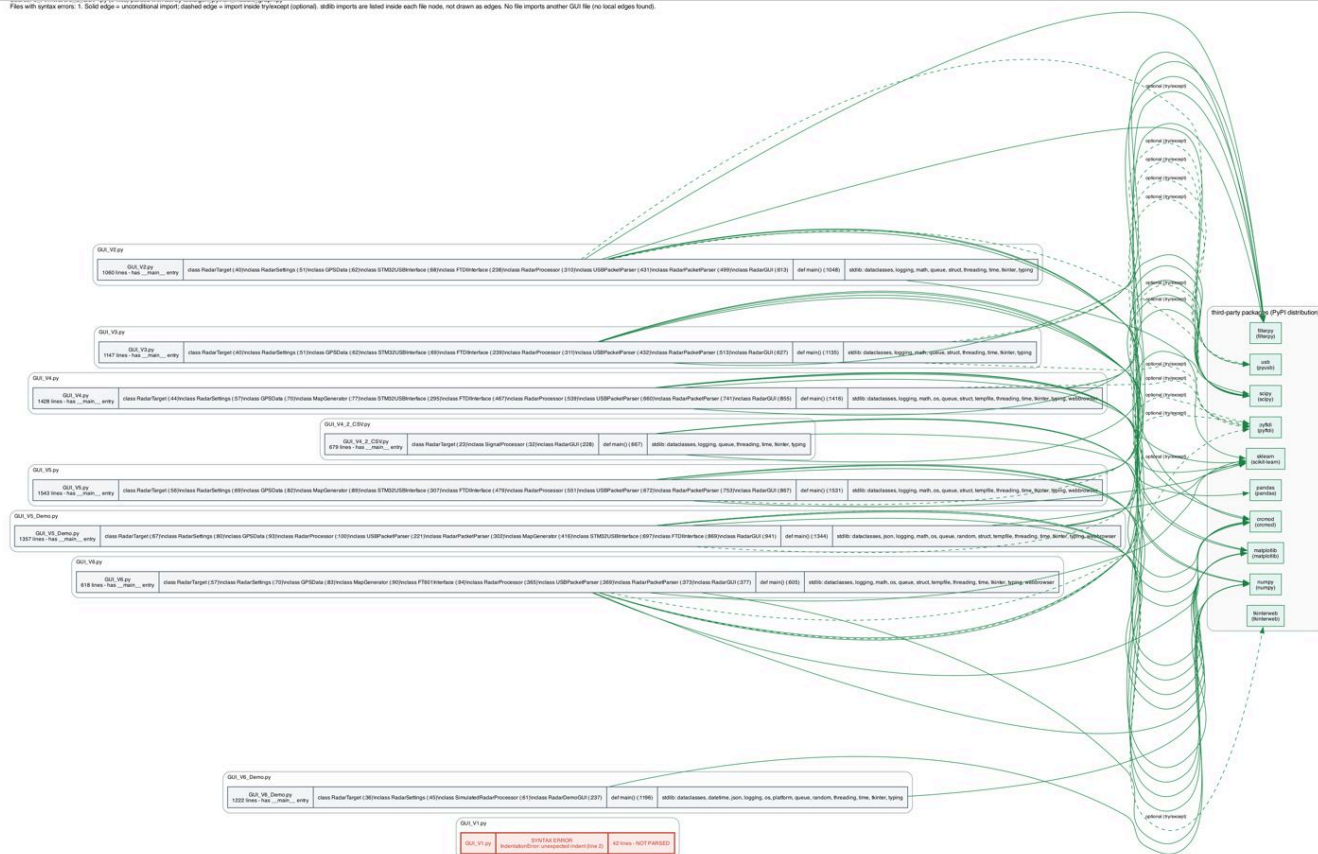


Figure 59 — F13.1 — Python GUI modules: imports and definitions of the nine upstream scripts, classified stdlib / third-party / local (SD-05; auto-generated) — SOURCE-DERIVED (source: engineering/SOFTWARE_DIAGRAMS/PYTHON/python_gui_modules.png; produced by tools/gen_python_module_graph.py + Graphviz dot)

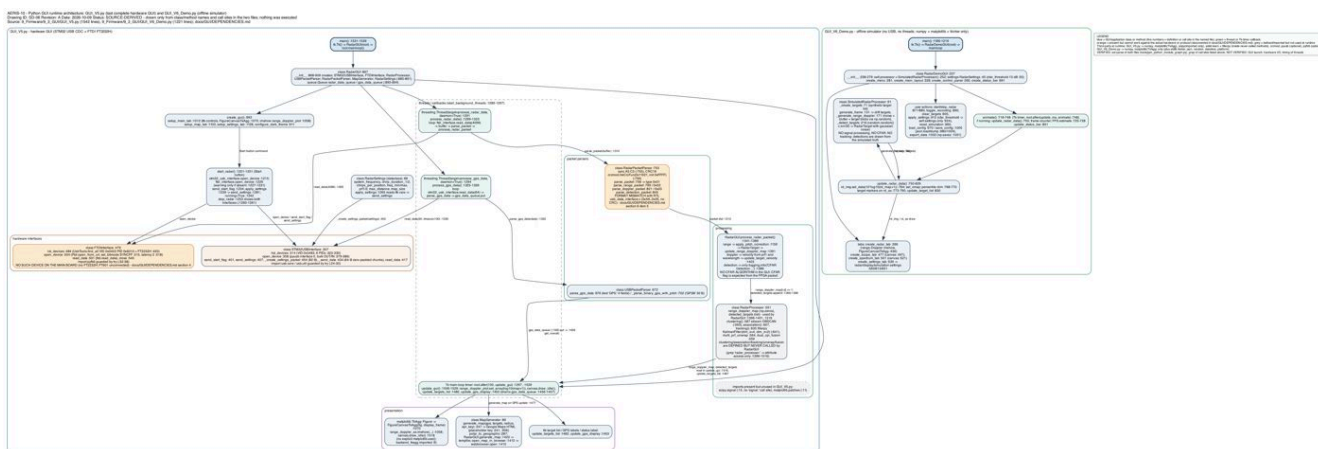


Figure 60 — F13.2 — Python GUI runtime architecture as found in GUI_V5.py / GUI_V6_Demo.py: threads, queues, parsers, processor, Tk timer; clustering/tracking defined but never called in V5 (SD-06) — SOURCE-DERIVED (source: engineering/SOFTWARE_DIAGRAMS/PYTHON/python_gui_architecture.png; produced by hand-authored DOT + Graphviz)

13.2 BETA package: what it implements

(source: beta/gui/README.md, "BETA statement", condensed)

- **FPGA → host data:** default path is the **SPI bridge (option B, DSN-LINK-01)**: the FPGA serves 64×32 log-magnitude frames over SPI, the STM32 forwards them unchanged over USB CDC interleaved with status strings and REG replies (frame layout in chapter 12 §12.7). Simulated and unit-tested only. The raw 35-byte RTL packet path (option A, FT601) remains behind `--raw-ft601`; "on the board the FT601 is not wired".
- **FPGA register access:** REG W/R over CDC and the "FPGA registers / ADC calibration" tab follow `beta/fpga/rtl/radar_control_regs.v`; the text protocol "is the same as the firmware's" (`beta/stm32/Core/Src/host_bridge_proto.c`): one command per USB transfer, replies `REG 0x%04X 0x%08X\r\n / REG ERR\r\n`, a write echoes the written value, single firmware command slot → the GUI sends one command at a time and waits (timeout + retransmit). "Not bench-tested; the top-level RTL still ties the register write port off."
- **STM32 CDC path** (settings upload, status/GPS reception): implemented from the firmware sources, "unverified on a board".
- **Range/velocity scaling** of the display rests on assumptions (§13.7).

Protocol facts the beta relies on (source: `beta/gui/README.md`, "Protocol facts", copied):

Item	Source	Beta
Start flag 17 2E 9E ED	<code>USBHandler.cpp:38</code>	<code>settings_packet.START_FLAG</code>
82-byte SET +3x >d +>I +6x >d + END, big-endian	<code>RadarSettings.cpp:23-120</code>	<code>build_settings_packet</code> , frozen vector in tests
SET must be at buffer offset 0 after the flag	<code>USBHandler.cpp:70</code>	no zero padding by default (<code>pad_to_64=False</code>)
Firmware value limits	<code>RadarSettings.cpp:86-101</code>	<code>validate_settings</code> (UI refuses values the firmware would reject)
FPGA packet = 11 FT601 words, 0xAA, 4 range words, 4 Doppler words, detection, 0x55	<code>usb_data_interface.v:39-160</code>	<code>fpga_packet.encode_words/decode_packet</code>
range_profile = {Q, I} Doppler word; doppler_real = bits 15:0	<code>radar_system_top.v:294-331</code> , <code>doppler_processor.v:244</code>	<code>encode_cell</code> , <code>check_top_level_consistency</code>
Emission order range-major, 64 x 32 cells	<code>doppler_processor.v:244-266</code> , <code>radar_receiver_final.v:291-293</code>	<code>processing.FrameAssembler</code>
Detection bit = `	I	+
Status string fields, ChirpCount last	<code>main.cpp:807-877</code>	<code>status_text.parse_status_line</code>
GPS: %.8f, %.8f, %.2f\r\n (UART3) and 30-byte GPSB (CDC)	<code>gps_handler.cpp:45-119</code>	<code>parse_gps_text</code> , <code>parse_gpsb</code>

Q	> 10000	<code>radar_system_top.v:318</code>	simulator
---	---------	-------------------------------------	-----------

Package modules (source: `beta/gui/README.md` mapping table; `CHANGELOG.md` "Added"): `model.py` (dataclasses, firmware defaults), `protocol/settings_packet.py`, `protocol/status_text.py`, `protocol/fpga_packet.py`, `protocol/bridge_frame.py`, `protocol/register_cmd.py`, `protocol/register_map.py`, `io/usb_cdc.py` (pyserial CDC-ACM primary, pyusb fallback on the CDC *data* interface), `io/ftdi_ft601.py` (stub raising `NotImplementedError`), `dsp/cfar.py` (new CA-CFAR), `dsp/clustering.py` (DBSCAN, now actually called), `dsp/tracking.py` (filterpy Kalman), `sim/simulator.py`, `sim/replay.py`, `sim/register_file.py` (in-memory model of `radar_control_regs.v`), `processing.py`, `ui/main_window.py`, `ui/theme.py`, `ui/sources.py` (LinkDecoder, default link bridge), `ui/register_panel.py`, `app.py/ __main__.py`.

13.3 Installation

Step 13.1 — Create the environment and install [BETA, executed]

- **Purpose:** reproducible interpreter with the pinned dependencies.
- **Parts & tools:** Python ≥ 3.10 with Tk (`python3 -c "import tkinter"` must work); tested with CPython 3.14.7 + Tk 9.0 on macOS arm64 (source: `beta/gui/README.md`, "Install"); `beta/gui/requirements.txt` ("pinned to the tested versions"), `pyproject.toml` (extras `dev = pytest ≥ 7`, `packaging = pyinstaller ≥ 6.0`).
- **Action:**

```
cd beta/gui
python3 -m venv .venv
.venv/bin/pip install -r requirements.txt      # pinned to the tested versions
# or, for development: .venv/bin/pip install -e .[dev,packaging]
```

pyusb (optional raw-USB fallback) needs the libusb-1.0 system library (`brew install libusb` / `apt install libusb-1.0-0`); nothing else needs system packages.

- **Check:** `.venv/bin/python -m aeris10_gui --version` prints `aeris10-gui 0.7.0b1` (version string from `app.py --version`; package version in `CHANGELOG.md` "0.7.0b1").
- **Figure:** —
- **Decision:** Tk 9.1 upgrade note — "Homebrew had upgraded `tk-tk` to 9.1, which broke `_tkinter` ... Fixed with `brew reinstall python-tk@3.14`" (source: `beta/gui/CHANGELOG.md`, "Environment note").

Dependency ranges (source: `docs/GUI/DEPENDENCIES.md` §2, for the upstream scripts; the beta pins the tested versions in `requirements.txt`): numpy 2.5.3 (≥1.24, <3, UNVERIFIED below 2.5.3), scipy 1.18.1, matplotlib 3.11.2 (TkAgg), scikit-learn 1.9.1, filterpy 1.4.5 ("last release 2018 ... runs on numpy 2.5.3 here; long-term maintenance risk" — `beta/gui/README.md` assumption 10), crcmod 1.7, pyusb 1.3.1, pyserial (added by the beta as the primary CDC transport), pyinstaller 6.22.3. `pyftdi` "does not support FT601" (`docs/GUI/DEPENDENCIES.md` §2) and is not used by the beta.

13.4 Running

Step 13.2 — Run in demo, self-test or hardware mode [BETA]

- **Purpose:** start the application against the simulator (no hardware) or against the STM32 CDC port.
- **Parts & tools:** the venv of Step 13.1; for hardware mode the STM32 CDC port (after chapter 16 Step 16.1).
- **Action:** (source: beta/gui/README.md, "Run", copied)

```
.venv/bin/python -m aeris10_gui --demo          # simulator, no hardware
.venv/bin/python -m aeris10_gui                # hardware mode: pick the STM32 CDC port, press Start
.venv/bin/python -m aeris10_gui --selftest      # hidden window, 3 simulated frames, exit 0 on success
.venv/bin/python -m aeris10_gui --port /dev/cu.usbmodemXXX  # hidden window, 3 simulated frames, exit 0 on success
.venv/bin/python -m aeris10_gui --demo --raw-ft601 # option A: raw RTL packets instead of bridge frames
```

Other options (source: beta/gui/aeris10_gui/app.py argparse): `--frames N` (frames for `--selftest`, default 3), `--update-ms` (UI poll interval, default 100), `--log-level`. After `pip install -e .` the same is available as `aeris10_gui [--demo]`. In demo mode the simulator emits per frame one **bridge frame** (build_frame, packed exactly like `rd_map_packer.v`) plus a status string in the firmware format, in one CDC byte stream, and "also answers REG commands from an in-memory model of `radar_control_regs.v`, so the register panel works without hardware".

- **Check:** `--selftest` exit 0. Executed for this edition on 2026-10-09 (`cd beta/gui && .venv/bin/python -m aeris10_gui --selftest`), output: `selftest: link=bridge frames=31 link_stats={'status': {'status': 31, 'gpsb': 0, 'reg': 31, 'errors': 0, 'dropped_bytes': 0}, 'bridge': {'frames': 31, 'crc_errors': 0, 'resyncs': 0}} reg_read_all=OK (sent=31, dropped_by_slot=0) -> OK` (the self-test keeps stepping until the register read-all of 31 single-slot commands is answered, hence 31 frames).
- **Figure:** F13.3.
- **Decision:** hardware mode depends on the firmware CDC path (chapter 12, C4/C5 fixed in the beta, never enumerated on a board) and on the placeholder VID/PID 0x0483:0x5740 (D-09).

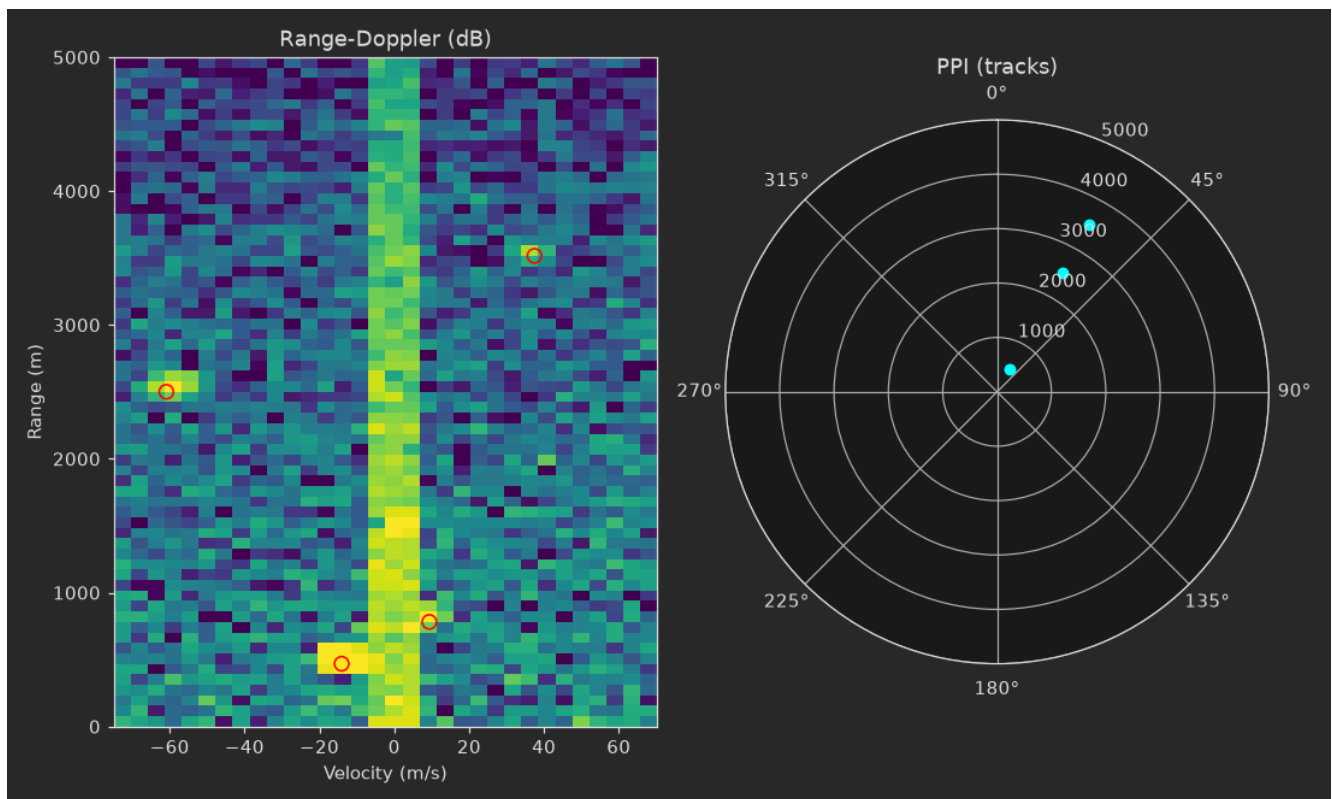


Figure 61 — F13.3 — Demo-mode plot area of the beta GUI after 5 simulated bridge frames: left range-Doppler map (dB) with CFAR detections (red circles), right PPI with Kalman tracks (cyan); range/velocity scaling UNVERIFIED (assumptions 4–6 of beta/gui/README.md). Rendered off-screen on 2026-10-09 by driving `MainWindow.step()` with the Tk root withdrawn and saving the matplotlib Figure (`<code>fig.savefig</code>`, 110 dpi) — only the plot canvas is captured; the Tk control row, notebook tabs and register panel are not in this image because no screen grab was taken — BETA (source: beta/gui/aeris10_gui/ui/main_window.py, sim/simulator.py; produced by a 20-line driver script equivalent to `app.py --selftest` plus `fig.savefig` into `manual/figures/gui_demo_plots.png`)

13.5 Tests

Step 13.3 — Run the test suite [BETA, 76 passed on 2026-10-09]

- **Purpose:** verify parsers, DSP, simulator, register client and UI construction without hardware.
- **Parts & tools:** `.venv/bin/python -m pytest -q in beta/gui` (pytest configured with `testpaths = ["tests"]`, `addopts = "-q"` in `pyproject.toml`).

- **Action:** run the command.
- **Check:** executed for this edition: **76 passed in 3.62s** (the README text still says 72 and docs/TESTING/ACCEPTANCE_CRITERIA.md AC-X3 says 55 — both older counts; the suite grew with the register-command rewrite, CHANGELOG.md "Tests"). Coverage as described (source: beta/gui/README.md, "Test"): bridge frames (RTL testbench vector tests/vectors/bridge_frame_from_rtl_tb.hex, a hardware source fed a synthetic CDC stream interleaving frames, status strings, GPS and REG replies), register commands (codec, RTL register semantics, client ↔ demo register file round trip, firmware single-slot model), settings packet (byte-exact firmware vector, round trip, a model of the firmware receiver proving unpadded framing is accepted and GUI_V5's 64-byte zero padding is rejected), FPGA packet (word- and byte-level vectors from the Verilog, corruption, truncation, resynchronisation), status/GPS/GPSB parsers, CA-CFAR, DBSCAN, Kalman tracker, simulator → parser → assembler exactness, pipeline target recovery, CSV loading, headless Tk smoke tests (skipped if no display).
- **Figure:** —
- **Decision:** none.

13.6 FPGA register panel and ADC calibration

The panel "FPGA registers / ADC calibration" (ui/register_panel.py) exposes (source: beta/gui/CHANGELOG.md, "Added"): CONTROL bits, CFAR threshold, decimation, start bin; auto (pattern) calibration, pattern-check enable, manual tap + lane load, bitflip + lane, pattern A/B; decoded CAL_STAT lock mask / done / busy / align_fail / fifo_ovf, chosen tap + pass window for all 8 lanes, CAL_ERR, CAL_UNDET, ID; "Read all / refresh". Inputs are range-checked against the RTL field widths. Register map transcribed from the RTL (source: beta/gui/aeris10_gui/protocol/register_map.py REGISTERS, which cites beta/fpga/rtl/radar_control_regs.v; registers are 16 bits wide, 4-bit word addresses as seen by the GUI):

Addr	Name	Access	Reset	Fields				
0x0	CONTROL	rw	0b101	bit0 use_long_chirp (1), bit1 adc_pwdn (0), bit2 usb_enable (1)	I	+	Q	threshold, 16 bit (10000 = original placeholder)
0x1	CFAR_THR	rw	10000	`				
0x2	DECIM	rw	0b01	bits[1:0] range decimation mode (01 = peak)				
0x3	START_BIN	rw	0	bits[9:0] first range bin passed to the decimator				
0x4	CAL_CTRL	rw	0	bit0 auto_start (write 1: start auto calibration, toggle), bit1 manual_load (toggle), bit2 bitslip_load (toggle), bit3 check_en (level: pattern-check error counting)				
0x5	CAL_LANE	rw	0	bits[2:0] lane 0..7 for CAL_TAP/CAL_SLIP writes and CAL_LANE_INFO read				
0x6	CAL_TAP	rw	16	bits[4:0] manual IDELAY tap 0..31 (78 ps each)				
0x7	CAL_SLIP	rw	0	bits[1:0] BITSLIP pulses for a manual bitslip load				
0x8	CAL_PATT	rw	0x55AA	{pattern_b[7:0], pattern_a[7:0]} expected alternating ADC test codes				
0x9	CAL_STAT	ro	0	{fifo_ovf, 4'b0, align_fail, busy, done, lock[7:0]}				
0xA	CAL_LANE_INFO	ro	0	{1'b0, win_hi[4:0], win_lo[4:0], tap[4:0]} of lane CAL_LANE				
0xB	CAL_ERR	ro	0	pattern-check error counter (saturating)				
0xC	CAL_UNDET	ro	0	{8'b0, undetermined[7:0]} lanes constant in pattern				
0xF	ID	ro	0xBE7A	beta build identifier				

The RTL additionally defines 0x0D CAL_BLIND_COEF, 0x0E CAL_BLIND_MARGIN and 0x10 CAL_BLIND_MIN for the blind method (source: beta/fpga/rtl/radar_control_regs.v address-map comment, lines 31-37); the GUI's 4-bit address map does not include them and the panel "shows it [blind calibration] as unavailable" (source: beta/gui/CHANGELOG.md, "Discrepancies" item 2). The calibration procedure using these registers is Step 16.5.

Recorded discrepancies (source: beta/gui/CHANGELOG.md, "Discrepancies / unresolved", condensed): (1) HOST_LINK_DESIGN.md §7 describes 32-bit registers with a different layout; the beta follows the RTL — run, mixers enable and the NCO word "are not offered because no RTL implements them"; (2) blind calibration not driven by any register in radar_control_regs.v (see above); (3) radar_system_top.v:321-323 ties reg_we/reg_addr/reg_wdata of ctl_regs to constants, so "register writes do not reach the register file on the current RTL"; (4) the firmware REG implementation exists (chapter 12 §12.7) and the GUI was re-aligned to it line for line; (5) no request ID — the single in-flight command relies on timeout/retransmit.

13.7 Assumptions and known limitations

(source: beta/gui/README.md, "Assumptions and known limitations", copied in condensed form, numbering kept)

1. **A1 — FT601 byte-lane order** little-endian; the RTL declares a 2-bit ft601_be for a 32-bit bus — an RTL defect to resolve at bring-up. Packet length 35 bytes follows from A1.
2. **No CRC in the RTL packet**; integrity uses header, footer and the redundant shifted copies.
3. **No bin indices in the packet**; any lost packet shifts the whole frame until the next resync (option A only).
4. **Range scaling UNVERIFIED**: cell $r \rightarrow r * \text{max_distance} / 64$ m; the physical spacing of the 64 decimated range bins is not documented anywhere in the repository.
5. **Velocity scaling** uses the standard pulse-Doppler relation with prf1 and system_frequency; not confirmed against the RTL.
6. **Azimuth angle** = (Azimuth-1) * 360/50 degrees from main.cpp:188-189; BeamPos is shown as an index.

8. The status string has no terminator; the parser uses `ChirpCount:<n>|` as end marker.
9. Google Maps export dropped.
10. `filterpy` 1.4.5 (2018) on numpy 2.5.3: maintenance risk.
11. Only CPython 3.14.7 (Tk 9.0, then 9.1) on macOS was exercised.
12. **Register map mismatch** between `HOST_LINK_DESIGN.md` §7 and the RTL; the GUI follows the RTL.
13. **REG text protocol** matches the firmware line for line, but neither side has been exercised on hardware.

Behavioural changes versus the originals that matter on the bench (source: `beta/gui/CHANGELOG.md`, "Behavioural changes", condensed): CDC writes are **not** zero-padded by default (checkbox "Zero-pad CDC writes" reproduces the legacy behaviour that the firmware rejects); firmware value limits are enforced in the UI before sending; the STM32 CDC port is opened with `pyserial` as an OS serial port (no mock devices on enumeration failure); the RTL's real packet format is decoded; GPS text with 3 fields (firmware) or 4 (legacy) is accepted; no background threads — a single Tk timer polls the source (`--selftest` is deterministic).

13.8 Packaging

Step 13.4 — Build the stand-alone bundle [BETA, executed by the beta author]

- **Purpose:** run the host application on a machine without Python.
- **Parts & tools:** `.venv/bin/pip install pyinstaller` (already in `requirements.txt`); `beta/gui/build_app.sh`; `pyinstaller_launcher.py` (because `aeris10_gui/__main__.py` uses a relative import).
- **Action:** `./build_app.sh` — `PyInstaller --onedir` → `dist/aeris10-gui/`, then runs `--demo --selftest`.
- **Check:** result recorded in the source: "PyInstaller 6.22.3 on Python 3.14.7 built a 134 MB `--onedir` bundle and `dist/aeris10-gui/aeris10-gui --demo --selftest` exited 0" (README "Package"); after the register work "135 MB ... passes `--demo --selftest` (bridge) and `--demo --selftest --raw-ft001`, both exit 0" (CHANGELOG). Not re-run for this edition.
- **Figure:** —
- **⚠ Decision:** AC-P7 ("Packaged demo runs on a machine without Python") remains NOT RUN — the bundle was only executed on the build machine.

Acceptance criteria touched by this chapter (source: docs/TESTING/ACCEPTANCE_CRITERIA.md): AC-P1, AC-P2 MET (upstream dependency install/import); AC-P3 NOT MET (gui_v1.py syntax); AC-P4, AC-P7 NOT RUN; AC-P5 "A unit-test suite exists and passes" is recorded NOT MET against the upstream tree and **MET (BETA)** as AC-X3 for beta/qui (count now 76); AC-P6 (hardware end-to-end) NOT RUN.

```
<!-- chapter 14: Manufacturing packages per board, stack-ups, BOMs with MPN confidence, fab checklist -->
```

14. Manufacturing packages

Chapter status summary: generated manufacturing packages per board — SOURCE-DERIVED (Frequency Synthesizer, RF PA) / PARTIAL (Main Board, Power Supply) in `engineering/PCB/<BOARD>/`, BETA in `beta/pcb/<BOARD>/exports/`; stack-ups — PARTIAL with PROPOSED fabrication notes; BOMs — PARTIAL (0 MPN in the source, BETA proposals with confidence classes); EAGLE-native export procedures — documented, not executed (EAGLE is not installed on the authoring machine). **The KiCad conversion is not the designer-released package:** the EAGLE files in `4_Schematics` and `Boards Layout/` remain the design master, and no package in this chapter may be sent to a fabricator without the checklist in §14.6 being closed. Compiled by Antidrone Ukraine · antidrone.cc.

Sources: `engineering/PCB/<BOARD>/README.md` (§1, §5, §8, §9) and `STACKUP.md` for the four boards; `beta/pcb/README.md`, `CHANGELOG.md`, `beta/pcb/<BOARD>/FAB_NOTES.md`, `BOM_<BOARD>_beta.csv`, `exports/EXPORT_LOG.md`; `docs/PCB/00_COMMON_EAGLE_PROCEDURES.md`; `docs/PCB/<BOARD>.md` §10; `docs/BOM/README.md`.

14.1 What exists, and where

Two generated package trees exist per board. Both were produced with `kicad-cli 10.0.6` from a KiCad import of the EAGLE board (`kicad-cli pcb import --format eagle, EAGLE layer 47 Measures` remapped to `Dwgs.User`, EAGLE design rules transferred to the project file — source: `engineering/PCB/<BOARD>/README.md` preserve-original paragraph). The conversion approximates EAGLE's polygon pours and thermals.

Board	Layers	Outline (mm)	Reference conversion (unmodified import)	BETA package (routed/cleaned copy)
Main Board (RADAR_Main_Board)	10	260 × 300	<code>engineering/PCB/MAIN_BOARD/</code> (gerber/, drill/, drawings/, svg/, assembly/, mechanical/, ipc/, 3d/, reports/, kicad/)	<code>beta/pcb/MAIN_BOARD/exports/</code> (same sub-directories) + <code>beta/pcb/MAIN_BOARD/MAIN_BOARD.kicad_pcb</code> , <code>FAB_NOTES.md</code> , <code>BOM_MAIN_BOARD_beta.csv</code>
Power Supply Board (PowerBoard)	2	280 × 300	<code>engineering/PCB/POWER_SUPPLY/</code> (gerber/, drill/, drawings/, svg/, assembly/, mechanical/, ipc/, 3d/, reports/, kicad/)	<code>beta/pcb/POWER_SUPPLY/exports/</code> (same sub-directories) + <code>beta/pcb/POWER_SUPPLY/POWER_SUPPLY.kicad_pcb</code> , <code>FAB_NOTES.md</code> , <code>BOM_POWER_SUPPLY_beta.csv</code>
Frequency Synthesizer (Clocks_Freq_Synth_board)	6	100 × 100	<code>engineering/PCB/FREQUENCY_SYNTHESIZER/</code> (gerber/, drill/, drawings/, svg/, assembly/, mechanical/, ipc/, 3d/, reports/, kicad/)	<code>beta/pcb/FREQUENCY_SYNTHESIZER/exports/</code> (same sub-directories) + <code>beta/pcb/FREQUENCY_SYNTHESIZER/FREQUENCY_SYNTHESIZER.kicad_pcb</code> , <code>FAB_NOTES.md</code> , <code>BOM_FREQUENY_SYNTHESIZER_beta.csv</code>
RF Power Amplifier (RF_PA)	4	35 × 60	<code>engineering/PCB/RF_PA/</code> (gerber/, drill/, drawings/, svg/, assembly/, mechanical/, ipc/, 3d/, reports/, kicad/)	<code>beta/pcb/RF_PA/exports/</code> (same sub-directories) + <code>beta/pcb/RF_PA/RF_PA.kicad_pcb</code> , <code>FAB_NOTES.md</code> , <code>BOM_RF_PA_beta.csv</code>

The "DRC before" figures are `engineering/PCB/<BOARD>/reports/DRC_report.json`, "after" `beta/pcb/<BOARD>/exports/reports/DRC_report.json`; KiCad caps each violation type at 199 entries (source: `beta/pcb/README.md`). The BETA export set is the same as that of `tools/kicad_pcb_pipeline.sh`, produced by `beta/pcb/tools/beta_export_package.sh` with the commands logged in `beta/pcb/<BOARD>/exports/EXPORT_LOG.md` (source: `beta/pcb/CHANGELOG.md` §All boards).

Files per package directory (file names for the Main Board; the other boards follow the same pattern with their own prefix and layer count — listing of `engineering/PCB/MAIN_BOARD/`):

Sub-directory	Files	Format / note
gerber/	<code>MAIN_BOARD-F_Cu.gtl</code> , <code>-In1_Cu.g1</code> ... <code>-In8_Cu.g8</code> , <code>-B_Cu.gbl</code> , <code>-F_Mask.gts</code> , <code>-B_Mask.gbs</code> , <code>-F_Paste.gtp</code> , <code>-B_Paste.gbp</code> , <code>-F_Silkscreen.gto</code> , <code>-B_Silkscreen.gbo</code> , <code>-F_Fab.gbr</code> , <code>-B_Fab.gbr</code> , <code>-Edge_Cuts.gm1</code> , <code>-job.gbrjob</code>	RS-274X X2, Protel extensions, 6 decimals, mm
drill/	<code>MAIN_BOARD-PTH.drl</code> , <code>-NPTH.drl</code> , <code>-PTH-drl_map.pdf</code> , <code>-NPTH-drl_map.pdf</code> , <code>drill_report.txt</code>	Excellon mm, PTH/NPTH separate
drawings/	<code>MAIN_BOARD_top_layer.pdf</code> , <code>_bottom_layer_mirrored.pdf</code> , <code>_copper_layers.pdf</code> (one page per copper layer), <code>_outline.pdf</code> , <code>_assembly_top.pdf</code> , <code>_assembly_bottom_mirrored.pdf</code>	PDF; the two assembly PDFs are clipped by the A4 page frame (chapter 4 §4.5)
svg/	<code>MAIN_BOARD-F_Cu.svg</code> ... <code>-B_Cu.svg</code> , <code>-F_Mask/-B_Mask</code> , <code>-F_Silkscreen/-B_Silkscreen</code> , <code>-F_Fab/-B_Fab</code> , <code>-Edge_Cuts.svg</code> , <code>MAIN_BOARD_top_composite.svg</code> , <code>MAIN_BOARD_bottom_composite_mirrored.svg</code>	per-layer plots (the layer-plot figures of chapters 4–7 are rendered from the two composites)
assembly/	<code>MAIN_BOARD_BOM.csv</code> , <code>MAIN_BOARD_pick_and_place.csv</code>	BOM from the schematic (values only); P&P both sides, mm
mechanical/	<code>MAIN_BOARD_outline.dxf</code> , <code>MAIN_BOARD_top_fab.dxf</code> , <code>MAIN_BOARD_board_only.step</code>	STEP is board body only (no component models)
ipc/	<code>MAIN_BOARD.xml.gz</code> (IPC-2581), <code>MAIN_BOARD_netlist.d356</code> (IPC-D-356)	for fabricator netlist test
3d/	<code>MAIN_BOARD_render_top.png</code> , <code>_bottom.png</code> , <code>_isometric.png</code>	KiCad renders, no component models

Sub-directory	Files	Format / note
reports/	DRC_report.txt, DRC_report.json, board_statistics.md, design_rules_mapping.md, kicad_import_report.txt	reference package only has the import report and the DRU mapping
kicad/	MAIN_BOARD.kicad_pcb, .kicad_pro, .kicad_prl	editable conversion (reference package); the BETA board is at beta/pcb/MAIN_BOARD/MAIN_BOARD.kicad_pcb

14.2 Deliverable tables (claude.md addendum §18), copied from the package READMEs

Main Board (RADAR_Main_Board) — engineering/PCB/MAIN_BOARD/README.md §1 [PARTIAL]

#	Item	File	Status
1	PCB top-layer drawing	drawings/MAIN_BOARD_top_layer.pdf	GENERATED
2	PCB bottom-layer drawing (mirrored)	drawings/MAIN_BOARD_bottom_layer_mirrored.pdf	GENERATED
3	Copper-layer views (one page per copper layer)	drawings/MAIN_BOARD_copper_layers.pdf	GENERATED
4	Board-outline drawing	drawings/MAIN_BOARD_outline.pdf	GENERATED
5	Mechanical dimensions (outline + holes, DXF / STEP)	mechanical/MAIN_BOARD_outline.dxf	GENERATED
6	Drill map	drill/MAIN_BOARD-PTH-drll_map.pdf	GENERATED
7	Hole table	README.md (§5) and drill/drill_report.txt	GENERATED
8	Component placement drawing (fab layer with pad outlines)	drawings/MAIN_BOARD_assembly_top.pdf	GENERATED
9	Top assembly drawing	drawings/MAIN_BOARD_assembly_top.pdf	GENERATED
10	Bottom assembly drawing (mirrored)	drawings/MAIN_BOARD_assembly_bottom_mirrored.pdf	GENERATED
11	Fabrication drawing	drawings/MAIN_BOARD_outline.pdf + drill maps + STACKUP.md	PARTIAL — no vendor notes (material, finish, tolerances) in the source
12	Layer stack-up documentation	STACKUP.md	PARTIAL — layer order from DRU; materials/thickness UNVERIFIED
13	BOM with reference designators	assembly/MAIN_BOARD_BOM.csv	GENERATED — 0 MPN attributes in the source (values only)
14	Pick-and-place file	assembly/MAIN_BOARD_pick_and_place.csv	GENERATED
15	Manufacturing export checklist	README.md (§8)	GENERATED

Additional exports: gerber/ (RS-274X X2, Protel extensions, *-job.gbrjob*), drill/ .drl (Excellon, PTH/NPTH separate), svg/ per-layer SVG, mechanical/MAIN_BOARD_board_only.step, mechanical/MAIN_BOARD_top_fab.dxf, ipc/MAIN_BOARD.xml (IPC-2581), ipc/MAIN_BOARD_netlist.d356 (IPC-D-356), 3d/*.png (KiCad 3-D renders, no component models), kicad/MAIN_BOARD.kicad_pcb + .kicad_pro.

Power Supply Board (PowerBoard) — engineering/PCB/POWER_SUPPLY/README.md §1 [PARTIAL]

#	Item	File	Status
1	PCB top-layer drawing	drawings/POWER_SUPPLY_top_layer.pdf	GENERATED
2	PCB bottom-layer drawing (mirrored)	drawings/POWER_SUPPLY_bottom_layer_mirrored.pdf	GENERATED
3	Copper-layer views (one page per copper layer)	drawings/POWER_SUPPLY_copper_layers.pdf	GENERATED
4	Board-outline drawing	drawings/POWER_SUPPLY_outline.pdf	GENERATED
5	Mechanical dimensions (outline + holes, DXF / STEP)	mechanical/POWER_SUPPLY_outline.dxf	GENERATED
6	Drill map	drill/POWER_SUPPLY-PTH-drll_map.pdf	GENERATED
7	Hole table	README.md (§5) and drill/drill_report.txt	GENERATED
8	Component placement drawing (fab layer with pad outlines)	drawings/POWER_SUPPLY_assembly_top.pdf	GENERATED
9	Top assembly drawing	drawings/POWER_SUPPLY_assembly_top.pdf	GENERATED
10	Bottom assembly drawing (mirrored)	drawings/POWER_SUPPLY_assembly_bottom_mirrored.pdf	GENERATED
11	Fabrication drawing	drawings/POWER_SUPPLY_outline.pdf + drill maps + STACKUP.md	PARTIAL — no vendor notes (material, finish, tolerances) in the source
12	Layer stack-up documentation	STACKUP.md	PARTIAL — layer order from DRU; materials/thickness UNVERIFIED
13	BOM with reference designators	assembly/POWER_SUPPLY_BOM.csv	GENERATED — 0 MPN attributes in the source (values only)
14	Pick-and-place file	assembly/POWER_SUPPLY_pick_and_place.csv	GENERATED
15	Manufacturing export checklist	README.md (§8)	GENERATED

Additional exports: gerber/ (RS-274X X2, Protel extensions, *-job.gbrjob*), drill/ .drl (Excellon, PTH/NPTH separate), svg/ per-layer SVG, mechanical/POWER_SUPPLY_board_only.step, mechanical/POWER_SUPPLY_top_fab.dxf, ipc/POWER_SUPPLY.xml (IPC-2581),

ipc/POWER_SUPPLY_netlist.d356 (IPC-D-356), 3d/*.png (KiCad 3-D renders, no component models), kicad/POWER_SUPPLY.kicad_pcb + .kicad_pro.

Frequency Synthesizer (Clocks_Freq_Synth_board) — engineering/PCB/FREQUENCY_SYNTHESIZER/README.md §1 [SOURCE-DERIVED]

#	Item	File	Status
1	PCB top-layer drawing	drawings/FREQUENCY_SYNTHESIZER_top_layer.pdf	GENERATED
2	PCB bottom-layer drawing (mirrored)	drawings/FREQUENCY_SYNTHESIZER_bottom_layer_mirrored.pdf	GENERATED
3	Copper-layer views (one page per copper layer)	drawings/FREQUENCY_SYNTHESIZER_copper_layers.pdf	GENERATED
4	Board-outline drawing	drawings/FREQUENCY_SYNTHESIZER_outline.pdf	GENERATED
5	Mechanical dimensions (outline + holes, DXF / STEP)	mechanical/FREQUENCY_SYNTHESIZER_outline.dxf	GENERATED
6	Drill map	drill/FREQUENCY_SYNTHESIZER-PTH-dr1_map.pdf	GENERATED
7	Hole table	README.md (§5) and drill/drill_report.txt	GENERATED
8	Component placement drawing (fab layer with pad outlines)	drawings/FREQUENCY_SYNTHESIZER_assembly_top.pdf	GENERATED
9	Top assembly drawing	drawings/FREQUENCY_SYNTHESIZER_assembly_top.pdf	GENERATED
10	Bottom assembly drawing (mirrored)	drawings/FREQUENCY_SYNTHESIZER_assembly_bottom_mirrored.pdf	GENERATED
11	Fabrication drawing	drawings/FREQUENCY_SYNTHESIZER_outline.pdf + drill maps + STACKUP.md	PARTIAL — no vendor notes (material, finish, tolerances) in the source
12	Layer stack-up documentation	STACKUP.md	PARTIAL — layer order from DRU; materials/thickness UNVERIFIED
13	BOM with reference designators	assembly/FREQUENCY_SYNTHESIZER_BOM.csv	GENERATED — 0 MPN attributes in the source (values only)
14	Pick-and-place file	assembly/FREQUENCY_SYNTHESIZER_pick_and_place.csv	GENERATED
15	Manufacturing export checklist	README.md (§8)	GENERATED

Additional exports: gerber/ (RS-274X X2, Protel extensions, `-job.gbrjob`), drill/ .dr1 (Excellon, PTH/NPTH separate), svg/ per-layer SVG, mechanical/FREQUENCY_SYNTHESIZER_board_only.step, mechanical/FREQUENCY_SYNTHESIZER_top_fab.dxf, ipc/FREQUENCY_SYNTHESIZER.xml (IPC-2581), ipc/FREQUENCY_SYNTHESIZER_netlist.d356 (IPC-D-356), 3d/*.png (KiCad 3-D renders, no component models), kicad/FREQUENCY_SYNTHESIZER.kicad_pcb + .kicad_pro.

RF Power Amplifier (RF_PA) — engineering/PCB/RF_PA/README.md §1 [SOURCE-DERIVED]

#	Item	File	Status
1	PCB top-layer drawing	drawings/RF_PA_top_layer.pdf	GENERATED
2	PCB bottom-layer drawing (mirrored)	drawings/RF_PA_bottom_layer_mirrored.pdf	GENERATED
3	Copper-layer views (one page per copper layer)	drawings/RF_PA_copper_layers.pdf	GENERATED
4	Board-outline drawing	drawings/RF_PA_outline.pdf	GENERATED
5	Mechanical dimensions (outline + holes, DXF / STEP)	mechanical/RF_PA_outline.dxf	GENERATED
6	Drill map	drill/RF_PA-PTH-dr1_map.pdf	GENERATED
7	Hole table	README.md (§5) and drill/drill_report.txt	GENERATED
8	Component placement drawing (fab layer with pad outlines)	drawings/RF_PA_assembly_top.pdf	GENERATED
9	Top assembly drawing	drawings/RF_PA_assembly_top.pdf	GENERATED
10	Bottom assembly drawing (mirrored)	drawings/RF_PA_assembly_bottom_mirrored.pdf	GENERATED
11	Fabrication drawing	drawings/RF_PA_outline.pdf + drill maps + STACKUP.md	PARTIAL — no vendor notes (material, finish, tolerances) in the source
12	Layer stack-up documentation	STACKUP.md	PARTIAL — layer order from DRU; materials/thickness UNVERIFIED
13	BOM with reference designators	assembly/RF_PA_BOM.csv	GENERATED — 0 MPN attributes in the source (values only)
14	Pick-and-place file	assembly/RF_PA_pick_and_place.csv	GENERATED
15	Manufacturing export checklist	README.md (§8)	GENERATED

Additional exports: gerber/ (RS-274X X2, Protel extensions, `-job.gbrjob`), drill/ .dr1 (Excellon, PTH/NPTH separate), svg/ per-layer SVG, mechanical/RF_PA_board_only.step, mechanical/RF_PA_top_fab.dxf, ipc/RF_PA.xml (IPC-2581), ipc/RF_PA_netlist.d356 (IPC-D-356), 3d/*.png (KiCad 3-D renders, no component models), kicad/RF_PA.kicad_pcb + .kicad_pro.

14.3 Stack-ups

Layer order and copper thickness in every STACKUP.md come from the EAGLE DRU; dielectric thicknesses are the EAGLE DRU table and

not a vendor stack-up; total thickness is not defined in any source (KiCad assumed 1.6 mm); material, prepreg/core assignment, finish and

impedance targets must be supplied by the designer/fabricator (source: `engineering/PCB/<BOARD>/STACKUP.md` caveats;

`engineering/MISSING_DRAWINGS_RECOVERY_PLAN.md` MDR-07). The only material evidence in the upstream project is `4_Schematics and Boards Layout/4_4_Board Stack-up/Stack_Hybrid.png` (a 6-layer picture, unlabelled, matching no board exactly) and the PCBWay impedance note delivered with the Frequency Synthesizer production files (RO4350B $h = 0.102$ mm, $50\ \Omega$ $w = 0.204$ mm, $100\ \Omega$ $w/s = 0.204/0.26$ mm, maskless RF, coupons).

Board	DRU	Layer setup (EAGLE)	DRU dielectrics (mm, top → bottom)	PROPOSED fabrication stack (BETA FAB_NOTES.md §2)	PROPOSED finished thickness
Main Board	PCBWay_8L_100um-Track (name says 8, board has 10 layers)	(1*2+3*4+5*12+13*14+15*16)	0.102, 0.2, 0.2, 0.2, 0.2, 0.15, 0.2, 0.2, 0.102	L1 and L9 on RO4350B 4 mil, FR-4 inside; ENIG; maskless RF; 1 oz all layers	2.0 mm ± 10 % (DRU sum 1.554 + copper ≈ 1.9 mm)
Power Supply	PCBWay_2L_100um-Track	(1*16)	1.5	FR-4 Tg 150 °C core; 35 µm, PROPOSED 70 µm / 2 oz; no impedance control	1.6 mm ± 10 %
Frequency Synthesizer	PCBWay_6L_100um-Track	(1+2*3+14*15+16)	0.11, 0.6, 0.11, 0.6, 0.11	L1 on RO4350B 4 mil (DRU 0.11 vs note 0.102 — reconcile), FR-4 cores 0.6, L5/6 RO4350B PROPOSED (B.Cu LVDS pairs); ENIG; maskless RF	1.6 mm ± 10 % with thinned cores, or 2.0 mm
RF PA	PCBWay_4L_100um-Track (unexplained 70 µm third copper slot)	(1+2*15+16)	0.11, 1.2, 0.11	F.Cu on RO4350B 4 mil, FR-4 core 1.2; paddle vias filled and capped PROPOSED; ENIG; maskless RF	1.6 mm ± 10 % (DRU sum 1.42 + copper ≈ 1.56 mm)

Full per-layer tables are in chapters 4–7 §Stack-up. Common proposal items for all boards (source: `beta/pcb/<BOARD>/FAB_NOTES.md` §3): ENIG 2–5 µm Au over 120–240 µm Ni; green LPI mask both sides, white silk; vias tented except RF-layer vias near maskless traces (fab to confirm); 100 % electrical test against the supplied IPC-D-356 netlist; IPC-A-600 Class 2; single boards, fab may panelise with rails, no V-cut through RF areas. Open items for every board (same source §6): confirm stack-up materials and total thickness; confirm the RF-layer dielectric (0.102 vs 0.11 mm) and re-solve w/s with coupons; confirm finish and mask-opening policy on RF traces; review the DRC disposition before CAM.

14.4 BOMs with MPN confidence

The source schematics carry **no MPN attribute on any part** (0/98, 0/28, 0/40, 0/11 lines; source: `docs/BOM/README.md`). `docs/BOM/BOM-<BOARD>.csv` is generated by `tools/gen_eagle_bom.py` directly from the `.sch` files; `beta/pcb/<BOARD>/BOM-<BOARD>-beta.csv` adds manufacturer, mpn, mpn_confidence, dnp, note with proposals from `beta/pcb/tools/beta_bom_mpn.py` (Murata GRM / Yageo RC / Murata LQP03 / TDK VLP families for passives, orderable IC codes for the device sets — source: `beta/pcb/README.md`).

Board	Physical references	Line items	References without value	HIGH (deviceset = MPN)	MEDIUM (standard passive from value+package)	LOW (guess / non-standard value / conflict)	EMPTY (no value)
Main Board	776	98	244	23 lines / 204 pcs	41 / 461	30 / 93	4 / 18
Power Supply Board	312	28	80	7 lines / 79 pcs	18 / 225	3 / 8	0 / 0
Frequency Synthesizer	184	40	47	10 lines / 37 pcs	22 / 110	6 / 29	2 / 8
RF Power Amplifier	25	11	6	5 lines / 6 pcs	6 / 19	0 / 0	0 / 0

Line counts obtained with `python3 -c "import csv,collections;print(collections.Counter(r['mpn_confidence'] for r in csv.DictReader(open('beta/pcb/<BOARD>/BOM-<BOARD>-beta.csv'))))"` on 2026-10-09; quantities from `beta/pcb/README.md` §BOM confidence; reference/line/value counts from `docs/BOM/README.md`. Conflicts flagged in the BETA note column that only the designer can resolve (source: `beta/pcb/README.md`): voltage ratings absent; non-E-series values (2.443 kΩ, 103 pF, 107.3 nH); 47 µF in 0201; NX3225 footprint with a 32.768 kHz value; 5 mΩ shunt with a 0.1 Ω part number; Main Board R60/R61/R83/R84/R145/R146 marked `dnp = yes` (no net). BOM validation procedure (source: `docs/BOM/README.md` §Validation): regenerate; compare `qty` sum with the element counts 776 / 312 / 25 / 184; look every UNVERIFIED line up at a distributor and write the MPN into the schematic part attribute `MPN`, then regenerate; fill every empty value (244/80/6/47); add a `DNP` attribute per variant; acceptance = 0 UNVERIFIED, 0 empty values, DNP column present, part count identical to the `.mnt` pick-and-place.

14.5 EAGLE-native export procedures (summary of `docs/PCB/00_COMMON_EAGLE_PROCEDURES.md`)

All four boards are EAGLE XML files: Main Board 7.4.0 (sch + brd), Frequency Synthesizer 9.6.2, RF PA 9.6.2, Power Board sch 9.6.2 / brd 7.4.0 (mismatch); libraries are embedded. EAGLE 9.6.2 opens all of them (and upgrades 7.4.0 files on save — keep the originals), as does Fusion 360 Electronics; KiCad 7/8/9 imports EAGLE 6+ XML. Output file names below are the tools' defaults and are **expected, not observed** — the upstream project never produced them (source: `00_COMMON_EAGLE_PROCEDURES.md` preamble). EAGLE is not installed on the authoring machine; none of these procedures has been executed here.

ID	Procedure	Tool / menu	Expected output	Acceptance
P-	Open and	EAGLE 9.6.2 File → Open →	both windows open	"Board and schematic are consistent"
EAGLE	consistency	2026-10-09; Schematic board opens with; ERC →		

ID	Procedure	Tool / menu	Expected output	Acceptance
01	check a board pair	<i>Consistency check.</i> Power Board: expect the "severed annotation" dialog, do not accept automatic fixes; export both part lists and reconcile		
P-EAGLE-02	ERC	schematic → Tools → ERC; reopen every <code><approved></code> entry (Main Board: 4 on STM32_MISO_1V8)	screenshot/transcript into <code>docs/PCB/reports/<board>_ERC_<date>.txt</code> (directory to create)	0 errors; every warning dispositioned
P-EAGLE-03	DRC with the stored rules	board → type RATSNEST (expect "Nothing to do" on Synth/PA; 2 390 airwires Main, 309 Power) → Tools → DRC → <i>Check</i> with the stored DRU	DRC error list, exported	0 unapproved errors; each approved error justified
P-EAGLE-04	Gerber + drill (EAGLE 9.6.2 CAM)	File → CAM Processor → Templates 2 Layer / 4 Layer / 6 Layer (Main Board: start from 6 Layer, add layers 3, 4, 5, 12, 13, 14 as inner copper); RS-274X, mm, 4.4, ZIP; Excellon mm, PTH/NPTH separate	<code>CAMOutputs/GerberFiles/copper_top.gbr</code> , <code>copper_inner_<n>.gbr</code> , <code>copper_bottom.gbr</code> , <code>soldermask_*</code> , <code>silkscreen_*</code> , <code>solderpaste_*</code> , <code>profile.gbr</code> , <code>gerber_job.gbrjob</code> ; <code>DrillFiles/drill_1_16.xln</code> , <code>drill_npth.xln</code>	viewer check: layer count and outline match <code>docs/MECHANICAL/drawings/<BOARD>_outline.svg</code> ; drill counts match <code>docs/PCB/<BOARD>.md</code>
P-EAGLE-05	Gerber + drill (EAGLE 7.4 CAM, for the 7.4.0 files)	<code>gerb274x.cam</code> / <code>excellon.cam</code> jobs	<code><board>.cmp</code> , <code>.sol</code> , <code>.ly2..</code> , <code>.plc/.pls</code> , <code>.stc/.sts</code> , <code>.crc/.crs</code> , <code>.drc</code> + <code>.dri</code>	prefer 9.6.2 unless the designer requires 7.4
P-KICAD-01	Alternative open-source export	KiCad File → Import → Non-KiCad Project → EAGLE; re-enter the 10-layer stack for the Main Board in Board Setup; <code>kiCad-cli pcb export gerbers / export drill</code>	<code>board-F_Cu.gbr</code> ... <code>board-PTH.drl</code> , <code>board-NPTH.drl</code> , <code>board-job.gbrjob</code>	re-DRC and compare with the EAGLE output before use (importer approximates polygons/thermals)
P-EAGLE-06	BOM export	schematic → File → Run ULP → <code>bom.ulp</code> → Parts/Values, CSV, all attributes; compare with <code>docs/BOM/BOM_<BOARD>.csv</code>	<code><Board>_BOM.csv</code>	every line has MPN + package; passives valued (244/80/6/47 missing today); DNP column
P-EAGLE-07	Pick-and-place and drawings	board → Run ULP → <code>mountsmd.ulp</code> (+ <code>mount.ulp</code> for THT); assembly drawing = layers 20, 21, 25, 51 (+ bottom) → Print → PDF 1:1; fab drawing = layers 20 + 44 + 45 + dimensions on 48 + text block (layer count, material, thickness, copper, finish, colours, impedance table from the PCBWay note, min track 0.1 mm, min drill 0.15 mm Main/Synth/PA, 0.3 mm Power)	<code><board>-smd.mnt</code> , <code>-tht.mnt</code> (RefDes, Value, Package, X, Y, Rot, Side), <code><board>_assembly_top/bottom.pdf</code> , <code><board>_fab.pdf</code>	Synth: regenerate the <code>.mnt</code> pair to supersede the four duplicate files
P-EAGLE-08	Schematic PDF	schematic → File → Print → PDF, all sheets, fit to page, black, caption	<code><board>_schematic.pdf</code> under <code>docs/PCB/schematics/</code> (directory to be created)	— (SOURCE-DERIVED renders exist in <code>engineering/ELECTRICAL/schematics/</code> meanwhile)
P-EAGLE-09	Netlist export without EAGLE	python3 <code>tools/extract_eagle_netlist.py <sch> --part <REF> --out <csv> / --list-parts</code> ; EAGLE alternative File → Export → Netlist	CSV net ↔ pin ↔ pad	used for the connection reports of chapters 4–7

Expected manufacturing package structure per board (source: `00_COMMON_EAGLE_PROCEDURES.md` §Manufacturing package structure):

```

4_7_Production Files/<Board>/
├─ Gerber/          copper_*.gbr, soldermask_*.gbr, silkscreen_*.gbr, solderpaste_*.gbr, profile.gbr, *.gbrjob
├─ Drill/          drill_*.pth.xln, drill_npth.xln, drill map pdf
├─ <Board>_fab.pdf
├─ <Board>_assembly_top.pdf, <Board>_assembly_bottom.pdf
├─ <Board>_BOM.csv   (with MPN, manufacturer, DNP)
├─ <Board>-smd.mnt, <Board>-tht.mnt
├─ <Board>_schematic.pdf
├─ <Board>_netlist.ipc (IPC-D-356, optional)
└─ reports/ ERC, DRC, impedance/stack-up confirmation from the fab

```

Board-specific notes for the EAGLE path: Main Board — P-EAGLE-04 with a **10-layer** job; reopen the 211 approved DRC entries; decide 8 vs 10 layers (copper exists on all 10) (source: `docs/PCB/MAIN_BOARD.md` §5, §7). Power Board — complete placement and routing first (designer); RATSNEST expects 309 airwires until then (source: `docs/PCB/POWER_SUPPLY.md` §5). Frequency Synthesizer — template *6 Layer* with inner layers 2, 3, 14, 15; keep `PCBWay_Impedance_Note_R04350B_h0p102mm.pdf` under `reports/` (source: `docs/PCB/FREQUENCY_SYNTHESIZER.md` §5, §9). RF PA — template *4 Layer* with inner layers 2 and 15; add the impedance-note text to the fab drawing only after the designer confirms it applies to this stack (source: `docs/PCB/RF_PA.md` §5).

14.6 Fabrication checklist (README §8 of every package)

AERONAUTICS Engineering Subsystems Manual The checklist is identical for the four boards except for the routing line; state on 2026-10-09 (source: engineering/PCB/<BOARD>/README.md

§8):

Item	Main Board	Power Supply	Frequency Synthesizer	RF PA
Gerber set (all copper, mask, paste, silk, outline, fab) exported	done	done	done	done
Excellon drill files PTH/NPTH + drill map + report	done	done	done	done
Pick-and-place (both sides, mm, CSV)	done	done	done	done
BOM with reference designators	done	done	done	done
BOM with manufacturer part numbers for 100 % of lines	open	open	open	open
Fabrication drawing with vendor notes (material, thickness, finish, mask/silk colours, tolerances, impedance)	open	open	open	open
Stack-up confirmed by the fabricator	open	open	open	open
Layout routed to completion (0 airwires / 0 unconnected)	open in the source (BETA: 0 unconnected, netlist issues remain)	open (BETA: 89 open)	done	open in the source (1 unconnected after import; BETA 0)
DRC with 0 unapproved errors, waivers documented	open	open	open	open
Designer review of the KiCad conversion against EAGLE (pours, thermals, text)	open	open	open	open
Gerbers viewed in an independent viewer and compared with the outline drawing	open	open	open	open
IPC-2581 / IPC-D-356 netlist test accepted by the fabricator	open	open	open	open

Known gaps common to all boards (source: engineering/PCB/<BOARD>/README.md §9, BLOCKED — MISSING DATA): board material, finished thickness, copper weight per layer, surface finish, mask/silk colours and impedance targets are not recorded in the EAGLE files; no MPN attributes; 3-D component models not embedded (Autodesk online URNs) so every STEP is board-only.

14.7 Acceptance criteria per board (copied from docs/PCB/<BOARD>.md §10)

Main Board (RADAR_Main_Board) — docs/PCB/MAIN_BOARD.md §10

Criterion	Measure
Layout complete	0 airwires; 0 elements outside 0..260 × 0..300 mm
DRC	0 unapproved errors with PCBWay 10-layer (or revised) rules
ERC	0 errors
Gerber set	10 copper + 2 mask + 2 silk + 2 paste + profile + job file; viewer check matches MAIN_BOARD_outline.svg (260 × 300 mm, 8 × Ø3.2 holes)
Drill	PTH count = 2 893 vias + plated pads; NPTH = 8 × 3.2 mm + 2 × 0.9 mm
Stack-up	vendor drawing for 10 layers, RO4350B 0.102 mm outer cores, 50 Ω at 0.204 mm, 100 Ω diff 0.204/0.26 mm confirmed
BOM	100 % MPN coverage; DNP column; quantity sum = 776 (minus DNP)
Design decisions closed	FT601 wired or removed; FPGA part; HSE crystal; XADC reference wiring

Power Supply Board (PowerBoard) — docs/PCB/POWER_SUPPLY.md §10

Criterion	Measure
Consistency	ERC consistency check passes after re-linking sch/brd in one EAGLE version
Placement/routing	0 airwires, 0 off-board parts, final outline recorded
DRC/ERC	0 errors
Gerber/drill	2 Cu layers + mask/silk/paste/profile; NPTH 8 × Ø3.2; viewer matches POWER_SUPPLY_outline.svg (280 × 300 mm, or the revised outline)
BOM	100 % MPN; all passives valued; datasheets for TPS562208 and LM2662 added to 7_Components Datasheets
Design closure	PA supply (22 V rail) architecture decided and consistent with Power Management V6.xlsx and RF_PA.sch

Frequency Synthesizer (Clocks_Freq_Synth_board) — docs/PCB/FREQUENCY_SYNTHESIZER.md §10

Criterion	Measure
DRC/ERC	0 errors, reports archived
Gerber/drill	6 Cu + mask/silk/paste/profile; NPTH 4 × Ø3.2; PTH = 769 vias + pads; viewer matches FREQUENCY_SYNTHESIZER_outline.svg (100 × 100 mm)
Stack-up	one vendor-confirmed 6-layer stack consistent with the impedance note
BOM	100 % MPN, 0 value-less references, oscillator conflict resolved
P&P	single regenerated pair of .mnt files, duplicates removed
Function (post-fab)	AD9523 lock (STATUS0/1 high), OUT6 = 100 MHz, OUT4/5 = 400 MHz, OUT10/11 = 120 MHz measured — hardware test, not a file check

RF Power Amplifier (RF_PA) — docs/PCB/RF_PA.md §10

Criterion	Measure
DRC/ERC	0 errors (already 0 approved in file; must be re-run)

Criterion	Measure
Gerber/drill	4 Cu + mask/silk/paste/profile; NPTH 7 × Ø3.2; PTH count = 342 vias + pads; viewer matches RF_PA_outline.svg (35 × 60 mm)
Stack-up	vendor drawing confirmed; RF layer mask-free as per impedance note
BOM	100 % MPN; all 25 references valued/identified
System	PA supply voltage/current source decided; quantity per variant (16 for AERIS-10X) confirmed; RF performance (gain, P1dB at 10.5 GHz) remains unverified until measured

Do not declare any board fabrication-ready on the basis of the existing .sch/.brd files, of the KiCad conversion or of the BETA packages (source: docs/PCB/MAIN_BOARD.md §10; beta/pcb/README.md).

14.8 Antenna panel package (PROPOSED DESIGN)

The proposed patch panel has its own KiCad-native package in engineering/DESIGN/ANTENNA/kicad_exports/ (Gerber F_Cu/B_Cu/F_Mask/B_Mask/F_Silkscreen/Edge_Cuts + job file, drill, PDF, SVG, STEP, 3-D render, statistics, DRC — all steps exit 0, source: kicad_exports/EXPORT_LOG.md). It is a design proposal awaiting simulation closure and a coupon measurement (chapter 8 §8.6) and is **not released** for fabrication.

<!-- chapter 15: Step-by-step mechanical assembly, board installation, harness, integration order, inspection checkpoints -->

Assembly procedure — PCB assemblies, head, harness, pedestal

Author of this chapter: Antidrone Ukraine · antidrone.cc.

Status summary: the whole procedure is **PROPOSED / BETA — never executed** (source: `manual/ASSEMBLY_STEPS_SOURCE.md`, header). Mechanical steps rest on the PROPOSED DESIGN of chapter 10 (D-07...D-13); electrical integration order is SOURCE-DERIVED from the connector matrix and the firmware power sequence (`engineering/ASSEMBLY/ASSEMBLY_SEQUENCE.md`, status PARTIAL); cable lengths are PROPOSED (`engineering/DESIGN/HARNESS/HARNESS_SCHEDULE.md`); Molex pin order, the RFIN/RFOUT side of the Main Board SMA pairs and the PA-instance assignment are UNVERIFIED (`engineering/SYSTEM/interfaces/interconnection_table.md` §2, §7). Steps that depend on an open decision carry $\triangle$ and name it. Torque values quoted from the step source have no deeper repository source and are marked ASSUMED.

Sources: `manual/ASSEMBLY_STEPS_SOURCE.md` (phases A–D), `engineering/ASSEMBLY/ASSEMBLY_SEQUENCE.md` (ASM-SEQ-01), `engineering/DESIGN/MECHANICAL/MECHANICAL_PARTS_LIST.md` (part # and fasteners), `engineering/DESIGN/MECHANICAL/drawings/DSN-MECH-04...07`, `engineering/DESIGN/HARNESS/HARNESS_SCHEDULE.md` (CBL-001...144), `engineering/SYSTEM/interfaces/interconnection_table.md`, `engineering/ELECTRICAL/power_distribution/power_rails.md`, `engineering/MECHANICAL/dimensions/*.md`, `tools/design_layout.py` (stations), `beta/pcb/<BOARD>/BOM-<BOARD>-beta.csv`.

Figures in this chapter: F15.1 exploded view, F15.2–F15.5 per-board assembly drawings, F15.6 DSN-MECH-07, F15.7 DSN-MECH-06, F15.8 DSN-MECH-04, harness schedule excerpt (table).

15.0 Safety preamble — read before any step

Copied from the step source (source: `manual/ASSEMBLY_STEPS_SOURCE.md`, "Safety preamble"), with the evidence behind each hazard:

- **S1 — 22 V PA drain supply (up to 45 A pulsed):** never connect PA boards with VG at 0 V; VG –4 V first (firmware sequence). Evidence: QPA2962 bias-up "VG –4 V → VD +22 V → raise VG until IDQ 1680 mA" (source: `engineering/ELECTRICAL/power_distribution/power_rails.md` §1 row +22V0/VD, xlsx L58-L62); 45 A pulse peak from D-14 (source: `engineering/DESIGN/00_DESIGN_BASIS.md` §2). The 22 V source itself is **not in any CAD file** (conflict K4).
- **S2 — RF radiation:** no transmission without a terminated or connected antenna/PA path; 16 × 10 W peak (source: QPA2962 PSAT 40 dBm, `engineering/DESIGN/00_DESIGN_BASIS.md` §1).
- **S3 — Rotating pedestal:** keep hands clear; set the stepper-driver current limit before first motion (TB6600-class, 9–42 V, 4 A, 20 V selected — source: `engineering/DESIGN/00_DESIGN_BASIS.md` §1, xlsx row 63).
- **S4 — ESD:** all boards are ESD-sensitive (GaN PA, ADC, FPGA).

Also applicable throughout: $\triangle$ **K2** — the beta firmware assumes the 8 MHz HSE crystal found on the schematic (`beta/stm32/DECISIONS.md` D-01); $\triangle$ **K4** — 22 V PA supply not in CAD; $\triangle$ **D-14** — per-pulse drain gating required by the thermal design (chapter 10 §10.7) has no hardware yet.

15.1 Pre-assembly inspection checkpoints

Copied (source: `engineering/ASSEMBLY/ASSEMBLY_SEQUENCE.md` §A):

CP	Check	Acceptance	Evidence/tool
CP-1	Bare board matches outline and hole table	dimensions per <code>engineering/MECHANICAL/dimensions/<BOARD>-dimensions.md</code> , ±0.2 mm (vendor tolerance UNSPECIFIED)	caliper; fab CoC
CP-2	Assembled board matches BOM + pick-and-place	every reference populated with the BOM value; polarity per <code>drawings/<BOARD>-assembly_top.pdf</code>	AOI / visual
CP-3	No shorts between rails	> 1 kΩ between every rail and GND before power-up (rail list <code>engineering/ELECTRICAL/power_distribution/power_rails.md</code>)	DMM
CP-4	Power Board rails at nominal with no load	each X2...X35 output within the net-name voltage (currents UNKNOWN)	bench PSU + DMM

- CBL-40..55 RF, CBL-56..71 VG, CBL-72..87 sense: Main→PA (cable IDs are proposals; types/lengths UNVERIFIED)
- CBL-30.. clocks/LO Synth→Main (SMA coax) (cable IDs are proposals; types/lengths UNVERIFIED)
- CBL-20 enable bus SV1→SV1 (20-way) (cable IDs are proposals; types/lengths UNVERIFIED)
- CBL-1 CBL-15 rails Power→Main (Molex 22-23-2021) (cable IDs are proposals; types/lengths UNVERIFIED)

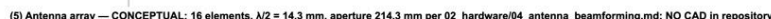


Figure 62 — F15.1 — Conceptual exploded view of the electronics set used as the orientation reference for phases A–C; board outlines verified from the .brd files, arrangement conceptual (ASM-EXP-01) — CONCEPTUAL (source: engineering/ASSEMBLY/EXPLODED_VIEWS/aeris10_exploded_conceptual.png; produced by tools/gen assembly exploded view.py)

15.2 Phase A — PCB assemblies (repeat per board: Main, Power, Synth, 16 × RF PA)

Step 15.1 — Inspect the bare boards [PROPOSED / SOURCE-DERIVED acceptance values]

- **Purpose:** confirm the fabricated board matches the verified outline and hole table before anything is mounted (CP-1).
- **Parts & tools:** bare PCBs; caliper; `engineering/MECHANICAL/dimensions/<BOARD>_dimensions.md`; fab certificate of conformance.
- **Action:** measure the outline bounding box and every mounting hole; compare with chapter 10 §10.1 (Main 260.00 × 300.00, 10 holes; Power 280.00 × 300.00, 8 holes; Synth 100.00 × 100.00, 4 holes; RF PA 35.00 × 60.00, 7 holes — all Ø3.20 NPTH).
- **Check:** dimensions within ±0.2 mm (vendor tolerance UNSPECIFIED); hole count exact; board thickness recorded (it is ASSUMED 1.6 mm everywhere in the design — G-01).
- **Figure:** F10.10a (plan view of the four outlines).
- **⚠ Decision:** G-10 — the Power Board's inner hole pattern "suggests a smaller board than 280 × 300"; if the delivered outline differs, the carrier rails #29/#30 (300 mm) and standoffs #31–#38 must be re-checked.

Step 15.2 — Populate per BOM and pick-and-place [BETA BOM, SOURCE-DERIVED drawings]

- **Purpose:** assembled board matches the BOM and the placement data (CP-2).
- **Parts & tools:** `beta/pcb/<BOARD>/BOM_<BOARD>_beta.csv` (with MPN confidence column) and `beta/pcb/<BOARD>/exports/pos` placement files; reflow/hand assembly equipment; AOI or magnifier.
- **Action:** place every reference listed in the BOM at the position and rotation of the P&P file; orient polarised parts per the assembly drawing (F15.2–F15.5).
- **Check:** every reference populated; no reference left with an empty value (the original EAGLE BOMs have 0 MPN attributes — AC-B5 NOT MET, so the beta MPN proposals must be reviewed by the owner before purchase); polarity per the assembly drawing.
- **Figure:** F15.2–F15.5.
- **⚠ Decision:** K1 (FPGA part XC7A50T on the schematic vs XC7A100T in the FPGA README) — the Main Board U42 placement follows the schematic part; the Main Board rev. B (FT601 wired) is a separate BETA layout (`beta/pcb/MAIN_BOARD_REVb/`), not the baseline of this step.

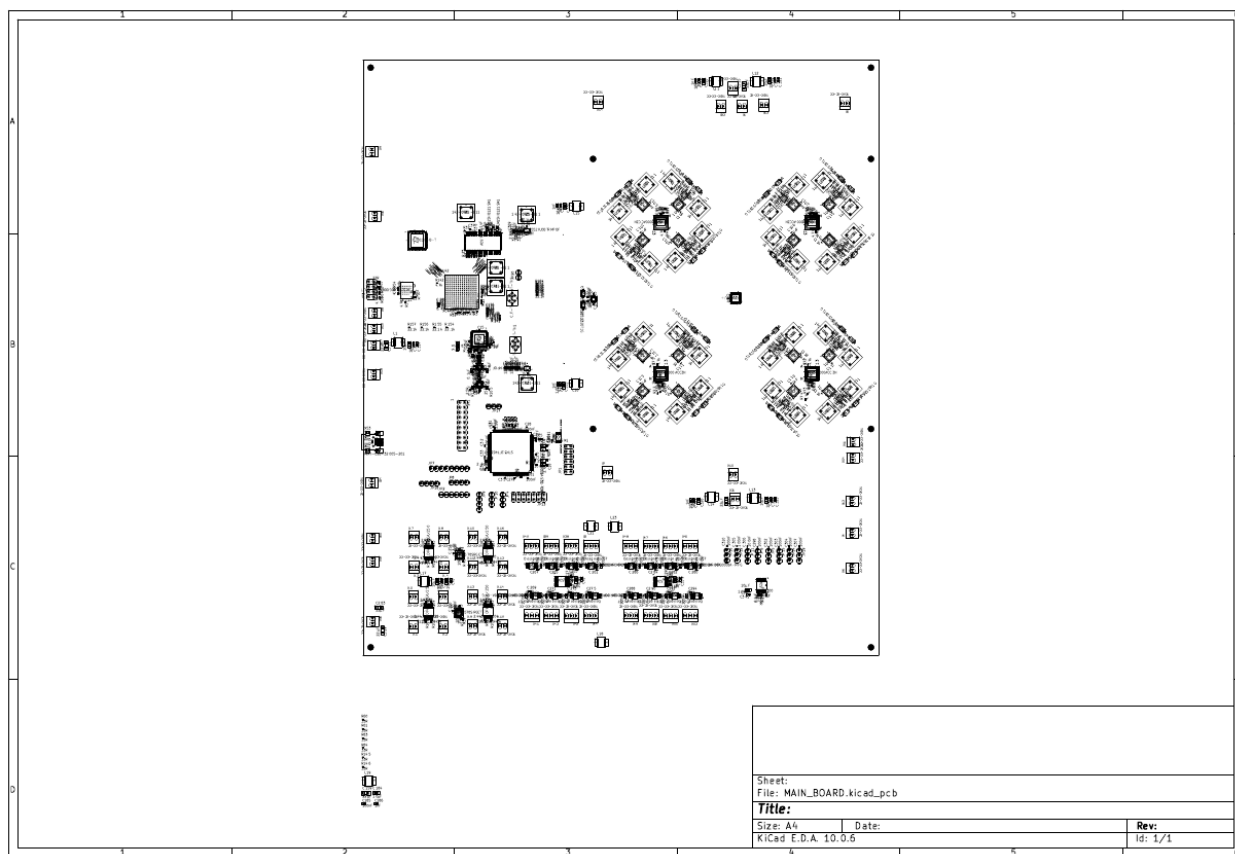


Figure 63 — F15.2 — Main Board, top assembly drawing (component placement, references), page 1 of engineering/PCB/MAIN_BOARD/drawings/MAIN_BOARD_assembly_top.pdf; layout state PARTIAL (2 390 airwires in the EAGLE source) — SOURCE-DERIVED / PARTIAL (source: engineering/PCB/MAIN_BOARD/drawings/MAIN_BOARD_assembly_top.pdf; produced by tools/kicad_pcb_pipeline.sh (kicad-cli 10.0.6), rendered to PNG by `pdftoppm -png -r 110 -f 1 -l 1`)

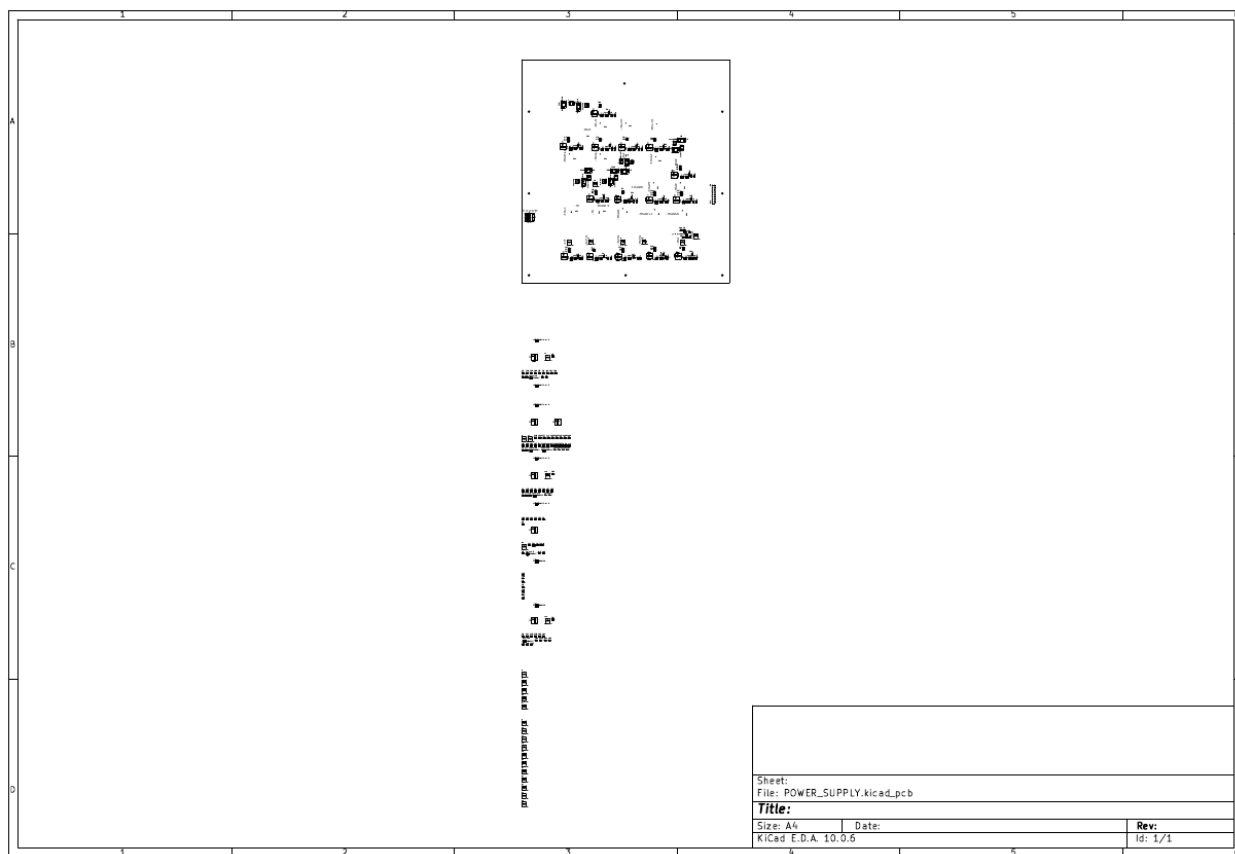


Figure 64 — F15.3 — Power Supply Board, top assembly drawing, page 1; layout state PARTIAL (309 airwires in the EAGLE source) — SOURCE-DERIVED / PARTIAL (source: engineering/PCB/POWER_SUPPLY/drawings/POWER_SUPPLY_assembly_top.pdf; produced by tools/kicad_pcb_pipeline.sh, rendered by pdftoppm)

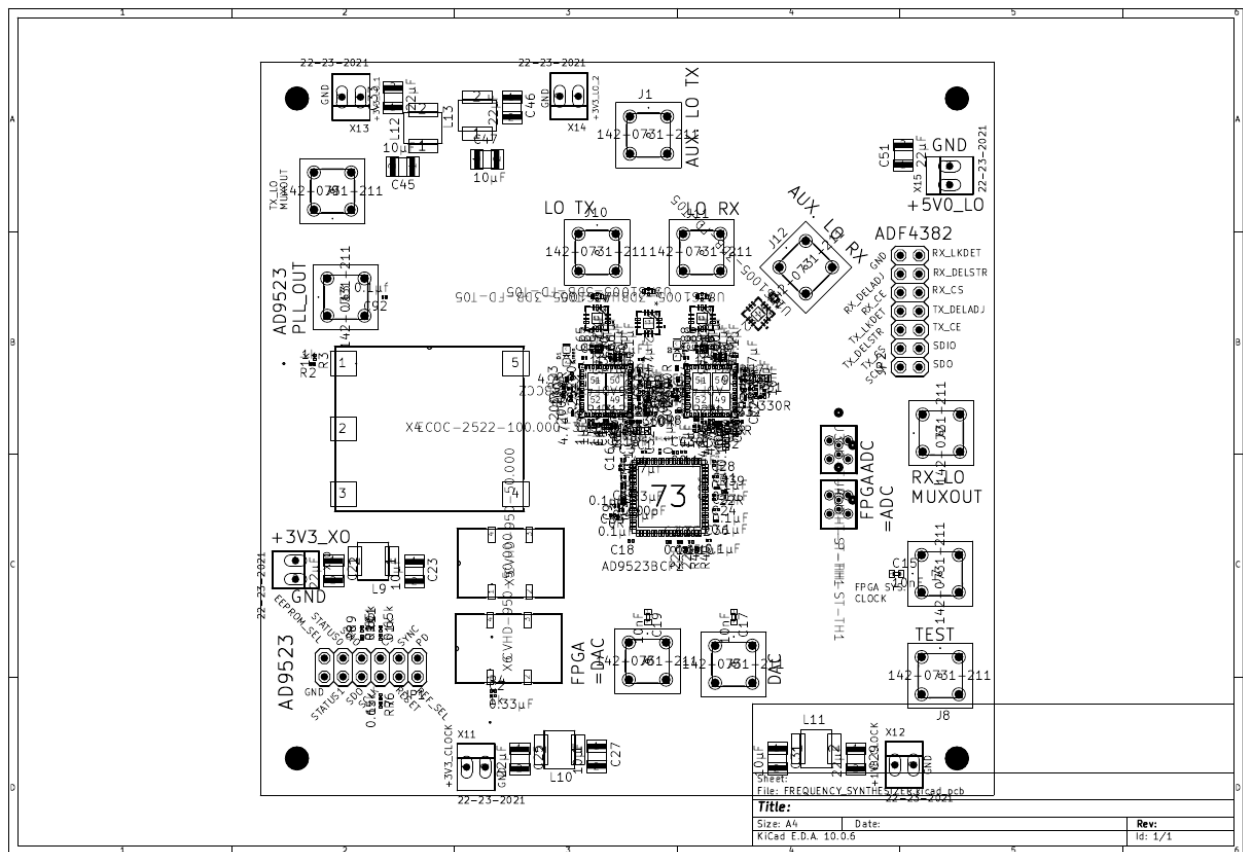


Figure 65 — F15.4 — Frequency Synthesizer Board, top assembly drawing, page 1 — SOURCE-DERIVED (source: engineering/PCB/FREQUENCY_SYNTHESIZER/drawings/FREQUENCY_SYNTHESIZER_assembly_top.pdf; produced by tools/kicad_pcb_pipeline.sh, rendered by pdftoppm)

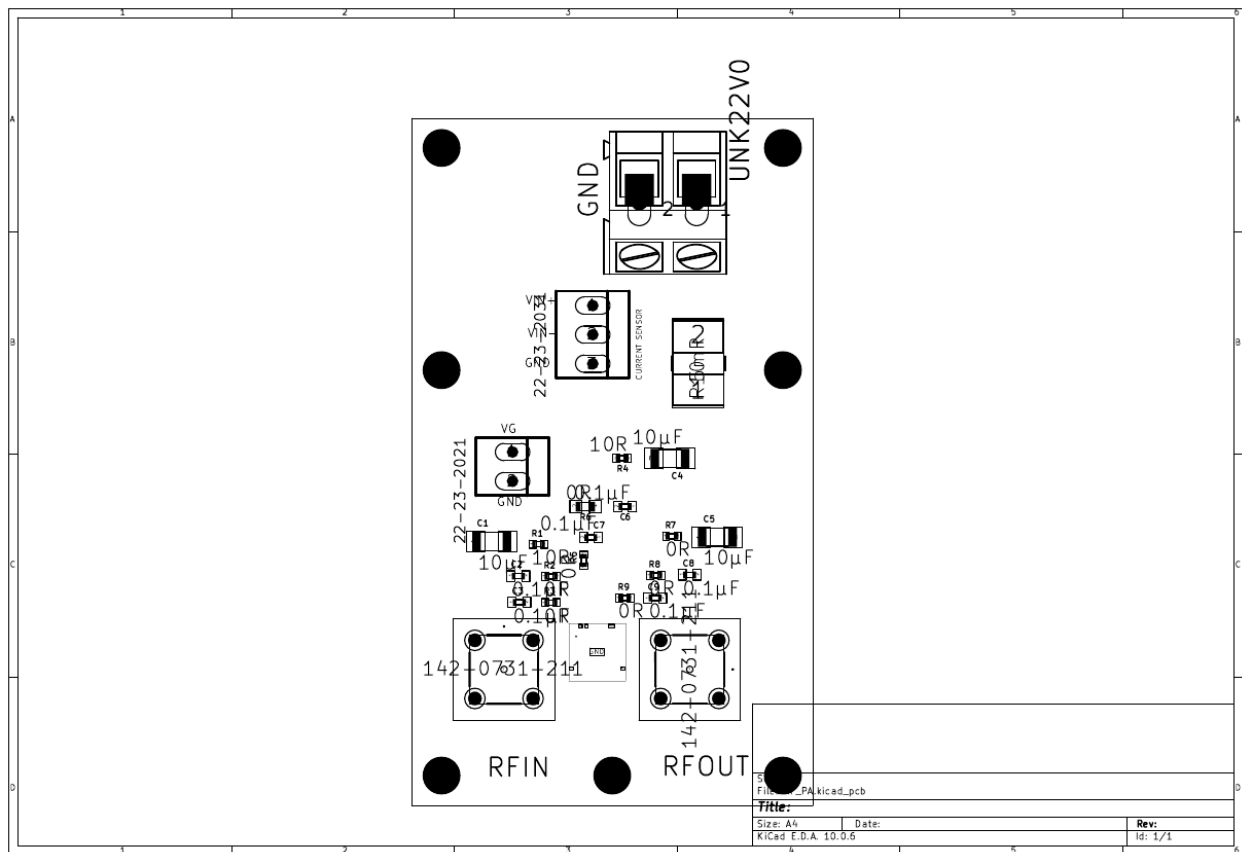


Figure 66 — F15.5 — RF PA board, top assembly drawing, page 1 (one of 16 identical boards) — SOURCE-DERIVED (source: engineering/PCB/RF_PA/drawings/RF_PA_assembly_top.pdf; produced by tools/kicad_pcb_pipeline.sh, rendered by pdftoppm)

Step 15.3 — Rail-to-ground resistance before power [SOURCE-DERIVED rail list]

- **Purpose:** detect solder shorts before any rail is energised (CP-3).
- **Parts & tools:** DMM; rail register engineering/ELECTRICAL/power_distribution/power_rails.md §1 (36 rows).
- **Action:** on every assembled board measure resistance between each rail net and GND at its connector (Power Board outputs X2... X35; Main Board inputs X1, X4...X24, X55, X56; Synth inputs X10...X15; PA 22v, X2 VG).
- **Check:** > 1 kΩ on every rail (source: engineering/ASSEMBLY/ASSEMBLY_SEQUENCE.md CP-3). Record the values.
- **Figure:** F3.2 (power distribution, chapter 3).
- **Decision:** none.

Step 15.4 — Power Board alone on the bench supply [SOURCE-DERIVED, design review item]

- **Purpose:** verify every regulator output with no load (CP-4) before the enable bus is ever driven.
- **Parts & tools:** bench PSU 12–17 V (VIN range from the board silkscreen V_{in} [12–17]V, source: power_rails.md §1 row VIN), DMM.
- **Action:** connect VIN/GND to X1 (AK300/2, KL1 VIN, KL2 GND — source: interconnection_table.md §1). Measure each output X2... X35 against its net-name nominal: always-on rails (+3V3 X16, +3V3_AN X12, +3V3_XO X35, +5V0_LO X6, +3V3_LO_1 X8, +3V3_LO_2 X7, +3V4 X23, -3V4 X24, +5V0_0 X22, +5V0_1...5 X2/X9/X17/X25/X28, +5V0_ADTR X26) are present with no enable; enabled rails stay at 0 V until the SV1 enable lines are driven (all EN nets are MCU outputs initialised LOW).
- **Check:** each present output within the net-name voltage (currents UNKNOWN — not documented anywhere, power_rails.md §4 item 5); the enabled rails read 0 V with SV1 open.
- **Figure:** F3.2.
- **Decision:** +5V0_LO LDO U30 (ADM7151) is fed directly from the 12–17 V bus — "CONFLICT / REQUIRES DESIGN REVIEW ... check against the ADM7151 input-voltage rating" (source: power_rails.md §1 row +5V0_LO, §4 item 2). Do not exceed the LDO's rating until the owner resolves it.

15.3 Phase B — Mechanical assembly of the radar head

Part numbers (#) are those of engineering/DESIGN/MECHANICAL/MECHANICAL_PARTS_LIST.md (copied in chapter 10 §10.5). Station coordinates (Y, mm from the inner face of the front wall) are the values printed by python3 tools/design_layout.py (chapter 10 §10.3). The step source (manual/ASSEMBLY_STEPS_SOURCE.md, phase B) uses an older part numbering in some rows; the numbers below follow the parts list, and each divergence is noted.

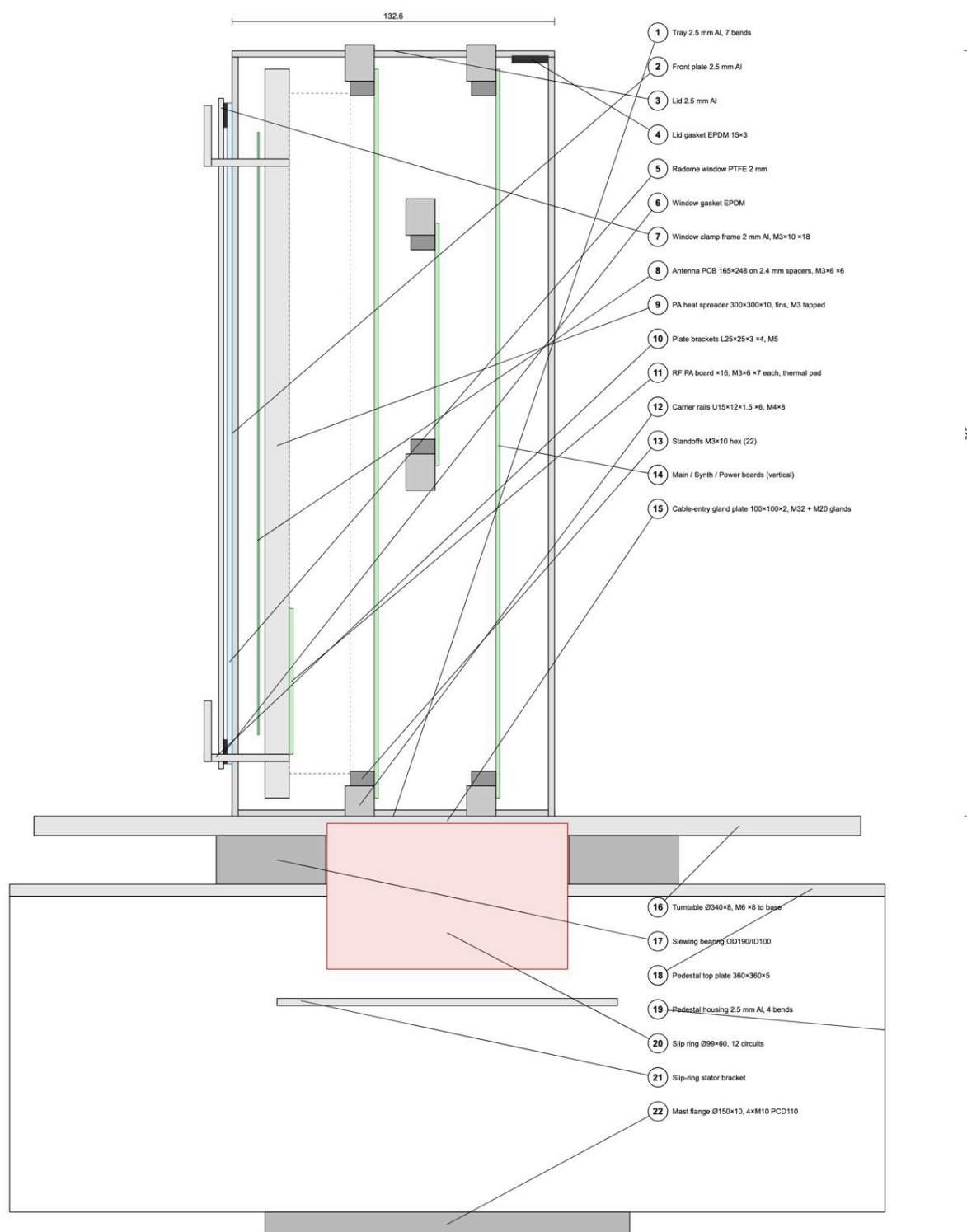


Figure 67 — F15.6 — Assembly section with fastener balloons: balloon number = part # of the mechanical parts list; defines which fastener goes where in Steps 15.5–15.13, DSN-MECH-07 Rev A — PROPOSED DESIGN (source: engineering/DESIGN/MECHANICAL/drawings/png/DSN-MECH-07_assembly_section_fasteners.png; produced by tools/design_enclosure_drawings.py)

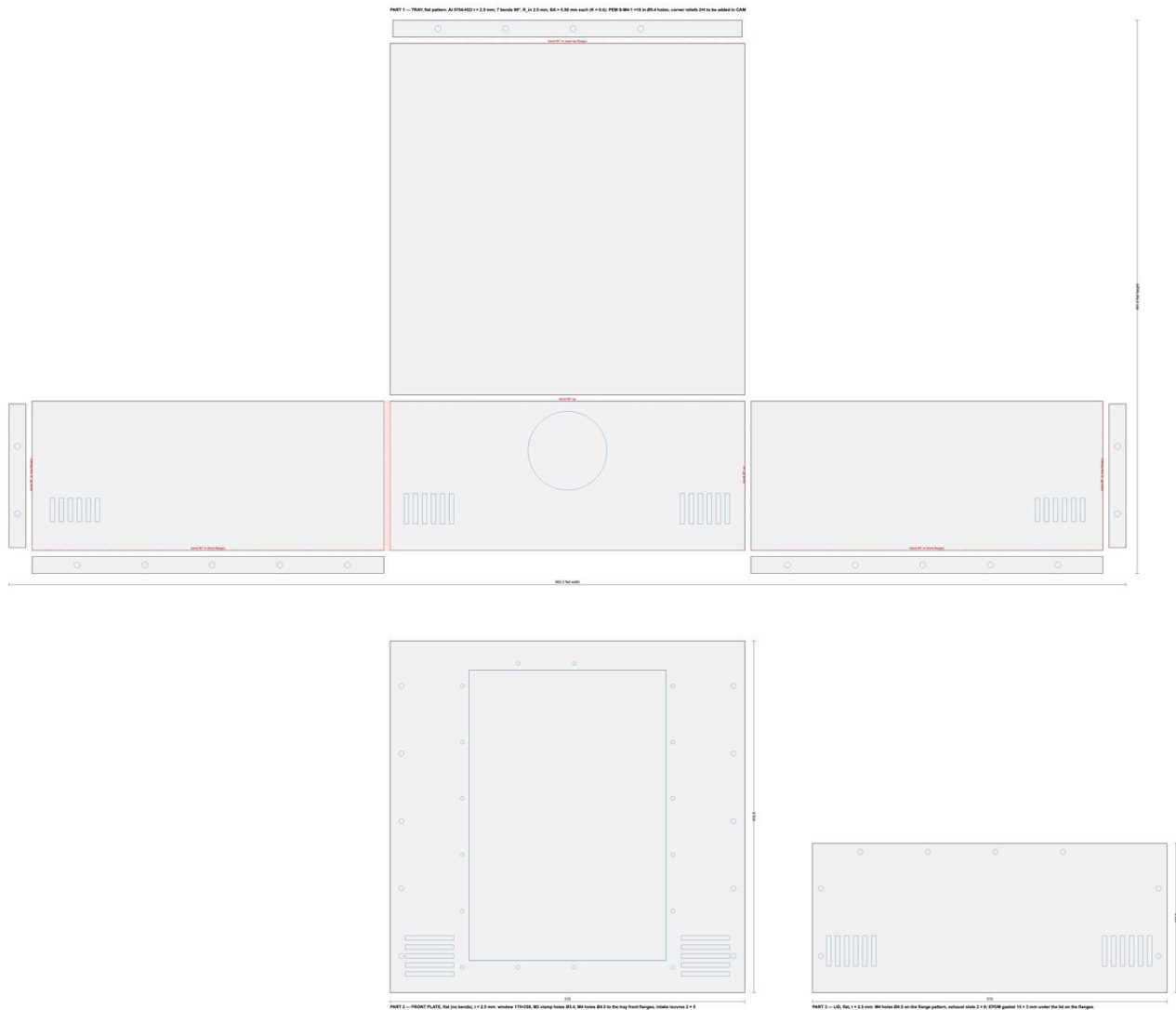


Figure 68 — F15.7 — Sheet-metal flat patterns (tray, front plate, lid) that the fabricator folds before Step 15.5; bend lines and allowance per the generator formula, DSN-MECH-06 Rev A — PROPOSED DESIGN (source: engineering/DESIGN/MECHANICAL/drawings/png/DSN-MECH-06_sheet_metal_flat_patterns.png; produced by tools/design_enclosure_drawings.py)

Step 15.5 — Fit PEM nuts, lid gasket and front gasket to the tray [PROPOSED DESIGN]

- **Purpose:** prepare the folded tray so that every later fastener has a nut and the IP54 sealing surfaces exist before parts block access.
- **Parts & tools:** tray #1 (2.5 mm Al 5754, 3 bends R2.5, 15 mm flanges); PEM nuts S-M4-1 × 18 (parts list #1 and fastener totals); lid gasket #7 (EPDM 3 mm self-adhesive, 15 mm wide, on the three top flanges); front-flange gasket: 1.5 mm EPDM strip on the two front flanges (parts list "Sealing and finish", no part number); PEM press tool.
- **Action:** press the 18 PEM nuts into the lid and front-plate flange holes (8 lid holes M4 + 10 front-plate holes M4, parts list #1 note); lay the lid gasket on the three top flanges and the front strip on the two front flanges, continuous around the corners.
- **Check:** gasket continuous, no gaps at corners; every flange hole has a nut; the Ø70 cable-entry hole and intake slots are free.
- **Figure:** F15.6, F15.7.
- **⚠ Decision:** the step source quotes "PEM S-M4 ×38" (source: manual/ASSEMBLY_STEPS_SOURCE.md B1) whereas the parts list and its fastener totals give S-M4-1 ×18; the parts list count is used here (observation MAN-15-1). Bend reliefs and corner welds of the tray are listed as open in the parts list.

Step 15.6 — Mount the PA heat-spreader brackets and the plate [PROPOSED DESIGN]

- **Purpose:** fix the 300 × 300 × 10 mm plate (#8) vertically at the front of the tray so that its front face is at station Y = 11.0 mm (plate front) and its rear face at Y = 21.0 mm.
- **Parts & tools:** plate #8 (Al 6061, fin fields machined, M3 tapped 16×7 + 6); brackets #9, #10, #11, #12 (L 25×25×3 Al at x = 5/280, z = 20/265); per bracket M5×12 ×2 + M5×16 ×1 (totals M5×12 ×12 incl. the stepper, M5×16 ×4); square, 5 mm hex key.
- **Action:** bolt each bracket to the side wall and to the front flange (parts list note), then bolt the plate to the four brackets; set the plate front face at Y = 11.0 mm from the inner face of the front wall (source: tools/design_layout.py, stations_y.plate_front).
- **Check:** plate square to the base ±0.5 mm (source: step source B2); front face flush for the antenna spacers; the fin fields face the rear (towards the Main Board tier).

- **Decision:** D-08/D-11 (chassis material and spreader concept); step source names "brackets #10" — the parts list has four brackets #9–#12.

Step 15.7 — Mount the 16 PA boards on the rear of the plate [PROPOSED DESIGN]

- **Purpose:** put every QPA2962 on the plate through a thermal pad so that the conduction path of the thermal design (chapter 10 §10.7) exists.
- **Parts & tools:** 16 assembled RF PA boards (Step 15.2); thermal pads 5 × 5 mm, one under each QPA2962 (parts list #8 note); M3×6 ×112 (parts list #8: "PA 112"); torque screwdriver.
- **Action:** place the boards on the tapped field in the 4 × 4 grid of the layout (pitch x 40.0 mm, z 68.0 mm; field origin x 77.5, z 23.0; source: `tools/design_layout.py pa_grid, pa_field`) with the pad centred on the QPA2962; fasten with 7 × M3×6 per board through the Ø3.20 holes at (2.6, 2.6), (17.5, 2.6), (32.4, 2.6), (2.6, 38), (32.4, 38), (2.6, 57.4), (32.4, 57.4) (source: `engineering/MECHANICAL/dimensions/RF_PA_dimensions.md §2`); number the boards PA1...PA16 as drawn on DSN-MECH-04 tier 1.
- **Check:** torque 0.5 N·m (step source B3 — ASSUMED, no repository torque specification); pads centred; numbering PA1...PA16 matches F15.8 tier 1.
- **Figure:** F15.8 (tier 1), F15.5.
- **Decision:** the QPA2962 is assumed at the board centre in the thermal model ("verify in `engineering/PCB/RF_PA", DESIGN_CALCULATIONS.md §1) — confirm the pad position on the real board before fastening.`

Step 15.8 — Mount the antenna panel on the plate front [PROPOSED DESIGN, D-01...D-06]

- **Purpose:** fix the proposed 16-row patch panel at station Y = 8.0 mm, 2.4 mm in front of the plate, with its row-1 connector at the bottom.
- **Parts & tools:** antenna PCB DSN-ANT-01 (165.0 × 248.0 × 0.6 mm, position x 72.5, z 31.0; source: `tools/design_layout.py antenna`); 6 × 2.4 mm nylon spacers (parts list #8 note; spacer material listed as open); M3×6 ×6 (parts list #8: "antenna 6").
- **Action:** stand the panel on the six spacers over the plate's six antenna tap holes; fasten; keep the 16 end-launch 2.92 mm connectors on the left edge accessible (D-05).
- **Check:** row 1 at the bottom; connectors accessible; gap to plate 2.4 mm; radome gap 8.0 mm remains in front (layout `antenna.radome_gap`).
- **Figure:** F10.2 (front elevation), F8.1 (chapter 8).
- **Decision:** D-01 (patch array vs slotted waveguide) is an owner decision; the panel is PROPOSED and only simulated (AC-E7 NOT RUN).

Step 15.9 — Fit the Main Board tier (Y = 56.0 mm) [PROPOSED DESIGN, D-07, D-09]

- **Purpose:** carry the Main Board vertically, components to the rear, with its SMA field reachable for the Synth coax (D-09).
- **Parts & tools:** carrier rails #13 (bottom) and #14 (top), U15×12×1.5 Al, 280 mm, M4×8 ×2 each to the side walls; standoffs #15–#22 (M3×10 hex) at the eight Ø3.20 holes (4, 4), (256, 4), (116, 114), (256, 114), (116, 250), (256, 250), (4, 296), (256, 296) (source: `MAIN_BOARD_dimensions.md §2`) with M3×6 ×1 each; assembled Main Board (Step 15.2).
- **Action:** bolt the rails to the side walls at station Y = 56.0 mm (`stations_y.main`); screw the eight standoffs into the rails; mount the board at (x 25.0, z 5.0) (`main_pos`) component side to the rear.
- **Check:** connector field positions match DSN-MECH-04 tier 2 (J1, J18...J55 on the rear face; X_n Molex, JP and SV1 reachable); board does not touch the PA boards (component heights ASSUMED 15 mm top / 4 mm bottom — G-02).
- **Figure:** F15.8 (tier 2), F15.2.
- **Decision:** D-07 — the 25 mm tier pitch is an assumption until component heights are measured (G-02).

Step 15.10 — Fit the Synth tier (Y = 81.0 mm) and the Power tier (Y = 106.0 mm) [PROPOSED DESIGN]

- **Purpose:** complete the board stack behind the Main Board.
- **Parts & tools:** Synth rails #23/#24 (120 mm, M4×8 ×2 each), standoffs #25–#28 at (5, 5), (95, 5), (5, 95), (95, 95) (source: `FREQUENCY_SYNTHESIZER_dimensions.md §2`), M3×6 ×4; Power rails #29/#30 (300 mm, M4×8 ×2 each), standoffs #31–#38 at (10, 10), (140, 10), (270, 10), (10, 120), (270, 120), (10, 230), (270, 230), (138, 268) (source: `POWER_SUPPLY_dimensions.md §2`, rounded as in the parts list), M3×6 ×8; assembled boards.
- **Action:** mount as in Step 15.9: Synth at station Y = 81.0 (`stations_y.synth`), position (x 46.36, z 141.60) — "centred on Main Board J1,J18,J20,J21,J22,J23 (P&P)" (`synth_pos.basis`); Power Board at Y = 106.0 (`stations_y.power`), position (x 15.0, z 5.0).
- **Check:** Synth J-ports face the Main Board SMA field (shortest coax, D-09); Power Board X1 input and X2...X35 outputs reachable from the rear; rear inner wall at Y = 127.6 leaves clearance for the Power Board components (ASSUMED heights).
- **Figure:** F15.8 (tiers 3–4), F15.3, F15.4.
- **Decision:** G-10 (Power Board outline), G-02 (heights).

Step 15.11 — Mount the 22 V PA supply / gate module on the rear wall [PROPOSED DESIGN — module not built]

- **Purpose:** provide the 22 V drain source, the high-side switch driven by `EN/DIS_RFPA_VDD` and the per-PA pulse-gating FETs (D-14).

- **Parts & tools:** DSN-PSU-01 module (block schematic, BOM and netlist only: `engineering/DESIGN/ELECTRICAL/PA_SUPPLY_22V/`; component-level capture is recovery item MDR-12); fasteners not defined.
- **Action:** fix the module to the rear wall between the Power Board tier and the rear inner wall (DSN-MECH-01 shows the location); route its 16 outputs OUT1...OUT16 towards the PA field (cables CBL-050, -056, -062, -068, -074, -080, -086, -092, -098, -104, -110, -116, -122, -128, -134, -140 in Step 15.17).
- **Check:** none possible — the module does not exist.
- **Figure:** F10.1, F9.1 (chapter 9).
- **Decision:** $\triangle$ D-14 — module not built; the `TX_GATE` FPGA/MCU pin is undecided; $\triangle$ K4 — no 22 V rail in any CAD file.

Step 15.12 — Fit the fans and the gland plate [PROPOSED DESIGN — fans not in the parts list]

- **Purpose:** establish the bottom-to-top forced airflow of the cooling concept (D-11) and seal the cable entry.
- **Parts & tools:** 2 × 60 mm fans (step source B8; **no fan part, bracket or airflow figure exists in the parts list** — "fan mounting brackets" is an open item there); gland plate #39 (100 × 100 × 2 Al) with M4×8 ×4 and IP68 glands M32 (power) + M20 (USB).
- **Action:** fit the fans at the duct intakes shown on DSN-MECH-01/03; bolt the gland plate under the base over the Ø70 entry hole.
- **Check:** airflow direction bottom → top (intake louvres in the front plate, exhaust slots in the lid — parts list #2, #6); glands tightened on the harness only after Step 15.18.
- **Figure:** F10.1, F10.3.
- **Decision:** G-08 — fan selection and airflow are not designed; the thermal map assumes an effective h over the fin strips (chapter 10 §10.7).

Step 15.13 — Fit the front plate, PTFE window, gasket and clamp frame [PROPOSED DESIGN]

- **Purpose:** close the front with the radome window in front of the antenna (radome gap 8.0 mm).
- **Parts & tools:** front plate #2 (window opening 175 × 258, intake louvres, 16 M3 clamp holes) with M4×8 ×10 to the tray front flanges (into the PEM nuts of Step 15.5); window #3 (PTFE 2 mm, outside the front plate); window gasket #4 (EPDM 1.5 mm, 10 mm ring); clamp frame #5 (2 mm Al, 14 mm wide) with M3×10 ×16 + nyloc.
- **Action:** bolt the front plate; lay the gasket, the window and the frame outside the plate; fasten the 16 M3×10 with nyloc nuts evenly in a cross pattern.
- **Check:** window flat; gasket compressed evenly; louvres unobstructed.
- **Figure:** F15.6, F10.2.
- **Decision:** observation MAN-10-1 (16 vs 18 clamp screws in the parts-list text); step source B9 numbers the parts "#2, #5, #6, #7" (older numbering) — the parts-list numbers #2, #3, #4, #5 apply. PTFE loss "≈ 0.1 dB at 10.5 GHz ... to be confirmed" (parts list #3).

The lid (#6, M4×8 ×8, gasket #7) stays off until the bring-up of chapter 16 is complete (Step 16.10 closes it).

15.4 Phase C — Harness

Lengths are PROPOSED: "Manhattan distance between connector positions in the head + service allowance (40 mm coax / 60 mm wire), rounded up to 10 mm ... cut after a first fit" (source: `engineering/DESIGN/HARNESS/HARNESS_SCHEDULE.md`, header). Totals: 144 cables; coax 55; wire 89; ≈ 41.6 m. $\triangle$ The pin order of every Molex 22-23-20x1 connector is UNVERIFIED because the EAGLE symbols name all pads S (source: `interconnection_table.md`, header) — buzz out before mating.

Harness schedule excerpt — the cables named in this phase (source: `engineering/DESIGN/HARNESS/HARNESS_SCHEDULE.md`, rows copied; the full 144-row table is reproduced in Appendix B):

ID	From	To	Signal	Cable	Length (mm)	Note
CBL-035	POWER_SUPPLY SV1	MAIN_BOARD SV1	enable bus (15 EN + GND)	20-way 1.27 mm IDC ribbon	320	
CBL-002	POWER_SUPPLY X3	MAIN_BOARD X55	+5V5_PA	2-wire 20 AWG, Molex 22-01-2027 both ends	780	rail name matched in both netlists
CBL-003	POWER_SUPPLY X4	MAIN_BOARD X8	+1V0_FPGA	2-wire 20 AWG, Molex 22-01-2027 both ends	310	rail name matched in both netlists
CBL-009	POWER_SUPPLY X10	MAIN_BOARD X17	+1V8_CLOCK	2-wire 20 AWG, Molex 22-01-2027 both ends	840	rail name matched in both netlists
CBL-001	POWER_SUPPLY X2	? ?	+5V0_1	2-wire 20 AWG	TBD	no connector with this net name on Main/Synth (rail renamed through a filter/inductor or spare output) — resolve with interconnection_table.md
CBL-036	FREQUENCY_SYNTHESIZER J7	MAIN_BOARD J1	100 MHz FPGA sys clk	coax RG-405 (0.086") SMA-SMA, phase-stable	140	mapping from interconnection_table.md §6
CBL-037	FREQUENCY_SYNTHESIZER J5	MAIN_BOARD J20	120 MHz DAC	coax RG-405 (0.086") SMA-SMA, phase-stable	170	mapping from interconnection_table.md §6
CBL-038	FREQUENCY_SYNTHESIZER J6	MAIN_BOARD J18	120 MHz DAC	coax RG-405 (0.086") SMA-SMA, phase-stable	120	mapping from interconnection_table.md §6
CBL-039	FREQUENCY_SYNTHESIZER J3	MAIN_BOARD J21	400 MHz ADC	coax RG-405 (0.086") SMA-SMA, phase-stable	120	mapping from interconnection_table.md §6
CBL-040	FREQUENCY_SYNTHESIZER J10	MAIN_BOARD J23	LO TX	coax RG-405 (0.086") SMA-SMA, phase-stable	100	mapping from interconnection_table.md §6

ID	From	To	Signal	Cable	Length (mm)	Note
CBL-041	FREQUENCY_SYNTHESIZER J11	MAIN_BOARD J22	LO RX	coax RG-405 (0.086") SMA-SMA, phase-stable	140	mapping from interconnection_table.md §6
CBL-043	FREQUENCY_SYNTHESIZER JP1	MAIN_BOARD JP1	control header	ribbon/discrete 26 AWG, 2.54 mm housings	210	
CBL-044	FREQUENCY_SYNTHESIZER JP2	MAIN_BOARD JP13	control header	ribbon/discrete 26 AWG, 2.54 mm housings	240	
CBL-045	MAIN_BOARD X_7	RF_PA PA1 X2	VG_1	2-wire 24 AWG shielded, Molex 22-01-2027	150	PA instance ↔ VG_n assignment PROPOSED (n = n); not documented in CAD
CBL-046	MAIN_BOARD X3	RF_PA PA1 X3	drain current sense	3-wire 24 AWG twisted, Molex 22-01-3037	150	Main sense connector ↔ PA n PROPOSED
CBL-047	MAIN_BOARD J27	RF_PA PA1 J1	RF 10.5 GHz (to PA RFIN)	coax RG-405 SMA-SMA, EQUAL LENGTH set	330	which SMA of the pair is RFIN/RFOUT is UNVERIFIED (interconnection_table §7)
CBL-048	MAIN_BOARD J26	RF_PA PA1 J2	RF 10.5 GHz (from PA RFOUT)	coax RG-405 SMA-SMA, EQUAL LENGTH set	320	which SMA of the pair is RFIN/RFOUT is UNVERIFIED (interconnection_table §7)
CBL-049	RF_PA PA1 J2	ANTENNA ROW1	RF to antenna row n	coax RG-405 SMA-2.92 mm, EQUAL LENGTH set	100	antenna row n ↔ PA n PROPOSED; equal length mandatory for elevation phase
CBL-050	DSN-PSU-01 OUT1	RF_PA PA1 22V	22 V drain, pulsed	2-wire 18 AWG twisted, AK300/2 screw terminal	510	
CBL-141	SLIP RING VIN	POWER_SUPPLY X1	VIN 12-17 V	2 × 2-wire 16 AWG	340	
CBL-142	SLIP RING VIN	DSN-PSU-01 IN	VIN to 22 V boost	2 × 2-wire 16 AWG	470	
CBL-143	SLIP RING USB	MAIN_BOARD X53	USB 2.0 FS to host	USB 2.0 shielded, mini-B	360	
CBL-144	MAIN_BOARD stepper pins	PEDESTAL TB6600 (via slip ring)	STEP/DIR/EN	3-wire 24 AWG shielded	TBD	the driver sits on the fixed base; needs 3 more slip-ring circuits OR move the driver into the head (then only motor phases cross: 4 circuits) — DECISION NEEDED

Step 15.14 — Enable bus SV1 ↔ SV1 [SOURCE-DERIVED pinout, PROPOSED cable]

- **Purpose:** connect the 15 enable lines from the STM32 to the Power Board regulators.
- **Parts & tools:** CBL-035, 20-way 1.27 mm IDC ribbon, 320 mm; two MA10-2 IDC connectors.
- **Action:** mate Main SV1 to Power SV1 pin-for-pin (pin map: 1 EN_+1V0_FPGA PE7 → U1, 2 EN_+5V0_PA2 PG1 → U15, 3 EN_+1V8_FPGA PE8 → U2, 4 EN_+5V0_PA3 PG2 → U16, 5 EN_+3V3_FPGA PE9 → U4, 6 EN_+5V5_PA PG3 → U17, 7 EN_+5V0_ADAR PE10 → U13, 8 EN_+1V8_CLOCK PG4 → U25, 9 EN_+3V3_ADAR12 PE11 → U6, 10 EN_+3V3_CLOCK PG5 → U23, 11 EN_+3V3_ADAR34 PE12 → U7, 13 EN_+3V3_ADTR PE13 → U32, 15 EN_+3V3_SW PE14 → U10, 17 EN_+3V3_VDD_SW PE15 → U8, 19 EN_+5V0_PA1 PG0 → U14; 12/14/16/18/20 GND — source: power_rails.md §2).
- **Check:** pin-1 orientation identical at both ends; continuity on all 20 positions.
- **Figure:** F15.8 (SV1 positions on tiers 2 and 4).
- **Decision:** none (the Power Board has no visible pull-downs on the EN nets — "REQUIRES VERIFICATION", power_rails.md §2).

Step 15.15 — Rail cables Power → Main / Synth [SOURCE-DERIVED mapping, △ pin order UNVERIFIED]

- **Purpose:** bring every supply rail to its consumer board.
- **Parts & tools:** the 2-wire 20 AWG Molex 22-01-2027 cables CBL-002...034 whose rail name "matched in both netlists" (X3→X55, X4→X8, X5→X10, X6→Synth X15, X10→X17, X12→X56, X16→X24, X18→X6, X19→X19, X20→X15, X21→X13, X22→X18, X23→X1, X24→X11, X27→X16, X30→X12, X31→X14, X32→X5, X33→X7, X34→X4, X35→Synth X4 — source: HARNESS_SCHEDULE.md rows CBL-002...034), crimp tool, DMM.
- **Action:** buzz out the polarity of each Molex pad pair on both boards against the netlist before crimping (pads are named S); crimp and mate.
- **Check:** continuity and polarity per power_rails.md §1 (consumer connector column).
- **Figure:** F3.2, F15.8.
- **Decision:** (a) rails whose destination connector could not be matched by net name — CBL-001, -006, -007, -008, -010, -012, -013, -014, -016, -024, -025, -027, -028 (TBD rows of the harness schedule; interconnection table §2/§4 resolves most of them: e.g. +3V3_ADAR_12 X14 → Main X20 +3V3_ADAR12, +3V3_ADAR_34 X15 → Main X21, +3V3_CLOCK X11 → Synth X11, +3V3_LO_1 X8 → Synth X13, +3V3_LO_2 X7 → Synth X14; +5V0_1..5, +5V0_ADAR X13, +5V0_ADTR X26, +3V3_SW X29 have **no consumer** and stay unconnected — power_rails.md §4 item 4); (b) +1V8_CLOCK X10 must feed both Main X17 and Synth X12 — "add a second output or document a Y-cable" (power_rails.md §4 item 3).

Step 15.16 — Synth ↔ Main control headers and coax clocks/LO [SOURCE-DERIVED mapping]

- **Purpose:** deliver the 100 / 120 / 400 MHz clocks and the TX/RX LO to the Main Board and connect the SPI4/control lines.
- **Parts & tools:** CBL-043 (JP1 ↔ JP1, 210 mm), CBL-044 (JP2 ↔ JP13, 240 mm); coax set CBL-036...041 (J7→J1 100 MHz FPGA, J5→J20 and J6→J18 120 MHz DAC, J3→J21 400 MHz ADC, J10→J23 LO TX, J11→J22 LO RX; RG-405 phase-stable, 100–170 mm) and CBL-042 (J4→J19 test, 100 mm); SMA torque wrench.
- **Action:** connect the headers; form and connect the coax; label each cable with its CBL-ID.
- **Check:** SMA torque 0.9 N·m (step source C3 — ASSUMED, no repository torque specification); labels present; no coax below its minimum bend radius.
- **Figure:** F15.8 (tiers 2–3).
- **Decision:** none beyond the clock-frequency intent being firmware-derived (`main.cpp:933–1072`), to be verified in Step 16.4.

Step 15.17 — PA harness: VG, sense, RF pairs, antenna coax, 22 V leads [PROPOSED mapping, △ several UNVERIFIED]

- **Purpose:** connect all 16 PA boards to the Main Board, the antenna rows and the 22 V module.
- **Parts & tools:** per PA n: VG cable (Main X_k → PA n X2, 24 AWG shielded, Molex 22-01-2027), sense cable (Main X3/X38...X52 → PA X3, 3-wire 24 AWG twisted, Molex 22-01-3037), RF pair (Main J-pair → PA J1/J2, RG-405 EQUAL LENGTH set), antenna coax (PA J2 → antenna row n, RG-405 SMA–2.92 mm, EQUAL LENGTH set), 22 V lead (DSN-PSU-01 OUT_n → PA 22v AK300/2, 2-wire 18 AWG twisted) — rows CBL-045...140 of the harness schedule. VG_n ↔ Main connector map from the schematic: X₇=VG₁, X₁₆=VG₂, X₈=VG₃, X₁₅=VG₄, X₄=VG₅, X₁₁=VG₆, X₃=VG₇, X₁₂=VG₈, X₅=VG₉, X₁₄=VG₁₀, X₆=VG₁₁, X₁₃=VG₁₂, X₂=VG₁₃, X₉=VG₁₄, X₁=VG₁₅, X₁₀=VG₁₆ (source: `interconnection_table.md` §7). RF pairs per n: J27/J26 (1), J29/J28 (2), J25/J24 (3), J31/J30 (4), J35/J34 (5), J37/J36 (6), J33/J32 (7), J39/J38 (8), J47/J46 (9), J41/J40 (10), J45/J44 (11), J43/J42 (12), J55/J54 (13), J49/J48 (14), J53/J52 (15), J51/J50 (16) (same source).
- **Action:** cable PA n to VG_n, sense connector n, RF pair n and antenna row n exactly as labelled; cut the 16 antenna coax and the 16 RF pairs as equal-length sets; connect the 22 V leads **last** and only with VG wiring complete (S1).
- **Check:** equal length within ±2 mm on the 16 antenna coax (step source C4 — ASSUMED tolerance; the harness schedule only states "equal length mandatory for elevation phase"); labels PA1...PA16; 22 V polarity (KL1 VD, KL2 GND, `interconnection_table.md` §7); screw-terminal torque of the AK300/2 per its datasheet (not in the repository).
- **Figure:** F15.8 (tier 1 PA numbering, tier 2 SMA field), F15.5.
- **Decision:** which SMA of each Main Board pair is RFIN vs RFOUT is UNVERIFIED — "determine from the Main Board switch RF_SW_n routing before cabling" (`ASSEMBLY_SEQUENCE.md` §B.7); PA-instance ↔ VG_n / sense-connector mapping is PROPOSED (n = n), not documented in CAD; △ K4 / D-14 for the 22 V leads.

Step 15.18 — Slip-ring harness through the base gland [PROPOSED DESIGN, D-13]

- **Purpose:** bring VIN, 22 V-boost input and USB from the pedestal into the rotating head.
- **Parts & tools:** CBL-141 (slip ring VIN → Power X1, 2 × 2-wire 16 AWG, 340 mm), CBL-142 (slip ring VIN → DSN-PSU-01 IN, 470 mm), CBL-143 (slip ring USB → Main X53 mini-B, 360 mm); gland plate #39 glands M32 (power) and M20 (USB).
- **Action:** route the three cables through the Ø70 base hole and the glands; tighten the glands.
- **Check:** continuity after a full 360° rotation of the turntable (Step 15.22); no cable strain.
- **Figure:** F10.3, F10.5.
- **Decision:** CBL-144 (STEP/DIR/EN to the TB6600) — "the driver sits on the fixed base; needs 3 more slip-ring circuits OR move the driver into the head ... DECISION NEEDED" (harness schedule); D-13 channel count (12) assumes USB stays the host link.

15.5 Phase D — Pedestal**Step 15.19 — Slewing bearing, turntable and ring pulley [PROPOSED DESIGN]**

- **Purpose:** build the rotating joint.
- **Parts & tools:** pedestal top plate #43 (360 × 360 × 5 Al, bore Ø90) with M6×20 ×12 to the bearing inner ring; slewing bearing #41 (OD 190 / ID 100 × 20, "part number to be selected (axial ≥ 50 kg, moment ≥ 60 N·m)"); turntable #40 (Ø340 × 8 Al) with M6×25 ×12 to the bearing outer ring; ring pulley #42 (GT3 180 T Ø172) with M4×10 ×6 clamped under the turntable; pedestal housing #44 with M5×10 ×12 to the top plate.
- **Action:** bolt the bearing inner ring to the top plate (12 × M6×20), the turntable to the outer ring (12 × M6×25), the ring pulley under the turntable (6 × M4×10); fit the top plate on the housing (12 × M5×10).
- **Check:** turntable runs freely; axial play < 0.1 mm (step source D1 — ASSUMED acceptance value).
- **Figure:** F10.5, F10.3.
- **Decision:** bearing part number not selected (parts list #41 note).

Step 15.20 — Stepper, bracket, pulley, belt and driver [PROPOSED DESIGN, △ D-12 revised]

- **Purpose:** install the azimuth drive.

- **Parts & tools:** stepper #45 (NEMA 23, 76 mm, M5×12 ×4); motor bracket #46 (80 × 80 × 3 Al, slots ±5 mm, M5×10 ×4 to the top plate); motor pulley #47 (GT3 60 T Ø57, bore 6.35, grub M4 ×2); belt GT3 9 mm (ratio 3, `design_layout.py` pedestal); driver #50 (TB6600 on DIN rail).
- **Action:** bolt the bracket, mount the motor and pulley, fit the belt and tension it with the bracket slots; mount the driver; set the driver current limit before any motion (S3).
- **Check:** belt deflection within the belt maker's figure (not in the repository); driver current limit set and recorded.
- **Figure:** F10.5.
- **Decision:** D-12 — DSN-CALC-01 §2 shows a NEMA 23 needs ≥ 200 ms per 7.2° step (≈ 19 s per revolution) for the ≈ 10 kg head; a NEMA 34 (3 N·m class) allows ≈ 100 ms; alternatives 1:6 ratio (`Stepper_steps = 1200`) or a lighter head. The firmware constant must match the ratio (`Stepper_steps = 600` for 1:3, `main.cpp:195`). Choose motor class and ratio **before** this step.

Step 15.21 — Slip ring, stator bracket, mast flange [PROPOSED DESIGN, **△** mast interface undefined]

- **Purpose:** fix the slip-ring stator to the pedestal with its rotor turning with the turntable; prepare the mast interface.
- **Parts & tools:** slip ring #48 (through-bore Ø99 × 60, bore 60, 12 circuits, "e.g. Senring H3899 class — select"); stator bracket #49 (140 × 140 × 3 Al, M4×8 ×4); mast flange #51 (Ø150 × 10 steel, 4 × M10 PCD 110, M10×30 ×4 — "ASSUMPTION — mast interface undefined").
- **Action:** mount the stator bracket under the top plate, the slip ring in the bore, the rotor flange to the turntable; route the harness of Step 15.18 through the bore; bolt the mast flange under the housing.
- **Check:** rotor turns with the turntable without cable strain over 360° .
- **Figure:** F10.5, F10.3.
- **Decision:** slip-ring part not selected; mast flange is an ASSUMPTION (parts list #51; `design_layout.py` mast_flange); D-13 channel count.

Step 15.22 — Bolt the head to the turntable [PROPOSED DESIGN]

- **Purpose:** join the head and the pedestal with the cable passage aligned.
- **Parts & tools:** M6×16 ×8 (parts list #40: "8×M6 to the head base"); head assembly of phase B; harness of Step 15.18 already through the Ø70 base hole.
- **Action:** align the Ø70 base passage with the slip-ring bore and bolt the tray base to the turntable with the 8 × M6×16.
- **Check:** head square to the turntable (no tolerance in the source); harness continuity after a 360° rotation (Step 15.18 check); lid still open for chapter 16.
- **Figure:** F10.3, F10.8a.
- **Decision:** lifting points and earthing stud are open items (parts list "Open").

15.6 Hand-over to bring-up

Phases A–D leave the head open (lid #6 off), the pedestal stationary, the 22 V leads connected only if the DSN-PSU-01 module exists, and every cable labelled with its CBL-ID. Chapter 16 continues with Step 16.1 (firmware flash). Evidence to collect during the first build (source: `engineering/ASSEMBLY/ASSEMBLY_SEQUENCE.md` §D): rail voltages (CP-4), photos of the first harness with cable labels, the measured component heights (closes G-02), the measured board thickness (G-01) and the real cut lengths of all 144 cables (replaces the PROPOSED lengths) — store under `engineering/VALIDATION/`.

<!-- chapter 16: Power-up, firmware flashing, FPGA programming, calibration (ADC taps, beam), bench tests, acceptance criteria -->

Bring-up and test — power-up, flashing, FPGA programming, calibration, bench tests, acceptance

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Status summary: every procedure in this chapter is **PROPOSED / BETA — never executed on hardware** (source: `manual/ASSEMBLY_STEPS_SOURCE.md`, header; `engineering/ASSEMBLY/ASSEMBLY_SEQUENCE.md`, "Nothing in this sequence has been executed on hardware"). The test plan and acceptance tables are copied from `docs/TESTING/`; a test is reported PASSED only where the source records an executed run, and physical tests are NOT RUN everywhere. The FPGA bitstream does not exist (no Vivado on the authoring machine), so Steps 16.3 and 16.5–16.9 cannot be started today.

Sources: `manual/ASSEMBLY_STEPS_SOURCE.md` (phase E), `engineering/ASSEMBLY/ASSEMBLY_SEQUENCE.md` §B/§D, `docs/TESTING/VALIDATION_PLAN.md`, `docs/TESTING/ACCEPTANCE_CRITERIA.md` (sections A–H), `beta/fpga/README.md` (calibration registers, Vivado scripts), `beta/gui/README.md` and `beta/gui/aeris10_gui/protocol/register_map.py` (register panel), `beta/stm32/README.md` §7, `engineering/ELECTRICAL/power_distribution/power_rails.md` §3, `engineering/DESIGN/CALCS/DESIGN_CALCULATIONS.md` §1–§2.

Figures in this chapter: none mandatory (tables); the register map is in chapter 13 §13.6, the enable sequence in chapter 12 §12.6.

16.0 Safety and preconditions

The safety preamble of chapter 15 §15.0 applies (S1 22 V drain, S2 RF, S3 pedestal, S4 ESD). Preconditions: phases A–D of chapter 15 complete with the lid off and the pedestal stationary; bench PSU 12–17 V with current limit; the 22 V leads disconnected until Step 16.6; a 50 Ω dummy load or anechoic setup for Step 16.9; instruments listed per step. Three decisions gate the chapter: $\triangle$ **K1** (FPGA part XC7A50T on the schematic vs XC7A100T in the FPGA README — `tools/vivado/create_project.tcl` header), $\triangle$ **K2** (HSE 8 MHz assumed by the beta firmware, D-01), $\triangle$ **K4** (22 V PA supply not in CAD).

16.1 Phase E — bench bring-up steps

Step 16.1 — Flash the STM32 beta image via SWD [BETA image, $\triangle$ K2]

- **Purpose:** load `aeris10_fw.elf` and prove the clock tree and CDC enumeration.
- **Parts & tools:** ST-LINK on SWD (PA13 SWDIO / PA14 SWCLK; SWO PB3 — `beta/stm32/CUBEMX_SETTINGS.md` §3); `beta/stm32/build_out/aeris10_fw.elf` (build of Step 12.3); USART3 terminal 115200 8N1 (PB10/PB11); host PC.
- **Action:** `STM32_Programmer_CLI -c port=SWD -w beta/stm32/build_out/aeris10_fw.elf -v -rst` (source: `beta/stm32/README.md` §7 item 2).
- **Check:** USART3 banner appears (SWO/USART3 output at 115200 "proves the clock" — same source); the host enumerates a CDC ACM device with the placeholder VID/PID 0x0483:0x5740 (D-09). Note the 180 s OCXO wait at boot (`HAL_Delay(180000)`), F0 in `power_rails.md` §3) before any rail is enabled.
- **Figure:** F12.2.
- $\triangle$ **Decision:** K2 — if the board's crystal is not 8 MHz the PLL constants of D-01 are wrong and `HAL_RCC_OscConfig` fails into `Error_Handler()`; verify XTAL1 on the assembled board first.

Step 16.2 — Observe the power-enable sequence F0–F10 on a scope [SOURCE-DERIVED order, BETA delays]

- **Purpose:** confirm that the rails appear in the coded order with the coded delays (chapter 12 §12.6) and that the D-11 additions (+5V5_PA before VG, VD last off) are present.
- **Parts & tools:** oscilloscope ≥ 4 channels; probe points at the Power Board outputs X2...X35 and at SV1 pins 1...19.
- **Action:** capture from reset: F1 `EN_+1V8_CLOCK` (SV1-8) → F2 `EN_+3V3_CLOCK` (SV1-10) → F3/F4/F5 FPGA rails (SV1-1/3/5) → F6 ADAR 3.3 V (SV1-9/11) → F7 `EN_+5V0_ADAR` (SV1-7) → F8 `EN_+3V3_VDD_SW` (SV1-17) → F9 `EN_+3V3_SW` (SV1-15) → `EN_+5V5_PA` (SV1-6, D-11 addition) → F10 VG DAC update then `EN/DIS_RFPA_VDD` (PD6 → JP10).
- **Check:** order matches `power_rails.md` §3; delays 100 ms (F1–F5), 500 ms (F6, F7), 2 ms (F8, F9), 100 ms `+5V5_PA` → DAC, 20 ms VG → VD (D-11); the xlsx rules "1.8 V before 3.3 V for the AD9523" and "AVDD3 before AVDD1 for the ADAR1000" are respected (F1→F2, F6→F7). Record the captures (AC-S7).
- **Figure:** F3.2, chapter 12 §12.6 tables.
- $\triangle$ **Decision:** whether `systemPowerUpSequence()` reaches F8/F9 via `initializeADTR1107Sequence()` is UNVERIFIED (`power_rails.md` §3 row F8); the actual assertion of `EN_+3V3_ADTR` / `EN_+5V0_PA1..3` settles observation MAN-12-1.

Step 16.3 — Program the FPGA via JP3 JTAG [BLOCKED — no bitstream; $\triangle$ K1]

- **Purpose:** load the radar RTL and prove the STM32 ↔ FPGA handshake.
- **Parts & tools:** Xilinx platform cable on Main Board JP3 (JTAG); Vivado with `beta/fpga/vivado/create_project.tcl` + `vivado/build.tcl` (NOT executed — `beta/fpga/README.md`); the two FFT IP cores per `beta/fpga/ip/README.md` (not generated).

- **Action:** when a bitstream exists: `vivado -mode batch -source tools/vivado/create_project.tcl -tclargs xc7a50tftg256-2` (part must be passed explicitly — `tools/vivado/create_project.tcl` header), then `build.tcl`, then program via Hardware Manager.
- **Check:** DONE pin high; STM32 ↔ FPGA handshake DIG_0..4 (PD8..PD12: new chirp, new elevation, new azimuth, mixers enable, FPGA reset — `CUBEMX_SETTINGS.md` §6) toggles.
- **Figure:** F11.1 (chapter 11).
- **Decision:** K1 part number; the beta constraints leave 116 port bits unconstrained ("`write_bitstream` will refuse the unconstrained I/O unless they are removed from the top or `UNCONSTRAINEDPINS_ALLOW` is set" — `beta/fpga/README.md`, "Constraints"); bank-14 LVDS termination (`DIFF_TERM FALSE`, external 100 Ω — option A, REQUIRES VERIFICATION).

Step 16.4 — AD9523 lock and output frequencies [SOURCE-DERIVED intent]

- **Purpose:** verify the clock distribution before any sampling or DAC test.
- **Parts & tools:** frequency counter (± 1 ppm or better); coax set CBL-036...041 in place (Step 15.16).
- **Action:** read the AD9523 status pins (PF8/PF9 `AD9523_STATUS0/1`) and measure Synth J7 (100 MHz FPGA), J5/J6 (120 MHz DAC), J3 (400 MHz ADC) at the Main Board ends.
- **Check:** 100 / 120 / 400 MHz ± 1 ppm (acceptance from `ASSEMBLY_SEQUENCE.md` §B.4; "firmware intent `main.cpp:933-1072`"; register table verified on the host against the driver's encoding, "not on silicon" — `beta/stm32/DECISIONS.md` D-12).
- **Figure:** F2.1 (chapter 2).
- **Decision:** the beta ADC capture (mode 1) requires BUFR (DCO/4) and `clk_100m` (AD9523 OUT6) to be frequency-locked — "if the two clocks are not frequency-locked the design needs a re-sampler (not present)" (`beta/fpga/README.md`, "Remaining risks").

Step 16.5 — ADC capture calibration (IDELAY taps) [BETA register procedure; $\triangle$ ADC SPI unwired]

- **Purpose:** centre each of the 8 LVDS lanes of the AD9484 in its valid window and align the lane rotation (the `adc_capture_calib` module of the beta RTL, `ADC_CAPTURE_MODE = 1`).
- **Parts & tools:** GUI register panel (chapter 13 §13.6) over CDC REG commands (firmware D-17) or any host that can issue the SPI bridge commands 0x02/0x03; the AD9484 test-pattern registers (datasheet: register 0x0D = 0x48 user pattern, P1/P2 = 0x19..0x1C = pattern A/B; or 0x04 checkerboard / 0x07 toggle — `beta/fpga/README.md`, "ADC capture and DDC front end"); logic analyser optional.
- **Action — auto (pattern) method:** (1) put the ADC in the 2-code test pattern matching `CAL_PATT` (reset 0x55AA: pattern_a 0xAA, pattern_b 0x55); (2) REG W 0x4 0x1 (`CAL_CTRL` auto_start, write-1-to-toggle); (3) poll REG R 0x9 (`CAL_STAT`) until busy = 0 and done = 1; (4) for lane = 0...7: REG W 0x5 <lane>, REG R 0xA (`CAL_LANE_INFO`: chosen tap, win_lo, win_hi) — the panel's "Read all / refresh" does this loop and restores `CAL_LANE`; (5) REG R 0xC (`CAL_UNDET`); (6) with the pattern still applied set `check_en` (REG W 0x4 0x8) and read REG R 0xB (`CAL_ERR`) after a few seconds; (7) switch the ADC to normal data, clear `check_en`.
- **Action — manual method (fallback):** REG W 0x5 <lane>, REG W 0x6 <tap> (`CAL_TAP` 0..31, 78 ps each, reset 16), REG W 0x4 0x2 (`manual_load`); for a lane rotation REG W 0x7 <n> (`CAL_SLIP` 0..3) then REG W 0x4 0x4 (`bitslip_load`).
- **Check:** `CAL_STAT` lock[7:0] = 0xFF, align_fail = 0, fifo_ovf (bit 15) = 0; per-lane pass windows "should be ~27 of 32 taps wide at 400 MSPS"; `CAL_ERR` stays 0 with the pattern applied and `check_en` set; `CAL_ERR` stays 0 in normal data with the pattern check off (source: `beta/fpga/README.md`, "Remaining risks / what to check in Vivado (mode 1)"). Simulation reference: `testbench tb_adc_iserdes_capture` — "8/8 lanes locked, centre tap 12 (skewed lane 5 ... compensated), bitslip detected/realigned, 4000 samples exact" (same README, table row 6g).
- **Figure:** chapter 13 §13.6 register table; F2.2.
- **Decision:** (a) the ADC test pattern needs the AD9484 SPI, which the step source marks " $\triangle$ SPI unwired" (`manual/ASSEMBLY_STEPS_SOURCE.md` E5) — if no SPI path to the ADC exists only the manual method and the blind method remain; (b) the **blind method** (`adc_capture_calib.v` `ctrl_blind`, registers 0x0D/0x0E/0x10 `CAL_BLIND_*`) is in the RTL but "no register in `radar_control_regs.v` drives it" from the GUI's 4-bit map — the panel shows it as unavailable (`beta/gui/CHANGELOG.md` "Discrepancies" 2); (c) on the current top level `reg_we/reg_addr/reg_wdata` of `ctl_regs` are tied to constants (`radar_system_top.v:321-323`) and the bridge RTL does not implement 0x02..0x04 — **register writes cannot reach the FPGA until that RTL work is done**; (d) ISERDES Q1..Q4 bit order "UNVERIFIED against UG471 ... run the ADC PN9 pattern (0x0D = 0x06) and compare against a PN9 generator before trusting the data" (`beta/fpga/README.md`).

Step 16.6 — PA bias-up, one board at a time [SOURCE-DERIVED sequence, PROPOSED thermal limit; $\triangle$ K4, D-14]

- **Purpose:** bring each QPA2962 to its quiescent point safely and confirm the thermal design case.
- **Parts & tools:** 22 V source with current limit (xlsx: ID limit 2840 mA — `power_rails.md` §1 row +22V0 / VD); GUI status tab (INA241 `PA_AvgCurrent` field of the status string); thermocouple on the heat-spreader plate; the DSN-PSU-01 module if built.
- **Action:** per board n: VG -4 V (firmware sets -3.98 V = DAC code 126, `main.cpp:1583`) → VD 22 V (`EN/DIS_RFPA_VDD`) → raise VG until IDQ 1.68 A (IDQ servo `main.cpp:1628-1656`) → only then RF (xlsx L58-L62 as quoted in `power_rails.md`).
- **Check:** IDQ within ± 10 % of 1.68 A (step source E6 — ASSUMED tolerance); plate temperature < 70 °C after 10 min **with drain gating** (design case B: plate max 64.3 °C, PA base $\approx$ 73 °C at 45 °C ambient — `DESIGN_CALCULATIONS.md` §1); never run continuous bias on all 16 boards (case A: plate > 213 °C).
- **Figure:** F10.9a/b.
- **Decision:** K4 (no 22 V source in CAD), $\triangle$ D-14 (drain gating not implemented — without it only short single-board tests are thermally admissible), ADAR1000/ADTR1107 LNA-PKA rails per MAN-12-1.

Step 16.7 — Host link [BETA]

- **Purpose:** prove the GUI ↔ firmware ↔ FPGA stream.
- **Parts & tools:** host PC with the `beta/gui` venv (chapter 13); USB cable to X53 (through the slip ring once Step 15.18 is done).
- **Action:** first `python -m aeris10_gui --selftest` on the host (no hardware); then `python -m aeris10_gui --port <CDC port>` and press Start: the GUI sends the start flag and the 82-byte settings packet, then expects status strings and, with the FPGA bridge running, bridge frames.
- **Check:** `--selftest` exit 0 (executed 2026-10-09, chapter 13 Step 13.2); with hardware: status strings parsed, bridge frames arrive with `seq` incrementing and `crc_errors` 0 in the GUI status bar; `REG R 0xF` returns 0xBE7A.
- **Figure:** F13.3.
- **Decision:** AC-S6 (CDC enumerates; start flag + settings accepted; status received) NOT RUN; the settings packet has no acknowledgement (`beta/gui/README.md` assumption 7) — confirmation is only indirect via the status string.

Step 16.8 — Beam steering [BETA tables, ⚠ no hardware verification]

- **Purpose:** verify that the firmware writes the ADAR1000 phase tables for the 31 elevation positions.
- **Parts & tools:** logic analyser on the 1.8 V side of the FPGA level shifters (ADAR1000 SPI1: SCLK/MOSI/MISO + CS PA0..PA3 → FPGA pass-through); optionally a VNA/phase measurement on one row.
- **Action:** step the beam position (firmware sequence); capture the SPI traffic.
- **Check:** CS toggles per device (D-05); SCLK 6.75 MHz (D-06); phase words follow `degreesTo7BitPhase` → data-sheet Tables 10-13 → `CHx_RX/TX_PHASE_I/Q` with register offset `((channel - 1) & 3)` (C10 fix); "data-sheet tables realise the nominal phase within ~3°; array calibration needed" (`beta/stm32/README.md` §7 item 3).
- **Figure:** —
- **Decision:** ADAR1000 SCLK maximum and level-shifter bandwidth at 6.75 MHz "REQUIRES DATASHEET / BOARD VERIFICATION" (D-06); the SPI pass-through adds one `clk_100m` cycle to SCLK/MOSI/CS and two to MISO (`beta/fpga/README.md`, "Known limitations").

Step 16.9 — First RF test into a dummy load, then antenna [PROPOSED]

- **Purpose:** confirm the transmit waveform at 10.5 GHz before radiating.
- **Parts & tools:** 50 Ω high-power load or anechoic setup (S2); spectrum analyser / fast power detector; the PA chain of Step 16.6.
- **Action:** transmit the long-chirp set into the load; observe the spectrum around 10.5 GHz and the pulse envelope.
- **Check:** carrier at 10.5 GHz; pulse envelope 30 μs (long chirp, PRI 167 μs; `main.cpp:180-186` as quoted in `engineering/DESIGN/00_DESIGN_BASIS.md` §1); chirp bandwidth B is TBD in the parameter table (`DESIGN_CALCULATIONS.md` §3 assumes 50 MHz) — record what is measured.
- **Figure:** —
- **Decision:** D-01 antenna variant; the chirp direction at baseband decides `CONJUGATE_REF` in the FPGA matched filter (`beta/fpga/README.md`, note 1).

Step 16.10 — Close the lid; pedestal rotation test [PROPOSED, ⚠ D-12]

- **Purpose:** complete the head and verify the azimuth scan timing.
- **Parts & tools:** lid #6 with gasket #7 and M4×8 ×8 (parts list); stopwatch / log of the firmware azimuth counter.
- **Action:** fit the lid (pitch 60 mm, gasket compressed to 2 mm — parts list "Sealing"); run 50 azimuth positions; log the scan time.
- **Check:** ≈ 19 s per revolution with the NEMA 23 at 200 ms per step (DSN-CALC-01 §2: 18.8 s for 50 × (175 ms dwell + 200 ms move)); harness continuity through the slip ring maintained over 360°.
- **Figure:** F10.5.
- **Decision:** D-12 motor class / ratio and the matching `Stepper_steps` constant (600 for 1:3, 1200 for 1:6); CBL-144 driver location.

16.2 Validation plan (software-based design checks and physical tests)

Copied from `docs/TESTING/VALIDATION_PLAN.md` (status date 2026-10-08 unless a later executed result is noted; the authoring machine had iverilog 13.0 and Verilator 5.052 but **no Vivado**, **no EAGLE**; the Arm GCC and KiCad used later by the beta trees are recorded in `beta/*/README.md`). "A test is reported PASSED only if it was executed and succeeded. 'Software-based design check' means a file/tool check; 'physical verification' needs hardware and is never claimed here." Run everything runnable with `bash tools/run_all_checks.sh` (exit code = number of failed checks).

FPGA (section 1)

ID	Test	Type	Command	Result on the original tree (2026-10-08)	BETA tree result (<code>beta/fpga/README.md</code> , 2026-10-09)
F-01	Syntax / parse of all RTL	design check	<code>tools/fpga_lint.sh</code> (iverilog-g2012)	FAIL — <code>radar_system_top.v:312</code> syntax error	PASS (iverilog, both views)

ID	Test	Type	Command	Result on the original tree (2026-10-08)	BETA tree result (beta/fpga/README.md , 2026-10-09)
F-02	Elaboration of radar_system_top	design check	same	FAIL — 6 missing modules on a patched copy	PASS
F-03	Lint	design check	verilator --lint-only -Wall	FAIL — 207 warnings incl. MULTIDRIVEN ×4, UNDRIVEN ×14 (with stubs)	PASS, 0 %Error, 165/159 %Warning (synth/sim view)
F-04	Constraint completeness	design check	python3 tools/check_fpga_constraints.py	FAIL (exit 1) — 142 placeholder lines, 180/180 bits without PACKAGE_PIN	beta XDC: 0 placeholders, 67/183 port bits constrained (116 UNRESOLVED)
F-05	Schematic-derived pin map generation	design check	python3 tools/gen_xdc_from_schematic.py	PASSED (exit 0) — 64 ports mapped, 25 unresolved (generation step, not a correctness proof)	used verbatim in the beta XDC
F-06	Simulation radar_system_tb	design check	Vivado xsim (SVA needed)	NOT RUN	9 open-source testbench runs PASS (build.sh : 0 failures)
F-07	Synthesis utilisation on xc7a50tftg256-2	design check	tools/vivado/create_project.tcl + launch_runs synth_1	NOT RUN	NOT RUN
F-08	Timing summary / clock interaction / CDC report	design check	report_timing_summary, report_cdc	NOT RUN	NOT RUN
F-09	Bitstream generation	design check	write_bitstream	NOT RUN	NOT RUN
F-10	ADC LVDS capture on hardware (eye/valid data)	physical	ILA + known tone into AD9484	NOT RUN	NOT RUN

STM32 (section 2)

ID	Test	Type	Command	Result on the original tree	BETA tree result (beta/stm32/README.md)
S-01	Dependency / include check	design check	python3 tools/check_stm32_includes.py	FAIL (exit 1) — 53 absent HAL/CMSIS/USB headers, 6 case mismatches	0 case mismatches; 11 UNKNOWN headers only in unbuilt iio*.c
S-02	Cube package completeness	design check	tools/stm32_check_cube_package.sh <CubeF7>	NOT RUN (no package)	"missing 0 of 34", exit 0
S-03	Host syntax probe (non-target)	design check	clang -fsyntax-only ...	29/60 TUs fail on missing stm32f7xx_hal.h (informational)	superseded by S-04
S-04	Cross-compile	design check	arm-none-eabi-g++	NOT RUN	build.sh exit 0, 0 errors, 5 pre-existing warnings
S-05	Link + size	design check	arm-none-eabi-size	NOT RUN	RAM 17 480 B (5.33 %), FLASH 93 276 B (8.90 %)
S-06	Static analysis	design check	cppcheck --enable=warning,style	NOT RUN (not installed)	NOT RUN
S-07	Pin-map cross-check firmware ↔ schematic	design check	tools/extract_eagle_netlist.py --part U2 vs main.h	PASSED for the 60 named GPIO macros; conflict found for HSE (8 vs 25 MHz) and ADF4382 pins	conflicts closed by D-01, D-04
S-08	Flash + CDC enumeration	physical	ST-LINK, lsusb / Device Manager	NOT RUN	NOT RUN (Step 16.1)
S-09	Settings packet accepted (state READY_FOR_DATA)	physical	send flag + 82-byte packet	NOT RUN (static analysis predicted failure: C4, C5)	NOT RUN; C4/C5 fixed and host-tested
S-10	Power-sequence timing on EN_* rails	physical	oscilloscope	NOT RUN	NOT RUN (Step 16.2)

Python (section 3)

ID	Test	Type	Command	Result on the original scripts	BETA package result
P-01	Dependency resolution	design check	pip install -r requirements.txt in a venv	PASSED (10 packages)	pinned beta/gui/requirements.txt installed
P-02	Syntax (ast) of 27 scripts	design check	python3 tools/check_python_imports.py	FAIL (exit 1) — GUI_V1.py IndentationError; 26/27 OK	n/a
P-03	Import test of third-party modules	design check	... --try-import (venv)	PASSED for GUI modules; FAIL for openEMS / CSXCAD (not on PyPI)	n/a
P-04	Module import of each GUI file	design check	python -c "import GUI_V6_Demo" etc.	PASSED V2...V6_Demo (import only); V1 FAIL	n/a
P-05	pyflakes	design check	pyflakes 9_Firmware/9_3_GUI/*.py	undefined names in GUI_V6.py:392 , GUI_V4_2_CSV.py:430 (informational)	n/a
P-06	Unit tests	design check	pytest	NOT RUN — no tests exist upstream	76 passed in 3.62s (executed 2026-10-09 for this edition)
P-07	GUI startup smoke	design check	procedure GUI-T05	NOT RUN	--selftest exit 0 (executed 2026-10-09); headless Tk smoke tests in the suite
P-08	Demo-data verification	design check	column/row check	PASSED (16 384 rows = 32 chirps × 512)	used by tests/test_sim_pipeline.py

ID	Test	Type	Command	Result on the original scripts	BETA package result
P-09	Hardware link (CDC / FTDI)	physical	—	NOT RUN	NOT RUN (Step 16.7)

PCB documentation (section 4)

ID	Test	Type	Command	Result
B-01	ERC (fresh run)	design check	EAGLE ERC	NOT RUN — stored approved ERC items: Main 4, others 0
B-02	DRC (fresh run)	design check	EAGLE RATSNEST; DRC	NOT RUN — stored: Main 211 approved + 2 390 airwires; Power 309 airwires; Synth 0; PA 0 (KiCad DRC of the beta boards: Main/RF PA 0 unconnected, Power 89 — beta/pcb/README.md)
B-03	Missing-library detection	design check	XML: all <library> embedded	PASSED — no external library references for any board
B-04	sch/brd consistency	design check	part/net set comparison (XML)	PASSED Synth, PA; FAIL Main (10 board-only signals, 210 value mismatches) and Power (version mismatch 9.6.2/7.4.0, 1 board-only signal)
B-05	BOM completeness	design check	python3 tools/gen_eagle_bom.py ... → docs/BOM/	FAIL — 0 MPN attributes on all boards; value-less references 244/80/6/47
B-06	Manufacturing-export completeness	design check	python3 tools/check_manufacturing_files.py	FAIL (exit 1) for the designer-released export; --include-generated exit 0 for the generated engineering/PCB/ packages (AC-B4)
B-07	Outline/hole extraction	design check	python3 tools/gen_board_outline_svg.py ...	PASSED for 4 boards
B-08	Stack-up confirmation	physical/vendor	fab quote with stack-up drawing	NOT RUN
B-09	Impedance coupon measurement	physical	TDR at fab	NOT RUN

Repository / documentation (section 5)

ID	Test	Command	Result
R-01	Inventory generation	python3 tools/gen_inventory_doc.py	PASSED (482 rows)
R-02	Documentation link validation	python3 tools/check_doc_links.py	FAIL — 34 broken references before the reconstruction (README 10_docs/* etc.); re-run for the residual list
R-03	Expected-artifact manifest	python3 tools/check_missing_files.py	FAIL — 26 P0 items missing (see docs/03_MISSING_COMPONENTS.md)

16.3 Acceptance criteria (sections A–H)

Copied from docs/TESTING/ACCEPTANCE_CRITERIA.md ("Each criterion is objective, references the test ID ... Current status is given honestly: nothing hardware-related is accepted"). Status column as recorded in the source on 2026-10-08/09; the BETA tree (section H) is evaluated on beta/, not on the originals.

ID	Criterion	Evidence required	Status
AC-F1	RTL parses and elaborates with iverilog and verilator with zero errors (F-01..F-03)	build/lint/iverilog_top.log empty of errors; verilator 0 %Error	NOT MET (original); see AC-X1
AC-F2	No module instantiated without definition; all Xilinx IP as committed .xci	Vivado elaboration log; ls 9_Firmware/9_2_FPGA/ip/*.xci	NOT MET (6 missing)
AC-F3	cntrt.xdc (or successor) has zero placeholders and every top-level port bit constrained or removed; F-04 exit 0	tool output	NOT MET
AC-F4	Pin map confirmed by the designer against the AMD FTG256 package file and the routed board	signed-off PIN_MAP_FROM_SCHEMATIC.md with "VERIFIED" column	NOT MET
AC-F5	Synthesis on the confirmed part with ≤ 90 % of LUT/FF/DSP/BRAM and zero unresolved CRITICAL WARNINGS	build/vivado/util.rpt	NOT MET
AC-F6	Timing closure WNS ≥ 0, WHS ≥ 0 on all constrained clocks; report_cdc no critical	timing.rpt, cdc.rpt	NOT MET
AC-F7	Testbench runs on xsim with pass/fail gating and passes	sim log with 0 assertion failures	NOT MET
AC-F8	Bitstream generated and archived with its build log	build/vivado/*.bit + log	NOT MET
AC-F9	(Physical) ADC capture validated with a CW tone	ILA capture / host data	NOT RUN
AC-S1	Cube package check exit 0 (S-02)	script output	NOT MET (no package; beta: exit 0)
AC-S2	.ioc regenerated; generated MSP equals repository MSP	diff	NOT MET
AC-S3	tools/check_stm32_includes.py exit 0 with Cube sources on the include path and zero case mismatches	tool output	NOT MET

ID	Engineering & Assembly Manual	Criterion	Evidence required	Status
AC-S4		Firmware compiles and links with <code>arm-none-eabi-g++</code> , zero errors; size < 1 MB flash / 320 KB RAM	map, size output	NOT MET (original); see AC-X2
AC-S5		Conflicts C1–C7 closed with documented decisions	docs/04_RECOVERY_TASKS.md R-STM-xx DONE	NOT MET
AC-S6		(Physical) CDC enumerates; start flag + settings accepted; status string received	host log	NOT RUN
AC-S7		(Physical) rail sequencing order and delays match Power Management V6.xlsx	scope captures	NOT RUN
AC-P1		<code>pip install -r requirements.txt</code> succeeds on a clean venv	pip log	MET (2026-10-08, Python 3.14.7)
AC-P2		All third-party imports succeed	tool output	MET for GUI modules
AC-P3		Every file in <code>9_Firmware/9_3_GUI</code> parses	tool exit 0	NOT MET (<code>GUI_V1.py</code>)
AC-P4		<code>GUI_V6_Demo.py</code> starts, shows moving targets, closes with exit 0	screenshot + exit code	NOT RUN
AC-P5		A unit-test suite exists and passes	pytest log	NOT MET (upstream); see AC-X3
AC-P6		One hardware GUI decodes the real firmware/FPGA packet format end-to-end	capture + decoded targets	NOT RUN
AC-P7		Packaged demo runs on a machine without Python	installer test log	NOT RUN
AC-B1		Fresh ERC and DRC reports with 0 unapproved errors for each board	reports under <code>docs/PCB/reports/</code> (directory to be created)	NOT MET
AC-B2		RATSNEST "Nothing to do" and 0 elements outside the outline for Main and Power	EAGLE status line	NOT MET
AC-B3		sch/brd consistent, single EAGLE version per board	consistency check pass	NOT MET (Main, Power)
AC-B4		Gerber + drill + fab + assembly + P&P + BOM(MPN) + schematic PDF for all four boards	tools/check_manufacturing_files.py exit 0	PARTIALLY MET (generated packages exist for all four boards; designer-released EAGLE export, MPNs, vendor fab notes NOT MET)
AC-B5		BOMs 100 % MPN, 0 empty values, DNP column	docs/BOM/* with <code>mpn_status</code> all VERIFIED	NOT MET
AC-B6		Vendor stack-up drawing per board consistent with DRU and impedance note	fab documents	NOT MET
AC-B7		(Physical) impedance coupons 50 $\Omega \pm 10\%$, 100 $\Omega \pm 8\%$	TDR report	NOT RUN
AC-B8		Design conflicts K1, K2, K3, K4, K7 closed	decision records	NOT MET
AC-M1		Outline/hole drawings for all PCBs	engineering/MECHANICAL/ DXF, STEP, plan view, dimension sheets	MET (thickness ASSUMED)
AC-M2		Enclosure, antenna and pedestal CAD committed with drawings	STEP + PDF under <code>10_docs/Hardware/</code> (missing in the repository)	NOT MET (PROPOSED designs exist under <code>engineering/DESIGN/</code> , chapter 10)
AC-M3		Assembly guide <code>10_docs/assembly_guide.md</code> (missing in the repository)	file exists, link check passes	NOT MET (this manual's chapter 15 is the proposed content)
AC-M4		Exploded view with balloons matching BOM	PDF	PARTIAL (CONCEPTUAL view with balloons; real geometry BLOCKED, MDR-06)
AC-M5		Mass table per assembly	measured or CAD-derived values	NOT MET (estimates only)
AC-D1		Documentation tree of <code>claudes.md</code> §13 present	tools/check_missing_files.py DOC-* all present	MET
AC-D2		<code>tools/check_doc_links.py</code> exit 0	tool output	NOT MET
AC-D3		README corrected (Gerber claim, FPGA part, <code>10_docs</code> references)	diff	NOT MET (needs the owner's decision)
AC-D4		LICENSE file consistent with the MIT badge; <code>.gitignore</code>	files	NOT MET
AC-D5		Every P0 item in <code>docs/03_MISSING_COMPONENTS.md</code> has an owner and a recovery task	docs/04_RECOVERY_TASKS.md	MET (tasks defined; none closed)
AC-E1		Every registered drawing has native file and exports present and well-formed	python3 tools/gen_drawing_register.py --check exit 0	MET (re-run 2026-10-09: 89 drawings, 333 files, 0 problems)
AC-E2		EAGLE XML ↔ KiCad conversion counts agree for all boards	engineering/VALIDATION/PCB_CROSS_CHECK.md	MET (32/32 rows OK)
AC-E3		No missing symbols/footprints; schematic ↔ board part lists identical	tools/gen_schematic_reports.py exit 0	MET
AC-E4		Any drawing VERIFIED against EAGLE output or hardware	VALIDATION/DRAWING_CHECKS.md §2	NOT MET (0 VERIFIED)
AC-E5		Enclosure/antenna/pedestal/cooling drawings exist	register MECH-ENC/ANT/PED/COOL	NOT MET (BLOCKED; DSN-* proposals exist)

id	Engineering & Assembly Manual	Criterion	Evidence required	Status
AC-E6		Proposed designs (DSN-*) registered with native editable files and exports present	tools/gen_drawing_register.py --check	MET (16 PROPOSED entries after regeneration, 0 file problems)
AC-E7		Antenna proposal simulated (S11 < -10 dB at f0 ± B/2) and coupon measured	openEMS log + VNA data	simulation MET per AC-X6; coupon NOT RUN
AC-E8		Owner approval of decisions D-01...D-15 recorded	signed decision log	NOT MET
AC-X1		beta RTL parses/elaborates/lints with 0 errors and all testbenches pass	beta/fpga/logs/	MET (BETA)
AC-X2		beta firmware compiles and links; host tests pass	beta/stm32/build_out/ , logs/	MET (BETA)
AC-X3		beta GUI test suite passes; packaged app starts	beta/gui pytest (source says 55 passed; 76 passed on 2026-10-09)	MET (BETA)
AC-X4		beta boards: 0 unconnected on Main/RF PA/Synth; Power ≤ 100 open with documented reasons	beta/pcb/*/exports/reports/DRC_report.json	MET (BETA; Power 89 listed)
AC-X5		host-link bridge testbench, firmware build and GUI parser agree on one frame vector	tb_frame.hex shared	MET (BETA)
AC-X6		antenna row S11 < -10 dB at f0 in simulation	TUNING_LOG.md	MET (-18 dB; band 128 MHz)

Observation MAN-16-1: docs/TESTING/ACCEPTANCE_CRITERIA.md AC-E1/AC-E6 still quote "73 drawings" and "11 PROPOSED entries"; the register regenerated for this edition has 89 drawings and 16 PROPOSED entries (Appendix A). The acceptance file was not modified (it is outside manual/).

16.4 Evidence collection

Rules (source: docs/TESTING/VALIDATION_PLAN.md §6, copied): every executed command's output is saved under build/ (lint logs in build/lint/, Vivado in build/vivado/, STM32 in build/stm32/) and referenced by test ID in docs/TESTING/ACCEPTANCE_CRITERIA.md when reporting; physical tests record instrument, serial number, date and raw capture file; a test result changes from NOT RUN to PASSED/FAILED only with an attached log. BETA logs already exist under beta/fpga/logs/, beta/stm32/logs/ and are referenced by the tables above.

Evidence to collect during the first integration (source: engineering/ASSEMBLY/ASSEMBLY_SEQUENCE.md §D, extended by the steps above), to be stored under engineering/VALIDATION/ with the acceptance ID it closes:

Item	Step	Closes
Rail voltages, no load, per X2...X35 (CP-4)	15.4	AC-S7 input
Scope captures of the enable sequence F1-F10 with delays	16.2	AC-S7
USART3 banner and host CDC enumeration log (VID/PID, serial = UID)	16.1	AC-S6
Settings packet accepted → status string received	16.7	AC-S6, AC-P6
AD9523 lock status and counter readings 100 / 120 / 400 MHz	16.4	Step 16.4 acceptance
CAL_STAT / CAL_LANE_INFO / CAL_ERR / CAL_UNDET per lane, ADC pattern used	16.5	AC-F9
FPGA DONE, DIG_0..4 handshake capture; Vivado utilisation/timing/CDC reports and bitstream log	16.3	AC-F5...AC-F8
PA IDQ per board and plate temperature vs time (drain gating on/off noted)	16.6	thermal design case B
ADAR1000 SPI capture (CS, SCLK, phase words)	16.8	D-06, D-19
Spectrum and envelope at 10.5 GHz into the load; measured chirp bandwidth B	16.9	parameter-table TBD
Scan time per revolution, motor class and Stepper_steps used	16.10	D-12
Photos of the first harness with cable labels; real cut lengths of the 144 cables	15.14-15.18	DSN-HAR-01
Measured board thickness and component heights	15.1, 15.9	G-01, G-02

<!-- chapter 17: K1-K8, D-01...D-19, G-01...G-12, MDR-*, register of unresolved items -->

Open issues and decisions — consolidated registers

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Status summary: this chapter is a register compilation. Conflicts K1–K8 are copied from docs/SYSTEM/BLOCK_DIAGRAM.md §4 (SOURCE-DERIVED findings, all OPEN); decisions D-01...D-15 from engineering/DESIGN/00_DESIGN_BASIS.md §2 and D-16...D-19 from engineering/DESIGN/HOST_LINK/HOST_LINK_DESIGN.md §3–4 (PROPOSED DESIGN, none approved by the owner); geometry gaps G-01...G-12 from engineering/VALIDATION/UNRESOLVED_GEOMETRY.md (BLOCKED / UNRESOLVED, with PROPOSED values where noted); recovery guides MDR-01...13 from engineering/MISSING_DRAWINGS_RECOVERY_PLAN.md; task states from docs/04_RECOVERY_TASKS.md. Four observations raised while compiling this manual are added as MAN-01...MAN-04. Nothing here is closed.

Sources: docs/SYSTEM/BLOCK_DIAGRAM.md §4, docs/AERIS10_BETA_ENGINEERING_REPORT.md §12, engineering/DESIGN/00_DESIGN_BASIS.md §2–3, engineering/DESIGN/HOST_LINK/HOST_LINK_DESIGN.md §3–4, §6, engineering/VALIDATION/UNRESOLVED_GEOMETRY.md, engineering/MISSING_DRAWINGS_RECOVERY_PLAN.md, docs/04_RECOVERY_TASKS.md, docs/03_MISSING_COMPONENTS.md, docs/TESTING/ACCEPTANCE_CRITERIA.md.

Planned figures: none.

1. Configuration conflicts K1–K8

Eight conflicts between the CAD, the firmware, the RTL, the GUI and the documentation; each has an owner and none is closed (source: docs/SYSTEM/BLOCK_DIAGRAM.md §4, copied verbatim):

#	Topic	A	B	Owner
K1	FPGA part	CAD/xlsx/drawio XC7A50T-2FTG256I	README/docs/XDC XC7A100T	hardware designer
K2	STM32 HSE	schematic 8 MHz crystal	firmware 25 MHz	hardware designer
K3	Host data path	CAD: STM32 USB-FS only	RTL/GUI V6: FT601 USB 3.0; GUI V2–V5: FT2232H	system architect
K4	PA supply	xlsx 16 × 22 V / 2 A	Power Board Vin [12–17] V, no 22 V rail	power designer
K5	FPGA packet format	RTL 0xAA ... 0x55	GUI A5C3 ... CRC16	software
K6	ADF4382 pins	main.h PG6..PG15 (= schematic)	adf4382a_manager.h PG0..PG9	firmware
K7	Synth oscillators	schematic OCXO 100 MHz + 2 × VCXO 50 MHz	xlsx VCXO 100 MHz; firmware vcxo_freq = 100 MHz	RF designer
K8	Antenna	README patch 8×16 / slotted 32×16	two different waveguide simulations, no CAD	antenna designer

Interim positions taken by the BETA work, which the owner must confirm or overturn (source: docs/AERIS10_BETA_ENGINEERING_REPORT.md §4, §7, §8, §12; beta/fpga/README.md "Remaining work" item 2; engineering/DESIGN/00_DESIGN_BASIS.md §2):

#	BETA / PROPOSED interim position	Decision needed from	Blocking for
K1	beta/fpga/vivado/create_project.tcl default part xc7a50tftg256-2 = schematic U42	hardware designer: confirm the mounted part	R-FPGA-01, AC-F5 (resource fit of two 1024-point FFT IPs and 64 DSP48E1 FIRs on the 50T)
K2	beta/stm32 clock tree written for the real 8 MHz crystal	hardware designer: board revision check of XTAL1	R-STM-05, AC-S7
K3	option B (SPI bridge through the STM32 CDC) implemented as BETA; option A (FT601 on Main Board rev. B, bank 35) prepared as pin plan and beta/pcb/MAIN_BOARD_REVB	system architect: D-16 / D-17 / D-18	R-SYS-01, AC-P6, slip-ring channel count D-13
K4	DSN-PSU-01 22 V module proposed; needs a TX_GATE line (FPGA spare pin UNRESOLVED)	power designer: D-14	MDR-12, thermal case B (drain gating)
K5	bridge frame (0xA5 0x5A sync, CRC-16/CCITT-FALSE) implemented in RTL, firmware and GUI	software: adopt the frame as the one format	R-SYS-01, AC-P6
K6	ADF4382 pin collision fixed in beta/stm32 to the main.h / schematic assignment	firmware: confirm	R-STM-06
K7	none — the BETA firmware keeps vcxo_freq = 100 MHz; schematic part values are 50 MHz VCXO / 100 MHz OCXO	RF designer: identify the mounted oscillators	clock tree of the AD9523, all programmed frequencies in chapter 2 §4
K8	PROPOSED 16-row × 8-patch panel per D-01; waveguide variant only as a sizing sheet	antenna designer: D-01 and the variant	R-MECH-01, MDR-03, AC-E7, range performance (chapter 1 §5)

2. Design decisions D-01...D-19

All decisions are PROPOSED DESIGN content; none has been approved by the owner (AC-E8 "Owner approval of decisions D-01...D-15 recorded" is NOT MET; source: docs/TESTING/ACCEPTANCE_CRITERIA.md, row AC-E8). D-01...D-15 (source: engineering/DESIGN/00_DESIGN_BASIS.md §2, columns Decision / Rationale / Alternatives copied verbatim; the "Owner approval" column is compiled from 00_DESIGN_BASIS.md §3, docs/04_RECOVERY_TASKS.md R-DSN-06 and docs/AERIS10_BETA_ENGINEERING_REPORT.md §12):

ID	Decision	Rationale	Alternatives	Owner approval
D-01	Design the patch-array (Nexus) antenna as the primary antenna; provide only a sizing sheet for the slotted-waveguide (Extended) variant	A patch array is manufacturable with the same PCB workflow and verifiable with the openEMS/Qucs tools already used in	waveguide array (CONCEPTUAL sizing sheet in ANTENNA/README.md)	REQUIRED — antenna designer (R-DSN-06 BLOCKED); rev. B of D-01 pending; if B exceeds ≈ 100

AERIS10	Engineering & Assembly Manual	Decision	Rationale	Alternatives	Owner approval
			5_Simulations ; slotted waveguides need machining and a full EM design		MHz the series row feed must change (report §8)
D-02	The 16 radiating "elements" are 16 horizontal rows stacked vertically at 14.3 mm; each row is a series-fed resonant array of 8 patches	Matches "8×16", the elevation-scanned ULA of the firmware and a fixed narrow azimuth beam rotated mechanically	corporate feed per row (lower squint, more layout area)	REQUIRED (AC-E8)	
D-03	8 patches per row	parameter table row 77	—	REQUIRED (AC-E8)	
D-04	Substrate RO4350B, h = 0.508 mm, 35 μm Cu, single dielectric layer, full back ground	datasheet in repo; PCBWay impedance note is for RO4350B; 20 mil gives good patch bandwidth (~2 %) with manageable surface waves	0.762 mm (wider band, more surface wave)	REQUIRED (AC-E8)	
D-05	50 Ω end-launch 2.92 mm connectors on the left edge, one per row, 14.3 mm pitch (body ≤ 12 mm wide)	simplest verifiable feed; keeps the ground plane continuous	probe feed from the back (allows SMA on rear)	REQUIRED (AC-E8)	
D-06	Row phase centre spacing is 1/2; each row's feed line length is equal so the ADAR1000 calibration tables remain valid	elevation scanning depends on equal electrical length to each row	—	REQUIRED (AC-E8)	
D-07	Board stack pitch 25 mm (component heights assumed 15 mm top / 4 mm bottom until measured)	no 3-D models; standard 25 mm M3 standoffs	measure, then shrink	REQUIRED — mechanical owner; depends on G-02 measurement (R-DSN-06, R-DSN-07)	
D-08	Enclosure: folded 2.5 mm aluminium chassis + lid, 10 mm board-to-wall clearance, IP54 target, antenna panel in the front face behind a PTFE/ABS radome window	aluminium gives the heat path for the PA spreader; sheet metal keeps cost low	machined frame	REQUIRED (AC-E8; MDR-01 "owner approval of D-07...D-09")	
D-09	Board arrangement: Power Board on the base, Main Board above it (25 mm), Synth on the Main-Board tier beside the SMA field; PA spreader vertical behind the antenna panel at the front; stepper/slip ring below the base	shortest coax from Synth J-ports to Main J1/J18–J23; rails cables short; RF to the front	side-by-side single tier (larger footprint)	REQUIRED (AC-E8; MDR-01)	
D-10	Thermal design case: ambient 45 °C, TBASE(PA) ≤ 85 °C, drain gated with the pulse train (duty computed from the timing); continuous-bias case reported as infeasible without liquid cooling	16 × 37 W = 591 W quiescent is not coolable in a sealed rotating head; gating requires a hardware change (D-14)	—	REQUIRED (AC-E8); consequence of K4	
D-11	10 mm aluminium heat spreader common to all 16 PA boards, fan-cooled fin stack at the rear, 1.5× airflow margin	conduction from 16 hot spots into one plate, then forced air	per-PA heatsinks	REQUIRED (AC-E8)	
D-12	Azimuth drive: stepper → GT3 belt 1:3 → turntable; revised by DSN-CALC-01 §2 : with a ≈ 10 kg head a NEMA 23 needs ≥ 200 ms per 7.2° step (revolution ≈ 19 s); a NEMA 34 (3 N·m class) allows ≈ 100 ms; firmware <code>Stepper_steps = 600</code> for 1:3	torque calculation from the detailed mass table; belt isolates the motor from the slewing ring	ratio 1:6 (<code>Stepper_steps = 1200</code>), lighter head	REQUIRED — motor class / ratio (R-DSN-06 BLOCKED; report §12)	
D-13	Slip ring 12 channels (VIN ×4 @10 A, 22 V ×4 @10 A, USB 2.0 ×4) through a 60 mm bore slewing bearing; everything else rotates with the head	only DC in and the host link cross the rotation	Ethernet/wireless host link (removes 4 channels)	REQUIRED — channel count depends on K3 and on CBL-144 (stepper driver location, "DECISION NEEDED") (R-DSN-06)	
D-14	22 V PA supply: synchronous boost 12–17 V → 22 V, 150 W average / bulk capacitance for 45 A pulse peaks, high-side eFuse/MOSFET switch from <code>EN/DIS_RFPA_VDD</code> , plus a per-pulse drain-gating FET per PA board (TTL from the FPGA/MCU DIG lines — pin to be allocated)	closes conflict K4; the firmware already sequences VG before VD	separate 22 V mains PSU on the pedestal (bigger slip ring)	REQUIRED — gating line allocation (R-DSN-06; MDR-12)	
D-15	Harness: cable IDs from <code>interconnection_table.md</code> ; coax RG-405 (0.086") hand-formable equal-length for Main↔PA and Synth↔Main; Molex 22-01-3027/3037 crimp housings for the 2-/3-pin headers; 20-way IDC ribbon for SV1	standard parts matching the footprints in the schematics	—	REQUIRED (AC-E8); see MAN-03 on cable-ID schemes	

Host-link decisions D-16...D-19 (source: `engineering/DESIGN/HOST_LINK/HOST_LINK_DESIGN.md` §3 option table and §4; owner per `docs/SYSTEM/BLOCK_DIAGRAM.md` §4 row K3 and `docs/AERIS10_BETA_ENGINEERING_REPORT.md` §12):

ID	Decision	Content	Status	Owner approval
D-16	Option A — FT601 on Main Board rev. B is the target for rev. B	route U6 to bank 35 (46 I/Os), add USB 3 connector, crystal, RREF, ESD (<code>ft601_added_parts_B0M.csv</code>); throughput 400 MB/s; RTL already written (fix 2-bit BE → 4-bit; honour TXE_N); host driver FTDI D3XX; risk: 10-layer board respin, USB 3 SI	pin plan <code>ft601_pin_assignment.csv</code> + <code>ft601_bank35.xdc</code> ; <code>beta/pcb/MAIN_BOARD_REVB</code> layout; FT601 datasheet not in the repository — AC timing, RREF, VBUS, 1.0 V core supply to be verified before the schematic is edited	REQUIRED — system architect (K3)
D-17	Option B — SPI bridge via the STM32, implement now (BETA)	no PCB change: <code>DIG_5/6/7</code> + SPI1 already routed to the FPGA; throughput ≤ 1 MB/s (CDC-bound); new RTL <code>host_bridge_spi.v</code> + packer, STM32 <code>host_bridge.c</code> , GUI parser; risk: protocol only, SPI1 shared with the ADAR1000 (time-multiplexed)	implemented in <code>beta/fpga</code> , <code>beta/stm32</code> , <code>beta/gui</code> ; <code>tb_host_bridge_top</code> PASS; bench test open (§6)	REQUIRED — system architect (K3); firmware rule "no ADAR1000 transaction overlaps a bridge read" to be enforced
D-18	Option C — Ethernet mezzanine on bank 35, documented only	new PCB with RGMII PHY on the 50 free bank-35 pins; 100 MB/s; new MAC/UDP stack; highest risk	not pursued	—
D-19	RTL changes for a real FT601	widen <code>ft601_be</code> to 4 bits, drive <code>BE = 4'b1111</code> for full words, respect <code>TXE_N</code> back-pressure, add <code>ft601_reset_n</code> / <code>wakeup_n</code> / <code>siwu_n</code> as outputs	recorded for <code>beta/fpga</code> , not yet applied there	follows D-16

What the proposals do **not** settle (source: `engineering/DESIGN/00_DESIGN_BASIS.md §3`, copied): conflicts K1–K8, component heights (G-02), final Power Board outline (G-10), whether the host link stays USB (affects D-13), antenna variant (D-01), the PA pulse-gating implementation (D-14 needs a spare control line), and all RF performance targets (gain, bandwidth B is TBD in the parameter table).

3. Unresolved geometry G-01...G-12

Every mechanical quantity that the drawings need but no repository file provides; nothing below was estimated into a drawing (source: `engineering/VALIDATION/UNRESOLVED_GEOMETRY.md`, VAL-GEO-01 Rev A, copied verbatim):

ID	Quantity	Needed by	Available evidence	Missing	Status
G-01	PCB thickness (all 4 boards)	STEP bodies, stack-up, stand-offs	none (KiCad assumed 1.6 mm)	vendor stack-up / designer decision	BLOCKED — MISSING DATA
G-02	Component heights (both sides)	enclosure clearances, stand-off heights, exploded view	packages only; 3-D URNs not embedded	3-D models or measurement of assembled boards	BLOCKED
G-03	Board masses (bare and assembled)	pedestal/motor sizing, mass table	area × assumed thickness (ESTIMATE in <code>MECHANICAL/dimensions/</code>)	weighing or CAD with densities	BLOCKED (estimate only)
G-04	Enclosure: envelope, wall thickness, mounting bosses, connector cut-outs	ASM-EXP-01, M1–M6	README text references only (<code>10_docs/Hardware/Enclosure</code> absent)	enclosure CAD	BLOCKED
G-05	Relative board placement and stacking height	internal layout drawing	none	layout decision	BLOCKED (CONCEPTUAL drawing only)
G-06	Antenna element geometry, substrate, feed network, radome	antenna assembly drawing	<code>02_hardware/04_antenna_beamforming.md</code> ($\lambda/2 = 14.3$ mm, aperture 214.3 mm), <code>8_Utils/Antenna_Array.jpg</code> (undimensioned photo), two contradictory waveguide simulations	antenna CAD and measurements	BLOCKED
G-07	Pedestal, stepper, slip ring, bearing envelope	pedestal drawing	firmware constants only (200 steps/rev, 50 positions)	mechanical design	BLOCKED
G-08	Heatsink / fan geometry and airflow	cooling drawing	PA dissipation implied by xlsx IDQ	thermal design	BLOCKED
G-09	Cable lengths, routing, bend radii (SMA coax ×34, Molex ×~50)	harness drawing	connector list only	harness design	BLOCKED
G-10	Power Board final outline (hole pattern suggests a smaller board than 280 × 300)	fab drawing	<code>POWER_SUPPLY_dimensions.md</code> (holes at inner positions)	designer confirmation	UNRESOLVED
G-11	Fastener sizes (Ø3.2 holes → M3 inferred)	parts list	hole diameters	specification	UNRESOLVED (inferred)
G-12	Keep-out zones around RF connectors and the QPA2962	layout/enclosure	none	designer input	BLOCKED

Update 2026-10-09 (source: same file, "Update 2026-10-09 — proposals", copied): G-04, G-05, G-07, G-08, G-09, G-12 now have PROPOSED values in `engineering/DESIGN/` (D-07...D-15); they stay open until the owner accepts the decisions and the assumed inputs (G-01 thickness, G-02 component heights, G-11 fasteners) are measured/specified. G-03 mass: FreeCAD volume-based estimate head ≈ 9.9 kg (includes component envelopes as solid plastic — overestimate), pedestal ≈ 12 kg + 1.1 kg stepper (`MECHANICAL/CAD/aeris10_mass_table.json`). G-06 antenna: PROPOSED patch panel 165 × 248 mm; G-10 Power Board outline unchanged. Photographs in `8_Utils/` show a prototype but carry no scale reference and were not used to derive dimensions.

ID	Owner / decision needed
G-01	PCB fabricator stack-up or designer decision per board (MDR-07)
G-02	measurement of assembled boards or 3-D models assigned in KiCad (MDR-09); gates D-07
G-03	weighing of assembled boards; replaces the ESTIMATE and the FreeCAD overestimate; feeds D-12
G-04, G-05, G-08, G-09, G-12	owner acceptance of D-07...D-11, D-15 (PROPOSED values exist)
G-06	antenna designer: D-01 and K8
G-07	owner acceptance of D-12, D-13 and part selection (bearing, slip ring, motor; MDR-05)
G-10	Power Board designer confirmation of the outline (also MDR-08)
G-11	fastener specification (M3 inferred from Ø3.2)

4. Missing-drawing recovery guides MDR-01...MDR-13

One guide exists for each drawing that could not be generated from repository evidence; the generator scripts themselves are the procedure for the drawings that were generated (source: `engineering/MISSING_DRAWINGS_RECOVERY_PLAN.md`, ENG-MDR-01 Rev A, headings, "Missing" fields and the status table "Status update 2026-10-09 — proposals now exist"; MDR-13 is referenced only in `engineering/DESIGN/HOST_LINK/HOST_LINK_DESIGN.md §6`):

AERIS-10 - Engineering & Assembly Manual					
ID	Drawing purpose	Missing source information (not recoverable from the repository)	Proposal delivered 2026-10-09	Remaining to reach SOURCE-DERIVED / VERIFIED	Owner
MDR-01	MECH-ENC-01 Main enclosure (overall geometry, mounting, PCB supports, connector openings)	envelope, material, wall thickness, board stacking order and spacing (G-04, G-05), component heights (G-02), thermal path (G-08), ingress/EMC requirements	engineering/DESIGN/MECHANICAL/ (FreeCAD model DSN-MECH-3D, drawings DSN-MECH-01...03)	owner approval of D-07...D-09; sheet-metal detailing (bends, fasteners, gaskets); component heights	mechanical owner
MDR-02	MECH-INT-01 Internal component layout and cable routing	enclosure (MDR-01), cable types/lengths/gauges (G-09), bend-radius rules for the 10.5 GHz coax	DSN-MECH-04 (connector positions from P&P), DSN-HAR-01 (144 cables, lengths)	first-fit length check; unmatched rail pairs; stepper driver location	mechanical owner + harness
MDR-03	MECH-ANT-01 Antenna assembly	everything dimensional (G-06); which simulation reflects the built prototype; feed/transition to the PA SMA ports	DSN-ANT-01 (native KiCad board, calc sheet, openEMS model)	run openEMS (one row done, report §8); coupon; connector footprint	antenna designer (K8, D-01)
MDR-04	MECH-COOL-01 Cooling and power assembly	dissipation budget (no current budget per rail), ambient/enclosure conditions, heatsink part, fan part (G-08)	DSN-THM-01 + fin fields in DSN-MECH-3D	PA-board via-field Rth; fan part selection; test	thermal / power designer
MDR-05	MECH-PED-01 Pedestal / azimuth drive	all mechanics (G-07)	DSN-MECH-05 + pedestal in DSN-MECH-3D	bearing/slip-ring/motor part selection; mast interface	mechanical owner (D-12, D-13)
MDR-06	ASM-EXP-01 upgrade from CONCEPTUAL to SOURCE-DERIVED exploded view	MDR-01..05 outputs, fastener spec	still CONCEPTUAL (ASM-EXP-01); the FreeCAD assembly can now be exploded in the GUI	FreeCAD Assembly exploded view + balloons	documentation / mechanical
MDR-07	PCB fabrication drawings with vendor notes (PCB-*09/11 to reach VERIFIED)	material, finished thickness, copper weights, finish, mask/silk colours, impedance table, tolerances	—	EAGLE-native Gerbers compared with the KiCad set; fabricator DFM report; status VERIFIED only after both agree and the fabricator confirms the stack-up	PCB designer + fabricator
MDR-08	Completion of the Main and Power Board layouts	routing of the listed airwires in EAGLE (master); Power Board final outline (G-10)	BETA: Main 15 → 0 unconnected, Power 308 → 89 (listed in <code>beta/pcb/POWER_SUPPLY/UNROUTED.md</code>)	<code>unconnected_items = 0</code> in the DRC report; airwires = 0 in the EAGLE file; regenerate packages	PCB designer
MDR-09	3-D component models and assembled STEP	model files (EAGLE <code>package3d_urn</code> are Autodesk cloud IDs, not embedded)	—	assign 3-D models per package in KiCad, <code>kicad-cli pcb export step --subst-models</code> ; record max height → resolves G-02	PCB designer
MDR-10	Native KiCad schematics (editable twin of the EAGLE schematics)	KiCad's schematic importer is GUI-only	—	import each <code>.sch</code> via KiCad GUI, ERC, netlist comparison (same net and pin counts)	PCB designer
MDR-11	Hardware-verified drawings (status VERIFIED)	comparison with an EAGLE print/CAM output or with the physical board	—	record each comparison in <code>VALIDATION/DRAWING_CHECKS.md</code> §2 with date, method and result	hardware owner
MDR-12	22 V PA supply (new)	component-level schematic and layout; <code>TX_GATE</code> pin	DSN-PSU-01 block schematic, BOM, nets	KiCad component-level schematic + layout; <code>TX_GATE</code> pin allocation; bench test of one gate channel	power designer (D-14, K4)
MDR-13	Main Board rev. B schematic/layout for option A (FT601)	EAGLE schematic update for U6 (46 signals, VCC33/VCCIO, crystal, RREF, VBUS divider, ESD, USB 3 connector); FT601 datasheet checks	<code>beta/pcb/MAIN_BOARD_REVB/</code> KiCad project with <code>NETLIST_DELTA.csv</code> (new FT_* nets U6 ↔ U42.C4) and <code>BOM_MAIN_BOARD_REVB_beta.csv</code> (BETA; no README in the directory at the time of writing)	EAGLE-native schematic update; FT601 datasheet checks; FTDI D3XX host driver test	hardware designer (D-16, K3)

5. Open tasks

5.1 Reconstruction tasks (source: `docs/04_RECOVERY_TASKS.md`, section headings; status date 2026-10-08: DONE 7, OPEN 20, BLOCKED 9)

ID	Task	Status
R-SYS-01	Decide the host data path and packet format (K3/K5)	BLOCKED
R-FPGA-01	Confirm FPGA part	BLOCKED
R-FPGA-02	Verify the schematic-derived pin map	BLOCKED
R-FPGA-03	Fix syntax and declaration-order defects	OPEN (done in <code>beta/fpga</code> , not in the originals)

	Task	Status
R-FPGA-04	Create the Vivado project	OPEN
R-FPGA-05	Recover or re-implement missing modules and FFT IP	BLOCKED → OPEN if originals unavailable
R-FPGA-06	Memory files and LUT authority	BLOCKED
R-FPGA-07	Functional redesign items	OPEN, design work
R-FPGA-08	Constraints: clock groups, CDC, I/O delays	OPEN
R-FPGA-09	Testbench with pass/fail	OPEN
R-FPGA-10	Synthesis, implementation, bitstream	OPEN
R-STM-01	Install toolchain and STM32CubeF7	OPEN
R-STM-02	Regenerate CubeMX project and USB CDC files	OPEN
R-STM-03	Build-set cleanup	OPEN
R-STM-04	Fix USB RX binding and start-flag padding	OPEN
R-STM-05	Resolve HSE 8 MHz vs 25 MHz	BLOCKED
R-STM-06	Fix ADF4382 driver (pins + platform ops + CS)	OPEN
R-STM-07	GPS path	OPEN
R-STM-08	Compile, link, size, static analysis	OPEN
R-GUI-01	Consolidate GUI versions	OPEN
R-GUI-02	Unit tests	OPEN
R-GUI-03	Packaging and smoke test	OPEN
R-PCB-01	Complete Main Board layout	BLOCKED
R-PCB-02	Complete Power Board layout and fix version mismatch	BLOCKED
R-PCB-03	RF PA production export	PARTIALLY DONE 2026-10-09 (KiCad-converted package; EAGLE-native export and fabricator review OPEN)
R-PCB-04	Frequency Synthesizer production export	PARTIALLY DONE 2026-10-09 (same)
R-PCB-05	BOM completion with MPNs	OPEN
R-PCB-06	Stack-up confirmation	BLOCKED
R-MECH-01	Obtain enclosure, antenna, pedestal CAD and part numbers	BLOCKED
R-MECH-02	Write <code>10_docs/assembly_guide.md</code> (missing in the repository) and exploded view	PARTIALLY DONE 2026-10-09 (ASSEMBLY_SEQUENCE.md, PARTS_LIST.md, CONCEPTUAL exploded view; real geometry needs R-MECH-01)
R-DOC-01	Repository hygiene	OPEN, owner decisions
R-DOC-02	Datasheet collection	OPEN
R-DOC-03	Recover <code>STM32_ALG0.docx</code>	BLOCKED

The statuses of the R-FPGA, R-STM and R-GUI tasks refer to the original trees; the corresponding BETA work is tracked separately as R-BETA-01...09 (section 5.4). The missing-component register `docs/03_MISSING_COMPONENTS.md` (IDs FPGA-00...15, STM-01...19, GUI-01...08, MECH-01...06, REPO-01...05, with P0/P1/P2 priorities) maps every item to one of these tasks (source: `docs/03_MISSING_COMPONENTS.md`, lead-in and tables).

5.2 Engineering drawings package R-ENG (source: `docs/04_RECOVERY_TASKS.md`, "Engineering drawings package", copied verbatim)

Task	Status	Evidence
R-ENG-01 — Render all schematics to SVG/PDF/PNG with title blocks	DONE	<code>engineering/ELECTRICAL/schematics/*/</code> (7 sheets); <code>tools/render_eagle_schematic.py</code> , <code>tools/svg_sheets_to_pdf.py</code>
R-ENG-02 — Netlists, component-to-net, unresolved-connection and missing-symbol/footprint reports	DONE	<code>engineering/ELECTRICAL/netlists/</code> , <code>connection_diagrams/</code> ; <code>tools/gen_schematic_reports.py</code> exit 0
R-ENG-03 — Convert the four EAGLE boards to KiCad and export Gerber/drill/PDF/SVG/DXF/STEP/P&P/IPC-2581/IPC-D-356/DRC/3-D	DONE (generated; not designer-released)	<code>engineering/PCB/<BOARD>/</code> , <code>engineering/VALIDATION/CAD_EXPORT_LOG.md</code> (0 failed steps), <code>PCB_CROSS_CHECK.md</code> 32/32 OK

Task	Status	Evidence
R-ENG-04 — System block / interconnection / data-flow / power-distribution diagrams (editable DOT + Mermaid)	DONE	engineering/SYSTEM/ , engineering/ELECTRICAL/power_distribution/
R-ENG-05 — Software architecture diagrams (FPGA hierarchy, pipeline, STM32, USB CDC, Python, end-to-end)	DONE	engineering/SOFTWARE_DIAGRAMS/ (SD-01...SD-07)
R-ENG-06 — Mechanical package from PCB geometry (DXF ×2 sources, STEP, 1:1 plan view, dimension sheets, hole tables)	DONE	engineering/MECHANICAL/ ; tools/gen_mechanical_package.py
R-ENG-07 — Drawing register + file check, recovery plan, validation records	DONE	engineering/DRAWING_REGISTER.md (73 drawings, 0 file problems), MISSING_DRAWINGS_RECOVERY_PLAN.md (MDR-01...11), VALIDATION/*
R-ENG-08 — Enclosure, internal layout, antenna, cooling, pedestal drawings	BLOCKED (MDR-01...05)	no source geometry; VALIDATION/UNRESOLVED_GEOMETRY.md G-01...G-12
R-ENG-09 — Upgrade exploded view from CONCEPTUAL to SOURCE-DERIVED	BLOCKED (needs R-ENG-08)	MDR-06
R-ENG-10 — Vendor fabrication notes + stack-up → fabrication drawings; EAGLE-native Gerbers compared with the KiCad set	OPEN	MDR-07
R-ENG-11 — Finish Main/Power routing, regenerate packages	OPEN (designer)	MDR-08; KiCad unconnected_items 15 / 308
R-ENG-12 — 3-D component models → assembled STEP, component heights	OPEN	MDR-09
R-ENG-13 — Native KiCad schematic twins + ERC (GUI import)	OPEN	MDR-10

5.3 Proposed designs R-DSN (source: docs/04_RECOVERY_TASKS.md , "Proposed designs", copied verbatim)

Task	Status	Evidence
R-DSN-01 — Antenna panel proposal (KiCad, calc, openEMS model)	DONE (PROPOSED; not simulated)	engineering/DESIGN/ANTENNA/
R-DSN-02 — Thermal budget + drain-gating requirement	DONE (PROPOSED)	engineering/DESIGN/THERMAL/
R-DSN-03 — 22 V PA supply/gate module block design	DONE (PROPOSED) → KiCad capture OPEN (MDR-12)	engineering/DESIGN/ELECTRICAL/PA_SUPPLY_22V/
R-DSN-04 — Head + pedestal 3-D model and 2-D drawings	DONE (PROPOSED)	engineering/DESIGN/MECHANICAL/
R-DSN-05 — Harness schedule	DONE (PROPOSED)	engineering/DESIGN/HARNESS/
R-DSN-06 — Owner decisions D-01, D-07, D-12, D-13, D-14	BLOCKED (owner)	engineering/DESIGN/00_DESIGN_BASIS.md §2–3
R-DSN-07 — Simulate/measure the antenna, test one gate channel, measure component heights	OPEN	ANTENNA_DESIGN_CALC.md §7; PA_SUPPLY_22V/README.md

The R-DSN-01 status "not simulated" predates the one-row openEMS simulation recorded as R-BETA-06 below and in docs/AERIS10_BETA_ENGINEERING_REPORT.md §8; the 16-row coupled simulation and the coupon measurement remain open.

5.4 BETA completions R-BETA (source: docs/04_RECOVERY_TASKS.md , "BETA completions", copied verbatim)

Task	Status	Evidence
R-BETA-01 FPGA RTL builds + missing modules + testbenches	DONE (BETA)	beta/fpga/build.sh 0 failures
R-BETA-02 STM32 firmware builds + defect fixes + host tests	DONE (BETA)	beta/stm32/build.sh exit 0, tests 4/4
R-BETA-03 GUI package + tests + packaging	DONE (BETA)	beta/gui pytest 55 passed
R-BETA-04 PCB routing completion + BOM MPN + fab notes	DONE (BETA; Power 89 open)	beta/pcb/*/README.md
R-BETA-05 FPGA→host path (DSN-LINK-01)	DONE (option B code), OPEN (bench, rev. B)	engineering/DESIGN/HOST_LINK/
R-BETA-06 Antenna simulation	DONE (one row)	engineering/DESIGN/ANTENNA/simulation/
R-BETA-07 Enclosure detail (flat patterns, fasteners, parts list)	DONE (PROPOSED)	engineering/DESIGN/MECHANICAL/
R-BETA-09 ISERDES ADC capture + polyphase DDC at 100 MHz	DONE (BETA, bit-exact vs legacy in simulation)	beta/fpga/rtl/ad9484_iserdes_capture.v , adc_capture_calib.v , ddc_4x_100m.v ; tb_adc_iserdes_capture, tb_ddc_4x
R-BETA-08 Vivado synthesis, flashing, fabrication, bench tests	BLOCKED (tools/hardware not on this machine)	—

The test counts in this table are those recorded at the tracker's date; the later BETA report states host tests 6/6 and pytest 72 passed (source: docs/AERIS10_BETA_ENGINEERING_REPORT.md §4–5).

5.5 Items impossible on the authoring machine (source: docs/AERIS10_BETA_ENGINEERING_REPORT.md §2, copied)

Vivado synthesis/timing/bitstream (open-source flow attempted — report §3.4), flashing and bench tests, fabrication, measurement of component heights, antenna coupon on a VNA, PA thermal test, 89 Power Board pour connections (manual CAD), EAGLE schematic updates for rev. B, owner decisions K1–K8 and D-01...D-19.

6. Observations raised by this manual (MAN-01...MAN-04)

These are inconsistencies found while compiling the chapters; they are recorded for the owners, not resolved here.

ID	Observation	Evidence	Owner / decision needed
MAN-01	The electronic scan range is stated four different ways: README "±45°"; 00_notation/parameter_table.md "approximately ±33° at $\Delta\varphi = \pm 160^\circ$ " from $\theta = \arcsin(\Delta\varphi/\pi)$; 02_hardware/04_antenna_beamforming.md §3.2 evaluates the same relation (HW-ANT-3, $\theta_0 = \arcsin(\Delta\varphi_n/180^\circ)$) to $\approx \pm 62.7^\circ$ and then calls ±33° a grating-lobe limit; 01_physics/03_beamforming_theory.md §6 shows $d = \lambda/2$ is grating-lobe-free at all scan angles. $\arcsin(160/180)$ is not $\approx 33^\circ$.	README.md "Technical Specifications"; 00_notation/parameter_table.md "Inconsistency Resolutions" §4; 02_hardware/04_antenna_beamforming.md §3.2; 01_physics/03_beamforming_theory.md §6	antenna / systems designer: state the intended maximum elevation scan angle and correct the documents; affects the antenna row pattern requirement (D-02) and the range estimate at scan
MAN-02	The matched-filter reference memories encode a 10 → 30 MHz baseband up-chirp (20 MHz span, 3000 samples at 100 MSPS), while the chirp bandwidth B is TBD in the parameter table and 50 MHz is ASSUMED in the range calculation. No file states whether the 20 MHz span is the RF chirp bandwidth.	beta/fpga/README.md "What was found and decided" item 1; 00_notation/parameter_table.md "TBD Tracking"; engineering/DESIGN/CALCS/DESIGN_CALCULATIONS.md §3	RF / FPGA designer: define B (ADF4382 sweep or DAC LUT), then redo ΔR , G_p , $R_{\max}$ and the antenna bandwidth requirement (D-01 rev. B)
MAN-03	Two cable-ID schemes coexist: interconnection_table.md uses CBL-00...CBL-106 per signal group; HARNESS_SCHEDULE.md uses CBL-001...CBL-144 per physical cable with different assignments (e.g. CBL-02 = +1V8_FPGA vs CBL-002 = +5V5_PA; CBL-21 = SV1 ribbon vs CBL-035).	engineering/SYSTEM/interfaces/interconnection_table.md §2-3; engineering/DESIGN/HARNESS/HARNESS_SCHEDULE.md rows CBL-002, CBL-003, CBL-035	harness / documentation owner: adopt one scheme (D-15 says "cable IDs from interconnection_table.md", the generated schedule does not follow it) before cable labels are printed
MAN-04	Stale cross-tree statements about bridge command set v2: beta/stm32/README.md §6a (line 120) and beta/stm32/DECISIONS.md D-18 say "The FPGA side of 0x02..0x04 is not implemented in beta/fpga yet", whereas beta/fpga/README.md and CHANGELOG.md record the implementation, tb_host_bridge_top PASS with a register write taking effect, and HOST_LINK_DESIGN.md §7 synced to the RTL; conversely beta/fpga/README.md remaining-work item 8 still lists the GUI register_map.py 4-bit mask as open while the file has ADDR_MASK = 0x1F and the GUI changelog records the final map.	beta/stm32/README.md:120; beta/stm32/DECISIONS.md D-18; beta/fpga/README.md "Host path", "Remaining work" item 8; beta/fpga/CHANGELOG.md "Command set v2..."; beta/gui/CHANGELOG.md "final register map"; beta/gui/aeris10_gui/protocol/register_map.py:18	firmware and FPGA documentation owners: refresh the two firmware statements and FPGA item 8; then run the end-to-end REG test on hardware (open in all three trees)

<!-- chapter A: Drawing register (copy) -->

Appendix A — Drawing register (copy of engineering/DRAWING_REGISTER.md)

Author of this appendix: Antidrone Ukraine · antidrone.cc (compilation); the register itself is generated by tools/gen_drawing_register.py .

Status summary: the register lists 89 drawings — 39 SOURCE-DERIVED, 28 PARTIAL, 16 PROPOSED DESIGN, 1 CONCEPTUAL, 5 BLOCKED — MISSING DATA, 0 VERIFIED (source: engineering/DRAWING_REGISTER.md , "Summary"). It was regenerated for this edition with python3 tools/gen_drawing_register.py --check on 2026-10-09; the tool reported "register: 89 drawings; statuses {'SOURCE-DERIVED': 39, 'PARTIAL': 28, 'BLOCKED': 5, 'CONCEPTUAL': 1, 'PROPOSED': 16}; file check: 0 problems" (333 registered files checked for existence, non-emptiness and SVG/PDF/DOT well-formedness).

Sources: engineering/DRAWING_REGISTER.md (ENG-REG-01 Rev A), tools/gen_drawing_register.py .

This appendix reproduces the register verbatim so that every figure of this manual can be traced to its drawing ID, native file, exports and missing-data note. Paths in the "Native format" and "exports" columns are relative to the repository root or to the drawing's own directory as the register prints them; the figures used in chapters 1–16 reference the same files. The text below this paragraph is the generated register, unchanged.

AERIS-10 — Drawing register (generated)

Document ENG-REG-01 · Rev A · 2026-10-09 · generated by tools/gen_drawing_register.py (run with --check to validate files). Status vocabulary: VERIFIED (checked against hardware/EAGLE output), SOURCE-DERIVED (generated from the native design files, not reviewed), PARTIAL, CONCEPTUAL, BLOCKED — MISSING DATA. No drawing is VERIFIED: nothing was compared with an EAGLE/vendor output or with hardware.

ID	Drawing	Source	
ELEC-SCH-MAIN_BOARD	Schematic set, MAIN_BOARD (4 sheets)	4_Schematics and Boards Layout/4_6_Schematics/MainBoard/RADAR_Main_Board.sch (native EAGLE, editable)	4_Schematics and Boards Layout/4_6_Schematics/Mai
ELEC-NET-MAIN_BOARD	Netlist + component-to-net report, MAIN_BOARD	4_Schematics and Boards Layout/4_6_Schematics/MainBoard/RADAR_Main_Board.sch	engineering/ELECTRICAL/ne
ELEC-SCH-POWER_SUPPLY	Schematic set, POWER_SUPPLY (1 sheet)	4_Schematics and Boards Layout/4_6_Schematics/PowerBoard/PowerBoard.sch (native EAGLE, editable)	4_Schematics and Boards l
ELEC-NET-POWER_SUPPLY	Netlist + component-to-net report, POWER_SUPPLY	4_Schematics and Boards Layout/4_6_Schematics/PowerBoard/PowerBoard.sch	engineering/ELECTRICAL/ne
ELEC-SCH-RF_PA	Schematic set, RF_PA (1 sheet)	4_Schematics and Boards Layout/4_6_Schematics/PowerAmplifierBoard/RF_PA.sch (native EAGLE, editable)	4_Schematics and Boards l
ELEC-NET-RF_PA	Netlist + component-to-net report, RF_PA	4_Schematics and Boards Layout/4_6_Schematics/PowerAmplifierBoard/RF_PA.sch	engineering/ELECTRICAL/ne
ELEC-SCH-FREQUENCY_SYNTHESIZER	Schematic set, FREQUENCY_SYNTHESIZER (1 sheet)	4_Schematics and Boards Layout/4_6_Schematics/FrequencySynthesizerBoard/Clocks_Freq_Synth_board.sch (native EAGLE, editable)	4_Schematics and Boards Layout/4_6_Schematics/Fre
ELEC-NET-FREQUENCY_SYNTHESIZER	Netlist + component-to-net report, FREQUENCY_SYNTHESIZER	4_Schematics and Boards Layout/4_6_Schematics/FrequencySynthesizerBoard/Clocks_Freq_Synth_board.sch	engineering/ELECTRICAL/ne
ELEC-PWR-01	Power distribution diagram + rail register	PowerBoard.sch, Main/Synth/PA schematics, main.h/main.cpp, Power Management V6.xlsx	engineering/ELECTRICAL/pc

	Drawing	Source	
SYS-01	System block diagram	4 schematics, firmware, RADAR_V6.drawio, docs/SYSTEM/BLOCK_DIAGRAM.md	engineering/SYSTEM/block_
SYS-02	Hardware interconnection diagram + table	connector nets of 4 schematics, .brd silkscreen, main.h	engineering/SYSTEM/interf
SYS-03	Signal and data flow diagram	schematics traced through passives; main.cpp:933-1072; adf4382a_manager.h; radar_system_top.v	engineering/SYSTEM/data_f
SD-01	FPGA module hierarchy (auto)	9_Firmware/9_2_FPGA/*.v via tools/gen_verilog_hierarchy.py	engineering/SOFTWARE_DIA
SD-02	FPGA signal-processing pipeline	radar_system_top.v and submodules	engineering/SOFTWARE_DIA
SD-03	STM32 firmware architecture	main.cpp, main.h, LIB/*	engineering/SOFTWARE_DIA
SD-04	STM32 USB CDC flow	main.cpp, USBHandler.cpp, GUI_V5.py	engineering/SOFTWARE_DIA
SD-05	Python GUI modules (auto)	9_Firmware/9_3_GUI/*.py via tools/gen_python_module_graph.py	engineering/SOFTWARE_DIA
SD-06	Python GUI runtime architecture	GUI_V5.py, GUI_V6_Demo.py	engineering/SOFTWARE_DIA
SD-07	End-to-end data flow	docs/SYSTEM, docs/FPGA, docs/STM32, PIN_MAP_FROM_SCHEMATIC.md	engineering/SOFTWARE_DIA
PCB-MAIN_BOARD-01	Top-layer drawing	4_Schematics and Boards Layout/4_6_Schematics/MainBoard/RADAR_Main_Board.brd (native EAGLE, editable)	engineering/PCB/MAIN_BOA
PCB-MAIN_BOARD-02	Bottom-layer drawing (mirrored)	4_Schematics and Boards Layout/4_6_Schematics/MainBoard/RADAR_Main_Board.brd (native EAGLE, editable)	engineering/PCB/MAIN_BOA
PCB-MAIN_BOARD-03	Copper-layer views (one page per layer)	4_Schematics and Boards Layout/4_6_Schematics/MainBoard/RADAR_Main_Board.brd (native EAGLE, editable)	engineering/PCB/MAIN_BOA
PCB-MAIN_BOARD-04	Board-outline drawing	4_Schematics and Boards Layout/4_6_Schematics/MainBoard/RADAR_Main_Board.brd (native EAGLE, editable)	engineering/PCB/MAIN_BOA
PCB-MAIN_BOARD-05	Mechanical dimensions (DXF/STEP, hole table)	4_Schematics and Boards Layout/4_6_Schematics/MainBoard/RADAR_Main_Board.brd (native EAGLE, editable)	engineering/MECHANICAL/D
PCB-MAIN_BOARD-06	Drill map + hole table + Excellon	4_Schematics and Boards Layout/4_6_Schematics/MainBoard/RADAR_Main_Board.brd (native EAGLE, editable)	engineering/PCB/MAIN_BOA
PCB-MAIN_BOARD-07	Component placement / top assembly drawing	4_Schematics and Boards Layout/4_6_Schematics/MainBoard/RADAR_Main_Board.brd (native EAGLE, editable)	engineering/PCB/MAIN_BOA
PCB-MAIN_BOARD-08	Bottom assembly drawing (mirrored)	4_Schematics and Boards Layout/4_6_Schematics/MainBoard/RADAR_Main_Board.brd (native EAGLE, editable)	engineering/PCB/MAIN_BOA
PCB-MAIN_BOARD-09	Fabrication package: Gerber set + stack-up document	4_Schematics and Boards Layout/4_6_Schematics/MainBoard/RADAR_Main_Board.brd (native EAGLE, editable)	engineering/PCB/MAIN_BOA
PCB-MAIN_BOARD-10	BOM with references + pick-and-place	4_Schematics and Boards Layout/4_6_Schematics/MainBoard/RADAR_Main_Board.sch / .brd	docs/BOM/BOM_MAIN_BOARD.0
PCB-MAIN_BOARD-11	DRC / IPC-2581 / 3-D renders / package README	4_Schematics and Boards Layout/4_6_Schematics/MainBoard/RADAR_Main_Board.brd (native EAGLE, editable)	engineering/PCB/MAIN_BOA
PCB-POWER_SUPPLY-01	Top-layer drawing	4_Schematics and Boards Layout/4_6_Schematics/PowerBoard/PowerBoard.brd (native EAGLE, editable)	engineering/PCB/POWER_SUF
PCB-POWER_SUPPLY-02	Bottom-layer drawing (mirrored)	4_Schematics and Boards Layout/4_6_Schematics/PowerBoard/PowerBoard.brd (native EAGLE, editable)	engineering/PCB/POWER_SUF
PCB-POWER_SUPPLY-03	Copper-layer views (one page per layer)	4_Schematics and Boards Layout/4_6_Schematics/PowerBoard/PowerBoard.brd (native EAGLE, editable)	engineering/PCB/POWER_SUF

	Drawing	Source	
PCB-POWER_SUPPLY-04	Board-outline drawing	4_Schematics and Boards Layout/4_6_Schematics/PowerBoard/PowerBoard.brd (native EAGLE, editable)	engineering/PCB/POWER_SUPPLY
PCB-POWER_SUPPLY-05	Mechanical dimensions (DXF/STEP, hole table)	4_Schematics and Boards Layout/4_6_Schematics/PowerBoard/PowerBoard.brd (native EAGLE, editable)	engineering/MECHANICAL/DR
PCB-POWER_SUPPLY-06	Drill map + hole table + Excellon	4_Schematics and Boards Layout/4_6_Schematics/PowerBoard/PowerBoard.brd (native EAGLE, editable)	engineering/PCB/POWER_SUPPLY
PCB-POWER_SUPPLY-07	Component placement / top assembly drawing	4_Schematics and Boards Layout/4_6_Schematics/PowerBoard/PowerBoard.brd (native EAGLE, editable)	engineering/PCB/POWER_SUPPLY
PCB-POWER_SUPPLY-08	Bottom assembly drawing (mirrored)	4_Schematics and Boards Layout/4_6_Schematics/PowerBoard/PowerBoard.brd (native EAGLE, editable)	engineering/PCB/POWER_SUPPLY
PCB-POWER_SUPPLY-09	Fabrication package: Gerber set + stack-up document	4_Schematics and Boards Layout/4_6_Schematics/PowerBoard/PowerBoard.brd (native EAGLE, editable)	engineering/PCB/POWER_SUPPLY
PCB-POWER_SUPPLY-10	BOM with references + pick-and-place	4_Schematics and Boards Layout/4_6_Schematics/PowerBoard/PowerBoard.sch / .brd	docs/BOM/BOM_POWER_SUPPLY
PCB-POWER_SUPPLY-11	DRC / IPC-2581 / 3-D renders / package README	4_Schematics and Boards Layout/4_6_Schematics/PowerBoard/PowerBoard.brd (native EAGLE, editable)	engineering/PCB/POWER_SUPPLY
PCB-RF_PA-01	Top-layer drawing	4_Schematics and Boards Layout/4_6_Schematics/PowerAmplifierBoard/RF_PA.brd (native EAGLE, editable)	engineering/PCB/RF_PA/kit
PCB-RF_PA-02	Bottom-layer drawing (mirrored)	4_Schematics and Boards Layout/4_6_Schematics/PowerAmplifierBoard/RF_PA.brd (native EAGLE, editable)	engineering/PCB/RF_PA/kit
PCB-RF_PA-03	Copper-layer views (one page per layer)	4_Schematics and Boards Layout/4_6_Schematics/PowerAmplifierBoard/RF_PA.brd (native EAGLE, editable)	engineering/PCB/RF_PA/kit
PCB-RF_PA-04	Board-outline drawing	4_Schematics and Boards Layout/4_6_Schematics/PowerAmplifierBoard/RF_PA.brd (native EAGLE, editable)	engineering/PCB/RF_PA/kit
PCB-RF_PA-05	Mechanical dimensions (DXF/STEP, hole table)	4_Schematics and Boards Layout/4_6_Schematics/PowerAmplifierBoard/RF_PA.brd (native EAGLE, editable)	engineering/MECHANICAL/DR
PCB-RF_PA-06	Drill map + hole table + Excellon	4_Schematics and Boards Layout/4_6_Schematics/PowerAmplifierBoard/RF_PA.brd (native EAGLE, editable)	engineering/PCB/RF_PA/kit
PCB-RF_PA-07	Component placement / top assembly drawing	4_Schematics and Boards Layout/4_6_Schematics/PowerAmplifierBoard/RF_PA.brd (native EAGLE, editable)	engineering/PCB/RF_PA/kit
PCB-RF_PA-08	Bottom assembly drawing (mirrored)	4_Schematics and Boards Layout/4_6_Schematics/PowerAmplifierBoard/RF_PA.brd (native EAGLE, editable)	engineering/PCB/RF_PA/kit
PCB-RF_PA-09	Fabrication package: Gerber set + stack-up document	4_Schematics and Boards Layout/4_6_Schematics/PowerAmplifierBoard/RF_PA.brd (native EAGLE, editable)	engineering/PCB/RF_PA/kit
PCB-RF_PA-10	BOM with references + pick-and-place	4_Schematics and Boards Layout/4_6_Schematics/PowerAmplifierBoard/RF_PA.sch / .brd	docs/BOM/BOM_RF_PA.csv
PCB-RF_PA-11	DRC / IPC-2581 / 3-D renders / package README	4_Schematics and Boards Layout/4_6_Schematics/PowerAmplifierBoard/RF_PA.brd (native EAGLE, editable)	engineering/PCB/RF_PA/kit
PCB-FREQUENCY_SYNTHESIZER-01	Top-layer drawing	4_Schematics and Boards Layout/4_6_Schematics/FrequencySynthesizerBoard/Clocks_Freq_Synth_board.brd (native EAGLE, editable)	engineering/PCB/FREQUENCY
PCB-FREQUENCY_SYNTHESIZER-02	Bottom-layer drawing (mirrored)	4_Schematics and Boards Layout/4_6_Schematics/FrequencySynthesizerBoard/Clocks_Freq_Synth_board.brd (native EAGLE, editable)	engineering/PCB/FREQUENCY
PCB-FREQUENCY_SYNTHESIZER-03	Copper-layer views (one page per layer)	4_Schematics and Boards Layout/4_6_Schematics/FrequencySynthesizerBoard/Clocks_Freq_Synth_board.brd (native EAGLE, editable)	engineering/PCB/FREQUENCY
PCB-FREQUENCY_SYNTHESIZER-04	Board-outline drawing	4_Schematics and Boards Layout/4_6_Schematics/FrequencySynthesizerBoard/Clocks_Freq_Synth_board.brd (native EAGLE, editable)	engineering/PCB/FREQUENCY
PCB-FREQUENCY_SYNTHESIZER-05	Mechanical dimensions (DXF/STEP, hole table)	4_Schematics and Boards Layout/4_6_Schematics/FrequencySynthesizerBoard/Clocks_Freq_Synth_board.brd (native EAGLE, editable)	engineering/MECHANICAL/DR
PCB-FREQUENCY_SYNTHESIZER-06	Drill map + hole table + Excellon	4_Schematics and Boards Layout/4_6_Schematics/FrequencySynthesizerBoard/Clocks_Freq_Synth_board.brd (native EAGLE, editable)	engineering/PCB/FREQUENCY

	Drawing	Source	
PCB-FREQUENCY_SYNTHESIZER-07	Component placement / top assembly drawing	4_Schematics and Boards Layout/4_6_Schematics/FrequencySynthesizerBoard/Clocks_Freq_Synth_board.brd (native EAGLE, editable)	engineering/PCB/FREQUENCY_SYNTHESIZER/07
PCB-FREQUENCY_SYNTHESIZER-08	Bottom assembly drawing (mirrored)	4_Schematics and Boards Layout/4_6_Schematics/FrequencySynthesizerBoard/Clocks_Freq_Synth_board.brd (native EAGLE, editable)	engineering/PCB/FREQUENCY_SYNTHESIZER/08
PCB-FREQUENCY_SYNTHESIZER-09	Fabrication package: Gerber set + stack-up document	4_Schematics and Boards Layout/4_6_Schematics/FrequencySynthesizerBoard/Clocks_Freq_Synth_board.brd (native EAGLE, editable)	engineering/PCB/FREQUENCY_SYNTHESIZER/09
PCB-FREQUENCY_SYNTHESIZER-10	BOM with references + pick-and-place	4_Schematics and Boards Layout/4_6_Schematics/FrequencySynthesizerBoard/Clocks_Freq_Synth_board.sch / .brd	docs/BOM/BOM_FREQUENCY_SYNTHESIZER
PCB-FREQUENCY_SYNTHESIZER-11	DRC / IPC-2581 / 3-D renders / package README	4_Schematics and Boards Layout/4_6_Schematics/FrequencySynthesizerBoard/Clocks_Freq_Synth_board.brd (native EAGLE, editable)	engineering/PCB/FREQUENCY_SYNTHESIZER/11
MECH-PLAN-01	PCB set plan view 1:1	4 × .brd layer 20 + holes	engineering/MECHANICAL/PCB/01
MECH-ENC-01	Main enclosure drawings	none (README references a non-existent 10_docs/Hardware/Enclosure)	— (none)
MECH-INT-01	Internal component layout / cable routing	none	— (none)
MECH-ANT-01	Antenna assembly drawing	02_hardware/04_antenna_beamforming.md (spacing only), photo	— (none)
MECH-COOL-01	Cooling and power assembly drawing	none (PA dissipation implied by xlsx)	— (none)
MECH-PED-01	Pedestal / azimuth drive drawing	firmware constants only	— (none)
ASM-EXP-01	Exploded assembly view (electronics set)	4 × .brd outlines; interconnection table; antenna spacing doc	engineering/ASSEMBLY/EXPLODED_VIEW/01
ASM-PL-01	Parts list	BOMs + connector matrix + firmware	engineering/ASSEMBLY/PARTS_LIST/01
ASM-SEQ-01	Assembly / integration sequence	connector matrix, power sequence (power_rails.md)	engineering/ASSEMBLY/ASSEMBLY_SEQUENCE/01
DSN-00	Design basis, decisions D-01...D-15, parameter file	verified repo constraints + decisions	engineering/DESIGN/00_DESIGN_BASIS
DSN-ANT-01	16×8 microstrip patch array panel (native KiCad PCB, calc sheet, openEMS model)	f0, pitch, 8×16 from docs; RO4350B datasheet; decisions D-01...D-06	engineering/DESIGN/ANTENNA_ARRAY/01
DSN-THM-01	Thermal budget + 22 V supply sizing	firmware timing, QPA2962 datasheet, D-10/D-11/D-14	engineering/DESIGN/THERMAL_BUDGET/01
DSN-PSU-01	22 V PA drain supply, enable switch, per-PA pulse gating (block schematic, BOM, nets)	D-14; xlsx VIN; main.h EN pin	engineering/DESIGN/ELECTRIC_POWER_SUPPLY/01
DSN-MECH-3D	Radar head + pedestal parametric 3-D model (FreeCAD native, STEP, STL, DXF page, mass table)	board outlines/holes (verified), D-07...D-13	engineering/DESIGN/MECHANICAL/3D_MODEL/01
DSN-MECH-01	Head plan section	tools/design_layout.py (D-07...D-13) + KiCad P&P	engineering/DESIGN/MECHANICAL/01
DSN-MECH-02	Head front elevation	tools/design_layout.py (D-07...D-13) + KiCad P&P	engineering/DESIGN/MECHANICAL/02
DSN-MECH-03	Head side section incl. pedestal	tools/design_layout.py (D-07...D-13) + KiCad P&P	engineering/DESIGN/MECHANICAL/03
DSN-MECH-04	Internal layout per tier with connector positions	tools/design_layout.py (D-07...D-13) + KiCad P&P	engineering/DESIGN/MECHANICAL/04
DSN-MECH-05	Pedestal plan	tools/design_layout.py (D-07...D-13) + KiCad P&P	engineering/DESIGN/MECHANICAL/05
DSN-LINK-01	FPGA → host data path: option A FT601 pin plan (bank 35) + option B SPI bridge	Main Board netlist (free bank-35 pins, DIG_5..7, SPI1), RTL geometry, firmware timing	engineering/DESIGN/HOST_INTERFACE/01

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	Drawing	Source	
	(RTL, STM32 driver, GUI parser)		
DSN-ANT-01-SIM	openEMS simulation of one antenna row (S11, directivity, tuning log)	openems_patch_row.py run with openEMS built from source	engineering/DESIGN/ANTENNA
DSN-MECH-3D-DETAIL	Detailed enclosure + pedestal model: 51 parts (tray, front plate, lid, window + gasket + frame, PA plate with brackets, carrier rails, standoffs, gland plate, bearing, pulleys, slip ring, motor bracket, mast flange)	tools/design_layout.py + PCB hole tables + D-07...D-13	engineering/DESIGN/MECHANICAL
DSN-MECH-06	Sheet-metal flat patterns with bend lines (tray, front plate, lid)	detail model, $BA = \pi/2 \cdot (R + K \cdot t)$	engineering/DESIGN/MECHANICAL/06_sheet_metal_flat_patterns
DSN-MECH-07	Assembly section with fastener balloons	detail model	engineering/DESIGN/MECHANICAL/07_assembly_section_fasteners
DSN-HAR-01	Harness schedule with computed lengths (144 cables)	interconnection_table.md + P&P + proposed layout	engineering/DESIGN/HARNESS
ASM-DRW-01	Assembly drawings (per board)	KiCad fab/silk plots	engineering/ASSEMBLY/ASSEMBLY_DRAWINGS

Summary

Status	Drawings
BLOCKED	5
CONCEPTUAL	1
PARTIAL	28
PROPOSED	16
SOURCE-DERIVED	39
Total registered	89

Existing drawings found in the repository before this work: 2_Functional Diagrams/RADAR_V6.drawio (+ .jpg, .dwg) system block diagram; 4_Schematics and Boards Layout/4_4_Board Stack-up/Stack_Hybrid.png (unlabelled stack-up); 4_Schematics and Boards Layout/4_7_Production Files/Frequency_Synthesizer/ (P&P + BOM xlsx); 8_Utills/*.jpg photographs; docs/MECHANICAL/drawings/*_outline.svg (first reconstruction). None of them is a dimensioned engineering drawing.

File check (2026-10-09)

Checked 333 registered files (existence, non-empty, SVG/PDF/DOT well-formed): **333 OK, 0 problems**.

<!-- chapter B: Parts lists (PCB BOM summaries, mechanical parts, fasteners), cable schedule -->

B. Parts lists and cable schedule

Appendix status summary: PCB BOM summaries — PARTIAL (generated from the schematics, 0 MPN in the source; BETA MPN proposals with confidence classes); mechanical parts list — PROPOSED DESIGN (DSN-MECH-PL Rev A, masses are volume × density estimates); cable schedule — PROPOSED DESIGN (DSN-HAR-01 Rev A, lengths are Manhattan estimates to be cut after a first fit). No part in this appendix has been purchased, fitted or measured. Compiled by Antidrone Ukraine · antidrone.cc.

Sources: docs/BOM/README.md, docs/BOM/BOM_<BOARD>.csv; beta/pcb/README.md, beta/pcb/<BOARD>/BOM_<BOARD>_beta.csv; engineering/DESIGN/MECHANICAL/MECHANICAL_PARTS_LIST.md; engineering/DESIGN/HARNESS/HARNESS_SCHEDULE.md, harness_schedule.csv.

B.1 PCB bills of materials

Generated 2026-10-08 by `python3 tools/gen_eagle_bom.py` directly from the EAGLE schematics (no EAGLE run); regenerate after any schematic change with the commands in docs/BOM/README.md. Columns of BOM_<BOARD>.csv: item, qty, value, deviceset, device, package, library, mpn_attribute, manufacturer_attribute, mpn_candidate, mpn_status, references; REF5_<BOARD>.csv has one row per reference designator (source: docs/BOM/README.md §What is in each file).

Summary copied from docs/BOM/README.md §Summary:

Board	Physical references	Line items	Lines with MPN attribute	References without value	Status
Main Board	776	98	0	244	INCOMPLETE — no MPNs, FT601 (U6) unconnected but listed, 11 parts parked off-board
Power Supply	312	28	0	80	INCOMPLETE — no MPNs, 132 parts not placed on the board
RF PA	25	11	0	6	INCOMPLETE — no MPNs; QPA2962_B is a deviceset name
Frequency Synthesizer	184	40	0	47	INCOMPLETE — matches the existing Clocks_Freq_Synth_board_BOM.xlsx (40 lines) which also has empty MPN columns

mpn_status meaning (same source): VERIFIED (attribute) — an explicit MPN attribute exists in the schematic (none today); UNVERIFIED (deviceset name) — the EAGLE deviceset name looks like a part number but has not been checked against a distributor; GENERIC — value/package only — passive to be sourced by value/package/tolerance (tolerance and voltage rating are not in the schematics).

BETA MPN proposals (beta/pcb/<BOARD>/BOM_<BOARD>_beta.csv, columns manufacturer, mpn, mpn_confidence, dnp, note; lines / quantity; source: beta/pcb/README.md §BOM confidence, line counts re-counted from the CSV files on 2026-10-09):

Board	HIGH (deviceset = MPN)	MEDIUM (standard passive from value+package)	LOW (guess / non-standard value / conflict)	EMPTY (no value in the source)
Main Board	23 / 204	41 / 461	30 / 93	4 / 18
Power Supply	7 / 79	18 / 225	3 / 8	0 / 0
RF PA	5 / 6	6 / 19	0 / 0	0 / 0
Frequency Synthesizer	10 / 37	22 / 110	6 / 29	2 / 8

The RF PA quantities are per board; the AERIS-10X variant uses 16 boards (source: docs/PCB/RF_PA.md §2). Chapters 4–7 §BOM summary list the conflicts flagged per board.

System-level items not on any PCB (source: docs/BOM/README.md §System-level items; quantities and part numbers UNKNOWN unless stated):

Item	Evidence	Part number
GPS module (NMEA 9600 on UART5)	main.cpp:2143–2170; xlsx "NEO-6M"	NEO-6M (xlsx only)
IMU GY-85 (ADXL345/ITG3205/HMC5883L, I2C3)	GY_85_HAL.c; xlsx	GY-85 module
Barometer BMP180 (I2C3)	BMP180.cpp; xlsx	BMP180
Temperature sensors TMP37 ×8 via ADS7830	main.cpp:1752–1775; xlsx	TMP37
Stepper motor 200 steps/rev + driver	main.cpp:195; xlsx "TBS6600 [9-42 V]" (TB6600-class)	motor UNKNOWN
Slip ring	README.md:86; Project_Description.docx	UNKNOWN
Cooling fans	main.h:142 EN_DIS_COOLING; xlsx "COOLING SYSTEM"	UNKNOWN
Antenna array (8×16 patch or 32×16 slotted waveguide)	README.md:82–83	no CAD, no part
Enclosure	README.md:143 (missing path)	none
Inter-board cables: SMA (37 + 11 + 2 ports), Molex 22-23-20xx (56 + 34 + 6 + 2 headers)	schematics	cable assemblies not defined

The proposed designs of chapters 8–10 and the harness schedule below replace the last five rows with PROPOSED parts; the 22 V supply module BOM is engineering/DESIGN/ELECTRICAL/PA_SUPPLY_22V/ (chapter 9).

B.2 Mechanical parts list — DSN-MECH-PL Rev A (PROPOSED DESIGN)

Copied verbatim from `engineering/DESIGN/MECHANICAL/MECHANICAL_PARTS_LIST.md` (date 2026-10-09, from `CAD/detail/parts_list.json`, FreeCAD model `tools/design_enclosure_detail_freecad.py`; masses are volume × density estimates). Head ≈ 6.4 kg (structure only, PCBs/antenna not included except as noted), pedestal ≈ 12.4 kg (incl. stepper 1.1 kg, bearing/slip ring as modelled) (same source).

#	Group	Part	Material	Mass (g)	Fasteners	Note
1	HEAD	Tray — base + sides + rear, 2.5 mm Al 5754, 3 bends R2.5, flanges 15 mm with M4 PEM nuts	Al	1559	PEM S-M4-1 ×18	8 lid holes M4, 10 front-plate holes M4, Ø70 cable entry, intake slots
2	HEAD	Front plate 2.5 mm Al with radome window opening and intake louvres	Al	346	M4×8 ×10 to tray front flanges	window 175×258, 16 M3 clamp holes
3	HEAD	Radome window PTFE 2 mm (outside the front plate)	PTFE	246		RF loss ≈ 0.1 dB at 10.5 GHz (PTFE ε _r 2.1, tanδ 0.0002 — to be confirmed)
4	HEAD	Window gasket EPDM 1.5 mm (ring 10 mm)	EPDM	17		
5	HEAD	Window clamp frame 2 mm Al, 14 mm wide	Al	69	M3×10 ×16 + nyloc	
6	HEAD	Lid 2.5 mm Al with exhaust slots	Al	272	M4×8 ×8	
7	HEAD	Lid gasket EPDM 3 mm self-adhesive, 15 mm wide on the flanges	EPDM	44		compressed to 2 mm → IP54 target
8	HEAD	PA heat spreader 300×300×10 Al 6061, machined fin fields, M3 tapped (16×7 PA + 6 antenna)	Al	3400	M3×6 ×118 (PA 112 + antenna 6)	thermal pads 5×5 under each QPA2962; antenna on 2.4 mm nylon spacers
9	HEAD	Plate bracket L 25×25×3 Al at x=5 z=20	Al	26	M5×12 ×2 + M5×16 ×1	bolts the plate to the side wall and to the front flange
10	HEAD	Plate bracket L 25×25×3 Al at x=5 z=265.0	Al	26	M5×12 ×2 + M5×16 ×1	bolts the plate to the side wall and to the front flange
11	HEAD	Plate bracket L 25×25×3 Al at x=280 z=20	Al	26	M5×12 ×2 + M5×16 ×1	bolts the plate to the side wall and to the front flange
12	HEAD	Plate bracket L 25×25×3 Al at x=280 z=265.0	Al	26	M5×12 ×2 + M5×16 ×1	bolts the plate to the side wall and to the front flange
13	HEAD	Main Board carrier rail bottom U15×12×1.5 Al, 280 mm	Al	42	M4×8 ×2	bolted to the side walls (M4) — spans the full width when the board is narrower
14	HEAD	Main Board carrier rail top U15×12×1.5 Al, 280 mm	Al	42	M4×8 ×2	bolted to the side walls (M4) — spans the full width when the board is narrower
15	HEAD	Main Board standoff M3×10 hex at (4,4)	steel	2	M3×6 ×1	
16	HEAD	Main Board standoff M3×10 hex at (256,4)	steel	2	M3×6 ×1	
17	HEAD	Main Board standoff M3×10 hex at (116,114)	steel	2	M3×6 ×1	
18	HEAD	Main Board standoff M3×10 hex at (256,114)	steel	2	M3×6 ×1	
19	HEAD	Main Board standoff M3×10 hex at (116,250)	steel	2	M3×6 ×1	
20	HEAD	Main Board standoff M3×10 hex at (256,250)	steel	2	M3×6 ×1	
21	HEAD	Main Board standoff M3×10 hex at (4,296)	steel	2	M3×6 ×1	
22	HEAD	Main Board standoff M3×10 hex at (256,296)	steel	2	M3×6 ×1	
23	HEAD	Synth carrier rail bottom U15×12×1.5 Al, 120 mm	Al	18	M4×8 ×2	bolted to the side walls (M4) — spans the full width when the board is narrower
24	HEAD	Synth carrier rail top U15×12×1.5 Al, 120 mm	Al	18	M4×8 ×2	bolted to the side walls (M4) — spans the full width when the board is narrower
25	HEAD	Synth standoff M3×10 hex at (5,5)	steel	2	M3×6 ×1	
26	HEAD	Synth standoff M3×10 hex at (95,5)	steel	2	M3×6 ×1	
27	HEAD	Synth standoff M3×10 hex at (5,95)	steel	2	M3×6 ×1	
28	HEAD	Synth standoff M3×10 hex at (95,95)	steel	2	M3×6 ×1	
29	HEAD	Power Board carrier rail bottom U15×12×1.5 Al, 300 mm	Al	44	M4×8 ×2	bolted to the side walls (M4) — spans the full width when the board is narrower
30	HEAD	Power Board carrier rail top U15×12×1.5 Al, 300 mm	Al	44	M4×8 ×2	bolted to the side walls (M4) — spans the full width when the board is narrower
31	HEAD	Power Board standoff M3×10 hex at (10,10)	steel	2	M3×6 ×1	
32	HEAD	Power Board standoff M3×10 hex at (140,10)	steel	2	M3×6 ×1	
33	HEAD	Power Board standoff M3×10 hex at (270,10)	steel	2	M3×6 ×1	
34	HEAD	Power Board standoff M3×10 hex at (10,120)	steel	2	M3×6 ×1	
35	HEAD	Power Board standoff M3×10 hex at (270,120)	steel	2	M3×6 ×1	
36	HEAD	Power Board standoff M3×10 hex at (10,230)	steel	2	M3×6 ×1	
37	HEAD	Power Board standoff M3×10 hex at (270,230)	steel	2	M3×6 ×1	
38	HEAD	Power Board standoff M3×10 hex at (138,268)	steel	2	M3×6 ×1	
39	HEAD	Cable-entry gland plate 100×100×2 Al under the base (M32 + M20 glands)	Al	48	M4×8 ×4; glands M32 + M20 IP68	harness from the slip ring: VIN, 22 V, USB
40	PEDESTAL	Turntable plate Ø340×8 Al, 8×M6 to the head base, 12×M6 to the bearing outer ring	Al	1864	M6×16 ×8, M6×25 ×12	

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#	Group	Part	Material	Mass (g)	Fasteners	Note
41	PEDESTAL	Slewing bearing OD190/ID100×20 (4-point contact, e.g. igus PRT-04-100 class)	steel	3218		part number to be selected (axial ≥ 50 kg, moment ≥ 60 N·m)
42	PEDESTAL	Ring pulley GT3 180T Ø172 Al (machined/3D-printed), clamped under the turntable	Al	0	M4×10 ×6	
43	PEDESTAL	Pedestal top plate 360×360×5 Al, 12×M6 to the bearing inner ring, bore Ø90	Al	1658	M6×20 ×12	
44	PEDESTAL	Pedestal housing 360×360×125 folded 2.5 mm Al (4 bends), open top, connector panel cut-out	Al	1575	M5×10 ×12 to the top plate	DC input (XT60/M12), USB-B bulkhead, vent
45	PEDESTAL	Stepper NEMA 23 76 mm	steel	1898	M5×12 ×4	
46	PEDESTAL	Motor bracket 80×80×3 Al with slotted belt-tension holes	Al	41	M5×10 ×4 to the top plate	slots ±5 mm for GT3 tension
47	PEDESTAL	Motor pulley GT3 60T Ø57, bore 6.35	Al	84	grub M4 ×2	
48	PEDESTAL	Through-bore slip ring Ø99×60, bore 60, 12 circuits (4×10 A, 4×10 A, 4 signal)	plastic	351		e.g. Senring H3899 class — select
49	PEDESTAL	Slip-ring stator bracket 140×140×3 Al	Al	95	M4×8 ×4	stator fixed to the pedestal, rotor flange to the turntable
50	PEDESTAL	Stepper driver TB6600 on DIN rail	plastic	299		
51	PEDESTAL	Mast flange Ø150×10 steel, 4×M10 PCD 110 (ASSUMPTION — mast interface undefined)	steel	1360	M10×30 ×4	

Fastener totals from the per-part lists (same source):

Fastener	Qty
M10×30	4
M3×10	16
M3×6	138
M4	2
M4×10	6
M4×8	38
M5×10	16
M5×12	12
M5×16	4
M6×16	8
M6×20	12
M6×25	12
S-M4-1	18

Sealing and finish (same source): lid EPDM 15 × 3 mm self-adhesive gasket on the three top flanges, compressed to 2 mm by the M4 screws (pitch 60 mm); front plate 1.5 mm EPDM strip on the two front flanges; radome window PTFE 2 mm clamped by the 2 mm Al frame with EPDM 1.5 mm gasket, M3 × 18 (alternative 2 mm Rogers/ABS radome); cable entry 100 × 100 gland plate under the base with M32 (power) and M20 (USB) IP68 glands; finish chromate conversion (Alodine) + powder coat RAL 7035 outside, bare chromate inside for grounding at the flanges, PA plate bare 6061 with thermal pads. Open: bend reliefs and corner welds of the tray, stiffeners of the 315 mm lid, fan mounting brackets, antenna spacer material, earthing stud, lifting points, mast interface (row 51 is an ASSUMPTION).

B.3 Cable schedule — DSN-HAR-01 Rev A (PROPOSED DESIGN)

Generated by `tools/design_mechanical_drawings.py` from the proposed layout (`tools/design_layout.py`) and the connector positions in the KiCad P&P files. Length = Manhattan distance between connector positions in the head + service allowance (40 mm coax / 60 mm wire), rounded up to 10 mm; lengths are PROPOSED, cut after a first fit (source: `HARNESS_SCHEDULE.md` header). Totals stated there: **144 cables; coax 55; wire 89; total proposed length ≈ 41.6 m**. Re-counted from `harness_schedule.csv` on 2026-10-09: 144 rows, 55 coax / 89 wire, sum of numeric lengths 41600 mm, 14 rows with length TBD (CBL-001, CBL-006, CBL-007, CBL-008, CBL-010, CBL-012, CBL-013, CBL-014, CBL-016, CBL-024, CBL-025, CBL-027, CBL-028, CBL-144).

Cable types (count from `harness_schedule.csv`, column `cable_type`):

Count	Cable type
32	coax RG-405 SMA-SMA, EQUAL LENGTH set
21	2-wire 20 AWG, Molex 22-01-2027 both ends
16	2-wire 24 AWG shielded, Molex 22-01-2027
16	3-wire 24 AWG twisted, Molex 22-01-3037
16	coax RG-405 SMA-2.92 mm, EQUAL LENGTH set
16	2-wire 18 AWG twisted, AK300/2 screw terminal
13	2-wire 20 AWG
7	coax RG-405 (0.086") SMA-SMA, phase-stable

Count	Cable type
2	ribbon/discrete 26 AWG, 2.54 mm housings
2	2 × 2-wire 16 AWG
1	20-way 1.27 mm IDC ribbon
1	USB 2.0 shielded, mini-B
1	3-wire 24 AWG shielded

Signal groups: 16 × each of V_{G_n} gate bias, drain-current sense, RF to PA RFIN, RF from PA RFOUT, RF to antenna row n, 22 V pulsed drain (CBL-045...CBL-140, six cables per PA); 34 Power Board rail cables (CBL-001...CBL-034, 13 of them TBD because no connector on the Main/Synth board carries the same net name — resolve with `engineering/SYSTEM/interfaces/interconnection_table.md`); the enable bus (CBL-035, 20-way IDC); 7 Synth → Main coax (CBL-036...CBL-042, RG-405 phase-stable); 2 control-header cables (CBL-043/044); slip-ring VIN, 22 V boost input and USB (CBL-141...CBL-143); stepper STEP/DIR/EN (CBL-144, DECISION NEEDED: driver on the fixed base needs 3 more slip-ring circuits, or move the driver into the head).

Caveats carried in the `note` column: PA instance ↔ V_{G_n} assignment PROPOSED (n = n, not documented in CAD); Main sense connector ↔ PA n PROPOSED; which SMA of each Main Board pair is RFIN/RFOUT is UNVERIFIED (`interconnection_table.md` §7); antenna row n ↔ PA n PROPOSED, equal length mandatory for elevation phase. Note that the cable IDs of this schedule (CBL-001...CBL-144) are the DSN-HAR-01 numbering; the interconnection table of chapter 3 uses its own earlier CBL-xx numbering.

B.3.1 Full schedule (copied verbatim from `engineering/DESIGN/HARNESS/HARNESS_SCHEDULE.md`)

ID	From	To	Signal	Cable	Length (mm)	Note
CBL-001	POWER_SUPPLY X2	??	+5V0_1	2-wire 20 AWG	TBD	no connector with this net name on Main/Synth (rail renamed through a filter/inductor or spare output) — resolve with <code>interconnection_table.md</code>
CBL-002	POWER_SUPPLY X3	MAIN_BOARD X55	+5V5_PA	2-wire 20 AWG, Molex 22-01-2027 both ends	780	rail name matched in both netlists
CBL-003	POWER_SUPPLY X4	MAIN_BOARD X8	+1V0_FPGA	2-wire 20 AWG, Molex 22-01-2027 both ends	310	rail name matched in both netlists
CBL-004	POWER_SUPPLY X5	MAIN_BOARD X10	+1V8_FPGA	2-wire 20 AWG, Molex 22-01-2027 both ends	340	rail name matched in both netlists
CBL-005	POWER_SUPPLY X6	FREQUENCY_SYNTHESIZER X15	+5V0_LO	2-wire 20 AWG, Molex 22-01-2027 both ends	960	rail name matched in both netlists
CBL-006	POWER_SUPPLY X7	??	+3V3_LO_2	2-wire 20 AWG	TBD	no connector with this net name on Main/Synth (rail renamed through a filter/inductor or spare output) — resolve with <code>interconnection_table.md</code>
CBL-007	POWER_SUPPLY X8	??	+3V3_LO_1	2-wire 20 AWG	TBD	no connector with this net name on Main/Synth (rail renamed through a filter/inductor or spare output) — resolve with <code>interconnection_table.md</code>
CBL-008	POWER_SUPPLY X9	??	+5V0_2	2-wire 20 AWG	TBD	no connector with this net name on Main/Synth (rail renamed through a filter/inductor or spare output) — resolve with <code>interconnection_table.md</code>
CBL-009	POWER_SUPPLY X10	MAIN_BOARD X17	+1V8_CLOCK	2-wire 20 AWG, Molex 22-01-2027 both ends	840	rail name matched in both netlists
CBL-010	POWER_SUPPLY X11	??	+3V3_CLOCK	2-wire 20 AWG	TBD	no connector with this net name on Main/Synth (rail renamed through a filter/inductor or spare output) — resolve with <code>interconnection_table.md</code>
CBL-011	POWER_SUPPLY X12	MAIN_BOARD X56	+3V3_AN	2-wire 20 AWG, Molex 22-01-2027 both ends	380	rail name matched in both netlists
CBL-012	POWER_SUPPLY X13	??	+5V0_ADAR	2-wire 20 AWG	TBD	no connector with this net name on Main/Synth (rail renamed through a filter/inductor or spare output) — resolve with <code>interconnection_table.md</code>
CBL-013	POWER_SUPPLY X14	??	+3V3_ADAR_12	2-wire 20 AWG	TBD	no connector with this net name on Main/Synth (rail renamed through a filter/inductor or spare output) — resolve with <code>interconnection_table.md</code>
CBL-014	POWER_SUPPLY X15	??	+3V3_ADAR_34	2-wire 20 AWG	TBD	no connector with this net name on Main/Synth (rail renamed through a filter/inductor or spare output) — resolve with <code>interconnection_table.md</code>
CBL-015	POWER_SUPPLY X16	MAIN_BOARD X24	+3V3	2-wire 20 AWG, Molex 22-01-2027 both ends	200	rail name matched in both netlists
CBL-016	POWER_SUPPLY X17	??	+5V0_3	2-wire 20 AWG	TBD	no connector with this net name on Main/Synth (rail renamed through a filter/inductor or spare output) — resolve with <code>interconnection_table.md</code>

ID	From	To	Signal	Cable	Length (mm)	Note
CBL-017	POWER_SUPPLY X18	MAIN_BOARD X6	-3V3_SW	2-wire 20 AWG, Molex 22-01-2027 both ends	310	rail name matched in both netlists
CBL-018	POWER_SUPPLY X19	MAIN_BOARD X19	-5V5_PA	2-wire 20 AWG, Molex 22-01-2027 both ends	380	rail name matched in both netlists
CBL-019	POWER_SUPPLY X20	MAIN_BOARD X15	-5V0_ADAR34	2-wire 20 AWG, Molex 22-01-2027 both ends	240	rail name matched in both netlists
CBL-020	POWER_SUPPLY X21	MAIN_BOARD X13	-5V0_ADAR12	2-wire 20 AWG, Molex 22-01-2027 both ends	360	rail name matched in both netlists
CBL-021	POWER_SUPPLY X22	MAIN_BOARD X18	+5V0_0	2-wire 20 AWG, Molex 22-01-2027 both ends	400	rail name matched in both netlists
CBL-022	POWER_SUPPLY X23	MAIN_BOARD X1	+3V4	2-wire 20 AWG, Molex 22-01-2027 both ends	360	rail name matched in both netlists
CBL-023	POWER_SUPPLY X24	MAIN_BOARD X11	-3V4	2-wire 20 AWG, Molex 22-01-2027 both ends	330	rail name matched in both netlists
CBL-024	POWER_SUPPLY X25	? ?	+5V0_4	2-wire 20 AWG	TBD	no connector with this net name on Main/Synth (rail renamed through a filter/inductor or spare output) — resolve with interconnection_table.md
CBL-025	POWER_SUPPLY X26	? ?	+5V0_ADTR	2-wire 20 AWG	TBD	no connector with this net name on Main/Synth (rail renamed through a filter/inductor or spare output) — resolve with interconnection_table.md
CBL-026	POWER_SUPPLY X27	MAIN_BOARD X16	+3V3_FPGA	2-wire 20 AWG, Molex 22-01-2027 both ends	430	rail name matched in both netlists
CBL-027	POWER_SUPPLY X28	? ?	+5V0_5	2-wire 20 AWG	TBD	no connector with this net name on Main/Synth (rail renamed through a filter/inductor or spare output) — resolve with interconnection_table.md
CBL-028	POWER_SUPPLY X29	? ?	+3V3_SW	2-wire 20 AWG	TBD	no connector with this net name on Main/Synth (rail renamed through a filter/inductor or spare output) — resolve with interconnection_table.md
CBL-029	POWER_SUPPLY X30	MAIN_BOARD X12	+3V3_VDD_SW	2-wire 20 AWG, Molex 22-01-2027 both ends	1090	rail name matched in both netlists
CBL-030	POWER_SUPPLY X31	MAIN_BOARD X14	+5V0_PA_1	2-wire 20 AWG, Molex 22-01-2027 both ends	1180	rail name matched in both netlists
CBL-031	POWER_SUPPLY X32	MAIN_BOARD X5	+5V0_PA_2	2-wire 20 AWG, Molex 22-01-2027 both ends	1310	rail name matched in both netlists
CBL-032	POWER_SUPPLY X33	MAIN_BOARD X7	+5V0_PA_3	2-wire 20 AWG, Molex 22-01-2027 both ends	1020	rail name matched in both netlists
CBL-033	POWER_SUPPLY X34	MAIN_BOARD X4	+3V3_ADTR	2-wire 20 AWG, Molex 22-01-2027 both ends	1120	rail name matched in both netlists
CBL-034	POWER_SUPPLY X35	FREQUENCY_SYNTHESIZER X4	+3V3_XO	2-wire 20 AWG, Molex 22-01-2027 both ends	1030	rail name matched in both netlists
CBL-035	POWER_SUPPLY SV1	MAIN_BOARD SV1	enable bus (15 EN + GND)	20-way 1.27 mm IDC ribbon	320	
CBL-036	FREQUENCY_SYNTHESIZER J7	MAIN_BOARD J1	100 MHz FPGA sys clk	coax RG-405 (0.086") SMA-SMA, phase-stable	140	mapping from interconnection_table.md \$6
CBL-037	FREQUENCY_SYNTHESIZER J5	MAIN_BOARD J20	120 MHz DAC	coax RG-405 (0.086") SMA-SMA, phase-stable	170	mapping from interconnection_table.md \$6
CBL-038	FREQUENCY_SYNTHESIZER J6	MAIN_BOARD J18	120 MHz DAC	coax RG-405 (0.086") SMA-SMA, phase-stable	120	mapping from interconnection_table.md \$6
CBL-039	FREQUENCY_SYNTHESIZER J3	MAIN_BOARD J21	400 MHz ADC	coax RG-405 (0.086") SMA-SMA, phase-stable	120	mapping from interconnection_table.md \$6
CBL-040	FREQUENCY_SYNTHESIZER J10	MAIN_BOARD J23	LO TX	coax RG-405 (0.086") SMA-SMA, phase-stable	100	mapping from interconnection_table.md \$6
CBL-041	FREQUENCY_SYNTHESIZER J11	MAIN_BOARD J22	LO RX	coax RG-405 (0.086") SMA-SMA,	140	mapping from interconnection_table.md \$6

ID	From	To	Signal	Cable	Length (mm)	Note
				phase-stable		
CBL-042	FREQUENCY_SYNTHESIZER J4	MAIN_BOARD J19	test	coax RG-405 (0.086") SMA-SMA, phase-stable	100	mapping from interconnection_table.md §6
CBL-043	FREQUENCY_SYNTHESIZER JP1	MAIN_BOARD JP1	control header	ribbon/discrete 26 AWG, 2.54 mm housings	210	
CBL-044	FREQUENCY_SYNTHESIZER JP2	MAIN_BOARD JP13	control header	ribbon/discrete 26 AWG, 2.54 mm housings	240	
CBL-045	MAIN_BOARD X_7	RF_PA PA1 X2	VG_1	2-wire 24 AWG shielded, Molex 22-01-2027	150	PA instance ↔ VG_n assignment PROPOSED (n = n); not documented in CAD
CBL-046	MAIN_BOARD X3	RF_PA PA1 X3	drain current sense	3-wire 24 AWG twisted, Molex 22-01-3037	150	Main sense connector ↔ PA n PROPOSED
CBL-047	MAIN_BOARD J27	RF_PA PA1 J1	RF 10.5 GHz (to PA RFIN)	coax RG-405 SMA-SMA, EQUAL LENGTH set	330	which SMA of the pair is RFIN/RFOUT is UNVERIFIED (interconnection_table §7)
CBL-048	MAIN_BOARD J26	RF_PA PA1 J2	RF 10.5 GHz (from PA RFOUT)	coax RG-405 SMA-SMA, EQUAL LENGTH set	320	which SMA of the pair is RFIN/RFOUT is UNVERIFIED (interconnection_table §7)
CBL-049	RF_PA PA1 J2	ANTENNA ROW1	RF to antenna row n	coax RG-405 SMA-2.92 mm, EQUAL LENGTH set	100	antenna row n ↔ PA n PROPOSED; equal length mandatory for elevation phase
CBL-050	DSN-PSU-01 OUT1	RF_PA PA1 22V	22 V drain, pulsed	2-wire 18 AWG twisted, AK300/2 screw terminal	510	
CBL-051	MAIN_BOARD X_16	RF_PA PA2 X2	VG_2	2-wire 24 AWG shielded, Molex 22-01-2027	140	PA instance ↔ VG_n assignment PROPOSED (n = n); not documented in CAD
CBL-052	MAIN_BOARD X38	RF_PA PA2 X3	drain current sense	3-wire 24 AWG twisted, Molex 22-01-3037	110	Main sense connector ↔ PA n PROPOSED
CBL-053	MAIN_BOARD J29	RF_PA PA2 J1	RF 10.5 GHz (to PA RFIN)	coax RG-405 SMA-SMA, EQUAL LENGTH set	310	which SMA of the pair is RFIN/RFOUT is UNVERIFIED (interconnection_table §7)
CBL-054	MAIN_BOARD J28	RF_PA PA2 J2	RF 10.5 GHz (from PA RFOUT)	coax RG-405 SMA-SMA, EQUAL LENGTH set	320	which SMA of the pair is RFIN/RFOUT is UNVERIFIED (interconnection_table §7)
CBL-055	RF_PA PA2 J2	ANTENNA ROW2	RF to antenna row n	coax RG-405 SMA-2.92 mm, EQUAL LENGTH set	150	antenna row n ↔ PA n PROPOSED; equal length mandatory for elevation phase
CBL-056	DSN-PSU-01 OUT2	RF_PA PA2 22V	22 V drain, pulsed	2-wire 18 AWG twisted, AK300/2 screw terminal	470	
CBL-057	MAIN_BOARD X_8	RF_PA PA3 X2	VG_3	2-wire 24 AWG shielded, Molex 22-01-2027	210	PA instance ↔ VG_n assignment PROPOSED (n = n); not documented in CAD
CBL-058	MAIN_BOARD X39	RF_PA PA3 X3	drain current sense	3-wire 24 AWG twisted, Molex 22-01-3037	160	Main sense connector ↔ PA n PROPOSED
CBL-059	MAIN_BOARD J25	RF_PA PA3 J1	RF 10.5 GHz (to PA RFIN)	coax RG-405 SMA-SMA, EQUAL LENGTH set	290	which SMA of the pair is RFIN/RFOUT is UNVERIFIED (interconnection_table §7)
CBL-060	MAIN_BOARD J24	RF_PA PA3 J2	RF 10.5 GHz (from PA RFOUT)	coax RG-405 SMA-SMA, EQUAL LENGTH set	300	which SMA of the pair is RFIN/RFOUT is UNVERIFIED (interconnection_table §7)
CBL-061	RF_PA PA3 J2	ANTENNA ROW3	RF to antenna row n	coax RG-405 SMA-2.92 mm, EQUAL LENGTH set	210	antenna row n ↔ PA n PROPOSED; equal length mandatory for elevation phase
CBL-062	DSN-PSU-01 OUT3	RF_PA PA3 22V	22 V drain, pulsed	2-wire 18 AWG twisted, AK300/2 screw terminal	430	
CBL-063	MAIN_BOARD X_15	RF_PA PA4 X2	VG_4	2-wire 24 AWG shielded, Molex 22-01-2027	240	PA instance ↔ VG_n assignment PROPOSED (n = n); not documented in CAD
CBL-064	MAIN_BOARD X40	RF_PA PA4 X3	drain current sense	3-wire 24 AWG twisted, Molex 22-01-3037	210	Main sense connector ↔ PA n PROPOSED
CBL-065	MAIN_BOARD J31	RF_PA PA4 J1	RF 10.5 GHz (to PA RFIN)	coax RG-405 SMA-SMA, EQUAL LENGTH set	290	which SMA of the pair is RFIN/RFOUT is UNVERIFIED (interconnection_table §7)

ID	From	To	Signal	Cable	Length (mm)	Note
CBL-066	MAIN_BOARD J30	RF_PA PA4 J2	RF 10.5 GHz (from PA RFOUT)	coax RG-405 SMA-SMA, EQUAL LENGTH set	330	which SMA of the pair is RFIN/RFOUT is UNVERIFIED (interconnection_table §7)
CBL-067	RF_PA PA4 J2	ANTENNA ROW4	RF to antenna row n	coax RG-405 SMA-2.92 mm, EQUAL LENGTH set	260	antenna row n ↔ PA n PROPOSED; equal length mandatory for elevation phase
CBL-068	DSN-PSU-01 OUT4	RF_PA PA4 22V	22 V drain, pulsed	2-wire 18 AWG twisted, AK300/2 screw terminal	390	
CBL-069	MAIN_BOARD X_4	RF_PA PA5 X2	VG_5	2-wire 24 AWG shielded, Molex 22-01-2027	190	PA instance ↔ VG_n assignment PROPOSED (n = n); not documented in CAD
CBL-070	MAIN_BOARD X41	RF_PA PA5 X3	drain current sense	3-wire 24 AWG twisted, Molex 22-01-3037	220	Main sense connector ↔ PA n PROPOSED
CBL-071	MAIN_BOARD J35	RF_PA PA5 J1	RF 10.5 GHz (to PA RFIN)	coax RG-405 SMA-SMA, EQUAL LENGTH set	340	which SMA of the pair is RFIN/RFOUT is UNVERIFIED (interconnection_table §7)
CBL-072	MAIN_BOARD J34	RF_PA PA5 J2	RF 10.5 GHz (from PA RFOUT)	coax RG-405 SMA-SMA, EQUAL LENGTH set	320	which SMA of the pair is RFIN/RFOUT is UNVERIFIED (interconnection_table §7)
CBL-073	RF_PA PA5 J2	ANTENNA ROW5	RF to antenna row n	coax RG-405 SMA-2.92 mm, EQUAL LENGTH set	90	antenna row n ↔ PA n PROPOSED; equal length mandatory for elevation phase
CBL-074	DSN-PSU-01 OUT5	RF_PA PA5 22V	22 V drain, pulsed	2-wire 18 AWG twisted, AK300/2 screw terminal	440	
CBL-075	MAIN_BOARD X_11	RF_PA PA6 X2	VG_6	2-wire 24 AWG shielded, Molex 22-01-2027	220	PA instance ↔ VG_n assignment PROPOSED (n = n); not documented in CAD
CBL-076	MAIN_BOARD X42	RF_PA PA6 X3	drain current sense	3-wire 24 AWG twisted, Molex 22-01-3037	220	Main sense connector ↔ PA n PROPOSED
CBL-077	MAIN_BOARD J37	RF_PA PA6 J1	RF 10.5 GHz (to PA RFIN)	coax RG-405 SMA-SMA, EQUAL LENGTH set	310	which SMA of the pair is RFIN/RFOUT is UNVERIFIED (interconnection_table §7)
CBL-078	MAIN_BOARD J36	RF_PA PA6 J2	RF 10.5 GHz (from PA RFOUT)	coax RG-405 SMA-SMA, EQUAL LENGTH set	320	which SMA of the pair is RFIN/RFOUT is UNVERIFIED (interconnection_table §7)
CBL-079	RF_PA PA6 J2	ANTENNA ROW6	RF to antenna row n	coax RG-405 SMA-2.92 mm, EQUAL LENGTH set	140	antenna row n ↔ PA n PROPOSED; equal length mandatory for elevation phase
CBL-080	DSN-PSU-01 OUT6	RF_PA PA6 22V	22 V drain, pulsed	2-wire 18 AWG twisted, AK300/2 screw terminal	400	
CBL-081	MAIN_BOARD X_3	RF_PA PA7 X2	VG_7	2-wire 24 AWG shielded, Molex 22-01-2027	290	PA instance ↔ VG_n assignment PROPOSED (n = n); not documented in CAD
CBL-082	MAIN_BOARD X43	RF_PA PA7 X3	drain current sense	3-wire 24 AWG twisted, Molex 22-01-3037	250	Main sense connector ↔ PA n PROPOSED
CBL-083	MAIN_BOARD J33	RF_PA PA7 J1	RF 10.5 GHz (to PA RFIN)	coax RG-405 SMA-SMA, EQUAL LENGTH set	300	which SMA of the pair is RFIN/RFOUT is UNVERIFIED (interconnection_table §7)
CBL-084	MAIN_BOARD J32	RF_PA PA7 J2	RF 10.5 GHz (from PA RFOUT)	coax RG-405 SMA-SMA, EQUAL LENGTH set	250	which SMA of the pair is RFIN/RFOUT is UNVERIFIED (interconnection_table §7)
CBL-085	RF_PA PA7 J2	ANTENNA ROW7	RF to antenna row n	coax RG-405 SMA-2.92 mm, EQUAL LENGTH set	200	antenna row n ↔ PA n PROPOSED; equal length mandatory for elevation phase
CBL-086	DSN-PSU-01 OUT7	RF_PA PA7 22V	22 V drain, pulsed	2-wire 18 AWG twisted, AK300/2 screw terminal	360	
CBL-087	MAIN_BOARD X_12	RF_PA PA8 X2	VG_8	2-wire 24 AWG shielded, Molex 22-01-2027	280	PA instance ↔ VG_n assignment PROPOSED (n = n); not documented in CAD
CBL-088	MAIN_BOARD X44	RF_PA PA8 X3	drain current sense	3-wire 24 AWG twisted, Molex 22-01-3037	280	Main sense connector ↔ PA n PROPOSED
CBL-089	MAIN_BOARD J39	RF_PA PA8 J1	RF 10.5 GHz (to PA RFIN)	coax RG-405 SMA-SMA, EQUAL LENGTH set	270	which SMA of the pair is RFIN/RFOUT is UNVERIFIED (interconnection_table §7)

ID	From	To	Signal	Cable	Length (mm)	Note
CBL-090	MAIN_BOARD J38	RF_PA PA8 J2	RF 10.5 GHz (from PA RFOUT)	coax RG-405 SMA-SMA, EQUAL LENGTH set	260	which SMA of the pair is RFIN/RFOUT is UNVERIFIED (interconnection_table §7)
CBL-091	RF_PA PA8 J2	ANTENNA ROW8	RF to antenna row n	coax RG-405 SMA-2.92 mm, EQUAL LENGTH set	250	antenna row n ↔ PA n PROPOSED; equal length mandatory for elevation phase
CBL-092	DSN-PSU-01 OUT8	RF_PA PA8 22V	22 V drain, pulsed	2-wire 18 AWG twisted, AK300/2 screw terminal	320	
CBL-093	MAIN_BOARD X_5	RF_PA PA9 X2	VG_9	2-wire 24 AWG shielded, Molex 22-01-2027	290	PA instance ↔ VG_n assignment PROPOSED (n = n); not documented in CAD
CBL-094	MAIN_BOARD X45	RF_PA PA9 X3	drain current sense	3-wire 24 AWG twisted, Molex 22-01-3037	340	Main sense connector ↔ PA n PROPOSED
CBL-095	MAIN_BOARD J47	RF_PA PA9 J1	RF 10.5 GHz (to PA RFIN)	coax RG-405 SMA-SMA, EQUAL LENGTH set	260	which SMA of the pair is RFIN/RFOUT is UNVERIFIED (interconnection_table §7)
CBL-096	MAIN_BOARD J46	RF_PA PA9 J2	RF 10.5 GHz (from PA RFOUT)	coax RG-405 SMA-SMA, EQUAL LENGTH set	270	which SMA of the pair is RFIN/RFOUT is UNVERIFIED (interconnection_table §7)
CBL-097	RF_PA PA9 J2	ANTENNA ROW9	RF to antenna row n	coax RG-405 SMA-2.92 mm, EQUAL LENGTH set	100	antenna row n ↔ PA n PROPOSED; equal length mandatory for elevation phase
CBL-098	DSN-PSU-01 OUT9	RF_PA PA9 22V	22 V drain, pulsed	2-wire 18 AWG twisted, AK300/2 screw terminal	380	
CBL-099	MAIN_BOARD X_14	RF_PA PA10 X2	VG_10	2-wire 24 AWG shielded, Molex 22-01-2027	280	PA instance ↔ VG_n assignment PROPOSED (n = n); not documented in CAD
CBL-100	MAIN_BOARD X46	RF_PA PA10 X3	drain current sense	3-wire 24 AWG twisted, Molex 22-01-3037	290	Main sense connector ↔ PA n PROPOSED
CBL-101	MAIN_BOARD J41	RF_PA PA10 J1	RF 10.5 GHz (to PA RFIN)	coax RG-405 SMA-SMA, EQUAL LENGTH set	260	which SMA of the pair is RFIN/RFOUT is UNVERIFIED (interconnection_table §7)
CBL-102	MAIN_BOARD J40	RF_PA PA10 J2	RF 10.5 GHz (from PA RFOUT)	coax RG-405 SMA-SMA, EQUAL LENGTH set	240	which SMA of the pair is RFIN/RFOUT is UNVERIFIED (interconnection_table §7)
CBL-103	RF_PA PA10 J2	ANTENNA ROW10	RF to antenna row n	coax RG-405 SMA-2.92 mm, EQUAL LENGTH set	130	antenna row n ↔ PA n PROPOSED; equal length mandatory for elevation phase
CBL-104	DSN-PSU-01 OUT10	RF_PA PA10 22V	22 V drain, pulsed	2-wire 18 AWG twisted, AK300/2 screw terminal	340	
CBL-105	MAIN_BOARD X_6	RF_PA PA11 X2	VG_11	2-wire 24 AWG shielded, Molex 22-01-2027	350	PA instance ↔ VG_n assignment PROPOSED (n = n); not documented in CAD
CBL-106	MAIN_BOARD X47	RF_PA PA11 X3	drain current sense	3-wire 24 AWG twisted, Molex 22-01-3037	240	Main sense connector ↔ PA n PROPOSED
CBL-107	MAIN_BOARD J45	RF_PA PA11 J1	RF 10.5 GHz (to PA RFIN)	coax RG-405 SMA-SMA, EQUAL LENGTH set	160	which SMA of the pair is RFIN/RFOUT is UNVERIFIED (interconnection_table §7)
CBL-108	MAIN_BOARD J44	RF_PA PA11 J2	RF 10.5 GHz (from PA RFOUT)	coax RG-405 SMA-SMA, EQUAL LENGTH set	140	which SMA of the pair is RFIN/RFOUT is UNVERIFIED (interconnection_table §7)
CBL-109	RF_PA PA11 J2	ANTENNA ROW11	RF to antenna row n	coax RG-405 SMA-2.92 mm, EQUAL LENGTH set	190	antenna row n ↔ PA n PROPOSED; equal length mandatory for elevation phase
CBL-110	DSN-PSU-01 OUT11	RF_PA PA11 22V	22 V drain, pulsed	2-wire 18 AWG twisted, AK300/2 screw terminal	300	
CBL-111	MAIN_BOARD X_13	RF_PA PA12 X2	VG_12	2-wire 24 AWG shielded, Molex 22-01-2027	380	PA instance ↔ VG_n assignment PROPOSED (n = n); not documented in CAD
CBL-112	MAIN_BOARD X48	RF_PA PA12 X3	drain current sense	3-wire 24 AWG twisted, Molex 22-01-3037	290	Main sense connector ↔ PA n PROPOSED
CBL-113	MAIN_BOARD J43	RF_PA PA12 J1	RF 10.5 GHz (to PA RFIN)	coax RG-405 SMA-SMA, EQUAL LENGTH set	160	which SMA of the pair is RFIN/RFOUT is UNVERIFIED (interconnection_table §7)

ID	From	To	Signal	Cable	Length (mm)	Note
CBL-114	MAIN_BOARD J42	RF_PA PA12 J2	RF 10.5 GHz (from PA RFOUT)	coax RG-405 SMA-SMA, EQUAL LENGTH set	120	which SMA of the pair is RFIN/RFOUT is UNVERIFIED (interconnection_table §7)
CBL-115	RF_PA PA12 J2	ANTENNA ROW12	RF to antenna row n	coax RG-405 SMA-2.92 mm, EQUAL LENGTH set	240	antenna row n ↔ PA n PROPOSED; equal length mandatory for elevation phase
CBL-116	DSN-PSU-01 OUT12	RF_PA PA12 22V	22 V drain, pulsed	2-wire 18 AWG twisted, AK300/2 screw terminal	260	
CBL-117	MAIN_BOARD X_2	RF_PA PA13 X2	VG_13	2-wire 24 AWG shielded, Molex 22-01-2027	360	PA instance ↔ VG_n assignment PROPOSED (n = n); not documented in CAD
CBL-118	MAIN_BOARD X49	RF_PA PA13 X3	drain current sense	3-wire 24 AWG twisted, Molex 22-01-3037	410	Main sense connector ↔ PA n PROPOSED
CBL-119	MAIN_BOARD J55	RF_PA PA13 J1	RF 10.5 GHz (to PA RFIN)	coax RG-405 SMA-SMA, EQUAL LENGTH set	250	which SMA of the pair is RFIN/RFOUT is UNVERIFIED (interconnection_table §7)
CBL-120	MAIN_BOARD J54	RF_PA PA13 J2	RF 10.5 GHz (from PA RFOUT)	coax RG-405 SMA-SMA, EQUAL LENGTH set	260	which SMA of the pair is RFIN/RFOUT is UNVERIFIED (interconnection_table §7)
CBL-121	RF_PA PA13 J2	ANTENNA ROW13	RF to antenna row n	coax RG-405 SMA-2.92 mm, EQUAL LENGTH set	110	antenna row n ↔ PA n PROPOSED; equal length mandatory for elevation phase
CBL-122	DSN-PSU-01 OUT13	RF_PA PA13 22V	22 V drain, pulsed	2-wire 18 AWG twisted, AK300/2 screw terminal	310	
CBL-123	MAIN_BOARD X_9	RF_PA PA14 X2	VG_14	2-wire 24 AWG shielded, Molex 22-01-2027	380	PA instance ↔ VG_n assignment PROPOSED (n = n); not documented in CAD
CBL-124	MAIN_BOARD X50	RF_PA PA14 X3	drain current sense	3-wire 24 AWG twisted, Molex 22-01-3037	380	Main sense connector ↔ PA n PROPOSED
CBL-125	MAIN_BOARD J49	RF_PA PA14 J1	RF 10.5 GHz (to PA RFIN)	coax RG-405 SMA-SMA, EQUAL LENGTH set	250	which SMA of the pair is RFIN/RFOUT is UNVERIFIED (interconnection_table §7)
CBL-126	MAIN_BOARD J48	RF_PA PA14 J2	RF 10.5 GHz (from PA RFOUT)	coax RG-405 SMA-SMA, EQUAL LENGTH set	230	which SMA of the pair is RFIN/RFOUT is UNVERIFIED (interconnection_table §7)
CBL-127	RF_PA PA14 J2	ANTENNA ROW14	RF to antenna row n	coax RG-405 SMA-2.92 mm, EQUAL LENGTH set	130	antenna row n ↔ PA n PROPOSED; equal length mandatory for elevation phase
CBL-128	DSN-PSU-01 OUT14	RF_PA PA14 22V	22 V drain, pulsed	2-wire 18 AWG twisted, AK300/2 screw terminal	270	
CBL-129	MAIN_BOARD X_1	RF_PA PA15 X2	VG_15	2-wire 24 AWG shielded, Molex 22-01-2027	450	PA instance ↔ VG_n assignment PROPOSED (n = n); not documented in CAD
CBL-130	MAIN_BOARD X51	RF_PA PA15 X3	drain current sense	3-wire 24 AWG twisted, Molex 22-01-3037	350	Main sense connector ↔ PA n PROPOSED
CBL-131	MAIN_BOARD J53	RF_PA PA15 J1	RF 10.5 GHz (to PA RFIN)	coax RG-405 SMA-SMA, EQUAL LENGTH set	150	which SMA of the pair is RFIN/RFOUT is UNVERIFIED (interconnection_table §7)
CBL-132	MAIN_BOARD J52	RF_PA PA15 J2	RF 10.5 GHz (from PA RFOUT)	coax RG-405 SMA-SMA, EQUAL LENGTH set	190	which SMA of the pair is RFIN/RFOUT is UNVERIFIED (interconnection_table §7)
CBL-133	RF_PA PA15 J2	ANTENNA ROW15	RF to antenna row n	coax RG-405 SMA-2.92 mm, EQUAL LENGTH set	180	antenna row n ↔ PA n PROPOSED; equal length mandatory for elevation phase
CBL-134	DSN-PSU-01 OUT15	RF_PA PA15 22V	22 V drain, pulsed	2-wire 18 AWG twisted, AK300/2 screw terminal	230	
CBL-135	MAIN_BOARD X_10	RF_PA PA16 X2	VG_16	2-wire 24 AWG shielded, Molex 22-01-2027	450	PA instance ↔ VG_n assignment PROPOSED (n = n); not documented in CAD
CBL-136	MAIN_BOARD X52	RF_PA PA16 X3	drain current sense	3-wire 24 AWG twisted, Molex 22-01-3037	370	Main sense connector ↔ PA n PROPOSED
CBL-137	MAIN_BOARD J51	RF_PA PA16 J1	RF 10.5 GHz (to PA RFIN)	coax RG-405 SMA-SMA, EQUAL LENGTH set	170	which SMA of the pair is RFIN/RFOUT is UNVERIFIED (interconnection_table §7)

ID	From	To	Signal	Cable	Length (mm)	Note
CBL-138	MAIN_BOARD J50	RF_PA PA16 J2	RF 10.5 GHz (from PA RFOUT)	coax RG-405 SMA-SMA, EQUAL LENGTH set	190	which SMA of the pair is RFIN/RFOUT is UNVERIFIED (interconnection_table §7)
CBL-139	RF_PA PA16 J2	ANTENNA ROW16	RF to antenna row n	coax RG-405 SMA-2.92 mm, EQUAL LENGTH set	230	antenna row n ↔ PA n PROPOSED; equal length mandatory for elevation phase
CBL-140	DSN-PSU-01 OUT16	RF_PA PA16 22V	22 V drain, pulsed	2-wire 18 AWG twisted, AK300/2 screw terminal	190	
CBL-141	SLIP RING VIN	POWER_SUPPLY X1	VIN 12-17 V	2 × 2-wire 16 AWG	340	
CBL-142	SLIP RING VIN	DSN-PSU-01 IN	VIN to 22 V boost	2 × 2-wire 16 AWG	470	
CBL-143	SLIP RING USB	MAIN_BOARD X53	USB 2.0 FS to host	USB 2.0 shielded, mini-B	360	
CBL-144	MAIN_BOARD stepper pins	PEDESTAL TB6600 (via slip ring)	STEP/DIR/EN	3-wire 24 AWG shielded	TBD	the driver sits on the fixed base; needs 3 more slip-ring circuits OR move the driver into the head (then only motor phases cross: 4 circuits) — DECISION NEEDED

<!-- chapter C: Tools and scripts (what each generator does, how to regenerate) -->

Appendix C — Tools and scripts: what each generator does and how to regenerate

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Status summary: documentation of the repository's own automation (`tools/`) and of the external tools installed on the authoring workstation. Every purpose statement is taken from the script's module docstring or header comment (read on 2026-10-09 with a shell loop over `tools/*`); every external version is the output of the tool's own `--version` call on the same date. No script modifies the upstream design files; outputs go to `engineering/`, `docs/`, `beta/`, `manual/` or `build/`.

Sources: `tools/.py`, `tools/.sh`, `tools/vivado/create_project.tcl` (docstrings/headers), `engineering/README.md` ("Regenerate everything"), `engineering/DESIGN/README.md` ("Regenerate"), `beta/README.md`, `beta/stm32/README.md` §1–§3, `beta/pcb/README.md`, `beta/fpga/README.md`, `docs/TESTING/VALIDATION_PLAN.md`.

C.1 Repository scripts (`tools/`)

Script	Purpose (from its docstring)	Inputs	
<code>build_manual.py</code>	"Build the AERIS-10 manual: concatenate manual/chapters/*.md in the order of manual/00_OUTLINE.md, number figures, resolve relative image links, write manual/AERIS10_MANUAL.md, manual/build/AERIS10_MANUAL.html (images embedded as data URIs; SVG inline) and manual/build/AERIS10_MANUAL.pdf (headless Chrome). <code>--check</code> only validates ... Stdlib only (+ Google Chrome for the PDF). Exit 0 ok, 1 validation problems, 2 error."	<code>manual/00_OUTLINE.md</code> , <code>manual/chapters/*.md</code> , figure files	<code>manual/At</code>
<code>check_doc_links.py</code>	"Markdown link and path-reference validator ... Scans every *.md file for Markdown links and backticked repository paths and verifies that each target exists relative to the referencing file's directory"	<code>all *.md</code>	report, exit c
<code>check_fpga_constraints.py</code>	"FPGA constraint completeness checker ... reports unresolved placeholders, top-level ports with no PACKAGE_PIN constraint, ports with no IOSTANDARD"	top-level <code>.v</code> , <code>.xdc</code>	report, exit c
<code>check_manufacturing_files.py</code>	"PCB manufacturing-package inventory ... checks presence of schematic, board, Gerber set, NC drill, pick-and-place, BOM, assembly drawing, fabrication drawing, netlist, DRC/ERC reports. Reports a readiness table. Never creates files."	repository tree	readiness ta
<code>check_missing_files.py</code>	"Expected-artifact manifest checker ... Each entry has an ID that matches docs/03_MISSING_COMPONENTS.md. Never creates files."	manifest in the script	report, exit c
<code>check_python_imports.py</code>	"Python dependency extractor and import checker ... Parses every *.py file with <code>ast</code> (no execution), lists imported top-level modules, classifies them as stdlib / third-party / local, maps third-party modules to PyPI distribution names, and (optionally, <code>--try-import</code>) tries importing each"	<code>*.py</code> under <code>--dir</code>	report, exit c
<code>check_stm32_includes.py</code>	"STM32 firmware include-graph and missing-header analyser ... classifies each header as LOCAL / HAL / ..."	C/C++ sources under <code>9_Firmware/9_1_Microcontroller</code> (or <code>--root</code>)	report, exit c
<code>design_antenna_array.py</code>	"PROPOSED DESIGN — 16-row × 8-patch series-fed microstrip array for AERIS-10 (10.5 GHz). Reads <code>engineering/DESIGN/design_parameters.json</code> , computes first-order patch and microstrip dimensions ... and writes <code>engineering/DESIGN/ANTENNA/kicad/aeris10_patch_array.kicad_pcb</code> "	<code>design_parameters.json</code>	engineer: model)
<code>design_antenna_tune.py</code>	"Tune the patch length of the openEMS row model until the resonance sits at f0 (secant on L_SCALE). Runs ... <code>openems_patch_row.py</code> with the openEMS Python venv; writes ... <code>TUNING_LOG.md</code> and the final <code>s11.csv</code> / <code>s11</code> plot / pattern."	openEMS model	engineer:
<code>design_calcs.py</code>	"PROPOSED DESIGN calculations (BETA, no measurement): 2-D thermal map of the PA heat-spreader plate, pedestal drive torque, and the radar range equation. Stdlib only."	<code>design_parameters.json</code> , <code>THERMAL/thermal_summary.json</code> , <code>CAD/detail/parts_list.json</code> , <code>ANTENNA/simulation/tuning_result.json</code>	engineer: thermal_r
<code>design_enclosure_detail_freecad.py</code>	"PROPOSED DESIGN — detailed radar-head enclosure and pedestal (DSN-MECH-06...09), FreeCAD headless. Builds every mechanical part as its own solid ... writes to <code>engineering/DESIGN/MECHANICAL/CAD/detail</code> "	<code>tools/design_layout.py</code> , PCB hole tables	<code>aeris10_</code> PNG/PDF
<code>design_enclosure_drawings.py</code>	"PROPOSED DESIGN — enclosure detail drawings (no CAD needed): sheet-metal flat patterns with bend lines (DSN-MECH-06), assembly section with fastener balloons (DSN-MECH-07) and the mechanical parts list ... Bend allowance: 90°, inside radius $R = t$, $K = 0.4 \rightarrow BA = \pi/2 \cdot (R + K \cdot t)$."	<code>CAD/detail/parts_list.json</code>	DSN-MECH
<code>design_host_link.py</code>	"PROPOSED DESIGN — FPGA → host data path (DSN-LINK-01). Option A 'FT601 on Main Board rev. B' ... Option B 'SPI bridge, no PCB change'"	Main Board netlist, RTL geometry	engineer:
<code>design_layout.py</code>	"Shared PROPOSED head/pedestal layout for the AERIS-10 design generators (D-08, D-09, D-11, D-12, D-13). Coordinate system of the	<code>design_parameters.json</code>	prints the st

Script	Purpose (from its docstring)	Inputs	
	HEAD ... Boards are VERTICAL, parallel to the antenna, in depth order"		
design_mechanical_drawings.py	"PROPOSED DESIGN — 2-D engineering drawings (SVG, scale 1:1) of the AERIS-10 head and pedestal plus the harness schedule, from tools/design_layout.py (no CAD application needed). Outputs ... DSN-MECH-01 ... DSN-MECH-05"	design_layout.py, KiCad P&P files	engineer: engineer:
design_mechanical_freecad.py	"PROPOSED DESIGN — parametric 3-D model of the AERIS-10 radar head and pedestal in FreeCAD. Run with FreeCAD's Python (headless) ... Reads tools/design_layout.py"	design_layout.py	engineer: table, isome
design_pa_supply_schematic.py	"PROPOSED DESIGN — block-level schematic, netlist and BOM of the 22 V PA drain supply / switch module (DSN-PSU-01, decision D-14). Draws an editable SVG ... Component-level capture in KiCad is the next step (MDR-12); values come from tools/design_thermal.py."	THERMAL/thermal_summary.json	engineer:
design_thermal.py	"PROPOSED DESIGN — thermal budget of the 16 × QPA2962 PA set and sizing of the 22 V drain supply. Reads ... design_parameters.json and writes ... THERMAL_AND_PA_SUPPLY.md (+ thermal_summary.json). All formulas are first-order engineering estimates"	design_parameters.json	engineer:
eagle_svg_common.py	"Shared helpers for rendering Autodesk EAGLE XML (.sch/.brd) to SVG. Stdlib only. Used by render_eagle_board.py and render_eagle_schematic.py." Holds the ATTRIBUTION title-block line (Antidrone Ukraine · antidrone.cc) applied when figures are regenerated.	—	library
extract_eagle_netlist.py	"Extract per-part pin/net connectivity from an EAGLE 6/7 XML schematic ... provide verifiable evidence (net name <-> device pin <-> package pad) for FPGA/MCU pin-assignment reconstruction. Read-only"	.sch	CSV named
fpga_lint.sh	"syntax/elaboration check of the AERIS-10 FPGA RTL with open-source tools. Runs iverilog -g2012 elaboration of radar_system_top, verilator --lint-only, iverilog elaboration of the testbench (expected to fail: SystemVerilog assertions) ... Never modifies RTL. Exit code: 0 if step 1 AND step 2 succeed, 1 otherwise, 3 if no tool found. Dependencies: bash 3.2+, iverilog >= 11 (tested 13.0), verilator >= 5 (tested 5.052)"	9_Firmware/9_2_FPGA/*.v	logs in bui
gen_assembly_exploded_view.py	"CONCEPTUAL exploded assembly view of the AERIS-10 electronics set (SVG). Board outlines and hole positions are taken from the EAGLE .brd files (verified geometry ...). The vertical ARRANGEMENT, the spacing, the antenna panel and the host computer are CONCEPTUAL"	.brd files	engineer:
gen_board_outline_svg.py	"Draw an outline/mounting-hole dimension drawing (SVG) from an EAGLE .brd. Verified geometry only ... The drawing contains NO enclosure, NO component outlines and NO thickness"	.brd	SVG
gen_drawing_register.py	"Drawing register for engineering/ — definition table + existence/validity check. Writes engineering/DRAWING_REGISTER.md from the REGISTER table below and, with --check, verifies that every native file and export exists, is non-empty and (for SVG/DOT/PDF) is well-formed"	register table in the script, files	engineer:
gen_eagle_bom.py	"Extract a bill of materials from an EAGLE 6+/7/9 XML schematic ... Output: CSV (grouped), optional per-reference CSV"	.sch	docs/BOM,
gen_engineering_pcb_docs.py	"Generate the per-board manufacturing-package README (engineering/PCB/<BOARD>/README.md), copy the schematic-derived BOM next to the exports, and write the EAGLE↔KiCad cross-check (engineering/VALIDATION/PCB_CROSS_CHECK.md)"	.brd, KiCad pipeline outputs	README/S
gen_inventory_doc.py	"Generate docs/01_REPOSITORY_INVENTORY.md (Path / File type / Subsystem / Purpose / Dependencies / Status) from the live file tree plus a curated purpose/dependency map ... Writes ONLY the --out file."	file tree	docs/01_f
gen_manual_asset_index.py	"Scan the repository for figures usable in the manual (PNG/SVG/PDF/JPG/DOT/MMD renders) and write manual/ASSET_INDEX.md: one row per asset with path, type, size, origin, proposed caption and status"	repository tree, drawing register	manual/AS
gen_mechanical_package.py	"Build engineering/MECHANICAL/ from verified PCB geometry. For every board it extracts from the EAGLE .brd XML: the board outline (layer 20 Dimension), all non-plated holes and mounting pads ... and writes ... <BOARD>_dimensions.md"	.brd	engineer:
gen_python_module_graph.py	"Python GUI module/import graph. For every .py file under --src ... parses the file with ast and records imports, classified as stdlib / third-party / local"	9_Firmware/9_3_GUI/*.py	engineer:
gen_schematic_reports.py	"Schematic connectivity reports from the EAGLE XML files (no EAGLE needed). For every board ... writes netlist CSV (by net, by part), connection report ..."	.sch	engineer:
gen_verilog_hierarchy.py	"Verilog module hierarchy extractor. Parses every .v file under --rtl (recursively, nothing skipped), finds module definitions and module instantiations and writes" the DOT/Markdown hierarchy	RTL directory	fpga_mod
gen_xdc_from_schematic.py	"Build a candidate pin-constraint file for radar_system_top from the Main Board EAGLE schematic (RADAR_Main_Board.sch, FPGA part	RADAR_Main_Board.sch	candidate X

Script	Purpose (from its docstring)	Inputs	
	U42). Every PACKAGE_PIN written by this tool is taken verbatim from the schematic pad of the FPGA symbol; the mapping 'schematic net -> RTL port' is a documented, reviewable table with a confidence level."		
kiCAD_pcb_pipeline.sh	"import the EAGLE .brd files into KiCad and export the complete manufacturing / drawing package with kicad-cli (KiCad >= 10.0). Non-destructive: original EAGLE files are only read. Outputs go to engineering/PCB/<BOARD>/ and every command + exit code is appended to engineering/VALIDATION/CAD_EXPORT_LOG.md."	.brd files	engineering/PCB/<BOARD>/DXF/STEP, engineering/VALIDATION/CAD_EXPORT_LOG.md
kiCAD_project_from_dru.py	"Create a KiCad project file (.kicad_pro) whose design rules come from the EAGLE design rules (<designrules> block) stored in an EAGLE .brd file" — otherwise DRC runs with KiCad defaults	.brd	.kicad_pro
render_eagle_schematic.py	"Render an Autodesk EAGLE schematic (.sch, XML) to one SVG per sheet. Stdlib only; no EAGLE needed ... a documentation rendering (SOURCE-DERIVED), not a CAD export: EAGLE's vector font is replaced by a monospace system font"	.sch	engineering/PCB/<BOARD>/SVG
repo_inventory.py	"AERIS-10 repository inventory generator. Walks the repository, classifies every file by type and subsystem, and prints a Markdown or CSV table. Non-destructive"	file tree	table on stdlib
run_all_checks.sh	"runs every AERIS-10 static check and summarises PASS/FAIL. Non-destructive. Exit code = number of failed checks (capped at 125). Dependencies: bash, python3 (>=3.10). Optional: iverilog, verilator"	—	summary
stl_to_svg_iso.py	"Render binary/ASCII STL files to a flat-shaded isometric SVG (painter's algorithm). Stdlib only ... Intended for quick visual checks of FreeCAD/KiCad STL exports; not a CAD drawing."	STL	SVG
stm32_check_cube_package.sh	"verify that a STM32CubeF7 package tree contains every file the AERIS-10 firmware needs (HAL modules enabled in stm32f7xx_hal_conf.h, CMSIS device files, startup/linker templates, USB Device Library Core + CDC class). Read-only ... Exit: 0 all found, 1 some missing, 2 bad argument."	Cube tree path	report
svg_sheets_to_pdf.py	"Combine one or more SVG drawings into a multi-page PDF using headless Chrome/Chromium. Each SVG becomes one page whose size equals the SVG's width/height (mm), so the drawing prints at scale 1:1. No Python packages needed" (--png-dir also writes a PNG per sheet)	SVG files	PDF (+ PNG)
vivado/create_project.tcl	"create a Vivado project from the repository RTL. This script DOES NOT choose the FPGA part for you ... You must pass the verified part explicitly ... It never edits RTL. Output: build/vivado/aeris10/aeris10.xpr. Tested: NOT executed in this repository (Vivado is not installed on the authoring machine)."	RTL, XDC	Vivado project

Scripts living next to their sub-projects, documented in their own READMEs: `beta/fpga/build.sh`, `beta/fpga/gen_chirp_mem.py`, `beta/fpga/tb/gen_vectors.py`, `beta/fpga/vivado/.tcl` (`beta/fpga/README.md`); `beta/stm32/build.sh`, `setup_cube.sh`, `tests/run_tests.sh` (`beta/stm32/README.md`); `beta/gui/build_app.sh` (`beta/gui/README.md`); `beta/pcb/tools/.py` `pcbnew/kicad-cli` scripts (`beta/pcb/tools/README.md`).

C.2 External tools installed on the authoring workstation (macOS arm64)

Versions as printed on 2026-10-09 by the commands in the last column; where a version was only recorded in a README it is marked so.

Tool	Version found	Used for	Command / evidence
Python (system)	3.14.8	all tools/*.py, build_manual.py	python3 --version
Python venv beta/gui/.venv	CPython 3.14.7 + Tk 9.x	GUI, pytest, matplotlib (coupling plot F8.4)	beta/gui/README.md, CHANGELOG.md
KiCad / kicad-cli	10.0.6	EAGLE import, Gerber/drill/PDF/STEP exports, DRC (kiCAD_pcb_pipeline.sh, beta/pcb)	~/Applications/KiCad/KiCad.app/Contents/MacOS/kicad-cli version
FreeCAD	1.1.4 (Revision 20260928)	design_mechanical_freecad.py, design_enclosure_detail_freecad.py	~/Applications/FreeCAD.app/Contents/Resources/bin/freecad --version
openEMS (+ CSXCAD)	built from source, installed under ~/opt/openEMS with its own Python venv	antenna row simulation (design_antenna_tune.py, openems_patch_row.py)	beta/README.md ("built from source on this machine"); pip install openEMS fails (not on PyPI — docs/GUI/DEPENDENCIES.md §7)
Arm GNU Toolchain	14.2.Rel1 (arm-none-eabi-gcc 14.2.1 20241119)	beta/stm32/build.sh	arm-none-eabi-gcc --version
CMake	4.4.3	STM32 build	cmake --version
OpenJDK	27 (Homebrew openjdk 27) — not on the shell PATH: java -version printed "Unable to locate a Java Runtime" during the check; beta/pcb/README.md records "Freerouting 2.5.0 jar (OpenJDK 27)"	Freerouting	brew list --versions openjdk
Freerouting	2.5.0 (jar; log in ~/Library/Logs/freerouting/)	Power Supply board routing (beta/pcb)	beta/pcb/README.md

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Tool	Version found	Used for	Command / evidence
Yosys	0.69+post (git 143eb14f)	open-source synthesis trial (beta/fpga_synth/)	yosys -V
Icarus Verilog	13.0 (stable)	fpga_lint.sh, beta/fpga/build.sh	iverilog -V
Verilator	5.052 (2026-09-05)	lint, beta/fpga/build.sh	verilator --version
Graphviz	16.1.0 (dot)	SYS-/SD-/ELEC- diagrams from .dot	dot -V
poppler	pdftoppm 26.10.0	PDF → PNG renders for the manual (assembly drawings F15.2–F15.5)	pdftoppm -v
Google Chrome	present at /Applications/Google Chrome.app	svg_sheets_to_pdf.py, build_manual.py PDF	path check
Vivado	not installed	synthesis, timing, bitstream (F-07...F-09)	docs/TESTING/VALIDATION_PLAN.md; beta/fpga/README.md
STM32CubeMX / CubeIDE	not installed	.ioc regeneration (AC-S2)	beta/stm32/CUBEMX_SETTINGS.md
EAGLE	not installed	fresh ERC/DRC on the originals (B-01/B-02)	docs/TESTING/VALIDATION_PLAN.md

C.3 Regeneration order

Full engineering package (source: engineering/README.md, "Regenerate everything (≈ 10 min, needs KiCad 10 and Google Chrome on the machine; Graphviz for the DOT renders)", copied):

```
bash tools/kicad_pcb_pipeline.sh                # 4 boards → engineering/PCB/*
python3 tools/gen_engineering_pcb_docs.py        # README/STACKUP per board + PCB_CROSS_CHECK.md
python3 tools/gen_schematic_reports.py          # netlists + connection reports
for b in RF_PA FREQUENCY_SYNTHESIZER MAIN_BOARD POWER_SUPPLY; do
  python3 tools/render_eagle_schematic.py "<path to .sch>" --out engineering/ELECTRICAL/schematics/$b/svg --board $b --root .
  python3 tools/svg_sheets_to_pdf.py -o engineering/ELECTRICAL/schematics/$b/${b}_schematic.pdf --png-dir engineering/ELECTRICAL/schematics/$b
done
python3 tools/gen_mechanical_package.py && python3 tools/gen_assembly_exploded_view.py
python3 tools/gen_verilog_hierarchy.py --rtl 9_Firmware/9_2_FPGA --out engineering/SOFTWARE_DIAGRAMS/FPGA
python3 tools/gen_python_module_graph.py
for d in $(find engineering -name "*.dot"); do dot -Tsvg -o ${d%.dot}.svg $d; dot -Tpdf -o ${d%.dot}.pdf $d; dot -Tpng -Gdpi=1200 -o ${d%.dot}.png $d; done
python3 tools/gen_drawing_register.py --check
```

Proposed designs (source: engineering/DESIGN/README.md, "Regenerate", copied; every generator reads engineering/DESIGN/design_parameters.json):

```
python3 tools/design_thermal.py && python3 tools/design_antenna_array.py && python3 tools/design_pa_supply_schematic.py && python3 tools/design_enclosure_drawings.py && python3 tools/design_calcs.py
```

followed by python3 tools/design_enclosure_drawings.py and python3 tools/design_calcs.py (they consume CAD/detail/parts_list.json and THERMAL/thermal_summary.json, so they run after the FreeCAD detail model and the thermal sheet — chapter 10 §10.9), and python3 tools/design_antenna_tune.py for the openEMS run.

BETA trees (source: beta/README.md, table "Build/verify"): bash beta/fpga/build.sh (iverilog + verilator + 9 testbench runs); bash beta/stm32/setup_cube.sh then bash beta/stm32/build.sh and bash beta/stm32/tests/run_tests.sh; cd beta/gui && .venv/bin/python -m pytest -q and .venv/bin/python -m aeris10_gui --selftest; beta/pcb scripts per board as documented in beta/pcb/tools/README.md.

Manual (this document): render the figures listed in manual/FIGURE_PLAN.md ("to render") into manual/figures/, then python3 tools/build_manual.py --check (0 problems required), then python3 tools/build_manual.py (HTML + PDF). Figures rendered for the chapters of this edition and their exact commands are recorded in manual/FIGURE_LOG.md.

Static checks in one go: bash tools/run_all_checks.sh (exit code = number of failed checks; optional FPGA checks are skipped if iverilog/verilator are absent).

<!-- chapter D: Glossary, acronyms, status vocabulary -->

Appendix D — Glossary, acronyms, symbols, status vocabulary and identifier families

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Status summary: reference appendix. Symbols are transcribed to plain text from `00_notation/symbol_table.md` (ORIGINAL PROJECT FILE); acronym expansions are standard engineering usage with the repository location where each term is used; the status vocabulary and identifier families are those of `manual/STYLE_GUIDE.md` and the registers cited in chapter 17.

Sources: `00_notation/symbol_table.md`, `00_notation/parameter_table.md`, `00_notation/conventions.md`, `manual/STYLE_GUIDE.md`, `engineering/SYSTEM/architecture/README.md`, `engineering/SYSTEM/interfaces/interconnection_table.md`, `docs/04_RECOVERY_TASKS.md`, `docs/TESTING/ACCEPTANCE_CRITERIA.md`, `docs/TESTING/VALIDATION_PLAN.md`, `docs/03_MISSING_COMPONENTS.md`, `docs/STM32/STM32_PROJECT_RECONSTRUCTION.md`.

Planned figures: none.

1. Acronyms and terms

Term	Meaning in this manual	Where used (example)
ADC	analogue-to-digital converter; here the AD9484, 8-bit, 500 MSPS rated, operated at 400 MSPS	<code>00_notation/parameter_table.md</code> "RF Front-End"; chapter 2 §4
ADAR1000	four-channel X/Ku-band beamformer IC with 7-bit phase and gain control; four devices give 16 channels	<code>02_hardware/04_antenna_beamforming.md</code> §2
ADF4382	wideband PLL/VCO synthesizer used as TX LO (U1, 10.5 GHz) and RX LO (U6, 10.38 GHz)	chapter 3 §2.6
AD9523	low-jitter clock generator (IC1 on the Synth Board) producing the 400/120/100/20 MHz clocks and the 300 MHz ADF4382 reference	chapter 2 §4
AD9708	8-bit DAC (U3) generating the chirp at 120 MHz	chapter 2 §4
ADTR1107	integrated X-band T/R front end (PA + LNA + switch), 16 devices, Nexus variant output stage	<code>README.md</code> "Main Board"
AF	array factor, $AF(\theta)$	<code>01_physics/03_beamforming_theory.md</code> Eq. BF-3
AGC	automatic gain control	<code>README.md</code> "FPGA" list (design intent only)
BETA	status label: builds/simulates/tests on a workstation, not on hardware	<code>manual/STYLE_GUIDE.md</code>
BOM	bill of materials	<code>docs/BOM/</code> , <code>beta/pcb/*/BOM_*.beta.csv</code>
BPF	band-pass filter; U\$2/U\$3 "BPF2" on the Main Board, part number NOT IDENTIFIED	chapter 2 §4
BRAM	FPGA block RAM	<code>beta/fpga/README.md</code> resource estimate
BUFG / BUFIO / BUFR	Xilinx global / I/O / regional clock buffers	<code>beta/fpga/README.md</code> "ADC capture"
CA-CFAR	cell-averaging constant false alarm rate detector	<code>01_physics/04_detection_theory.md</code> §6
CDC (1)	USB Communications Device Class (virtual serial port) — the STM32 host link on X53	chapter 3 §2.8
CDC (2)	clock-domain crossing (FPGA)	<code>beta/fpga/README.md</code> item 4
CFAR	constant false alarm rate detection; a fixed-threshold PLACEHOLDER in the RTL	chapter 2 §2.3
CIC	cascaded integrator–comb decimation filter (5 stages, decimate by 4)	<code>00_notation/parameter_table.md</code> "Signal Processing"
CPI	coherent processing interval (M chirps per beam position)	<code>00_notation/symbol_table.md</code> §1
DAC	digital-to-analogue converter	chapter 2 §4
DAC5578	8-channel I2C DAC driving the PA gate voltages VG_1..16	chapter 3 §4.1 row VG_1..16
DDC	digital down-converter (NCO + mixer + CIC + FIR)	chapter 2 §5
DIG_0..7	STM32 PD8..PD15 handshake lines to the FPGA (new chirp/elevation/azimuth, mixers enable, reset; DIG_5..7 used by the option B bridge)	<code>docs/SYSTEM/BLOCK_DIAGRAM.md</code> §3
DRC / ERC	design-rule check (PCB) / electrical-rule check (schematic)	<code>docs/TESTING/ACCEPTANCE_CRITERIA.md</code> AC-B1
DRU	EAGLE design-rules file	<code>engineering/MISSING_DRAWINGS_RECOVERY_PLAN.md</code> MDR-07
DSP48E1	Xilinx 7-series DSP slice	<code>beta/fpga/README.md</code> resource estimate
DXF / STEP / STL	2-D drawing exchange / 3-D CAD exchange / mesh formats	<code>engineering/MECHANICAL/</code>
EAGLE	Autodesk EAGLE, the native schematic/board CAD of the upstream project (<code>.sch</code> , <code>.brd</code>)	<code>4_Schematics and Boards Layout/</code>
eFuse	electronic fuse / hot-swap controller (LM5069 in DSN-PSU-01)	<code>engineering/DESIGN/00_DESIGN_BASIS.md</code> D-14
ENOB / SQNR / SINAD	effective number of bits / signal-to-quantisation-noise ratio / signal-to-noise-and-distortion	<code>01_physics/05_noise_analysis.md</code> §5
FFT / IFFT	fast Fourier transform / inverse; 1024-point range, 32-point Doppler	<code>00_notation/parameter_table.md</code> "Signal Processing"
FIR	finite impulse response filter	chapter 2 §5

Term	Meaning in this manual	Where used (example)
FMCW	frequency-modulated continuous wave; the physics notes use the FMCW formulation for the dechirp relations	<code>01_physics/01_fmcw_theory.md</code>
FPGA	field-programmable gate array; U42, Xilinx Artix-7 (XC7A50T per CAD, XC7A100T per README — K1)	chapter 1 §4
FreeCAD	open-source MCAD used for the proposed head/pedestal model	<code>engineering/DESIGN/MECHANICAL/</code>
FT601	FTDI USB 3.0 FIFO bridge (U6), placed but unwired on the Main Board	chapter 2 §6
GaN	gallium nitride (QPA2962 PA technology)	README.md
Gerber	PCB photoplot data format	<code>engineering/PCB/<BOARD>/</code>
GT3	3 mm pitch timing belt profile (pedestal drive proposal)	<code>engineering/DESIGN/00_DESIGN_BASIS.md</code> D-12
GUI	graphical user interface (Python, Tk)	<code>beta/gui/</code>
HAL	STM32Cube hardware abstraction layer	<code>docs/STM32/STM32_PROJECT_RECONSTRUCTION.md</code>
HSE	STM32 high-speed external oscillator (8 MHz crystal vs 25 MHz firmware — K2)	chapter 17 §1
I/Q	in-phase / quadrature components of a complex baseband signal	chapter 2 §5
I2C / SPI / UART / USART	serial buses used by the STM32 to peripherals	chapter 3 §2.5, §2.8
IDELAYE2 / ISERDESE2 / IDELAYCTRL	Xilinx input delay element / input serialiser-deserialiser / delay-control block used in the BETA ADC capture	<code>beta/fpga/README.md</code> "ADC capture"
IF	intermediate frequency, 120 MHz	<code>main.cpp:190</code>
IMU	inertial measurement unit (GY-85 module on I2C3)	chapter 3 §2.8
INA241A3	current-sense amplifier, 16 devices reading the PA drain shunts	chapter 3 §2.7
IP core	vendor-supplied FPGA block (Xilinx xfft FFT cores, not generated)	<code>beta/fpga/ip/README.md</code>
IP54	ingress-protection target of the proposed enclosure	<code>engineering/DESIGN/00_DESIGN_BASIS.md</code> D-08
JTAG / SWD	FPGA test-access port (JP3) / ARM serial-wire debug (JP2)	chapter 3 §2.8
KiCad	open-source PCB CAD used for the converted manufacturing packages and the proposed antenna	<code>engineering/PCB/</code> , <code>beta/pcb/</code>
LFM / PLFM	linear frequency modulation / pulsed LFM	<code>01_physics/02_lfm_waveform_model.md</code>
LNA	low-noise amplifier	<code>00_notation/parameter_table.md</code> "RF Front-End"
LO	local oscillator (TX LO 10.5 GHz, RX LO 10.38 GHz)	chapter 2 §4
LTC5552 / LT5552	wideband mixer (U5 up-converter, U13 down-converter); the README writes LT5552, the schematic LTC5552	chapter 2 §4
LUT	look-up table (chirp LUT in the FPGA; also FPGA logic resource)	<code>beta/fpga/README.md</code>
LVDS / LVDS_25	low-voltage differential signalling; the ADC data standard and the FPGA I/O standard	chapter 3 §3
M3SWA2-34DR+	Mini-Circuits SPDT RF switch (17 devices)	chapter 2 §4
MCU	microcontroller unit (STM32F746ZGT7, U2)	chapter 1 §3
MMCM	Xilinx mixed-mode clock manager	<code>beta/fpga/README.md</code>
MPN	manufacturer part number	<code>docs/TESTING/ACCEPTANCE_CRITERIA.md</code> AC-B5
MTI	moving-target indication (README intent; not implemented)	README.md
NCO	numerically controlled oscillator	chapter 2 §5
NEMA 23 / NEMA 34	stepper motor frame sizes (57 mm / 86 mm class)	<code>engineering/DESIGN/CALCS/DESIGN_CALCULATIONS.md</code> §2
NF	noise figure (dB)	<code>00_notation/symbol_table.md</code> §4
OCCO / VCXO	oven-controlled / voltage-controlled crystal oscillator (X4; X5, X6 on the Synth Board — K7)	chapter 3 §4.1 row +3V3_X0
openEMS	open-source FDTD electromagnetic solver used for the antenna simulation	<code>engineering/DESIGN/ANTENNA/simulation/</code>
P&P	pick-and-place (component placement) file	<code>engineering/PCB/<BOARD>/assembly/</code>
PA	power amplifier (QPA2962 GaN 10 W on the RF PA board)	chapter 1 §3
PAE / PSAT / IDQ	power-added efficiency / saturated output power / quiescent drain current of the PA	<code>engineering/DESIGN/00_DESIGN_BASIS.md</code> §1 row "PA"
PLL	phase-locked loop	chapter 2 §4
PRF / PRI	pulse repetition frequency / interval	<code>00_notation/symbol_table.md</code> §1
PSL	peak sidelobe level	<code>01_physics/02_lfm_waveform_model.md</code> Eq. LFM-21
QPA2962	Qorvo GaN power amplifier, 10 W class, 22 V drain	<code>engineering/DESIGN/00_DESIGN_BASIS.md</code> §1
RCS	radar cross section σ	<code>00_notation/symbol_table.md</code> §4
RF	radio frequency	—
RTL	register-transfer level (Verilog) description of the FPGA design	<code>beta/fpga/rtl/</code>
SDR (1)	single data rate (AD9484 LVDS output timing)	<code>beta/fpga/README.md</code> item 2
SDR (2)	software-defined radio (README audience)	README.md
SMA / 2.92 mm	coaxial connector types (Main Board RF ports; proposed antenna connectors)	chapter 3 §2.7; D-05
SNR	signal-to-noise ratio	<code>00_notation/symbol_table.md</code> §4
STM32CubeMX / <code>.ioc</code>	ST configuration tool and its project file (absent from the repository)	<code>docs/STM32/STM32_PROJECT_RECONSTRUCTION.md</code>
T/R	transmit/receive	<code>02_hardware/01_system_overview.md</code> §1.1
TBP	time-bandwidth product B·T·c	<code>01_physics/02_lfm_waveform_model.md</code> Eq. LFM-5

Term	Meaning in this manual	Where used (example)
TBD	to be determined (value absent from every repository file)	00_notation/parameter_table.md "TBD Tracking"
TMP37	analogue temperature sensor (8 devices via ADS7830)	chapter 3 §2.8
TPS562208 / ADM7151 / TPS7A8300 / LM2662	buck regulator / LDO / LDO / charge-pump inverter families on the Power Board	chapter 3 §4.1
ULA	uniform linear array	01_physics/03_beamforming_theory.md §1
VD / VG / VIN_M	PA drain voltage / gate voltage / drain-current sense return	chapter 3 §2.7
Vivado	AMD/Xilinx FPGA toolchain (not available on the authoring machine)	beta/fpga/README.md
VNA	vector network analyser (antenna coupon measurement, NOT RUN)	docs/TESTING/ACCEPTANCE_CRITERIA.md AC-E7
XDC	Xilinx design constraints file	beta/fpga/constraints/

2. Mathematical symbols

Transcribed to plain text from 00_notation/symbol_table.md (notation authority IEEE 686-2024 per that file); subscripts are written after an underscore, e.g. `T_c,1` for the long chirp duration.

Symbol	Definition	Units
f_c	centre (carrier) frequency	Hz
B	chirp bandwidth (sweep range)	Hz
T_c	chirp duration (pulse width); T_c,1 long, T_c,2 short	s
μ	chirp rate, μ = B / T_c	Hz/s
f_b	beat frequency (IF after dechirp)	Hz
f_r	pulse repetition frequency; f_r,1 long mode, f_r,2 short mode	Hz
T_r	pulse repetition interval, T_r = 1/f_r; T_r,1, T_r,2	s
τ	round-trip delay, τ = 2R/c	s
T_guard	guard time between chirp sequences	s
M	number of chirps per CPI (per beam position)	—
R, R_max, ΔR	range, maximum unambiguous range, range resolution	m
v, Δv	target radial velocity, velocity resolution	m/s
f_d	Doppler frequency shift	Hz
c	speed of light, ≈ 2.998 × 10 ⁸	m/s
λ	wavelength, λ = c / f_c	m
N	number of array elements	—
N_el, N_az	beam elevation positions; azimuth positions per revolution	—
d	inter-element spacing	m
θ, θ_0	beam angle from broadside; desired steering angle	rad or deg
Δφ, Δφ_n	phase shift per element; phase difference for elevation position n	rad or deg
G, G_t, G_r	antenna gain (combined; transmit; receive)	dBi
k	wavenumber, k = 2π/λ	rad/m
ψ	electrical angle, ψ = k d sinθ + Δφ	rad
θ_3dB	half-power beamwidth	rad or deg
w_n, a_n	amplitude weight / nominal amplitude weight of element n	—
δφ_n, δa_n	phase / amplitude error of element n	rad / —
AF(θ)	array factor	—
P_t, P_r	transmit power (per element), received power	W
σ	radar cross section	m ²
L	total system losses (linear ratio)	—
F, NF	noise figure (linear; dB)	— / dB
T_0	reference noise temperature, 290 K	K
k_B	Boltzmann constant, 1.381 × 10 ⁻²³	J/K
P_fa, P_d	probability of false alarm, probability of detection	—
SNR, SNR_min	signal-to-noise ratio, minimum detectable SNR	dB
α	CFAR threshold multiplier	—
N_ref, N_guard	CFAR reference cells, guard cells (total)	—
T_e, B_n	equivalent noise temperature, noise bandwidth	K, Hz
f_s	ADC sampling frequency	Hz
f_IF	intermediate frequency	Hz
N_FFT, N_Doppler, N_R	FFT size (range), Doppler FFT size, number of range bins	—
N_CIC, D_CIC	CIC filter stages, CIC decimation factor	—
w[n]	window function (discrete)	—
χ(τ, v)	ambiguity function	—

Symbol	Definition	Units
$\Delta\phi_{\text{NCO}}$	NCO phase accumulator increment per clock cycle	—
G_{CIC}	CIC filter DC gain, $G_{\text{CIC}} = D_{\text{CIC}} \wedge N_{\text{CIC}}$	—
$N_{\text{seg}}, L_{\text{adv}}, L_{\text{overlap}}$	overlap-save segments, segment advance, overlap length (matched filter)	— / samples
$N_{\text{rb}}, D_{\text{rb}}$	output range bins after decimation, range-bin decimation factor	—
$V_{\text{rail}}, I_{\text{rail}}$	voltage rail value, current draw per rail	V, A
$P_{\text{diss}}, T_{\text{junction}}, \theta_{\text{JA}}$	power dissipation, junction temperature, thermal resistance junction-to-ambient	W, °C, °C/W
t_{lock}	PLL lock time	s
$L(f_{\text{m}})$	phase noise at offset f_{m} from carrier	dBc/Hz
t_{pipeline}	end-to-end pipeline latency	s
$N_{\text{LUT}}, N_{\text{FF}}, N_{\text{BRAM}}, N_{\text{DSP}}$	FPGA look-up tables, flip-flops, block RAMs, DSP48E1 slices	—

Equation tags (FMCW-n, LFM-n, BF-n, DET-n, NF-n, CAL-n, HW-ANT-n) refer to the display equations of the upstream physics and hardware notes under the document-prefix scheme of `00_notation/conventions.md §1`.

3. Status vocabulary

Labels on figures and procedures (source: `manual/STYLE_GUIDE.md`): ORIGINAL PROJECT FILE · SOURCE-DERIVED · PARTIAL · CONCEPTUAL · PROPOSED DESIGN · BETA · BLOCKED — MISSING DATA · VERIFIED (reserved; unused). Qualifiers on numbers: ASSUMED, ESTIMATE, TBD, UNKNOWN, UNRESOLVED, REQUIRES VERIFICATION.

Register-specific vocabularies kept as in their sources:

Register	Values	Source
Interconnection table, power-rail register	CONFIRMED / UNVERIFIED / MISSING SPEC; PARTIAL; CONFLICT	<code>engineering/SYSTEM/interfaces/interconnection_table.md</code> ; <code>engineering/ELECTRICAL/power_distribution/power_rails.md</code>
Drawing register	SOURCE-DERIVED / PARTIAL / CONCEPTUAL / PROPOSED / BLOCKED	<code>engineering/DRAWING_REGISTER.md</code>
Recovery tasks	DONE / OPEN / BLOCKED (designer input) / PARTIALLY DONE	<code>docs/04_RECOVERY_TASKS.md</code>
Acceptance criteria	MET / NOT MET / PARTIALLY MET / NOT RUN (physical tests)	<code>docs/TESTING/ACCEPTANCE_CRITERIA.md</code>
Missing components	priorities P0 (blocks a reproducible build or required artefact) / P1 (blocks a complete release or reliable validation) / P2 (documentation, maintainability)	<code>docs/03_MISSING_COMPONENTS.md</code>
Firmware defects	C1–C7 conflict rows of STM-T04	<code>docs/STM32/STM32_PROJECT_RECONSTRUCTION.md</code>
Interface confidence	MEDIUM (bit mapping inferred from source comments)	<code>docs/SYSTEM/BLOCK_DIAGRAM.md §3</code>

4. Identifier families

Family	Range	Meaning	Register
K	K1–K8	configuration conflicts between CAD, firmware, RTL, GUI and documentation	<code>docs/SYSTEM/BLOCK_DIAGRAM.md §4</code> ; chapter 17 §1
D	D-01...D-15; D-16...D-19	proposed-design decisions (mechanical/antenna/thermal/harness; host link)	<code>engineering/DESIGN/00_DESIGN_BASIS.md §2</code> ; <code>engineering/DESIGN/HOST_LINK/HOST_LINK_DESIGN.md §3–4</code> ; chapter 17 §2
G	G-01...G-12	unresolved mechanical geometry	<code>engineering/VALIDATION/UNRESOLVED_GEOMETRY.md</code> ; chapter 17 §3
MDR	MDR-01...MDR-13	missing-drawing recovery guides	<code>engineering/MISSING_DRAWINGS_RECOVERY_PLAN.md</code> ; chapter 17 §4
DSN	DSN-00, DSN-ANT-01, DSN-THM-01, DSN-PSU-01, DSN-MECH-01...07, DSN-MECH-3D, DSN-HAR-01, DSN-LINK-01, DSN-CALC-01	proposed-design documents and drawings	<code>engineering/DRAWING_REGISTER.md</code>
SYS / SD / ELEC-PWR	SYS-01...04; SD-01...07; ELEC-PWR-01	system-level and software diagrams (SOURCE-DERIVED / PARTIAL)	<code>engineering/SYSTEM/architecture/README.md</code> ; <code>engineering/DRAWING_REGISTER.md</code>
PCB-, MECH-, ASM-EXP-01, VAL-GEO-01, ENG-MDR-01	per board / per drawing	registered PCB drawings, mechanical drawings, exploded view, validation and recovery documents	<code>engineering/DRAWING_REGISTER.md</code>
CBL	CBL-00...CBL-106 (two-digit, per signal group) and CBL-001...CBL-144 (three-digit, per physical cable)	cable identifiers — two schemes, see MAN-03	<code>engineering/SYSTEM/interfaces/interconnection_table.md</code> ; <code>engineering/DESIGN/HARNESS/HARNESS_SCHEDULE.md</code>
AC	AC-F1...F9, AC-S1...S7, AC-P1...P7, AC-B1...B8, AC-M1...M5, AC-D1...D5, AC-E1...E8, AC-X1...	acceptance criteria per area (FPGA, STM32, Python, PCB, mechanical, documentation, engineering package, BETA tree)	<code>docs/TESTING/ACCEPTANCE_CRITERIA.md</code>

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Family	Range	Meaning	Register
F / S / P / B / R	F-01...F-10, S-01...S-10, P-01...P-09, B-01...B-09, R-01...R-03	validation-plan checks (FPGA, STM32, Python, PCB, repository) referenced by the acceptance criteria	docs/TESTING/VALIDATION_PLAN.md
R	R-SYS-01, R-FPGA-01...10, R-STM-01...08, R-GUI-01...03, R-PCB-01...06, R-MECH-01...02, R-DOC-01...03, R-ENG-01...13, R-DSN-01...07, R-BETA-01...09	recovery tasks	docs/04_RECOVERY_TASKS.md ; chapter 17 §5
FPGA- / STM- / GUI- / PCB- / MECH- / REPO-	FPGA-00...15, STM-01...19, GUI-01...08, MECH-01...06, REPO-01...05	missing-component manifest IDs checked by <code>tools/check_missing_files.py</code>	docs/03_MISSING_COMPONENTS.md
C	C1-C7	STM32 firmware conflicts/defects (HSE, ADF4382 pins, platform ops, USB RX, start-flag padding, GPS_Init, AD9523 CS/SPI speed)	docs/STM32/STM32_PROJECT_RECONSTRUCTION.md STM-T04
F0-F10	—	firmware power-enable sequence steps as coded	engineering/ELECTRICAL/power_distribution/power_rails.md §3; chapter 3 §4.2
MAN	MAN-01...04	observations raised by this manual while compiling the sources	chapter 17 §6
Figure numbers	F<chapter>.<n>	figure slots of <code>manual/FIGURE_PLAN.md</code> ; the build tool prefixes a running "Figure n —"	<code>manual/FIGURE_PLAN.md</code>